Quant Trading and Quantitative Development

From First Principles

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Part I

Preliminaries

Chapter 1

How to Read This Book

This book defines every finance term on first use, derives every result that teaches something, and stops short of derivations that do not. It is written for a strong mathematics and computer science student with *zero* prior exposure to finance. Existing resources either assume the reader already knows what a bid–ask spread is, or dilute the mathematics to the point where nothing can be derived; this book sits in the gap.

Two concrete goals share a deadline in mid-April 2026. **IMC Prosperity 4** is a two-week algorithmic trading competition starting 2026-04-14 in which teams write Python algorithms to trade simulated products against each other and scripted bots. Past editions rewarded market making, basket statistical arbitrage, options pricing, locational arbitrage, and exploitation of bot behaviour — all taught here from first principles. The **Goldman Sachs Quant Strats off-cycle placement** begins around 2026-04-20; the reader will sit on a trading desk as a strategist ("Strats"). Since the desk assignment is not yet known, Part VII covers every major asset class — rates, equities, FX, credit, commodities — at comparable depth, together with production risk, numerical methods, and firm-specific infrastructure (SecDB, Slang).

Parts I–V form the common core; Part VI is Prosperity-specific and Part VII Goldman-specific. A reader pursuing only one goal can skip the other Part entirely. The reading paths in Section 1.2 make this concrete.

1.1 Who this book is for

The reader is assumed to have the following background.

Mathematics. Undergraduate probability (random variables, expectation, conditional expectation, the central limit theorem, basic hypothesis testing), linear algebra (vector spaces, eigenvectors, SVD at the mechanical level), multivariable calculus, and some exposure to stochastic processes at the level of Brownian motion, Itô's lemma, and simple stochastic differential equations. Measure theory is *not* assumed; where a measure-theoretic statement would be cleaner, the book states it informally and cites the rigorous version. Basic optimisation (Lagrange multipliers, convex objectives) is used without reintroduction.

Computer science. Python fluency, including `numpy`, `pandas`, and standard data structures; time complexity; basic algorithms (sorting, hashing, dynamic programming); and the ability to read a moderately-sized Python codebase unaided. Code in this book is sparing — at most one reference skeleton per strategy, typically 15–30 lines, pedagogical rather than production-ready. Full runnable code lives outside the PDF.

Finance. Zero. The reader is not assumed to know what a stock is, what "going long" means, what an exchange does, what an option is, or what the terms "bid", "ask", "mid", "spread", "tick", "lot", "maker", "taker", "slippage", "delta", "vega", or "hedge" mean. Each is defined on first use; no finance jargon appears without a prior in-text definition. Appendix B collects the master notation table and a glossary of trading terms as a reference backstop.

Readers whose probability, linear algebra, or stochastic-calculus background is rusty should consult Appendix A before Part II (microstructure) and again before Part IV (options). Appendix A is a refresher, not a course: it restates the results used and points to standard sources for proofs. If the refresher is insufficient, spend a week with a probability textbook before continuing.

1.2 Three reading paths

The book has 39 numbered chapters across seven Parts. The three paths below are each *complete and self-contained* for their target use case: a chapter not on a path is safe to skip for that goal; a chapter on a path is load-bearing.

Path A: Prosperity speed-run

$$\text{Ch } 1 \to 2 \to 3 \to 8 \to 9 \to 11 \to 14 \to 15 \to 16$$
$$\to 23 \to 24 \to 25 \to 26.$$

For a reader whose sole immediate goal is Prosperity 4. It covers exchanges, participants, and price (Ch 1–3), then the minimum strategy vocabulary the competition historically rewards: market making (Ch 8), mean reversion (Ch 9), and basket statistical arbitrage (Ch 11). Enough options theory to price and delta-hedge the voucher products follows (Ch 14–16), then the full Prosperity playbook (Ch 23–26) — tooling, algorithmic round strategies, manual rounds, and lessons from top-team writeups. Omitted: Parts V and VII, most of the arbitrage zoo beyond baskets, and the full vol surface (Ch 17). Roughly 130 pages; intended for a reader starting a week before Prosperity 4 begins.

Path B: Goldman Sachs Strats prep

$$\text{Ch } 1 \to 2 \to 3 \to 14 \to 15 \to 16 \to 17 \to 27 \to 28 \to 29$$
$$\to 30 \to 31 \to 32 \to 33 \to 34 \to 35 \to 36 \to 37 \to 38.$$

For a reader showing up to a sell-side quant strats desk ready to contribute. It begins with the minimum market-structure vocabulary (Ch 1–3), then the full options stack (Ch 14–17) since every Part VII chapter cross-references it, then every Part VII chapter in order: the role itself (Ch 27), fixed income foundations — the largest knowledge gap for a CS reader (Ch 28), rates derivatives and SABR (Ch 29), equity derivatives and variance swaps (Ch 30), FX derivatives (Ch 31), credit derivatives (Ch 32), commodities (Ch 33), numerical methods (Ch 34), production risk and P&L Explain (Ch 35), SecDB and Slang (Ch 36), regulatory context (Ch 37), and interview-grade problems (Ch 38). Omitted: Parts III, V, and VI, which teach buy-side strategy construction — not the Strats job. Roughly 500 pages; intended for a reader with three to five weeks until the placement starts.

Path C: Full cover-to-cover

All 39 chapters in order, plus Appendices A–C. Intended for a reader building a permanent foundation rather than preparing for a deadline. Estimated effort: around 300 hours of focused reading, working every derivation and worked example by hand. Hard-flagged chapters (see Section 1.6) should be budgeted at roughly double the time of an average chapter.

1.3 Chapter dependency map

Any Part can be read after Parts I and II, but not before them. Figure 1.1 shows the coarse dependency structure; the prose below explains it.

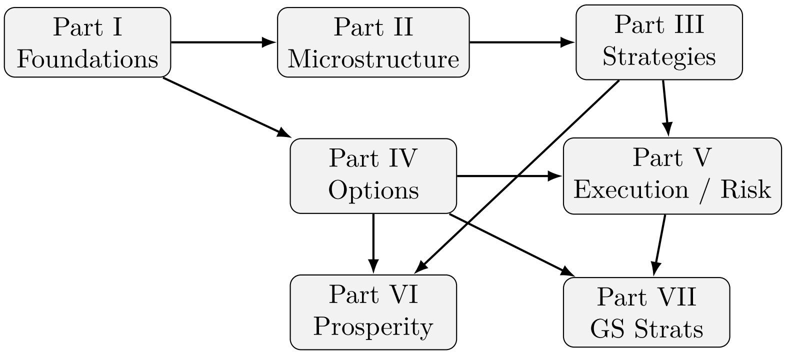


Figure 1.1: Coarse dependency graph between Parts. An arrow $X \to Y$ means Y assumes the content of X. Parts I and II are prerequisites for everything; Part IV (options) is a prerequisite for both Part VI and Part VII; Part V cross-references Parts III and IV.

At chapter granularity: Part I (Ch 1–3) introduces exchanges, the limit order book, market participants, and the many definitions of "the price"; every later chapter assumes this vocabulary. Part II (Ch 4–7) builds the theory of *why* spreads exist, underpinning every Part III strategy — Ch 8 (market making) in particular makes no sense without the adverse-selection argument of Ch 5. Inside Part IV, Ch 14 (payoffs, no-arbitrage bounds, put–call parity) precedes Ch 15 (Black–Scholes), which precedes Ch 16 (Greeks and delta hedging) and Ch 17 (the volatility surface and gamma scalping). Part VII assumes the Part IV options stack in full, then layers production, sell-side, and asset-class-specific content on top: Ch 29 (rates derivatives) cross-references Ch 15–17, Ch 30 extends Ch 17 with variance swaps, and Ch 34 (numerical methods) is where the pricing problems of Ch 29–32 get solved in practice. Part V (execution and risk) sits between the strategy and options material and the two target-specific Parts: it covers how trades are actually placed without moving the market against oneself, and how positions are sized and monitored so a losing day does not become a career-ending one. Part VI (Prosperity) is an applied layer over Parts I–IV rather than a prerequisite for anything — Part VI is the most self-contained Part in the book.

1.4 Notation

Every symbol used across multiple chapters is defined once and catalogued in Appendix B, alongside a glossary of trading terms. Appendix B is a reference — consult it whenever a symbol introduced many chapters earlier needs refreshing. No chapter redefines a symbol that appears in the master table, so a symbol's meaning is fixed the first time it is used.

A few symbols are introduced locally within a single chapter (for example, the half-life in the Ornstein–Uhlenbeck chapter, or the hazard rate in the credit chapter); these do not recur globally and are omitted from the master table.

1.5 Callout boxes

Seven coloured callout boxes recur throughout the book, introduced here so that a reader meeting one in Chapter 4 already knows what it signals. This chapter uses no boxes — they are reserved for technical content.

- **Background** (gray). Prerequisite reminders at the start of most chapters, listing the earlier-chapter results assumed. Skim if confident, or follow the back-references if not.
- **Key fact** (blue). A formal result worth memorising — the bid–ask spread formula, Kyle's λ, Avellaneda–Stoikov's reservation-price formula. Every major result appears in a key-fact box near where it is derived.
- **Intuition** (green). A plain-English restatement of what a formal result *means*, placed immediately after the derivation. When the reader has followed the algebra but is unsure what was shown, the intuition box is the answer. *Slow down and read these carefully.*
- **Prosperity example** (orange). A worked example using a specific Prosperity product from a prior edition. Appears inline in Parts I–V wherever theory maps cleanly onto a Prosperity product. The full playbook is Part VI.
- **On a Goldman Sachs desk** (teal). How the technique just taught is used on a sell-side trading desk: what the Strat writes, which system owns the calculation, and which desk cares most. Appears mostly in Parts IV and V; Part VII treats the same material in depth.
- **Common mistake / Pitfall** (red). A wrong intuition that sounds right, or a failure mode the reader will otherwise walk into. *Slow down and read these carefully.* These boxes are the highest value-per-word in the book — they encode mistakes that cost professionals money.
- **Historical note** (purple). An aside on the origin of a result, a biographical note, or a short story about how a model came to dominate practice. Skip on a first read without losing anything technical.

1.6 How to read a hard chapter

Five chapters are flagged as *hard*: Ch 5 (Glosten–Milgrom and Kyle), Ch 8 (Avellaneda–Stoikov market making), Ch 15 (Black–Scholes from first principles), Ch 22 (the "which strategy when" decision framework), and Ch 29 (SABR and the rates vol surface). Hard means the chapter contains a derivation or synthesis which, on first reading, feels dense regardless of the reader's mathematical background. These deserve extra time and, usually, a second pass.

Every hard chapter follows the same four-stage pattern, mirroring the worked-example style of the NLP lecture-notes reference this book takes its form from.

1. **Toy example first.** The chapter opens with a concrete numerical example worked by hand with small integers, before any symbol is defined abstractly — so the reader experiences the phenomenon before formalising it. Ch 5, for instance, opens with a Glosten–Milgrom game on two states $\{V = 100, V = 110\}$ with known probabilities; the reader computes the bid and ask by hand before seeing the general formula.
2. **Formalise.** The same problem is restated in general notation. No new results are proved here; the goal is to translate the toy example into the language the derivation will use.
3. **Derive.** The main result is derived line by line. Routine steps are flagged and skipped; illuminating ones are written out in full with underbraces labelling each term. Derivations follow rigour level B: rigorous where the algebra teaches something, stated-with-citation where it does not.
4. **Intuition box.** The chapter closes with an intuition box restating the result in one or two sentences of plain English, often with a picture. If the reader got lost in the derivation, the intuition box is the minimum they need to carry forward.

Recommended reading protocol: work stage 1 carefully with a pen, checking every number in the toy example is reproducible from the rules. Read stages 2 and 3 once at normal speed, then re-read while writing down each line of algebra on paper. Finally, read the intuition box and ask whether it would have been obvious from the toy example alone — if yes, the chapter has done its job. If no, re-read stages 3 and 4 together and look for what the intuition box points at that the derivation glossed over.

Budget roughly double the time for a hard chapter versus an average chapter in the same Part. The five hard chapters contain most of the load-bearing theory; time spent on them compounds because every later chapter assumes the result without rederiving it.

Part II

Foundations of Markets

Chapter 2

Exchanges, Order Types, and the Limit Order Book

A market is a crowd of buyers and sellers, each with a different view of what something is worth, trying to find each other without knowing who is on the other side, without wanting to wait, and without trusting each other to hold an order until it matches. *How do you engineer a fair, fast, rule-based mechanism that pairs them off without missing any willing trade?* The answer dominates this chapter and every chapter that follows: the **limit order book** (LOB), a simple data structure that every professional trading venue converges on, and the **matching engine** that operates on it. This chapter defines the vocabulary — bid, ask, spread, order types, maker/taker, tick size — that every later chapter assumes. Nothing here is mathematically deep, but every bold term is load-bearing.

Prerequisites

This is the first content chapter; no prior chapters are required. From outside the book the reader should bring:

- **Programming.** Ability to read a Python snippet using a sorted data structure and plain-old objects. No library-specific knowledge required.
- **Real world.** Passing awareness that stocks trade on exchanges. Zero finance background is assumed; every technical term below is defined on first use.

2.1 Double-sided auctions

An **exchange** is a venue — nowadays almost always software in a colocation data centre — whose only job is to accept messages called *orders*, match buyers with sellers according to a fixed published rule, and report the resulting trades. A **product** is whatever is being traded; think of a stock, but it could equally be a bond, a futures contract, a Prosperity `RAINFOREST_RESIN`, or a kilo of coffee. The exchange runs one such auction per product, in parallel, continuously. Because a product has both buyers and sellers at any instant, the mechanism is a *double-sided auction* — as opposed to a one-sided auction like eBay, where bids are collected against a single seller.

Historical note

Before the late 1990s a "double-sided auction" meant a physical *pit*: human traders shouting bids and asks in hand signals. The NYSE ran its floor this way from 1792; the CBOT and LIFFE kept their pits through the 1990s. Electronic matching arrived gradually — NASDAQ's SelectNet (1988), the Swiss Exchange fully electronic (1996), the LSE abandoning its floor (1997), Eurex absorbing the Deutsche Terminbörse (1998) — and by about 2010 equity and futures volume was essentially all electronic. The mental model of this chapter, orders as messages and the book as a data structure, only became universally applicable in the last two decades; before that, a crowd of humans approximated the same data structure by shouting. Harris (Harris, 2003) gives the long version.

The two sides of the auction get names:

- A **bid** is a willingness to buy at a specified price. Synonyms: "buy order", "bid order". The highest bid price currently resting in the market is the **best bid**, written p^{bid}.
- An **ask** (also **offer**) is a willingness to sell at a specified price. Synonyms: "sell order", "ask order". The lowest ask price currently resting in the market is the **best ask**, written p^{ask}.

The gap between them is the **spread**, $s = p^{\text{ask}} - p^{\text{bid}}$. The arithmetic average of the two is the **mid** (sometimes *mid-price*), $p^{\text{mid}} = (p^{\text{bid}} + p^{\text{ask}})/2$. A trade happens when someone sends a new order that *crosses* the book: a buy at or above p^{ask}, or a sell at or below p^{bid}. Until then, no trade; the orders simply sit and wait.

Why is $p^{\text{bid}} < p^{\text{ask}}$ at all? If someone will sell at 103.23 and someone else will buy at 102.54, isn't a trade missing? No: it means *no one* was willing to buy at 103.23 or higher, and *no one* was willing to sell at 102.54 or lower. The spread is the signature of that fact. Chapter 5 shows that a strictly positive spread is forced on any real market by inventory risk, order-processing cost, and adverse selection; for now, take a non-zero spread as an empirical fact.

Bid/ask/spread/mid

$$\begin{aligned} p^{\text{bid}} &= \text{highest buy price resting in the book}, \\ p^{\text{ask}} &= \text{lowest sell price resting in the book}. \end{aligned}$$

$$s := p^{\text{ask}} - p^{\text{bid}} \geq 0, \qquad p^{\text{mid}} := \tfrac{1}{2}(p^{\text{bid}} + p^{\text{ask}}).$$

These four numbers are the minimum summary of a market's current state. *None* of them is "the price" of the product — that ambiguity is the subject of Chapter 4, which introduces five more candidates.

Here is a toy limit order book with four price levels per side. The *bid side* is sorted with highest price at top (best for a seller); the *ask side* with lowest price at top (best for a buyer).

BID SIDE		**ASK SIDE**	
Qty	Price	Price	Qty
131	102.54	103.23	48
32	101.87	103.98	84
293	101.48	104.17	38
65	101.10	104.75	127

For this book, $p^{\text{bid}} = 102.54$, $p^{\text{ask}} = 103.23$, $s = 0.69$, $p^{\text{mid}} = 102.885$. A new limit buy at 102.00 does *not* cross (below the best ask) and rests on the bid side between 102.54 and 101.87. A new market buy executes immediately against the 48 shares at 103.23, then continues

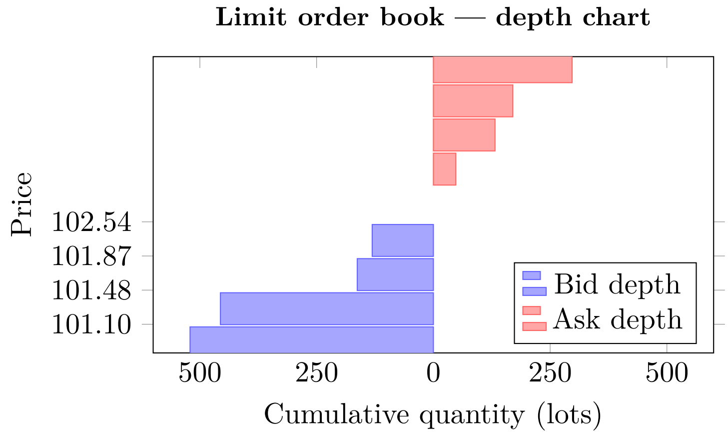


Figure 2.1: Limit order book depth chart for the toy book tabulated below. Bid-side bars (blue) extend leftward; ask-side bars (red) extend rightward. The best bid is 102.54 and the best ask is 103.23, giving a spread of 0.69. Each bar shows the cumulative resting quantity at that price level and all levels behind it.

into 103.98 if its size exceeds 48. That "eating into the book" behaviour is the central risk of Section 2.2, worked through below.

2.2 Order types

An **order** is a message to the exchange: "I want to trade, and here is how." Every order carries a *side* (buy or sell), a *quantity*, and a *product*. Beyond that, orders differ by *when* they execute, at *what* price, and for *how long* they remain active. The set below is the core vocabulary; every real exchange supports all of them, and Prosperity supports the first two.

Definition 2.1 (Limit order)**.** A *limit order* is an order to buy up to q units at a price no higher than p, or to sell up to q units at a price no lower than p. If it can be matched immediately against the resting book, it is; whatever quantity cannot be matched rests as a new entry in the book. A limit order with p set such that it crosses the best quote on the other side is called *marketable*.

Limit orders are the fundamental object; every other order type is a limit order with an extra constraint on price or time-in-force.

Definition 2.2 (Market order)**.** A *market order* is an order to buy or sell q units *immediately* at whatever prices are available, sweeping from the best quote outward until q units are filled. Equivalently — the cleanest implementation in the Python section below — a market buy is a limit buy with $p = +\infty$, and a market sell is a limit sell with $p = 0$.

Definition 2.3 (Immediate-or-cancel (IOC))**.** An *IOC* order behaves like a limit order, but any portion that cannot be matched immediately is cancelled rather than resting in the book. Used by takers who do not want to leave a resting quote behind. Typical use: a strategy sweeps the best ask with a capped price to take available size now, without risk of the unfilled remainder sitting in the book and being picked off if the mid moves against it a moment later.

Definition 2.4 (Fill-or-kill (FOK))**.** An *FOK* order executes in full immediately, or not at all. If the full quantity q cannot be filled against the current book, the exchange rejects the order entirely and no partial fill is reported. Used when partial fills are operationally costly — for example, a hedge that is only valid as one unit.

Three further order types complete the core vocabulary. They are less fundamental (you won't need them algorithmically in this book) but every practitioner is expected to know them. A **stop order** is dormant until the last traded price crosses a *trigger* p_s, at which point it becomes a market order (or a limit order, for a *stop-limit*); the classic example is a stop-loss sell — "if the price drops to p_s, sell me out." A **hidden order** is a resting limit order the exchange will match against but does not publish in public book data; the trader loses price–time priority against visible orders at the same level in exchange for invisibility. An **iceberg order** is a large limit order whose total size is hidden but which publishes a small *tip*; each time the tip is consumed, the exchange reveals another slice of the same size. Used by institutions that need to transact size without announcing it.

Orthogonal to the order type, every real exchange attaches a **time-in-force (TIF)** flag saying *how long* a resting order should live if it does not match immediately. The three you will see everywhere:

- **DAY.** Cancels at session end. The default on most equity venues.
- **GTC (Good-till-cancelled).** Rests across sessions until the participant cancels or the order trades. Used by passive strategies that persist through overnight breaks.
- **GTD (Good-till-date).** Cancels at a specified future timestamp. Useful when an order is conditional on an event whose relevance expires.

IOC and FOK are technically TIFs too: "cancel the unfilled remainder now" and "cancel the whole order now if it cannot fill in full". Prosperity orders behave as "immediate only" — they match in the current tick or are cancelled when the next `TradingState` arrives (Section 2.9).

The key distinction to internalise — limit vs market vs IOC/FOK — controls whether an order *provides* liquidity (rests) or *consumes* it (hits something resting). Section 2.7 turns that split into the maker/taker dichotomy that runs through the whole book.

Market orders can slip through a thin book

A market buy sent into a visible best ask of 103.23 does *not* guarantee an execution price of 103.23. Take the toy book from Section 2.1 verbatim:

BID SIDE		ASK SIDE	
Qty	Price	Price	Qty
131	102.54	103.23	48
32	101.87	103.98	84
293	101.48	104.17	38
65	101.10	104.75	127

Send a market buy for 200 shares. The ask side has only $48 + 84 + 38 = 170$ shares at or below 104.17, so the first three levels fill in full and the fourth absorbs the remaining 30 shares at 104.75:

$$\underbrace{48 \times 103.23}_{\text{level 1}} + \underbrace{84 \times 103.98}_{\text{level 2}} + \underbrace{38 \times 104.17}_{\text{level 3}} + \underbrace{30 \times 104.75}_{\text{level 4}} = 20{,}790.32.$$

Average execution price is $20{,}790.32/200 = 103.9516$, about 72 cents ($\sim 70$ basis points) above the best ask of 103.23. The gap between top-of-book quote and realised average fill is **slippage**. In a thin book, slippage from one aggressive market order can dwarf a strategy's edge. The standard defence is a marketable limit with a price cap (e.g. 103.50), accepting a partial fill rather than a bad full one. Chapter 7 treats slippage as a systematic cost and shows it scales empirically as $\sqrt{Q}$ on liquid venues.

2.3 The limit order book as a data structure

Stripped of exchange-specific packaging, a limit order book is two sorted lists of resting limit orders, one per side.

Definition 2.5 (Limit order book)**.** A *limit order book* (LOB) for a product is the pair of multisets

$$B_t = \{(\text{price}_i, \text{qty}_i, \text{time}_i)\}, \qquad A_t = \{(\text{price}_j, \text{qty}_j, \text{time}_j)\},$$

of resting buy (*bid*) and sell (*ask*) limit orders at time t, respectively. B_t is conventionally sorted by price descending then time ascending; A_t is sorted by price ascending then time ascending. We require B_t and A_t to be *uncrossed*: every bid price is strictly below every ask price.

Definition 2.6 (Price–time priority)**.** An exchange uses *price–time priority* if an incoming marketable order is filled first against the resting order with the most attractive price for the incoming side, and if several orders share that price, against the one that arrived first. In other words, the matching engine consumes orders in exactly the order they appear in the sorted LOB above.

Price–time priority is the dominant rule. The main alternative is **pro-rata matching**, where at a given price level an incoming order is split proportionally by size across resting orders: resting orders of 300 and 100 at 102.54 would receive 75% and 25% of any incoming fill. Pro-rata is used on some futures markets (notably CME short-term interest rate futures) because it encourages participants to post *large* resting orders; under price–time priority only the first resting order in queue gets filled, which rewards speed not size.

Two matching rules

Price–time priority. Best price wins; ties broken by arrival time. Rewards speed. Used by NASDAQ, NYSE, LSE, every equity ECN, and by Prosperity's exchange throughout all four years of the competition.

Pro-rata. Best price wins; ties split proportionally by resting size. Rewards posted size, not speed. Used on certain large-lot futures venues (CME Eurodollars, short sterling).

This book assumes price–time priority throughout unless a chapter explicitly says otherwise. Every result in Chapter 9 and Chapter 8 about queue position assumes price–time priority.

Intuition

A limit order book is two sorted lists. That's it. Everything people call "the market" — depth, spread, mid, slippage, impact, adverse selection, market making, queue position, BBO feeds, the entire vocabulary of HFT — is built from manipulations of those two lists. Parts II and III of this book are, at bottom, an attempt to explain what those two lists look like, why they look that way, and how to trade against them.

2.4 The matching engine: a worked example

We play out a sequence of events on a concrete book so the previous section's rules stop being abstract. This is the chapter's central worked example, reproduced with minor edits from Pelgrims's matching-engine blog post (Pelgrims, 2020). Read it with a pen.

Step 1: initial book

The book starts in the same state as the toy example from Section 2.1:

```
              LIMIT ORDER BOOK
         BID SIDE             ASK SIDE
    QUANTITY     PRICE   PRICE    QUANTITY
   [131.00  -  102.54 | 103.23  -   48.00]
   [ 32.00  -  101.87 | 103.98  -   84.00]
   [293.00  -  101.48 | 104.17  -   38.00]
   [ 65.00  -  101.10 | 104.75  -  127.00]
```

$p^{\text{bid}} = 102.54$, $p^{\text{ask}} = 103.23$, $s = 0.69$. Nothing is crossing; the matching engine has no work to do.

Step 2: two limit buys post

A first participant submits a limit buy for 24 shares at 102.55. That is better than the current best bid of 102.54 and does not cross the ask ($102.55 < 103.23$). The engine finds no cross and inserts the order at the top of the bid side.

A second participant submits a limit buy for 14 shares at 102.55 — same price, so it does not improve or cross. Under price–time priority it queues *behind* the 24-share order. The book is now:

```
         BID SIDE             ASK SIDE
    QUANTITY     PRICE   PRICE    QUANTITY
   [ 24.00  -  102.55 | 103.23  -   48.00]
   [ 14.00  -  102.55 | 103.98  -   84.00]
   [131.00  -  102.54 | 104.17  -   38.00]
   [ 32.00  -  101.87 | 104.75  -  127.00]
```

The two 102.55 entries are drawn on separate lines for clarity, but at the exchange they sit in a single FIFO queue at that price: the 24-share order first, the 14-share order second.

Step 3: a limit sell arrives and crosses

A third participant sends a limit sell for 40 shares at 102.55. This price is *at* the best bid of 102.55, so the order is marketable. The matching engine runs its fill loop:

1. **Level 1.** Top of the bid side: 24 at 102.55. The incoming sell wants 40; this order supplies 24. Print trade 24×102.55, remove. Remaining incoming: 16.

2. **Level 1 (continued).** Next in FIFO at 102.55: 14 shares. Print trade 14×102.55, remove. Remaining: 2.

3. **Level 2?** Next bid is 131 at 102.54. The sell's limit is $102.55 > 102.54$, so it will not cross further. Stop.

4. **Post remainder.** The 2 unfilled shares rest on the ask side at 102.55, a new best ask (below the previous 103.23).

Total traded in this event: 38 shares at 102.55 across two trades. The book state after the dust settles:

```
        BID SIDE              ASK SIDE
   QUANTITY     PRICE    PRICE    QUANTITY
  [131.00   -   102.54 | 102.55   -     2.00]
  [ 32.00   -   101.87 | 103.23   -    48.00]
  [293.00   -   101.48 | 103.98   -    84.00]
  [ 65.00   -   101.10 | 104.17   -    38.00]
```

Best bid slips back to 102.54, best ask jumps down to 102.55, spread collapses from 0.69 to 0.01. A snapshot feed with one-second resolution would show only that the mid dropped by about thirty-five cents, missing the four events that produced it.

Intuition

Every order-book event, however exotic, is some combination of three operations you just saw: *insert a new resting order*, *match against resting orders until quantity or price runs out*, and *remove consumed orders*. The matching engine does these three things deterministically, in a fixed priority order, millions of times per second, for every product listed.

2.5 A Python matching engine

The code below is a stripped-down version of the reference matching engine from Pelgrims's blog. It keeps only what's needed to reproduce the worked example: an `Order`, a `Trade`, an `OrderBook` with two sorted lists, and a `match` method implementing the Step 3 fill loop. Not production code (no threading, error handling, cross-trader accounting, or feed publication) — just enough to *run* the example.

```python
from dataclasses import dataclass
from sortedcontainers import SortedList

@dataclass
class Order:                       # +inf price => market buy; 0 => market sell
    side: str; price: float; qty: int

@dataclass
class Trade:
    price: float; qty: int

class OrderBook:
    def __init__(self):
        # Real engines add a sequence counter to break price ties by
        # arrival order; omitted here for brevity.
        self.bids = SortedList(key=lambda o: (-o.price,))    # high to low
        self.asks = SortedList(key=lambda o: ( o.price,))    # low  to high

    def match(self, incoming: Order) -> list[Trade]:
        book = self.asks if incoming.side == 'buy' else self.bids
        crosses = (lambda r: r.price <= incoming.price) if incoming.side == 'buy' \
                  else (lambda r: r.price >= incoming.price)
        trades, consumed = [], []
        for resting in book:
            if incoming.qty == 0 or not crosses(resting): break
```

```
            fill = min(incoming.qty, resting.qty)
            trades.append(Trade(resting.price, fill))
            incoming.qty -= fill; resting.qty -= fill
            if resting.qty == 0: consumed.append(resting)
        for r in consumed: book.remove(r)
        if incoming.qty > 0:                              # rest remainder
            (self.bids if incoming.side == 'buy' else
                self.asks).add(incoming)
        return trades
```

Listing 2.1: Minimal price–time-priority matching engine. Adapted from (Pelgrims, 2020).

Intuition

The `price` field comment — "+inf for a market buy, 0 for a market sell" — is the whole point of the listing. A market buy is a limit buy with $p = +\infty$: the crossing predicate "r.price $\leq$ incoming.price" is true for *every* resting sell, so the fill loop eats levels until quantity or book is exhausted. Market sells collapse to limit sells with $p = 0$ the same way. The seven-line `match` method therefore handles all four of $\{$limit, market$\} \times \{$buy, sell$\}$ with no special cases. A generic mechanism that degenerates into familiar special cases at the parameter limits is a pattern you will see again, especially in Chapter 16.

2.6 The order lifecycle

From the instant a participant hits "send" to the instant cash and shares change hands, an order passes through a fixed sequence of states. Different exchanges publish different subsets of these states and use slightly different names, but the skeleton is universal.

1. **Submit.** The trading system builds the order message (side, price, quantity, product, flags) and sends it to the exchange gateway.
2. **Accept.** The gateway performs cheap sanity checks (valid product? valid side? price within the day's collar? margin or position limit OK?) and either admits the order to the matching engine or rejects it.
3. **Rest or match.** The engine checks for a cross. If none, the order is inserted at its price–time position and *rests*. Otherwise, the Section 2.5 fill loop runs and produces trades.
4. **Partial or full fill.** A fully-filled order leaves the book. On a partial fill, the remainder rests (plain limit), cancels (IOC), or the whole order is retroactively cancelled (FOK).
5. **Clear.** The clearing system reports to a clearing house: "A bought q at p from B." The clearing house typically becomes the central counterparty, so A and B never need to trust each other directly.
6. **Settle.** One or two business days later (T+2 for most equities, T+1 for US Treasuries, T+0 for intraday futures), the clearing house moves cash and shares irrevocably. Only now does the trade economically *exist*.

For strategy content you will rarely care about steps 5 and 6: on Prosperity both happen instantly, and on a sell-side desk the Strat role touches post-trade infrastructure little. But know the names — every real trading system logs events at every stage, and you will read those logs when debugging.

Order lifecycle

Submit → participant sends the order message. **Accept** → gateway validates and admits it. **Rest or match** → engine either posts it or runs the fill loop. **Partial or full fill** → some or all of the quantity is traded; remainder handled per time-in-force flag. **Clear** → clearing house records the obligation. **Settle** → cash and shares actually move.

2.7 Makers and takers

Every trade has two sides: the **maker** of liquidity (whose order was already resting) and the **taker** of liquidity (whose incoming order crossed the book). These roles play different economic games, and exchanges price them differently to influence behaviour.

Definition 2.7 (Maker and taker)**.** In a trade, the *maker* is the participant whose limit order was already resting in the LOB when the match occurred; the *taker* is the participant whose marketable order (limit, market, IOC, or FOK) crossed the book and triggered the match. By construction, makers provide liquidity by posting quotes; takers consume liquidity by hitting them.

Exchanges want a deep, tight book — it attracts more traders, more volume, and more fees. But someone has to *post* the resting orders, and that someone is exposed to adverse selection, inventory risk, and opportunity cost. The standard solution is a **maker–taker fee schedule**: the exchange *charges* the taker and *pays* the maker (a "rebate"). Both are small fractions of notional; the difference is the exchange's cut.

Concrete example. An exchange charges takers 0.30 basis points (bps) of notional and pays makers a rebate of 0.20 bps (where 1 bp $= 10^{-4} = 0.01\%$). Alice rests a buy limit of 1,000 shares at \$50.00. Bob sends a marketable sell and fills it. Trade notional: $1{,}000 \times 50 = \$50{,}000$. Then:

- Bob (taker) pays a fee of $\$50{,}000 \times 0.30$ bps $= \$1.50$ plus the \$50,000 he owes Alice.
- Alice (maker) receives a rebate of $\$50{,}000 \times 0.20$ bps $= \$1.00$ plus the \$50,000 Bob owes.
- The exchange pockets the difference, \$0.50.

The rebate of \$1.00 on \$50,000 is tiny — two ten-thousandths of a percent. For a strategy that crosses the spread once per trade it is irrelevant. For a market maker who collects the rebate on every fill, turns over billions of notional daily, and barely breaks even on price, the rebate *is* the business model. An enormous fraction of US equity volume is precisely this: *rebate-harvesting market making.* Chapter 3 taxonomises the participants; Chapter 9 derives the optimal quoting problem that results.

Maker–taker economics

Exchanges charge takers and pay makers a small per-trade fee and rebate, typically in the single digits of basis points of notional. The difference is the exchange's revenue. A strategy that crosses the spread on every trade pays the taker fee; a strategy that patiently posts limit quotes on both sides of the market collects the maker rebate. The rebate alone can make an otherwise-flat strategy profitable, which is why modern equity exchanges run overwhelmingly on passive maker flow from market-making firms. See Chapter 3 for the full participant taxonomy and Chapter 9 for how the rebate shapes optimal quoting.

2.8 Tick sizes and lot sizes

Real exchanges quantise both order *prices* and *quantities*.

Definition 2.8 (Tick size)**.** The *tick size* is the smallest price increment allowed between orders. A US large-cap stock has a tick of $0.01; EUR/USD on EBS has a tick of 0.00001 (a "pip"); Prosperity 3 products had integer ticks of 1. Prices not at an integer number of ticks from a reference are rejected.

Definition 2.9 (Lot size)**.** The *lot size* is the smallest quantity increment. Most modern equity venues use 1 share, but futures and FX spot often trade in *round lots* of hundreds or thousands of units.

Ticks and lots impose discrete-lattice constraints on quoting. A model might say the optimal bid is 10,001.37; with tick 1, the maker must quote either 10,001 or 10,002, and that rounding affects fill probability and queue position. Worse, the tick imposes a *minimum spread*: at top of book with tick 1, you cannot improve 10,000 to 10,000.5, so the only way to grab priority is to take the other side of the market. For products with large ticks relative to volatility, the book is typically pinned at a one-tick spread and queue position becomes the dominant maker edge.

You cannot quote between ticks

Your fair-value model (derived in Chapter 9) says the optimal bid is 9,999.62 and the optimal ask is 10,000.38 on a product with tick size 1. You cannot post those quotes. Your options are exactly:

- Bid 9,999 / ask 10,000: 1-tick spread, both quotes worse than the model wants.
- Bid 10,000 / ask 10,001: 1-tick spread, symmetric on the other side.
- Bid 9,999 / ask 10,001: 2-tick spread, leaves edge on the table per trade but grabs top-of-book on both sides.

The pitfall: a model that ignores the tick lattice can confidently tell you to quote numbers that are illegal on the venue. Which of the three legal roundings is optimal depends on inventory, queue position, and adverse selection — Chapter 9 content. For now, just remember the lattice constrains every quoting strategy.

2.9 A first look at a real exchange API

Time to see what a real order book looks like in code. Prosperity exposes its LOB via a Python dataclass `OrderDepth`, inside a `TradingState` passed to your `Trader.run` method each simulation tick. The full API is Chapter 24; for now, see the "two sorted lists" of Section 2.3 in concrete Python.

Prosperity example

Each iteration of a Prosperity round, `Trader.run` gets a `TradingState` containing an `OrderDepth` per product. The resting book on each side is a plain Python dict mapping price to signed quantity: positive for bids, negative for asks.

```
class OrderDepth:
    buy_orders:  Dict[int, int]    # price -> +qty, sorted high to
        low
```

```python
    sell_orders: Dict[int, int]   # price -> -qty, sorted low to high

class Trader:
    def run(self, state: TradingState):
        result = {}
        for product, depth in state.order_depths.items():
            orders = []
            fair = 10  # placeholder; real code computes this
            if depth.sell_orders:
                best_ask = min(depth.sell_orders.keys())
                ask_qty  = depth.sell_orders[best_ask]     # already negative
                if best_ask < fair:                        # ask below fair -> BUY
                     orders.append(Order(product, best_ask, -ask_qty))
            if depth.buy_orders:
                best_bid = max(depth.buy_orders.keys())
                bid_qty  = depth.buy_orders[best_bid]      # already positive
                if best_bid > fair:                        # bid above fair -> SELL
                     orders.append(Order(product, best_bid, -bid_qty))
            result[product] = orders
        return result, 0, ""
```

Listing 2.2: The Prosperity `OrderDepth` interface and a one-line "buy if cheap" skeleton `Trader`. Adapted from `prosperity-wiki/writing-an-algorithm-in-python.md`.

This is *exactly* the LOB of Section 2.3 in Python primitives: `depth.buy_orders` is B_t and `depth.sell_orders` is A_t. Encoding ask quantities as *negative* integers is a Prosperity-specific convention (remember it in Part VI); otherwise the API is the data structure you just met. The `run` method posts a marketable limit whenever either best quote crosses a placeholder fair value of 10; replacing that placeholder with a defensible fair-value estimator is all that separates this skeleton from a first submissible algorithm, and is the subject of Chapter 4.

2.10 Sell-side quoting vs an exchange book

This chapter's LOB model is the *central-limit-order-book (CLOB) exchange*: one venue, one book, anonymous participants, price–time priority. The right mental model for equities, listed futures, and Prosperity. Not the right model for much of what a sell-side bank quotes.

On a Goldman Sachs desk

A Goldman Sachs rates desk that makes a market in, say, a 10-year USD interest rate swap does *not* post resting limit orders into a public CLOB. The desk receives RFQs (requests for quote) — "what's your price on \$500m 10y fixed-for-float?" — and responds with a bid and an ask bespoke to that client, size, and instant. There is a book of sorts — the Strat's pricing system maintains a theoretical mid from curve models plus an inventory-adjusted skew — but it is internal; client quotes are individually-priced two-way streams, not resting orders in a public queue. The CLOB model still applies in two places: the *hedge* leg (the desk offsets the swap against exchange-traded instruments like Treasury futures, a real CLOB), and the *inter-dealer* market (dealers quote each other anonymously through a CLOB-like venue). Chapter 28 maps the full sell-side floor.

2.11 Key facts to memorise

If you forget everything else, carry forward this list.

- A **limit order book** is two sorted lists of resting limit orders — bids descending, asks ascending. Everything later chapters call "the market" is a manipulation of those lists.
- **Best bid** p^{bid}, **best ask** p^{ask}, **spread** $s = p^{\text{ask}} - p^{\text{bid}}$, and **mid** $p^{\text{mid}} = (p^{\text{bid}} + p^{\text{ask}})/2$ summarise a market state. None of them is "the" price; see Chapter 4.
- **Limit orders** rest; **market orders** do not. A market order is a limit with $p = +\infty$ (buy) or $p = 0$ (sell) and collapses to a limit in one line of code.
- Under **price–time priority** (this book's default) marketable orders fill against the book in sorted order: best price first, ties broken by arrival time. Pro-rata matching is the main alternative.
- Every trade has a **maker** (resting) and a **taker** (crossing). Exchanges typically charge takers and pay makers a small fee; the rebate is the basis of rebate-harvesting market making.
- Order lifecycle: *submit* → *accept* → *rest or match* → *partial or full fill* → *clear* → *settle*. Strategy work cares about the first four.
- **Tick** and **lot size** impose a discrete lattice on prices and quantities. A quote between ticks is illegal; rounding policy is first-order for market making (Chapter 9).
- **Market orders slip**. Quoted best ask is not your realised fill on a size-200 market buy in a thin book; use marketable limits with a price cap whenever order size is comparable to top-of-book size.
- A Prosperity `OrderDepth` is exactly this chapter's LOB: `buy_orders` and `sell_orders` dicts keyed by price. Full Prosperity API: Chapter 24.

That is the vocabulary. The next two chapters add people and prices: Chapter 3 asks *who* posts the orders and what they are trying to accomplish; Chapter 4 asks what "the price" means when there are this many sensible candidates.

Harris's *Trading and Exchanges* (Harris, 2003) is the canonical encyclopaedic reference; Pelgrims (Pelgrims, 2020) is the source for the Python skeleton and matching example. Both are listed in Chapter C.

Chapter 3

Market Participants and Their Incentives

Every trade has two sides. When your buy fills, *someone* sold; when your sell fills, *someone* bought. That someone's motivation, information, and pressures determine whether your trade was a good idea. The same resting quote is a free gift if the counterparty is an inattentive retail trader dumping a position, and a lethal trap if it is an insider with next-quarter's earnings release. Until you can reason about *who is on the other side*, you cannot reason about your own edge.

This chapter introduces the seven categories of participant you will meet repeatedly: why they trade, how their orders appear in the limit order book, and how to recognise them in market data. The centrepiece is the distinction between **informed** and **uninformed** traders — the foundation of every microstructure model in Chapters 5 and 6, and the reason a market maker can earn money posting quotes, or lose it quickly if they misjudge who is hitting them.

Prerequisites

From Chapter 2:

- The **limit order book** as two sorted lists of resting buy/sell orders.
- **Limit** (post a price) vs **market** (cross the book) orders; the **order lifecycle**: submit, queue, match, report.
- **Makers** (resting order) and **takers** (marketable order crossing). Exchanges charge the taker a fee and pay the maker a rebate.
- **Bid**, **ask**, **spread**, **mid**, **tick size**.

No prior finance knowledge assumed beyond these mechanics.

3.1 The trader taxonomy

Larry Harris's *Trading and Exchanges* (Harris, 2003) organises market participants by why they trade, not what they trade. Motivation predicts behaviour: a participant trading on information acts differently from one rebalancing a pension fund, even sending identical orders to the same exchange. The seven categories below are not mutually exclusive — one firm can run desks in several roles — but they map cleanly onto the strategies in Chapters 9 and 14.

Market makers

Definition 3.1 (Market maker)**.** A *market maker* is a participant whose strategy is to continuously quote both a bid and an ask on a product, profiting from the spread $s = p^{\text{ask}} - p^{\text{bid}}$ between the two by being on the maker side of many trades in both directions. A pure market maker does not take a view on the direction of the underlying price; their goal is to end the trading session with roughly zero net position (*flat*) and cash equal to their aggregated spread capture minus losses.

The canonical mental picture is the airport currency booth: buy dollars from tourists at one rate, sell at a higher rate, and if flow balances, bank the difference. Electronically the market maker posts a buy limit at p^{bid} and a sell limit at p^{ask}, then waits. When a taker hits either side, position moves one unit and cash moves by the fill price. If a buy and a sell fill in close succession, they bought at p^{bid} and sold at p^{ask}, netting s per unit at zero end-of-session exposure.

The behaviour is recognisable in the LOB. A market maker refreshes quotes on both sides, clustered near the top of the book in competition with other market makers, cancelling and re-posting as the mid moves. They rarely send a *marketable* order (one that crosses the book) except to unwind a lopsided inventory. Chapter 9 derives the optimal quoting strategy under inventory risk; for now, the takeaway is that market makers are maker-side, two-sided, and almost always present.

Identification heuristic. Very high fill count, near-equal split between buy and sell fills, tiny average holding time, positions that oscillate near zero, quotes that appear within one tick of the best on both sides.

Arbitrageurs

Definition 3.2 (Arbitrageur)**.** An *arbitrageur* is a participant who trades simultaneously in two or more markets (or two or more products) to lock in a price discrepancy. A pure *riskless* arbitrage requires the discrepancy to close with certainty; a *statistical* arbitrage merely requires it to close in expectation.

The archetype is triangular FX arbitrage: if EUR/USD/JPY cross-rates imply a loop EUR $\to$ USD $\to$ JPY $\to$ EUR worth more than one euro, an arbitrageur sends three market orders in a race against everyone else doing the same. Each leg takes liquidity from a different book; the profit is the loop's excess; the risk is that a leg fails to fill at the expected price, leaving the position directionally exposed.

In the LOB, arbitrageurs are almost exclusively **takers**: the opportunity closes in milliseconds, too fast to rest a limit order. Their flow is *correlated across venues* — simultaneous hits on EUR/USD at two venues are likely one arbitrageur firing two legs. Chapter 14 catalogues the arbitrage types (triangular, cash-and-carry, put–call parity, ETF creation–redemption, and more); for now, note that arbitrageurs force prices across venues to stay in line.

Identification heuristic. Short bursts of taker orders, highly correlated timestamps across multiple venues or products, near-zero inventory held over the round, directionless aggregate P&L (they do not care which way the market moves).

Informed traders

Definition 3.3 (Informed trader)**.** An *informed trader* is a participant who possesses private information about the fair value of a product that is not yet reflected in the market price. They trade on that information: buying when their private signal says the product is underpriced, selling when it says overpriced, and exiting the position once the price moves to their target.

The prototypical informed trader in academic models is an *insider*: an executive who knows an earnings announcement is coming, an analyst who has done original research, a trader who has seen a regulatory filing minutes before it goes public. Harris's broader sense of "informed" covers any participant whose expected value for the product exceeds the market's information set — not only illegal insider trading but also superior models, proprietary data feeds, or faster news parsing.

From the market maker's point of view, they *cannot see the information nor the label on the order*. An informed buy and a noise buy are indistinguishable at the message level; only the subsequent price move reveals which hit them. This is the setup for both classical theories of the bid–ask spread in Chapter 6: Glosten–Milgrom models informed trading as an adverse-selection tax on the market maker, and Kyle's 1985 model derives the equilibrium in which a single strategic informed trader hides inside noise flow while the market maker learns from order flow alone.

Identification heuristic. Consistently on the same side of a subsequent price move; buys precede up-moves and sells precede down-moves with far higher hit-rate than chance. The hardest counterparty to recognise in real time; the easiest in hindsight.

Noise traders (liquidity traders)

Definition 3.4 (Noise trader)**.** A *noise trader*, also called a *liquidity trader* or *uninformed trader*, is a participant whose reason for trading is unrelated to private information about fair value. They trade because they need cash, because a pension fund is rebalancing on a calendar schedule, because a mutual fund is meeting redemptions, because someone pressed the wrong button, or because an automated strategy's signal was pure statistical nonsense.

The terminology is historical and slightly unfair: a pension fund rebalancing a multi-billion-dollar portfolio is not literally *noisy*, just orthogonal to information about the next price move. What matters is that the noise trader's decision is *uncorrelated with the product's true value*: on average they buy as often before a down-move as before an up-move. To the market maker, the noise trader is the perfect counterparty — every fill is a fair bet and the spread collected is profit.

Real markets always contain a mix of informed and noise flow, and every microstructure model in this book is built around that mix. The market maker's willingness to quote a tight spread depends on the *noise fraction* being high; when informed flow dominates, spreads widen or market makers withdraw. Chapter 5 makes this quantitative.

Identification heuristic. Uncorrelated with subsequent price moves; roughly 50% of trades face up-moves and 50% face down-moves. Often larger order sizes and longer holding horizons than information-driven flow. In Prosperity, the scripted "retail" bots that simply hit the top of the book at random intervals are pure noise traders.

High-frequency traders (HFTs)

Definition 3.5 (High-frequency trader)**.** A *high-frequency trader (HFT)* is a participant whose strategy operates at microsecond to sub-millisecond latencies, rewriting quotes hundreds or thousands of times per second in response to every incoming message on the book. HFTs are technologically, not fundamentally, defined: the category includes market-making HFTs, arbitrage HFTs, and short-horizon signal-driven HFTs.

HFTs sit on top of the taxonomy rather than inside it: most are a hybrid of market-making and arbitrage roles, operating at a latency regime only hardware-colocated strategies can reach. Their defining feature is intense concern with **queue position**: in a price–time priority market

(Chapter 2) the first order at a price level fills first, and moving from tenth- to first-in-queue can double a strategy's fill rate. HFTs pay to co-locate servers in the same data centre as the matching engine, measured in metres of fibre, so their messages arrive ahead of competitors'.

The economic contribution of HFTs is contested. The defensive case: HFT market making has driven spreads on liquid products to historic lows, benefiting every other participant. The critical case: HFTs impose an adverse-selection tax on slower participants by racing in front of institutional flow. Chapter 8 develops the order-flow imbalance framework underlying most short-horizon HFT signals.

Identification heuristic. Enormous message rates (cancels per second), near-zero average holding time, aggressive quote competition at the top of the book, sensitivity to microsecond-level latency events. In exchange-level data they show up as the firms submitting 95%+ of all cancellations.

Institutional asset managers

Definition 3.6 (Institutional asset manager)**.** An *institutional asset manager* is a firm that trades on behalf of long-horizon capital: pension funds, mutual funds, sovereign wealth funds, university endowments, insurance companies, and the like. Their positions are measured in hundreds of millions to hundreds of billions of dollars, their holding periods run from months to decades, and their trading is driven by portfolio construction mandates rather than by microstructure signals.

Institutional flow is the natural counterparty of the HFT market makers above: slow, size-constrained orders meet fast, two-sided quotes. The institutional manager consumes liquidity (takes); the HFT provides it (makes), with the maker–taker fee schedule (Chapter 2) aligning the incentives.

An institutional manager buying 10 million shares cannot send one market order for 10 million: it would eat the entire visible book and move price by percent. Instead they slice the parent order over hours or days using execution algorithms such as VWAP, TWAP, and participation-of-volume targets. Chapter 19 derives the optimal schedule under a linear impact model. For now: institutional trading is *size-constrained and execution-sensitive* — P&L is dominated by the price received on each slice, not by the raw direction of the trade.

To the market maker, institutional flow is *partially informed*: the rebalance decision is uncorrelated with short-horizon moves (looks like noise), but the signed persistence of a parent sliced over hours reveals its direction (looks like information). This is why institutional orders leak impact even when executed carefully — persistent one-sided flow is itself informative.

Identification heuristic. Persistent one-sided flow over hours with roughly constant slice size; use of VWAP/TWAP child orders that match the day's volume profile; the parent order ends when the target quantity is reached rather than when a price target is hit.

Retail traders

Definition 3.7 (Retail trader)**.** A *retail trader* is an individual trading their own personal account through a broker. Order sizes are small (typically tens to thousands of dollars per trade), the trader is usually less informed than professional participants, and their flow reaches the exchange only after the broker has first had an opportunity to internalise it or to sell it to a wholesaler.

Retail motivations are varied and typically non-professional: long-horizon saving for retirement, speculation on a view picked up from news or social media, hobbyist day-trading, or hedging a concentrated stock grant from an employer. What unites the category is that trading is not the participant's job and the capital at risk is personal.

The retail trader never reaches the exchange directly. Their order travels first to their broker, who in most modern markets either executes against the broker's own inventory (*internalisation*) or sells the order to a wholesale market maker for a small fee under *payment for order flow (PFOF)*. The wholesaler then matches it internally or routes to the exchange. By the time a retail order hits the public book, it has been inspected by at least one professional.

Wholesale market makers pay for retail flow because it is, on average, heavily noise-dominated. A retail trader is more likely selling to pay a bill than because they have private information about earnings. Paying for retail flow therefore buys the best counterparty in the market — the noise trader of Section 3.1. Retail flow is not uniformly uninformed: retail-level insider trading, meme-stock coordination, and reactions to public news all inject short bursts of information into the channel.

Identification heuristic. Small average order size, heavy concentration in a handful of popular tickers, bursts of correlated buying or selling around news events, and, in venues where it is observable, the characteristic latency signature of a broker's customer router rather than a direct exchange feed.

3.2 Informed vs uninformed: the single most important distinction

The seven categories collapse onto a single axis that dominates all others: is the counterparty *informed* or *uninformed*? A market maker trades with noise all day and flees informed flow; an execution algorithm disguises its parent order as noise; an HFT arbitrageur wants to act on information first. Every microstructure result in Chapters 5 and 6 is ultimately a statement about how this distinction shapes prices.

Definition 3.8 (Informed and uninformed flow)**.** Let V denote the unobserved fair value of a product a short time after the current instant. An order is *informed* if the participant submitting it has information that changes their conditional expectation of V relative to the public information set. An order is *uninformed* (equivalently *noise* or *liquidity*) if its submission is conditionally independent of V given public information: the participant is trading for reasons orthogonal to the product's short-horizon value.

The market maker *cannot see this label.* From the message stream, an informed buy and a noise buy are bit-for-bit identical: side, quantity, timestamp, maybe an anonymised account identifier. The only way to infer the label is to observe what the price does *after* the trade. Over many trades the market maker can build a statistical model of each counterparty's informativeness; on a single trade, they are flying blind.

This blindness is asymmetric against the market maker. Fills against noise traders are fair coin flips around the mid, and the spread turns them into profit. Fills against informed traders are systematically adverse: the informed buyer only hits the ask when fair value strictly exceeds the ask, leaving the market maker's post-trade inventory underwater. This is **adverse selection**.

A toy adverse-selection calculation

Consider the simplest model. A product has two possible fair values, $V = 100$ and $V = 110$, equally likely a priori. Fraction $\alpha = 0.3$ of arriving traders are *informed*: they know V and buy when $V = 110$, sell when $V = 100$. The other $1 - \alpha = 0.7$ are *noise*: they buy or sell with probability $\frac{1}{2}$ each, independent of V. The market maker posts a single price at the mid, $p^{\text{mid}} = 105$, and trades with whoever arrives.

Suppose a buy order arrives and fills at 105. What does the market maker expect to lose?

Step 1: P(buy | V).

$$P(\text{buy} \mid V = 110) \;=\; \underbrace{\alpha \cdot 1}_{\text{informed, buys}} \;+\; \underbrace{(1-\alpha)\cdot \tfrac{1}{2}}_{\text{noise, random}} \;=\; 0.3 + 0.7 \cdot 0.5 \;=\; 0.65.$$

$$P(\text{buy} \mid V = 100) \;=\; \underbrace{\alpha \cdot 0}_{\text{informed, sells}} \;+\; \underbrace{(1-\alpha)\cdot \tfrac{1}{2}}_{\text{noise, random}} \;=\; 0 + 0.35 \;=\; 0.35.$$

Step 2: P(buy).

$$\begin{aligned} P(\text{buy}) \;&=\; P(V=110)\,P(\text{buy}\mid V=110) + P(V=100)\,P(\text{buy}\mid V=100) \\ &=\; 0.5\cdot 0.65 + 0.5 \cdot 0.35 \;=\; 0.50. \end{aligned}$$

Step 3: posterior over V given a buy, by Bayes.

$$P(V=110 \mid \text{buy}) \;=\; \frac{P(V=110)\,P(\text{buy}\mid V=110)}{P(\text{buy})} \;=\; \frac{0.5\cdot 0.65}{0.50} \;=\; 0.65.$$

$$P(V=100\mid \text{buy}) \;=\; 1 - 0.65 \;=\; 0.35.$$

Step 4: expected fair value given a buy.

$$\mathbb{E}[V \mid \text{buy}] \;=\; 0.65\cdot 110 + 0.35 \cdot 100 \;=\; 71.5 + 35 \;=\; 106.5.$$

Step 5: market maker's expected P&L on this fill. The market maker sold at $p^{\text{mid}} = 105$ and now holds -1 unit of a product whose conditional fair value is 106.5. The expected loss per unit is

$$\underbrace{105}_{\text{price received}} \;-\; \underbrace{106.5}_{\mathbb{E}[V|\text{buy}]} \;=\; -1.5.$$

The market maker loses 1.5 on average on every buy that hits their ask, because the buy is informationally tilted toward the high-value state. By symmetry, every sell that hits their bid loses 1.5 as well: sells come preferentially from informed traders who know $V = 100$, so the market maker buys at 105 a product worth 103.5, giving P&L $= -1.5$ per unit.

The market maker cannot stay at the mid. Allowing a genuine spread — ask strictly above 105, bid strictly below — shrinks losses, and at a wide enough spread turns them positive. "How wide must the spread be to offset adverse selection?" is the Glosten–Milgrom problem, derived in full in Chapter 6. The toy calculation here shows that *a nonzero spread is forced by the mere existence of informed flow*, even without order-processing costs or inventory risk. The other two spread drivers are added in Chapter 5.

The market maker's fundamental problem

The market maker sees only anonymous order flow and cannot tell informed from noise. Informed flow is systematically adverse (buyers buy only when fair value exceeds the ask, sellers sell only when fair value is below the bid), so fills against it lose money in expectation. The whole point of quoting a spread rather than a single price is to charge noise flow enough to cover losses to informed flow. This tension is **adverse selection**, the first-order phenomenon every microstructure model quantifies.

Intuition

You would never play poker against someone who can see your hand. A market maker plays a worse game: against a table of strangers, some of whom see the hand and some of whom do not, and the market maker is not told which. Their only tool is to charge enough per hand to break even averaged over the whole table. That charge is the spread.

"Small spread

Novices assume a tight spread means cheap trading. True for a *taker* (Chapter 2): tight spread means small crossing distance. False for a *maker*: tight spread means small reward per fill while adverse-selection losses continue unchanged. On a product with heavy informed flow, a tight spread is a warning, not an invitation. Chapter 9 shows how a market maker skews quotes *inside* or *outside* the top of book depending on their view of adverse selection.

3.3 Maker/taker economics in full

Chapter 2 introduced the maker–taker fee schedule mechanically: the exchange charges the taker a small fee, pays the maker a small rebate, and keeps the difference. This chapter can now explain *why*, and why the asymmetry is necessary.

The maker of every trade is someone whose limit order was resting when the trade happened. That order has been exposed to **adverse selection** for every microsecond it sat: any informed trader arriving during that window would have picked off the stale quote. The longer a quote rests, the more likely the next fill is adverse. Posting is costly, and someone must compensate the posters or they will stop posting.

The taker, in the same trade, crossed the book. They paid the spread — the gap between fill price and mid — and received the one thing the market maker cannot promise: **immediacy**. The taker traded right now, at the quoted price, without risk of the opportunity disappearing. This is valuable to any participant with a short-shelf-life trade: an arbitrageur whose loop closes in milliseconds, an execution algorithm needing to complete by the close, a retail trader reacting to a news event. Takers pay for immediacy because without it their strategy does not exist.

The economic symmetry of the maker/taker fee

The maker bears adverse-selection risk, inventory risk, and the opportunity cost of capital in resting orders. In return they get (a) the spread and (b) an exchange rebate. The taker gets immediacy. In return they pay (a) the spread and (b) an exchange fee. The exchange clips the difference to fund matching, compliance, and margin. Without the rebate, no posted liquidity; without the fee, no exchange.

A market charging neither side would still function in principle — participants would trade for the spread alone — but the exchange would have no revenue, and without a rebate the incentive to post rather than wait-and-take disappears. Rebates are typically a few tenths of a basis point, but at the scale of rebate-harvesting market making (Chapter 9) they are the entire business model.

Not every venue uses the same schedule. A **maker–maker model** rebates both sides and funds the exchange from subscription and data fees. Some derivatives exchanges use **pro-rata allocation** instead of price–time priority, devaluing queue position and shifting market-making economics. Dark pools charge neither side and exist as venues for institutional block crosses that avoid signalling. Chapter 9 treats the full range.

3.4 Dealer markets vs exchange markets

So far this book has described an **order-driven market**: a central limit order book with anonymous participants, price–time priority, and a matching engine that pairs crossing orders. This is the dominant structure for listed equities, listed futures, major cryptocurrencies, and all of Prosperity. It is not the only architecture, and several of the world's largest asset classes do *not* trade this way.

Definition 3.9 (Order-driven market)**.** An *order-driven market* is one in which all participants submit limit and market orders to a single shared limit order book, trades are matched by an anonymous matching engine under fixed priority rules, and every participant faces the same book. Prices emerge from the interaction of orders; no single participant is privileged. Examples: equities on NASDAQ and NYSE, futures on CME, cryptocurrencies on Binance, and Prosperity.

Definition 3.10 (Quote-driven or dealer market)**.** A *quote-driven market* (also called a *dealer market*) is one in which a participant wishing to trade does not send an order to a central book but instead sends a *request for quote (RFQ)* to one or more dealers, typically large banks. Each dealer responds with a bid and an ask at which they are willing to trade; the client picks the best one and trades against that dealer's inventory. The dealer takes the position onto their own book and is responsible for hedging it. There is no anonymous matching; the dealer is a *counterparty*, not a neutral intermediary. Examples: most over- the-counter (OTC) bond markets, interest rate swaps, FX spot for institutional size, OTC derivatives.

The difference is fundamental enough that this book's intuitions transfer only partially. In an order-driven market the sell-side bank is just another participant; in a quote-driven market the bank is the *only* participant on one side, with entirely different incentives. A dealer pricing an interest rate swap is not a neutral matching engine: they price against their own inventory, their fair value, their hedging costs, and their assessment of how informed the client is. A macro hedge fund gets quoted wider; a corporate treasurer rolling a hedge gets quoted tighter.

The microstructure models of Chapters 5 and 6 were developed for order-driven markets, but the core intuition — adverse-selection pressure on the liquidity provider — carries over. A dealer faces the same problem as a market maker with a different interface: they see the client's identity (helps: classify the flow) but commit to a price before the client commits to trade (hurts: the client shops around). Chapter 28 returns to the day-to-day life of a Strat on the quote-driven side.

On a Goldman Sachs desk

On a Goldman rates desk, a pension fund calls asking a price on a \$500m 10-year interest rate swap. There is no central book for swaps in that size. You — the Strat — receive the RFQ; your system computes a fair mid from the current swap curve; you add a spread reflecting your inventory (already long duration? charge more), your view of the client's informedness (routine corporate hedge or macro fund taking a directional view?), and your hedging cost in the linear futures market where you will offset delta within minutes. The client picks your price or someone else's; if yours, the position lands on your book and you immediately sell Eurodollar futures to neutralise first-order rate exposure. Chapter 28 teaches how every piece of this workflow is implemented.

The boundary is not watertight. Some equity venues run both: a central book by day and an RFQ facility for block trades. Some crypto products trade on central books and via OTC dealers. Some order-driven exchanges host dealer-style market makers ("lead market makers" in ETFs, designated market makers on NYSE, specialists in older designs). What does not change is the economic pressure: the liquidity provider — central-book market maker or sell-side dealer

— is paid to absorb risk from flow they cannot fully classify. The game is the adverse-selection game of Chapter 6.

3.5 The Olivia story: an informed trader in the wild

The informed/uninformed distinction can feel abstract. Prosperity 3 provided a vivid real example showing the informed-trader category is a recognisable pattern in actual order flow — and exploiting it paid.

Prosperity example

In the 2025 IMC Prosperity 3 competition, participants traded against scripted bots whose trades were *anonymous* for the first four rounds: you saw trade size, price, and direction but not which bot acted. In Round 5 IMC made historical **trader IDs** public, lifting the anonymity curtain and letting teams tag every past trade with its submitting bot.

One bot, **Olivia**, had followed a strikingly simple rule throughout: on Squid Ink, buy 15 lots at the daily low, sell 15 lots at the daily high. In the vocabulary of Section 3.2 she was an archetypal informed trader, acting on private information (the day's price extrema) no one else could see until after the fact.

The Frankfurt Hedgehogs (Frankfurt Hedgehogs, 2025a), who finished 2nd overall, spotted this during the first four rounds without knowing the bot's name: a particular anonymous counterparty consistently bought at what turned out to be the daily minimum and sold at the daily maximum, a pattern their running-extremum detector could latch onto. The optimal strategy on Squid Ink became "mimic the informed trader": buy when Olivia buys, sell when Olivia sells. This single signal generated roughly 8,000 SeaShells on Squid Ink in Round 1. In Round 2 they used Olivia's inferred Croissants position (a picnic-basket constituent) to tilt basket-spread thresholds — the cross-product extension of Chapter 12.

When Round 5 revealed trader IDs, the Hedgehogs already had a working detector; the reveal merely let them swap heuristic matching for a literal trader-ID check, eliminating false positives. Several other top-ten teams converged on the same strategy.

The story maps onto this chapter's theory. Olivia was informed in the strict sense: a private signal, acted on. Any team running pure market-making on Squid Ink was unwittingly the market maker of Section 3.2, paying for adversely-selected flow. Teams that identified and copied Olivia were on the informed side, and P&L flowed accordingly.

Lesson for Prosperity 4 (and for any real exchange with mixed counterparties): the informed/uninformed distinction is not abstraction. It is the difference between making and losing money, visible in the order flow if you know where to look.

3.6 Key facts to memorise

- The seven participant categories — **market makers**, **arbitrageurs**, **informed traders**, **noise (liquidity) traders**, **HFTs**, **institutional asset managers**, **retail traders** — are distinguished by motivation, not by the orders they send.

- The most consequential split is **informed vs uninformed**: flow correlated with subsequent price moves vs flow orthogonal to them. Every result in Chapters 5 and 6 rests on this distinction.

- **Adverse selection**: a market maker's fills against informed counterparties systematically lose money because the informed trader crosses only when fair value is favourable. In the toy model of Section 3.2 ($\alpha = 0.3$, $V \in \{100, 110\}$, mid 105), expected loss per fill is \$1.5 per unit, forcing a nonzero spread.

- The **maker–taker fee schedule** is economic symmetry: makers get a rebate for bearing adverse selection and inventory risk; takers pay a fee for immediacy. Without the rebate, no one posts; without the fee, the exchange has no revenue.
- **Order-driven** (CLOB, anonymous, price–time priority) and **quote-driven / dealer** (RFQ, dealer holds inventory) markets are architecturally distinct. Equities, futures, crypto, Prosperity sit in the first; bonds, swaps, institutional FX spot, OTC derivatives sit in the second. The adverse-selection game is the same.
- **Queue position** matters under price–time priority; this is the economic reason HFTs co-locate.
- **Institutional flow** is partially informed: slices are uncorrelated with the next tick, but parent-order persistence leaks direction. Chapter 19 covers large-order execution.
- **Retail flow is valuable because it is uninformed**: wholesalers pay brokers for PFOF access, because a noisy counterparty is the best counterparty.
- The **Olivia story** is empirical ground truth for the informed-trader category — detectable from order flow alone, and top-ten teams copied her for direct P&L. The informed/uninformed distinction is a pattern in real data, not an abstraction.

Chapter 4

Price, Value, and the Many Definitions of the Price

Ask a newcomer what the price of a stock is, and the answer is almost always "the last trade price." That answer is not wrong, only insufficient. On any real market, at any instant, several equally valid numbers could be called "the price," and a strategy that picks the wrong one bleeds edge invisibly until the end-of-day report. A market maker computing skew off the raw mid when one side has a one-lot stub at top of book is quoting against noise, not fair value. An execution algorithm benchmarking against a last trade from four minutes ago looks good on paper and loses money in practice. There is no single number; the skill this chapter builds is knowing *which number to use for which purpose*.

This chapter replaces the unqualified phrase "the price" with a menu of well-defined price *measures*, each with a formal definition, a use case, and a failure mode: best bid, best ask, spread, mid, microprice, volume-weighted average price, last-trade price, and one Prosperity-specific construction called the Wall Mid. By the end you will be able to compute every one on a given book and state which answers the question you are actually asking.

Prerequisites

From Chapter 2:

- The **limit order book (LOB)** as two sorted lists of resting limit orders, and the definitions of **bid**, **ask**, **spread**, and **mid** introduced there as quick working definitions. This chapter is the promised formal treatment.
- **Maker** vs **taker**: in every trade the maker was the resting side, the taker was the crossing side.
- **Tick size** and **lot size** as the price and quantity grids on which orders live.

From Chapter 3:

- **Informed** vs **uninformed** traders, and the adverse-selection intuition: posted quotes are systematically hit by the side that knows more. This motivates why the microprice is typically a better short-horizon predictor than the mid.

No prior finance knowledge assumed beyond these.

4.1 Bid, ask, spread, mid — formally

Chapter 2 introduced p^{bid}, p^{ask}, s, and p^{mid} as working definitions in a single paragraph. The rest of this chapter hangs off these four symbols, so we restate them formally here.

Definition 4.1 (Best bid)**.** The *best bid* p^{bid} at time t is the highest price among all resting limit buy orders currently in the LOB:

$$p^{\text{bid}} \;:=\; \max\{\,\text{price}(o) : o \in B_t\,\},$$

where B_t is the bid side of the book at time t. If B_t is empty, p^{bid} is undefined.

Definition 4.2 (Best ask)**.** The *best ask* (or *best offer*) p^{ask} at time t is the lowest price among all resting limit sell orders currently in the LOB:

$$p^{\text{ask}} \;:=\; \min\{\,\text{price}(o) : o \in A_t\,\}.$$

If A_t is empty, p^{ask} is undefined.

Definition 4.3 (Spread)**.** The *bid–ask spread* s is the gap between the best ask and the best bid, $s := p^{\text{ask}} - p^{\text{bid}}$. On an uncrossed book $s \geq 0$; we have $s = 0$ only in the degenerate instant when the book is *locked* mid-match — that is, the best bid and best ask are momentarily equal because the matching engine has not yet cleared the crossing orders.

Definition 4.4 (Mid-price)**.** The *mid-price* p^{mid} is the arithmetic average of the best bid and the best ask:

$$p^{\text{mid}} \;:=\; \tfrac{1}{2}\big(p^{\text{bid}} + p^{\text{ask}}\big).$$

The mid is the most common "summary" of where a market is trading, and p^{mid} appears in almost every subsequent chapter. It is not "the price" in any rigorous sense; it is a convenient average that pretends the two sides of the book are symmetric. When they are not — almost always — there are better choices.

"The mid is the price" is wrong

The mid is easy to compute, plot, and defend in a meeting. It also implicitly asserts that the best bid and best ask carry *equal* information about where the next trade happens. This is almost never true. If the bid is backed by 131 lots and the ask by only 48, the ask clears first under a modest market buy, and the post-trade mid sits strictly above the pre-trade mid — meaning the expected next traded price was above the mid all along. The microprice (Section 4.2) is the correction. Use the mid as an *input* to strategies, not as a belief about fair value.

4.2 Beyond the mid: three more price measures

The mid treats the two sides as interchangeable. In reality, the side with less resting quantity is more likely to be consumed next, so the short-horizon expected price sits closer to the *opposite* side. The **microprice** makes this precise. The **volume-weighted average price (VWAP)** serves a different purpose: not "what is fair value now" but "what was the average traded price over some window." The **last-trade price** is the cheapest summary, and the one most easily manipulated and most prone to staleness. We take them in turn.

The microprice

Definition 4.5 (Microprice)**.** Let p^{bid} and p^{ask} be the best bid and best ask, and let Q_b and Q_a be the total resting quantities at those two top-of-book levels. The *microprice* is

$$p^{\text{micro}} := \frac{Q_a \cdot p^{\text{bid}} + Q_b \cdot p^{\text{ask}}}{Q_a + Q_b}.$$

It is a weighted average of the two top quotes in which each side's *own* quantity is the *opposite* side's weight.

The cross-weighting is the whole point. The side with smaller size gets the larger weight: if Q_a is tiny, the microprice leans hard toward p^{ask}, because an incoming market buy of modest size consumes Q_a and lifts the ask. Symmetrically, small Q_b pulls it toward p^{bid}. When $Q_a = Q_b$ the weights are equal and $p^{\text{micro}} = p^{\text{mid}}$; in every other case p^{micro} and p^{mid} disagree, and the disagreement is informative.

Bouchaud, Bonart, Donier, and Gould (Bouchaud et al., 2018) give the formal treatment and show the microprice is a markedly better short-horizon predictor of the next mid movement than the mid — it anticipates the mid rather than lagging it. The underlying reason is order-flow imbalance, developed in Chapter 8.

Intuition

The side with less size breaks first: a smaller buffer against the next market order means a higher probability that side gets wiped and its top quote moves. If you believe that — and the data in Chapter 8 say you should — then the near-future price sits closer to the *other* side. The microprice is the cheapest implementation of that belief: one weighted average, no model, no parameter to fit. Whenever a market-making chapter refers to "fair value" as an input to a quoting formula, assume the author means microprice unless stated otherwise.

VWAP

Definition 4.6 (Volume-weighted average price (VWAP))**.** Given a time window $[t_0, t_1]$ over which n trades $(p_1, q_1), \ldots, (p_n, q_n)$ have printed on a product, the *volume-weighted average price* over that window is

$$\text{VWAP}_{[t_0,t_1]} := \frac{\sum_{i=1}^{n} p_i\, q_i}{\sum_{i=1}^{n} q_i}.$$

VWAP is not a fair-value estimate; it is a *benchmark*. The standard use case is execution measurement. Suppose a pension fund buys a million shares over a trading day. The fund does not care about the current mid — it cares whether its average acquisition price beat or missed the day's VWAP. Beating VWAP by a basis point on a million-share parent order is a valuable sell-side relationship; missing by five basis points is an excuse to find a different broker. (One *basis point*, or *bp*, is one hundredth of one percent: $1\,\text{bp} = 0.01\% = 0.0001$. It is the standard unit for quoting small return differences on a trading floor.) Almgren and Chriss's optimal execution framework (Chapter 19) is ultimately an argument about trading off market impact against the risk of drifting from a VWAP-like benchmark.

Two practical notes. First, VWAP depends on the window: "day VWAP" is the common version, but "trailing one-hour VWAP" and "VWAP since open" appear routinely on execution dashboards. Second, VWAP is backward-looking — it exists only after the trades print — so it cannot be a forward-looking fair value. Do not confuse the two.

Last-trade price

Definition 4.7 (Last-trade price)**.** The *last-trade price* p_t^{last} is the price of the most recent trade to have printed on the product strictly before time t. If no trades have yet occurred in the current session, p_t^{last} is undefined.

The last-trade price is the folk-meaning of "the price" — what retail websites display as "current" and what the closing-auction print becomes for end-of-day accounting. Its use case is narrow but important: **marking a position to market**. At session end every firm must assign a dollar value to every position it holds (for profit-and-loss reporting, margin, and risk limits), and that dollar value has to come from *somewhere*. For liquid products trading every second, the last-trade price is cheap, well-defined, and consistent across firms. For illiquid products trading every half hour, it is stale and can be far from the current market; the mid, or an agreed closing-auction print, is preferred.

Its weakness: a single number from a noisy stream. One small trade at an odd price moves it; one unwatched cancelled trade leaves it stale for an hour. It is also the cheapest price to manipulate near the close — the regulatory phrase *"marking the close"* refers to exactly this — which is why closing auctions exist as a separate mechanism.

4.3 Wall Mid: a Prosperity-specific fair value

Prosperity publishes a limited-depth snapshot each tick, typically three or four levels per side. Most visible flow comes from a few bot market makers whose quotes sit on the *outside* of the visible book; the levels *between* top of book and those outer bot quotes are transient orders from competing human-written algorithms, including one-lot "stub" orders posted for signalling or to momentarily narrow the top. Using raw top-of-book as a fair-value proxy therefore reacts to every skirmish there. The Frankfurt Hedgehogs noticed that the *outer* visible levels — the lowest bid and highest ask in each snapshot — were consistently posted by the same designated market-maker bots, quoting rounded versions of the true underlying price (typically ± 2 ticks). They named those outer levels the *bid wall* and *ask wall*, and defined the **Wall Mid** as their midpoint.

Definition 4.8 (Bid wall, ask wall, Wall Mid)**.** Given the visible depth of a Prosperity LOB snapshot, the *bid wall* is the bid price of the deepest (lowest-priced) visible bid level, and the *ask wall* is the ask price of the deepest (highest-priced) visible ask level:

$$\begin{aligned} p^{\text{bid,wall}} &:= \min\{\, p : p \in \text{visible bid levels}\,\}, \\ p^{\text{ask,wall}} &:= \max\{\, p : p \in \text{visible ask levels}\,\}. \end{aligned}$$

The *Wall Mid* is the midpoint between the two walls:

$$p^{\text{wall}} := \tfrac{1}{2}\big(p^{\text{bid,wall}} + p^{\text{ask,wall}}\big).$$

The construction is deliberately crude: a one-line, parameter-free formula computable in constant time from any snapshot. Its only claim is that on the Prosperity platform the outer visible levels were posted by informed market-maker bots whose quotes were near-unbiased estimates of a hidden "true" price, so averaging the two outer levels gave a far more stable estimate than any top-of-book quantity.

Prosperity example

Wall Mid — the Frankfurt Hedgehogs' signature construction. Below is the exact helper used by the 2nd-place Frankfurt Hedgehogs algorithm in Prosperity 3 to compute the Wall Mid from a `TradingState`. `mkt_buy_orders` and `mkt_sell_orders` are price→size dictionaries for the two sides of the visible depth; `bid_wall` is the *lowest* visible bid and `ask_wall` the *highest* visible ask, matching the definition above. The defensive `try/except` blocks handle ticks where one side is empty.

```
def get_walls(self):

    bid_wall = wall_mid = ask_wall = None

    try: bid_wall = min([x for x,_ in
        self.mkt_buy_orders.items()])
    except: pass

    try: ask_wall = max([x for x,_ in
        self.mkt_sell_orders.items()])
    except: pass

    try: wall_mid = (bid_wall + ask_wall) / 2
    except: pass

    return bid_wall, wall_mid, ask_wall
```

Listing 4.1: Wall Mid helper from `FrankfurtHedgehogs_polished.py`, lines 153–166, verbatim.

Every strategy in the Frankfurt Hedgehogs writeup (`imc-prosperity-3/README.md`) feeds its taking decisions and quote skew off this single `wall_mid` number. It is the one custom construction separating their 2nd-place algorithm from a generic top-of-book market maker, and the single piece of Prosperity-specific insight most worth stealing for a 2026 entry.

4.4 A running numerical example

Take the same four-level toy LOB carried through Chapter 2, reproduced with sizes made explicit — for this chapter every size matters.

BID SIDE		**ASK SIDE**	
Qty	Price	Price	Qty
131	102.54	103.23	48
32	101.87	103.98	84
293	101.48	104.17	38
65	101.10	104.75	127

Assume ten trades have just printed in the VWAP window:

Trade	Price	Qty
T_1, T_2, T_3, T_4, T_5	102.80	10 each
T_6, T_7, T_8	103.00	20 each
T_9, T_{10}	103.20	15 each

The most recent trade T_{10} printed at 103.20. We now compute every price measure on this one book.

Bid, ask, spread, mid. By inspection, $p^{\text{bid}} = 102.54$ and $p^{\text{ask}} = 103.23$, so

$$s = p^{\text{ask}} - p^{\text{bid}} = 103.23 - 102.54 = 0.69,$$
$$p^{\text{mid}} = \tfrac{1}{2}(102.54 + 103.23) = 102.885.$$

Microprice. Top-of-book sizes are $Q_b = 131$, $Q_a = 48$, so p^{ask} gets weight $Q_b/(Q_a + Q_b)$ and p^{bid} weight $Q_a/(Q_a + Q_b)$:

$$p^{\text{micro}} = \frac{Q_a \cdot p^{\text{bid}} + Q_b \cdot p^{\text{ask}}}{Q_a + Q_b} = \frac{48 \cdot 102.54 + 131 \cdot 103.23}{48 + 131}.$$

Numerator $48 \cdot 102.54 + 131 \cdot 103.23 = 4{,}921.92 + 13{,}523.13 = 18{,}445.05$; denominator 179; so

$$p^{\text{micro}} = \frac{18{,}445.05}{179} \approx 103.0450.$$

The microprice sits $103.0450 - 102.8850 = 0.1600$ *above* the mid. That is the signal: because the bid is stacked roughly $131/48 \approx 2.7$ times deeper than the ask, the next trade is more likely to be a buy lifting the ask, and the microprice expects the top of book to drift upward.

VWAP. Using the ten trades in the window, the size-weighted numerator is

$$\sum_i p_i q_i = \underbrace{50 \cdot 102.80}_{T_1..T_5} + \underbrace{60 \cdot 103.00}_{T_6..T_8} + \underbrace{30 \cdot 103.20}_{T_9, T_{10}}$$
$$= 5{,}140 + 6{,}180 + 3{,}096 = 14{,}416,$$

and the total traded quantity is $\sum_i q_i = 50 + 60 + 30 = 140$, giving

$$\text{VWAP} = \frac{14{,}416}{140} \approx 102.9714.$$

Last-trade price. T_{10} printed at 103.20, so $p^{\text{last}} = 103.20$.

Wall Mid. The bid wall is the lowest visible bid, $p^{\text{bid,wall}} = 101.10$; the ask wall the highest visible ask, $p^{\text{ask,wall}} = 104.75$. So

$$p^{\text{wall}} = \tfrac{1}{2}(101.10 + 104.75) = \tfrac{1}{2} \cdot 205.85 = 102.925.$$

Summary table. All six price measures on one book:

Symbol	Measure	Value
p^{bid}	Best bid	102.54
p^{ask}	Best ask	103.23
s	Spread	0.69
p^{mid}	Mid-price	102.885
p^{micro}	Microprice (Q_b=131, Q_a=48)	103.0450
VWAP	VWAP (10-trade window)	102.9714
p^{last}	Last-trade price	103.20
p^{wall}	Wall Mid	102.925

Six numbers, all nominally near "one hundred and three," spread across a range of $103.20 - 102.54 = 0.66$ — almost exactly the quoted spread. This dispersion is the point. Each number answers a different question, and confusing them is the cheapest way a new quant loses money.

Which price when?

- **Quoting a passive limit order** (market making) $\rightarrow p^{\text{micro}}$ or p^{wall}, not p^{mid}. Use the one the book's top-of-book noise behaviour prefers.
- **Marking a position to market** at session end $\rightarrow p^{\text{last}}$ for liquid products, p^{mid} for illiquid ones where the last trade is stale.
- **Benchmarking an execution** over a time window $\rightarrow$ VWAP over that window. Nothing else.
- **Estimating "fair value"** as an input to a taking or hedging decision $\rightarrow p^{\text{micro}}$ on a real exchange, p^{wall} on Prosperity.

Never use a single one of the six numbers for all four purposes.

4.5 Fair value, informally

Every price measure in this chapter is a *statistic*: a deterministic function of the LOB or the recent trade tape. None estimates a latent quantity. The word quants use for that latent quantity is **fair value**; it is worth saying clearly what it means, since the term is thrown around casually on trading desks.

Definition 4.9 (Fair value, informally)**.** The *fair value* of an asset at time t is, informally, the price at which the asset would trade in a frictionless market in which all currently-available information had already been fully incorporated into prices. It is not an observable number; it is a model-dependent *estimate*, and different models will give different fair values for the same asset at the same instant.

The informal definition suffices here. Two rigorous instances appear later. Chapter 5 covers Roll's 1984 bid–ask spread model, which defines the *efficient price* as a latent random walk of unobservable innovations and extracts it from trade-price autocovariance alone. Chapter 16 covers Black–Scholes, where the fair value of a European option is the unique no-arbitrage price implied by the underlying's dynamics — rigorous once inputs are fixed, but inheriting all the fragility of those inputs. In both cases fair value is the output of a model, and the model is only as good as its inputs and assumptions.

The fair-value trap

The most seductive mistake in quant trading is to build a fair-value model, believe its output, and trade size against it. A fair-value model can be wrong in at least three uncorrelated ways: (i) its functional form is misspecified; (ii) its inputs are stale, noisy, or wrong; (iii) its calibration data come from a regime that no longer applies. Any one alone produces a confident estimate dollars away from where the market trades. Chapter 21 treats the calibration trap: a model fit on a window with survivorship bias or look-ahead agrees with that history by construction and disagrees with live markets exactly when it matters. The discipline is to treat every fair-value number as one signal among many and to size positions against the *dispersion* of candidate fair values, not the one you like best.

4.6 Key facts to memorise

- p^{bid} and p^{ask} are the top of the two sides of the LOB; $s = p^{\text{ask}} - p^{\text{bid}}$ and $p^{\text{mid}} = (p^{\text{bid}} + p^{\text{ask}})/2$. These four numbers are the minimum summary of a market's state, but *none* of them is "the price."

- The **microprice** is the cross-size-weighted mid: $p^{\text{micro}} = (Q_a p^{\text{bid}} + Q_b p^{\text{ask}})/(Q_a + Q_b)$. The side with less size gets the larger weight, because that side will break first.

- The microprice is a short-horizon predictor of the next mid move; its motivating quantity is *order-flow imbalance*, developed in Chapter 8.

- **VWAP** is a *backward-looking* benchmark over a window, not a fair-value estimate. It is the standard yardstick for institutional execution quality and the reference point for Chapter 19.

- The **last-trade price** is the cheapest summary of where the market is trading, used for end-of-day marking in liquid products, but it is noisy, stale in illiquid products, and manipulable near the close.

- The **Wall Mid** p^{wall} is the Frankfurt Hedgehogs' Prosperity-specific fair-value proxy: the midpoint of the outermost visible bid and ask levels, which in Prosperity correspond to the designated-market-maker bot quotes.

- **Fair value** is not a statistic; it is a model-dependent estimate of a latent frictionless price, and different models give different numbers. Every model is only as good as its inputs and its calibration data (Chapter 21).

- The one-sentence rule: *there is no single price, only price measures, and the skill is knowing which measure answers which question.*

Part III

Market Microstructure

Chapter 5

Why Spreads Exist: Three Causes and the Roll Model

A typical US large-cap stock has a bid–ask spread as tight as one cent on a share worth two hundred dollars — five basis points. Why not zero? Why not ten cents? No regulator, exchange, or single market maker sets the width. The spread is **endogenous**: the equilibrium output of competition between market makers posting quotes and the order flow crossing them, determined by three economic forces.

This chapter has two jobs. First, it names the three forces: *order-processing costs*, *inventory risk*, and *adverse selection*. That three-cause framework (Hasbrouck, 2007) organises Chapters 5 to 7 and 9. Second, it derives the **Roll (1984) spread estimator** (Roll, 1984) from first principles: given only a tape of trade prices — no quotes, no volumes — the spread leaves a recoverable fingerprint in the lag-1 autocovariance of price changes. Four lines of algebra and one square root suffice.

Prerequisites

From earlier chapters:

- Chapter 2: the limit order book, resting limit orders, market orders versus limit orders, makers versus takers, tick size and lot size. The working definitions of p^{bid}, p^{ask}, s, and p^{mid} as quoted quantities.
- Chapter 3: the **informed** vs **uninformed** split, and the adverse-selection toy calculation of Section 3.2 in which a market maker who quotes a zero-spread mid loses an expected \$1.5 per fill against informed counterparties.
- Chapter 4: the formal definitions of best bid, best ask, spread, mid, and microprice.

From outside the book:

- Random variables, expectation, variance, covariance, independence; in particular the identities $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X, Y)$ and $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.
- The notion of an **iid sequence** and of a **random walk**. Appendix A has a one-page refresher if any of these are rusty.

Programming is optional: the Roll estimator is presented as pure arithmetic in the worked example and as a five-line Python snippet in the Prosperity box. Either route teaches it.

5.1 Three reasons the spread is positive

A market maker who quotes a positive spread is compensated for three distinct costs. If any one were removed, the others would still keep the spread above zero. The decomposition is conceptual, not mechanical, and different products weight the three components differently. Recognising which dominates in a given market is most of what a microstructure researcher does.

Order-processing costs

The most mundane component. Running a market-making operation costs money: exchange connectivity fees, colocation rent, clearing and settlement charges per trade, salaries of the quants and developers maintaining the quoting system, and per-message exchange fees for every quote update. These costs must be amortised across the trades the market maker fills. If the spread is narrower than the per-trade cost floor, the market maker runs at a loss on every fill. Call this minimum width the **cost floor**: the reason spreads on even the most contested products do not collapse to zero.

Two counterintuitive points. First, order-processing costs are *small* on the most liquid products — US large-cap stocks, major currency pairs, top futures — because volume is enormous and fixed costs amortise over billions of trades. Second, despite this, they often *dominate* the spread on exactly those products. Inventory and adverse selection are both much smaller on a product like SPY (positions unwind in milliseconds; informed flow has little edge) so the order-processing floor is what remains, and that is why SPY's spread is stuck at one cent. On a thinly-traded small-cap the spread is wide, but order-processing is the *smallest* of its three components.

Inventory risk

A market maker who fills a buy now holds a long position, whether they wanted one or not. That position is exposed to price risk until unwound — seconds on a liquid product, minutes on a slower one, hours on an illiquid one. During that interval the position can move against them: a downtick in the efficient price on a long inventory is a real dollar loss. A rational market maker demands compensation for that exposure, and the compensation shows up in the spread as an additive term on both sides of the mid.

Formally, if the market maker holds inventory q over a window during which the efficient price has variance $\sigma^2 t$, the variance of P&L is $q^2\sigma^2 t$, and a mean-variance-optimising maker's compensation scales with that variance. The Avellaneda–Stoikov treatment in Chapter 9 derives the optimal quoting formulas and shows how the market maker *skews* quotes asymmetrically when inventory drifts away from zero. For this chapter the qualitative picture suffices: the spread includes a risk premium that grows with (i) efficient-price volatility, (ii) expected holding time, and (iii) risk aversion.

Adverse selection

In plain English: *adverse selection is the cost of getting picked off by a counterparty who knows something you don't.* A market maker who quotes two-sided will be preferentially hit by traders whose information makes the quote look mispriced, so the average fill is systematically a loser unless the quote is wide enough to charge for the leak. This is the largest of the three components on mid-liquidity products and the theoretical meat of modern microstructure. Recall Section 3.2: a market maker faces flow with fraction $\alpha = 0.3$ *informed* traders who know the true fair value ($V = 100$ or $V = 110$), the rest uninformed. Quoting at the unconditional mid of 105, expected profit per fill is $-\$1.5$: fills against informed flow are systematically on the side of the trade that will turn out correct, so conditional on a fill the market maker's expected P&L is strictly negative.

The fix is to quote a *spread*. An ask above the mid and a bid below it is a tax levied on every trade. The tax must be set high enough that gains on noise trades offset losses on informed ones. That equilibrium spread is what the Glosten–Milgrom (1985) and Kyle (1985) models derive in Chapter 6. The structural observation: informed flow forces the spread strictly above zero, and wider informed fraction means wider spread.

Adverse selection is hardest to measure directly and easiest to get wrong in backtests. A strategy that looks profitable on historical mid-to-mid data may pay a live adverse-selection premium that historical mids never showed — historical mids are where the market quoted, not where the strategy would have been filled. Chapter 21 returns to this failure mode.

The three-causes decomposition

The quoted bid–ask spread on any market decomposes conceptually into three strictly positive components:

$$s = \underbrace{c_{\text{proc}}}_{\text{order-processing cost}} + \underbrace{c_{\text{inv}}}_{\text{inventory-risk premium}} + \underbrace{c_{\text{adv}}}_{\text{adverse-selection premium}}.$$

All three are strictly positive in equilibrium. Their relative weights depend on the product: on the most liquid products the *order-processing* floor dominates; on mid-liquidity and illiquid products *adverse selection* dominates; inventory risk is a second-order contributor on liquid products and a first-order one on illiquid ones. No rational market maker quotes a spread narrower than the sum of the three.

Intuition

The spread is rent the market maker collects for three services. First, they run the plumbing (exchange connections, software) which costs money. Second, they hold a position for you when nobody else will, charging a premium for the risk. Third, they insure everyone against adverse selection by better-informed counterparties. Three costs, three premiums, one observable width. The second half of this chapter shows that *even without seeing the quotes* you can recover the total width, because all three premiums leave the same statistical fingerprint on the trade tape.

5.2 The Roll (1984) model

Roll's puzzle, set in the early 1980s: intraday tick data was expensive and often impossible to obtain retrospectively. Researchers studying historical transaction costs had access only to the *tape* of printed trades — a time-ordered sequence of trade prices, with no quotes, quantities, or trade-direction labels. Can you estimate the effective bid–ask spread using only the trade-price time series?

Roll's answer: yes. The key insight is the **bid–ask bounce**: because trades alternate between hitting the bid and hitting the ask, trade prices zig-zag around the efficient price even when no new information arrives, and that zig-zag leaves a distinctive negative correlation between consecutive price changes. Under two assumptions — a latent random-walk efficient price and iid trade directions — the trade-price series has a negative lag-1 autocovariance whose magnitude depends only on the spread. Invert the relationship and you have a spread estimator. The argument is four lines long.

Setup

We model two quantities. The first is an unobservable **efficient price** m_t — the hypothetical price at which the product would trade in a frictionless market where all currently available information has already been incorporated. We assume it follows a random walk:

$$m_t \;=\; m_{t-1} + u_t, \tag{5.1}$$

where u_t is an i.i.d. shock sequence with $\mathbb{E}[u_t] = 0$ and $\mathrm{Var}(u_t) = \sigma_u^2$. Shocks represent new information moving the fundamental price; mean-zero means no drift on average, i.i.d. means m_{t-1} tells you nothing about u_t.

The second quantity is the **observable trade price** p_t — what actually prints on the tape. Each trade happens either at the bid (if it was a customer sell hitting a resting buy quote) or at the ask (if it was a customer buy hitting a resting sell quote). Introduce a **trade-direction indicator** q_t taking the value

$$q_t \;=\; \begin{cases} +1 & \text{if the } t\text{-th trade was a customer buy (filled at the ask),} \\ -1 & \text{if the } t\text{-th trade was a customer sell (filled at the bid).} \end{cases}$$

The market maker posts symmetric quotes around the efficient price: a bid at $m_t - s/2$ and an ask at $m_t + s/2$, where s is the (constant) spread. The observable trade price is therefore

$$p_t \;=\; m_t + q_t \cdot \frac{s}{2}. \tag{5.2}$$

This is the key structural equation. The trade price is the efficient price plus a *bounce* of half a spread, with the sign of the bounce determined by which side of the book the trade hit.

We make three further assumptions, each of which is the formal version of a convenient fiction:

1. q_t is symmetric iid with $\mathbb{P}(q_t = +1) = \mathbb{P}(q_t = -1) = 1/2$. In particular $\mathbb{E}[q_t] = 0$ and $\mathbb{E}[q_t^2] = 1$, so $\mathrm{Var}(q_t) = 1$.

2. $\{u_t\}$ and $\{q_t\}$ are independent of each other and of their own pasts — i.e. the efficient-price shocks carry no information about the direction of the current or future trades, and trade directions carry no information about the efficient-price shocks.

3. The spread s is constant across the sample window.

All three are false in real markets (Section 5.3 catalogues how). But the estimator they produce remains the workhorse for spread estimation from end-of-day data four decades later.

The question: *given only $\{p_t\}$, can we recover s?*

Worked toy example, before any derivation

Before the algebra, do the experiment. Pick constant efficient price $m_t = 100.00$ (set $\sigma_u = 0$, so price changes come from the bounce alone), spread $s = 0.10$ (so $s/2 = 0.05$), and a sequence of 13 trade directions

$$q_1, \ldots, q_{13} \;=\; -1, +1, -1, -1, +1, -1, -1, +1, -1, -1, +1, +1, -1$$

(5 customer buys, 8 customer sells, not alternating). Apply Equation (5.2) to get the observable prices:

$$p_t \;=\; 100.00 + q_t \cdot 0.05,$$

which gives

t	1	2	3	4	5	6	7	8	9	10	11	12	13
q_t	-1	$+1$	-1	-1	$+1$	-1	-1	$+1$	-1	-1	$+1$	$+1$	-1
p_t	99.95	100.05	99.95	99.95	100.05	99.95	99.95	100.05	99.95	99.95	100.05	100.05	99.95

A tape of thirteen trade prices, all either 99.95 or 100.05 — looks like noise around a round number. The differences $\Delta p_t = p_t - p_{t-1}$ are

t	2	3	4	5	6	7	8	9	10	11	12	13
Δp_t	$+0.10$	-0.10	0	$+0.10$	-0.10	0	$+0.10$	-0.10	0	$+0.10$	0	-0.10

Twelve price changes. Half are zero (two fills on the same side); the rest are ± 0.10. The mean of the twelve is

$$\overline{\Delta p} \;=\; \frac{1}{12}\big(0.1 - 0.1 + 0 + 0.1 - 0.1 + 0 + 0.1 - 0.1 + 0 + 0.1 + 0 - 0.1\big) \;=\; \frac{0}{12} \;=\; 0,$$

matching Equation (5.1) with $\sigma_u = 0$: no drift. Now look at pairs of consecutive price changes. No $(+0.10, +0.10)$ or $(-0.10, -0.10)$ pairs appear; products $\Delta p_t \cdot \Delta p_{t-1}$ are either 0 (one side zero) or -0.01 (opposite signs). *None is positive.* The sample lag-1 autocovariance will be strictly negative — Roll's point, because a pure random walk would predict zero.

We will compute the exact number in Section 5.2 and recover the spread. But first, the general derivation.

Derivation

Return to the general model, with σ_u possibly nonzero, and compute the first two moments of Δp_t. Each step substitutes Equations (5.1) and (5.2) and expands.

Step 1: rewrite Δp_t. From Equation (5.2),

$$\begin{aligned}
\Delta p_t &= p_t - p_{t-1} \\
&= \big(m_t + q_t \cdot \tfrac{s}{2}\big) - \big(m_{t-1} + q_{t-1} \cdot \tfrac{s}{2}\big) \\
&= \underbrace{(m_t - m_{t-1})}_{= u_t \text{ by Equation (5.1)}} + \frac{s}{2}\,(q_t - q_{t-1}) \\
&= u_t + \frac{s}{2}\,(q_t - q_{t-1}).
\end{aligned}$$

Price changes decompose into an efficient-price shock plus a *bounce* term proportional to the change in trade direction. The bounce is zero when two consecutive trades hit the same side and $\pm s$ when they hit opposite sides.

Step 2: compute $\mathrm{Var}(\Delta p_t)$. Using the identity $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$ and the independence of $\{u_t\}$ and $\{q_t\}$,

$$\begin{aligned}
\mathrm{Var}(\Delta p_t) &= \mathrm{Var}(u_t) + \left(\frac{s}{2}\right)^2 \mathrm{Var}(q_t - q_{t-1}) + 2 \cdot \frac{s}{2} \cdot \underbrace{\mathrm{Cov}(u_t,\, q_t - q_{t-1})}_{=0} \\
&= \sigma_u^2 + \frac{s^2}{4}\,\mathrm{Var}(q_t - q_{t-1}).
\end{aligned}$$

The cross-covariance vanishes because u_t is independent of the q's by assumption. Now expand $\mathrm{Var}(q_t - q_{t-1})$ using the same identity and the independence of q_t and q_{t-1}:

$$\mathrm{Var}(q_t - q_{t-1}) \;=\; \mathrm{Var}(q_t) + \mathrm{Var}(q_{t-1}) - 2\,\underbrace{\mathrm{Cov}(q_t, q_{t-1})}_{=0 \text{ (iid)}} \;=\; 1 + 1 - 0 \;=\; 2,$$

using $\mathrm{Var}(q_t) = \mathbb{E}[q_t^2] - (\mathbb{E}[q_t])^2 = 1 - 0 = 1$ from the symmetric-± 1 assumption. Therefore

$$\mathrm{Var}(\Delta p_t) \;=\; \sigma_u^2 \;+\; \frac{s^2}{2}. \tag{5.3}$$

Instructive already: the variance of observable price changes sums an information piece (σ_u^2) and a bounce piece ($s^2/2$). Two unknowns; we need a second equation.

Step 3: compute $\mathrm{Cov}(\Delta p_t, \Delta p_{t-1})$. Substitute Step 1 for both Δp_t and Δp_{t-1}:

$$\mathrm{Cov}(\Delta p_t, \Delta p_{t-1}) \;=\; \mathrm{Cov}\big(\, u_t + \tfrac{s}{2}(q_t - q_{t-1}),\; u_{t-1} + \tfrac{s}{2}(q_{t-1} - q_{t-2}) \,\big)\,.$$

Expand using bilinearity of covariance. This gives four terms:

$$\begin{aligned} \mathrm{Cov}(\Delta p_t, \Delta p_{t-1}) \;=\; & \mathrm{Cov}(u_t, u_{t-1}) \\ & +\; \tfrac{s}{2}\,\mathrm{Cov}\big(u_t,\, q_{t-1} - q_{t-2}\big) \\ & +\; \tfrac{s}{2}\,\mathrm{Cov}\big(q_t - q_{t-1},\, u_{t-1}\big) \\ & +\; \tfrac{s^2}{4}\,\mathrm{Cov}\big(q_t - q_{t-1},\, q_{t-1} - q_{t-2}\big). \end{aligned}$$

The first term vanishes: $\mathrm{Cov}(u_t, u_{t-1}) = 0$ because $\{u_t\}$ is i.i.d.. The second and third terms vanish because u and q are independent. Only the fourth term survives:

$$\mathrm{Cov}(\Delta p_t, \Delta p_{t-1}) \;=\; \frac{s^2}{4}\,\mathrm{Cov}\big(q_t - q_{t-1},\, q_{t-1} - q_{t-2}\big). \tag{5.4}$$

Step 4: expand the surviving q-covariance. Use $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$. Since $\mathbb{E}[q_t] = 0$ for all t, we have $\mathbb{E}[q_t - q_{t-1}] = 0$ and $\mathbb{E}[q_{t-1} - q_{t-2}] = 0$, so both $\mathbb{E}$'s vanish and we need only

$$\mathbb{E}\big[(q_t - q_{t-1})(q_{t-1} - q_{t-2})\big].$$

Multiply out the brackets:

$$(q_t - q_{t-1})(q_{t-1} - q_{t-2}) \;=\; q_t q_{t-1} \;-\; q_t q_{t-2} \;-\; q_{t-1}^2 \;+\; q_{t-1} q_{t-2}.$$

Now take expectations term by term, using iid of the q's:

$$\begin{aligned} \mathbb{E}[q_t q_{t-1}] \;&=\; \mathbb{E}[q_t] \cdot \mathbb{E}[q_{t-1}] \;=\; 0 \cdot 0 \;=\; 0, \\ \mathbb{E}[q_t q_{t-2}] \;&=\; \mathbb{E}[q_t] \cdot \mathbb{E}[q_{t-2}] \;=\; 0 \cdot 0 \;=\; 0, \\ \mathbb{E}[q_{t-1}^2] \;&=\; 1 \quad (\text{since } q_{t-1} \in \{-1, +1\} \text{ always}), \\ \mathbb{E}[q_{t-1} q_{t-2}] \;&=\; \mathbb{E}[q_{t-1}] \cdot \mathbb{E}[q_{t-2}] \;=\; 0 \cdot 0 \;=\; 0. \end{aligned}$$

Summing with the signs from the product expansion:

$$\mathbb{E}\big[(q_t - q_{t-1})(q_{t-1} - q_{t-2})\big] \;=\; 0 \;-\; 0 \;-\; 1 \;+\; 0 \;=\; -1.$$

Substituting back into Equation (5.4):

$$\boxed{\mathrm{Cov}(\Delta p_t, \Delta p_{t-1}) \;=\; -\frac{s^2}{4}.} \tag{5.5}$$

Step 5: invert to get the estimator. The population lag-1 autocovariance of Δp_t is exactly $-s^2/4$ — a negative number whose magnitude is one quarter of the squared spread. Solving for s:

$$\boxed{s \;=\; 2\sqrt{-\,\mathrm{Cov}(\Delta p_t, \Delta p_{t-1})}.} \tag{5.6}$$

Replace the population covariance with its sample analogue. If the tape gives N price changes $\Delta p_1, \ldots, \Delta p_N$, define $\hat{c}_1 := \frac{1}{N}\sum_{i=1}^{N-1}(\Delta p_{i+1} - \overline{\Delta p})(\Delta p_i - \overline{\Delta p})$ and you have the **Roll spread estimator**

$$\widehat{s}_{\mathrm{Roll}} \;:=\; 2\sqrt{-\,\hat{c}_1} \qquad \text{(undefined when } \hat{c}_1 \ge 0\text{)}.$$

Everything needed is in the trade-price series: no quotes, sizes, or direction labels required.

Roll spread estimator

Under the Roll model: (i) m_t is an efficient price following $m_t = m_{t-1} + u_t$ with u_t i.i.d. mean zero; (ii) the observable trade price is $p_t = m_t + q_t s/2$ with $q_t \in \{+1, -1\}$ symmetric i.i.d.; (iii) $\{u_t\}$ and $\{q_t\}$ are independent and the spread s is constant. Then

$$\mathrm{Cov}(\Delta p_t, \Delta p_{t-1}) \;=\; -\frac{s^2}{4}, \qquad s \;=\; 2\sqrt{-\,\mathrm{Cov}(\Delta p_t, \Delta p_{t-1})}.$$

The Roll estimator replaces the population covariance with its sample analogue; it is well-defined whenever the sample lag-1 autocovariance of price changes is negative.

Three remarks about this result before we feed it numbers.

Remark 5.1 (The efficient-price variance cancels). Equations (5.3) and (5.5) together say: variance of price changes has two contributions (information and bounce) but the lag-1 autocovariance has one — the bounce alone. i.i.d. information innovations contribute zero lag-1 autocorrelation; only the bounce survives at lag one, and it depends only on the spread. Hence the Roll estimator's robustness to efficient-price volatility: σ_u^2 dominates $\mathrm{Var}(\Delta p_t)$ but does not leak into the autocovariance.

Remark 5.2 (Sign). The population autocovariance is *always* negative in this model. A positive sample autocovariance signals a model-assumption violation; the estimator is inapplicable. Section 5.3 lists the usual culprits.

Remark 5.3 (Units). The estimator returns s in the same units as p_t. Tape in seashells gives spread in seashells; Roll is scale-covariant.

Applying Roll to the toy example

Return to the thirteen trades of Section 5.2. The twelve price changes were

$$\Delta p_2, \ldots, \Delta p_{13} \;=\; (+0.10,\ -0.10,\ 0,\ +0.10,\ -0.10,\ 0,\ +0.10,\ -0.10,\ 0,\ +0.10,\ 0,\ -0.10).$$

The sample mean is $\overline{\Delta p} = 0$ (already computed). Because the mean is exactly zero, the sample lag-1 autocovariance simplifies to

$$\hat{c}_1 \;=\; \frac{1}{N}\sum_{i=1}^{N-1} \Delta p_{t_{i+1}} \cdot \Delta p_{t_i},$$

where $N = 12$ is the number of price changes and the 11 products are listed below.

Write out the eleven consecutive-pair products $\Delta p_{t+1} \cdot \Delta p_t$ explicitly:

Pair	Values	Product
$(\Delta p_2, \Delta p_3)$	$(+0.10, -0.10)$	-0.01
$(\Delta p_3, \Delta p_4)$	$(-0.10, 0)$	0
$(\Delta p_4, \Delta p_5)$	$(0, +0.10)$	0
$(\Delta p_5, \Delta p_6)$	$(+0.10, -0.10)$	-0.01
$(\Delta p_6, \Delta p_7)$	$(-0.10, 0)$	0
$(\Delta p_7, \Delta p_8)$	$(0, +0.10)$	0
$(\Delta p_8, \Delta p_9)$	$(+0.10, -0.10)$	-0.01
$(\Delta p_9, \Delta p_{10})$	$(-0.10, 0)$	0
$(\Delta p_{10}, \Delta p_{11})$	$(0, +0.10)$	0
$(\Delta p_{11}, \Delta p_{12})$	$(+0.10, 0)$	0
$(\Delta p_{12}, \Delta p_{13})$	$(0, -0.10)$	0
	Sum	-0.03

Three pairs contributed -0.01, the other eight contributed 0; no positive products. Sum: -0.03. Divide by $N = 12$:

$$\hat{c}_1 \;=\; \frac{-0.03}{12} \;=\; -0.0025.$$

Apply Equation (5.6):

$$\widehat{s}_{\text{Roll}} \;=\; 2\sqrt{-\hat{c}_1} \;=\; 2\sqrt{0.0025} \;=\; 2 \cdot 0.05 \;=\; 0.10.$$

Recovered spread is exactly 0.10, matching the input at Section 5.2. Job done, tape alone.

Intuition

The spread as a statistical signature of trading. Shown a tape of thirteen prices cold, most observers would call it noise. Roll's observation: the noise has *structure*. Every time a trade hits the opposite side of the book, the price travels the full spread; same side, no move (under $\sigma_u = 0$). The bounce is a deterministic function of q_t, and consecutive bounces are negatively correlated because $q_t - q_{t-1}$ is overwhelmingly likely to reverse sign on the next step.

The fingerprint survives a genuinely random efficient price. i.i.d. shocks u_t inflate $\mathrm{Var}(\Delta p_t)$ but contribute nothing at lag one. Spread-signal and information-noise live in orthogonal statistics; Roll's estimator reads the statistic that isolates the spread.

5.3 When Roll works, and when it doesn't

Roll's derivation relied on three assumptions: independent-increment efficient price, iid symmetric trade directions, constant spread. Each fails in real markets in a specific way, biasing the estimator in a specific direction.

Assumption 1: trade directions are not iid

In real markets, trade directions cluster. A parent order worked through a VWAP algorithm generates a long run of buys, followed by a long run of sells when the next institutional order arrives. Runs produce long sequences of zero Δp punctuated by jumps; the sample lag-1 autocovariance is pushed toward zero, biasing the Roll estimate downward. The Hawkes-process treatment in Chapter 8 formalises this and gives corrections.

Assumption 2: the efficient price is not independent of order flow

Informed flow moves the efficient price in the direction of the trade: an informed buy at the ask is followed, on average, by a permanent upward drift as beliefs update. This permanent impact creates positive correlation between the bounce and the next efficient-price shock, spilling into the lag-1 autocovariance with opposite sign. The effects partly cancel, and Roll under-reports the true spread whenever adverse selection is material. The Glosten–Milgrom and Kyle models of Chapter 6 formalise this, and the Hasbrouck VAR decomposition corrects for it given quote-level data.

Assumption 3: the spread is not constant

Spreads widen around news, open, and close; narrow midday; widen when large makers step back. Roll applied across a regime change returns a time-average, biased toward whichever regime dominates the sample.

Don't apply Roll to illiquid data

Roll fails worst on illiquid products with few prints. A stock trading three times an hour on a day with real-information drift has Δp_t observations dominated by u_t shocks, not the bounce — so $\sigma_u^2 \gg s^2/2$ in Equation (5.3). Population lag-1 autocovariance is still $-s^2/4$, but the sample version from a handful of drift-dominated changes can easily come out positive by sampling noise, leaving the estimator undefined (`NaN`).
Worse, an analyst who flips the sign by hand — "surely it must be negative, take the absolute value" — computes a statistic with no theoretical backing. Roll is a *model*, and on illiquid data the model is misspecified. When sample autocov comes out positive, discard, don't patch. For illiquid products use the Corwin–Schultz high-low estimator, or direct quote data if available.

These caveats make Roll a *starting point*, not the end. Model-light, quick, requires only trade prices. Modern practice uses it as a sanity check and default when nothing better is available, cross-validated against Hasbrouck or direct quotes when those exist.

5.4 A Prosperity application

The Roll estimator transfers cleanly to Prosperity, where trades are the simplest data channel and quotes (though available through `TradingState`) can be noisy with competitor-algo signalling. Below: Roll on a simulated `RAINFOREST_RESIN` tape (success) and a drifting-`KELP` tape (failure).

Prosperity example

Roll on `RAINFOREST_RESIN` (stationary, iid directions). Prosperity's `RAINFOREST_RESIN` trades around a near-constant fair value of about 10,000 seashells, quoted with a small market-maker spread of $s = 0.40$ seashells ($s/2 = 0.20$). Take a stretch of 13 consecutive trade prints with an efficient price of 10,000 throughout ($\sigma_u = 0$) and the same trade-direction sequence as Section 5.2:

$$q = (-1, +1, -1, -1, +1, -1, -1, +1, -1, -1, +1, +1, -1).$$

Using $p_t = 10{,}000 + q_t \cdot 0.20$, the trade prices are

$$9999.8,\ 10000.2,\ 9999.8,\ 9999.8,\ 10000.2,\ 9999.8,\ 9999.8,$$

$$10000.2,\ 9999.8,\ 9999.8,\ 10000.2,\ 10000.2,\ 9999.8.$$

The twelve price changes are

$$\Delta p \;=\; (+0.4, -0.4,\, 0, +0.4, -0.4,\, 0, +0.4, -0.4,\, 0, +0.4,\, 0, -0.4).$$

The mean price change is 0 exactly (the sequence is balanced). Each non-zero product of consecutive price changes is -0.16; three such pairs contribute, all others are zero, so

$$\sum_{t=2}^{N-1} \Delta p_{t+1}\,\Delta p_t \;=\; 3\cdot(-0.16) \;=\; -0.48, \qquad \hat{c}_1 \;=\; \frac{-0.48}{12} \;=\; -0.04.$$

The Roll estimator returns

$$\widehat{s}_{\text{Roll}} \;=\; 2\sqrt{0.04} \;=\; 2\cdot 0.2 \;=\; 0.40,$$

recovering the true spread exactly. The happy case: `RAINFOREST_RESIN` is stationary with a tight, near-constant spread and balanced flow. Roll is the right tool.

Roll on a drifting `KELP` tape (fails). Now suppose `KELP` is trending upward during the sample window — its efficient price drifting by +0.5 seashells per trade — and nine consecutive taker orders arrive on the ask (so $q_t = +1$ throughout). With a true spread of $s = 1$, the trade prices are

$$\begin{gathered} 2000.5,\ 2001.0,\ 2001.5,\ 2002.0,\ 2002.5, \\ 2003.0,\ 2003.5,\ 2004.0,\ 2004.5,\ 2005.0, \end{gathered}$$

giving nine price changes $\Delta p_t = +0.5$ for every t. The sample mean of Δp is 0.5, and every centred change $(\Delta p_t - 0.5)$ is exactly zero, so

$$\hat{c}_1 \;=\; \frac{1}{N}\sum_t (\Delta p_{t+1} - 0.5)(\Delta p_t - 0.5) \;=\; 0.$$

Plugging into Equation (5.6): $\widehat{s}_{\text{Roll}} = 0$, on a product with true spread 1. Trending plus one-sided flow destroyed both assumptions at once; Roll returned garbage.

Lesson. Roll fits stationary, two-sided, uninformed-flow regimes like `RAINFOREST_RESIN`; it fails on trending one-sided regimes like `KELP` during a directional burst. For the latter, compute the spread directly from the top of book in `TradingState`.

Reference implementation. A five-line NumPy version of the Roll estimator, suitable for dropping into any Prosperity research notebook:

```python
import numpy as np

def roll_spread(prices: np.ndarray) -> float:
    """Estimate the effective spread from a trade-price series."""
    dp = np.diff(prices)                        # N price changes
    dp_c = dp - dp.mean()                       # centre once, full-sample
    c1 = (dp_c[1:] * dp_c[:-1]).sum() / len(dp)    # lag-1 autocov, /N
    if c1 >= 0:
        return float("nan")                     # Roll undefined
    return 2.0 * np.sqrt(-c1)
```

Listing 5.1: Roll spread estimator from trade prices.

Feed it the RESIN array and it returns 0.40; feed it KELP and it returns `nan`, matching the hand arithmetic. Subtlety: `np.cov` with `bias=True` would differ because it re-centres each slice by its own mean and divides by $N-1$. The implementation above matches the chapter's derivation exactly.

On a Goldman Sachs desk

On a sell-side desk, Roll is a workhorse for historical spread estimation from end-of-day OHLC on products without licensed intraday tick data. A PM asking "what was the effective spread on the Mexican peso non-deliverable forward in February 2019?" typically gets a Roll-based estimate unless the desk paid for the tick archive — and for exotic products, usually they haven't.

On HFT desks with abundant tick data, Roll is superseded by the Hasbrouck VAR decomposition, which separates permanent (information) and transient (noise) impacts and recovers Kyle's λ plus the adverse-selection share of the spread. But for daily/weekly backtests and execution-assumption validation, Roll remains the default. Chapter 21 covers walk-forward validation: estimate the spread on the trailing window, apply it as cost on the next, iterating through the sample.

5.5 Key facts to memorise

- **Three-causes decomposition.** The bid–ask spread decomposes into *order-processing costs* (the mechanical floor), *inventory risk* (compensation for holding a position), and *adverse selection* (insurance against informed flow). All three strictly positive; their relative weights depend on the product's liquidity.

- **Roll's model.** Trade prices $p_t = m_t + q_t s/2$, where m_t is a latent random-walk efficient price and $q_t \in \{+1, -1\}$ is an iid trade-direction indicator.

- **Roll's result.** $\operatorname{Cov}(\Delta p_t, \Delta p_{t-1}) = -s^2/4$, so $s = 2\sqrt{-\operatorname{Cov}(\Delta p_t, \Delta p_{t-1})}$. Derivable from the tape alone; no quote data required.

- **Why it works.** The σ_u^2 information-shock variance dominates $\operatorname{Var}(\Delta p_t)$ but contributes *zero* to the lag-1 autocovariance; the spread dominates the lag-1 autocovariance and nothing else. The two statistics are orthogonal.

- **Assumptions required.** Random-walk efficient price, iid symmetric trade directions, constant spread. All three are wrong in real markets in specific, correctable ways.

- **When to use Roll.** Stationary markets, balanced two-sided order flow, tape data only. Default for daily-bar backtests on products without tick data.

- **When not to use Roll.** Trending markets, illiquid products with few prints, one-sided order flow, news-event windows, and *any* sample in which the computed $\hat{c}_1$ comes out positive (which means the model is misspecified; the correct response is to discard, not to patch).

- **Connection to Chapter 3.** The adverse-selection component of the three-cause decomposition is exactly the \$1.5-per-fill loss derived in the toy calculation of Section 3.2: informed flow imposes a tax, and the spread is the market maker's mechanism for collecting that tax back from everyone.

- **Cross-reference map.** Adverse selection in detail: Chapter 6. Inventory-risk premium in detail: Chapter 9. Clustered order flow and the first assumption: Chapter 8. Using Roll inside a historical backtest: Chapter 21.

Chapter 6

Informed Trading: Glosten–Milgrom and Kyle

Why does the market maker charge a spread even when they have zero costs, zero inventory, and zero holding period?

Strip the frictions away. No fees, no clearing charges, no per-message costs; a book-flat starting position; every fill unwound for free at the same mid-price an instant later. The three-cause decomposition of Chapter 5 has silenced two of its three sources. The naive prediction is a zero spread.

The prediction is wrong. Even in this frictionless world the rational market maker quotes a strictly positive spread, because of *adverse selection*: some counterparties know something the market maker does not, and every fill against such a counterparty is a small expected loss. The market maker cannot see who is informed, so they charge everyone a tax wide enough that the gains on the uninformed cover the losses on the informed. That tax is the spread.

This chapter formalises two models of the phenomenon. Glosten–Milgrom (1985) (Glosten and Milgrom, 1985) is sequential: trades arrive one at a time, the market maker updates beliefs by Bayes, and the equilibrium spread falls out of zero expected profit on each trade. Kyle (1985) (Kyle, 1985) is batched and strategic: a single informed trader picks trade size optimally knowing their orders move prices, and a competitive market maker prices the aggregate order flow by projection. Both models will be referenced repeatedly in Chapters 7, 9 and 19.

Prerequisites

From earlier in the book:

- Chapter 2: the limit order book, limit vs market orders, makers and takers, the idea that a market maker's job is to quote a bid and an ask simultaneously and wait for the flow to come to them.
- Chapter 3: the **informed vs uninformed** taxonomy, the toy adverse-selection calculation of Section 3.2 with $\alpha = 0.3$, $V \in \{100, 110\}$ and a zero-spread market maker losing \$1.5 per fill, and the Prosperity 3 Olivia story as ground truth for the existence of informed flow in the wild.
- Chapter 4: the formal definitions of p^{bid}, p^{ask}, s, and p^{mid}.
- Chapter 5: the three-cause decomposition of the spread, the Roll estimator, and the explicit forward references that pointed to *this* chapter for the formal treatment of the adverse-selection component.

From outside the book:

- **Bayes' theorem** and **conditional expectation**: given a prior $\mathbb{P}(V = v)$ and a likelihood $\mathbb{P}(\text{observation} \mid V = v)$, the posterior is $\mathbb{P}(V = v \mid \text{observation}) = \mathbb{P}(\text{observation} \mid V = v)\mathbb{P}(V = v)/\mathbb{P}(\text{observation})$, and $\mathbb{E}[f(V) \mid \text{observation}] = \sum_v f(v)\mathbb{P}(V = v \mid \text{observation})$.
- Elementary random walks and the fact that conditioning on new information updates beliefs about a latent variable.
- For Kyle: the **projection formula** for bivariate Gaussians. If X, Y are jointly normal with $\mathbb{E}[X] = \mathbb{E}[Y] = 0$, then $\mathbb{E}[X \mid Y] = (\operatorname{Cov}(X, Y)/\operatorname{Var}(Y)) \cdot Y$. This is a one-line result; Appendix A has the derivation if it is rusty.
- Basic calculus of one-variable optimisation (taking a derivative and setting it to zero) for the insider's profit-maximisation step.

6.1 Glosten–Milgrom: the sequential trade model

How little machinery is enough to derive a positive spread? The Glosten–Milgrom model uses four ingredients: (i) a binary latent asset value, (ii) informed and uninformed traders in a fixed mix, (iii) a risk-neutral market maker who quotes bid and ask so that each trade has zero expected profit, and (iv) Bayes' theorem. Out of those drops the prediction that the spread is strictly positive whenever $\alpha > 0$, and narrows as the market maker learns from the trade sequence.

Setup

There is a single risky asset with fundamental value V, either high or low:

$$V \in \{V_L, V_H\}, \qquad V_L < V_H.$$

The prior is $\theta_0 := \mathbb{P}(V = V_H) \in (0, 1)$. The true value V is drawn once at the start of the game; only informed traders learn it.

A fraction $\alpha \in [0, 1]$ of arriving traders are **informed**: they know V and trade in the profitable direction.

$$\text{informed trader} \implies \begin{cases} \text{buys} & \text{if } V = V_H, \\ \text{sells} & \text{if } V = V_L. \end{cases}$$

The remaining $1 - \alpha$ are **uninformed** (also **noise** or **liquidity** traders): they trade for reasons exogenous to price (rebalancing, cash needs, hedging). Their buy/sell decisions are independent of V and symmetric:

$$\text{uninformed trader} \implies \begin{cases} \text{buys} & \text{with probability } 1/2, \\ \text{sells} & \text{with probability } 1/2. \end{cases}$$

Each trader arrives, looks at the posted quotes, and trades exactly one unit (GM has nothing to say about order sizing; Kyle below does).

The market maker is **risk-neutral** (cares about expected profit, not variance) and **competitive** (quotes at the break-even level: wider and a competitor undercuts, narrower and they lose money). Competitive risk-neutral pricing means the quoted ask and bid equal the expected value conditional on the next trade being a buy or a sell:

$$p_t^{\text{ask}} = \mathbb{E}[V \mid \text{next trade is a buy}, \mathcal{F}_t], \qquad p_t^{\text{bid}} = \mathbb{E}[V \mid \text{next trade is a sell}, \mathcal{F}_t], \tag{6.1}$$

where $\mathcal{F}_t$ is the history of trades up to time t. Everybody observes each realised trade direction, and the market maker updates $\theta_t := \mathbb{P}(V = V_H \mid \mathcal{F}_t)$ by Bayes before re-posting quotes. Only the ordering of trades matters, not the physical waits between them.

Toy example, worked by hand

Before the general formula, do one Bayesian update with concrete numbers, chosen to match the adverse-selection toy of Section 3.2. Take

$$V_L = 100, \qquad V_H = 110, \qquad \theta_0 = 0.5, \qquad \alpha = 0.3.$$

Five numbered steps.

Step 1: setup.

At $t = 0$, $\mathbb{P}(V = 110) = 0.5$, so the unconditional expectation is

$$\mathbb{E}[V] \;=\; 0.5 \cdot 110 + 0.5 \cdot 100 \;=\; 105.$$

Quoting both bid and ask at 105 loses $-\$1.5$ per fill (Section 3.2). The question is what to quote instead.

Step 2: compute $\mathbb{P}(\text{buy} \mid V)$.

With probability α the arriving trader is informed (buys iff $V = V_H$); with probability $1 - \alpha$ uninformed (buys with probability 1/2). Therefore

$$\begin{aligned}
\mathbb{P}(\text{buy} \mid V = V_H) \;&=\; \underbrace{\alpha \cdot 1}_{\text{informed, sees } V=V_H\text{, buys}} \;+\; \underbrace{(1-\alpha)\cdot \tfrac{1}{2}}_{\text{uninformed, flips a coin}} \\
&=\; 0.3 + 0.7 \cdot 0.5 \;=\; 0.3 + 0.35 \;=\; 0.65,
\end{aligned}$$

and by symmetry

$$\begin{aligned}
\mathbb{P}(\text{buy} \mid V = V_L) \;&=\; \underbrace{\alpha \cdot 0}_{\text{informed, sees } V=V_L\text{, sells}} \;+\; \underbrace{(1-\alpha)\cdot \tfrac{1}{2}}_{\text{uninformed, flips a coin}} \\
&=\; 0 + 0.35 \;=\; 0.35.
\end{aligned}$$

A buy is more likely in the high state ($0.65 > 0.35$), which is what makes a buy informative about V. By symmetry, $\mathbb{P}(\text{sell} \mid V_H) = 0.35$ and $\mathbb{P}(\text{sell} \mid V_L) = 0.65$.

Step 3: compute the posterior $\mathbb{P}(V = V_H \mid \text{buy})$.

The prior is $\theta_0 = 0.5$. The unconditional probability of a buy under the mixture is

$$\mathbb{P}(\text{buy}) \;=\; \theta_0 \cdot \mathbb{P}(\text{buy} \mid V_H) \;+\; (1-\theta_0) \cdot \mathbb{P}(\text{buy} \mid V_L) \;=\; 0.5 \cdot 0.65 + 0.5 \cdot 0.35 \;=\; 0.5.$$

At a symmetric prior the aggregate buy rate is 1/2. Bayes gives

$$\mathbb{P}(V = V_H \mid \text{buy}) \;=\; \frac{\mathbb{P}(\text{buy} \mid V_H) \cdot \mathbb{P}(V_H)}{\mathbb{P}(\text{buy})} \;=\; \frac{0.65 \cdot 0.5}{0.5} \;=\; 0.65.$$

The buy updates the high-state posterior from 0.5 to 0.65; by symmetry $\mathbb{P}(V = V_H \mid \text{sell}) = 0.35$.

Step 4: compute $\mathbb{E}[V \mid \text{buy}]$ and $\mathbb{E}[V \mid \text{sell}]$.

$$\mathbb{E}[V \mid \text{buy}] \;=\; \mathbb{P}(V_H \mid \text{buy}) \cdot V_H \;+\; \mathbb{P}(V_L \mid \text{buy}) \cdot V_L \;=\; 0.65 \cdot 110 + 0.35 \cdot 100 \;=\; 106.5.$$

By Equation (6.1), this is the market maker's ask:

$$p_0^{\text{ask}} \;=\; 106.5.$$

Symmetrically, the bid is

$$p_0^{\text{bid}} \;=\; \mathbb{E}[V \mid \text{sell}] \;=\; 0.35 \cdot 110 + 0.65 \cdot 100 \;=\; 103.5.$$

The implied spread and mid at the start of the game are

$$s_0 \;=\; p_0^{\text{ask}} - p_0^{\text{bid}} \;=\; 106.5 - 103.5 \;=\; 3.0, \qquad p_0^{\text{mid}} \;=\; \frac{p_0^{\text{ask}} + p_0^{\text{bid}}}{2} \;=\; 105.0.$$

Step 5: interpret.

The mid is 105.0, equal to the prior expected value. The spread of 3.0 is positive despite zero costs, zero inventory, and zero holding period — attributable entirely to adverse selection.

Check break-even: the ask equals $\mathbb{E}[V \mid \text{buy}]$, so the expected P&L on the next buy is $p_0^{\text{ask}} - \mathbb{E}[V \mid \text{buy}] = 0$, and by symmetry zero on a sell. The \$1.5-per-fill loss of the naive 105-quote (Section 3.2) has been priced away.

Intuition

The ask 106.5 sits above the prior mean 105 because *"the next trade is a buy" is itself information about V*: a buy is more likely when $V = V_H$, so conditioning on a buy pushes $\mathbb{P}(V_H)$ from 0.5 to 0.65 and the expected value from 105 to 106.5. The bid at 103.5 is below 105 by symmetry.

The spread is no free lunch. The market maker's expected profit per trade is zero. Uninformed traders pay 1.5 away from fair value on each trade; that loss, summed over the $(1-\alpha)$ uninformed fraction, exactly covers the $\alpha = 0.3$ of informed trades on which the market maker loses roughly 5 per trade (half the width $V_H - V_L$). The spread is a **tax on the entire population** used to insure the market maker against the informed minority.

The general GM spread formula

Let $\theta_t := \mathbb{P}(V = V_H \mid \mathcal{F}_t)$ be the posterior after t trades. The likelihoods of the next trade direction, in each state, are functions of α alone:

$$\mathbb{P}(\text{buy} \mid V_H) \;=\; \alpha + \tfrac{1-\alpha}{2} \;=\; \tfrac{1+\alpha}{2}, \qquad \mathbb{P}(\text{buy} \mid V_L) \;=\; \tfrac{1-\alpha}{2}.$$

Symmetrically $\mathbb{P}(\text{sell} \mid V_H) = (1-\alpha)/2$ and $\mathbb{P}(\text{sell} \mid V_L) = (1+\alpha)/2$. The unconditional buy probability at time t is

$$\mathbb{P}_t(\text{buy}) \;=\; \theta_t \cdot \tfrac{1+\alpha}{2} \;+\; (1-\theta_t) \cdot \tfrac{1-\alpha}{2} \;=\; \tfrac{1}{2} + \tfrac{\alpha}{2}(2\theta_t - 1). \tag{6.2}$$

The posterior after a buy at time $t+1$ is

$$\theta_{t+1}^{\text{buy}} \;=\; \frac{\mathbb{P}(\text{buy} \mid V_H)\,\theta_t}{\mathbb{P}_t(\text{buy})} \;=\; \frac{(1+\alpha)\,\theta_t}{1 + \alpha(2\theta_t - 1)}, \tag{6.3}$$

and after a sell

$$\theta_{t+1}^{\text{sell}} \;=\; \frac{(1-\alpha)\,\theta_t}{1 - \alpha(2\theta_t - 1)}. \tag{6.4}$$

The ask and bid for the $(t+1)$-st trade are the conditional expectations of V under these two posteriors:

$$p_t^{\text{ask}} = \mathbb{E}[V \mid \text{buy}, \mathcal{F}_t] = \theta_{t+1}^{\text{buy}} V_H + (1-\theta_{t+1}^{\text{buy}}) V_L, \tag{6.5}$$
$$p_t^{\text{bid}} = \mathbb{E}[V \mid \text{sell}, \mathcal{F}_t] = \theta_{t+1}^{\text{sell}} V_H + (1-\theta_{t+1}^{\text{sell}}) V_L. \tag{6.6}$$

Substituting Equations (6.3) and (6.4) gives the spread as an explicit function of $(\alpha, \theta_t, V_H, V_L)$:

$$s_t = p_t^{\text{ask}} - p_t^{\text{bid}} = (V_H - V_L) \cdot \left[\frac{(1+\alpha)\theta_t}{1+\alpha(2\theta_t - 1)} - \frac{(1-\alpha)\theta_t}{1-\alpha(2\theta_t-1)} \right]. \tag{6.7}$$

The expression has four clean limiting properties.

Limit 1: $\alpha \to 0$. With no informed traders, $\mathbb{P}(\text{buy} \mid V_H) = \mathbb{P}(\text{buy} \mid V_L) = 1/2$, so a buy (or a sell) is uninformative about V, the posteriors $\theta_{t+1}^{\text{buy}} = \theta_{t+1}^{\text{sell}} = \theta_t$, and $p_t^{\text{ask}} = p_t^{\text{bid}}$. The spread collapses to zero. No informed flow $\Rightarrow$ no adverse selection $\Rightarrow$ no spread, exactly as intuition demands.

Limit 2: $\alpha \to 1$. Every trader is informed, so a buy perfectly signals $V = V_H$ and $\theta_{t+1}^{\text{buy}} = 1$, $\theta_{t+1}^{\text{sell}} = 0$. The quotes collapse onto the revealed value, giving

$$s_t = V_H - V_L,$$

the maximum. The market maker quotes only at fair value: any tighter loses, any wider goes unfilled. Dark pools populated entirely by informed hedge funds exhibit this pathology — spreads there are large or the venue goes unused.

Limit 3: symmetric prior $\theta_t = 1/2$. At a symmetric prior the posteriors simplify. From Equations (6.3) and (6.4) with $\theta_t = 1/2$ and $2\theta_t - 1 = 0$,

$$\theta_{t+1}^{\text{buy}} = \frac{(1+\alpha)\cdot 1/2}{1+\alpha\cdot 0} = \frac{1+\alpha}{2}, \qquad \theta_{t+1}^{\text{sell}} = \frac{1-\alpha}{2}.$$

Plugging into Equations (6.5) and (6.6):

$$p_t^{\text{ask}} = \frac{1+\alpha}{2} V_H + \frac{1-\alpha}{2} V_L, \qquad p_t^{\text{bid}} = \frac{1-\alpha}{2} V_H + \frac{1+\alpha}{2} V_L,$$

and

$$s_t = p_t^{\text{ask}} - p_t^{\text{bid}} = \left[\frac{1+\alpha}{2} - \frac{1-\alpha}{2} \right] (V_H - V_L) = \alpha (V_H - V_L).$$

Exactly, not approximately. For $\alpha = 0.3$, $V_H - V_L = 10$ this gives $s_0 = 3$, matching Section 6.1. The symmetric-prior spread $\alpha(V_H - V_L)$ is GM's cleanest formula: linear in the informed share, linear in the value range.

Limit 4: $\theta_t \to 0$ or 1. As the posterior concentrates, both quotes converge to the certain value and the spread collapses. Full learning $\Rightarrow$ zero spread $\Rightarrow$ trading stops.

Glosten–Milgrom spread formula

In the Glosten–Milgrom (1985) model (Glosten and Milgrom, 1985), a risk-neutral, competitive market maker facing a mixture of an α-fraction of informed traders (who know $V \in \{V_L, V_H\}$) and a $(1-\alpha)$-fraction of symmetric uninformed traders posts

$$\begin{aligned} p_t^{\text{ask}} &= \mathbb{E}[V \mid \text{next trade is a buy}, \mathcal{F}_t], \\ p_t^{\text{bid}} &= \mathbb{E}[V \mid \text{next trade is a sell}, \mathcal{F}_t]. \end{aligned}$$

The equilibrium spread is strictly positive whenever $\alpha > 0$, and at a symmetric prior $\theta_0 = 1/2$ with small α it satisfies

$$s_0 = \alpha(V_H - V_L) > 0.$$

The market maker incurs *no* costs, bears *no* inventory risk, and holds positions for *no* time. The entire spread is compensation for adverse selection alone.

How the market maker learns

After each trade the market maker updates θ via Equation (6.3) or Equation (6.4), re-computes quotes, and posts them for the next trader. Over many trades θ_t wanders toward the truth and the quotes track it.

A concrete trajectory with $V_L = 100$, $V_H = 110$, $\alpha = 0.3$, $\theta_0 = 0.5$, and first five trades B, B, B, S, S:

t	Trade	θ_t	p_t^{bid}	p_t^{ask}	s_t
0	—	0.5000	103.500	106.500	3.000
1	B	0.6500	105.000	107.752	2.752
2	B	0.7752	106.500	108.650	2.150
3	B	0.8650	107.752	109.225	1.472
4	S	0.7752	106.500	108.650	2.150
5	S	0.6500	105.000	107.752	2.752

For comparison, a longer mostly-buy trajectory BBBBB-S-BBBB (9 buys, one sell at position 6) from the same prior evolves as

t	Trade	θ_t	p_t^{bid}	p_t^{ask}	s_t
0	—	0.5000	103.500	106.500	3.000
1	B	0.6500	105.000	107.752	2.752
2	B	0.7752	106.500	108.650	2.150
3	B	0.8650	107.752	109.225	1.472
4	B	0.9225	108.650	109.567	0.917
5	B	0.9567	109.225	109.762	0.538
6	S	0.9225	108.650	109.567	0.917
7	B	0.9567	109.225	109.762	0.538
8	B	0.9762	109.567	109.870	0.304
9	B	0.9870	109.762	109.930	0.168
10	B	0.9930	109.870	109.962	0.092

By $t = 10$, $\theta = 0.993$ and the spread has collapsed from 3 to below 0.1 — a 30× compression.

Three observations. First, both mid and spread change after every trade: the mid tracks θ_t ($\theta_t V_H + (1-\theta_t)V_L$) and the spread narrows as θ_t moves away from 1/2, since a more confident posterior makes additional trades less informative.

Second, the posterior trajectory is a **martingale**: $\mathbb{E}[\theta_{t+1} \mid \mathcal{F}_t] = \theta_t$. This is a general fact about Bayesian updating and will be used throughout Chapters 10 and 16; it is worth proving in the GM setting.

Proposition 6.1 (The GM posterior is a martingale)**.** *In the Glosten–Milgrom model, the sequence $\{\theta_t\}_{t\geq 0}$ of posteriors on $V = V_H$ is a martingale with respect to the filtration $\mathcal{F}_t$ generated by the observed trades:*

$$\mathbb{E}[\theta_{t+1} \mid \mathcal{F}_t] \;=\; \theta_t.$$

Proof. Condition on $\mathcal{F}_t$; $\mathbb{P}_t(\text{buy}) + \mathbb{P}_t(\text{sell}) = 1$. Write the expected posterior as a weighted sum of the two possible updates:

$$\begin{aligned}
\mathbb{E}[\theta_{t+1} \mid \mathcal{F}_t] &= \mathbb{P}_t(\text{buy}) \cdot \theta_{t+1}^{\text{buy}} \;+\; \mathbb{P}_t(\text{sell}) \cdot \theta_{t+1}^{\text{sell}} \\
&= \mathbb{P}_t(\text{buy}) \cdot \frac{\mathbb{P}(\text{buy} \mid V_H)\theta_t}{\mathbb{P}_t(\text{buy})} \;+\; \mathbb{P}_t(\text{sell}) \cdot \frac{\mathbb{P}(\text{sell} \mid V_H)\theta_t}{\mathbb{P}_t(\text{sell})} \\
&= \mathbb{P}(\text{buy} \mid V_H)\theta_t \;+\; \mathbb{P}(\text{sell} \mid V_H)\theta_t \\
&= \theta_t \cdot \underbrace{\big(\mathbb{P}(\text{buy} \mid V_H) + \mathbb{P}(\text{sell} \mid V_H)\big)}_{=1} \\
&= \theta_t.
\end{aligned}$$

The $\mathbb{P}_t(\cdot)$ factors cancel in each update ratio, leaving a sum that telescopes to $\theta_t \cdot 1 = \theta_t$. □

The quoted mid $p_t^{\text{mid}} = \theta_t V_H + (1-\theta_t)V_L$ is therefore also a martingale: the market maker has no ex-ante directional bias. Tradable prices in efficient markets are martingales for the same reason: any non-martingale forecast is tradable until the profit disappears.

Third, the trajectory $(0.5, 0.65, 0.7752, 0.8650, 0.7752, 0.65)$ ends where it was four steps earlier. In the binary GM model a BS-pair (in either order) returns the posterior to its starting value — a useful sanity check on hand-computed trajectories. (This is special to the binary case; multi-state models have more elaborate path dependence.)

Intuition

The spread as the price of ignorance. A zero-spread GM market maker loses $\alpha \cdot (V_H - V_L)/2$ per trade to the informed. The spread $\alpha(V_H - V_L)$ (at $\theta = 1/2$) is exactly the width so uninformed traders, paying $\alpha(V_H - V_L)/2$ each way, collectively cover the loss. The spread is the **expected loss to the informed, amortised across the entire population** — a tax, not a profit.

Poker analogy (Chapter 3): one card-counter at a table of nine tourists, nobody can tell them apart. The house charges a rake that covers the card-counter's expected winnings. Rake is the spread; card-counter is the informed trader; tourists are noise; house is the GM market maker. Nobody gets cheated, nobody gets rich, the house stays in business.

A short historical and modelling aside

Historical note

Glosten (Columbia) and Milgrom (Stanford) published "Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders" in the *JFE* in 1985 (Glosten and Milgrom, 1985). Against a decade of debate over positive spreads — the received answer was order-processing and inventory costs — GM (and Kyle, in the same year's *Econometrica*) showed adverse selection alone produces a spread. The technical innovation was bringing Bayes' theorem to quote-setting. Milgrom later won the 2020 Nobel for his auction work.

6.2 Kyle (1985): the one-shot strategic insider

What changes when the informed trader chooses their size? GM's informed traders are non-strategic: they always trade one unit in the profitable direction. A real informed trader chooses *how much*, trading less aggressively when their order would move the price against them. Kyle (1985) captures this as a fixed point: the insider's optimal size depends on the market maker's price-impact coefficient, and the market maker's impact depends on how aggressively the insider trades.

Setup

The model is one-shot: a single round, after which V is revealed and everybody settles up. (Kyle's original paper has N equispaced auctions limiting to Brownian motion; the single-period version captures the intuition and suffices here.)

There is a single risky asset with fundamental value V. Unlike GM, V is continuous: drawn once from a Gaussian prior

$$V \;\sim\; \mathcal{N}(0, \sigma_V^2).$$

Three types of agents observe or trade in this market.

First, a single **strategic insider** privately observes V before the auction and submits a signed market order x (positive = buy), observing V but not u. We restrict to *linear* strategies

$$x(V) \;=\; \beta\, V,$$

where $\beta > 0$ is to be solved for. (Linearity is an equilibrium property, not an assumption: the unique Kyle equilibrium has linear strategies. We state this as a fact.)

Second, **noise traders** (liquidity traders in Kyle's paper) submit an aggregate net order

$$u \;\sim\; \mathcal{N}(0, \sigma_u^2), \qquad u \perp V.$$

Only their aggregate u is observable, not the individual trades. The noise traders are indispensable: without them every order would equal βV, the market maker could invert to recover V exactly, and the insider's edge would collapse to zero before any profit was realised — so the insider would not trade and the market would unravel.

Third, a **risk-neutral, competitive market maker** observes total order flow $y \coloneqq x + u$ (she cannot separate insider from noise) and prices each unit of flow to break even in expectation. Competition drives the markup to the conditional expectation of V (any higher invites undercutting; any lower loses money on average):

$$p_1(y) \;=\; \mathbb{E}[V \mid y].$$

Linear pricing rules give

$$p_1(y) \;=\; \lambda\, y$$

for constant $\lambda > 0$, **Kyle's lambda**. (Again, linearity is an equilibrium property, taken as given.)

λ and β must be mutually consistent:

- Given λ, β maximises insider profit.
- Given β, λ is the coefficient of the linear projection of V onto y.

We solve the fixed point in two steps: insider best response to arbitrary λ, then Bayes-consistency of λ given β.

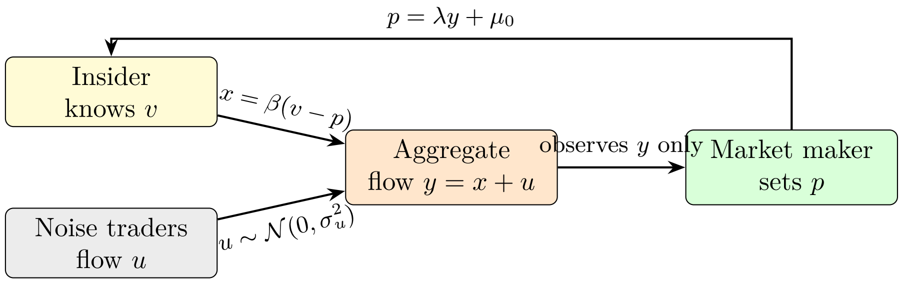


Figure 6.1: Kyle (1985) information flow. The insider submits order size x proportional to the gap between fundamental value v and the current price. Noise-trader flow u provides cover. The market maker, observing only aggregate flow y, sets price via the linear rule $p = \lambda y + \mu_0$. The feedback arrow shows the price updating as the market maker extracts the signal.

Worked toy example, before the derivation

Before the general derivation, work through one realisation. Take

$$\sigma_V \;=\; 1, \qquad \sigma_u \;=\; 1.$$

Section 6.2 will show the unique linear equilibrium has

$$\lambda^* \;=\; \frac{\sigma_V}{2\sigma_u} \;=\; \frac{1}{2}, \qquad \beta^* \;=\; \frac{\sigma_u}{\sigma_V} \;=\; 1.$$

Accept these pending derivation: $x = V$ and $p_1 = 0.5y$. Suppose $V = 0.5$ and $u = 0.2$. Then

$$\begin{aligned} x &\;=\; \beta^* V \;=\; 1 \cdot 0.5 \;=\; 0.5, \\ y &\;=\; x + u \;=\; 0.5 + 0.2 \;=\; 0.7, \\ p_1 &\;=\; \lambda^* \cdot y \;=\; 0.5 \cdot 0.7 \;=\; 0.35. \end{aligned}$$

The market maker clears at 0.35 (above the prior mean 0, since positive flow signals high V). Insider profit is

$$(V - p_1) \cdot x \;=\; (0.5 - 0.35) \cdot 0.5 \;=\; 0.075.$$

A few more realisations:

V	u	$x = V$	$y = x + u$	$p_1 = 0.5y$	$(V - p_1)x$
+0.5	+0.2	+0.50	+0.70	+0.350	+0.0750
+0.5	−0.3	+0.50	+0.20	+0.100	+0.2000
−1.0	+0.4	−1.00	−0.60	−0.300	+0.7000
+1.5	−0.6	+1.50	+0.90	+0.450	+1.5750
0.0	+1.0	0.00	+1.00	+0.500	0.0000

Every realised profit here is non-negative because the insider trades with V. The derivation below (Section 6.2) gives expected profit

$$\mathbb{E}[\pi] \;=\; \frac{\sigma_V \sigma_u}{2} \;=\; 0.5$$

per round, verified numerically at the end of the chapter.

Derivation of the linear equilibrium

Three steps: (i) insider expected profit as a function of β and λ, (ii) maximise over β, (iii) impose Bayes-consistency of λ and solve for (β^*, λ^*).

Step 1: insider expected profit. Given $p_1 = \lambda y$, realised profit on a trade of size x is

$$\pi \;=\; (V - p_1)\cdot x \;=\; (V - \lambda(x+u))\cdot x \;=\; Vx - \lambda x^2 - \lambda ux.$$

The insider chooses $x = \beta V$. Conditional on V, using $\mathbb{E}[u] = 0$:

$$\mathbb{E}[\pi \mid V] \;=\; V\cdot\beta V \;-\; \lambda(\beta V)^2 \;-\; \lambda\cdot\underbrace{\mathbb{E}[u]}_{=0}\cdot\beta V \;=\; \beta V^2 - \lambda\beta^2 V^2.$$

Now take expectation over V, using $\mathbb{E}[V^2] = \sigma_V^2$:

$$\mathbb{E}[\pi] \;=\; \beta\sigma_V^2 \;-\; \lambda\beta^2\sigma_V^2 \;=\; (\beta - \lambda\beta^2)\,\sigma_V^2. \tag{6.8}$$

Quadratic in β with positive linear and negative quadratic terms: unique maximum at finite positive β.

Step 2: insider best response. Differentiate Equation (6.8) in β and set to zero:

$$\frac{\partial\mathbb{E}[\pi]}{\partial\beta} \;=\; \sigma_V^2 - 2\lambda\beta\sigma_V^2 \;=\; 0.$$

Dividing by σ_V^2 (strictly positive by assumption),

$$1 - 2\lambda\beta \;=\; 0 \quad\Longrightarrow\quad \boxed{\beta \;=\; \frac{1}{2\lambda}.}$$

The SOC $\partial^2\mathbb{E}[\pi]/\partial\beta^2 = -2\lambda\sigma_V^2 < 0$ confirms a maximum. Double the impact, halve the order size.

Step 3: market maker's Bayes-consistent λ. The market maker observes $y = \beta V + u$ and posts $p_1 = \mathbb{E}[V \mid y]$. For bivariate zero-mean Gaussians,

$$\mathbb{E}[V \mid y] \;=\; \frac{\mathrm{Cov}(V,y)}{\mathrm{Var}(y)}\cdot y.$$

Compute (using $u \perp V$):

$$\mathrm{Cov}(V,y) \;=\; \beta\sigma_V^2, \qquad \mathrm{Var}(y) \;=\; \beta^2\sigma_V^2 + \sigma_u^2.$$

Therefore

$$\lambda \;=\; \frac{\mathrm{Cov}(V,y)}{\mathrm{Var}(y)} \;=\; \frac{\beta\sigma_V^2}{\beta^2\sigma_V^2 + \sigma_u^2}. \tag{6.9}$$

Step 4: solve the fixed point. Two equations, two unknowns:

$$\beta \;=\; \frac{1}{2\lambda}, \tag{6.10}$$

$$\lambda \;=\; \frac{\beta\sigma_V^2}{\beta^2\sigma_V^2 + \sigma_u^2}. \tag{6.11}$$

Substitute $\beta = 1/(2\lambda)$ into Equation (6.11). Numerator: $\sigma_V^2/(2\lambda)$. Denominator:

$$\beta^2\sigma_V^2 + \sigma_u^2 \;=\; \frac{\sigma_V^2}{4\lambda^2} + \sigma_u^2 \;=\; \frac{\sigma_V^2 + 4\lambda^2\sigma_u^2}{4\lambda^2}.$$

So the RHS is

$$\frac{\sigma_V^2/(2\lambda)}{(\sigma_V^2 + 4\lambda^2\sigma_u^2)/(4\lambda^2)} \;=\; \frac{2\lambda\sigma_V^2}{\sigma_V^2 + 4\lambda^2\sigma_u^2}.$$

Setting this equal to λ and cancelling $\lambda > 0$:

$$1 \;=\; \frac{2\sigma_V^2}{\sigma_V^2 + 4\lambda^2\sigma_u^2} \quad\Longrightarrow\quad \sigma_V^2 + 4\lambda^2\sigma_u^2 \;=\; 2\sigma_V^2 \quad\Longrightarrow\quad 4\lambda^2\sigma_u^2 \;=\; \sigma_V^2.$$

Therefore

$$\lambda^2 \;=\; \frac{\sigma_V^2}{4\sigma_u^2} \quad\Longrightarrow\quad \boxed{\lambda^* \;=\; \frac{\sigma_V}{2\sigma_u}}$$

(positive root). Then

$$\boxed{\beta^* \;=\; \frac{1}{2\lambda^*} \;=\; \frac{1}{2\cdot\sigma_V/(2\sigma_u)} \;=\; \frac{\sigma_u}{\sigma_V}.}$$

The equilibrium: insider trades $x^*(V) = (\sigma_u/\sigma_V)V$, market maker prices $p_1(y) = (\sigma_V/2\sigma_u)y$.

Step 5: expected insider profit. Plug into Equation (6.8):

$$\begin{aligned}
\mathbb{E}[\pi^*] &= (\beta^* - \lambda^*(\beta^*)^2)\,\sigma_V^2 \\
&= \left(\frac{\sigma_u}{\sigma_V} - \frac{\sigma_V}{2\sigma_u}\cdot\frac{\sigma_u^2}{\sigma_V^2}\right)\sigma_V^2 \\
&= \left(\frac{\sigma_u}{\sigma_V} - \frac{\sigma_u}{2\sigma_V}\right)\sigma_V^2 \\
&= \frac{\sigma_u}{2\sigma_V}\,\sigma_V^2 \;=\; \frac{\sigma_V\sigma_u}{2}.
\end{aligned}$$

Expected profit per round is $\sigma_V\sigma_u/2$, matching Section 6.2.

Step 6: how much does the market maker learn? Residual uncertainty about V after observing y:

$$\mathrm{Var}(V \mid y) \;=\; \sigma_V^2 - \frac{\mathrm{Cov}(V,y)^2}{\mathrm{Var}(y)} \;=\; \sigma_V^2 - \frac{(\beta^*\sigma_V^2)^2}{(\beta^*)^2\sigma_V^2 + \sigma_u^2}.$$

With $\beta^* = \sigma_u/\sigma_V$: numerator $\sigma_u^2\sigma_V^2$, denominator $2\sigma_u^2$, so

$$\mathrm{Var}(V \mid y) \;=\; \sigma_V^2 - \frac{\sigma_u^2\sigma_V^2}{2\sigma_u^2} \;=\; \sigma_V^2 - \frac{\sigma_V^2}{2} \;=\; \frac{\sigma_V^2}{2}.$$

A single Kyle round erases exactly half the prior variance. The market maker learns half the truth per round; the insider's private information dissipates at a corresponding rate.

Applying Kyle to the toy example

With $\sigma_V = \sigma_u = 1$ from Section 6.2:

$$\lambda^* \;=\; \frac{1}{2\cdot 1} \;=\; \tfrac{1}{2}, \qquad \beta^* \;=\; \frac{1}{1} \;=\; 1,$$

with expected profit 1/2. For $V = 0.5$, $u = 0.2$:

$$\begin{aligned}
x &= 1\cdot 0.5 \;=\; 0.5, \\
y &= 0.5 + 0.2 \;=\; 0.7, \\
p_1 &= 0.5\cdot 0.7 \;=\; 0.35, \\
(V - p_1)x &= (0.5 - 0.35)\cdot 0.5 \;=\; 0.075.
\end{aligned}$$

Matching Section 6.2. A Monte Carlo over 2×10^5 draws from $V, u \sim \mathcal{N}(0,1)$ gives sample mean profit 0.501, within 0.2% of the analytical 0.5.

Kyle's linear equilibrium

In the Kyle (1985) model (Kyle, 1985) with prior $V \sim \mathcal{N}(0, \sigma_V^2)$, noise-trade aggregate $u \sim \mathcal{N}(0, \sigma_u^2)$ independent of V, a single strategic insider submitting $x = \beta V$, and a risk-neutral competitive market maker observing total order flow $y = x + u$ and quoting $p_1 = \lambda y$, the unique linear equilibrium is

$$\boxed{\lambda^* = \frac{\sigma_V}{2\sigma_u}, \qquad \beta^* = \frac{\sigma_u}{\sigma_V}.}$$

The insider's expected profit per round is $\sigma_V \sigma_u / 2$, and the post-trade residual variance $\mathrm{Var}(V \mid y) = \sigma_V^2/2$, exactly half the prior. Kyle's λ^* has units of price per unit order flow and is the natural measure of **price impact** in the model. It is what every market-impact study of the last forty years tries to estimate (Hasbrouck, 2007), (Bouchaud et al., 2018).

Intuition

The insider hides inside the noise. $\beta^* = \sigma_u/\sigma_V$ says insider size is proportional to noise volatility and inversely proportional to signal volatility. Both are right on reflection. Loud noise (σ_u large): the insider's βV gets absorbed into the dispersion of y and barely moves p_1, so trade large. Quiet noise ($\sigma_u \to 0$): $y \approx \beta V$ and the market maker inverts to V, collapsing the edge into p_1; trade small.
A stronger signal (σ_V large) means each unit of V is a bigger secret: trading one extra unit gives away more of V, so trade less. The equilibrium $\beta^* = \sigma_u/\sigma_V$ is the compromise between profiting on a correct V and avoiding self-impact.
The lesson: **never trade large in a quiet market**. In Prosperity, "Olivia"-style informed flow in a sleepy product is the small-σ_u Kyle case. A team following her aggressively advertises the signal; following quietly captures the edge.

A sketch of the multi-period Kyle model

The one-shot model omits how private information is gradually incorporated into the price across rounds. Kyle's 1985 paper (Kyle, 1985) works out the N-round version and takes the continuous-time limit; a qualitative sketch suffices here.

In N sequential auctions, the insider submits x_t linear in the residual information $V - p_{t-1}$, and the market maker posts a price with coefficient λ_t. Trade too aggressively early and the signal reveals; too timidly and the rounds run out.

The equilibrium has deterministic sequences $\{\beta_t\}, \{\lambda_t\}$ (see (Cartea et al., 2015) for a modern treatment). Headlines:

1. Information is revealed gradually: $\mathrm{Var}(V \mid \mathcal{F}_t)$ decreases geometrically to zero, linearly in the continuous-time limit.

2. λ_t is *constant in time* to leading order: a marginal share moves the price the same in round 1 as in round N. The insider trades more in later rounds to compensate.

3. The aggregate flow y_t converges to a Brownian motion in the continuous-time limit, and the price walks smoothly to the true V.

The constant-λ result justifies using a single Kyle-style scalar in optimal-execution models (Chapter 19).

Historical note

Albert "Pete" Kyle wrote his 1985 *Econometrica* paper "Continuous Auctions and Insider Trading" (Kyle, 1985) at Princeton, appearing alongside Glosten–Milgrom. Kyle's headline result is the continuous-time limit of his N-auction game, where the insider trades continuously at a rate proportional to $(V - p_t)\,dt$. The single-period version is the classroom workhorse. The coefficient λ is universally known as "Kyle's lambda," and market-impact software across the industry contains a function for estimating it.

6.3 Connecting Glosten–Milgrom and Kyle

GM and Kyle approach the same phenomenon from opposite sides:

Feature	Glosten–Milgrom	Kyle (1985)
Latent asset value	Binary $V \in \{V_L, V_H\}$	Gaussian $V \sim \mathcal{N}(0, \sigma_V^2)$
Trading protocol	Sequential, one-at-a-time	Batched, single auction
Informed trader	Myopic, non-strategic	Strategic, size-optimising
Noise component	Symmetric Bernoulli coin-flip	Gaussian aggregate u
Market maker action	Quote bid and ask each period	Clear a single price at y
Price impact summary	s in price units	λ in price per unit flow
Learning per trade	Bayesian update of θ_t	Single-shot Gaussian projection
Limit shapes	$\alpha \to 0 \Rightarrow$ spread $\to 0$	$\sigma_u \to 0 \Rightarrow \lambda \to \infty$
Primary output	Bid/ask quotes as functions of history	Price impact coefficient

Both start with an informed/uninformed mixture and a zero-expected-profit market maker, and both derive a strictly positive price-impact function whose width depends on the informed share and on signal/noise volatilities. They differ in protocol: GM one-at-a-time, Kyle simultaneous batch. Real markets live between: exchanges match trades tick-by-tick (GM-flavoured) but institutional flow arrives in blocks (Kyle-flavoured). A full treatment uses GM for tick-level updates and Kyle for block-sizing.

Generalised, both predict a **square-root law**: $\Delta p \propto \sqrt{Q}$ for a large trade of size Q. Kyle's single-round result is linear ($\Delta p = \lambda Q$), but multi-period multi-agent versions collapse to the empirical square-root shape — Chapter 7 covers this.

Kyle's λ is not the GM spread

A common first-reading error is to identify λ with the spread. They have different units. GM's spread s has units of *price*: the bid–ask gap, independent of trade size (GM assumes one unit per trade).
Kyle's λ has units of *price per unit order flow*: a sensitivity. There is no bid/ask in pure Kyle; every trade clears at $p_1(y) = \lambda y$.
Dimensionally, to compare, multiply λ by a typical order size Q:

$$\text{spread} \;\sim\; 2\lambda \cdot Q,$$

a mnemonic, not a theorem. Do not write "λ is the spread" — it is a category error comparing a sensitivity to a price.

Prosperity example

Olivia maps to GM. The Prosperity 3 Olivia story (Section 3.2) is a GM-world phenomenon. Olivia was a bot on `SQUID_INK` whose buys/sells preceded fair-value moves in her direction. Top-ten teams identified her via the trader-ID field in historical logs and ran the GM market maker's optimal strategy *in reverse*: instead of widening the spread, they copied her and became informed themselves.
Calibrate: $V \in [1990, 2010]$ (so $V_H - V_L \approx 20$) and $\alpha \approx 0.05$ (one bot among twenty). At symmetric prior,

$$s_{\text{GM}} \;\approx\; \alpha(V_H - V_L) \;\approx\; 1.0 \text{ seashells},$$

near the observed Round 1 `SQUID_INK` spreads. Teams copying Olivia recouped this tax; teams ignoring her paid it.
A large basket order maps to Kyle. Trading a 100-unit basket in Round 2 with Kyle impact $\lambda \approx 0.01$ shifts the executed price by

$$\Delta p \;\approx\; \lambda \cdot 100 \;=\; 1 \text{ seashell}$$

against the mid. The expected edge must exceed 1 seashell per unit or Kyle impact eats it. This is why top teams slice large orders into small blocks.

On a Goldman Sachs desk

On a Goldman Sachs sell-side desk, Kyle's λ is an everyday object.
Execution-algorithm calibration. Execution algos (VWAP, POV, IS; derived from first principles in Chapter 19) rest on a linear-impact model $\Delta p = \eta \cdot (\text{trade rate})$, where η generalises λ. An execution Strat re-estimates it daily from parent-order fills and monitors residuals for regime changes.
Client-quote pricing on large trades. A block-desk client asks "sell 5M XYZ, what's my price?" The pricing tool runs a Kyle-flavoured impact model, estimates the expected shift over the unwind horizon, and adds it as a markup/markdown to the mid — exactly the $\lambda \cdot Q$ arithmetic. Informed clients (hedge funds with history) get λ adjusted upward; rebalancing pension funds get it adjusted downward.
Delta-hedging equity-derivative positions. Scheduling an N-share delta hedge balances speed against self-impact. With $N = 200{,}000$ and $\lambda \approx 0.0002$, the naive one-shot hedge moves the price by ≈ 40 bp — enough to turn a structured-product trade from a winner into a loser. This is why Chapter 19 is a central Strats chapter.
Research output. Junior Strats deliver monthly price-impact refreshes: last six weeks of parent-order logs, a Kyle-style linear (or square-root) regression on post-trade reversion, sensitivities reported to senior traders and risk. The form is textbook Kyle; the values are data-driven.

6.4 Key facts to memorise

- **Glosten–Milgrom setup.** Binary $V \in \{V_L, V_H\}$, prior $\theta_0 = \mathbb{P}(V = V_H)$, informed fraction α, uninformed traders flip symmetric coins, risk-neutral competitive market maker quotes bid and ask as conditional expectations.
- **Glosten–Milgrom spread formula.** $p_t^{\text{ask}} = \mathbb{E}[V \mid \text{buy}, \mathcal{F}_t]$, $p_t^{\text{bid}} = \mathbb{E}[V \mid \text{sell}, \mathcal{F}_t]$. At symmetric prior with small α, $s_0 = \alpha(V_H - V_L) > 0$, and the spread is strictly positive for every $\alpha > 0$.

- **Glosten–Milgrom toy numbers.** With $V_L = 100$, $V_H = 110$, $\theta_0 = 0.5$, $\alpha = 0.3$: $p_0^{\text{bid}} = 103.5$, $p_0^{\text{ask}} = 106.5$, $s_0 = 3.0$. Zero costs, zero inventory, $s > 0$.
- **Glosten–Milgrom Bayesian update.** After a buy, $\theta_{t+1} = (1+\alpha)\theta_t/(1+\alpha(2\theta_t - 1))$; after a sell, $\theta_{t+1} = (1-\alpha)\theta_t/(1-\alpha(2\theta_t - 1))$. The posterior is a martingale.
- **Spread as tax.** The Glosten–Milgrom spread is the expected loss to informed flow, amortised across all traders. Uninformed traders cover, collectively, the market maker's expected losses to the informed.
- **Kyle setup.** $V \sim \mathcal{N}(0, \sigma_V^2)$, $u \sim \mathcal{N}(0, \sigma_u^2)$ independent, strategic insider submits $x = \beta V$, risk-neutral competitive market maker observes $y = x + u$ and quotes $p_1 = \lambda y$.
- **Kyle linear equilibrium.** $\lambda^* = \sigma_V/(2\sigma_u)$ and $\beta^* = \sigma_u/\sigma_V$.
- **Kyle at $\sigma_V = \sigma_u = 1$.** $\lambda^* = 1/2$, $\beta^* = 1$. Realisation $V = 0.5$, $u = 0.2$: insider trades 0.5, market maker sees $y = 0.7$, clears at $p_1 = 0.35$, insider profit $= 0.075$.
- **Expected insider profit.** $\mathbb{E}[\pi^*] = \sigma_V\sigma_u/2$ per round, proportional to the product of the two volatilities. (Monte Carlo check: 0.501 over 2×10^5 draws, within 0.2%.)
- **Residual variance.** $\operatorname{Var}(V \mid y)^* = \sigma_V^2/2$. A single Kyle round erases exactly half the prior uncertainty about V.
- **Kyle intuition.** $\beta^* = \sigma_u/\sigma_V$ means the insider hides inside the noise. Larger noise $\Rightarrow$ larger insider trades; larger signal $\Rightarrow$ smaller insider trades. Never trade large in a quiet market.
- **Category error.** Kyle's λ has units price / order flow; the Glosten–Milgrom spread has units of price. They are not the same quantity. Related by the dimensional heuristic $s \sim 2\lambda \cdot Q$ for typical order size Q, but not equal.
- **Cross-reference map.** Empirical generalisation and the square-root impact law: Chapter 7. Inventory-aware market-making on top of adverse selection: Chapter 9. Optimal execution with Kyle-style impact: Chapter 19.

GM and Kyle together take "spreads exist because of adverse selection" from rhetoric to arithmetic, deriving a positive price of liquidity purely from informational asymmetry. The next chapter turns to the empirical footprint: the square-root market-impact law, the most robust regularity in microstructure and cleanly generated by Kyle's λ.

Chapter 7

Market Impact and the Square-Root Law

Buy 1000 shares of a liquid stock; the price moves up. Buy 2000; it moves up further. Kyle's linear equilibrium (Chapter 6) says twice the order, twice the impact. Real markets disagree: thirty years of empirical evidence says impact grows as $\sqrt{Q}$, not Q. Double the order, and impact grows by $\sqrt{2} \approx 1.41$.

This is not a small correction. A trader executing 5M shares who predicts cost from a linear model calibrated on 100,000-share trades overestimates the cost by roughly $\sqrt{5{,}000{,}000/100{,}000} = \sqrt{50} \approx 7$. On a \$500m parent order that is tens of basis points. Execution desks live and die on this one law.

The **square-root law of market impact** is one of the most robust quantitative regularities in finance. It survives across equities, futures, FX, and options, across decades, across trade-size scales spanning four or five orders of magnitude, and across the break between open-outcry and electronic trading. It also contradicts the linear impact derived in Chapter 6 and assumed (for tractability) in Chapter 19. This chapter covers (i) what the law is, (ii) three theoretical justifications, (iii) how it fails to match the linear textbook models, and (iv) how practitioners live with the contradiction.

Prerequisites

From earlier in the book:

- Chapter 2: the limit order book, limit orders, market orders, the "walking the book" calculation for a market order that consumes multiple price levels, the specific toy four-level LOB with best bid $p^{\text{bid}} = 102.54$ (quantity 131) and best ask $p^{\text{ask}} = 103.23$ (quantity 48) used throughout Part I.

- Chapter 5: the three-cause decomposition of the bid–ask spread into order-processing cost, inventory risk, and adverse selection.

- Chapter 6: Kyle's (1985) linear equilibrium and the price-impact coefficient $\lambda = \sigma_V/(2\sigma_u)$, which posits that the market clearing price moves *linearly* in the net order flow y. This chapter contradicts that linearity on empirical grounds, so the reader must know what is being contradicted.

From outside the book:

- Basic calculus and square-root arithmetic (if the reader finds $\sqrt{50} \approx 7.07$ surprising, start here).

- Standard descriptive statistics — mean, standard deviation, daily volatility. The "σ" in this chapter is always a daily return volatility unless otherwise stated.
- The notion of a **basis point** (bp): one hundredth of a percentage point, i.e. 1 bp $= 10^{-4}$. "63 bps" means 0.63%. Every trading-cost number in this chapter is given in basis points so that magnitudes across asset classes are comparable.

7.1 Defining market impact

Market impact has three distinct meanings; muddling them is a common source of confusion. This section pins each one down before stating the law that connects them.

The basic picture

A trader wants to buy a quantity Q. Sending it as a single market order walks up the ask (Chapter 2); instead, the trader slices the parent order into many small **child orders** executed over minutes or hours, letting the market refill in between.

Definition 7.1 (Metaorder)**.** A **metaorder** (also called a **parent order**) is a trading decision taken at one moment by one agent — "buy 1,000,000 shares of AAPL by the close" — that is then implemented by a sequence of smaller child orders sent to the market over an execution window of finite duration T. The total executed size is Q. A metaorder is atomic from the point of view of the trader who made the decision; it is dozens or thousands of orders from the point of view of the exchange.

Three timestamps matter. Before the metaorder starts at t_0 the mid is $p^{\text{mid}}(t_0)$. At the end of execution ($t_0 + T$) it is $p^{\text{mid}}(t_0 + T)$. Some time later (say $t_0 + 2T$, or next morning, after the market has absorbed the trade) the mid settles at $p^{\text{mid}}(t_\infty)$. The quantities of interest are the differences between these three prices.

Definition 7.2 (Temporary impact)**.** The **temporary impact** of a buy metaorder of size Q is the portion of the price move during execution that subsequently reverts. Formally,

$$I_{\text{temp}}(Q) \;\coloneqq\; p^{\text{mid}}(t_0 + T) - p^{\text{mid}}(t_\infty).$$

This is the short-lived liquidity premium the trader paid for consuming visible liquidity faster than the market could replenish it, which the market then unwinds. Temporary impact is a cost to the initiator but not a lasting move in the underlying price.

Definition 7.3 (Permanent impact)**.** The **permanent impact** of a buy metaorder of size Q is the portion of the execution-window price move that remains after the market has had time to absorb the trade:

$$I_{\text{perm}}(Q) \;\coloneqq\; p^{\text{mid}}(t_\infty) - p^{\text{mid}}(t_0).$$

This is the information-revealing component: the market treats the trade as a signal that fair value is higher, and updates the mid permanently. Same mechanism as Kyle / Glosten–Milgrom (Chapter 6): informed order flow moves the mid.

Definition 7.4 (Total (execution) impact)**.** The **total impact**, or *execution cost*, of a buy metaorder of size Q is the difference between the *average fill price* the trader achieved and the mid-price at the moment the metaorder started:

$$I_{\text{tot}}(Q) \;\coloneqq\; \bar{p}_{\text{fill}}(Q) - p^{\text{mid}}(t_0).$$

This is the P&L-relevant cost: what the trader paid relative to the price they saw when they decided to trade. The square-root law is a statement about I_{tot}.

Picture the mid as a function of time during a buy metaorder: flat before, rising during, then partly reverting after. The peak decomposes as

$$\underbrace{p^{\text{mid}}(t_0+T)-p^{\text{mid}}(t_0)}_{\text{peak impact}} = \underbrace{I_{\text{perm}}(Q)}_{\text{what stays}} + \underbrace{I_{\text{temp}}(Q)}_{\text{what reverts}}.$$

On equity metaorders the permanent component is typically one-half to two-thirds of the peak. The trader prices the metaorder using I_{tot} (it hits their P&L); the risk manager uses I_{perm} (it sets tomorrow's mark).

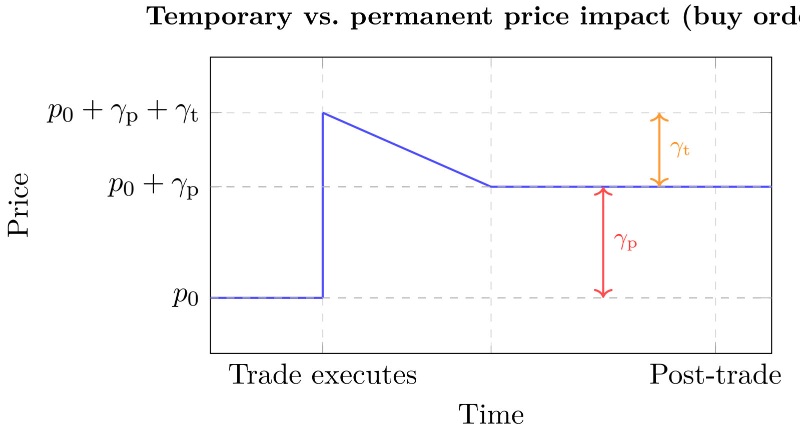


Figure 7.1: Schematic of market impact for a buy order executed at $t = 2$. The price spikes by $\gamma_\text{p} + \gamma_\text{t}$ at execution. The temporary component γ_t decays as the book replenishes. The permanent component γ_p (information leakage or supply shift) persists indefinitely.

Concave, linear, convex

How does impact depend on trade size?

Definition 7.5 (Concave impact function)**.** An impact function $I(Q)$ is **concave** in Q if, for any $Q_1, Q_2 > 0$ and any $\lambda \in [0,1]$,

$$I(\lambda Q_1 + (1-\lambda)Q_2) \geq \lambda I(Q_1) + (1-\lambda)I(Q_2).$$

Equivalently, its graph lies above every chord. The key consequence: for concave I, doubling the order size *less than* doubles the cost. Liquidity has decreasing marginal cost.

Three possibilities, three pictures:

Shape	Functional form	What it says
Linear	$I(Q) = aQ$	Twice the order, twice the cost.
Concave	$I(Q) = a\sqrt{Q}$ or $aQ^{0.5\text{–}0.7}$	Twice the order, $\sqrt{2}\times$ cost.
Convex	$I(Q) = aQ^2$	Twice the order, four times the cost.

Kyle (1985), reviewed in Chapter 6, derives the linear case under Gaussian values, Gaussian noise trading, and a one-shot batch auction. Almgren–Chriss (Chapter 19) assumes linearity for tractability of the calculus-of-variations problem. The convex case has no empirical support: no asset class shows large trades costing disproportionately more per share than small ones.

Data shows the concave case: impact scales as the square root of trade size.

7.2 The square-root law

The formula and what each symbol means

The square-root law states that the total impact of a metaorder of size Q scales as

$$I_{\text{tot}}(Q) \approx Y \cdot \sigma \cdot \sqrt{\frac{Q}{V_{\text{daily}}}}. \tag{7.1}$$

Each symbol has a precise meaning:

- $I_{\text{tot}}(Q)$ is the total impact in *fractional price units* — i.e. a dimensionless return. To convert to absolute currency per share, multiply by the price p^{mid}. To convert to basis points, multiply by 10,000.
- Y is a dimensionless constant depending on asset class and calibration window but not on Q. Empirical estimates fall in $Y \in [0.3, 1.5]$, with $Y \approx 1$ a default for mid-cap equities.
- σ is the **daily return volatility** — the standard deviation of daily close-to-close log returns, computed over a recent rolling window (say 60 trading days). "$\sigma = 2\%$" means a 2% daily standard deviation.
- Q is the executed size in shares (or contracts, or notional).
- V_{daily} is the **average daily trading volume** in the same units as Q. The ratio Q/V_{daily} — the **participation rate** or **volume fraction** — is the scale-free measure of trade size relative to the market.

The law is attributed to Almgren et al. (2005) (Almgren et al., 2005), who fit a large Citigroup parent-order database and found the Q/V_{daily} power consistent with 0.5 at high precision. The same scaling had been reported on futures by Torre and Ferrari (BARRA, 1999). It has since been re-verified on dozens of datasets and summarised by Bouchaud, Bonart, Donier, and Gould (Bouchaud et al., 2018).

The square-root law of market impact

The total impact of a metaorder of size Q, executed over a reasonable fraction of a trading day, scales as

$$\boxed{I_{\text{tot}}(Q) \approx Y \cdot \sigma \cdot \sqrt{\frac{Q}{V_{\text{daily}}}},}$$

where Y is an asset-class-specific dimensionless constant (typically $0.3 \leq Y \leq 1.5$), σ is the daily volatility, and V_{daily} is the daily traded volume. This is the central empirical regularity of execution cost modelling. It is concave in Q, holds across equities, futures, FX, and options, and contradicts the linear-impact predictions of Kyle (1985) and the linear-impact assumption of Almgren–Chriss (2000).

Three features of Equation (7.1):

1. **Scale-free in Q/V.** Impact depends on the participation rate, not Q absolutely. A 10k-share trade in a stock with 10M daily volume has the same impact as a 100k-share trade in a stock with 100M daily volume (all else equal) — execution cost in a single liquidity-normalised coordinate across all stocks.
2. **Proportional to σ.** Impact scales with the asset's own volatility, not its price-per-share. A penny stock and a blue chip at the same σ and participation rate have the same impact in bps. This is the empirical analogue of "trade where you can hide in the noise" (Chapter 6).

3. **Exponent is exactly 0.5.** Every high-quality fit on equity metaorder data produces a 95% confidence interval containing 0.5. The exponent is 0.5 to within the resolution of the data.

Worked numerical example

Take a concrete stock. Suppose:

- $\sigma = 2\%$ per day (a middle-of-the-road US mid-cap),
- $V_{\text{daily}} = 10$ million shares per day,
- the trader's parent order is $Q = 1$ million shares (a sizeable institutional trade), and
- $Y = 1$, our reasonable mid-cap default.

Plug in:

$$\begin{aligned} \frac{Q}{V_{\text{daily}}} &= \frac{10^6}{10^7} = 0.10, \\ \sqrt{\frac{Q}{V_{\text{daily}}}} &= \sqrt{0.10} = 0.31623, \\ I_{\text{tot}} &= 1 \cdot 0.02 \cdot 0.31623 = 0.006325. \end{aligned}$$

That is a fractional impact of 0.006325, i.e. 63.25 basis points. If the stock trades at \$50, the trader pays about $\$50 \times 0.006325 \approx \$0.32$ per share above the pre-trade mid, for a total cost of \$316,000 on the metaorder.

A 63-bp cost on a 1M-share order is what a sell-side execution desk quotes to a client before accepting the order, and what they are measured against afterwards. Getting the scaling wrong by a factor of 2 is the difference between a profitable and an unprofitable execution business.

Table 7.1 tabulates the predicted impact at several trade sizes, alongside a linear-impact model calibrated to match at $Q = 1$M shares.

Q (shares)	Q/V_{daily}	Sqrt-law impact (bps)	Linear-impact prediction (bps)
10,000	0.001	6.32	0.63
100,000	0.010	20.00	6.32
1,000,000	0.100	63.25	63.25
5,000,000	0.500	141.42	316.23

Table 7.1: Square-root versus linear-impact predictions as a function of trade size, for a stock with $\sigma = 2\%$, $V_{\text{daily}} = 10$M shares, $Y = 1$. The linear model is calibrated to match the square-root prediction at $Q = 1$M shares. The two agree at the calibration point and diverge violently away from it.

Read Table 7.1 line by line. At 10k shares, the linear model predicts 10× less cost than reality. At 5M shares, it predicts 5× more cost; a trader using it would refuse a reasonable metaorder or budget for wildly pessimistic slippage. Every row except the calibration point is wrong in the wrong direction. Hence the practitioner compromise of Section 7.4: use a calibrated linear model, but only near the calibration point.

Intuition

The square-root law says *liquidity has decreasing marginal cost.* The first shares are expensive per unit: you blow through tight top-of-book liquidity. The last shares are cheap: participants have had time to refill the book, bring in latent orders, and absorb your flow into the background noise.

7.3 Three theoretical justifications

The law is empirically rock-solid but theoretically awkward. No model derives it cleanly the way Kyle (1985) derives linear impact. Three heuristic arguments reach $\sqrt{Q}$ from different starting points; none is a proof, but collectively they suggest the law is a genuine emergent feature of liquidity formation.

Bouchaud's latent liquidity argument

Bouchaud et al. (Bouchaud et al., 2018) argue as follows. At any moment the limit order book shows only a tiny fraction — maybe 1% — of the true liquidity that would match a buy order. The rest is **latent liquidity**: orders not yet in the book because agents are waiting, not because they do not intend to sell. Think of a hedge fund willing to sell at the right price but unwilling to sit with a resting limit all day.

A metaorder eats through the thin visible liquidity almost immediately. The resulting price move is the signal that latent sellers use to enter the book. Latent liquidity refills at a rate proportional to the price move; steady state is reached when consumption rate (metaorder intensity) matches backfill rate (price move).

The algebra — a diffusion-reaction model for the latent supply density — yields a steady-state impact scaling as $\sqrt{Q}$. Supply is not fixed: new sellers enter *because* the price has moved, keeping marginal cost below linear.

Read in reverse, this explains why Kyle gets linear impact: the one-shot auction has no time for replenishment. Bouchaud lives in real time; replenishment is the whole story, and replenishment concaves the impact function.

Farmer–Gerig dimensional analysis

A shorter route: dimensional analysis. Impact is a fractional price (dimensionless). The available quantities are:

$$Q \text{ (shares)}, \qquad V_{\text{daily}} \text{ (shares/day)}, \qquad \sigma \text{ (returns/day}^{1/2}), \qquad T \text{ (execution time, days)}.$$

The only scale-free dimensionless combinations of Q, V_{daily}, and T are $Q/(V_{\text{daily}}T)$ (the participation rate) and Q/V_{daily} (the volume fraction). The only way to build a *return* out of these dimensionless combinations and σ (which has units of day$^{-1/2}$) is to multiply σ by the square root of a time. The relevant time is the execution duration, which, for a metaorder that trades at a constant rate, satisfies $T \propto Q/V_{\text{daily}}$. Thus

$$I \;\propto\; \sigma\sqrt{T} \;\propto\; \sigma\sqrt{Q/V_{\text{daily}}},$$

which is the square-root law up to the dimensionless prefactor Y.

This argument is due to Gabaix, Gopikrishnan, Plerou and Stanley, developed further by Farmer, Gerig and colleagues. It does not prove the law; it says that dimensional consistency forces it on any impact function depending only on Q, V_{daily}, and σ. A different exponent would require an additional dimensional scale, and no such scale appears in the data.

Gabaix power-law informed demand

A third route, due to Gabaix et al., builds the square-root law from the empirical fact that large institutional trade sizes follow a power law with tail exponent $\approx 3/2$: $\mathbb{P}(\text{size} > q) \sim q^{-3/2}$ for large q. These large investors are also the Kyle-informed ones; their aggregate demand dominates realised order flow.

Given the 3/2 tail, what impact function makes an individual investor's expected profit scale-invariant under rescaling of the size distribution? Under risk-aversion and market-efficiency assumptions detailed in the paper, the answer is $\sqrt{Q}$. The exponent 1/2 is tied by a self-consistency condition to the 3/2 tail of the institutional-trade-size distribution.

Less intuitive than Bouchaud, but appealing because it predicts *two* empirical laws (trade-size distribution and impact) and ties them together. The 3/2 tail is itself well-established.

Intuition

Common theme: *the market is not a fixed stack of liquidity but a living system that partially refills as you consume it.* Trading triggers other participants — latent sellers, front-runners, HFTs, mean-reversion bots — to absorb part of your flow. Your cost is the fraction you had to push through before replenishment kicked in. Under broad assumptions, that fraction scales as $\sqrt{Q}$.

7.4 The contradiction with Kyle and Almgren–Chriss

How does $\sqrt{Q}$ reconcile with the two central microstructure models — Kyle (Chapter 6) and Almgren–Chriss (Chapter 19) — that both assume linear impact?

Kyle: linear by construction

In Kyle (1985), the market clearing price satisfies

$$p_1 \;=\; \lambda \cdot y \;=\; \lambda \cdot (x + u),$$

with equilibrium $\lambda^* = \sigma_V/(2\sigma_u)$ a strictly positive linear coefficient on total order flow. Linearity is built in: the ansatz restricts the market-maker's pricing rule to $p_1 = \lambda y$, and joint normality of V and u makes this the unique best response. Gaussian projection cannot produce a $\sqrt{|y|}$ rule.

Is Kyle's linearity wrong? Kyle is a *single-period* model: one auction, one price. The square-root metaorder lives over many periods, and repeated interaction generates concavity — each child order moves the price, latent liquidity enters in response, the next child faces a partially-replenished book. Kyle is silent on multi-period dynamics, not wrong.

Continuous-time generalisations (Kyle's own follow-up, Back, Baruch) give a time-varying λ_t but still linear-in-flow impact at each instant. They do not naturally produce $\sqrt{Q}$ cumulative cost. The distinction: λ is an instantaneous slope; the square-root law is an integrated cost.

Almgren–Chriss: linear by choice

Almgren and Chriss (Almgren and Chriss, 2000) set up the optimal-liquidation problem: a trader holds X shares, must liquidate over $[0, T]$, and chooses a trading rate $v(t)$ balancing expected impact cost against the inventory-variance penalty (Chapter 19 gives the full solution). For the variational calculation to yield a closed form, impact must be linear in the trading rate:

$$\begin{aligned} \text{permanent impact per share} \;&=\; \gamma \cdot (\text{total traded}), \\ \text{temporary impact per share} \;&=\; \eta \cdot v(t), \end{aligned}$$

for constants $\gamma, \eta > 0$ (η here is the Almgren–Chriss temporary-impact coefficient, unrelated to the Y in the square-root law). This is a modelling choice. The paper is candid: linearity makes the Euler–Lagrange equation separable and produces the hyperbolic-sine optimal trajectory. Square-root impact has no clean closed form.

So Almgren–Chriss is linear *for tractability*, not because the authors believed in linear impact — the same compromise as pricing a vanilla option with Black–Scholes (Chapter 16) despite fat tails and smiles.

The practitioner's reconciliation

The industry response is the **calibration approach**. A sell-side execution desk will:

1. Use an Almgren–Chriss-style framework (closed-form optima are worth a lot in production).
2. Accept that the underlying impact is square-root, not linear.
3. Calibrate η so the linear model matches the square-root law at the typical trade size. If most parent orders are 100k VWAP executions, set η so linear impact at 100k equals the square-root prediction at 100k.
4. Trust the model only within a factor of ~ 3 of the calibration point; outside, use a direct $\sqrt{Q}$ estimate.

The model's form is chosen for tractability, the parameter for empirical agreement near typical trade sizes. The trader gets the closed-form trajectory and approximately correct costs at the sizes they execute — but no accuracy on order-of-magnitude extrapolation. Chapter 19 returns to this calibration point.

Don't extrapolate linear impact outside its calibration range

A desk calibrates its linear-impact parameter on 10k–100k-share orders, fits at the 10k end, and ships. A client arrives with a 5M-share parent order. The calibrated linear model reports:

$$I_{\text{linear}}(5\text{M}) \;=\; \frac{5\text{M}}{10\text{k}} \cdot I_{\text{linear}}(10\text{k}) \;=\; 500 \cdot 6.32 \text{ bps} \;=\; 3160 \text{ bps}.$$

Three thousand bps. The square-root prediction is 141 bps. The desk quotes 22× too high, loses the trade, and the client goes to a competitor.
Symmetrically: calibrate on large trades, send a tiny one, and the linear model predicts near-zero cost and *underestimates* aggressive-child slippage on a thin book. Extrapolating linear impact outside its calibration range is a stealth error. Always report the calibration range alongside the model, and switch to $\sqrt{Q}$ outside it.

7.5 Temporary versus permanent impact and the reversion profile

The square-root law tells us how total cost scales with Q but not how it decomposes into temporary and permanent parts, nor the shape of post-trade reversion. A few robust claims hold up.

Permanent component is roughly two-thirds. Almgren et al. (2005), on a Citigroup equity dataset, found 60–70% of peak impact persists indefinitely; the remaining 30–40% reverts within hours. A metaorder pushing the mid up 100 bps at execution end leaves 60–70 bps permanent and the rest decays.

Reversion is slow. Not a sharp snap-back: the temporary component decays as a power law (not exponential), on a time scale comparable to execution duration. A two-hour metaorder sees temporary impact decay over several hours afterwards. Any honest cost model has to wait for reversion to finish before measuring permanent impact.

Permanent impact scales the same way. The permanent component scales as $\sqrt{Q}$ in participation rate, with the permanent/temporary ratio roughly constant across trade sizes (a property of the asset, not the trade):

$$I_{\text{perm}}(Q) \;\approx\; \alpha \cdot Y \cdot \sigma \sqrt{Q/V_{\text{daily}}},$$

with $\alpha \in [0.5, 0.7]$. Permanent impact is a robust multiple of the total cost, not an independent quantity.

The information interpretation. Permanent impact measures how much information the market read into the metaorder (Chapter 6). High permanent/temporary ratio: the trade was read as a signal and the mid updated. Low ratio: the trade was read as liquidity-demanding noise and the dislocation unwound. The decomposition distinguishes "paying for information leakage" (preventable by better tactics) from "paying for instantaneous liquidity" (the cost of wanting to trade now).

Prosperity example

Return to the canonical four-level toy book from Chapter 2:

BID SIDE		ASK SIDE	
Qty	Price	Price	Qty
131	102.54	103.23	48
32	101.87	103.98	84
293	101.48	104.17	38
65	101.10	104.75	127

Mid $p^{\text{mid}} = 102.885$. You send a market buy for $Q = 50$ shares. The ask side's top level has only 48 units, so the order walks into the second level and fills 2 shares at 103.98. The average fill price is

$$\bar{p}_{\text{fill}} \;=\; \frac{48 \cdot 103.23 + 2 \cdot 103.98}{50} \;=\; \frac{5163.00}{50} \;=\; 103.26.$$

The two concrete numbers to compare are:

- **Naive top-of-book impact.** Had the trader read only the best ask and assumed they would get all 50 shares there, they would predict an impact of $103.23 - 102.885 = 0.345$ price units, or about 33.5 bps.

- **Actual walking-the-book impact.** The true average fill of 103.26 sits 0.375 price units above the mid, or about 36.5 bps.

The 3-bp gap is the visible concavity of this book: 50 clears the thin top level and nicks the second.

Compare to a square-root estimate. Take $\sigma = 1\%$ and $V_{\text{daily}} = 2000$ shares (reasonable for a Prosperity-scale product: rounds run 10^4–10^5 ticks, per-tick volumes are 50–200 shares). With $Y = 1$:

$$I_{\text{sqrt}} \;=\; Y \cdot \sigma \cdot \sqrt{Q/V_{\text{daily}}} \cdot p^{\text{mid}} \;=\; 1 \cdot 0.01 \cdot \sqrt{50/2000} \cdot 102.885 \;\approx\; 0.163,$$

about 15.8 bps — less than half the walking-the-book estimate. Normal at Prosperity scale: the square-root law is a large-sample fit, and a single 50-share trade against a thin four-level book is dominated by the idiosyncratic geometry of the top two levels. Two lessons: (i) for market orders into a thin book, do a by-hand walking-the-book calculation; the square-root estimate is an order-of-magnitude sanity check. (ii) When possible, post a limit order one tick inside the spread: the first 3 bps of this 36-bp estimate vanishes and is collected as fill-improvement.

7.6 Measuring impact: a short Python sketch

Measuring impact is harder than it looks: the realised move from t_0 to $t_0 + T$ mixes your impact with everything else the market was doing. The standard fix is to benchmark against an index (or the market-wide open-to-close drift) and then average over many trades of similar participation rate, fitting $I = Y\sigma\sqrt{Q/V_{\text{daily}}}$ by non-linear least squares.

On a parent-order log, that is a handful of lines. Assume a DataFrame with columns `size`, `avg_fill`, `pre_mid`, `sigma_daily`, `volume_daily`, and `side` ($+1$ buys, -1 sells).

```
import numpy as np
import pandas as pd
from scipy.optimize import curve_fit

def signed_impact_bps(row):
    # Side +1 for buys, -1 for sells; impact is positive when the
    # trade pushed the price against the trader's own side.
    return 1e4 * row["side"] * (row["avg_fill"] - row["pre_mid"]) /
        row["pre_mid"]

df["impact_bps"] = df.apply(signed_impact_bps, axis=1)
df["part_rate"]  = df["size"] / df["volume_daily"]

# Drop the tiny and the catastrophic for robust fitting.
mask = (df["part_rate"] > 1e-4) & (df["part_rate"] < 0.5)
fit_df = df[mask]

def sqrt_law(part_rate, Y):
    return 1e4 * Y * fit_df["sigma_daily"] * np.sqrt(part_rate)

popt, pcov = curve_fit(
    sqrt_law,
    fit_df["part_rate"].values,
    fit_df["impact_bps"].values,
)
Y_hat = popt[0]
print(f"Estimated Y = {Y_hat:.3f} (asset-class constant)")
```

Listing 7.1: Fitting the square-root law to a parent-order log.

Comments, all mandatory in production:

- The `side` multiplier ensures buys and sells are both measured as cost to the trader: a sell that moves price down is still positive impact.
- The participation-rate filter drops the smallest trades (measurement noise) and the biggest (finite-sample tail; the law is reliable up to $Q/V_{\text{daily}} \approx 0.3$).

- Fit the exponent as well: $I = Y\sigma(Q/V)^{\delta}$ and report the 95% CI on δ. Every large-scale study gives a CI containing 0.5.

On a Goldman Sachs desk

On a sell-side execution desk, impact models are the central input to algorithmic execution. A junior Strat's first rotation is typically the **transaction cost analysis** (TCA) pipeline: ingest last week's fills, match each child to its parent-order decision, compute realised impact against the pre-trade mid, and compare to model prediction. Systematic biases ("we underestimate impact on low-float stocks by 15 bps") feed back into monthly recalibration.

The numbers are enormous. A rates desk hedging a newly-issued structured note (Chapter 30) might execute a $500m swap in an afternoon; a 2-bp mis-estimate is $100,000 of P&L on one trade. Over a year, impact-model accuracy can swing seven-figure P&L. This is why every bank has several people working full-time on the Y constant per asset class, and why "we match the square-root law to within 5% out of sample" is a promotable line.

Workflow: (i) pull last month of parent-order fills from the OMS, (ii) compute signed impact in bps vs pre-trade mid, (iii) bucket by product, participation rate, and volatility regime, (iv) re-fit Y per bucket, (v) write a short memo on any step-change from the previous calibration, (vi) ship the new Y's to the execution algorithm's config. Everything in this chapter is a prerequisite.

7.7 Key facts to memorise

- **Market impact decomposes into temporary and permanent components.** The temporary component is the portion of the execution-window price move that reverts after the metaorder ends; the permanent component is what stays. Both scale as $\sqrt{Q/V_{\text{daily}}}$. Empirical estimates put permanent impact at roughly 60–70% of the peak impact on equities.

- **The square-root law.** $I_{\text{tot}}(Q) \approx Y \cdot \sigma \cdot \sqrt{Q/V_{\text{daily}}}$, with $Y \in [0.3, 1.5]$ a dimensionless asset-class constant, σ the daily return volatility, and Q/V_{daily} the participation rate. One of the most robust empirical regularities in all of microstructure.

- **Worked example to hold in memory.** $\sigma = 2\%$, $V_{\text{daily}} = 10\text{M}$, $Q = 1\text{M}$, $Y = 1$. Impact $= 0.02 \cdot \sqrt{0.1} \approx 0.006325 = 63.25$ bps. On a $50 stock, that is about $316k of cost on a $50m parent order.

- **Concavity is the point.** Doubling the metaorder size multiplies the *impact* (not cost) by $\sqrt{2} \approx 1.41$, not 2. Equivalently, the marginal cost of the last share is *lower* than the marginal cost of the first share.

- **Three theoretical stories.** Bouchaud's latent liquidity (replenishment-driven), Farmer–Gerig dimensional analysis (scale-free from Q, V, σ), and Gabaix power-law informed demand (self-consistency with $q^{-3/2}$ tail). None is a proof, but all point to the same exponent.

- **Kyle gives linear impact by construction.** The Kyle (1985) λ of Chapter 6 is an instantaneous slope in a one-shot auction. Integrated over many child orders of a metaorder, it does *not* produce $\sqrt{Q}$; the real-time replenishment of liquidity is what converts the linearity of a single-auction price rule into a concave cumulative cost.

- **Almgren–Chriss assumes linear impact by choice.** The linearity is a tractability assumption for the calculus-of-variations derivation in Chapter 19, not a claim about nature. Practitioners calibrate the linear coefficient so that the Almgren–Chriss model agrees with the square-root law at the typical trade size they execute.

- **Do not extrapolate linear impact outside its calibration window.** A linear model calibrated on small trades wildly *overestimates* cost on large trades; one calibrated on large trades *underestimates* cost on small aggressive child orders. Switch to a direct $\sqrt{Q}$ estimate outside a factor-of-three band around the calibration point.

- **Impact is scale-free in volume fraction, not shares.** A 10k-share trade in a stock with 10M daily volume has the same predicted impact as a 100k-share trade in a stock with 100M daily volume, at the same σ. This is why institutional traders think in participation rates, not absolute share counts.

- **On a sell-side desk, impact modelling is the core transaction-cost-analysis task.** Monthly recalibration of the asset-class Y constant, bucketed by product and participation rate, is a typical Strats workflow and a standard junior deliverable. The tolerance on the Y is measured in percent; the cost of being wrong is measured in 100-thousand-dollar increments per large trade.

- **Cross-reference map.** The Kyle linear impact this chapter contradicts: Chapter 6. The linear-impact assumption this chapter flags for later repair: Chapter 19. The broader story of short-horizon price changes driven by order flow: Chapter 8.

Together with Kyle's linear impact (Chapter 6) and the Roll decomposition (Chapter 5), the square-root law forms a triangle: Kyle gives the instantaneous cost of trading on information; the square-root law gives the integrated cost of a metaorder; Roll gives the spread decomposition into inventory, processing, and informed flow. Holding all three at once is what lets a quant strat read a trade log usefully. Chapter 8 turns to short-horizon order-flow dynamics — imbalance, queue position, and order-flow signals as return predictors — tick-by-tick analogues of the metaorder-scale story here.

Chapter 8

Order Flow Dynamics: Imbalance, Queue, Latency

You have a snapshot of the limit order book and a few seconds of trade history. In thirty seconds' time, will the mid-price have moved up or down? No crystal ball, no news feed — just what has been happening in the book for the last few hundred milliseconds. Can you do better than a coin flip?

Yes, and by a surprising margin. The single most informative number is the **order-flow imbalance** (OFI): the net aggressive buying pressure over a short window. Over sub-minute horizons, OFI explains a larger share of the variation in mid-price changes than any other easily-computed signal — larger than lagged returns, larger than volume alone, and larger than the raw bid–ask depth (Bouchaud et al., 2018). Read the book, compute OFI, and act on it faster than the next trader: this is close to the only honest edge at the microsecond scale.

This chapter builds three related pieces of structure around that claim. First, OFI itself: definition, empirical regularity, and mechanism. Second, **queue position** — where in the line of resting orders you sit — and why being near the front is worth measurably more than being near the back. Third, two machinery observations: arrivals *cluster in time* (the Hawkes observation) rather than being independent, and *latency is a second-derivative game* — being fast is only useful up to the point where nobody faster can pick you off.

Hawkes processes and latency are treated qualitatively; a full quantitative treatment of either would fill its own chapter. The Avellaneda–Stoikov market-making model of Chapter 9 is built around an assumed arrival intensity κ that comes straight out of this chapter.

Prerequisites

From earlier in the book:

- Chapter 2: the limit order book as two sorted sides, the specific four-level toy LOB with best bid $p^{\text{bid}} = 102.54$ (quantity 131) and best ask $p^{\text{ask}} = 103.23$ (quantity 48), mid $p^{\text{mid}} = 102.885$, spread $s = 0.69$. Price–time priority for the matching engine, i.e. first-in-first-out within a given price level. The "walking the book" calculation for a market order that consumes multiple levels.

- Chapter 4: the **microprice** $p^{\text{micro}} = (Q_a p^{\text{bid}} + Q_b p^{\text{ask}})/(Q_a + Q_b)$, introduced there with a promise that the underlying reason is order-flow imbalance and that the present chapter would develop it. This is where that promise is cashed in.

- Chapter 6: adverse selection and the idea that incoming order flow carries information. Kyle's linear equilibrium in particular gave a price response $\Delta p^{\text{mid}} = \lambda\, y$ where y is the aggregate order flow. OFI is the empirical version of that y.

From outside the book:

- Elementary probability: Poisson processes (arrival intensity λ, exponential inter-arrival times, the memoryless property), expectations, independence.
- Enough calculus to read an integral with a kernel in it. No Itô, no measure theory.

8.1 Order flow imbalance (OFI)

Intuition before the definition

Watch the top-of-book quotes. Each aggressive buyer (a market buy or a marketable limit buy crossing the best ask) removes quantity from the ask; each aggressive seller removes quantity from the bid. Passive traders post inside the existing quotes (tightening the spread, adding depth); resting orders cancel (removing depth).

If over the last few hundred milliseconds the ask is being eaten faster than the bid — six market buys for every sell, lots of cancels on the bid and none on the ask — buyers are the urgent side: willing to pay the half-tick for immediacy. An urgent side is almost always an *informed* side (Chapter 6-sense), because informed traders cannot afford to wait. Urgent buying tends to mean the next few ticks of discovery move up. OFI quantifies how urgent one side is relative to the other.

Definition 8.1 (Order-flow imbalance)**.** Let $[t, t + \Delta t]$ be a short time window. For each event on the top of the LOB during that window, assign a signed quantity e_k as follows:

- a new limit buy at the best bid, or an increase in best-bid depth: $e_k = +q_k$;
- an aggressive sell, or a cancel at the best bid, or a decrease in best-bid depth: $e_k = -q_k$;
- a new limit sell at the best ask, or an increase in best-ask depth: $e_k = -q_k$;
- an aggressive buy, or a cancel at the best ask, or a decrease in best-ask depth: $e_k = +q_k$.

The **order-flow imbalance** over the window is

$$\mathrm{OFI}(t, t + \Delta t) \;:=\; \sum_{k\,:\,t_k \in (t, t+\Delta t]} e_k \;=\; \underbrace{\Delta Q_b}_{\substack{\text{net increase}\\ \text{of best-bid depth}}} \;-\; \underbrace{\Delta Q_a}_{\substack{\text{net increase}\\ \text{of best-ask depth}}} . \tag{8.1}$$

A positive OFI means buy-side pressure; a negative OFI means sell-side pressure.

Check the sign convention in Equation (8.1). A buyer hitting the ask *removes* quantity, so $\Delta Q_a < 0$ and $-\Delta Q_a$ contributes positively to OFI. A buyer posting a new limit at the bid *adds* quantity, so $\Delta Q_b > 0$ contributes positively too. Both are buy-pressure events; both push OFI up. OFI is the net “demand minus supply” that hit the top of the book over the window.

Two standard subtleties (Bouchaud et al., 2018):

1. *OFI is computed at the top of the book, not aggregated over all levels.* Deeper levels change less often and carry less information. The top-of-book version predicts short-horizon returns.
2. *OFI counts quantity, not messages.* Ten aggressive buys of size 1 and one aggressive buy of size 10 contribute the same +10 to OFI. The econometric justification: the price-impact kernel is linear in quantity at the single-trade scale, not in message count.

Worked toy example on the four-level LOB

We use the toy LOB from Chapter 2, reproduced here to anchor the numbers:

Bid side		**Ask side**	
Qty	Price	Price	Qty
131	102.54	103.23	48
32	101.87	103.98	84
293	101.48	104.17	38
65	101.10	104.75	127

So $p^{\text{bid}} = 102.54$, $p^{\text{ask}} = 103.23$, $p^{\text{mid}} = 102.885$, and the top-of-book quantities are $Q_b = 131$ and $Q_a = 48$.

Consider a window of ten consecutive timesteps during which the following events arrive at the top of the book:

- 3 aggressive market buys of size 10 each (each removes 10 from the best ask);
- 1 aggressive market sell of size 5 (removes 5 from the best bid);
- no new limit orders or cancels at the top of book during the window.

Apply Theorem 8.1 event-by-event. Each buy consumes 10 from the ask, so $\Delta Q_a = -10$ on that event, contributing $-\Delta Q_a = +10$ to OFI. The three buys together contribute +30. The single sell consumes 5 from the bid, so $\Delta Q_b = -5$ on that event, contributing $+\Delta Q_b = -5$ to OFI. Summing:

$$\text{OFI} \;=\; \underbrace{(+10) + (+10) + (+10)}_{\text{three buys}} \;+\; \underbrace{(-5)}_{\text{one sell}} \;=\; +25. \tag{8.2}$$

The value 25 is an urgency meter for the window: sign says buyer-dominated, magnitude says 25 more units of aggressive buy pressure than sell pressure.

What does OFI predict about the mid? The empirical fact (Bouchaud et al., 2018), replicated on equities, futures, and FX, is that the change in the mid over the *next* window of similar length is approximately linear in OFI:

$$\Delta p^{\text{mid}} \;\approx\; \beta \cdot \text{OFI} + \varepsilon, \tag{8.3}$$

with $\beta > 0$, and R^2 on sub-minute bars typically several percent — small-sounding but an enormous signal-to-noise ratio for short-horizon forecasting. A slope $\beta \approx 0.01$ ticks per unit of OFI is typical on a mid-cap equity; for the toy example, that gives an expected next-window mid move of $\beta \cdot 25 = 0.25$ ticks — a quarter-tick drift in the direction of the buying pressure. A strategy trading on this signal is *not* predicting the direction of this window's flow; it uses this window's flow to predict the next window's mid.

Table 8.1 shows a synthetic bin table in the style of Cont, Kukanov, and Stoikov's OFI paper and matching the shape reported in (Bouchaud et al., 2018): slope approximately one-hundredth of a tick per unit of OFI, monotonic and linear across a wide range of OFI values.

OFI predicts short-horizon mid-price returns

Over sub-minute windows, the mid-price change is approximately linear in the preceding-window order-flow imbalance:

$$\Delta p^{\text{mid}}(t + \Delta t) \;\approx\; \beta \cdot \text{OFI}(t - \Delta t, t), \qquad \beta > 0.$$

This is the most robust short-horizon signal in empirical microstructure. It holds across assets, venues, and decades, and is the workhorse feature of every HFT alpha model. The

OFI bin	$\mathbb{E}[\Delta p^{\text{mid}} \mid \text{OFI}]$ (ticks)
-40	-0.38
-25	-0.27
-10	-0.08
0	-0.02
$+5$	$+0.07$
$+15$	$+0.13$
$+30$	$+0.32$
$+45$	$+0.43$

Table 8.1: Synthetic OFI-to-next-mid-move table, in the shape predicted by the linear model Equation (8.3) with slope $\beta \approx 0.0098$ and near-zero intercept. Real equity-futures data produces curves that look like this (Bouchaud et al., 2018).

slope β is calibrated per product; the sign is always positive.

Intuition

Why does OFI work? Traders with private information cannot afford to wait. If you know the fair value has risen, the cheapest way to exploit that knowledge is to post a passive bid and wait for the next aggressive seller; the expensive way is to cross the spread and pay the half-spread for immediacy. Choosing the expensive way reveals that the information decays faster than the time a passive order would need to fill. The *willingness to pay* the spread is itself evidence. OFI aggregates that willingness. Chapter 6 framed informed order flow as the channel by which information reaches prices; OFI is the model-free statistic that measures how much of the channel was used.

OFI is the microprice, reorganised

Chapter 4 defined the microprice as the cross-weighted combination of the two top quotes,

$$p^{\text{micro}} = \frac{Q_a p^{\text{bid}} + Q_b p^{\text{ask}}}{Q_a + Q_b},$$

and claimed p^{micro} is a better short-horizon predictor of the next mid than p^{mid}, promising an explanation here. The OFI story is the explanation.

Define the *queue imbalance ratio*

$$\rho := \frac{Q_b}{Q_a + Q_b} \in (0, 1).$$

This is the fraction of the total top-of-book quantity sitting on the bid side. A little algebra gives an alternative form for the microprice:

$$p^{\text{micro}} = \frac{Q_a p^{\text{bid}} + Q_b p^{\text{ask}}}{Q_a + Q_b} \tag{8.4}$$

$$= \frac{Q_a + Q_b}{Q_a + Q_b} \cdot p^{\text{mid}} + \Big(\frac{Q_b}{Q_a + Q_b} - \frac{1}{2}\Big)(p^{\text{ask}} - p^{\text{bid}}) \tag{8.5}$$

$$= p^{\text{mid}} + \Big(\rho - \tfrac{1}{2}\Big)s. \tag{8.6}$$

(Readers should verify Equation (8.6) by substituting $\rho = Q_b/(Q_a+Q_b)$ into the right-hand side; a single line of algebra recovers the original definition. For the toy LOB we have $Q_a = 48$, $Q_b = 131$,

$\rho = 131/179 \approx 0.7318$, and the formula gives $p^{\text{mid}} + (0.7318 - 0.5)(0.69) = 102.885 + 0.1599 = 103.0449$, which matches the explicit 103.0450 computed in Chapter 4.)

The microprice equals the mid plus a correction proportional to the *static depth imbalance*: if $\rho > 1/2$, the bid has more resting depth than the ask, which under Chapter 6-style reasoning should pressure the ask and lift the mid. The correction pushes the microprice *towards* the ask, anticipating the lift.

Static depth imbalance is the *level* of which OFI is the *first difference*. Between two snapshots, the change in top-of-book quantities is exactly $\Delta Q_b - \Delta Q_a = \text{OFI}$ by Theorem 8.1.

Microprice and OFI are the same object on two different time scales. The microprice is the instantaneous depth-imbalance-corrected mid; OFI is the change in signed top-of-book depth over a short window. Both are positive when buyers dominate and both predict the same sign of the next mid move. A market maker quoting around p^{micro} is implicitly using the same signal as an alpha trader predicting returns from OFI; the two differ only in how they convert the signal into action.

8.2 Queue position and the value of priority

Chapter 2 established that the matching engine runs price–time priority within a level: among resting orders at the same price, the first-in is filled first. That rule is why queue position — where in the FIFO line your order sits — is a dimension of the book you can extract edge from.

Front of the queue is not the same as back of the queue

Consider two market makers, Alice and Bob, each posting a passive limit bid at p^{bid}, same quantity, same product, same instant they decide to quote. Alice posted five milliseconds earlier. With 131 units of resting depth before they arrived, Alice is at queue position 132 (the 132nd unit an aggressive seller would hit); Bob is at position 132+ (Alice's size).

The next aggressive sell of size 135 fills the first 131 existing units, then all of Alice's order, then a small piece of Bob's. Alice collects the full half-spread; Bob gets a slice. Over the next minute, both want their full order filled: what are the respective probabilities?

The empirical shape is stark. On the big equity futures — Moallemi–Yuan and Huang–Polak on CME products — *front-of-queue* (first 10% of level depth) minute-horizon fill probabilities run 50–70%, while *back-of-queue* (last 10%) sit at 5–15% (Bouchaud et al., 2018). The ratio is typically four to seven. Queue position matters more than price level — since the two orders are, by construction, at the same price.

Pricing the queue position

Put a dollar number on this. Fix a tick size Δp and assume a filled quote captures, on average, half a tick of value relative to the true fair at fill. (A market maker quoting p^{bid} who gets filled buys below p^{mid}; captured value is $\frac{1}{2}s$, half a tick for a single-tick spread. This is the Roll-model half-spread of Chapter 5.)

Let p_{front} be the probability that a front-of-queue order fills over the next minute, and p_{back} the same for a back-of-queue order. The expected value of holding the quote for that minute is

$$\text{EV} \;=\; p_{\text{fill}} \cdot \tfrac{1}{2}\Delta p,$$

ignoring inventory and adverse-selection risk (both of which Chapter 9 will restore). Plug in representative numbers $p_{\text{front}} = 0.60$ and $p_{\text{back}} = 0.10$:

$$\text{EV}_{\text{front}} \;=\; 0.60 \cdot \tfrac{1}{2} \;=\; 0.30 \text{ ticks}, \tag{8.7}$$

$$\text{EV}_{\text{back}} \;=\; 0.10 \cdot \tfrac{1}{2} \;=\; 0.05 \text{ ticks}. \tag{8.8}$$

The ratio $0.30/0.05 = 6$: front-of-queue is worth six times back-of-queue on the same instrument at the same price. Being in front is a first-order consideration.

Queue position is worth a factor of six

Under representative fill probabilities, a quote at the front of the queue is worth roughly 6× a quote at the back — on the same product, at the same price level, at the same time. The only thing separating the two numbers is when each order arrived.

Intuition

Time in the queue is free optionality. Your position was set when you posted; from then on, every new same-price order is queued behind you, and every aggressive hit works through the queue in arrival order. An earlier order has been "converting aggressive flow into fills" for longer and has, on expectation, more of the minute left to spend at the front. A queue position is a credit you earn by posting early and consume by getting filled.

Consequences for strategy design

Four consequences follow from the factor-of-six result, all formalised later in the Avellaneda–Stoikov context (Chapter 9).

1. *Cancelling a resting order is expensive even when the order was wrong.* Cancel-and-repost loses all accrued queue credit. A one-tick re-quote is not worth doing unless the expected benefit exceeds the lost queue value.

2. *Posting deeper than best is often better than posting at best.* Deeper levels have shorter queues; a post at the second-best price might start near the front of its short queue, whereas a post at the best price might end up buried. The calculation is "price × fill probability", not price alone.

3. *Queue estimates must be tracked per-venue, per-time-of-day, per-product.* The front-of-queue advantage is 6× on one product, 2× on another, 15× on a third. Calibrate each from historical fill data.

4. *The reservation price of a market-making strategy should depend on queue position.* A front-of-queue quote needs a smaller spread to break even than a back-of-queue one, because $\mathbb{E}[\text{value per minute}]$ is higher. Chapter 9 folds this into the Avellaneda–Stoikov optimal-spread formula as an effective arrival-intensity correction.

The Frankfurt Hedgehogs' Prosperity 3 RAINFOREST_RESIN quoter is an explicit example of this logic being applied by hand: the bot did not cancel on small price noise because, as their writeup notes, "the queue was worth more than the tick" (Frankfurt Hedgehogs, 2025a). A Prosperity algorithm can estimate its own queue position directly from the `TradingState.order_depths` object each tick and from its own position delta, then estimate fill probability as $1 - e^{-\lambda t_{\text{remain}}}$ under a thin-Poisson approximation to the aggressive-sell arrival rate. Cancelling only when the adverse-selection cost of holding exceeds the lost queue value is the one-line summary of the strategy.

8.3 Hawkes processes: clustered arrivals

So far we have treated order flow as a stream of events with a well-defined rate, without specifying how the rate varies in time. **Poisson** arrivals with constant intensity λ get two things right (inter-arrival times $\mathrm{Exp}(\lambda)$; independent counts on disjoint intervals) and one thing catastrophically wrong: real order flow is *clustered*. A market order tends to be followed by another on the same side within a few hundred milliseconds, far more often than Poisson allows. Quiet periods follow quiet periods; bursty periods follow bursty periods.

Measurements on equity order flow show a sharp excess of short inter-arrival gaps relative to the Poisson null, and a conditional intensity given a recent arrival five to ten times the long-run average for several seconds (Bouchaud et al., 2018). A Poisson-based market-making model is unprepared for bursts: quotes too wide in quiet periods (over-estimated adverse selection from the constant mean) and too tight in bursts (local rate far above long-run mean). Both directions lose money.

The Hawkes model in one line

The standard fix, used somewhere in every HFT stack, is the **Hawkes process**, introduced by Alan Hawkes in 1971 (Hawkes, 1971). A Hawkes process is a point process whose intensity at time t depends on past arrivals:

$$\lambda(t) \;=\; \mu \;+\; \sum_{t_i<t} \phi(t-t_i), \tag{8.9}$$

where $\mu \geq 0$ is baseline intensity and $\phi : [0,\infty) \to [0,\infty)$ is a non-negative decay kernel, typically exponential $\phi(\tau) = \alpha e^{-\beta\tau}$ with $\alpha, \beta > 0$. Each past event t_i adds a time-decaying "kick" to the intensity: an arrival makes the next arrival more likely for a short window, then the memory decays.

Two facts to carry forward:

- *Stationarity.* With $\phi(\tau) = \alpha e^{-\beta\tau}$, the *branching ratio* α/β must be strictly less than one for stationarity. A branching ratio near one is near-critical self-excitation (arrivals driven almost entirely by other arrivals rather than by μ), characteristic of modern liquid equity markets (Bouchaud et al., 2018).

- *Effect on strategy.* Any model using a single average arrival rate (Avellaneda–Stoikov, Glosten–Milgrom, Roll) is making a Poisson approximation. It is harmless for coarse reasoning; wrong for burst handling. Practical strategies use either a short-window rolling average intensity (cheap, most of the benefit) or a Hawkes estimator (more complex, slightly better burst response).

Clustering matters in two places the book will treat elsewhere: market making and mean reversion.

In market making, a burst of same-side aggressive flow signals that you are about to be *picked off* on the opposite side: more flow on the same side is coming, and your resting quote looks stale. A good maker widens, skews, or cancels the at-risk side. All three depend on recognising the burst quickly; a rolling Hawkes-style intensity estimate is the standard trigger. Chapter 9 shows how the Avellaneda–Stoikov arrival-intensity κ is in practice an estimator of this instantaneous rate, even though the original (Avellaneda and Stoikov, 2008) derivation assumed constant rate.

In mean-reversion and stat arb, clustering matters for the opposite reason: Roll (Chapter 5) assumed iid trade-direction signs, and most pairs-trading P&L analysis assumes iid residual shocks. Clustered direction violates iid: a naive variance estimator under-reports residual variance because shocks bunch rather than average out. Failing to correct is a standard cause of "backtested Sharpe 3, live 0.8" surprises. Chapter 10 flags this in the Ornstein–Uhlenbeck context.

Hawkes, seismology, and the migration to finance

Hawkes introduced the self-exciting point process in 1971 (Hawkes, 1971) for *aftershocks*: an earthquake raises the probability of another in the same region for hours or days. The exponentially-decaying kernel in Equation (8.9) captured the empirical decay: each earthquake is a "parent" generating a Poisson number of "aftershock children", each child can generate its own aftershocks, and stationarity requires the expected children per parent (branching ratio α/β) below one. The order-flow analogy was noticed in the mid-2000s: trade arrivals look like aftershocks on a seismograph. Bacry, Delattre, Hoffmann, and Muzy were among the first to calibrate Hawkes processes to HF financial data, and by the late 2010s a Hawkes kernel somewhere in the stack was standard on sell-side HFT desks. Quant trading has always imported tools from other data-rich sciences — signal processing, ML, fluid dynamics, and here, seismology.

8.4 Latency economics

Everything so far — OFI, queue position, intensity estimation — has a clock attached. OFI is computed over hundreds of milliseconds; queue position changes on the timescale of individual orders; Hawkes kicks decay over seconds. If your trading loop runs slower than those clocks, you are reading stale state.

What latency is, and what it isn't

Definition 8.2 (Latency)**.** The **latency** of a trading system is the wall-clock time between a triggering event on the exchange and the trader's response being acted on at the matching engine. It decomposes into (i) *market-data latency* (event to decision module), (ii) *decision latency* (strategy code computing an action), (iii) *order-entry latency* (action reaching and being sequenced by the matching engine).

The three numbers are measured separately because each has its own cost structure. Market-data latency is dominated by network propagation and matching-engine publish pipelines; decision latency is a software problem; order-entry latency is again network and, on exchanges that allow it, FPGA. On modern US equity exchanges each component is in microseconds; Prosperity intentionally removes latency competition by simulating discrete ticks.

Every signal in this chapter is *time-stamped* and valid only for a short window. If OFI over the past 200 ms predicts the next 200 ms, a strategy taking 300 ms to compute and send an order arrives too late. If a fill-priority quote must be posted within a few milliseconds of the level becoming attractive, a 10 ms round-trip loses all the queue value Section 8.2 just priced at 6×.

Cost-benefit of speed

Latency-reduction costs are lumpy. In descending order on a real desk:

- **Colocation.** Rent rack space in the exchange's data centre so market data and order entry traverse a short local network instead of the open Internet. A rack at a major US exchange costs tens of thousands per month; saving is the end-to-end latency to your alternative site: 10–50 ms on US-east-coast-to-east-coast, 50–200 ms intercontinental.

- **FPGA order entry.** Replacing a software path (microseconds of kernel/NIC overhead) with hardware (hundreds of nanoseconds) saves 1–5 μs per round trip. Cost: specialised hardware and FPGA engineers. Only shops fighting for explicit speed rank do this.

- **Optical fibre routes.** The Chicago–New-Jersey fibre project (Spread Networks, 2010) shaved $\sim 3\,$ms off the round trip between CME and NASDAQ versus incumbent telecom routes, used by a few HFT firms before microwave and laser relays surpassed it. Construction ran into the hundreds of millions for that $3\,$ms saving.

Benefits decompose along three axes: (i) better queue position (you post first), (ii) fewer stale-quote fills (you cancel before pickoff), (iii) ability to act on faster-decaying signals (OFI on shorter windows has larger slope). "How fast do I need to be?" has no universal answer; it depends on who else is in the product.

Latency is a second-derivative game

The common mistake is treating latency as an arms race to be the absolute fastest. What matters is whether you are slow enough to be *picked off* by the faster traders in the same product. If the fastest HFT has a $10\,\mu$s round trip and you have $25\,\mu$s, the $15\,\mu$s gap lets them consume your stale quote before you update. You lose on every signal flip. At $10\,\mu$s vs $12\,\mu$s, the gap is two microseconds and the HFT usually cannot exploit it before you react.
The damage function is nonlinear in the latency gap: last to middle of the pack is worth more than middle to first. A $10\,\mu$s improvement can save a strategy; a $1\,$ms improvement rarely matters outside explicit latency-arb. The question is not "am I the fastest?" but "am I fast enough that nobody can systematically pick me off?". On Prosperity, which fixes tick granularity and processes all bots simultaneously, the question does not arise; on real venues it is the difference between a profitable and unprofitable market-making book.

Chapter 9 closes the loop: Avellaneda–Stoikov treats arrival intensity κ as a parameter tuned by the trader's own latency. A slower trader assumes a lower effective arrival rate (slow quotes get fewer fills) and quotes wider to break even. Latency is one of the inputs to market-making theory, not a separate topic.

8.5 Putting it together: an OFI-driven trading loop

A single picture connects the four sections before the summary:

1. **Maintain the book and the top-of-book quantities.** Every tick, record the current p^{bid}, p^{ask}, Q_b, Q_a, and microprice $p^{\text{micro}} = p^{\text{mid}} + (\rho - \frac{1}{2})s$.

2. **Compute OFI over a rolling window.** For the last Δt of order-book events, accumulate the signed contributions of Theorem 8.1. This is a single running sum; it is trivial in code.

3. **Estimate the instantaneous arrival rate.** Over the same window, count aggressive events. A simple exponentially-weighted moving average of the count is an adequate proxy for the Hawkes intensity for most practical purposes.

4. **Derive a short-horizon fair value.** The preferred estimator is the microprice, optionally shifted by $\beta \cdot \text{OFI}$ for a few seconds of look-ahead. This is the "fair value" that later chapters will refer to when they write "quote around fair value" in a market-making algorithm.

5. **Act subject to queue constraints.** Before cancelling a resting order, check its queue position; back-of-queue orders have low fill value and can be moved cheaply, front-of-queue orders hold edge and should only be moved when the signal change is larger than the cost of losing queue credit.

6. **Bound every step by latency.** Each of the above reads a time-stamped piece of state. If the wall-clock time between observation and action exceeds the timescale over which the signal is valid, the output is actively misleading; better to not act at all than to act on stale OFI. The latency budget is as much a part of the algorithm as the signal itself.

The shape: a high-frequency inner loop feeding a strategy-level outer loop. The outer loop is the market-maker or stat-arb trader (Chapter 9 onwards); the inner loop is shared infrastructure across every HFT strategy in the book. Two strategies with identical outer loops can have radically different P&L because one has a better inner loop. That is what "trading infrastructure" means in practice.

Prosperity example

An OFI-based alpha is easy to prototype on Prosperity's KELP product (Prosperity 3's trending fish price). Each `TradingState` tick exposes `order_depths`; differencing consecutive ticks gives ΔQ_b and ΔQ_a, and the Trader can accumulate OFI over the last five ticks. Synthetic tests on Prosperity-style KELP data with moderate autocorrelation in aggressive-side flow show the sign of rolling OFI agrees with the next-tick mid change around 70% of the time — well above the 50% coin flip. Because KELP is *trending* (unlike mean-reverting RAINFOREST_RESIN), the natural strategy is *directional following*: take the side indicated by positive OFI, exit on OFI sign flip or position limit. This is exactly the structure the Frankfurt Hedgehogs used on KELP in their Prosperity 3 run, and it produced the bulk of their KELP P&L.

On a Goldman Sachs desk

On a sell-side Goldman equities HFT desk, OFI is the dominant short-horizon alpha signal and one of the first features every new strat builds from tick data. Two production uses:

- **Systematic internaliser (SI) routing.** When Goldman receives a client equity order above a size threshold, the SI engine chooses between filling from Goldman's book ("internalise") or routing to an exchange. The decision depends on whether the short-horizon expected move — estimated from OFI and related signals — is larger than the internalisation fee. If OFI says the market is about to rally, the engine prefers to fill from inventory at the current mid rather than pay up on the open market.

- **Execution algos.** In-house VWAP, TWAP, and implementation-shortfall algorithms consult real-time OFI when scheduling the next child order. In high-positive-OFI windows, a buy child is held back (price drifting against the client; wait); in negative-OFI windows, accelerated (price coming to you). "Improve the OFI estimator" is a common junior-strat project.

Hasbrouck (Hasbrouck, 2007) gives the econometric framing: the signed trade series (or OFI) is treated as a covariance-stationary process whose impulse response on the mid is the quantity of interest; the research question is robust estimation of that impulse response. This framing — and VAR-style decompositions of price and trade series — is a standard interview topic for equities-desk strat placements.

8.6 Key facts to memorise

- **Order-flow imbalance** is the net signed top-of-book depth change over a short window: $\text{OFI} = \Delta Q_b - \Delta Q_a$, positive for buy pressure and negative for sell pressure. It is the workhorse short-horizon return predictor.
- **OFI–return regression.** Over sub-minute windows, $\Delta p^{\text{mid}} \approx \beta \cdot \text{OFI}$ with positive slope β. Typical R^2 is a few percent, which is enormous by short-horizon alpha standards.
- **Microprice ≡ OFI.** The microprice equals $p^{\text{mid}} + (\rho - 1/2)s$ where $\rho = Q_b/(Q_a + Q_b)$. Its change over a short window is OFI. The two signals are the same object at different time scales.
- **Queue position carries a factor-of-six value premium.** Front-of-queue expected fill value is about 0.30 ticks per minute versus 0.05 at the back under representative fill probabilities (60% vs 10%); the ratio is six, on the same instrument at the same price.
- **Cancelling destroys queue credit.** Cancel-and-repost only makes sense if expected gain exceeds lost queue value. Hence "stay in the queue" is the universal passive-MM maxim.
- **Arrivals are clustered, not Poisson.** Hawkes (1971) self-exciting processes, $\lambda(t) = \mu + \sum_{t_i<t} \phi(t - t_i)$, are the standard model. The branching ratio $\alpha/\beta < 1$ is the stationarity condition; modern liquid equity markets sit close to criticality.
- **Latency is nonlinear in its effect.** Being $10\,\mu$s slower than the fastest HFT on a venue can destroy a market-making strategy via pickoff; being 1 ms slower is usually survivable outside of explicit latency-arb plays. The question is never "am I the fastest?"; it is "am I fast enough to avoid being picked off?"
- **Every model using a single arrival rate is making a Poisson approximation.** That includes Glosten–Milgrom, Roll, and Avellaneda–Stoikov. Acceptable for coarse reasoning; in production, replace with a rolling intensity estimate.
- **The inner trading loop.** Maintain the book, compute OFI, estimate an intensity, form a microprice-based fair value, respect queue constraints, and bound every step by latency. This inner loop is common infrastructure across the market-making and alpha strategies in the rest of the book.

This chapter closes Part II. Chapters 5 to 7 gave *cross-sectional* results — why spreads exist, how information reaches prices, how impact scales with size; this chapter added the *time-series* layer: how flow looks on a clock, what short-horizon signal it carries, what constraints the clock imposes. Part III turns to strategies that use them, starting with market making in Chapter 9, where Avellaneda–Stoikov stitches together the spread economics of Chapter 5, the inventory risk of Chapter 6, the impact model of Chapter 7, and the OFI / queue / latency infrastructure of this chapter into a single optimisation.

Part IV

Core Strategies

Chapter 9

Market Making: Quoting, Inventory, and Avellaneda–Stoikov

You are quoting RAINFOREST_RESIN on the Prosperity exchange. The product's fair value sits around 10 000 seashells, barely moves, and every other team on the leaderboard is printing steady profit off it. You can post any limit bid at any tick and any limit ask at any tick. What do you quote?

A first guess: bid one tick below 10 000, ask one tick above, collect the spread forever. The guess is almost right — top Prosperity teams quote something close to it — but it has to survive two adversaries already met in Chapters 3 and 6: a counterparty who knows more than you, and flow that drifts the price against whatever inventory you hold. The bid-ask spread a maker earns is precisely the compensation demanded for bearing those two risks — *adverse selection* (trading against better-informed counterparties) and *inventory risk* (carrying unwanted positions through a volatile mid). A naive symmetric quoter beats neither. This chapter builds from first principles the machinery that fixes it: the **Avellaneda–Stoikov (A–S) market-making model** (Avellaneda and Stoikov, 2008), the closed-form rule for how a risk-averse quoter skews bid and ask as inventory drifts. The route is: diagnose the naive quote against the Chapter 2 toy book; derive the **reservation price** $r = S - q\gamma\sigma^2(T-t)$ from an indifference argument (stating the Hamilton–Jacobi–Bellman equation but not solving it); state the **optimal spread** $\delta^* = \gamma\sigma^2(T-t) + (2/\gamma)\ln(1+\gamma/\kappa)$ with its two-term decomposition and sanity limits; and close with the failure modes, where A–S does *not* apply.

Prerequisites

From earlier in the book:

- Chapter 2: the limit order book, limit versus market orders, the maker/taker distinction, the matching engine, and the specific four-level toy LOB with best bid $p^{\text{bid}} = 102.54$ (quantity 131), best ask $p^{\text{ask}} = 103.23$ (quantity 48), next bid 101.87 (quantity 32), next ask 103.98 (quantity 84), and mid $p^{\text{mid}} = 102.885$ with spread $s = 0.69$.
- Chapter 3: the trader taxonomy, the informed/uninformed split, and Olivia the informed bot with $\alpha = 0.3$ and $V \in \{100, 110\}$.
- Chapter 4: the formal definitions of bid, ask, mid, microprice, and the distinction between a quoted mid and a true fair value.
- Chapter 5: the three-cause decomposition of the spread (order-processing, inventory, adverse-selection). Each of these three causes will reappear as a parameter in A–S, which is this chapter's unifying theme.

- Chapter 6: Kyle's linear equilibrium with price impact λ and the Glosten–Milgrom equilibrium spread as an adverse-selection premium — the formal machinery behind why a zero-spread market maker loses money.
- Chapter 7: the empirical square-root impact law and the distinction between permanent and temporary impact.
- Chapter 8: order-flow imbalance, queue position as a source of alpha, the qualitative Hawkes clustered-arrivals picture, and the economic view of latency as a second-derivative game. A–S's arrival intensity κ is calibrated directly from order-flow data of the kind that chapter introduced.

From outside the book:

- Elementary probability: expectation, variance, Poisson processes with rate λ, the memoryless property.
- Elementary calculus: second-order Taylor expansion, and basic single-variable differentiation for optimisation. No stochastic calculus needed beyond the statement of the value function; the HJB equation is stated, not solved.
- Basic utility theory: exponential (CARA) utility $U(X) = -\exp(-\gamma X)$, with $\gamma > 0$ the risk aversion. The reader only needs to know it is a concave function of wealth.

9.1 The naive symmetric market maker

Start with the simplest policy: pick a half-width δ, quote a bid at $p^{\text{mid}} - \delta$ and an ask at $p^{\text{mid}} + \delta$, wait. Each arriving buyer lifting the ask earns δ above the mid; each seller hitting the bid earns δ below. The round-trip gain per matched buy–sell pair is 2δ, the *spread capture*.

In a world of symmetric uninformed flow — equal Poisson buyers and sellers, no mid drift, no memory — this policy is optimal and profits grow linearly. Chapter 3 called that world fiction. The real world has two structural deviations, either of which can dominate spread capture on any single run; the numerical example below, on the toy LOB of Chapter 2, makes both concrete.

What the naive maker earns: a benign run

Take the Chapter 2 LOB. The best bid is $p^{\text{bid}} = 102.54$ (quantity 131), the best ask is $p^{\text{ask}} = 103.23$ (quantity 48), and the mid is $p^{\text{mid}} = 102.885$. The round-trip spread is $s = 0.69$, giving a half-spread of $\delta = 0.345$. Set the market maker's quotes to coincide with the top of book: $p^{\text{bid}}_{\text{mm}} = 102.54$ and $p^{\text{ask}}_{\text{mm}} = 103.23$. (The round numbers are deliberately chosen so the toy sits exactly at the inside of the book; for this illustration we assume the market maker always wins queue priority on their side.)

Ten customer orders arrive. Six are buys (each lifts one share of the market maker's ask); four are sells (each hits one share of the market maker's bid). After the ten arrivals, the market maker has:

$$
\begin{aligned}
\text{sold at ask: } & 6 \text{ shares} \times 103.23 && = +\$619.38 \text{ in cash} \\
\text{bought at bid: } & 4 \text{ shares} \times 102.54 && = -\$410.16 \text{ in cash} \\
\text{net cash: } & && + \$209.22 \\
\text{net inventory: } & \underbrace{-6}_{\text{sold}} + \underbrace{+4}_{\text{bought}} && = -2 \text{ shares (net short)}
\end{aligned}
$$

The market maker ends the batch *short two shares*. To compare this to the spread capture alone, subtract off the mark-to-market value of the residual inventory at the mid:

$$\begin{aligned} \text{edge earned} &= \text{cash} - q \cdot p^{\text{mid}} &(9.1)\\ &= 209.22 - (-2)(102.885) &(9.2)\\ &= 209.22 + 205.77 = 414.99. &(9.3) \end{aligned}$$

That is wealth in absolute cash, which does not match the intuitive "earned the spread" number. The cleaner decomposition measures earnings *per share crossed*: every customer crossing earns the maker $p^{\text{mid}} - p^{\text{bid}}_{\text{mm}} = 0.345$ or $p^{\text{ask}}_{\text{mm}} - p^{\text{mid}} = 0.345$ depending on direction. Across the ten arrivals,

$$\text{edge} = 6 \cdot (p^{\text{ask}}_{\text{mm}} - p^{\text{mid}}) + 4 \cdot (p^{\text{mid}} - p^{\text{bid}}_{\text{mm}}) = 10 \cdot 0.345 = 3.45. \quad (9.4)$$

The maker has earned 3.45 seashells of edge and is carrying a residual short of -2 shares. If the mid drifts up by 0.5, the inventory marks down by $-2 \cdot (+0.5) = -1.0$ seashells. Total batch P&L:

$$\underbrace{3.45}_{\text{edge}} + \underbrace{(-2)(0.5)}_{\text{inventory drift}} = 2.45 \text{ seashells}. \quad (9.5)$$

The market maker keeps most of what they earned. On any single benign run the model looks fine.

What the naive maker loses: a directional run

Rerun the same policy on less friendly flow. Ten orders arrive, all buys: every one lifts the ask, none hits the bid. The maker sells ten shares into the buying crowd and ends ten shares *short*. Meanwhile — the key point — the mid drifts up by 2 seashells as buying pressure carries the market with it. The decomposition:

$$\begin{aligned} \text{edge} &= 10 \cdot (p^{\text{ask}}_{\text{mm}} - p^{\text{mid}}) = 10 \cdot 0.345 = 3.45, &(9.6)\\ \text{inventory drift} &= (-10)(+2.0) = -20.0, &(9.7)\\ \text{net P\&L} &= 3.45 - 20.0 = -16.55. &(9.8) \end{aligned}$$

The maker collected 3.45 seashells of edge and was handed back -20 in mark-to-market loss on the unhedged short: -16.55 underwater *and* still carrying the exposure. The arithmetic is symmetric under a sign flip: ten sells and a downward drift would leave the maker net long and equally underwater.

Symmetric quoting kills you on trend

The naive symmetric market maker's P&L has two terms. The *edge* term scales linearly in the number of fills and in the half-spread δ: it grows at most like $O(\text{fills} \cdot \delta)$. The *inventory-drift* term scales like $q \cdot \Delta p^{\text{mid}}$, which can grow as fast as (number of fills) $\times$ (total mid move) when flow is one-sided. On any persistent directional run — trend, newsbreak, stop cascade, or informed flow in the Chapter 6 sense — the second term wins and the maker loses more on the inventory than they collected on the spread. This is the entire economic content of the observation "market makers lose to trends." The fix is not to widen δ (which reduces fill rate linearly); it is to *skew* the quotes against inventory so the fills themselves mean-revert the position.

The run also exposes adverse selection in the sense of Chapter 6: the ten buyers were not drawn from a zero-drift distribution — they knew something the maker did not. The Glosten–Milgrom calculation of Section 6.1 predicts a symmetric quoter loses at least $\alpha(V_H - V_L)/2$ per fill against informed counterparties; on this run every counterparty was informed. The loss shows up as the −20 seashells of drift, Kyle's λ acting on the flow.

A maker who refuses to skew against inventory is exposed to both failures. A–S fixes both through a single knob — the risk-aversion coefficient γ.

9.2 Inventory-aware quoting: the reservation price

The previous section's lesson: quotes should not be centred on the market mid, but on a price that *depends on current inventory*. Long inventory: shift both quotes down so the next matching flow is more likely a buy (selling the long down). Short inventory: shift both up so the next flow is more likely a sell (buying the short back). In A–S language, quotes are centred on a private, inventory-dependent **reservation price**.

Setup

A single risky asset has mid S_t evolving as driftless Brownian motion

$$\mathrm{d}S_t \;=\; \sigma\,\mathrm{d}W_t, \tag{9.9}$$

with constant $\sigma > 0$ and W_t a standard Brownian motion. The maker quotes bid p_t^{bid} and ask p_t^{ask}, holds inventory $q_t \in \mathbb{Z}$, and has cash X_t. Fills arrive as Poisson processes with intensity specified in Section 9.3; the reservation-price derivation does not yet need the arrival rate. Trading horizon $[0, T]$, with terminal inventory marked to the mid:

$$\text{terminal wealth} \;=\; X_T \;+\; q_T \cdot S_T. \tag{9.10}$$

The market maker is risk-averse with exponential (CARA) utility

$$U(W) \;=\; -\exp(-\gamma W), \qquad \gamma > 0. \tag{9.11}$$

CARA utility has the convenient property that optimal behaviour depends only on risk aversion γ, not on initial wealth. Define the **value function**

$$v(t, x, S, q) \;:=\; \sup_{p^{\mathrm{bid}}, p^{\mathrm{ask}}} \mathbb{E}\Big[-\exp\big(-\gamma(X_T + q_T S_T)\big) \;\Big|\; X_t = x,\, S_t = S,\, q_t = q\Big], \tag{9.12}$$

with supremum over admissible quoting policies. The value function encodes "best expected utility achievable from state (x, S, q) at time t."

The HJB equation, stated

Stochastic optimal control turns (9.12) into a PDE. Formally (Avellaneda and Stoikov, 2008), §3, (Cartea et al., 2015), Ch 10.2, v satisfies the Hamilton–Jacobi–Bellman equation

$$\begin{aligned} \partial_t v \;+\; \tfrac{1}{2}\sigma^2\,\partial_S^2 v \;+\; \max_{p^{\mathrm{bid}}} \lambda^b(S - p^{\mathrm{bid}})\big[\, v(t, x - p^{\mathrm{bid}}, S, q+1) - v \,\big] \\ +\; \max_{p^{\mathrm{ask}}} \lambda^a(p^{\mathrm{ask}} - S)\big[\, v(t, x + p^{\mathrm{ask}}, S, q-1) - v \,\big] \;=\; 0, \end{aligned} \tag{9.13}$$

with terminal condition

$$v(T, x, S, q) \;=\; -\exp\big(-\gamma(x + qS)\big). \tag{9.14}$$

The first line is mid diffusion; the next two are the control terms, one per side, each reading "pick the quote maximising expected utility gain from a fill, weighted by Poisson fill intensity." The intensities λ^b, λ^a depend on distance from the mid and are parameterised in Section 9.3.

This chapter will **not** solve (9.13). Avellaneda and Stoikov (2008) solve it via a substitution separating the (x, S) dependence from the inventory dependence, yielding a one-dimensional ODE in q with an exact solution. The three pages of algebra add nothing to intuition; only the output matters here.

Remark 9.1 (Why state an unsolved HJB?). The reservation-price and optimal-spread formulas are genuine byproducts of a stochastic-control problem, not identities one can guess. Stating the HJB shows where the closed form comes from and points readers at (Avellaneda and Stoikov, 2008) and (Cartea et al., 2015) for the algebra. The same pattern recurs in Chapter 19 for optimal execution.

Reservation price via indifference

A shorter route avoids the HJB entirely and is the one worth internalising. Define the reservation price by an **indifference** condition: $r(t, S, q)$ is the price at which a maker holding q shares is indifferent between the current portfolio and the portfolio after buying one more share at r. Equivalently, it is the private mid the maker would use if forced to quote a single number and trade both sides at it.

Write the indifference condition in utility-adjusted wealth. With cash x and inventory q, the certainty-equivalent portfolio value is

$$\mathrm{CE}(t, x, S, q) \;=\; -\frac{1}{\gamma}\ln\bigl(-\mathbb{E}[\,U(X_T + q_T S_T)\,]\bigr), \tag{9.15}$$

with expectation under the pure-hold strategy: no trading between t and T, just mark q to the terminal mid. Under (9.9) the terminal mid is $S_T \sim \mathcal{N}(S, \sigma^2(T-t))$, so terminal wealth is normal with mean $x + qS$ and variance $q^2\sigma^2(T-t)$. For CARA utility and a normal argument, the standard identity

$$\mathbb{E}[-\exp(-\gamma Y)] \;=\; -\exp\bigl(-\gamma\,\mathbb{E}[Y] + \tfrac{1}{2}\gamma^2\,\mathrm{Var}[Y]\bigr) \tag{9.16}$$

applies, so the certainty equivalent collapses to

$$\mathrm{CE}(t, x, S, q) \;=\; x \;+\; qS \;-\; \tfrac{1}{2}\gamma q^2\sigma^2(T-t). \tag{9.17}$$

The first two terms are the mark-to-market value; the third is the *risk penalty* the inventory-q holder pays for mark-to-market variance between now and T. It is negative-definite in q: bigger positions hurt more.

The reservation price is now defined by the indifference equation: the maker is indifferent between (x, q) and $(x - r, q + 1)$, i.e. paying r to acquire one more share:

$$\mathrm{CE}(t, x, S, q) \;=\; \mathrm{CE}(t, x - r, S, q + 1). \tag{9.18}$$

Plug (9.17) into both sides. All linear-in-S and linear-in-x terms cancel except the payment r and the shift of the quadratic penalty. After cancellation,

$$qS - \tfrac{1}{2}\gamma q^2\sigma^2(T-t) \;=\; -r + (q+1)S - \tfrac{1}{2}\gamma(q+1)^2\sigma^2(T-t), \tag{9.19}$$

$$\begin{aligned} r \;&=\; S \;-\; \tfrac{1}{2}\gamma\sigma^2(T-t)\,\bigl[(q+1)^2 - q^2\bigr] \\ &=\; S \;-\; \tfrac{1}{2}\gamma\sigma^2(T-t)(2q+1). \end{aligned} \tag{9.20}$$

For the marginal (infinitesimal) indifference defining the A–S continuous-share reservation price, drop the $+1$ relative to $2q$ (equivalent to differentiating the certainty equivalent in q):

$$\boxed{r(t, S, q) \;=\; S \;-\; q\,\gamma\,\sigma^2\,(T-t)} \tag{9.21}$$

which is the Avellaneda–Stoikov reservation price (Avellaneda and Stoikov, 2008), eq. (9). The full HJB solution reproduces this formula with no correction terms in the large-$\kappa\sigma^2 T/\gamma$ regime that (Cartea et al., 2015) flag as practically-relevant. The indifference derivation above is not an approximation — it is the exact marginal statement.

Avellaneda–Stoikov reservation price

The A–S reservation price is

$$r(t, S, q) \;=\; S \;-\; q\,\gamma\,\sigma^2\,(T - t).$$

Reading the parameters:

- S is the current market mid.
- q is the market maker's current inventory in shares (positive = long, negative = short).
- $\gamma > 0$ is the market maker's risk aversion.
- σ is the mid's volatility (same units as S, per $\sqrt{\text{time}}$).
- $T - t$ is the time remaining until the risk horizon (end of session, end of trading day).

The second term is the *inventory skew*: the shift of the reservation price away from the market mid in response to an open position. Long inventory ($q > 0$) pushes r *down* relative to S; short inventory ($q < 0$) pushes r *up*. The magnitude of the shift grows with risk aversion, with variance per unit time σ^2, and with time remaining $(T - t)$.

Intuition

The reservation price is your *private* fair value. It is what you would accept to trade one more share in either direction, given that you already hold q shares and are averse to carrying position risk over the remaining horizon. A risk-neutral trader ($\gamma = 0$) would have $r = S$: they are equally happy to hold five shares or fifty. A risk-averse trader with non-zero inventory knows the variance of their mark-to-market is $q^2\sigma^2(T - t)$, and shifts their private mid in the direction that makes the next trade reduce q^2. A–S centres the market maker's *quotes* on this private mid, not on the market mid.

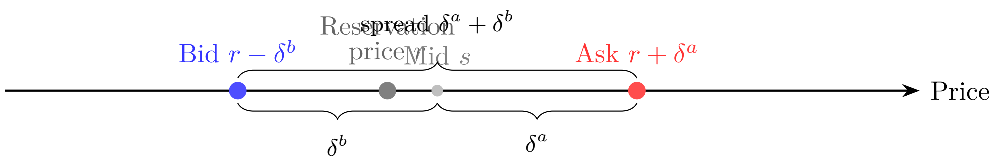


Figure 9.1: Avellaneda–Stoikov quoting structure. The maker posts bid at $r - \delta^b$ and ask at $r + \delta^a$, where r is the *reservation price* — the mid adjusted downward (upward) when inventory is long (short). When the maker holds excess long inventory, r shifts below the market mid, making the bid less competitive and the ask more competitive, steering inventory back to zero.

A worked reservation-price example, and a calibration lesson

Take a concrete Prosperity-style setup. The product is RAINFOREST_RESIN, the market mid is $S = 10\,000$ seashells, the market maker currently holds $q = +10$ shares (a long position), volatility is $\sigma = 5$ seashells per unit time, and $T - t = 100$ units. A careless first choice of the risk aversion parameter is $\gamma = 0.1$. Plug in:

$$\begin{aligned} r &= 10\,000 - (+10)\cdot 0.1\cdot 25\cdot 100 \\ &= 10\,000 - 2500 = 7500. \end{aligned} \tag{9.22}$$

The formula says a maker holding ten shares long should shift the private mid to 7500 — a 2500-seashell shift. That is absurd: no maker would lower quotes by that much to offload ten shares on a product whose round-trip spread is a few seashells.

The absurdity is a calibration lesson, not a bug. The dimensioned quantity controlling inventory skew is

$$\gamma\sigma^2(T-t), \tag{9.23}$$

which has units of (price per share). A–S parameterises the penalty through γ; σ and $T-t$ are given by the problem (σ measured from data, horizon fixed by session). The only dial the practitioner turns is γ, and it must be tuned so that $\gamma\sigma^2(T-t)$ has the same order of magnitude as the spread, not orders of magnitude larger.

Rerun the same example with $\gamma = 0.001$:

$$\begin{aligned} r &= 10\,000 - 10\cdot 0.001\cdot 25\cdot 100 \\ &= 10\,000 - 25 = 9975. \end{aligned} \tag{9.24}$$

The reservation price shifts by 25 seashells — enough to lean noticeably against the long inventory, not so much that the quotes walk off one side of the book.

The product $\gamma\sigma^2(T-t)$ is what matters

γ is dimensioned (1/currency); only the product $\gamma\sigma^2(T-t)$ is economically meaningful. σ and $T-t$ are given by the market and session, so γ is the only real dial. Rule of thumb (Hummingbot Foundation, 2021): tune γ so that $\gamma\sigma^2(T-t)$ is comparable to the typical per-share spread capture of the symmetric policy at $q=0$. That keeps the inventory skew large enough to control position drift but small enough that quotes stay on the book. Copying $\gamma = 0.1$ from another asset's config and applying it on a flat-valued product like RAINFOREST_RESIN is the fastest way to get a bot that never trades.

9.3 The optimal bid and ask

The reservation price says *where* to centre quotes, not *how wide*. Quotes near r fill on almost every arrival and collect little per fill; wide quotes fill rarely but collect more per fill. The right half-width maximises the product of fill probability and edge per fill.

A–S model the fill probability through a Poisson arrival whose intensity depends on distance from the *market mid*. Empirically (Avellaneda and Stoikov, 2008), §4, the fraction of arriving market orders reaching a quote at distance δ decays exponentially. Writing $\delta^b, \delta^a > 0$ for the distances from mid to bid and ask,

$$\lambda^b(\delta^b) = A\exp(-\kappa\,\delta^b), \qquad \lambda^a(\delta^a) = A\exp(-\kappa\,\delta^a), \tag{9.25}$$

where $A > 0$ is the overall market-order arrival rate and $\kappa > 0$ is the **decay constant** of fill probability with distance. Larger κ corresponds to a deeper, more liquid book: moving the quote outward is penalised harder.

Remark 9.2 (Origin of the exponential fit). The $\exp(-\kappa\delta)$ fit is not ad-hoc. A–S derive it from (i) power-law order sizes (consistent with the impact data of Chapter 7) and (ii) approximately constant LOB density around the mid. Under both, the fraction of arriving market orders reaching a quote at distance δ decays exponentially, with κ = (power-law exponent) / (mean order size in ticks).

Setup and the optimisation problem

Extend the Section 9.2 setup with fills governed by (9.25). A bid fill jumps $(x,q) \to (x-p^{\text{bid}}, q+1)$; an ask fill jumps $(x,q) \to (x+p^{\text{ask}}, q-1)$. Between fills, the mid diffuses. The maximisation over $p^{\text{bid}}, p^{\text{ask}}$ inside the HJB (9.13) defines the optimal spread.

Avellaneda and Stoikov (2008) solve the HJB via the substitution $v(t,x,S,q) = -\exp(-\gamma(x+qS))\,\theta(t,q)^{-\gamma/\kappa}$, turning the PDE into a linear ODE in θ; Cartea et al. (2015) give the same derivation in cleaner notation. The output is closed-form optimal quotes and — crucial for teaching — an asymptotic formula for the *total* bid-ask spread $\delta^* = \delta^a + \delta^b$ holding in the regime $\kappa\sigma^2 T/\gamma \gg 1$.

Avellaneda–Stoikov optimal spread

Under the setup of Section 9.2, with Brownian mid, exponential fill intensities $\lambda^{a,b}(\delta) = Ae^{-\kappa\delta}$, and CARA utility with risk aversion γ, the optimal total quoted spread (bid to ask) is (Avellaneda and Stoikov, 2008), eq. (17), (Cartea et al., 2015), §10.4

$$\delta^* \;=\; \gamma\,\sigma^2\,(T-t) \;+\; \frac{2}{\gamma}\,\ln\!\left(1+\frac{\gamma}{\kappa}\right).$$

The optimal quotes are placed symmetrically around the reservation price of (9.21):

$$p^{*,\,\text{bid}} \;=\; r - \tfrac{1}{2}\delta^*, \qquad p^{*,\,\text{ask}} \;=\; r + \tfrac{1}{2}\delta^*.$$

Note the convention: δ^* is the *total* bid-ask spread. The half-spread used on each side is $\delta^*/2$.

Reading the two terms

The formula has two separable components.

Inventory-risk term, $\gamma\sigma^2(T-t)$. Independent of κ; widens with risk aversion, volatility, and time remaining. Economic meaning: the more variance per unit time and the more time remaining, the more the maker demands for holding any position across the uncertain window. At $\gamma = 0$ (risk-neutral) the term vanishes. This is the descendant of the "inventory" cause of the Chapter 5 decomposition.

Fill-rate term, $(2/\gamma)\ln(1+\gamma/\kappa)$. Independent of σ and the horizon; widens as κ *decreases* (thinner book). As $\kappa \to \infty$ (infinitely liquid), the term vanishes — the maker earns the same expected edge at $\delta = 0$. As $\kappa \to 0$, the term grows without bound. This is the descendant of the "order-processing / fill-rate" cause: to get filled at all, the maker must quote at positive distance from the mid. The Roll and GM decompositions in Chapters 5 and 6 implicitly contain this effect in their reduced-form "spread" parameter.

Sanity check: the risk-neutral limit

Check the fill-rate term as $\gamma \to 0$. Expand $\ln(1+\gamma/\kappa)$ in powers of γ/κ:

$$\ln\!\left(1+\frac{\gamma}{\kappa}\right) \;=\; \frac{\gamma}{\kappa} \;-\; \tfrac{1}{2}\left(\frac{\gamma}{\kappa}\right)^2 \;+\; O\!\left(\frac{\gamma^3}{\kappa^3}\right). \tag{9.26}$$

Multiply by $2/\gamma$:

$$\frac{2}{\gamma}\,\ln\!\left(1+\frac{\gamma}{\kappa}\right) \;=\; \frac{2}{\kappa} \;-\; \frac{\gamma}{\kappa^2} \;+\; O(\gamma^2). \tag{9.27}$$

The $\gamma \to 0$ limit gives $2/\kappa$; the inventory-risk term vanishes too. So a risk-neutral maker quotes

$$\delta^*_{\text{risk-neutral}} = \frac{2}{\kappa}. \tag{9.28}$$

Not a coincidence: in the risk-neutral benchmark the maker independently optimises each side by maximising $\delta \cdot A\exp(-\kappa\delta)$ in δ, whose unique maximum is at $\delta = 1/\kappa$; two sides give $2/\kappa$ total. A–S correctly reduces to the independent-sides profit-maximiser when risk aversion is turned off.

Intuition

The A–S total spread is the sum of two terms you would write down anyway. The fill-rate term is how wide you must quote to extract profit from a one-sided auction ($2/\kappa$, tweaked by γ). The inventory term is the extra width for carrying inventory across a volatile horizon. At $\gamma = 0$ the inventory term vanishes; at $\sigma = 0$ or $T = t$ it vanishes too. The real work the model does is the *ratio* between the two terms: it tells you which failure mode — narrow spreads eaten by variance, or wide spreads going unfilled — dominates, so you size them against each other.

Connecting the two terms to the three causes of the spread

Chapter 5 decomposed the observed spread into three causes: *order processing*, *inventory*, *adverse selection*. A–S maps cleanly onto two of them; the third is ceded to other models.

Inventory cause. The A–S inventory-risk term $\gamma\sigma^2(T-t)$ is the descendant of Roll's inventory cause. Roll only observed that the spread must contain an inventory-risk component; A–S pins down the size, proportional to variance per unit time and risk aversion. (Cartea et al., 2015) call this the "internal hedging cost": what the maker would pay to instantaneously offload one unit of inventory.

Order-processing cause (fill-rate term). The A–S fill-rate term $(2/\gamma)\ln(1+\gamma/\kappa)$ descends from the order-processing cause. The modern reading is broad: order processing is the cost of being in the game at all. Quote at the mid and you fill too often to monetise edge; quote far and you do not fill. The single-sided optimum is $1/\kappa$; two sides give $2/\kappa$ as the risk-neutral order-processing contribution.

Adverse-selection cause. Not in the A–S formula. The symmetric fill intensities (9.25) rule it out by construction. A–S concedes the adverse-selection component to Glosten–Milgrom (Glosten and Milgrom, 1985) and Kyle (Kyle, 1985) (see Chapter 6); the practical fix is to shift the reservation price with an OFI-based drift estimator (see Section 9.8). (Bouchaud et al., 2018) gives the combined inventory-plus-adverse-selection quoting rule at length.

Together, the three-cause decomposition and A–S answer "what is the spread for?": the maker must recover order processing ($2/\kappa$), inventory risk ($\gamma\sigma^2(T-t)$), and adverse selection (ceded to GM/Kyle). Each component has a separate analytic form, so a well-engineered system treats them as three modules combined at quoting time.

A numerical example, step by step

Take the same Prosperity-style setup as in Section 9.2: $S = 10\,000$, $q = +10$, $\sigma = 5$, $T - t = 100$, and now $\kappa = 1.5$. Use the calibrated $\gamma = 0.001$ so the reservation price comes out to the plausible $r = 9975$ computed in (9.24).

Step 1: inventory-risk term.

$$\gamma\sigma^2(T-t) \;=\; 0.001\cdot 5^2\cdot 100 \;=\; 0.001\cdot 2500 \;=\; 2.5. \tag{9.29}$$

Step 2: fill-rate term.

$$\begin{aligned}\frac{2}{\gamma}\ln\!\left(1+\frac{\gamma}{\kappa}\right) \;&=\; \frac{2}{0.001}\ln\!\left(1+\frac{0.001}{1.5}\right)\\ &=\; 2000\cdot\ln(1.000\overline{6})\\ &=\; 2000\cdot 0.00066644\ldots\\ &\approx\; 1.33289.\end{aligned} \tag{9.30}$$

Step 3: total spread.

$$\delta^* \;=\; 2.5 \;+\; 1.33289 \;=\; 3.83289\ldots \tag{9.31}$$

Step 4: half-spread.

$$\delta^*/2 \;\approx\; 1.91644. \tag{9.32}$$

Step 5: quotes. With $r = 9975$ (long inventory has shifted the reservation price down by 25 seashells from the mid 10 000),

$$p^{*,\,\mathrm{bid}} \;=\; r-\delta^*/2 \;=\; 9975-1.91644 \;\approx\; 9973.08, \tag{9.33}$$

$$p^{*,\,\mathrm{ask}} \;=\; r+\delta^*/2 \;=\; 9975+1.91644 \;\approx\; 9976.92. \tag{9.34}$$

Compare against a symmetric maker centred on the mid 10 000 with the same width 3.833: that maker quotes 9998.08/10 001.92. The A–S quote sits about 25 seashells lower on both sides, so the next buy is far more likely to lift our ask at 9976.92 than a competitor's at 10 001.92 — each such fill reduces the long by one. Meanwhile our bid at 9973.08 sits well below the mid, so the next sell is less likely to hit it. That asymmetry is the skew Section 9.1 asked for.

9.4 The full A–S quoting rule

Combine (9.21) with the optimal-spread formula from the The optimal bid and ask `keyfactbox`:

$$p^{*,\,\mathrm{bid}}(t,S,q) \;=\; S \;-\; q\,\gamma\sigma^2(T-t) \;-\; \tfrac{1}{2}\delta^*, \tag{9.35}$$

$$p^{*,\,\mathrm{ask}}(t,S,q) \;=\; S \;-\; q\,\gamma\sigma^2(T-t) \;+\; \tfrac{1}{2}\delta^*, \tag{9.36}$$

with δ^* from the keyfactbox in Section 9.3; (Cartea et al., 2015) present the same combined rule in cleaner notation. Read (9.35)–(9.36) carefully. The **centre** $S - q\gamma\sigma^2(T-t)$ depends on inventory: long shifts it down, short shifts it up. The **width** δ^* does not depend on inventory at all. The two pieces are fully separable.

Intuition

A–S skews your *centre*, never your *width*. The edge per trade is constant; only the price those trades are centred on moves. Running long makes both bid and ask cheaper, so the next buy is more likely to lift the ask (pulling inventory toward zero) and the next sell less likely to hit the bid (growing it further). Running short does the opposite. Inventory is controlled by biasing which side gets hit, not by widening the edge demanded.

9.5 Parameter intuition and tuning

Four parameters, four different sources, four different tuning stories.

σ — **from the data.** Mid volatility is not a free knob; it is whatever a short-window estimator (e.g. an EWMA of $(\Delta p^{\text{mid}})^2$) reports. Production systems re-estimate σ at high frequency. On Prosperity estimate it from the previous few hundred ticks of mid. Units matter: σ is price per square root of time, in whichever clock $T - t$ is measured. Confusing per-tick with per-session time makes the inventory-risk term land orders of magnitude off.

κ — **fit from fill data.** The decay κ is a property of the book, not a knob. Fit it by logging every quote at distance δ and whether it filled before cancellation, then regressing log(fill prob) on δ with slope $-\kappa$. Liquid crypto markets report $\kappa \in [1, 5]$ per inverse tick; Prosperity is typically 0.5–1.5 because bot fills are less price-sensitive than human ones. κ is asset-specific and should be refit when the regime changes.

$T - t$ — **given by the session.** On Prosperity a "session" is one submitted episode (a few hundred to a few thousand timesteps). In live equity making it is usually one trading day; the inventory-risk term shrinks into the close. Trap: as $T - t \to 0$, both the inventory term and the reservation-price skew vanish, so the model loses inventory control in the final minutes. Production systems either force a flat position or cap $(T - t)$ from below.

γ — **the only real knob.** Per Section 9.2, γ is the only freely tunable parameter. Operational rule: pick γ so that the inventory skew at maximum *tolerable* inventory is a meaningful fraction of δ^* at that inventory. On Prosperity, with position limit 50 and a flat-valued product, start at $\gamma = 10^{-3}$ and backtest.

Historical note: from Ho–Stoll to Avellaneda–Stoikov

Inventory-dependent dealer quotes predate A–S. Ho and Stoll (1981) wrote a two-period model where a risk-averse dealer with inventory q quotes asymmetrically to drive expected post-trade inventory to zero — the same skewing logic, without continuous-time machinery. (Glosten and Milgrom, 1985) added the informational (adverse-selection) side four years later, deriving a spread from Bayesian updating alone. A–S (Avellaneda and Stoikov, 2008) is the first closed-form unification: continuous-time, risk-averse, Poisson fills with distance-dependent intensity. Every subsequent inventory-aware treatment — including Cartea et al. (2015) — builds on the same two-term decomposition.

9.6 A short reference implementation

The whole A–S rule fits in under a dozen lines of Python. The reference implementation below is a *pure function* from $(p^{\text{mid}}, q, \gamma, \sigma, \kappa, T - t)$ to $(p^{*,\text{bid}}, p^{*,\text{ask}})$ — the shape the Prosperity `Trader.run(state)` loop wants to call.

```python
import math

def as_quotes(mid: float, q: int, gamma: float, sigma: float,
              kappa: float, T_minus_t: float) -> tuple[float, float]:
    """Return (bid, ask) using the Avellaneda-Stoikov formulas.

    mid        : current market mid price
    q          : current inventory (positive=long, negative=short)
    gamma      : risk aversion (1/currency)
```

```
    sigma       : volatility of the mid (price / sqrt(time))
    kappa       : exponential decay of fill intensity with distance
    T_minus_t   : remaining session time, same units as sigma
    """
    r = mid - q * gamma * sigma**2 * T_minus_t
    inv_term  = gamma * sigma**2 * T_minus_t
    fill_term = (2.0 / gamma) * math.log(1.0 + gamma / kappa)
    delta_star = inv_term + fill_term
    half = 0.5 * delta_star
    return r - half, r + half
```

Listing 9.1: Avellaneda–Stoikov quoting as a pure function.

Two production-use remarks the skeleton does not encode. First, κ should be refit online from the last few hundred fills — the function assumes the caller passes a fresh value. Second, q should already be clipped to the position limit (e.g. ± 50 for RAINFOREST_RESIN). A well-calibrated bot rarely hits the clip; a miscalibrated one will produce quotes so skewed that one side walks off the book. Top Prosperity submissions (see Section 9.7) combine a lightly-skewed base quote with a hard inventory clip rather than letting A–S do the whole job.

9.7 A–S on Prosperity and on a Goldman desk

Prosperity example: RAINFOREST_RESIN

RAINFOREST_RESIN is the canonical "static" Prosperity product: the fair value of every round is pinned near 10 000 seashells, empirical volatility of the mid over a single episode sits in $\sigma \in [1.5, 2.0]$, and a position limit of 50 shares caps directional risk to at most $\pm 50 \cdot 2 \approx 100$ seashells of mark-to-market swing over $\pm 1\sigma$ moves. Calibrate A–S on a running inventory $q = +4$ with $\kappa = 0.8$, $\gamma = 0.001$, $T - t = 500$:

- **Reservation price** at $\sigma = 2$: $r = 10\,000 - 4 \cdot 0.001 \cdot 4 \cdot 500 = 9992$. At $\sigma = 1.5$: $r = 10\,000 - 4 \cdot 0.001 \cdot 2.25 \cdot 500 = 9995.5$.
- **Inventory-risk term** at $\sigma = 2$: $0.001 \cdot 4 \cdot 500 = 2.0$. At $\sigma = 1.5$: $0.001 \cdot 2.25 \cdot 500 = 1.125$.
- **Fill-rate term**: $(2/0.001)\ln(1 + 0.001/0.8) = 2000 \cdot \ln(1.00125) \approx 2.498$.
- **Total spread**: ≈ 4.498 at $\sigma = 2$, ≈ 3.623 at $\sigma = 1.5$.
- **Half-spread**: ≈ 2.249 or ≈ 1.812.
- **Quotes at** $\sigma = 2$: bid ≈ 9989.75, ask ≈ 9994.25 — both *below* the true fair value 10 000, skewed down by the $+4$ inventory so that the maker is more likely to sell than buy.

Now compare against the actual top-team policy. The Frankfurt Hedgehogs' RAINFOREST_RESIN trader (`imc-prosperity-3/FrankfurtHedgehogs_polished.py`, `StaticTrader.get_orders()`) quotes `bid_wall` $+ 1$ and `ask_wall` $- 1$ — exactly one tick inside the deepest visible wall prices. The wall mid is always 10 000, so the base-case quotes are the fixed offsets 9 999 bid / 10 001 ask (or very near it). Inventory control is done by a hard clip at the position limit, not by shifting the centre of the quotes.

The comparison is instructive. Pure A–S on RAINFOREST_RESIN recommends wider quotes than the top teams use (≈ 3.6–4.5 vs a fixed 2-seashell round-trip). A–S is not wrong; its inventory-risk term is conservatively provisioned for a product whose true risk is dominated by a known-flat fair value and a hard position limit, not by mid variance. On RAINFOREST_RESIN the mark-to-market variance comes almost entirely from reversals within a tight range, not accumulating drift — so A–S overstates the risk and prices it into an overly-wide spread.

The lesson that transfers off Prosperity is the decomposition, not the literal formula: A–S's reservation-price skew is the right control knob for inventory management, but when the price process is not driftless Brownian (flat-value, mean-reverting, news-driven), the variance term must be recalibrated against the actual source of risk — or, as top Prosperity teams do, replaced with an external device like a hard position clip or an EMA fair-value re-anchor.

On a Goldman Sachs desk

A–S (or a close relative) is the backbone of automated internal market making for single-stock direct-market-access (DMA) flow on sell-side equity desks. Strats own the calibration pipeline: σ refit every five minutes from a rolling realised-vol window, κ refit continuously from live fill rates at each offset, γ set by the risk committee from the desk's daily VaR (Value at Risk) budget. The trader has one override: a "desk-breaker" raising γ in real time, tightening inventory control and widening the fill-rate term at once — pressed when a name's inventory drifts past an internal soft limit (set lower than the hard regulatory limit). As a Strats intern, expect to write the code that reestimates κ from overnight fill logs and to explain why a name's A–S spread widened from 3 bps to 5 bps — answer almost always: a σ spike from overnight news, or κ drop because a flow provider pulled out and the book thinned. See Chapter 28 for the role itself.

9.8 Extensions, limitations, and when A–S breaks

A–S's closed-form optimality rests on four assumptions, each violated by some real markets. This section flags the four failure modes and points at the fixes.

Asymmetric quotes near the position limit. The Section 9.3 derivation assumes q_t is unconstrained. Real makers have hard limits $\bar{q}$, and as $|q|$ approaches $\bar{q}$ the quote that would grow $|q|$ further must be suppressed (or moved so far it will not fill). (Cartea et al., 2015) derive the constrained version: near the long limit, the bid is widened dramatically while the ask stays close to r, so the next fill almost certainly reduces inventory. Practitioners implement this as a soft wall: multiply the one-sided half-spread by a factor that grows as $|q|/\bar{q} \to 1$.

Jump risk and the Brownian assumption. (9.9) makes the mid driftless Brownian. Real mids jump, cluster (Chapter 8 Hawkes), and mean-revert at short horizons. With drift or jumps, the cancellation of S-linear terms in Section 9.2 fails and the reservation price picks up a drift term. Fix: replace bare S with $S + \hat{\mu}(T - t)$ where $\hat{\mu}$ is a short-horizon drift estimator (EMA of recent returns).

Informed flow. (9.25) makes fill intensities symmetric in distance to mid — a strong simplification. With informed traders in the Chapter 6 sense, fill intensities are *anti*-correlated with the direction the mid is about to move. A maker filled on the ask at $p^{\text{mid}} + \delta$ has just received a signal that the mid is more likely to rise — the adverse-selection effect Glosten–Milgrom formalised, ignored by A–S. Fix: use an OFI-based estimator (Chapter 8) and add a drift adjustment to r when OFI is persistently one-sided.

Queue position. Chapter 8 established that queue position is a substantial alpha source: a front-of-queue resting order at p^{bid} is worth meaningfully more than a back-of-queue one at the same price. A–S treats price as the only decision variable and is blind to queue position. On any price–time-priority venue with slow quote refresh, the right operational layer on top of A–S is a queue-aware policy that avoids cancelling a front-of-queue order unnecessarily, even when A–S's recommended quote has shifted slightly. (Bouchaud et al., 2018) covers this interaction in detail; queue priority is a separate concern, applied after the A–S rule.

Don't run vanilla A–S on a trending asset

A–S's reservation-price skew controls inventory *variance*, not *drift*. On a persistently drifting asset — a trending stock, a commodity in storage glut, Prosperity's SQUID_INK during an Olivia informed-trading window — the machinery will keep recentering quotes on a mid that is drifting the wrong way, and the maker accumulates an unhedged position at precisely the wrong time. Increasing γ does not cure it (it widens quotes without correcting drift); the cure is a separate drift estimator that shifts the whole quote pair, or refusing to quote when OFI is persistently one-sided. Vanilla A–S is a pure variance-control device — plugging it into a drifting environment without a drift signal is how Section 9.1's pathological run (9.8) comes back to bite.

9.9 Key facts to memorise

- Naive symmetric quoting earns δ per trade but is exposed to inventory drift $q \cdot \Delta p^{\text{mid}}$; on persistently one-sided flow, drift dominates spread capture.
- A–S **reservation price** $r = S - q\gamma\sigma^2(T - t)$ is the maker's private fair value; quotes are centred on r, not on the mid.
- A–S **optimal total spread** is the sum of an inventory-risk term $\gamma\sigma^2(T - t)$ and a fill-rate term $(2/\gamma)\ln(1 + \gamma/\kappa)$.
- As $\gamma \to 0$, the inventory term vanishes and the fill-rate term reduces to $2/\kappa$, the independent-sides profit-maximising spread.
- A–S skews the *centre*, never the *width*. Inventory is controlled by biasing which side gets filled.
- σ from data, κ from fills, $T - t$ from the session. γ is the only dial: tune so $\gamma\sigma^2(T - t)$ is comparable to per-trade spread capture. Copying from another asset gives a bot that never trades.
- A–S controls variance, not drift. On trending or informed-flow assets, pair with a drift estimator (EMA of returns, OFI from Chapter 8).
- Top Prosperity teams use fixed offsets from a known flat value on RAINFOREST_RESIN with a hard position clip — A–S is the right lens but overstates risk on that product.
- A–S is the default automated maker on live sell-side equity desks; Strats own σ, κ calibration, trader owns the γ override.
- Ho–Stoll (1981) is the predecessor; Glosten–Milgrom (1985) is the parallel adverse-selection development; A–S (2008) is the continuous-time unification.

Chapter 10

Mean Reversion and the Ornstein–Uhlenbeck Process

You have a time series that wanders around a constant value: strays high, comes back; strays low, comes back. Can you trade it? The naive answer is short the highs and buy the lows, but turning "it looks mean-reverting" into a strategy requires three quantitative answers in order:

1. Is the series *actually* mean-reverting, or does it only look that way? Most series that visually appear to revert are random walks that happened to visit the same level twice. Trading them on a mean-reversion thesis loses money.

2. If the series *is* mean-reverting, what is the half-life — how long does the average excursion take to fade? A half-life of one minute is tradeable on a one-day horizon; a half-life of six months is not.

3. Given the half-life, when do you enter, exit, and how do you size? The standard answer is the ***z*-score rule**: enter when the standardised deviation is large enough that the expected reversion is worth the capital, exit when it has decayed close to zero.

This chapter answers all three with the **Ornstein–Uhlenbeck (OU) process**, the canonical continuous-time model of a mean-reverting variable. The OU model solves (2) with an explicit half-life formula in one parameter; it solves (3) by turning that parameter into entry and exit thresholds; and it gives partial answers to (1) via three statistical tests (Augmented Dickey–Fuller, variance ratio, Hurst exponent). We follow Chan (Chan, 2013) and the more rigorous Cartea–Jaimungal–Penalva (Cartea et al., 2015), with one worked example carried through every section.

Context. The Glosten–Milgrom posterior of Chapter 6 is a *martingale*: no reversion, best prediction is the current value. OU is the opposite. Efficient prices of tradeable securities are martingales by no-arbitrage; *spreads*, ETF *premia*, and *deviations* from a slow fair value are mean-reverting. Recognising which is which is half the work; the rest is the OU machinery.

Prerequisites

From earlier in the book:

- Chapter 5: the Roll (1984) decomposition $\mathrm{Cov}(\Delta p_t, \Delta p_{t+1}) = -s^2/4$. Roll is the discrete-time trade-horizon mean-reverter; OU is the continuous-time generalisation that separates reversion speed from size.

- Chapter 6: the Glosten–Milgrom posterior is a *martingale*. Opposite worldview — contrast deliberately.

- Chapter 8: Hawkes processes modelled clustered *arrivals*. OU is the mean-reversion analogue for clustered *values* (drifting back to a mean rather than wandering off).
- Chapter 9: the Avellaneda–Stoikov σ for mid-price volatility is the same symbol and units as the OU driving-Brownian volatility here.

From outside the book:

- Stochastic calculus, informally: Brownian motion W_t, Itô integral $\int_0^t f(s)\,dW_s$, Itô isometry $\mathbb{E}[(\int_0^t f\,dW_s)^2] = \int_0^t f^2\,ds$. Full machinery is built in Chapter 16.
- OLS linear regression; the t-statistic, AR(1), and standard normal.

10.1 The Ornstein–Uhlenbeck process

Introduced in 1930 by Ornstein and Uhlenbeck for a particle subject to friction and thermal kicks, OU is the workhorse of stochastic mean reversion in finance: every short-rate model in Chapter 30 uses an OU-like SDE, every cointegrated pair in Chapter 11 treats its residual as OU, and every mean-reversion strategy here is parameterised by the OU half-life. (Symbols: θ mean-reversion rate, μ long-run mean, σ instantaneous volatility, W_t Brownian motion, $\tau_{1/2}$ half-life, z standardised deviation.)

Definition 10.1 (Ornstein–Uhlenbeck process)**.** A stochastic process $(X_t)_{t\geq 0}$ taking values in $\mathbb{R}$ is an *Ornstein–Uhlenbeck process* with parameters (θ, μ, σ) if it satisfies the stochastic differential equation

$$dX_t \;=\; \theta\,(\mu - X_t)\,dt \;+\; \sigma\,dW_t, \qquad \theta > 0,\ \mu \in \mathbb{R},\ \sigma > 0, \tag{10.1}$$

where W_t is a standard Brownian motion. The parameter θ is the *mean-reversion speed*, μ is the *long-run mean*, and σ is the *instantaneous volatility*.

The drift $\theta(\mu - X_t)$ does the work. When $X_t > \mu$ the drift is negative and pulls down; when $X_t < \mu$ it pulls up, with strength proportional to the distance from μ. Larger θ means faster reversion. The noise $\sigma\,dW_t$ kicks X_t around as it reverts; without it X_t would decay exponentially toward μ.

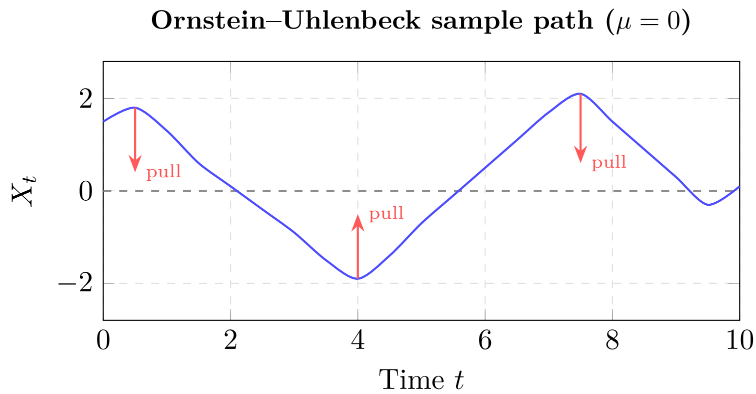


Figure 10.1: Simulated Ornstein–Uhlenbeck path. Red arrows show the mean-reversion force: whenever X_t strays from $\mu = 0$, the drift term $\theta(\mu - X_t)$ pulls it back. The half-life $\tau_{1/2} = \log(2)/\theta$ sets the typical return-to-mean timescale. Larger θ produces faster, tighter oscillations; $\theta \to 0$ recovers a random walk.

Solving the SDE

Equation (10.1) is a first-order linear SDE and solves in closed form by the integrating-factor trick. Multiply both sides by $e^{\theta t}$:

$$e^{\theta t}\,dX_t \;+\; \theta\,e^{\theta t}\,X_t\,dt \;=\; \theta\,e^{\theta t}\,\mu\,dt \;+\; \sigma\,e^{\theta t}\,dW_t. \tag{10.2}$$

The left-hand side is a perfect differential. Applying Itô's product rule to $f(t, X_t) = e^{\theta t} X_t$ (linear in X_t, so the second-order term vanishes),

$$\begin{aligned} d\Big(e^{\theta t} X_t\Big) \;&=\; \theta\,e^{\theta t}\,X_t\,dt \;+\; e^{\theta t}\,dX_t \\ &=\; e^{\theta t}\,dX_t \;+\; \theta\,e^{\theta t}\,X_t\,dt, \end{aligned} \tag{10.3}$$

which is exactly the left-hand side of (10.2). So (10.2) reads

$$d\Big(e^{\theta t} X_t\Big) \;=\; \theta\,\mu\,e^{\theta t}\,dt \;+\; \sigma\,e^{\theta t}\,dW_t. \tag{10.4}$$

Integrate both sides from 0 to t:

$$\begin{aligned} e^{\theta t}\,X_t \;-\; X_0 \;&=\; \theta\,\mu \int_0^t e^{\theta s}\,ds \;+\; \sigma \int_0^t e^{\theta s}\,dW_s \\ &=\; \mu\,(e^{\theta t} - 1) \;+\; \sigma \int_0^t e^{\theta s}\,dW_s. \end{aligned} \tag{10.5}$$

Divide through by $e^{\theta t}$ and rearrange:

$$\boxed{X_t \;=\; X_0\,e^{-\theta t} \;+\; \mu\,(1 - e^{-\theta t}) \;+\; \sigma \int_0^t e^{-\theta(t-s)}\,dW_s.} \tag{10.6}$$

Three pieces. $X_0 e^{-\theta t}$ is the *exponential decay of the initial condition*: the influence of X_0 shrinks at rate θ. $\mu(1 - e^{-\theta t})$ is the *exponential growth of the mean toward* μ: the deterministic part heads to μ regardless of start. The Itô integral is the *accumulated noise*, exponentially weighted so recent shocks count more — old kicks have already reverted away.

Mean and variance at finite time

Take expectations of (10.6). The Itô integral of a deterministic integrand is a zero-mean Gaussian, so it drops out of the expectation:

$$\mathbb{E}[X_t] \;=\; X_0\,e^{-\theta t} \;+\; \mu\,(1 - e^{-\theta t}). \tag{10.7}$$

The expected value moves exponentially from X_0 toward μ at rate θ.

For the variance, the deterministic terms contribute zero. The stochastic piece $\sigma \int_0^t e^{-\theta(t-s)}\,dW_s$ yields by Itô isometry

$$\mathrm{Var}\Big[\sigma \int_0^t e^{-\theta(t-s)}\,dW_s\Big] \;=\; \sigma^2 \int_0^t e^{-2\theta(t-s)}\,ds. \tag{10.8}$$

Substitute $u = t - s$ and flip the limits:

$$\int_0^t e^{-2\theta(t-s)}\,ds \;=\; \int_0^t e^{-2\theta u}\,du \;=\; \frac{1 - e^{-2\theta t}}{2\theta}. \tag{10.9}$$

Therefore

$$\mathrm{Var}[X_t] \;=\; \frac{\sigma^2}{2\theta}\,(1 - e^{-2\theta t}). \tag{10.10}$$

At $t = 0$ this is zero; as $t \to \infty$ it grows monotonically to a finite limit:

$$\mathrm{Var}[X_t] \xrightarrow{t\to\infty} \sigma_\infty^2 \;:=\; \frac{\sigma^2}{2\theta}. \tag{10.11}$$

The finite limit is why a mean-reverter can be *traded*. A random walk has variance growing linearly forever, so "it will come back" has unbounded downside. OU has *bounded* variance — values stay in a tight neighbourhood of μ with a finite typical excursion size.

The stationary distribution

For each finite t, X_t in (10.6) is Gaussian (linear combination of Gaussian initial condition and Gaussian Itô integral), with mean and variance (10.7)–(10.10). As $t \to \infty$ both converge: mean to μ, variance to $\sigma_\infty^2 = \sigma^2/(2\theta)$. The limiting distribution is

$$X_\infty \;\sim\; \mathcal{N}\left(\mu,\, \frac{\sigma^2}{2\theta}\right). \tag{10.12}$$

This is the *stationary distribution*: if X_0 is drawn from it, X_t has the same distribution for every $t \ge 0$. The process never settles to a single point — noise keeps kicking it — but it settles to a stable Gaussian envelope of typical width $\sigma_\infty = \sigma/\sqrt{2\theta}$ around μ.

Autocorrelation

Two snapshots of the OU process at times t and $t + h$ are correlated; the correlation decays exponentially with the lag h. In the stationary regime,

$$\mathrm{Corr}(X_t,\, X_{t+h}) \;=\; e^{-\theta h}. \tag{10.13}$$

The derivation starts from $\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(X_t, X_t e^{-\theta h} + \text{noise}) = e^{-\theta h}\sigma_\infty^2$ and divides by σ_∞^2; Cartea (Cartea et al., 2015) (Ch. 5) reproduces the steps. The key fact: the autocorrelation of a stationary OU process is a *single decaying exponential* with the same rate θ as the drift. The OU process has one timescale, $1/\theta$.

Half-life

The mean of X_t, given a starting point X_0, decays toward μ as $\mathbb{E}[X_t] - \mu = (X_0 - \mu)\, e^{-\theta t}$ from (10.7). The *half-life* of the process is the time it takes for the expected deviation from μ to shrink to half its initial size:

$$e^{-\theta\, h_{1/2}} \;=\; \tfrac{1}{2} \quad\Longleftrightarrow\quad \theta\, h_{1/2} \;=\; \ln 2 \quad\Longleftrightarrow\quad h_{1/2} \;=\; \frac{\ln 2}{\theta}. \tag{10.14}$$

The half-life is the single most important quantity for a trader: in the natural units of the series, how long on average you must wait for an excursion to revert. Everything else — entry threshold, exit threshold, trade duration, capital allocation — is downstream.

OU process summary

The Ornstein–Uhlenbeck SDE $dX_t = \theta(\mu - X_t)\, dt + \sigma\, dW_t$ with $\theta > 0$ has the following properties.

- **Closed-form solution** $X_t = X_0 e^{-\theta t} + \mu(1 - e^{-\theta t}) + \sigma \int_0^t e^{-\theta(t-s)}\, dW_s$.
- **Stationary distribution** $X_\infty \sim \mathcal{N}(\mu,\, \sigma^2/(2\theta))$. Stationary standard deviation $\sigma_\infty = \sigma/\sqrt{2\theta}$.

- **Autocorrelation** $\mathrm{Corr}(X_t, X_{t+h}) = e^{-\theta h}$. Single exponential timescale $1/\theta$.
- **Half-life** $h_{1/2} = (\ln 2)/\theta$. Time for an expected excursion to fade by half.

A worked numerical example

Take $\mu = 100$, $\theta = 0.3$, $\sigma = 2$, $X_0 = 105$ (five units above the mean). Time units are arbitrary — think "hours" or "Prosperity ticks".

Half-life. From (10.14),

$$h_{1/2} = \frac{\ln 2}{0.3} \approx \frac{0.693147}{0.3} \approx 2.310 \text{ time units.} \tag{10.15}$$

A deviation of size 5 above the mean is expected to shrink to 2.5 in about 2.31 time units, to 1.25 after another 2.31, and so on.

Stationary spread. The stationary standard deviation is

$$\sigma_\infty = \frac{\sigma}{\sqrt{2\theta}} = \frac{2}{\sqrt{0.6}} = \frac{2}{0.7746\ldots} \approx 2.582. \tag{10.16}$$

So X_t stays mostly within $\mu \pm 2\sigma_\infty \approx [94.84, 105.16]$ (95% mass by the Gaussian rule).

Expected trajectory and variance build-up. From (10.7) and (10.10), plug in:

t	$\mathbb{E}[X_t]$	$\mathrm{Var}[X_t]$
0	105.000	0.000
1	103.704	3.008
2	102.744	4.659
$h_{1/2} = 2.310$	102.500	5.000
5	101.116	6.335
10	100.249	6.650
20	100.012	6.667
∞	100.000	6.667

Sanity checks: at $t = h_{1/2}$ the expected deviation is $2.5 = (105 - 100)/2$, as the half-life formula demands; at $t = \infty$ the variance is $\sigma^2/(2\theta) = 4/0.6 = 6.667$, matching (10.11), with $\sqrt{6.667} \approx 2.582$, matching (10.16).

Intuition

The half-life is the only number that matters. If it is one day and your holding window is five days, you live through several revert-and-reset cycles — profitable in expectation. If it is six months and your window is five days, no excursion reverts in time; the strategy becomes a directional bet. Half-life $\ll$ horizon is the gate that separates "trade this" from "leave it alone". Everything else is calibration.

10.2 Testing for mean reversion

Most financial time series are *not* OU — they are random walks. Equities, FX rates, and commodity futures are martingale-like at any human-tradeable horizon. Trading a random walk on a mean-reversion thesis loses money: there is no reversion to trade against.

Before applying OU machinery you must *reject the random-walk null*. Three tests are standard and a sell-side stat-arb desk runs all three: they fail in different ways, so a series surviving all three is much more credible than one surviving only one.

The Augmented Dickey–Fuller test

The **Augmented Dickey–Fuller (ADF) test** is the canonical test of stationarity in a univariate time series. In plain terms it asks: "is there a statistically significant force pulling this series back toward a constant level, or does it wander freely?" The null is no such force (a random walk, technically a *unit root*); rejecting the null is evidence of mean reversion. It runs the regression

$$\Delta X_t \;=\; \alpha \;+\; \beta\, X_{t-1} \;+\; \sum_{i=1}^{p} \gamma_i\, \Delta X_{t-i} \;+\; \epsilon_t, \tag{10.17}$$

where $\Delta X_t \coloneqq X_t - X_{t-1}$ is the first difference, α an intercept (drift), and the p lagged differences are nuisance terms that soak up serial correlation in ϵ_t. The null is $H_0\colon \beta = 0$: under H_0 the level X_{t-1} has no predictive power for ΔX_t — X_t is a random walk. Under $H_1\colon \beta < 0$, a high X_{t-1} makes ΔX_t more negative on average — mean reversion in difference form.

The ADF test statistic is the t-statistic of $\hat{\beta}$. Under the null its distribution is *not* Student's t: the regressor X_{t-1} is itself a random walk and standard regression theory breaks. Correct critical values are tabulated by MacKinnon (cited via (Chan, 2013)); a typical 5% critical value at long sample sizes is ≈ -2.86, and the test rejects when the observed t-statistic is more *negative*. Choose lag count p by AIC or BIC; Chan recommends starting at $p = 1$ and increasing until residual autocorrelation disappears.

For the running example ($\theta = 0.3$, $\mu = 100$, $\sigma = 2$, $N = 1000$), ADF typically yields a statistic around -7, comfortably past -2.86, and rejects the random-walk null.

The ADF test has low power in short samples

The ADF test is conservative: it fails to reject the random-walk null much more often than it should when the true process is OU but the sample is short or θ is small. With $N = 200$ and $\theta = 0.05$ (half-life ≈ 14), simulated ADF rejects at 5% only $\sim 30\%$ of the time even though every simulation came from an OU. A non-rejecting ADF is consistent with both a random walk and a slow reverter. Corroborate with a half-life estimate and a variance-ratio test; never trust a single ADF reading on < 250 observations.

The variance ratio test

The **variance ratio (VR) test** of Lo and MacKinlay uses a simpler observation: under a random walk the variance of k-period changes equals k times the variance of one-period changes, because increments are independent. Define

$$VR(k) \;\coloneqq\; \frac{\mathrm{Var}(X_t - X_{t-k})}{k\;\mathrm{Var}(X_t - X_{t-1})}. \tag{10.18}$$

Under the random-walk null $VR(k) = 1$ for every k. Under mean reversion, negatively-correlated changes (a step up tends to be followed by a step down) give $VR(k) < 1$; under trending, $VR(k) > 1$.

For the running example, a direct calculation using the OU autocorrelation (10.13) gives, in the stationary regime,

$$VR(k) \;=\; \frac{1 - e^{-\theta k}}{k\,(1 - e^{-\theta})}, \tag{10.19}$$

which at $\theta = 0.3$ takes values

k	$VR(k)$
1	1.000
2	0.870
5	0.600
10	0.367
20	0.192

The ratio falls toward zero as k grows: long-horizon changes are bounded by the OU range, while one-period changes are dominated by noise. The Lo–MacKinlay test maps these ratios to a z-statistic with known asymptotic distribution; for a long-enough sample, rejection at any $k \in [5, 20]$ is almost mechanical for a true OU with non-trivial θ.

The Hurst exponent

The **Hurst exponent** H is a third diagnostic, defined so that the long-run scaling of the range of a process is $R_t \propto t^H$. Three regimes:

- $H = 0.5$ — pure random walk.
- $H < 0.5$ — mean-reverting (the range grows slower than $\sqrt{t}$, indicating that excursions get pulled back).
- $H > 0.5$ — trending (the range grows faster than $\sqrt{t}$, indicating persistent moves).

The classical estimator is the rescaled-range (R/S) procedure of Mandelbrot, reviewed in (Chan, 2013); modern practice uses a multifractal extension. H returns one number that classifies the regime, but it is the noisiest of the three tests — Chan recommends using it only as a tiebreaker after ADF and VR.

The three tests are not redundant: ADF asks "can a linear mean-reversion regression beat a random walk?", VR asks "do multi-period variances accumulate sublinearly?", Hurst asks "does the range scale slower than $\sqrt{t}$?". Different summaries of the same series; a stat-arb shop runs all three before committing capital. The Frankfurt Hedgehogs Prosperity 3 writeup (`imc-prosperity-3/README.md`) explicitly checked for OU-style mean reversion in PICNIC_BASKET premia and VOLCANIC_ROCK before turning on those strategies.

10.3 Estimating OU parameters from data

Once a series is classed as mean-reverting, estimate θ, μ, σ from data. The standard trick: the OU SDE has an exact discrete-time representation as an AR(1), whose parameters fall out of a single OLS regression.

The discrete-time AR(1) representation

Sample X_t at uniform intervals of length Δt and apply the closed-form solution (10.6) between t and $t + \Delta t$:

$$X_{t+\Delta t} \;=\; X_t\, e^{-\theta \Delta t} \;+\; \mu\,(1 - e^{-\theta \Delta t}) \;+\; \sigma \int_t^{t+\Delta t} e^{-\theta(t+\Delta t - s)}\, dW_s. \tag{10.20}$$

The integral is a zero-mean Gaussian with variance (by the same Itô-isometry argument)

$$\mathrm{Var}\!\left[\sigma \int_t^{t+\Delta t} e^{-\theta(t+\Delta t-s)}\, dW_s\right] \;=\; \frac{\sigma^2}{2\theta}\,(1 - e^{-2\theta\Delta t}). \tag{10.21}$$

Define

$$\phi \;\coloneqq\; e^{-\theta\Delta t}, \qquad a \;\coloneqq\; \mu\,(1-\phi), \qquad \sigma_\epsilon^2 \;\coloneqq\; \frac{\sigma^2}{2\theta}\,(1-\phi^2), \tag{10.22}$$

and the discrete-time recursion (10.20) becomes

$$X_{t+\Delta t} \;=\; \phi\, X_t \;+\; a \;+\; \epsilon_t, \qquad \epsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\, \sigma_\epsilon^2). \tag{10.23}$$

An exact AR(1). Under uniform sampling the AR(1) is not an approximation: the OU SDE and the AR(1) are literally the same model.

OLS recovery of θ, μ, σ

Regress $X_{t+\Delta t}$ on X_t by OLS. This returns slope $\hat\phi$, intercept $\hat a$, and residual std $\hat\sigma_\epsilon$. Invert (10.22):

$$\hat\theta \;=\; -\frac{\ln\hat\phi}{\Delta t}, \tag{10.24}$$

$$\hat\mu \;=\; \frac{\hat a}{1-\hat\phi}, \tag{10.25}$$

$$\hat\sigma \;=\; \hat\sigma_\epsilon\sqrt{\frac{2\,\hat\theta}{1-\hat\phi^2}}. \tag{10.26}$$

The first inverts $\phi = e^{-\theta\Delta t}$; the second inverts $a = \mu(1-\phi)$; the third solves (10.21) for σ.

Worked verification. Simulate an OU trajectory with $\theta = 0.3$, $\mu = 100$, $\sigma = 2$, $\Delta t = 1$, $N = 5000$; true $\phi = e^{-0.3} \approx 0.7408$. OLS returns $\hat\phi \approx 0.734$, $\hat\theta \approx 0.310$, $\hat\mu \approx 100.04$, $\hat\sigma \approx 2.00$, within sampling error. On $N = 500$ points the same procedure returns $\hat\theta \in [0.2, 0.4]$ depending on seed — θ is the hardest of the three to pin down, and narrow confidence intervals on the half-life require a lot of data.

OU fitting via AR(1)

To fit (θ, μ, σ) to a uniformly sampled time series $\{X_t\}$ at spacing Δt:

1. Regress $X_{t+\Delta t}$ on X_t by OLS, obtaining slope $\hat\phi$, intercept $\hat a$, residual standard deviation $\hat\sigma_\epsilon$.
2. Recover $\hat\theta = -\ln(\hat\phi)/\Delta t$.
3. Recover $\hat\mu = \hat a/(1-\hat\phi)$.
4. Recover $\hat\sigma = \hat\sigma_\epsilon\sqrt{2\hat\theta/(1-\hat\phi^2)}$.

The half-life follows immediately: $\hat h_{1/2} = (\ln 2)/\hat\theta$.

10.4 The z-score entry/exit rule

With $(\hat\theta, \hat\mu, \hat\sigma)$ in hand, define the *z-score* of X_t:

$$z_t \;\coloneqq\; \frac{X_t - \hat\mu}{\hat\sigma_\infty}, \qquad \hat\sigma_\infty \;=\; \frac{\hat\sigma}{\sqrt{2\hat\theta}}. \tag{10.27}$$

The z-score is the current deviation from the long-run mean in stationary-std units. Under OU in steady state $z_t \sim \mathcal{N}(0,1)$, so "$z = 2$" is a $\sim 2.5\%$ tail event. The threshold $z_{\text{entry}} = 2$ is therefore picked not arbitrarily but because a stationary OU process spends only $\sim 5\%$ of its time outside $\pm 2\sigma_\infty$: when it does stray that far, reversion is highly probable and the expected move back toward μ is large enough to pay for transaction costs.

The standard **z-score trading rule**, used by practitioners since the 1980s and codified by Chan (Chan, 2013), is:

- **Enter short**: when $z_t > z_{\text{entry}}$ (e.g. $z_{\text{entry}} = 2$). The series is high relative to its long-run mean and is expected to fall back.
- **Enter long**: when $z_t < -z_{\text{entry}}$. Symmetrically.
- **Exit**: when $|z_t| < z_{\text{exit}}$ (e.g. $z_{\text{exit}} = 0.5$). The deviation has decayed back to near zero and there is no further reversion to harvest.
- **Stop out**: when $|z_t| > z_{\text{stop}}$ (e.g. $z_{\text{stop}} = 4$). Four standard deviations from the supposed mean means the OU model is wrong, the parameters are stale, or the regime has changed. Exit at a controlled loss before the position blows up.

The asymmetry between z_{entry} and z_{exit} is deliberate: requiring $|z_t| < z_{\text{entry}}$ to exit would never exit on a deviation that decayed to the entry level and bounced. The slack between z_{exit} and z_{entry} ensures most trades close on a meaningful reversion.

Expected trade duration

How long does the typical trade last? Use the OU mean trajectory $\mathbb{E}[X_t] - \mu = (X_0 - \mu)e^{-\theta t}$. In z-units, an excursion of size z_{entry} decays in expectation to z_{exit} after time h:

$$z_{\text{entry}}\, e^{-\theta h} \;=\; z_{\text{exit}} \quad\Longrightarrow\quad h \;=\; \frac{1}{\theta}\ln\!\left(\frac{z_{\text{entry}}}{z_{\text{exit}}}\right). \tag{10.28}$$

The expected trade duration. With $z_{\text{entry}} = 2$, $z_{\text{exit}} = 0.5$, $\theta = 0.3$,

$$h \;=\; \frac{\ln 4}{0.3} \;\approx\; \frac{1.3863}{0.3} \;\approx\; 4.621 \text{ time units.} \tag{10.29}$$

Sanity check: about twice the half-life $h_{1/2} \approx 2.310$, as expected — two halvings ($z = 2 \to 1 \to 0.5$) rather than one.

Equation (10.28) is a deterministic-trajectory estimate, ignoring the noise around the mean trajectory. The exact mean first-passage time involves an integral over the OU generator with no closed form for general thresholds; for threshold-setting and Kelly sizing (Chapter 20) the deterministic approximation is adequate.

The z-score OU strategy

Given OU parameter estimates $(\hat\theta, \hat\mu, \hat\sigma)$ and stationary std $\hat\sigma_\infty = \hat\sigma/\sqrt{2\hat\theta}$:

- Compute the live z-score $z_t = (X_t - \hat\mu)/\hat\sigma_\infty$.
- **Enter** a position of size $-\operatorname{sgn}(z_t)$ when $|z_t| > z_{\text{entry}}$.
- **Exit** when $|z_t| < z_{\text{exit}}$.
- **Stop out** when $|z_t| > z_{\text{stop}}$.
- **Expected duration** of a trade is $h \approx \theta^{-1}\ln(z_{\text{entry}}/z_{\text{exit}})$.

Refit $(\hat{\theta}, \hat{\mu}, \hat{\sigma})$ on a rolling window (see Section 10.5) so the parameters track the current regime.

Intuition

A z-score entry is a bet that the half-life is shorter than your deadline. With trading horizon H and half-life $h_{1/2}$, you get $H/h_{1/2}$ revert-and-reset cycles in expectation. Ratio $\gg 1$: deviations come back inside the window and the strategy harvests them. Ratio $\ll 1$: you mark to market for many half-lives without reversion — functionally a directional bet. The thresholds set how far from the mean you wait before entering, but they do not change the half-life. Pick the asset first $(h_{1/2} < H)$, then the thresholds.

10.5 Walk-forward estimation

OU parameters drift. The mean of a stat-arb residual shifts with the company's business; the mean-reversion speed shifts with regime; the volatility shifts continuously. A model fit in January is substantially wrong by July, and a model fit on the entire historical record uses post-trade information to compute pre-trade z-scores — look-ahead bias that inflates simulated profits and disappears live.

The fix is **walk-forward estimation**. Pick a fitting window W and refit cadence R. At each time t:

1. Fit the OU model on the most recent W observations $X_{t-W}, \ldots, X_{t-1}$, using only data *strictly before* time t.
2. Compute the live z-score $z_t = (X_t - \hat{\mu}_t)/\hat{\sigma}_{\infty,t}$ from the parameters fit on the prior window.
3. Generate the trading signal from z_t.
4. Every R steps, repeat the fit on the latest window.

Typical Prosperity-scale numbers: $W \in [500, 5000]$, $R \in [50, 1000]$, scaled by half-life (shorter half-life $\Rightarrow$ shorter W and R). On a sell-side single-name desk, W is one to two trading years and R is one trading day.

Look-ahead bias in OU fitting

The most common mistake on a first OU strategy: fit $\hat{\mu}$ on the entire dataset and use it for every historical z-score. This feeds *future* data into the strategy. Simulated z-scores are systematically better-centred than in live trading, entries are timed more cleverly than possible, and the backtest shows fictional profit. Same trap for $\hat{\theta}$ and $\hat{\sigma}$. The cure: parameter values used at time t must depend only on X_s for $s < t$. Chapter 21 returns to look-ahead bias as a category.

10.6 A short reference implementation

The OU fitting pipeline fits in under two dozen lines of Python. The reference below is a *pure function* from a numpy array of past observations (with optional sampling interval) to $(\hat{\theta}, \hat{\mu}, \hat{\sigma})$. The signal generator on top is one line per the keyfactbox in Section 10.4.

```python
import numpy as np

```

```python
def fit_ou(x: np.ndarray,
           dt: float = 1.0) -> tuple[float, float, float]:
    """Fit OU parameters (theta, mu, sigma) via the AR(1) mapping."""
    y, xp = x[1:], x[:-1]
    # np.polyfit returns highest-order coef first => (slope, intercept)
    phi, a = np.polyfit(xp, y, 1)
    theta = -np.log(phi) / dt
    mu    = a / (1.0 - phi)
    resid = y - (phi * xp + a)
    var_eps = float(np.var(resid, ddof=2))
    sigma = float(np.sqrt(2.0 * theta * var_eps / (1.0 - phi**2)))
    return float(theta), float(mu), sigma
```

Listing 10.1: OU fitting via AR(1) as a pure function.

Three production remarks not encoded above. First, the function assumes $\hat{\phi} \in (0, 1)$; $\hat{\phi} \leq 0$ does not satisfy OU and $-\ln(\hat{\phi})$ is undefined, so the caller should guard and refuse to trade. Second, `ddof=2` accounts for the two OLS-estimated parameters (ϕ, a); without it $\hat{\sigma}$ is slightly biased downward on small samples. Third, this function is meant to be called inside a walk-forward loop — calling it once on full history is the look-ahead trap above.

Prosperity example: KELP

KELP is the canonical mildly-mean-reverting Prosperity product. Per Frankfurt Hedgehogs (`imc-prosperity-3/README.md`), KELP's true price follows a slow random walk with a small mean-reverting overlay from integer-tick rounding of an internal floating-point fair value: "a minor technical edge stemming from the fact that the true price was internally a floating-point value and orders could only be posted at integer levels (creating slight mean-reversion tendencies after ticks)." KELP is the ideal teaching example for OU fitting on Prosperity data.

Procedure.

1. Pull the KELP wall-mid (Chapter 4) from the first 1000 ticks of a round — the fitting window W.
2. Run `fit_ou` from Section 10.6 with $\Delta t = 1$.
3. Compute $h_{1/2} = (\ln 2)/\hat{\theta}$ and $\hat{\sigma}_\infty = \hat{\sigma}/\sqrt{2\hat{\theta}}$.
4. Typically $\hat{\theta} \in [0.02, 0.05]$, giving a half-life of 15–30 ticks.
5. Set $z_{\text{entry}} = 1.5$, $z_{\text{exit}} = 0.3$ (tighter than the section default because fills are quick and waiting for $|z| > 2$ misses trades). Expected duration $h = \ln(5)/\hat{\theta} \approx 1.61/\hat{\theta}$, so 30–80 ticks per trade. Over a 10 000-tick episode, comfortably more than 50 trades.
6. Refit every 250 ticks on a rolling 1000-tick window. $\hat{\mu}$ tracks the slow random walk; $\hat{\theta}$ stays approximately constant (the rounding mechanism does not change).

What the top teams actually did. The Frankfurt Hedgehogs' final KELP handler (`imc-prosperity-3/FrankfurtHedgehogs_polished.py`, `DynamicTrader.get_orders()`) does *not* run a literal z-score OU strategy on KELP. It quotes one tick inside the wall bid/ask — the fixed-offset rule used on RAINFOREST_RESIN in Chapter 9 — and lets the wall-mid update absorb the slow random walk. They treated KELP as "unpredictable apart from a tiny rounding effect" and concluded the effect was too small to extract reliably ($\sim 5\,000$ seashells per round, mostly market making). The lesson: OU is the right tool to *diagnose* that KELP's reversion is too weak to trade as a pure stat-arb signal, and the right tool to *trade* a stronger mean-reverter like the PICNIC_BASKET premium of Chapter 12, where the half-life is minutes and $|z| > 2$ happens regularly.

On a Goldman Sachs desk

On a sell-side equity stat-arb desk, every single-name mean-reversion signal is fitted by an overnight walk-forward OU pipeline owned end-to-end by the Strats group. A typical morning workflow on name `XYZ`:

1. **Stationarity gate.** Run an ADF test on the fitting window. If the test fails to reject the random-walk null, the name does not trade today — the z-score signal is meaningless on a non-stationary series.

2. **Parameter estimation.** Fit $(\hat{\theta}, \hat{\mu}, \hat{\sigma})$ on the fitting window via the AR(1) mapping of Section 10.3.

3. **Half-life sanity check.** Compute $\hat{h}_{1/2} = (\ln 2)/\hat{\theta}$ and confirm it falls in an asset-class plausible range (intraday: minutes to hours; multi-day: days to weeks). Out-of-range flags the Strat to inspect.

4. **z-score signal.** Stream live $z_t = (X_t - \hat{\mu})/\hat{\sigma}_\infty$ to the trader's monitor and the automated execution layer.

5. **Kelly sizing.** Convert z_t into position size via Kelly (Chapter 20), capped at the desk's per-name VaR contribution.

A standard GS/quant interview question: *"Your OU fit reports a half-life of* 0.5 *days on this name, but the average holding period is* 3 *days. What do you investigate?"* Three buckets in order: (i) a stale or mis-applied $\hat{\mu}$ that has drifted from the live mean, so z_t takes longer to revert because the supposed mean is not where the series is heading; (ii) a regime change that has lengthened true θ since the fit, often visible by re-fitting on the most recent sub-window; (iii) overfitting, where $\hat{\theta}$ from a short window is biased toward larger values than the truth. All three are diagnosable; walking through them in order signals familiarity. Chapter 28 returns to this kind of diagnostic conversation.

10.7 Limitations and when OU breaks

The OU model is the simplest non-trivial mean-reverter and the default tool of practitioners. Like every default, it has failure modes worth knowing before pointing it at production money.

The mean is not constant. OU assumes a *single* long-run mean μ. Real series typically have a slow-moving level on top of the fast-reverting noise — for an equity, the company's growing earnings; for an FX rate, the inflation differential; for a Prosperity ETF premium, whatever the simulator calibrated. Fix: de-trend before fitting (subtract a moving average or a regression against a slow factor) or refit $\hat{\mu}$ on a rolling window short enough that the drift inside is small. Walk-forward handles this implicitly when the refit cadence is fast enough.

The half-life is not constant. OU assumes a single θ. Real θ varies with regime: faster in calm markets, slower in stressed ones. Walk-forward picks this up to first order, but a calm-regime strategy quietly stops working when the half-life lengthens beyond the holding window.

The series is not stationary in level. Many "mean-reverting" series are stationary only after transformation: log, first difference, or residual against a slow factor. Classic case: two cointegrated stocks — each price is a random walk, but their difference (with the right hedge ratio) is OU. Trading the OU residual is Chapter 11; the technique here applies directly once cointegration is established. A related worry is *over-differencing*: aggressive differencing destroys the long-run information carrying the signal. (López de Prado, 2018) (Ch. 5) treats this via *fractional differentiation*. Rule of thumb: use the *lowest* order of transformation that makes the series stationary.

Jumps and fat tails. OU has Gaussian increments. Real series jump (news, earnings, central-bank decisions, or Prosperity's Olivia informed-trading windows) and the Gaussian envelope underestimates the tails. A live OU strategy should always layer a stop-out ($|z| > z_{\text{stop}}$, per Section 10.4) to bail out before a jump turns a signal into a runaway position.

Cointegration without OU. Chapter 11 covers the next layer: many tradeable opportunities live in *linear combinations* of series, where components are random walks but the combination reverts. OU applies as soon as the combination is identified; finding it, picking the hedge ratio, and testing it are the next chapter.

10.8 Key facts to memorise

- The Ornstein–Uhlenbeck SDE is $dX_t = \theta(\mu - X_t)\,dt + \sigma\,dW_t$, with $\theta > 0$, $\mu \in \mathbb{R}$, $\sigma > 0$. It has a unique stationary distribution $\mathcal{N}(\mu,\, \sigma^2/(2\theta))$.
- The closed-form solution $X_t = X_0 e^{-\theta t} + \mu(1 - e^{-\theta t}) + \sigma \int_0^t e^{-\theta(t-s)}\,dW_s$ is obtained by multiplying the SDE through by the integrating factor $e^{\theta t}$ and integrating.
- **Half-life.** $h_{1/2} = (\ln 2)/\theta$. The single most important number for trading. If $h_{1/2} \ll H$ (your trading horizon), the strategy is viable; otherwise it is not.
- **Stationary standard deviation.** $\sigma_\infty = \sigma/\sqrt{2\theta}$. The natural unit of the z-score.
- **Autocorrelation.** $\mathrm{Corr}(X_t, X_{t+h}) = e^{-\theta h}$. A single exponential with the same rate as the drift.
- **Discrete-time mapping.** OU sampled at spacing Δt is exactly an AR(1) with $\phi = e^{-\theta\Delta t}$, intercept $\mu(1-\phi)$, and innovation variance $\sigma^2(1-\phi^2)/(2\theta)$. Fit by OLS, recover $\hat{\theta} = -\ln(\hat{\phi})/\Delta t$.
- **Stationarity tests.** ADF (regress ΔX_t on X_{t-1} plus lags), variance ratio (compare $\mathrm{Var}(X_t - X_{t-k})$ against $k\ \mathrm{Var}(X_t - X_{t-1})$), and Hurst exponent ($H < 0.5$ mean reverting). Run all three; ADF alone has low power on short samples or slow reverters.
- **z-score rule.** Enter when $|z| > z_{\text{entry}}$, exit when $|z| < z_{\text{exit}}$, stop out when $|z| > z_{\text{stop}}$. Asymmetry between entry and exit thresholds is essential. Expected trade duration is $h \approx \theta^{-1} \ln(z_{\text{entry}}/z_{\text{exit}})$.
- **Walk-forward fitting is non-negotiable.** Fitting $\hat{\mu}$ once on the entire dataset and then using it for every historical z-score is the canonical look-ahead bias. Always refit on a rolling window using only data strictly before the trade time.
- OU is a *variance-control* mean-reverter with Gaussian increments. It cannot model fat tails, jumps, or regime changes by itself — always pair with a hard stop-out for protection.

- On Prosperity, KELP is mildly mean-reverting via a rounding artifact and is best traded with a fixed-offset market-maker (Frankfurt Hedgehogs); the OU z-score machinery shines on the PICNIC_BASKET premium of Chapter 12, where half-lives are short and excursions large.

- On a sell-side stat-arb desk every single-name reversion signal is fitted by walk-forward OU; the Strat owns the pipeline ADF gate $\rightarrow$ estimation $\rightarrow$ half-life check $\rightarrow$ live z-score $\rightarrow$ Kelly sizing (Chapter 20). Standard interview probe: "half-life says X, average holding is Y, what do you investigate?" — three buckets: stale $\hat{\mu}$, regime change, walk-forward overfitting.

Chapter 11

Pairs Trading and Cointegration

Two stocks move together. When the spread widens, short the winner and buy the loser; when it narrows, close the position and pocket the difference. That is *pairs trading* in one sentence, and the strategy is seductive because it is market-neutral by construction (long one name, short another) and the thesis ("these two belong together") is weaker than predicting direction.

This chapter shows when the strategy works and why *correlation is not enough*. Screening for high-correlation pairs and trading the spread is the single most expensive beginner mistake; the correct test, **cointegration**, differs from correlation in a way invisible to the naked eye but routinely takes whole strategies offline. We develop both concepts, give the standard econometric machinery (Engle–Granger, Johansen), and show that once a cointegrated pair is found the trading rule is the Ornstein–Uhlenbeck z-score machinery of Chapter 10: the cointegration residual is an OU process, and every formula from that chapter transfers unmodified.

Symbol anchors. β — cointegrating coefficient (hedge ratio); $\hat{\epsilon}_t$ — cointegration residual (spread); z_t — spread z-score; $h_{1/2}$ — half-life; θ — mean-reversion speed.

Prerequisites

From earlier in the book:

- Chapter 10: the OU machinery. Closed-form solution, stationary standard deviation $\sigma_\infty = \sigma/\sqrt{2\theta}$, half-life $h_{1/2} = (\ln 2)/\theta$, AR(1) discrete-time mapping, augmented Dickey–Fuller (ADF) stationarity test, z-score entry/exit rule. All of it carries over.
- Chapter 5: the Roll model — first appearance of a stationary linear combination (successive trade-price differences) inside a non-stationary process (trade prices). Cointegration is the multivariate generalisation.
- Chapter 4: wall-mid is the reference price for any Prosperity spread, since marketable orders execute against the far side of the wall.
- Forward reference: Chapter 21 for walk-forward discipline that prevents look-ahead bias when the cointegration coefficient is refit in time.

From outside the book:

- OLS regression $y_t = \alpha + \beta z_t + \epsilon_t$, and the fact that swapping y and z produces a different slope.
- Random walks: $X_t = X_{t-1} + \eta_t$ with iid η_t, $\mathrm{Var}[X_t - X_0] = t\sigma_\eta^2$. Every individual price in this chapter is a random walk; the magic is in linear combinations.

11.1 Correlation vs cointegration: the canonical counter-example

In casual use, "correlated" is a synonym for "related"; in pairs trading the technical meaning is narrower and misleading if treated as a substitute for the right test, which is *cointegration.* The two concepts do not measure the same thing.

Definition 11.1 (Sample correlation of returns)**.** For two time series Y_t, Z_t sampled at uniform intervals, the *return correlation* is the Pearson correlation of the one-period changes,

$$\mathrm{Corr}(\Delta Y,\, \Delta Z) \;=\; \frac{\mathrm{Cov}(\Delta Y_t,\, \Delta Z_t)}{\sqrt{\mathrm{Var}[\Delta Y_t]\ \mathrm{Var}[\Delta Z_t]}}, \qquad \Delta Y_t \coloneqq Y_t - Y_{t-1}. \tag{11.1}$$

This is a *short-horizon* same-direction-of-motion measure: "when one goes up, does the other go up at the same instant?"

Definition 11.2 (Cointegration of two series)**.** Two time series Y_t, Z_t are *cointegrated* if there exists a constant $\beta \in \mathbb{R}$ such that the linear combination $Y_t - \beta Z_t$ is *stationary*, even though Y_t and Z_t themselves are non-stationary (typically random walks). The constant β is the *cointegration coefficient* or *hedge ratio.* The stationary combination $\hat{\epsilon}_t \coloneqq Y_t - \beta Z_t$ is the *cointegration residual* or *spread.*

The definitions live on different time horizons. Return correlation asks about *steps* (changes from one observation to the next); cointegration asks about *levels* (the long-run tendency of a linear combination to stay near a constant). A pair can have either property without the other. Plain-English version: two stocks can be highly correlated — both rise in a bull market, both fall in a crash — without being cointegrated. Cointegration is the stronger claim that their *spread* is stationary: any deviation from the long-run level is temporary and mean-reverts. Correlation is about co-movement; cointegration is about a tether. Two examples drive the point home.

Counter-example 1: spurious correlation of two random walks

Take two independent Gaussian random walks starting at the same point,

$$Y_t \;=\; Y_{t-1} \;+\; \eta_t^Y, \qquad Z_t \;=\; Z_{t-1} \;+\; \eta_t^Z, \qquad Y_0 = Z_0 = 100, \tag{11.2}$$

with $\eta_t^Y, \eta_t^Z \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ independent. The two series share no common noise, so the population return correlation is zero.

But the sample correlation of the *levels* Y_t, Z_t over a finite window is nearly uniform on $[-1, +1]$. This is Yule's (1926) *spurious correlation*: any two random walks drift, and over a finite window those drifts line up by accident.

Numerical demonstration. Simulate 20 independent pairs of 1000-step Gaussian random walks (different seeds; both starting at 100; iid $\mathcal{N}(0,1)$ innovations). Across 20 seeds the return correlation stays in $[-0.04, +0.08]$ around zero, but the level correlation ranges from -0.80 to $+0.88$, typical absolute value above 0.4. A credulous screen would flag these independent walks as "strongly correlated." Trading the spread $Y_t - Z_t$ is throwing money away: the spread is itself a random walk (variance doubled) with no tendency to revert.

Historical note

Yule's "Why do we sometimes get nonsense-correlations between time series?" (1926) is the first formal warning. Granger and Newbold (1974, "Spurious regressions in econometrics") sharpened the point: t-statistics from a regression of one random walk on another are misleadingly large, and naive hypothesis tests reject the (true) null far more often than nominal size suggests. The 1980s cointegration literature grew out of fixing this.

Counter-example 2: cointegration without obvious return correlation

Now flip it around. Construct

$$Y_t \;=\; Y_{t-1} \;+\; \eta_t, \qquad Z_t \;=\; Y_t \;+\; e_t, \tag{11.3}$$

where $\eta_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\,1)$ is Y's random walk innovation, and e_t is an independent *Ornstein–Uhlenbeck process* with mean 0, mean-reversion speed $\theta = 0.3$, and instantaneous volatility $\sigma_e = 1$. Y_t is a random walk; Z_t is the same random walk plus a small mean-reverting overlay. The spread $Y_t - Z_t = -e_t$ is OU, hence stationary by construction.

Return correlation is high but not unity: $\Delta Y_t = \eta_t$, $\Delta Z_t = \eta_t + (e_t - e_{t-1})$, so they share η_t but ΔZ_t also picks up the OU innovation. A 5000-step simulation gives return correlation ≈ 0.74 and level correlation ≈ 0.999. The spread has stationary variance $\sigma_e^2/(2\theta) = 1/0.6 \approx 1.67$, and ADF rejects the random-walk null with $t \approx -26$ and p-value ≈ 0. The pair is cointegrated; trading the spread as an OU process makes money in expectation.

Contrast with Section 11.1: independent random walks can exhibit level correlation $+0.85$ by accident, while the constructed cointegrated pair has return correlation 0.74 from a genuinely shared walk. *Neither correlation number tells you which case you are in.* The only distinguishing test takes the residual of the linear combination and ADF-tests it for stationarity.

The 2x2 table

The four combinations of (high/low return correlation) $\times$ (cointegrated/not) are below. Top-left is the target; top-right is the trap naive screens fall into; bottom-left is the unusual counter-example we just constructed; bottom-right is the default.

	Cointegrated	**Not cointegrated**
High return correlation	Move together *and* revert to each other. The pairs trader's target. Spread has a half-life $\sim$ days.	Move together but drift apart over time. The spread diverges. Looks like a pair, is not. *The trap.*
Low return correlation	Steps look unrelated, but a particular linear combination is pinned. Less common; arises e.g. when the OU overlay dominates the shared random walk on a per-step basis.	Independent series. No pair to trade. The default case for unrelated names.

"Move together and drift apart" (top right) is what happens to two trending stocks in the same sector riding the market upward: daily returns highly correlated (same beta), levels diverge (idiosyncratic drift). Selling the spread on a correlation screen leaves you short the winner and long the loser for as long as the trend continues — months.

Don't pairs-trade on correlation alone

Any two well-known equities — two gold miners, two US banks, two oil majors — can show return correlation above 0.9 over a multi-year sample (same factor exposure). Yet the spread can drift to infinity if earnings or operating leverage diverge. A trader who screens for "correlation > 0.9", sells the high one and buys the low one is *trend-fighting*, not mean-reverting — the spread has no level to revert to. The right test: regress, take the residual, ADF-test it; only if ADF rejects is the pair tradeable. (Chan, 2013) (Chapter 2) is the practitioner reference.

11.2 The Engle–Granger two-step

Engle and Granger (1987) proposed the *Engle–Granger two-step* test for cointegration. Five lines of Python, and it remains the standard first-pass test on a stat-arb desk.

Step 1: estimate the hedge ratio by OLS

Run OLS of Y_t on Z_t with an intercept,

$$Y_t \;=\; \alpha \;+\; \beta\, Z_t \;+\; \epsilon_t, \tag{11.4}$$

where (α, β) minimise the sum of squared residuals over the fitting window. OLS returns

$$\hat\beta \;=\; \frac{\mathrm{Cov}(Y_t,\, Z_t)}{\mathrm{Var}(Z_t)}, \qquad \hat\alpha \;=\; \bar Y - \hat\beta\, \bar Z, \tag{11.5}$$

where bars denote sample means over the fitting window.

The slope $\hat\beta$ is the *cointegration coefficient* or *hedge ratio*: the units of Z to short per unit of Y long to make the spread $Y - \beta Z$ as quiet as possible. The intercept $\hat\alpha$ absorbs the average level difference.

Step 2: test the residual for stationarity

Compute the residual

$$\hat\epsilon_t \;\coloneqq\; Y_t - \hat\alpha - \hat\beta\, Z_t, \tag{11.6}$$

and run an ADF test (Chapter 10) on $\hat\epsilon_t$. The logic: Y_t and Z_t are each random walks (non-stationary), so OLS can *always* find coefficients that fit them against each other; the question is whether the leftover residual $\hat\epsilon_t$ is itself stationary (pair is cointegrated, spread is tradeable) or another random walk (regression was spurious, pair is not cointegrated). One wrinkle: because $\hat\beta$ was estimated, the standard ADF critical values are wrong. Corrected values tabulated by Engle–Granger (1987) and updated by MacKinnon (1991) are modestly more demanding; the corrected test is the *Engle–Granger CADF*. Every package implements both; call the right one.

If (C)ADF rejects the random-walk null, a stationary linear combination exists — the pair is cointegrated. Failure to reject (given ADF's low power) is inconclusive; the prudent action is not to trade the pair.

Engle–Granger two-step

Given two candidate price series $\{Y_t\}$, $\{Z_t\}$ over a fitting window:

1. Run OLS on (11.4): $Y_t = \alpha + \beta Z_t + \epsilon_t$. Read off $(\hat\alpha,\, \hat\beta)$ from (11.5).
2. Compute residuals $\hat\epsilon_t = Y_t - \hat\alpha - \hat\beta Z_t$.

3. Run an ADF test (with cointegration-corrected critical values) on $\hat{\epsilon}_t$. Reject the random-walk null at 5% $\Rightarrow$ the pair is cointegrated.

4. If cointegrated, the hedge ratio is $\hat{\beta}$ and the spread is $\hat{\epsilon}_t$. Treat $\hat{\epsilon}_t$ as an OU process and apply the Chapter 10 machinery.

Worked example: EWA / EWC

The canonical practitioner pair is **EWA** (iShares MSCI Australia) and **EWC** (iShares MSCI Canada). These ETFs (exchange-traded funds: securities holding a basket of stocks and trading like a single share; formal definition in Chapter 12) track two commodity-exporter economies with similar industrial composition (mining, energy, banks), so they move together. Chan ((Chan, 2013), Chapter 2) runs the cointegration analysis on early-2000s daily data:

- OLS hedge ratio $\hat{\beta} \approx 0.75$ (EWA on EWC): one unit long EWA hedges with 0.75 units short EWC.
- CADF $t \approx -3.7$, $p \approx 0.005$ — cointegrated.
- Residual half-life ≈ 7 trading days.
- σ_∞ a few percent of spot ETF price.

Plug these into the OU machinery from Chapter 10. The half-life is $h_{1/2} \approx 7$ days, so $\theta = (\ln 2)/7 \approx 0.099$ per day. With entry threshold $z_{\text{entry}} = 2$ and exit threshold $z_{\text{exit}} = 0.5$ the expected duration of a single trade is, by (10.28) of Chapter 10,

$$h = \frac{1}{\theta} \ln\left(\frac{2}{0.5}\right) = \frac{\ln 4}{0.099} \approx 14 \text{ trading days}, \tag{11.7}$$

i.e. a typical EWA/EWC round trip lasts a couple of weeks. With a couple of trades per quarter and target reversion of three to four residual standard deviations, historical Sharpe is 1–1.5 (Chan) — modest, consistent, and uncorrelated with market direction. Modern alpha on this exact pair has decayed (Section 11.6), but the geometry is the canonical teaching example.

The cointegration vector is not symmetric

Which series goes on the left (Y_t, dependent) vs right (Z_t, regressor) matters. OLS minimises squared residuals along the Y axis only, so swapping changes the slope. Regressing EWA on EWC might give $\hat{\beta}_{Y|Z} \approx 0.75$, EWC on EWA $\hat{\beta}_{Z|Y} \approx 1.10$ ($1/0.75 = 1.33$, so $\hat{\beta}_{Y|Z} \cdot \hat{\beta}_{Z|Y} \neq 1$ unless the regression is noiseless — the regression-to-the-mean fallacy). The two regressions produce different residuals and, near the 5% threshold, different test outcomes.

Fix, in order of preference: (i) put the higher-variance series on the right; (ii) use total least squares (orthogonal regression); (iii) use Johansen (Section 11.3), invariant to ordering. Chan ((Chan, 2013), Chapter 2) recommends running both directions; if they disagree, the pair is not robustly cointegrated.

11.3 The Johansen test (stated)

For two assets, Engle–Granger is adequate. For three or more, two problems appear: (i) the choice of left-hand-side asset becomes a combinatorial mess; (ii) a basket of n assets can have *multiple* cointegration relationships (up to $n - 1$), but Engle–Granger finds at most one. The **Johansen test** (Johansen, 1988, 1991) solves both in vector form.

The setting is a vector *error-correction model* (VECM) for an n-dimensional process $\boldsymbol{X}_t = (X_{1,t}, \ldots, X_{n,t})^\top$,

$$\Delta \boldsymbol{X}_t \;=\; \Pi\, \boldsymbol{X}_{t-1} \;+\; \sum_{i=1}^{p-1} \Gamma_i\, \Delta \boldsymbol{X}_{t-i} \;+\; \boldsymbol{\mu} \;+\; \boldsymbol{\epsilon}_t, \tag{11.8}$$

with Π, Γ_i as $n \times n$ matrices, $\boldsymbol{\mu}$ an intercept, $\boldsymbol{\epsilon}_t$ multivariate noise. The matrix Π is the long-run impact. Three cases for its rank $r := \operatorname{rank}\Pi$:

- $r = 0$: $\Pi = 0$, the VECM collapses to a pure VAR in differences, no stationary linear combination exists, no cointegration.
- $r = n$: Π is full rank, $\boldsymbol{X}_t$ is stationary, no random-walk component.
- $0 < r < n$: exactly r independent stationary linear combinations. Π factors as $\Pi = \alpha\beta^\top$ where α, β are $n \times r$; columns of β are the *cointegration vectors* (each a stationary combination of $\boldsymbol{X}_t$), and α is the *adjustment matrix* (how strongly each component pulls back toward the stationary level).

Johansen estimates r via likelihood-ratio tests on the eigenvalues of Π. Two statistics: *trace* tests $H_0 : r \le r_0$ for $r_0 = 0, \ldots, n-1$; *maximum-eigenvalue* tests $H_0 : r = r_0$ against $H_1 : r = r_0 + 1$. Both have non-standard distributions tabulated by Johansen. Derivation omitted; (Hasbrouck, 2007) treats the vector-cointegration setting, and (Cartea et al., 2015) (Chapter 13) gives an algorithmic treatment.

The pipeline is one library call: `statsmodels.tsa.vector_ar.vecm.coint_johansen` takes a $T \times n$ price matrix and returns trace statistic, max-eigenvalue statistic, and estimated cointegration vectors. Read the rank by comparing each statistic to its critical value; the columns of the cointegration matrix are the basket weights.

Intuition

Johansen finds every stationary linear combination at once. For two assets, it matches Engle–Granger. For n assets, it reports how many *independent* stationary combinations exist: five sector stocks might yield three mispricing factors to trade, or zero. Engle–Granger cannot discover this multi-relationship structure because it bakes in a single left-hand-side regression.

11.4 Trading the residual as an OU process

Once a test has produced $\hat\beta$ and residuals $\hat\epsilon_t$, the trading rule is the z-score OU strategy from Chapter 10 applied to $\hat\epsilon_t$ — every formula transfers unmodified.

Recap of the OU machinery on the residual

Treat $\hat\epsilon_t$ as OU. By the AR(1) mapping of Chapter 10 (equation (10.22)), regressing $\hat\epsilon_{t+1}$ on $\hat\epsilon_t$ recovers $(\hat\theta_\epsilon,\, \hat\mu_\epsilon,\, \hat\sigma_\epsilon)$. Then $h_{1/2} = (\ln 2)/\hat\theta_\epsilon$, $\hat\sigma_\infty = \hat\sigma_\epsilon/\sqrt{2\hat\theta_\epsilon}$, and the live z-score

$$z_t \;:=\; \frac{\hat\epsilon_t - \hat\mu_\epsilon}{\hat\sigma_\infty}. \tag{11.9}$$

Apply the entry / exit / stop-out rule of Chapter 10: enter when $|z_t| > z_{\text{entry}}$, exit when $|z_t| < z_{\text{exit}}$, stop out when $|z_t| > z_{\text{stop}}$. Expected trade duration is $h \approx \theta_\epsilon^{-1} \ln(z_{\text{entry}}/z_{\text{exit}})$.

Pairs trading z-score signal with entry/exit bands

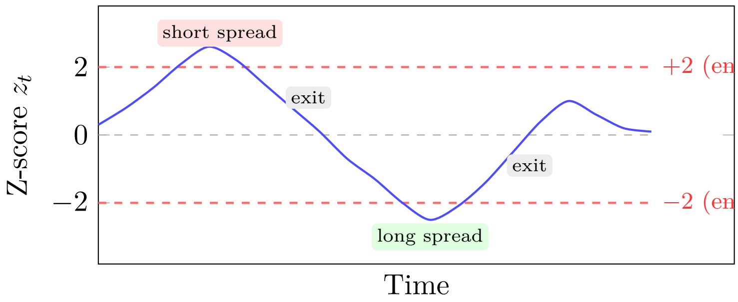


Figure 11.1: Pairs-trading z-score signal. The strategy shorts the spread when $z_t > +2$ (spread too wide relative to history) and goes long when $z_t < -2$ (spread too narrow). Both positions are closed when z_t reverts to zero. The ± 2 thresholds are chosen to balance entry frequency against the probability of a false signal.

What "a position" means in a pair trade

The trade is on the spread, not on either name:

- **Long the spread** ($z_t < -z_{\text{entry}}$, residual too low): buy 1 unit of Y, sell $\hat{\beta}$ units of Z. Zero net dollar exposure to a parallel rally.
- **Short the spread** ($z_t > +z_{\text{entry}}$, residual too high): sell 1 unit of Y, buy $\hat{\beta}$ units of Z.

$\hat{\beta}$ is the literal short-leg sizing. At $\hat{\beta} = 0.75$ (EWA/EWC), every \$100 long EWA pairs with \$75 short EWC: notional \$175, net-market exposure \$25.

Refitting the hedge ratio: how often?

$\hat{\beta}$ is not fixed. Like the OU parameters, it drifts as businesses evolve: a mining company shifts asset mix, an ETF rebalances, a rate-regime change shifts two banks' sensitivities. The fix is walk-forward fitting from Section 10.5: pick a fitting window W and refit cadence R, recompute $\hat{\beta}$ on the most recent W observations using only data strictly before each refit timestamp.

W and R trade off. Long W gives stable $\hat{\beta}$ but adapts slowly. Short W adapts fast but produces noisy slopes. Short R refits often but turns $\hat{\beta}$ into an overfittable parameter. Rules of thumb (Chan, (Chan, 2013) Chapter 3) for daily equity pairs: $W = 250$ days, $R = 21$ (monthly), or $W = 90$, $R = 5$ for faster signals. For Prosperity tick data, scale both by ticks-per-round.

Don't over-refit $\hat{\beta}$

Refitting $\hat{\beta}$ every bar introduces look-ahead and overfitting two ways.
First, if refit at t uses $[t - W, t]$ (*including* Y_t, Z_t), then $\hat{\beta}_t$ depends on the current observation, and $\hat{\epsilon}_t = Y_t - \hat{\alpha}_t - \hat{\beta}_t Z_t$ is mechanically biased toward zero. The strategy then buys low and sells high against a mean that already knows the current realisation. Backtests look spectacular; live is noise.
Second, even using only data strictly before t, a one-bar refit cadence wastes degrees of freedom: each fit yields a different hedge ratio, the spread fragments into short stationary pieces, and OU parameter estimates on each are noisier than on the full window.
Fix: refit weekly or monthly, hold $\hat{\beta}$ constant out of sample until the next refit. Only the z-score moves bar by bar.

Cointegration + OU is one algorithm

Pairs trading and OU mean reversion (Chapter 10) are the same algorithm with different inputs. The z-score code from Section 10.6 wraps unchanged around a cointegration residual; the only addition is an upstream Engle–Granger (or Johansen) fit producing $\hat{\beta}$ and the residual. Every downstream formula — half-life, σ_∞, z-thresholds, expected duration, walk-forward refit — transfers unmodified.

11.5 A short reference implementation

The Engle–Granger pipeline fits in one pure function: two equal-length numpy arrays in; OLS hedge ratio, intercept, residual mean, residual standard deviation, and ADF p-value out. The OU machinery is `fit_ou` from Section 10.6 applied to the residual.

```python
import numpy as np
from statsmodels.tsa.stattools import adfuller

def engle_granger(y: np.ndarray, z: np.ndarray) -> dict:
    """Engle-Granger two-step. Returns hedge ratio, residual stats,
       ADF p-value."""
    # Step 1: OLS regression  y = alpha + beta * z + eps
    beta, alpha = np.polyfit(z, y, 1)
    resid = y - (alpha + beta * z)
    # Step 2: ADF test on the residual
    adf_p = adfuller(resid, autolag='AIC')[1]
    return {
        'alpha': float(alpha),
        'beta':  float(beta),
        'mean':  float(resid.mean()),
        'std':   float(resid.std(ddof=1)),
        'adf_p': float(adf_p),
    }
```

Listing 11.1: Engle–Granger two-step as a pure function.

Three production notes. First, `adfuller` returns the univariate ADF p-value, not the CADF-corrected value; for production use `statsmodels.tsa.stattools.coint`. Second, the skeleton silently assumes $\hat{\beta} > 0$; $\hat{\beta} \leq 0$ signals a broken regression (can occur for a long ETF and its inverse, but rare). Third, call this inside a walk-forward loop. One call on the entire history with cached residual is the look-ahead trap above.

Prosperity example: PICNIC_BASKET as a cointegration trade

The cleanest Prosperity cointegration setup is the **picnic basket**: PICNIC_BASKET contains a fixed multiple of three underlying products (Prosperity 3: six CROISSANTS, three JAMS, one DJEMBE). It trades against a "synthetic basket" replicated from constituents. Define

$$\text{spread}_t := P_t^{\text{basket}} - \left(6\,P_t^{\text{cr}} + 3\,P_t^{\text{jam}} + 1\,P_t^{\text{dj}}\right). \tag{11.10}$$

Each underlying is a random walk; the spread is stationary (mean reflects basket miscalibration, volatility reflects quote noise). The hedge ratio is *known*: $\boldsymbol{\beta} = (1, -6, -3, -1)$, no fitting needed. Engle–Granger is overkill, but OU on the residual is essential. Chapter 12 treats the full strategy; Frankfurt Hedgehogs' Prosperity 3 writeup (`imc-prosperity-3/README.md`) trades it on a z-score rule.

Numerical sanity check. Frankfurt reported stationary standard deviation ≈ 40 seashells and half-life ≈ 50 Prosperity timesteps. Plugging $h_{1/2} = 50$ into $\theta = (\ln 2)/h_{1/2}$ gives $\theta \approx 0.01386$. With $z_{\text{entry}} = 1.5$, $z_{\text{exit}} = 0.3$, expected round-trip duration, by (10.28) of Chapter 10, is

$$h = \frac{\ln(1.5/0.3)}{\theta} = \frac{\ln 5}{0.01386} \approx \frac{1.6094}{0.01386} \approx 116 \text{ timesteps.} \tag{11.11}$$

More than twice the half-life ($z = 1.5 \to 0.3$ is a factor-5 shrink, more than two halvings). Over a 10 000-timestep round expect 80–100 round-trip basket trades on this signal alone, consistent with Frankfurt's reported counts.

On a Goldman Sachs desk

On a cash-equities pairs desk at Goldman (or any sell-side stat-arb shop), Engle–Granger and Johansen run as a daily *sweep* over every plausible pair/triple in a sector universe. Morning workflow:

1. **Universe screen.** One sector at a time (US banks, European energy majors, gold miners) — cross-sector cointegration is rare and usually spurious.
2. **Johansen sweep.** Johansen on every pair (sometimes triple) in the sector basket, rolling 250–500 day window. Record trace statistic, rank, cointegration vectors.
3. **Half-life filter.** Reject residual half-lives shorter than a few days (costs eat the edge) or longer than a few weeks (capital tied up). Sweet spot: 3–20 days.
4. **Cost gate.** Stationary spread SD must be a few multiples of round-trip cost on both legs. A spread reverting by 20 bps is not tradeable if round-trip costs 15 bps.
5. **Sizing.** Convert live z-scores into Kelly-fraction positions (Chapter 20), cap at per-pair VaR, feed to the execution algo (Chapter 19).

Division of labour: the *Strat* owns sweep code, the Engle–Granger/Johansen library, the half-life filter, and Kelly sizing. The *trader* owns exit-on-fundamentals: an M&A, index reweighting, or major analyst downgrade flags a pair as "cointegration broken" and the trader pulls the position regardless of z-score. Strats handle statistics and sizing; traders handle structural breaks the model cannot see.

11.6 Failure modes and when pairs trading dies

Pairs trading is not free money. The failure modes, in order of how often they bite, plus LTCM as the canonical reminder:

Structural breaks in $\hat{\beta}$. A merger, index inclusion or exclusion, regulatory change, or analyst event can shift the Y–Z relationship discontinuously. $\hat{\beta}$ jumps, the historical residual stops being stationary at the old level, and the spread trends to a new equilibrium the model cannot see. Walk-forward refit catches this only after a loss equal to the gap between old and new equilibrium. Discipline: hard stop-out on $|z| > z_{\text{stop}}$, plus a manual "no-trade" watchlist for pairs with imminent earnings, M&A rumours, or index reweightings.

Transaction costs overwhelm small reversion. A half-life of one day with $\sigma_\infty = 20$ bps sounds great until a US-equity round trip costs 5–10 bps in commissions and bid-ask, plus the borrow on the short leg. A $z = 2 \to 0.5$ reversion harvests $1.5\sigma_\infty = 30$ bps, more than half of which goes to costs. The GS cost gate above exists for this.

Convergence timing is unbounded. The OU model gives the *expected* reversion time, but the variance is large. A 3σ deviation might revert in a day or in six months, and the position marks to market every step. With leverage — most real pairs books use it, since P&L per unit notional is small — a long convergence triggers a margin call before the model says to give up. Discipline: hard stop-out (Chapter 10), *conservative* Kelly fraction (Chapter 20), and unwind any position that has not reverted within a few half-lives.

Look-ahead through the refit. Covered in the pitfall above: a pipeline that fits $\hat{\beta}$ on data through t to compute the residual at t is silently cheating and reveals itself only in live trading.

Decay of historical alpha. Canonical pairs (EWA/EWC, Coke/Pepsi, Shell/BP) have had their margins compressed to where they barely cover costs. Strategies that work in 2005 papers do not work in 2026 on the same pairs at the same parameters. Not a bug in cointegration — standard edge erosion. Discipline: keep finding new candidates, on underexplored sector slices or in markets with structural barriers to high-frequency entry.

Pair trades are not riskless arbitrage

The cautionary tale is **Long-Term Capital Management (LTCM)**, founded 1994 by John Meriwether with Nobel laureates Scholes and Merton on the board. LTCM ran a vast cointegration-based portfolio — on-the-run vs off-the-run Treasuries, swap spreads across countries, convertible-bond arbitrage — with peak notional around \$1.25 trillion against about \$5 billion of equity. Their analysis was correct: the relationships *were* cointegrated and *would* have reverted given time.

In August 1998 Russia defaulted, investors fled to safety, and every spread LTCM held widened simultaneously. The relationships were not broken — spreads did eventually revert — but LTCM's leverage meant marked-to-market losses triggered margin calls they could not meet. The fund collapsed in September 1998; a consortium of banks injected \$3.6 billion to unwind positions in orderly fashion. Lowenstein's *When Genius Failed* (2000) is the canonical narrative.

Lesson: cointegration tells you about the long-run mean. *It tells you nothing about the worst drawdown along the way.* The position-sizing rules of Chapter 20 and the stop-out rules of Chapter 10 are the only thing standing between "profitable in expectation" and "insolvent before convergence."

11.7 Key facts to memorise

- **Correlation $\neq$ cointegration.** Return correlation is on *differences*; cointegration is a long-run-stationary property of a linear combination of *levels*. Only cointegration is the right test for pairs trading.
- **Spurious correlation.** Independent random walks routinely exhibit sample level correlation across $[-1, +1]$ by chance. Screening on level correlation alone throws money away.
- **Engle–Granger two-step.** (1) OLS $Y_t = \alpha + \beta Z_t + \epsilon_t$ gives the hedge ratio; (2) ADF (with cointegration-corrected critical values) on the residual tests stationarity. Reject the random-walk null $\Rightarrow$ cointegrated.

- **Asymmetric regression.** OLS in (11.4) is not symmetric in Y and Z. Fix: total least squares, or Johansen.
- **Johansen test.** For $n \geq 2$ assets, estimates the rank r of Π in a VECM and returns r independent cointegration vectors at once. Engle–Granger is the $n = 2$ case.
- **Trading rule = OU z-score on the residual.** Fit $(\hat{\theta}, \hat{\mu}, \hat{\sigma})$ via AR(1) mapping, apply the entry/exit/stop rules of Chapter 10. No new logic.
- **Position sizing.** Long the spread $\Rightarrow$ buy 1 unit of Y, sell $\hat{\beta}$ units of Z.
- **Walk-forward refit.** Refit $\hat{\beta}$ on a rolling W-window every R steps, using only data strictly before the refit timestamp. Hold $\hat{\beta}$ constant out of sample. Refitting every bar is the canonical look-ahead trap.
- **Failure modes.** Structural breaks, transaction costs, unbounded convergence timing, look-ahead, edge decay.
- **LTCM (1998).** Cointegration correct, convergence timing wrong, leverage too high. Risk management is the strategy.
- **Prosperity link.** PICNIC_BASKET premium is a cointegrated spread with known hedge ratio $(1, -6, -3, -1)$; Chapter 12 treats it in detail.
- **Goldman link.** Sell-side stat-arb desks run a daily Johansen sweep, gate by half-life and cost, size by Kelly; trader handles structural-break exits.

Chapter 12

Basket and ETF Arbitrage

An exchange-traded fund (ETF) on the S&P 500 holds 500 underlying stocks in declared proportions. Summing those prices at the declared weights gives the *synthetic* value of one ETF share — what the wrapper should cost if it equalled its contents. The ETF's market price can drift from synthetic value, but not far: *authorised participants* (a class of large dealer) can mechanically convert constituents into ETF shares (and back) through the issuer, pocketing the discrepancy. This chapter is about what happens when that mechanism is, and is not, available.

In Prosperity 3 the basket trade was the largest single source of PnL for many top teams. `PICNIC_BASKET1` contained six `CROISSANTS`, three `JAMS`, and one `DJEMBE` per share; `PICNIC_BASKET2` contained four `CROISSANTS` and two `JAMS`. The key caveat: the Prosperity simulator exposes *no* creation / redemption mechanism. PICNIC_BASKET prices mean-revert toward synthetic value only because the simulator's data-generating process is built that way, not because anyone can force convergence. The Prosperity basket trade is a *pure cointegration play with a known hedge ratio* — the cleanest application of Chapter 11 — but not, strictly, an arbitrage.

The chapter has two halves. The first develops the theory: creation / redemption, the synthetic-value formula, static-vs-dynamic hedge ratios, and Avellaneda and Lee's 2010 PCA-residual framework. The second half is a case study of the Frankfurt Hedgehogs' 2nd-place PICNIC_BASKET strategy from Prosperity 3. Every numerical claim traces to `imc-prosperity-3/README.md` or `imc-prosperity-3/FrankfurtHedgehogs_polished.py`.

Prerequisites

From earlier:

- Chapter 10: OU z-score machinery, half-life $h_{1/2} = (\ln 2)/\theta$, stationary std $\sigma_\infty = \sigma/\sqrt{2\theta}$, entry/exit/stop-out rule, AR(1) estimator. The basket-basis rule is the same z-score applied to the residual.
- Chapter 11: cointegration and cointegration vectors, Engle–Granger two-step, and *declared* (given by the basket constructor) vs *estimated* (must fit and walk-forward refit) hedge ratios.
- Chapter 4: wall-mid as the reference for Prosperity spreads; the Chapter 3 taxonomy for who the APs of a real ETF are.
- Forward: Chapter 20 for the variance-vs-expected-return reasoning behind Round 5 half-hedging; Chapter 22 for the ML view of the Avellaneda–Lee residual.

From outside the book:

- Linear combinations of random variables: if $B = \sum_i w_i C_i + \epsilon$ then $\mathrm{Var}[B] = \sum_i w_i^2 \mathrm{Var}[C_i] + \mathrm{Var}[\epsilon] + 2\sum_{i<j} w_i w_j \mathrm{Cov}[C_i, C_j]$; a portfolio +1 basket, $-w_i$ of each constituent has variance $\mathrm{Var}[\epsilon]$ alone.
- Fund-structure vocabulary: a fund is a legal vehicle, the issuer runs it, its "shares" trade on an exchange. ETFs differ from mutual funds in trading intraday rather than only at end-of-day net asset value (NAV) through the issuer.

12.1 ETFs and the creation/redemption mechanism

An ETF alone is not an arbitrage instrument — it is a passively-managed fund whose shares trade on an exchange. The arbitrage angle comes from a parallel mechanism in the prospectus that lets a specific class of dealer convert wrapper to contents and back at near-zero cost.

Definition 12.1 (Exchange-traded fund)**.** An *exchange-traded fund (ETF)* is a passively-managed investment fund whose shares trade intraday on a securities exchange. The fund holds a basket of underlying assets (stocks, bonds, commodities, futures) in declared proportions, the *creation basket*, and the published net asset value (NAV) per ETF share equals the sum of the constituents' market prices weighted by those proportions, less management fees. SPY (the SPDR S&P 500 ETF), QQQ (Invesco Nasdaq-100), and IWM (iShares Russell 2000) are the three most-traded examples in US equities.

Definition 12.2 (Authorised participant)**.** An *authorised participant (AP)* is a large broker-dealer contractually permitted by the ETF issuer to deliver the creation basket of constituents to the fund and receive new ETF shares in return (*creation*), or to deliver ETF shares back to the fund and receive the constituents (*redemption*). APs typically pay a small fixed fee per creation / redemption unit (thousands of US dollars on a US-equity ETF, irrespective of notional). For SPY the AP is one of a handful of US bank broker-dealers; the contract is a private bilateral agreement between AP and issuer. A retail trader cannot create or redeem: the barrier is both legal (no AP contract) and operational (delivering a 500-name basket in the correct proportions, to the issuer's custodian, on the right settlement cycle, requires the clearing and prime-brokerage plumbing only large dealers have).

The mechanism gives the AP an automatic arbitrage. When the ETF trades *above* synthetic value the wrapper is said to be at a *premium*; *below* synthetic value, at a *discount*. A premium typically arises when end-investor demand for ETF shares outruns the pace at which APs are creating new units; the arbitrage below is precisely the mechanism that collapses it. Suppose the ETF trades at P_t^{ETF} with synthetic value $P_t^{\mathrm{synth}} = \sum_i w_i C_{i,t} < P_t^{\mathrm{ETF}}$ (a premium). The AP can:

1. Buy the constituents at cost P_t^{synth} per creation unit (plus bid–ask and impact, small for liquid names).
2. Deliver one creation unit to the fund, plus the creation fee.
3. Receive newly-issued ETF shares.
4. Sell them on the exchange at P_t^{ETF}.

Profit per creation unit is $P_t^{\mathrm{ETF}} - P_t^{\mathrm{synth}} - (\text{fee} + \text{costs})$. Step 4's selling pressure pushes P_t^{ETF} back toward P_t^{synth}; the incentive vanishes once the gap falls below the fee-plus-cost floor. Redemption is the mirror image: when $P_t^{\mathrm{ETF}} < P_t^{\mathrm{synth}}$ the AP buys cheap ETF shares, hands them to the fund, receives constituents, and sells them at the higher synthetic value.

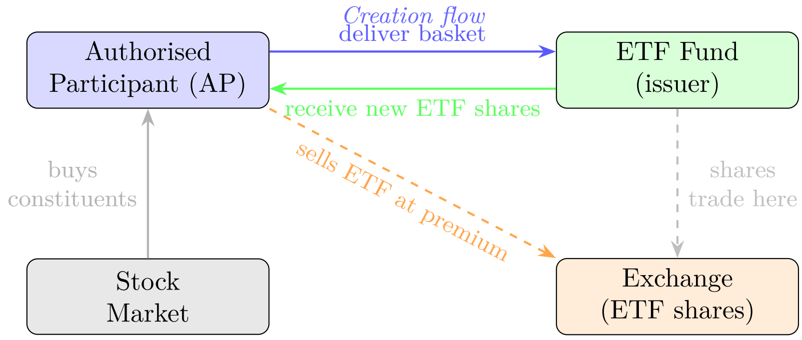


Figure 12.1: ETF creation mechanism. When the ETF trades at a *premium* to NAV, the authorised participant buys the constituent basket on the stock market, delivers it to the fund, and receives freshly minted ETF shares to sell on the exchange — locking in the premium as riskless profit. Redemption is the reverse: the AP buys ETF shares cheaply, redeems them for the basket, and sells the constituents. This in-kind arbitrage keeps the ETF price anchored to its net asset value.

The ETF basis is bounded by the round-trip creation/redemption fee

Define the *ETF basis* as

$$\mathrm{Basis}_t := P_t^{\mathrm{ETF}} - P_t^{\mathrm{synth}}, \qquad P_t^{\mathrm{synth}} := \sum_{i=1}^{n} w_i\, C_{i,t}. \tag{12.1}$$

On a real, liquid ETF, the absolute basis is bounded above by the round-trip cost an AP would pay to close the gap:

$$|\mathrm{Basis}_t| \lesssim \underbrace{\text{creation fee}}_{\text{fixed per unit}} + \underbrace{\text{constituent half-spread}}_{\text{liquidity cost}} + \underbrace{\text{market impact}}_{\text{size-dependent}}. \tag{12.2}$$

SPY and QQQ in normal markets: 1–5 bps. Illiquid sector or emerging-market ETFs whose constituents trade in a different time zone: tens of bps or more.

Intuition

Creation / redemption is a *hard floor* on what would otherwise be a soft mean-reversion trade. The OU residual of Chapter 10 was bounded only by the stationary distribution: the spread could wander out arbitrarily long before reverting, stopped only by statistics and a stop-out. The cointegration spread of Chapter 11 has the same problem — the LTCM box there is the canonical example.

The ETF basis on a liquid AP-served ETF is different. Once it exceeds the fee-plus-cost floor, APs converge within minutes (often seconds); the gap closes as mechanical engineering, not statistical drift. ETF basis trading is among the safest stat-arb variants — and, as the case study shows, one of the lowest-margin, because the floor is well-known and well-policed.

12.2 The basket spread formula

Strip away the wrapper and the trade is a linear combination of prices. Let B be the basket, $C_1, \ldots, C_n$ its constituents, with declared weights $w_1, \ldots, w_n$ (non-negative integers in Prosperity; fractional on a real ETF).

Definition 12.3 (Synthetic basket value and basket basis)**.** The *synthetic value* of one basket share at time t is

$$B_t^{\text{synth}} := \sum_{i=1}^{n} w_i\, C_{i,t}, \tag{12.3}$$

and the *basket basis* (or simply *basket spread*) is

$$\text{Basis}_t := B_t - B_t^{\text{synth}} = B_t - \sum_{i=1}^{n} w_i\, C_{i,t}. \tag{12.4}$$

The vector $\boldsymbol{\beta} = (1, -w_1, -w_2, \ldots, -w_n) \in \mathbb{R}^{n+1}$ is the *declared cointegration vector*: the $\mathbb{R}^{n+1}$ portfolio $+1$ basket, $-w_i$ of each constituent, has time-t value exactly Basis_t.

Remark 12.4. The basket's cointegration vector is *declared by construction*: the issuer publishes the weights, so there is no Engle–Granger or Johansen fit. The ADF test on the residual stays informative — it signals whether wrapper and contents have decoupled — but the slope is given.

Under cointegration (enforced by creation / redemption on a real ETF; by the simulator's data-generating process in Prosperity) the basis is stationary around some mean, often but not always zero, so the OU z-score machinery of Chapter 10 applies directly. Compute sample mean $\hat{\mu}_{\text{Basis}}$ and standard deviation $\hat{\sigma}_{\text{Basis}}$, define the running z-score

$$z_t := \frac{\text{Basis}_t - \hat{\mu}_{\text{Basis}}}{\hat{\sigma}_{\text{Basis}}}, \tag{12.5}$$

and apply the threshold rule:

Basket basis trading rule

With thresholds $z_{\text{entry}} > z_{\text{exit}} \geq 0$, apply the Chapter 10 rule:

- $z_t > +z_{\text{entry}}$: *short the basis* — short 1 basket, long w_i of each constituent. Basket is rich.
- $z_t < -z_{\text{entry}}$: *long the basis* — long 1 basket, short w_i of each constituent. Basket is cheap.
- $|z_t| < z_{\text{exit}}$: close; the basis has reverted.

The hedge leg is sized *at the declared weights* because $\boldsymbol{\beta}$ is declared: no slope refit, only the running mean and variance of the residual need updating.

Expected round-trip duration and PnL follow Chapter 10 applied to the basis residual: the basket spread strategy is the OU z-score strategy with a multi-asset hedge leg.

A toy numerical example

Take a basket B with two constituents, weights $w_1 = 3$, $w_2 = 2$, and wall-mid prices at three consecutive timesteps:

t	B_t	$C_{1,t}$	$C_{2,t}$
1	510	100	105
2	508	101	103
3	516	100	106

With $B_t^{\text{synth}} = 3C_{1,t} + 2C_{2,t}$ and $\text{Basis}_t = B_t - B_t^{\text{synth}}$:

$$\begin{aligned} B_1^{\text{synth}} &= 3 \cdot 100 + 2 \cdot 105 = 510, & \text{Basis}_1 &= 510 - 510 = 0, \\ B_2^{\text{synth}} &= 3 \cdot 101 + 2 \cdot 103 = 509, & \text{Basis}_2 &= 508 - 509 = -1, \\ B_3^{\text{synth}} &= 3 \cdot 100 + 2 \cdot 106 = 512, & \text{Basis}_3 &= 516 - 512 = +4. \end{aligned}$$

The sample mean of $\{0, -1, +4\}$ is $\hat{\mu} = 1$ and the sample standard deviation (unbiased $n-1$ divisor) is $\hat{\sigma} = \sqrt{((0-1)^2 + (-1-1)^2 + (4-1)^2)/2} = \sqrt{7} \approx 2.65$. The z-score at $t = 3$ is

$$z_3 = \frac{4-1}{2.65} \approx 1.13.$$

With $z_{\text{entry}} = 1.0$ this crosses and the rule fires *short basis*: sell one B, buy three C_1 and two C_2. If the basis reverts to its mean of 1 next bar (constituents unchanged), PnL is $(\text{Basis}_3 - \text{Basis}_4) \cdot 1 = 3$ seashells per basket short, hedge leg breaking even.

Two takeaways. First, the *hedge ratio is the literal share count* — no fitting. Second, $\hat{\mu} = 1 \neq 0$; the z-score is taken against it, not against zero (the running-mean correction of Section 12.4). Shorting whenever $\text{Basis}_t > 0$ would have shorted at $t = 1$ at zero, sat through -1, and unwound at $+4$ for a four-seashell loss per unit.

12.3 Static vs dynamic hedge ratios

One degree of freedom hides in (12.3): how do you know the weights w_i? Real ETF: the issuer publishes, the AP trades. Synthetic basket without a mechanical wrapper-contents link (proprietary index, factor-mimicking portfolio): murkier.

Definition 12.5 (Static hedge ratio)**.** A *static hedge ratio* fixes w_i once at the constructor's declared values (or, on a real ETF, the prospectus) and never re-estimates.

Definition 12.6 (Dynamic hedge ratio)**.** A *dynamic hedge ratio* estimates w_i empirically via Engle–Granger or Johansen (Chapter 11) on a rolling window, refit on a fixed cadence ($W = 250$, $R = 21$ trading-day defaults). The hedge ratio at time t is the latest fit, held out of sample until the next refit.

The right choice depends on whether wrapper and contents are mechanically coupled.

Static hedge: when the wrapper is the contents. On a real ETF the AP loop closes the basis at the published weights; any departure leaves exposure the AP will not bail out. Same for Prosperity PICNIC_BASKETs: weights $(6, 3, 1)$ and $(4, 2, 0)$ are declared by the simulator; any re-estimate adds noise.

Dynamic hedge: when the wrapper is loosely coupled. A proprietary "smart-beta" factor index with no AP, rebalancing at month-end, can exhibit a drifting empirical hedge ratio even with published methodology. A dynamic hedge on rolling windows tracks the drift; a static hedge anchored to the month-end snapshot accumulates unhedged factor exposure the trader mistakes for alpha.

Static hedge is right only when you can actually do creation/redemption

The temptation is to plug any basket's published weights into (12.3) and call it done. Correct for SPY, QQQ, and Prosperity PICNIC_BASKETs, where creation/redemption or the simulator enforces the weights. *Wrong* for a non-tradable proprietary index whose weights drift, for an actively-managed mutual fund with monthly disclosures, or for any

basket whose "published weights" are marketing, not a tradable contract.
Diagnostic: *can someone, somewhere, do creation / redemption against this basket at the published weights?* If yes, static hedge. If no, fit with Engle–Granger or Johansen (Chapter 11) and walk-forward refit. The half-second check catches this trap before shipping a strategy silently long an unintended factor.

12.4 Case study: Frankfurt Hedgehogs and PICNIC_BASKET

The Frankfurt Hedgehogs placed 2nd in IMC Prosperity 3 (2025); their PICNIC_BASKET strategy (Frankfurt Hedgehogs, 2025b, `imc-prosperity-3/README.md`, § Round 2) was the leaderboard's largest basket-PnL source. Their public source `FrankfurtHedgehogs_polished.py` lets us cross-check every parameter; every numerical claim below traces to the writeup or to a source line.

The setup

Round 2 introduced three individual products (`CROISSANTS`, `JAMS`, `DJEMBES`) and two baskets:

- `PICNIC_BASKET1`: 6 `CROISSANTS` + 3 `JAMS` + 1 `DJEMBE`.
- `PICNIC_BASKET2`: 4 `CROISSANTS` + 2 `JAMS`.

`DJEMBES` appears only in basket 1, which matters for the inter-basket spread below. Position limits (`POS_LIMITS` dict, source lines 30–44): 60 basket 1, 100 basket 2, 250 `CROISSANTS`, 350 `JAMS`, 60 `DJEMBES`.

Prosperity has no creation / redemption mechanism

The PICNIC_BASKET in Prosperity 3 cannot be unwrapped into constituents: no AP, no issuer, no creation fee. The basket is a tradable instrument the simulator's bots quote; its price tracks synthetic value only because the data-generating process adds a mean-reverting noise term on top of the constituent-implied price (Frankfurt writeup: "*the most natural generation process seemed to be that the three constituents' prices were independently randomized, and a mean-reverting noise sequence was then added on top to produce the basket price*").
The basis is therefore bounded only by the *statistical* mean reversion of that noise term, not by the hard floor (12.2) sets on a real ETF. The Hedgehogs' strategy is a pure cointegration play (Chapter 11), not a hard arbitrage (Chapter 14); every LTCM-box risk lesson carries over.

Spread construction

The Hedgehogs computed two separate spreads, one per basket. With P_t^{B1} and P_t^{B2} the basket prices and P_t^{cr}, P_t^{jam}, P_t^{dj} the constituent prices, define

$$S_t^{B1} := P_t^{B1} - \left(6\,P_t^{\text{cr}} + 3\,P_t^{\text{jam}} + 1\,P_t^{\text{dj}}\right), \tag{12.6}$$
$$S_t^{B2} := P_t^{B2} - \left(4\,P_t^{\text{cr}} + 2\,P_t^{\text{jam}}\right). \tag{12.7}$$

This is the static-hedge basket basis of Theorem 12.3 with simulator-declared weights. The source hard-codes them as `ETF_CONSTITUENT_FACTORS = [[6, 3, 1], [4, 2, 0]]` (line 57); line 437 computes the spread as a numpy dot of constituent wall-mids with the factor vector, subtracted from the basket wall-mid (line 440). The wall-mid is the Chapter 4 wall-mid (midpoint of deepest visible bid and ask), not top-of-book mid.

The Hedgehogs also considered an *inter-basket* spread

$$S_t^{\text{inter}} \;:=\; P_t^{B1} \;-\; 1.5\,P_t^{B2} \;-\; P_t^{\text{dj}}, \tag{12.8}$$

which removes the constituents by combining the two baskets. Since $1.5 \times (4,2,0) = (6,3,0)$, the difference $(6,3,1) - 1.5 \times (4,2,0) = (0,0,1)$ leaves one `DJEMBE` unhedged — hence the $-P_t^{\text{dj}}$. The writeup is explicit (§ Round 2): "*we quickly identified that comparing baskets directly to their constituent sums was the stronger and more reliable path.*" Production trades only S_t^{B1} and S_t^{B2}. Likely reason: constituent legs are deep while `PICNIC_BASKET2` has thin top-of-book volume.

Empirical spread behaviour

Figure 4 (writeup, § Round 2) plots S_t^{B1} and S_t^{B2} across a round. Two dominant features:

1. Short-term mean reversion at a "noise around a level" frequency, consistent with an OU residual on top of a stationary mean.
2. A persistent positive premium: S_t^{B1}'s running mean is non-zero. The wrapper trades above the sum of its parts by tens of seashells, the level drifting slowly across a round.

The source initialises premiums as `INITIAL_ETF_PREMIUMS = [5, 53]` (line 62, a prior from Round 1 back-test medians), then updates online (Section 12.4).

The writeup publishes no values for OU half-life or stationary std; Figure 4 suggests a half-life of tens of timesteps and a stationary std on the order of the premium itself.

The static threshold rule

The Hedgehogs applied a fixed-threshold rule. With base thresholds $T^{B1}, T^{B2} > 0$ and running premium $\hat{\mu}_t^{Bk}$, the de-meaned spread is

$$\tilde{S}_t^{Bk} \;:=\; S_t^{Bk} - \hat{\mu}_t^{Bk}, \tag{12.9}$$

and:

- $\tilde{S}_t^{Bk} > +T^{Bk}$: short the basket (sell bid wall, position-limit sized);
- $\tilde{S}_t^{Bk} < -T^{Bk}$: long the basket (buy ask wall, position-limit sized);
- $\tilde{S}_t^{Bk}$ crosses zero through an open position: close.

The zero-crossing close is gated by `ETF_CLOSE_AT_ZERO = True` (line 67), the closing branch being the "`elif ETF_CLOSE_AT_ZERO`" block on lines 493–501. Base thresholds are `BASKET_THRESHOLDS = [80, 50]` (line 59): enter basket 1 at ±80 from the running mean, basket 2 at ±50 — *absolute-seashell* thresholds, not z-scores. The Hedgehogs chose raw thresholds because basis volatility did not appear to vary meaningfully — "*using a z-score normalizes the spread by recent volatility, but unless volatility is known to vary meaningfully over time . . . this introduces unnecessary complexity and risk of overfitting*" (writeup, § Round 2).

Why 80 and 50, not 200 and 100? A small grid search (Figure 5, the "Optimal Parameter Search Grid" heat map) led the Hedgehogs to *not pick the best back-test parameter* but the centre of a flat plateau on the PnL surface — the canonical defence against small-sample overfitting. The chrispyroberts P3 writeup[1] reaches a similar conclusion.

[1]See `research/writeups/chrispyroberts_P3/ROUND 2/basket_gridsearch_results.csv`; an independent grid search confirms a broad optimum, not a sharp peak.

Dynamic threshold adjustment using Olivia

The strategy's most distinctive feature — the one that materially separated its PnL from a vanilla threshold rule — is the use of an external trading-bot signal to bias the thresholds. The bot is *Olivia*, already detected on `SQUID_INK` in Round 1 as an "informed-buyer / informed-seller" whose trades preceded short-term moves. In Round 2 the same pattern applied to `CROISSANTS`: Olivia long predicted a bullish `CROISSANTS` move and hence, since `CROISSANTS` is the largest constituent of both baskets, on the basis.

The source mechanic (lines 480–491) is a threshold *shift*, not a z-score adjustment. With $d_t \in \{-1, 0, +1\}$ for Olivia's direction and magnitude A (`ETF_THR_INFORMED_ADJS = [90, 90]` line 65), the modified entry rule is

$$\text{enter short basket if} \quad \tilde{S}_t^{Bk} \; > \; +T^{Bk} + d_t \cdot A, \tag{12.10}$$

$$\text{enter long basket if} \quad \tilde{S}_t^{Bk} \; < \; -T^{Bk} + d_t \cdot A. \tag{12.11}$$

The single offset $d_t \cdot A$ shifts *both* thresholds the same way. Olivia long ($d_t = +1$) slides the band up: more positive deviation for a short, less negative for a long — the strategy biases toward long-basket. Olivia short ($d_t = -1$) biases toward short-basket. $d_t = 0$ reduces to Section 12.4.

Prosperity example: the Olivia cross-product adjustment

The clearest illustration is in the Frankfurt writeup itself (§ Round 2): *"if our base threshold was* ± 50*, detecting Olivia as short would shift the long entry to* -80 *and the short entry to* $+20$*."* Plug $T = 50$, $d_t = -1$, $A = 30$ into (12.10)–(12.11):

$$\begin{aligned} \text{long entry} = -T + d_t A \;&=\; -50 + (-1)(30) = -80, \\ \text{short entry} = +T + d_t A \;&=\; +50 + (-1)(30) = +20. \end{aligned}$$

Matches the writeup. An Olivia long symmetrically shifts long entry to -20, short entry to $+80$. Production $A = 90$ is larger than the didactic 30 because production base thresholds are 80 and 50 (not 50 and 50); A was re-tuned on the larger back-test grid.

The economic content is cross-product information leakage. `CROISSANTS` is the dominant constituent in both baskets (6 in B1, 4 in B2), so any informed signal on `CROISSANTS` translates almost mechanically into one on the basis. Shifting the threshold band rather than the running-mean estimate keeps the two estimators separate: $\hat{\mu}_t^{Bk}$ updates from price data alone (Section 12.4); offset $d_t \cdot A$ comes from Olivia alone; they combine at execution.

The contribution is large. The Hedgehogs report (writeup, § Round 2 conclusion): "*about* 40,000–60,000 *SeaShells per round on baskets, plus another* 20,000 *SeaShells per round directly from trading Croissants individually.*" Attribution in Section 12.4.

The drifting premium and the running-mean correction

S_t^{B1}'s running mean was non-zero: a persistent positive premium drifting slowly across a round. A zero-centred rule is biased — the basis spends more time positive, the strategy runs net short on average, long-entry never fires. Fix: estimate $\hat{\mu}_t^{Bk}$ online and subtract before thresholding.

The estimator (line 448) is an *incremental running mean*:

$$\hat{\mu}_n^{Bk} \;=\; \hat{\mu}_{n-1}^{Bk} \;+\; \frac{S_n^{Bk} - \hat{\mu}_{n-1}^{Bk}}{n} \tag{12.12}$$

with n incrementing from `n_hist_samples = 60,000` (line 61) and $\hat{\mu}_0^{Bk}$ = `INITIAL_ETF_PREMIUMS = [5, 53]` (line 62). Not an EMA (the most-recent-observation weight is $1/n$, not a fixed α); the 60,000-sample prior dominates, so live updates move the estimate slowly.

`CALCULATE_RUNNING_ETF_PREMIUM = True` (line 68) gates the live update; False keeps only the prior. Production ran with it on.

The basket premium is not constant

The most expensive beginner failure: compute spread as $B - \sum_i w_i C_i$, centre the threshold at zero, trade. If the basket carries a persistent premium — Prosperity's PICNIC_BASKET1 did, by tens of seashells — the spread is almost always positive and only the "short basket" branch fires. The trader becomes systematically short, correct in expectation *only if* the premium is about to revert to zero, which it is not.
Fix: the running-mean update (12.12), subtracted *before* thresholding. Chapter 10's z-score does this implicitly via $\hat{\mu}$ subtraction; raw thresholds make the step explicit.
Deeper lesson: a basket basis with non-zero mean can still be cointegrated. The residual's mean encodes fees, microstructure costs, and adverse-selection premia — non-zero in general. Centre on the running mean, not structural zero.

PnL attribution

The headline PnL is from the writeup's Round 2 conclusion: "*about* 40,000–60,000 *SeaShells per round on baskets, plus another* 20,000 *SeaShells per round directly from trading Croissants individually.*" We treat it as ground truth — the only number published, not re-attributable without rerunning the simulator.

Qualitative observations:

- Basket-leg to `CROISSANTS`-only PnL ratio $\approx$ 2–3:1. Basket dominates; Olivia on `CROISSANTS` alone is still a meaningful book.
- The 40k–60k range spans the five rounds, re-tuned each round.
- PnL is seashells per 10,000-timestep round; position limits above.

Economic anchor: with stationary basis range $\pm O(100)$ around the premium, capturing a fraction of each round-trip over several dozen cycles yields $\sim 10^4$ seashells per round per basket. Olivia contributes the *bias* that tilts each round-trip toward the predicted side.

The Round 5 half-hedging decision

In the final round the Hedgehogs had a ~190,000-seashell lead; full-aggression upside was small versus the downside of a spread excursion closing the gap. Their decision (writeup, § Round 5): *half-hedge*. Buy $\frac{1}{2}w_i$ of each constituent against each long basket (sell $\frac{1}{2}w_i$ against each short).

Source: `ETF_HEDGE_FACTOR = 0.5` (line 70), fed into line 532 as `expected_hedge_position += -basket.expected_position * etf_const_factor * ETF_HEDGE_FACTOR`.

The reasoning is the linear-combination identity of Chapter 20. Let X be unhedged PnL (basis signal, no hedge) and Y fully-hedged PnL (w_i of each constituent). Any convex combination $\alpha X + (1-\alpha)Y$ is a valid strategy. The Hedgehogs chose $\alpha = \frac{1}{2}$. Then:

- **Expected return** is linear: $\mathbb{E}[\alpha X + (1-\alpha)Y] = \alpha\mathbb{E}[X] + (1-\alpha)\mathbb{E}[Y]$. Both books trade the same signal with positive expected PnL, so the half-hedged book splits the difference. Crucially $\mathbb{E}[Y] > 0$: hedging removes constituent-direction exposure; the basis — the alpha — is preserved.

- **Variance** is not linear:

$$\mathrm{Var}[\alpha X + (1-\alpha)Y] = \alpha^2 \mathrm{Var}[X] + (1-\alpha)^2 \mathrm{Var}[Y] + 2\alpha(1-\alpha)\,\mathrm{Cov}[X,Y]. \tag{12.13}$$

  At $\alpha = \frac{1}{2}$ this is $\frac{1}{4}(\mathrm{Var}[X] + \mathrm{Var}[Y] + 2\,\mathrm{Cov}[X,Y])$, smaller than $\mathrm{Var}[X]$ whenever $\mathrm{Var}[Y]$ and $\mathrm{Cov}[X,Y]$ are not too large.

A Kelly-style trade-off: give up PnL for lower variance because the marginal seashell of PnL is worth less than the marginal seashell of downside protection. $\alpha = \frac{1}{2}$ (not Kelly-optimal) because it is robust to misestimation of $\mathbb{E}[X]$ and $\mathrm{Var}[X]$, highly uncertain in one round. Round 5 also tightened the zero-crossing close for the same reason.

Why half, not zero? $\alpha = 0$ removes all constituent-direction risk, but the legs eat trading costs and position limits, reducing throughput. Half-hedge is the compromise.

Why not dynamic? Variance-minimising α solves a quadratic in live $\mathrm{Var}[X]$, $\mathrm{Var}[Y]$, $\mathrm{Cov}[X,Y]$; the round-sample is too small to fit. A static α is robust — same discipline as the threshold choice.

12.5 Avellaneda–Lee PCA stat-arb

The Hedgehogs' strategy is a special case. Two assets: residual is the spread (Chapter 11). $n+1$ assets with a wrapper: residual is the basis (Theorem 12.3). The natural extension — the one that took academic stat-arb from pair trades to a thousand-name portfolio — is a large constituent universe with no single wrapper, where the residual is computed by first removing a few common factors from data.

Avellaneda and Lee (2010) formalised this in their 2010 paper *Statistical arbitrage in the U.S. equities market* ((Avellaneda and Lee, 2010)). The construction:

1. Take the constituent daily-return panel $R \in \mathbb{R}^{T\times N}$ for $N \approx 500$ S&P names.
2. PCA on $R^\top R$; retain the first k eigenvectors (k a small single digit: $k = 1$ for the market factor, $k = 3$–5 for sector factors).
3. For each stock i, regress returns on the principal-component series: $\epsilon_{i,t} = R_{i,t} - \sum_{j=1}^{k} \beta_{i,j} F_{j,t}$ is the part unexplained by common factors.
4. Cumulate: $X_{i,t} = \sum_{s\le t} \epsilon_{i,s}$. Under Avellaneda–Lee each $X_{i,t}$ is approximately OU around zero.
5. Apply the Chapter 10 z-score rule to each $X_{i,t}$. Each name is its own basket trade, with the factor portfolio as basket and stock i as wrapper.

The framework generalises both: pairs trading (residual from many constituents rather than one partner) and the basket basis trade (the wrapper is extracted statistically, not declared). We do not derive the PCA estimator here — see the paper and (Chan, 2013) (Chapter 5, on ETF-vs-constituent strategies); Chapter 22 returns to the broader ML view of factor residuals. The point: the whole chapter so far is the special case of Avellaneda–Lee with one factor (the wrapper), declared loadings, and the basis as residual.

Historical note

Common-factor residuals as a trading signal predate Avellaneda and Lee (2010) by two decades. Programme trading desks at Salomon Brothers, Morgan Stanley, and D.E. Shaw in the late 1980s / early 1990s ran "equity statistical arbitrage" on residuals against single-factor (CAPM) and later multi-factor (Fama–French, BARRA) regressions, with the same OU-residual machinery the literature later formalised. Avellaneda and Lee (2010) is the first public-domain account, collecting 1995–2008 results and showing the strategy was profitable until the August 2007 "quant quake," when stat-arb funds simultaneously deleveraged and residuals decoupled from pre-2007 OU behaviour for several days — the canonical reminder that "statistically stationary" is an empirical property, not a guarantee.

On a Goldman Sachs desk

On a sell-side ETF arbitrage desk at Goldman Sachs, the creation / redemption loop of Section 12.1 is the core business. The desk holds constituent inventory and issuer connectivity (typically a daily cut-off window for AP creations against the next NAV strike). The Strat layer owns:

- **Real-time basis calculation.** A service reading constituent quotes, FX, dividend accruals, and borrow cost on any short-leg redemption hedge. Output: a live *theoretical NAV*; its difference from ETF market price is the basis.
- **Creation / redemption cost model.** A function $C(\text{size, side, time of day})$ accounting for settlement lag (T+2 on US equities through 2024, T+1 from 2025), borrow, and constituent-leg impact. Basis above cost plus margin fires the AP engine.
- **The automatic AP engine.** State machine that, on creation trigger, sends constituent buys, instructs back-office to create units, and queues resulting shares for market-making sales. Redemption mirrors. Autonomous with a one-line override.

The trader decides *when to suppress*. Two canonical cases: around the NAV-striking window (fast constituents, stale leg), and on a name with dried-up borrow (shorting the redemption leg is expensive). The trader pulls via an exception report. Otherwise the engine runs the flow: SPY alone generates hundreds of round-trips per day across the AP community; PnL is a fraction of a basis point on each multiplied by very large notional. ETF arb is among the most systematically run and reliably profitable desks at a large bank — edge has compressed as APs have proliferated, but the fee floor remains and inventory turnover is enormous.

12.6 A short reference implementation

The whole basket basis strategy is one short pure function. Input: wall-mid prices for basket and three constituents, running mean and std of the basis, and Olivia's `CROISSANTS` signal. Output: one of `short_basket`, `long_basket`, `close`, `hold`. Drop-in for `Trader.run(state)`. `olivia_position` is Olivia's inferred `CROISSANTS` position (positive/negative/zero); `olivia_bias` is offset A.

```python
def picnic_basket_spread(basket_price, croissant_price, jam_price,
    djembe_price,
                         running_mean, running_std, olivia_position,
                         z_entry=0.7, z_exit=0.2, olivia_bias=15):
    """Basket spread + z-score + Olivia threshold bias. Returns one
        action."""
    synthetic = 6 * croissant_price + 3 * jam_price + djembe_price
```

```python
    basis     = basket_price - synthetic
    # Olivia adjustment: shift entry thresholds based on her croissant side.
    direction = (1 if olivia_position > 0
                 else -1 if olivia_position < 0
                 else 0)
    bias = olivia_bias * direction
    z = (basis - running_mean - bias) / max(running_std, 1e-6)
    if z >  z_entry:  return 'short_basket'   # basis is rich
    if z < -z_entry:  return 'long_basket'    # basis is cheap
    if abs(z) < z_exit: return 'close'
    return 'hold'
```

Listing 12.1: Frankfurt-style basket spread signal with Olivia bias.

Three production notes. First, `running_mean` and `running_std` are stateful and must be updated each tick via (12.12) or an EMA — forgetting is the Section 12.4 failure. Second, production uses absolute-seashell thresholds, not z-scores ($T^{B1} = 80$, $T^{B2} = 50$, $A = 90$ on lines 59 and 65); $z_{\text{entry}} = 0.7$ above is the order-of-magnitude equivalent divided by stationary std. Third, the returned label must be translated by the caller into `Order` objects (typically sell the basket's bid wall sized to the position-limit remaining; if half-hedge Section 12.4 is on, simultaneously buy $\frac{1}{2}w_i$ of each constituent at ask).

12.7 Key facts to memorise

- **Synthetic value.** Basket B, weights $(w_1, \ldots, w_n)$ on $C_1, \ldots, C_n$: synthetic value $B^{\text{synth}} = \sum_i w_i C_i$, basis $\text{Basis}_t = B_t - B_t^{\text{synth}}$. A linear combination, nothing more.

- **Creation/redemption is the floor.** On a liquid AP-served ETF the basis is bounded by round-trip fee plus constituent liquidity costs — single-digit bps on SPY. One of the safest stat-arb variants because the floor is hard.

- **No AP $\Rightarrow$ no floor.** A basket without creation/redemption (proprietary indices, all Prosperity baskets) has no hard floor: pure cointegration play with the LTCM-box caveats of Chapter 11.

- **Cointegration vector is declared, not fitted.** With published weights, $\boldsymbol{\beta} = (1, -w_1, \ldots, -w_n)$ is known: no Engle–Granger/Johansen. Walk-forward refit applies to running mean and variance, not to the slope.

- **Static vs dynamic hedge.** Static = published weights (AP or simulator enforces). Dynamic = fitted weights (loose coupling, drifting effective ratio).

- **Trading rule.** Short basis at $z_t > +z_{\text{entry}}$, long at $z_t < -z_{\text{entry}}$, close at $|z_t| < z_{\text{exit}}$ — same OU z-score machinery as Chapter 10.

- **Frankfurt PICNIC compositions.** `PICNIC_BASKET1` = 6 `CROISSANTS` +3 `JAMS` +1 `DJEMBE`; `PICNIC_BASKET2` = 4 `CROISSANTS` +2 `JAMS`. `DJEMBES` only in basket 1. Source: `ETF_CONSTITUENT_FACTORS = [[6, 3, 1], [4, 2, 0]]`, line 57 of `FrankfurtHedgehogs_polished.py`.

- **Frankfurt thresholds.** `BASKET_THRESHOLDS = [80, 50]` (raw seashells) and Olivia adjustment `ETF_THR_INFORMED_ADJS = [90, 90]`; both thresholds shift by $d_t \cdot A$.

- **Premium drift.** Basis mean is non-zero and slowly drifting. Fix: online running-mean update (12.12) before thresholding; else the rule is systematically biased.

- **Olivia cross-product.** The same bot drives `CROISSANTS` and the basket basis. Shifting the band by $\pm A$ was worth a material fraction of basket PnL.
- **Headline PnL.** 40k–60k seashells/round on baskets, plus 20k/round on `CROISSANTS` alone.
- **Round 5 half-hedge.** Convex combination of unhedged and fully-hedged strategies: linear in return, sub-linear in variance via (12.13). Kelly-style trade-off (Chapter 20).
- **Avellaneda–Lee.** Basket basis is the one-factor special case of PCA stat-arb: with k PCA factors removed on the constituent return panel, every stock has its own residual basis and trades on the same OU z-score rule (Chapter 22).

Chapter 13

Momentum and Trend Following

Every chapter so far has bet *against* the last move. The Roll estimator (Chapter 5) infers a bid–ask from negative autocovariance of trade-to-trade changes; the Glosten–Milgrom posterior (Chapter 6) is a martingale converging toward fair value as informed and uninformed flow average out; the OU machinery (Chapter 10) trades the standardised deviation of a stationary process back to its long-run mean; the pairs and basket spreads (Chapters 11 and 12) are cointegrating residuals, traded with the same z-score rule. Five chapters, one instinct: when a price has moved, lean against the move.

This chapter takes the opposite bet: the last move will *continue*. The signal is the raw past return itself, used as a forecast of the next return with a positive coefficient. Mean reversion and momentum are statements about different time scales. On microsecond horizons, trends are absorbed by market-maker quoting within milliseconds; on monthly horizons, the opposite is empirically true. Time series momentum is one of the most robust anomalies in modern asset pricing, documented across 58 liquid futures contracts in four asset classes over two decades by Moskowitz, Ooi, and Pedersen (Moskowitz et al., 2012), and its canonical formulation fits on a single line. The chapter's job is to make the coexistence of momentum and mean reversion precise: when each, and what changes between them.

Prerequisites

From earlier in the book:

- Chapter 10: OU and the mean-reversion worldview. Same machinery, sign of the coefficient flipped. A return process can have one statistical signature on one horizon and the opposite on another.
- Chapter 4: the distinction between the efficient price (a martingale) and a deviation from slow fair value. Momentum is a claim about the efficient price itself, not about a spread.
- Chapter 3: who trades and why. TSMOM's behavioural explanation rests on slow-moving allocators (pension funds, central banks) whose flow is predictable on monthly horizons but invisible at the trade level.
- Chapter 8: Hawkes arrival clustering as the microsecond version of trend persistence. "Momentum" applies to the millisecond and the month, but the mechanism, the signal, and the trade differ.
- Elementary statistics: standard deviation, annualisation ($\sigma_{\text{ann}} = \sigma_{\text{monthly}}\sqrt{12}$), the t-statistic, sign function.

13.1 Time series momentum (TSMOM)

Time series momentum's signal is the simplest non-trivial function of past prices: the *sign* of the past twelve-month excess return. Up over the past year: go long. Down: go short. Flat: do nothing. The position is sized inversely to recent realised volatility, so a volatile market gets a small position and a calm one a large position; every contract contributes the same ex-ante risk. That is the entire strategy.

Definition 13.1 (Time series momentum signal)**.** Let $r_{t-12m,\,t}$ denote the excess return of a tradeable asset over the twelve months ending at time t. The *time series momentum signal* at time t is

$$\text{signal}_t \;:=\; \text{sign}\big(r_{t-12m,\,t}\big) \;\in\; \{-1,\,0,\,+1\}, \tag{13.1}$$

where $\text{sign}(x) = +1$ for $x > 0$, -1 for $x < 0$, and 0 for $x = 0$.

The signal is binary in direction; the position is not. To make positions comparable across assets, scale notional inversely to a recent realised volatility estimate $\hat{\sigma}_t$, targeting an annualised volatility σ_{target}. (Moskowitz et al., 2012) use $\sigma_{\text{target}} = 40\%$ annualised per instrument — the level at which a forty-instrument diversified portfolio with realistic cross-instrument correlations reaches an aggregate target of about 12% at the strategy level. It is an aggregate-vol decision projected back onto per-instrument level given diversification. Their volatility estimator is an exponentially weighted average of squared daily log-returns with centre-of-mass about sixty days; any recent realised-vol estimate will do. The worked example below uses an equally-weighted estimator; treat the choice as a tuning knob, not a load-bearing assumption.

TSMOM position-sizing formula

The time series momentum position in a single instrument at time t, in units of notional exposure (positive for long, negative for short), is

$$w_t \;=\; \text{sign}\big(r_{t-12m,\,t}\big)\cdot\frac{\sigma_{\text{target}}}{\hat{\sigma}_t}, \qquad \sigma_{\text{target}} = 0.40 \text{ (annualised)}. \tag{13.2}$$

The signal contributes the direction; the volatility ratio contributes the magnitude. A volatile asset ($\hat{\sigma}_t$ large) gets a small position; a calm asset gets a large position. The position is rebalanced monthly as both the signal and the volatility estimate update.

First, what the formula does *not* say. Not "buy the asset that went up the most," and not a cross-asset ranking. The signal for gold depends only on gold's own past return; an asset can be TSMOM-long even if it underperformed every other asset in the universe. Each instrument's signal is self-contained — the contrast with cross-sectional momentum is the next section's subject.

Second, what volatility scaling buys. Without it, one unit of S&P futures and one unit of two-year Treasury futures contribute wildly different risk — the equity index is 5 to 10 times more volatile than the short bond. A naive "one unit per long-signal asset" aggregate would be dominated by the most volatile contracts, and the portfolio would behave as its equity sleeve. Inverse-vol scaling forces every asset to contribute equal ex-ante risk, so diversification works *across* asset classes.

A worked numerical example: gold futures

Take gold futures at month-end. Past twelve-month return is +15%; recent realised volatility is $\hat{\sigma}_t = 12\%$ annualised. Plug into (13.2):

$$w_t = \underbrace{\mathrm{sign}(+0.15)}_{=+1} \cdot \underbrace{\frac{0.40}{0.12}}_{=3.333\ldots} = +3.33.$$

The position is +3.33 units of notional — 3.33 times the dollar size of one "baseline" unit. The multiplier exceeds one because gold's 12% volatility is below the 40% target, so the strategy levers up by $0.40/0.12 \approx 3.333$ to reach target ex-ante volatility per instrument. At 40% realised vol the multiplier would be 1; at 80%, 0.5. A -15% twelve-month return would flip the sign, giving -3.33 (short).

Intuition

Read TSMOM as "which way did the slow money push, and how confident are we given how noisy the asset is?" The sign of the past return is the direction slow allocators have collectively pushed over the past year. Vol-scaling is confidence weighting: when recent moves are small relative to the trend, the trend stands out above noise and the bet is bigger; otherwise smaller. Behavioural story: when a large institution decides to overweight a sector — say, a sovereign wealth fund adding two percentage points of NAV to commodities — it does not buy in an hour. It buys over weeks or months, for impact (Chapter 7) and governance reasons (approval, cash clearing, benchmark updates). That flow pushes prices in one direction over a long horizon; TSMOM rides the wave.

The empirical result

TSMOM earns its place here on robustness. (Moskowitz et al., 2012) study 58 liquid futures contracts across four asset classes — 24 commodity, 9 equity-index, 13 bond, 12 currency forwards — from January 1985 to December 2009. Their pooled regression of next-month vol-scaled return on the sign of the past twelve-month vol-scaled return,

$$\frac{r_t^s}{\sigma_{t-1}^s} = \alpha + \beta_h\, \mathrm{sign}(r_{t-h}^s) + \varepsilon_t^s,$$

yields a positive, statistically significant β_h for $h = 1, 2, \ldots, 12$ months pooled, and within each asset class individually. The t-statistic on β_{12} in the all-asset pooled regression exceeds 5 with month-clustered standard errors.

The diversified TSMOM strategy, equal-vol-weighting all 58 contracts, delivers a gross Sharpe around 1.5 and monthly alpha of 1.58% ($t \approx 8.0$) against the Fama–French three-factor model augmented with the cross-sectional momentum factor *UMD* of Jegadeesh and Titman (1993). The alpha shrinks to 1.09% ($t \approx 5.4$) with added controls for bond, commodity, and currency cross-sectional momentum, but survives. TSMOM is not a relabelling of cross-sectional momentum; it captures a distinct anomaly. The next section makes the distinction precise.

13.2 Cross-sectional vs time-series momentum

The momentum literature has two parents; do not confuse them. Jegadeesh and Titman (1993) introduced *cross-sectional* momentum (XSMOM) on US equities: rank stocks by past-window return (three to twelve months), go long the top decile and short the bottom decile, hold for a similar window. The signal is a *rank*. The strategy is dollar-neutral and asset-class-neutral by construction. It works in equities, with variants in country indices, currencies, and commodities.

Concretely, the two ask different questions. TSMOM asks "has *this* asset gone up over the past year?" — if yes, buy it; if no, short it. XSMOM asks "has this asset gone up *more than its peers*?" — buy the winners, short the laggards, regardless of absolute direction. In a year when every stock is up 20%, TSMOM is long everything; XSMOM is long the top-decile +40% names and short the bottom-decile +5% names for zero net exposure.

The younger parent is (Moskowitz et al., 2012)(2012) time series momentum. The signal is each asset's own past return, not its rank: the strategy can simultaneously be long every asset (if all 58 are up on the year) or short every asset (if all are down). There is no mechanical dollar- or asset-class-neutrality. Inverse-vol scaling equalises risk across instruments, but net market exposure is whatever the signs collectively dictate.

Two kinds of momentum

	Cross-sectional (XSMOM)	**Time series (TSMOM)**
Signal	$\text{rank}(r^s_{t-12,t})$ within universe	$\text{sign}(r^s_{t-12,t})$ of own return
Positions	Long top decile, short bottom decile (dollar neutral)	Long if own past return > 0, short if < 0 (any net)
Net exposure	Zero by construction	Time-varying; can be net long, net short, or zero
Canonical paper	Jegadeesh–Titman (1993) on US equities	Moskowitz–Ooi–Pedersen (2012) on 58 futures
Universe	Often single-asset-class (e.g. equities)	Multi-asset-class futures
Captured by	Fama–French *UMD* factor	*Not* fully captured by *UMD*; has additional alpha

The two are positively correlated — if equities rally across the board, both XSMOM and TSMOM tend to be long equities — but not identical. (Moskowitz et al., 2012) regress TSMOM returns on *UMD* and find significant positive loading with residual alpha around 1.09% per month ($t > 5$). Even after subtracting what cross-sectional momentum explains, TSMOM still earns money; time-series direction adds incremental information.

Three operational consequences. First, Prosperity's typical universe is a handful of products with no sensible cross-section, so only the time-series formulation translates. Second, a Goldman macro desk's trend pod is multi-asset (currencies, rates, equities, commodities); even with XSMOM-style ranking within an asset class, the aggregate portfolio combines both kinds of momentum. Third, the two effects are independent, so blending yields a small Sharpe boost over either alone.

13.3 Why momentum works on monthly horizons but not microseconds

Five chapters have argued mean reversion; this one argues the opposite. Both cannot hold on the same time scale for the same object, so what is going on?

They are true on *different* time scales, with a different mechanism dominating each. Time scale, not asset class, is the key axis. From fastest to slowest:

Microsecond horizon (10^{-6} to 10^{-3} seconds). A market-making algorithm sees an aggressive buy, infers a small adverse-selection signal, widens its ask and lifts its bid (Chapter 9, Avellaneda–Stoikov skew). The next aggressive buy meets a higher quote; within milliseconds the quoting algorithm reverts price toward fair value. Market-makers *eat* momentum on this horizon: any short-burst trend in trade prices is absorbed into the spread within milliseconds. The Roll spread of Chapter 5 is this effect viewed through autocovariance: $\text{Cov}(\Delta p_t, \Delta p_{t+1}) = -s^2/4 < 0$; trade-level returns are *negatively* autocorrelated.

Sub-second to one-minute horizon (10^{-3} to 60 seconds). Mixed regime. Hawkes-style arrival clustering (Chapter 8) creates positive autocorrelation in *order flow*, and imbalance predicts the next-second mid move with a small positive coefficient. It is the closest thing to "momentum" in the millisecond–second band, but the prediction is about *order flow* continuing, not trends. The signal decays within seconds and is invisible on monthly bars.

Minutes to hours. Genuinely mixed. Some markets trend because execution algorithms break parent orders into child orders (the Chapter 19 machinery spreads trades over an hour, generating predictable drift in the execution window); others mean-revert under contrarian market-maker inventory pressure. Tradeability depends on liquidity profile and participant mix.

Days to weeks. Mixed but momentum-leaning for liquid macro futures. The MOP regressions give positive coefficients down to one-month lookbacks; Chan (Chan, 2013) (Ch 6) reports practitioner experience with 1–5 day momentum on FX carry pairs and commodity futures. Equity pairs and ETF baskets mean-revert on the same horizon because the dominant force is cointegration convergence. Chapter 23 diagnoses which regime a product is in; the rule of thumb is that single instruments with macro drivers trend, and spreads of related instruments revert.

One to twelve months. Strongly momentum-favouring across all four asset classes (Moskowitz et al., 2012). Three non-exclusive explanations coexist in the literature: (i) *underreaction* to gradual fundamental news — a slow inflation trend, a multi-quarter earnings-revision drift — means prices move in steps rather than jumps; (ii) *investor herding* and delegated-manager career concerns draw late capital toward already-moving assets; (iii) *slow capital allocation* (sovereign funds, pension plans, insurance asset-liability desks) moves on monthly-to-quarterly cycles, and once a flow direction is established it persists because allocator decisions are rebalanced infrequently. All three produce the same statistical signature — positive autocorrelation of monthly returns — and all three decay: the edge is strong over weeks to months but flips sign beyond about a year, as the next paragraph shows.

Beyond one year. The MOP regressions show *negative* coefficients beyond about thirteen months — long-horizon mean reversion that DeBondt and Thaler documented for equities in the 1980s and that has been reproduced for commodities and currencies. The five-year reversal reads as slow correction of multi-year mispricing, as opposed to slow propagation of new information driving 1–12 month momentum.

Don't run monthly momentum on a five-minute bar

The (Moskowitz et al., 2012) edge is about *monthly* horizons on *liquid macro futures*. It does not translate to higher-frequency data without substantial modification:

- At a five-minute lookback the sign-of-past-return signal is dominated by microstructure noise. Roll and Glosten–Milgrom both imply short-horizon *negative* autocorrelation at the trade level. Five-minute TSMOM buys every uptick, sells every downtick, pays spread on every flip, and loses to market-makers.

- The 40% annualised target was calibrated for monthly rebalancing on contracts with single-digit monthly volatilities. Five-minute realised vols give nonsense leverage: the five-minute-to-annualised conversion factor is an order of magnitude larger than the monthly one ($\sqrt{78 \cdot 252} \approx 140$ on a 6.5-hour equity session, $\sqrt{288 \cdot 252} \approx 269$ round-the-clock). Comparing scaled five-minute dispersion to $\sigma_{\text{target}} = 0.40$ gives ratios unrelated to the investment-horizon risk the formula controls.

- The behavioural mechanism — slow allocator flow — has no analogue at the second or minute horizon. High-frequency trend persistence comes from order-flow clustering (Chapter 8) or parent-order execution drift (Chapter 19), not allocator behaviour.

Rule: pick the model whose mechanism matches the horizon. Hawkes for milliseconds, Roll for trade-level, OU for spreads on hours and days, TSMOM for macro futures on weeks and months, five-year reversal beyond a year.

13.4 Momentum crashes

TSMOM works on average over long samples, but its return distribution has a heavy left tail. It is occasionally the worst-positioned strategy in the world, exactly at regime turns; the 2009 crash is the textbook case.

The mechanism is mechanical. Momentum is long what has gone up, short what has gone down. After a sustained bear market, the longs concentrate in defensives (bonds, gold, defensive sectors) and the shorts in cyclicals (small caps, banks, materials). When the market bottoms and reverses violently — as in March 2009 — the cyclical shorts rally hardest and the defensive longs lag. The portfolio is short the rally and long the laggards; both legs lose.

Momentum is *short gamma* with respect to regime turns. A gamma-positive position (a long option) profits from large moves in either direction; gamma-negative loses. Momentum is gamma-negative because it bets the recent direction continues: any large reversal moves the underlying against the position and forces a rebalance — both losses. Daniel and Moskowitz (2016) formalise this as "momentum is short a call option on the market when the market has been down, and short a put option when it has been up".

Historical note

The March 2009 momentum crash is the canonical failure mode. The S&P 500 bottomed at 666 on 6 March 2009 after a 57% peak-to-trough drop from the October 2007 high; by end-April it had rallied about 30%. Cross-sectional momentum in US equities — short banks and cyclicals, long defensives and gold — lost 25–30% of NAV in March and April alone. AQR Capital's flagship momentum fund (Cliff Asness) took a comparable drawdown. This single episode explains a meaningful fraction of post-crisis underperformance of academic momentum factors. Cross-asset TSMOM fared better than equity-only XSMOM that quarter — bond and currency sleeves diversified — but still recorded one of its worst quarters in the MOP sample. Live trend-following CTAs reported single-month drawdowns of 5–15% and several prominent funds closed. The industry lesson: *trend-following needs a tail-risk overlay* — stop-outs, cross-asset diversification, and (at the largest funds) options-based protection against violent reversals.

Practical mitigations fall into three buckets. *Cross-asset diversification* is cheapest: bond and currency legs do not crash on the same days as equities, so a diversified 58-instrument portfolio has shallower drawdowns. *Strategy-level vol targeting* (added on top of per-instrument targeting in the position formula) cuts exposure when realised vol spikes, which approximates cutting exposure near regime turns. *Explicit tail hedges* using OTM options on major equity indices are expensive in calm periods but pay off precisely when momentum crashes; the largest trend funds run a permanent options overlay. Option mechanics in Chapters 15 and 17; Kelly sizing of tail-protected portfolios in Chapter 20.

Prosperity example

The cleanest Prosperity TSMOM application is `SQUID_INK` in Round 1. The product was officially mean-reverting short-term, but the Frankfurt Hedgehogs' inspection (`imc-prosperity-3/README.md`, Round 1) found an anonymous bot, later identified as Olivia, consistently buying 15 lots at the daily low and selling 15 at the daily high. Whenever a trade printed at a daily extreme in a direction consistent with Olivia's pattern, the Hedgehogs flagged it as a directional signal and held to the opposing extreme. Their `InkTrader` class (`FrankfurtHedgehogs_polished.py`, lines 382–405) implements this: when `informed_direction == LONG`, set `expected_position = position_limit` and bid the ask wall toward the cap; symmetrically for SHORT.

A discrete cousin of TSMOM. The signal is not the sign of the past twelve-month return (twelve months exceeds a Prosperity round) but the sign of the inferred informed-trader direction — a proxy for "which way is the slow money pushing". Position sizing is hard-clipped to the limit rather than vol-scaled, a degenerate form of the same "size-to-target-risk" instinct. Once the trader is identified directly via trade ID in Round 5, false positives drop to near zero and the strategy becomes the dominant SQUID_INK PnL contributor. A pure TSMOM with a five-tick lookback without the Olivia overlay yields only a modest edge — noise-to-signal at five ticks is too high and the product only marginally trending — but combined with the informed-trader detector the same time-series-direction instinct produced a clean, high-Sharpe contributor that depended on detecting slow capital flow, not predicting it. The overall pattern matches this chapter's trend-following theme: small but consistent, with upside from disciplined risk control rather than clever signal engineering.

On a Goldman Sachs desk

On a Goldman systematic macro desk, TSMOM is one ingredient in a "trend pod" running on 50–100 liquid futures and forwards (G10 FX, on-the-run government bonds, major equity indices, energy and metals). The Strat owns the signal pipeline: return-windowing on adjusted-rolled futures series, the vol estimator (EWMA with a 60-day half-life), the 40% target scaling, and the monthly rebalance schedule. The desk trader owns execution, which on a monthly rebalance with hundreds of positions is a small Almgren–Chriss problem (Chapter 19) per name — the Strat estimates the target, the trader spreads the trade across hours to manage impact, and post-trade attribution reconciles realised against model entry. Risk is monitored separately: strategy-level VaR, gross and net leverage, pairwise correlation with peer CTAs (Man AHL, Winton, Aspect, Lynx). When all trend funds are positioned identically, the risk team imposes a temporary gross cap against crowded-trade unwinds. Drawdowns are correlated across managers, so the primary defence is not within-strategy diversification (which trend cannot supply through a regime turn) but position limits and options overlays on the equity-index sleeve, sized via the gamma-and-vega framework of Chapter 17.

13.5 A short reference implementation

The TSMOM signal fits in fewer than twenty lines of Python. The function below takes a 1-D array of historical prices (most recent last), the momentum lookback in months, the vol lookback, and the annualised target, and returns a signed notional position. No state, no portfolio code, no backtester — those belong in Chapter 21. The point is to make (13.2) unambiguous and reproducible.

```python
import numpy as np
def tsmom_signal(prices: np.ndarray,
                 lookback_months: int = 12,
                 vol_lookback: int = 60,
                 vol_target: float = 0.40) -> float:
    """Time series momentum signal with vol scaling.
    prices: array of monthly prices, most recent last.
    Returns position size in units of notional, signed.
    """
    if len(prices) < max(lookback_months + 1, vol_lookback + 1):
        return 0.0
    past_ret = prices[-1] / prices[-lookback_months - 1] - 1.0
    sign = 1.0 if past_ret > 0 else (-1.0 if past_ret < 0 else 0.0)
    log_rets = np.diff(np.log(prices[-vol_lookback - 1:]))
    vol_ann = log_rets.std(ddof=1) * np.sqrt(12.0)
    if vol_ann < 1e-8:
        return 0.0
    return sign * (vol_target / vol_ann)
```

Listing 13.1: Time series momentum signal with vol scaling.

Three production notes not encoded in the snippet. First, the vol estimator is an equally-weighted standard deviation of the last `vol_lookback` monthly log-returns; MOP use an EWMA with 60-day-equivalent centre of mass computed on *daily* returns and annualised. EWMA reacts faster to vol regime shifts, but the strategy is qualitatively robust. Second, the function is stateless and assumes a clean monthly price series; in practice the trader handles futures roll adjustments, missing data, and timing conventions. Third, the returned position is notional in an unspecified unit; the caller translates "notional +3.33" into contracts given the multiplier and current price. None of these production concerns changes the core signal of (13.2).

13.6 Key facts to memorise

- **TSMOM signal.** The sign of the past twelve-month excess return predicts the next-month return with a positive coefficient. Documented across 58 liquid futures in four asset classes over 1985–2009 by (Moskowitz et al., 2012).

- **Position formula.** $w_t = \text{sign}(r_{t-12m,t}) \cdot (\sigma_{\text{target}}/\hat{\sigma}_t)$ with $\sigma_{\text{target}} = 40\%$ annualised per instrument. Direction from sign; magnitude from inverse-vol.

- **Vol-scaling rationale.** Equalises ex-ante risk across instruments so a diversified portfolio is not dominated by the most volatile asset.

- **XSMOM vs TSMOM.** XSMOM ranks within a universe (dollar- and asset-class-neutral). TSMOM judges each asset against its own history (time-varying net exposure). Positively correlated but not identical; TSMOM retains alpha after controlling for *UMD*.

- **Horizon beats asset class.** Momentum monthly, mean reversion at milliseconds. Behavioural underreaction and slow capital drive monthly momentum; adverse selection and Glosten–Milgrom convergence drive trade-level reversion. Match model to horizon.

- **Crash risk.** Momentum is short gamma with respect to regime turns: a violent reversal loses on both legs. March 2009: equity momentum lost 25–30% in two months as cyclical shorts rallied and defensive longs lagged.
- **Mitigations.** Cross-asset diversification (cheapest), strategy-level vol targeting, and options-based tail hedges (most expensive, most effective in a crash).
- **Operational rule.** The MOP edge is monthly bars on liquid macro futures. It does *not* translate to five-minute bars without reformulation; intraday "trend" machinery is order-flow clustering (Chapter 8), not TSMOM.

Chapter 14

The Arbitrage Zoo

This chapter tours the zoo of textbook arbitrages: the strategies a quant looks for first when evaluating a new market. Each follows the template *observable*, *trade*, *P&L*, *why it usually doesn't exist*, *when it does*. The goal is to fix that template so firmly that, given any market data dump, the right diagnostic question fires automatically: when you see this shape in a market, set up this trade.

Seven arbitrages are presented in the same fixed format, two to three pages each. Once the template is internalised, encountering an unfamiliar instrument — a new commodity future, a token on an exotic crypto venue, a thinly-traded ETF — becomes a matter of ticking through the seven types to see which, if any, is open. An eighth section closes the chapter by stating that statistical arbitrage (Chapter 10, Chapter 11) is *not* a member of the zoo: it shares the name but not the no-risk guarantee, and conflating the two has wiped out more firms than the other six combined.

Prerequisites

From earlier:

- Chapter 2: the limit order book, the bid–ask spread, and *walking the book*. Several arbitrages here are nothing more than walking two books simultaneously, one as buyer and one as seller.
- Chapter 12: the ETF creation / redemption mechanism and authorised participants. The ETF c/r entry refers back rather than re-deriving.
- Forward: Chapter 15 for European call/put payoffs. The put-call parity entry uses only the payoff diagrams; readers unfamiliar with C, P, S, K should skim that section's opener first.
- Discounting: a riskless cashflow X at time T is worth Xe^{-rT} today, with r the risk-free rate. No stochastic calculus is used; every payoff is deterministic by construction.

No prior finance vocabulary is assumed beyond Chapter 2 through Chapter 13.

14.1 The arbitrage template

We fix the template first. The seven case studies that follow are instances of the same five-step pattern.

Definition 14.1 (Arbitrage)**.** An *arbitrage* is a trading strategy that, executed at market prices, has zero or negative initial cost, non-negative payoff in every state of the world, and strictly positive payoff in at least one state. Equivalently: a portfolio that costs nothing (or pays you to hold it) and cannot lose money. The Type I / Type II split in the technical literature does not matter operationally here.

"In every state of the world" does all the work. A strategy that wins on *average* but can lose is not an arbitrage; it is at best a positive-EV bet, requiring the sizing and risk apparatus of Chapter 20. A true arbitrage is a free lunch, and the central fact of mature markets is that free lunches are rare and short-lived. This chapter is about recognising the few that exist.

Every arbitrage in the zoo follows the same five-step pattern.

The arbitrage template

For any candidate arbitrage, answer the following five questions in order. If any answer is unsatisfactory, the arbitrage is not open and you move on.

1. **Observable**: What do you see in market data that tells you the arbitrage is open? This is the trigger condition — a numerical inequality between observable prices.
2. **Trade**: What is the explicit set of orders that captures the discrepancy? Specify each leg, on which venue, in which direction, in what size.
3. **P&L**: What is the *deterministic* profit in the ideal case — frictionless, fully filled, instant execution? This is a number, not a distribution.
4. **Why it usually doesn't exist**: Which forces in a normal, healthy market keep the inequality from opening? Knowing this is what tells you when the arb is real and when it is a data artefact or a stale quote.
5. **When it does**: What specific conditions temporarily disable those forces and let the arbitrage open? This is the catalogue entry — the situations in which to watch.

Intuition

The template does two jobs. Steps 1–3 are the trade: see the gap, do the orders, count the money. Steps 4–5 are the *theory*: an arbitrage is interesting only insofar as it is rare, because anything common is already arbed out by someone faster. Winning relative-value desks know the closures and openings of each arbitrage so intimately that they are positioned for the open before it is visible to everyone else.

In steady state every arbitrage here is closed, because if it were not, someone would close it. The interesting question is not "how do I capture this arb?" (mechanical once you know it is open) but "what would have to break for this arb to open, and how would I know?" Each entry answers both.

14.2 Triangular FX arbitrage

Currency pairs are quoted bilaterally — EUR/USD, USD/JPY, EUR/JPY — and the three prices are linked by a consistency requirement: in a frictionless market, walking the triangle EUR $\to$ USD $\to$ JPY $\to$ EUR returns exactly one EUR. A round trip above one is an arbitrage; below one, the reverse direction is.

Observable. Three FX rates whose product around a triangle differs from one. Concretely, denote the price of one unit of currency X in units of currency Y by $E_{X/Y}$. The arbitrage indicator is

$$\Delta \;:=\; E_{\mathrm{EUR/USD}} \cdot E_{\mathrm{USD/JPY}} \cdot E_{\mathrm{JPY/EUR}} \;-\; 1. \tag{14.1}$$

If $\Delta > 0$, the EUR $\to$ USD $\to$ JPY $\to$ EUR direction is profitable; if $\Delta < 0$, the reverse direction is. If $\Delta = 0$ the triangle is consistent and no arbitrage exists.

Trade. Starting with one EUR:

1. Sell EUR for USD at rate $E_{\mathrm{EUR/USD}}$, receiving $E_{\mathrm{EUR/USD}}$ USD.
2. Sell USD for JPY at rate $E_{\mathrm{USD/JPY}}$, receiving $E_{\mathrm{EUR/USD}} \cdot E_{\mathrm{USD/JPY}}$ JPY.
3. Sell JPY for EUR at rate $E_{\mathrm{JPY/EUR}}$, receiving the round-trip total $E_{\mathrm{EUR/USD}} \cdot E_{\mathrm{USD/JPY}} \cdot E_{\mathrm{JPY/EUR}}$ EUR.

The three trades must execute as simultaneously as the matching engines permit, because rates move while legs are pending. Real desks send all three orders in a single market message or use guaranteed-fill order types.

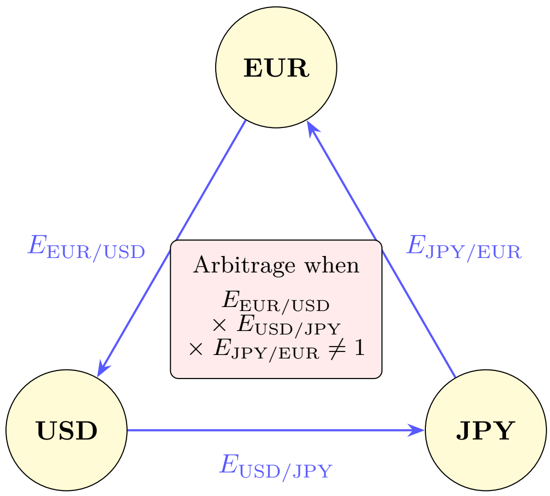


Figure 14.1: Triangular FX arbitrage. Starting with one EUR, convert to USD at $E_{\mathrm{EUR/USD}}$, then to JPY at $E_{\mathrm{USD/JPY}}$, then back to EUR at $E_{\mathrm{JPY/EUR}}$. If the product of the three rates exceeds one, a riskless profit exists. Competitive markets enforce the no-arbitrage constraint that the product equals one. In practice the window is milliseconds wide and requires co-located systems to exploit.

P&L. Per unit of starting EUR, profit $= \Delta$. Locked the moment all three legs fill; no holding period, no subsequent price exposure.

Worked numerical example. Suppose at some instant the three rates are

$$\begin{aligned} E_{\mathrm{EUR/USD}} &= 1.08, \\ E_{\mathrm{USD/JPY}} &= 148, \\ E_{\mathrm{JPY/EUR}} &= 0.0063. \end{aligned}$$

Then the round-trip is

$$\underbrace{1.08}_{\mathrm{EUR}\to\mathrm{USD}} \times \underbrace{148}_{\mathrm{USD}\to\mathrm{JPY}} \times \underbrace{0.0063}_{\mathrm{JPY}\to\mathrm{EUR}} = 1.006992,$$

so $\Delta \approx 0.006992$, about 0.70% per round trip — seven thousand EUR on one million, before fees and the cost of executing fast enough that no leg moves between trades.

The JPY/EUR quote of 0.0063 implies an EUR/JPY of ≈ 158.73. The no-arb inverse of the first two rates is $1/(1.08 \cdot 148) \approx 0.006257$; the quoted 0.0063 exceeds that by exactly what the round trip captures. In general: fix any two rates and solve for the third; any deviation is the locked-in profit.

Why it usually doesn't exist. The major-pair FX market is the most heavily arbitraged in the world. EBS, Reuters, and bank-internal ECNs run continuous triangular consistency monitors, and any basis-point opening closes in microseconds. (Bouchaud et al., 2018) report that on liquid majors the indicator lives almost entirely within the constituent bid–ask spreads — apparent gaps are not capturable after crossing the spread on each leg.

When it does. Three families. (i) *Fragmented liquidity*: a small ECN quoting all three pairs with fewer participants drifts off consensus, and a desk plugged into both the ECN and consensus venues captures the gap. (ii) *News-driven dislocations*: news hitting EUR/USD a millisecond before EUR/JPY leaves the triangle inconsistent for the propagation delay; the FX latency arms race is a race to close that gap. (iii) *Exotic pairs*: EUR → TRY → RUB → EUR is much less competed; the indicator opens further and stays open longer in thinner pairs.

Prosperity example

Round 1's manual challenge in Prosperity 3 was a literal triangular FX exercise: a 4×4 conversion matrix between `Snowballs`, `Pizza`, `Silicon Nuggets`, `SeaShells`, with the task of finding a sequence of at most five trades maximising final `SeaShells`. The Frankfurt Hedgehogs enumerated all sequences and found the optimum `SeaShells` → `Snowballs` → `Silicon Nuggets` → `Pizza` → `Snowballs` → `SeaShells`, yielding $\approx 8.9\%$ per round trip (Frankfurt Hedgehogs, 2025a). This is the brute-force version of triangular arbitrage: enumerate cycles in the currency graph, take the one whose product of edge weights most exceeds one. In the wild the market enumerates continuously and closes cycles within microseconds; in Prosperity the matrix was static for the round, making the "arbitrage" a one-shot optimisation. The pedagogical point survives: this is triangular FX arbitrage in miniature.

14.3 Locational arbitrage

Same trade, but across a pair of *venues* for the same asset rather than around a currency triangle.

Observable. Identical contract, identical settlement, identical quality, different prices on two exchanges. Classical example: BTC at \$50,000 on exchange A and \$50,050 on exchange B. Equity examples include dual-listed stocks on NYSE/Toronto or E-mini contracts on CME vs alternative venues. The observable is $\text{ask}_A < \text{bid}_B$ (crossable in the lift-A, hit-B direction).

Trade. Buy on the cheaper venue, sell on the dearer, in equal size. If the asset is fungible and venues share custody (or the trader holds inventory at both), the position closes immediately at the gross spread. If the asset must be *moved* between venues — as for crypto, where transfer takes minutes to hours and incurs network fees — it is not strictly an arbitrage during transit: price can move and the trader wears the risk.

P&L. Per unit, gross $= \text{bid}_B - \text{ask}_A$. Net subtracts (i) exchange fees on both venues, (ii) transfer cost, (iii) borrow cost if shorting one side, (iv) slippage beyond top of book. In the BTC example gross is \$50 per coin; net is typically a small fraction of gross.

Why it usually doesn't exist. Automated cross-venue bots continuously monitor every pair of liquid venues and close any gap beyond net costs in subsecond time. On BTC/USD between Coinbase and Binance the gap exceeds joint fees for less than a fraction of a percent of trading time, and closes within seconds when it does. The closing *is* the arb; its existence maintains convergence.

When it does. Three families:

- **Regulatory fragmentation.** Pre-2018 Japan vs US crypto: capital controls made moving BTC expensive or illegal, and BTC traded at a 5–10% premium on Japanese venues for sustained periods. The Coinbase / bitFlyer spread was a multi-month feature, because the closing trade was blocked by regulation.
- **Latency asymmetry.** CME E-mini S&P and less-liquid alternative venues briefly dislocate during fast moves: a market-maker quoting both from one feed updates one before the other. Co-located firms capture the gap in a few hundred microseconds.
- **Transient liquidity imbalance.** A large aggressive order drives the local bid–ask off consensus; the asymmetry persists for seconds before cross-venue MMs re-equalise. A bot positioned on both venues lifts the cheap side and hits the dear side during the window.

Prosperity example

Round 4 of Prosperity 3 introduced `MAGNIFICENT_MACARONS`, tradeable on the local order book and via a fixed-fee external "conversion" channel exposing an external bid/ask adjusted for import/export and transport fees (Frankfurt Hedgehogs, 2025a). Whenever local bid exceeded external ask (after fees), or local ask fell below external bid, a profitable conversion was available — textbook locational arbitrage with the local book as one venue and the external channel as the other. The Frankfurt Hedgehogs report that over 10,000 timesteps with a 10-unit per-tick conversion cap, the trade could theoretically yield up to 300,000 SeaShells per round, with realistic captures of 130,000–160,000. Their implementation placed limit sells on the local book at $\lfloor \text{externalBid} + 0.5 \rfloor$ each tick — the maximum price that still attracted local taker bots — and converted resulting inventory through the external channel. Locational arbitrage can be a stable, high-throughput source of profit when the closing mechanism is rate-limited.

14.4 Cash-and-carry arbitrage

Cash-and-carry links spot and futures on the same asset. Its no-arbitrage relation $F = Se^{(r-y)(T-t)}$ is one of the half-dozen most fundamental formulas in derivative pricing.

The cost-of-carry relation, derived from no-arbitrage

Consider a non-dividend-paying asset with spot S, risk-free rate r (borrow = lend), and a futures contract with delivery T and price F. We derive F by exhibiting a replicating portfolio.

Portfolio A: at $t = 0$, borrow S in cash, buy one unit at spot, and enter a short futures with delivery price F. Setup is cashless (borrowed cash pays for the asset; the short futures requires no upfront premium beyond margin, treated as zero-cost). At time T:

- Deliver into the short futures for F cash.
- Repay the loan, grown to Se^{rT} under continuous compounding.

The terminal cashflow $F - Se^{rT}$ is deterministic (both legs locked at $t = 0$), so no-arbitrage forces it to zero:

$$F = S\,e^{rT}, \tag{14.2}$$

which is the no-dividend, no-storage-cost version of the cost-of-carry formula.

If $F > Se^{rT}$, Portfolio A delivers positive cashflow at zero cost: borrow cash, buy spot, short futures, deliver, pocket the difference (*cash-and-carry arbitrage*). If $F < Se^{rT}$, the reverse trade — short spot, lend cash, long futures, take delivery — is the arbitrage (*reverse cash-and-carry*).

Adding a yield. If the asset pays a continuous yield at rate y (dividend yield for equities, coupons for bonds, convenience yield for commodities useful to hold), Portfolio A's holding leg receives extra $S(e^{yT} - 1)$. Re-running the argument:

$$F \;=\; S\, e^{(r-y)T}, \tag{14.3}$$

which is the general cost-of-carry forward formula.

Cost-of-carry forward formula

For an asset with spot price S, risk-free rate r, and continuous yield y (dividend yield for equities, convenience yield net of storage for commodities), the no-arbitrage forward / futures price for delivery at time T is

$$F \;=\; S\, e^{(r-y)T}. \tag{14.4}$$

A *cash-and-carry arbitrage* exists when the market forward F_{market} differs from this fair value:

- $F_{\text{market}} > Se^{(r-y)T}$: long spot, short futures, finance at r, deliver at T. Locks in $F_{\text{market}} - Se^{(r-y)T}$ per unit.
- $F_{\text{market}} < Se^{(r-y)T}$: short spot, long futures, lend at r, take delivery at T. Locks in $Se^{(r-y)T} - F_{\text{market}}$ per unit.

Worked numerical example

Consider an equity index with spot $S = 100$, risk-free rate $r = 5\%$, dividend yield $y = 1\%$, and a futures contract expiring in $T = 0.5$ years. The fair forward is

$$F \;=\; 100 \cdot e^{(0.05-0.01)\cdot 0.5} \;=\; 100 \cdot e^{0.02} \;=\; 100 \cdot 1.02020134\ldots \;\approx\; 102.02.$$

Suppose the market quotes the futures at $F_{\text{market}} = 103$. Then the cash-and-carry arbitrage is open at

$$103 - 102.02 \;\approx\; 0.98 \text{ per unit.}$$

At $t = 0$: borrow 100 at 5%, buy one index unit, sell one futures at 103. At $T = 0.5$: receive dividends $100(e^{0.005} - 1) \approx 0.5013$ (continuous compounding), deliver for 103, repay loan $100 \cdot e^{0.025} \approx 102.5316$. Net at T:

$$\underbrace{103}_{\text{delivery}} + \underbrace{0.5013}_{\text{dividend}} - \underbrace{102.5316}_{\text{loan repayment}} \;=\; 0.9697 \text{ per unit,}$$

matching the ~ 0.98 headline up to compounding rounding. The arb locks the moment loan, spot, and futures legs are in place.

Why it usually doesn't exist. Index-futures-vs-spot bases are watched continuously by every relative-value desk. r is well-known, y is forecast from public dividend schedules, and $F_{\text{market}} - Se^{(r-y)T}$ typically lives below joint round-trip cost. The basis is the cleanest and most heavily competed trade in equities.

When it does. Cash-and-carry opens when one formula input moves faster than the market can price it in:

- **Dividend-date dislocations.** On a large ex-div morning the cash market drops by the dividend; if futures lags, the basis opens by a comparable amount.
- **Forced unwinds.** A margin-called leveraged fund dumps spot; spot drops below fair value while futures stays anchored, opening a reverse cash-and-carry.

- **Commodity storage constraints.** For storage- constrained commodities (April 2020 WTI), the convenience yield y turns large and negative, and the futures curve contangoes beyond rates-implied fair value. The arbitrageur cannot find a tank to store physical oil — the "free" part of the free lunch is not free.
- **Funding-rate spikes.** If r jumps (year-end repo squeezes), Se^{rT} jumps with it; if the futures curve does not follow, the basis opens.

Hull (Hull, 2014) (Ch 5) is the standard reference.

14.5 Put-call parity arbitrage

Put-call parity is the no-arbitrage relation between a European call, a European put, and the underlying. It is the only formula here that requires knowing what calls and puts are; we state the payoffs and derive parity directly. Full treatment: Chapter 15.

Setup

A *European call* with strike K and maturity T on underlying S pays $\max(S_T - K, 0)$ at T; its price today is C. A *European put* with the same strike and maturity pays $\max(K - S_T, 0)$ and has price P. The risk-free rate is r. We assume no dividends between today and T; the dividend correction is in the pitfall box below.

Derivation from no-arbitrage

Consider the portfolio: long one call, short one put, lend Ke^{-rT} (equivalently, a zero-coupon bond paying K at T). Cost today:

$$V_0 \;=\; C - P + Ke^{-rT}.$$

At T the portfolio value depends on S_T:

Case 1: $S_T > K$**.** Call pays $S_T - K$; short put expires worthless; bond pays K. Total:

$$V_T \;=\; (S_T - K) + 0 + K \;=\; S_T.$$

Case 2: $S_T \leq K$**.** Call expires worthless; short put owes $K - S_T$; bond pays K. Total:

$$V_T \;=\; 0 - (K - S_T) + K \;=\; S_T.$$

Both cases give $V_T = S_T$, so the portfolio replicates one share and its cost today must equal the spot price:

$$C - P + Ke^{-rT} \;=\; S.$$

Rearranging gives the famous put-call parity identity.

Put-call parity

For European call C and put P with the same strike K and maturity T, on a non-dividend-paying underlying S at risk-free rate r,

$$C - P \;=\; S - Ke^{-rT}. \tag{14.5}$$

The identity holds at every instant before maturity by no-arbitrage alone, with no dynamic model, no distributional assumption — only the absence of free lunches.

Intuition

Picture the payoffs. Long call: zero below K, slopes up at 45° above. Short put: zero above K, slopes down at 45° below. Stacked, they form a single 45° line — a synthetic long forward struck at K. A forward to buy at K is worth $S - Ke^{-rT}$ by cost-of-carry, hence $C - P = S - Ke^{-rT}$. Parity and cost-of-carry are two views of the same no-arbitrage skeleton.

The arbitrage trade

Observable. A market in which C, P, S violate (14.5). Two cases:

- $C - P > S - Ke^{-rT}$: the call–put combination is overvalued relative to the synthetic-forward right-hand side.
- $C - P < S - Ke^{-rT}$: the call–put combination is undervalued.

Trade. Overvalued case (the other is symmetric): sell call, buy put, buy underlying, borrow Ke^{-rT}. Cash inflow at $t = 0$ is

$$+C - P - S + Ke^{-rT} \; = \; (C - P) - (S - Ke^{-rT}),$$

the parity violation, positive by hypothesis. At maturity the four legs cancel by the two-state argument, so the violation is pocketed and nothing is owed later.

P&L. The parity violation, locked at $t = 0$.

Why it usually doesn't exist. Listed options market makers enforce parity continuously: their quoting algorithms compute, per (strike, maturity) pair, the call implied by the put (or vice versa) via parity, and refuse to let quotes drift more than a tick from consistency. A visible parity gap on a liquid name is usually a stale quote.

When it does. Parity opens when the underlying leg becomes non-trivial:

- **Hard-to-borrow underlyings.** For heavily-shorted small-caps or squeezed meme stocks, the short-underlying leg carries real borrow cost, modelled as a positive y in the cost-of-carry relation. Assuming $y = 0$ produces fake violations that are not capturable.
- **Discrete dividends.** A dividend between t and T goes to the underlying holder but not to the synthetic combination; the formula subtracts $PV(\text{div})$. See the pitfall box.
- **Corporate actions.** Mergers, spin-offs, special dividends, splits disrupt the options chain while the exchange recalculates specs; parity can drift during the adjustment.
- **Early-exercise on American options.** Parity holds strictly only for European options; American options give a pair of inequalities, one of the six no-arb relations in (Damodaran, 2012).

Worked numerical example

Take $S = 100$, $K = 100$, $r = 5\%$, $T = 1$ year, and a non-dividend-paying underlying. The right-hand side of parity is

$$S - Ke^{-rT} \; = \; 100 - 100 \cdot e^{-0.05} \; = \; 100 \cdot (1 - e^{-0.05}) \; = \; 100 \cdot 0.04877\ldots \; \approx \; 4.877.$$

So put-call parity says $C - P \approx 4.877$.

Suppose the market quotes $C = 8$, $P = 2$. Then $C - P = 6$ exceeds the parity-implied 4.877 by $6 - 4.877 \approx 1.123$ per pair, which the arbitrageur locks by selling the call, buying the put, buying the stock at 100, and borrowing $100e^{-0.05} \approx 95.123$. Net inflow $t = 0$ is 1.123; at maturity the four legs cancel.

Parity breaks with discrete dividends

$C - P = S - Ke^{-rT}$ assumes no dividends. Real equities pay discrete dividends to whoever holds the share *on the ex-dividend date*, not to the holder of the synthetic call–put combination. Corrected parity:

$$C - P \;=\; S - PV(\text{div}) - Ke^{-rT},$$

where $PV(\text{div})$ is the present value of dividends scheduled between t and T. Forgetting this term is a common scanner failure: every quarter, dividend-paying stocks report a dividend-sized "violation" that is entirely fake. Always strip the dividend stream. The same correction applies to (14.3) for equity indices with non-zero y — index versions use a continuous-yield approximation, single-name equities need the discrete version.

The Damodaran catalogue (Damodaran, 2012) develops six no-arb relations on options — intrinsic-value bounds, strike and maturity monotonicity, and parity — each derived by the same pattern: two-state payoff at expiry, show portfolio payoff non-negative in both states, conclude cost today non-positive. Hull (Hull, 2014) (Ch 5) is the standard textbook treatment.

14.6 Index arbitrage

Index arbitrage is cost-of-carry on an index futures contract, where the underlying is a basket (for the S&P 500, five hundred stocks in declared proportions). The fair forward comes from (14.3) with S the weighted constituent sum and y the weighted-average dividend yield.

Observable. Index futures away from

$$F^*_{\text{index}} \;=\; S_{\text{SPX}} \cdot e^{(r-y)(T-t)},$$

with S_{SPX} the 500-stock synthetic at wall-mid prices, r the overnight rate (fed funds or SOFR), y the index dividend yield.

Trade. If $F_{\text{market}} > F^*_{\text{index}}$: short futures, buy all 500 constituents in index proportions. If $F_{\text{market}} < F^*_{\text{index}}$: reverse. Hold to expiry; deliver (or cover). Dedicated "program trading" or "index arb" desks exist to run this — the largest single trade by volume in equities.

P&L. Per index unit, $F_{\text{market}} - F^*_{\text{index}}$, times contract notional and size.

Why it usually doesn't exist. Index arb is the most heavily competed trade in the world. The basis typically lives inside the joint bid–ask of the futures and the cheapest tracking portfolio. Convergence is so tight that the basis is sometimes read as a real-time benchmark for r: given observable F_{market} and S_{SPX}, (14.3) is inverted for the market's funding cost.

When it does. Index arb opens at moments of constituent-level chaos:

- **Index-rebalance events.** When S&P changes membership on a known date, index funds must track the new composition; the basis can dislocate during the recompute window.
- **Quadruple witching.** Once a quarter, futures, options, and underlyings expire together, triggering massive program-trading flow and occasional dislocations beyond joint transaction cost.
- **Fast-market stress.** During extreme vol (March 2020 COVID, August 2024 Yen unwind), the 500-stock basket does not track futures in lock-step; the basis can blow out by percentage points for tens of minutes.

Connection to basket arbitrage. Mathematically identical to Chapter 12: both compare a synthetic weighted-constituent value to a single wrapper quote (futures or ETF). The structural difference is the closing mechanism — creation/redemption for ETFs, physical delivery at maturity for futures.

14.7 Calendar arbitrage

Calendar arbitrage exploits inconsistent pricing between two contracts on the same underlying with different expiries. The common venue is options, where the implied-vol term structure occasionally inverts by more than the calendar no-arb bound permits.

Observable. Two options on the same underlying, same strike, maturities $T_1 < T_2$. The no-arb bound says the longer-dated option is at least as valuable (it retains option value after the shorter expires): for European calls, $C(T_2) \geq C(T_1)$. A violation is the observable — equivalently, a term-structure inversion where short-dated vol exceeds long-dated vol by more than the bound permits. Hull (Hull, 2014) (Ch 17) covers calendar spreads.

Trade. Buy the cheap (long-dated), sell the expensive (short-dated), hold to short-dated expiry. The short expires; the trader is left holding the long, worth at least what the short was, plus the captured violation.

P&L. The violation net of joint transaction cost. This is a true Type I arbitrage only if the long is held to its own expiry and residual value works out; in practice desks unwind earlier and treat it as a relative-value position.

Why it usually doesn't exist. Options market makers smooth the vol surface by construction: quoting algorithms enforce monotonicity and convexity in strike and maturity, and errors correct within seconds. The calendar bound is one of the six monotonicity conditions in (Damodaran, 2012).

When it does. Around scheduled events between the two maturities — earnings, central bank meetings, FDA decisions — where short-dated implied vol spikes for event risk while long-dated does not move much (the event is a small fraction of its remaining variance). Biotech FDA decisions are the classic case: short-dated implied vol can grow by an order of magnitude in the days before announcement.

A second source is liquidity: for thin long-dated markets (exotic commodity options, deep-OTM LEAPS), the long quote goes stale relative to the short, inverting the term structure as a quoting artefact. The detection logic is the same though the capture looks more like market making.

14.8 ETF creation/redemption arbitrage

The mechanism is in Chapter 12; this section places the trade in the zoo and draws the line between it and the basket-spread stat-arb of the same chapter.

Observable. ETF price diverges from synthetic basket value by more than the creation/redemption fee plus joint transaction cost. With ETF mid P_{ETF} and synthetic NAV P_{NAV}:

$$|P_{\text{ETF}} - P_{\text{NAV}}| > \text{creation fee} + \text{transaction costs}.$$

Trade. If $P_{\text{ETF}} > P_{\text{NAV}}$: the AP shorts the ETF, buys the basket, uses the basket to *create* new ETF shares with the issuer, delivers against the short. If $P_{\text{ETF}} < P_{\text{NAV}}$: the AP buys the ETF, redeems with the issuer for the basket, sells the basket.

P&L. $|P_{\text{ETF}} - P_{\text{NAV}}|$ minus creation fee minus joint transaction cost, locked per creation unit ($\sim$ 50,000 shares).

Why it usually doesn't exist. APs run automated basis-monitoring on every liquid ETF and are contractually incentivised to keep the basis tight. On SPY, IVV, VOO the basis rarely exceeds a fraction of a basis point intraday, and openings close in seconds.

When it does. Three families:

- **Creation fee changes.** A fee increase widens the no-arb band; a trader who knows the new schedule before the basis adjusts captures the transition.
- **Settlement lag.** For ETFs with international constituents (EM ETFs spanning 20 time zones), constituent prices are stale when home markets are closed; the basis is a noisy stale-vs-current mix. Sophisticated APs use fair-value models; traders who model time zones better than consensus find real openings.
- **AP outages.** When an AP is unavailable (trading system down, credit line frozen), the closing mechanism is gone. The basis can blow out by percent until another AP picks up. The "flash crash" episodes are the dramatic cases: ETF prices collapse relative to NAV when AP-side and exchange-side liquidity vanish together.

The Prosperity 3 `PICNIC_BASKET` trade in Chapter 12 is *not* an instance: the simulator has no creation/redemption mechanism; basket-to-NAV convergence is a property of the data-generating process, not a closed loop, so the trade has no zero-risk completion. The next section is about that distinction.

14.9 Statistical arbitrage — not a true arbitrage

The seven preceding sections cover *true arbitrage*: deterministic, non-negative-in-every-state by construction, with a no-arb argument that pins fair value down.

The strategies of Chapter 10, Chapter 11, and Chapter 12 (outside the literal ETF c/r mechanism above) are different in kind. They share the word "arbitrage" for historical/marketing reasons but are not arbitrages in the sense of Theorem 14.1. A statistical arbitrage:

- Has positive *expected* payoff under an assumed data-generating process.
- Has non-zero variance of realised payoff.
- Can lose in any individual trade and over any finite window.
- Requires an assumption about the persistence of a statistical relationship, with no contractual no-arb guarantee enforcing it.

The OU z-score rule of Chapter 10 is positive-EV under the assumption that the spread follows OU with the calibrated parameters; if parameters change, the rule can be wrong for arbitrarily long and the position can lose arbitrarily much before convergence. The cointegration tests of Chapter 11 are statistical inferences that can be correct in expectation and wrong on the realised path. The `PICNIC_BASKET1` spread trade works in Prosperity because the quoting bots are programmed to mean-revert toward synthetic value — a property of the simulation, not the universe; a different parameterisation breaks it.

Sizing follows from the distinction. A true arb is sized to maximum capacity, because there is no downside; the only constraint is broker-deployable principal. A stat arb must be sized via Chapter 20: by the ratio of expected return to variance, with full Kelly as an upper bound and some fraction thereof in practice. Sizing a stat arb as a true arb is unbounded risk in disguise.

Statistical arbitrage is not arbitrage

"Arbitrage" in "statistical arbitrage" is a marketing label from the late 1980s/early 1990s, not the technical term of Theorem 14.1. A stat arb has positive expected return, not deterministic, and it can and does lose. LTCM (1998, (Lowenstein, 2000)), the 2007 quant quake that vaporised equity-market-neutral funds in three days, and a long list of smaller

blow-ups reduce to one error: a stat-arb book sized as if it could not lose, then realising cross-position-correlated losses far beyond the in-sample variance estimate.
Two rules:

1. Never call a bet an "arbitrage" or an arbitrage a "trade". Vocabulary discipline is the first defence against the sizing error.

2. Size stat arbs by Kelly (Chapter 20), never by "fully deploy capital". The bound is the risk model's worst plausible drawdown, not the capital balance.

Apply the five-step template of Section 14.1 and ask at step (3): *is the P&L deterministic?* For pairs trading, non-creatable basket spreads, and OU mean reversion the answer is no, and the trade moves out of the zoo and into the strategy bestiary.

On a Goldman Sachs desk

Every large bank runs a desk mandated to monitor and trade the seven arbitrages here — *relative value*, *program trading*, or simply *the arb desk*. On a Goldman Sachs floor in 2025, a junior strat's morning begins with a *zoo sweep*: a scripted walk-through of every arbitrage against intraday prices, with any opening flagged in a Slack channel the senior traders read over first coffee. On most mornings every entry returns "closed by less than round-trip cost" and nothing happens. On the rare morning one entry opens — a parity violation on a hard-to-borrow name, a basis dislocation around quadruple witching, an index rebalance, an FDA decision — the trader descends for an hour and the desk makes a meaningful share of its monthly P&L. The patience-and-automation culture flows from this rhythm: daily expected return is flat, but conditional expected return given an opening is huge, so the trader set up to capture openings the moment they appear is the one who eats. The strats building zoo-sweep monitors are accordingly some of the quietly important people on the floor: their dashboards are the institutional memory of when, how, and why each arbitrage has historically opened.

Historical note

Systematic study of no-arb relations dates to the early 1970s. Black–Scholes (1973 JPE) with Merton's companion (Merton, 1973) derived (14.5) as a corollary of the replicating-portfolio argument behind the Black–Scholes equation. Stoll (1969) is the canonical pre-1973 citation for the US-equity parity inequality, but the modern hierarchical view — parity as the simplest of a ladder topped by the Black–Scholes PDE — comes from the 1973 papers.
Institutional arbitrage on a trading floor is older. Gustave Levy ran Goldman Sachs' arb desk from the late 1940s and was Senior Partner by 1969, making his name on *merger arbitrage*: buying a target at a discount to the deal price, hedged by shorting the acquirer. Merger arb is not in the zoo — deals break, and the spread compensates deal-break risk — but Goldman's modern arb desk traces to Levy's 1950s practice.
The cautionary half is LTCM, founded 1994 and bankrupt 1998 (Lowenstein, 2000). Founded by Meriwether with Merton and Scholes themselves, LTCM ran statistical arbitrages dressed as textbook arbs — on-the-run/off-the-run Treasury spreads, swap spreads, sovereign convergence — all positive-EV with small normal-regime variance, all simultaneously losing during the August–September 1998 Russian-default flight to quality. A Fed-organised \$3.6B private-sector recapitalisation averted broader collapse. The lesson: a strategy whose name contains "arbitrage" is not, by that fact alone, an arbitrage.

14.10 The arb-diagnostic flowchart

The chapter ends where it began: with a diagnostic. Given an observable condition, which arbitrage fits? The table lists the seven zoo entries with their triggers, trade structures, and section references. Keep it in arm's reach when encountering an unfamiliar product and walk down row by row.

If you observe...	**...set up**	**See**
Three FX rates whose product around a triangle differs from one.	Triangular arbitrage: cycle through the three pairs in the profitable direction.	Section 14.2
The same asset quoted at materially different prices on two venues, after fees.	Locational arbitrage: buy on the cheap venue, sell on the dear venue.	Section 14.3
A futures price away from $Se^{(r-y)(T-t)}$, after financing and yield.	Cash-and-carry arbitrage: long spot, short futures (or the reverse), deliver at maturity.	Section 14.4
$C - P$ on a European pair away from $S - Ke^{-rT}$ (after dividends and borrow).	Put-call parity arbitrage: trade the four legs to lock in the violation.	Section 14.5
Index futures away from the synthetic basket value (after dividends and rates).	Index arbitrage: trade the futures against all constituents in index proportions.	Section 14.6
A short-dated and long-dated option on the same underlying with relative pricing violating monotonicity.	Calendar arbitrage: buy the cheap maturity, sell the expensive, hold to short-leg expiry.	Section 14.7
ETF wrapper trading away from synthetic NAV by more than the creation fee.	ETF creation / redemption arbitrage: create or redeem with the issuer to capture the basis.	Section 14.8
A spread between historically-correlated assets that has drifted from its long-run mean, with no contractual mechanism to force convergence.	*Statistical arbitrage* (NOT a true arb): a positive-EV bet to be sized via Kelly.	Section 14.9, Chapter 20

One row per zoo entry, plus the closing row explicitly excluding stat arb. A reader with this table internalised can walk into any market, look at available instruments, and within minutes have an opinion on which arbitrages are worth monitoring and which are inapplicable. That is the chapter's operational deliverable.

14.11 Key facts to memorise

- **Definition.** An arbitrage is a strategy with zero or negative initial cost, non-negative payoff in every state of the world, and strictly positive payoff in some state. Equivalently, a free lunch.

- **The five-step template.** Observable, trade, P&L, why it usually does not exist, when it does. Every entry in the zoo follows this template.
- **Triangular FX.** Three currency pairs whose product around a triangle differs from one; trade the cycle. Profit per unit is the gap; example $1.08 \cdot 148 \cdot 0.0063 - 1 \approx 0.70\%$.
- **Locational arbitrage.** Same asset, two venues, different prices after fees; buy cheap, sell dear. Opens on regulatory fragmentation, latency asymmetry, transient imbalance.
- **Cost of carry.** The no-arbitrage forward price is $F = Se^{(r-y)T}$. The cash-and-carry arbitrage closes any deviation from this; the example with $S = 100$, $r = 5\%$, $y = 1\%$, $T = 0.5$ gives fair forward ≈ 102.02.
- **Put-call parity.** For European options on a non-dividend underlying, $C - P = S - Ke^{-rT}$. With $S = K = 100$, $r = 5\%$, $T = 1$, parity gives $C - P \approx 4.877$. Add $-PV(\text{div})$ if the underlying pays dividends.
- **Index arbitrage.** Cost-of-carry applied to a basket: index futures vs the synthetic value of the 500-stock basket. Most heavily competed trade in equities; opens on rebalances, witching days, fast markets.
- **Calendar arbitrage.** Two maturities of the same option, monotonicity violated; buy the cheap one, sell the expensive one. Opens around scheduled events between the two maturities.
- **ETF creation / redemption arbitrage.** ETF wrapper price away from synthetic NAV by more than the creation fee; create or redeem to close. Distinct from the Prosperity `PICNIC_BASKET` stat arb of Chapter 12, which has no creation mechanism.
- **Statistical arbitrage is not arbitrage.** Positive expected value, non-zero variance, can and does lose. Size via Kelly (Chapter 20), not by capital deployment.
- **The desk rhythm.** Expected open arbitrages per day is small; the relative-value desk detects openings automatically the moment they appear. The patience-and-automation culture follows from the conditional-vs-unconditional expected-return asymmetry.

Part V

Options and Volatility

Chapter 15

Options: Payoffs, No-Arbitrage Bounds, and Put–Call Parity

You think Tesla might go up but are not sure, and do not want to lose more than $500 if wrong. How do you express that view? The naive answer is "buy a few shares and set a stop". The options answer is "buy a call". This chapter builds the call from first principles: its payoff, its price bounds, and its relationship to the put via the most famous identity in derivatives.

The treatment is pre–Black–Scholes. Nothing here requires a stochastic differential equation, an Itô lemma, or a probability measure. Every result is proved by writing down two portfolios, comparing their payoffs in every state, and applying the no-arbitrage axiom from Chapter 14: if portfolio A pays at least as much as B in every state, A's price today is at least B's. We need only the discount factor $e^{-r(T-t)}$ that converts a sure cash flow at maturity into its time-t value, plus algebra on $\max(x, 0)$. Volatility, the central ingredient of Chapter 16, does not appear in this chapter.

Symbol anchor. Throughout: S_t spot at time t; K strike; T maturity; r risk-free rate; C_t call price; P_t put price; σ volatility (absent in this chapter).

Prerequisites

From earlier in the book:

- Chapters 2 to 4: exchanges, the limit order book, the bid–ask–mid–microprice family. Options trade on order books too; everything earlier about quoting, queues, and execution applies unchanged.
- Chapter 14, especially §13.1's arbitrage template (*state the observable, exhibit the portfolio, lock in the violation*). That chapter derived put–call parity once already in a diagnostic-arbitrage context with the worked example $S = K = 100$, $r = 0.05$, $T = 1$ giving $C - P \approx 4.877$. Here we re-derive it as the foundational replication identity: the Chapter 14 version asks "how do you trade a parity violation?"; this one asks "what does parity *say* about how put and call relate?"
- Math: the expectation operator $\mathbb{E}[\cdot]$, continuous discounting $e^{-r(T-t)}$, the indicator $\mathbb{1}\{S_T \geq K\}$, and algebra on $\max(x, 0) = (x)^+$. No stochastic calculus.

15.1 Calls, puts, and payoffs

An option is a contract whose value at a future date is a known function of an *underlying* asset's value at that date. The simplest options, and the only ones in this chapter, are the European call and put on a single non-dividend-paying stock.

Before the formula, the one-line mental model. A *call* with strike $K = 100$ lets its holder buy the stock at \$100 at expiry. If the stock ends at \$110, she exercises and pockets \$10. If the stock ends at \$90, she walks away, losing only whatever premium she paid up front. A *put* is the reverse: the right to *sell* at the strike, valuable only when the stock is below it. Both contracts floor losses at the premium paid.

Definition 15.1 (European call option)**.** A *European call option* on an underlying asset S, with *strike* K and *maturity* (or *expiry*) T, is a contract that grants its holder the *right but not the obligation* to buy one unit of S at price K at time T. Its *payoff* at expiry is

$$C_T \;=\; \max(S_T - K,\, 0) \;=:\; (S_T - K)^+, \tag{15.1}$$

where S_T denotes the (random) value of the underlying at time T. The option's price at any time $t < T$ is written C_t (or simply C when the time is clear).

The mechanics: at time T, if $S_T > K$ the holder *exercises* — pays K, takes delivery of an asset worth S_T, and immediately sells for S_T, locking in $S_T - K$. If $S_T \le K$, exercise would be a loss (paying K for something she could buy for S_T), so the contract grants the right to walk away. The walking-away clause is the entire point: a forward obliges its holder to transact at the strike with linear payoff $S_T - K$ (potentially deeply negative); an option's piecewise-linear payoff $(S_T - K)^+$ is non-negative in every state.

Definition 15.2 (European put option)**.** A *European put option* on an underlying S, with strike K and maturity T, grants its holder the right but not the obligation to *sell* one unit of S at price K at time T. Its payoff at expiry is

$$P_T \;=\; \max(K - S_T,\, 0) \;=\; (K - S_T)^+. \tag{15.2}$$

A put is the reflection of a call across the strike. If $S_T < K$ the put-holder exercises (buys at S_T, sells to the writer for K, pockets $K - S_T$); if $S_T \ge K$ she walks away.

Non-negativity of both payoffs is the source of every other property below. Since $(S_T - K)^+$ and $(K - S_T)^+$ are both ≥ 0, no option price may be negative; if it were, buying it would be a free positive payoff plus a positive cash receipt at t — the simplest arbitrage.

Definition 15.3 (American option)**.** An *American* call (or put) is identical to its European counterpart except that the holder may exercise at *any* time $t \in [0, T]$, not only at expiry. American options are the dominant exchange-traded variant on US equities; European options dominate index and FX. The two coincide in payoff at expiry but differ in the early-exercise option, which we treat in §15.4.

The party who *sells* an option is said to *write* it. A call writer is obligated to deliver the underlying at K if the holder exercises; a put writer is obligated to take delivery at K. The writer's payoff is the negative of the holder's:

$$\text{short call payoff} \;=\; -(S_T - K)^+, \qquad \text{short put payoff} \;=\; -(K - S_T)^+,$$

both ≤ 0. The writer can never make money at expiry from the option alone, but she receives the *premium* (C_0 or P_0) at $t = 0$ and hopes it exceeds the eventual payout.

Moneyness

Three pieces of jargon describe the relationship between spot S_t and strike K, used in every options conversation.

Definition 15.4 (Moneyness)**.** A call option with strike K on underlying S_t is called

- *in the money (ITM)* if $S_t > K$,
- *at the money (ATM)* if $S_t = K$,
- *out of the money (OTM)* if $S_t < K$.

A put option is called ITM if $S_t < K$, ATM if $S_t = K$, and OTM if $S_t > K$. "ITM" always means *exercising right now would yield a positive payoff*; "OTM" means *exercising right now would yield zero payoff*; "ATM" is the borderline.

Conventionally the strike is named: "a 250-strike Tesla call" means $K = 250$. "Buy me ATM Tesla calls" means the listed strike closest to current spot.

Intrinsic value and time value

The price of an option before expiry is decomposed into two non-negative parts.

Definition 15.5 (Intrinsic and time value)**.** Let C_t be the price of a European call on non-dividend-paying S with strike K and maturity $T \geq t$. Define

$$\begin{aligned} \text{intrinsic value} &:= (S_t - K)^+, && (15.3)\\ \text{time value} &:= C_t - (S_t - K)^+. && (15.4)\end{aligned}$$

The same decomposition applies to a put with intrinsic value $(K - S_t)^+$.

Intrinsic value is what the option would pay if expiry were *now*. Time value is everything else — the market's price for the chance of going deeper in the money by T, plus the carry effect of deferring the strike payment. We prove below that time value is non-negative for European calls and peaks at the money; for now it is just notation C_t = intrinsic + time.

Hockey-stick payoff diagrams

Traders call the piecewise-linear payoff curve a *hockey stick*. Plotting the four single-leg payoffs in the S_T–P&L plane gives the visual grammar the rest of the options material builds on.

Combining single legs gives *spreads* and *combinations*. These reappear throughout Chapters 17 and 18.

Long call spread. Long one call at K_1, short one at $K_2 > K_1$, same maturity. Payoff $\min((S_T - K_1)^+, K_2 - K_1)$: zero below K_1, sloping up at 45° in $[K_1, K_2]$, capped at $K_2 - K_1$ above. Expresses "prices rise, but modestly".

Long put spread. Long put at K_2, short put at $K_1 < K_2$. Capped at $K_2 - K_1$ above, zero below K_1. Expresses "prices fall, but modestly" — the downside analogue of the call spread, and the canonical cheap hedge against a bounded drawdown.

Long straddle and long strangle. A *straddle* is a long call and long put at the same strike K and maturity, payoff $|S_T - K|$. A *strangle* uses two strikes (call at $K_2 > K_1$, put at K_1); cheaper, but pays nothing in $[K_1, K_2]$. Both express "large move, either direction" — a pure volatility bet, the topic of Chapter 18.

Tesla worked example

Tesla trades at $S_0 = \$240$. A 3-month European call with strike $K = \$250$ is quoted at premium $C_0 = \$12$. Track the holder's P&L in two expiry scenarios.

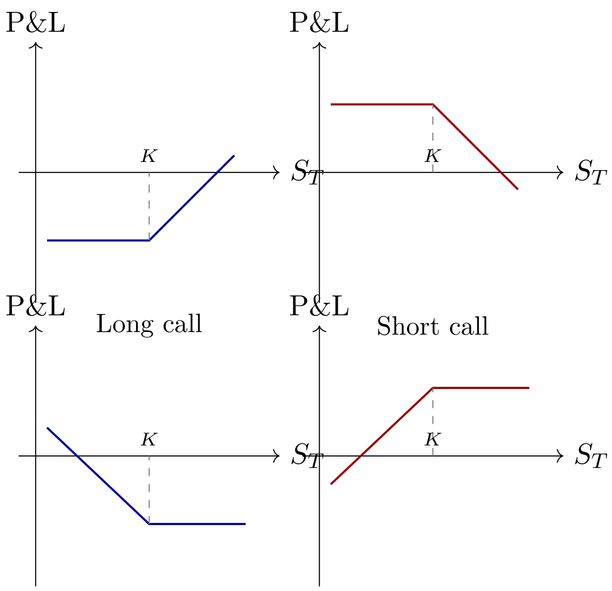


Figure 15.1: The four single-leg hockey-stick payoffs at expiry, P&L net of the premium paid or received at $t = 0$. Long positions are blue; short positions are red. The horizontal asymptote of each curve is the premium with the appropriate sign.

Scenario A: $S_T = \$280$. The call is ITM. The holder exercises, paying \$250 for stock worth \$280:

$$\text{call payoff at expiry} \;=\; (280 - 250)^+ \;=\; \$30.$$

Subtract the \$12 premium paid at $t = 0$. *Net P&L = \$18.*

Scenario B: $S_T = \$240$. The call is OTM. Exercising would mean paying \$250 for something worth \$240, so the holder walks away:

$$\text{call payoff at expiry} \;=\; (240 - 250)^+ \;=\; \$0.$$

Subtract the \$12 premium. *Net P&L* $= -\$12$.

Break-even. Net P&L crosses zero when $(S_T - K)^+ = C_0$, i.e. $S_T = K + C_0 = \$262$. Above \$262 the holder makes money; in $[\$250, \$262]$ she exercises but loses some premium; below \$250 she walks away and loses the full \$12.

The capped \$12 loss per share is the point of the contract: the holder has converted "buy 100 shares and pray" into "buy 100 calls, max loss \$1,200 regardless of crash". That is the answer to the chapter's opening question.

Option payoffs at expiry

For a non-dividend-paying underlying S:

$$\begin{aligned} \text{long call:} \quad & C_T = (S_T - K)^+, \\ \text{long put:} \quad & P_T = (K - S_T)^+, \\ \text{short call:} \quad & -(S_T - K)^+, \\ \text{short put:} \quad & -(K - S_T)^+. \end{aligned}$$

P&L at expiry equals the payoff above minus the premium paid (or plus the premium received). All four functions are piecewise linear in S_T with a single kink at $S_T = K$.

Intuition

Takeaway: *an option converts a linear view into a piecewise-linear one.* A forward or stock position has slope ± 1 everywhere, symmetric P&L, no downside cap without selling. An option's payoff has slope 0 on one side of the strike and ± 1 on the other. The kink is the entire content of the contract: it is what the premium buys, and what the next four chapters try to value.

15.2 No-arbitrage bounds by portfolio construction

Even before we can price options, no-arbitrage alone forces the price into a tight interval. We prove five such bounds, each following the same template.

The bound-by-replication template

To prove "price of X at time t is at least price of Y", construct portfolios A and B with $A_T \geq B_T$ in *every state*. By no-arbitrage, prices today satisfy $\text{cost}_t(A) \geq \text{cost}_t(B)$; otherwise buying A and selling B at t is a free lunch. For an equality, do it in both directions, or show $A_T = B_T$ pointwise.

We apply the template five times, with bounds increasing in strength.

Bound 1: a European call has non-negative price

Proposition 15.6. *For any European call on a non-dividend-paying underlying, at every $t \leq T$,*

$$C_t \;\geq\; 0.$$

Proof. Let A = "one call" and B = "nothing". At maturity $A_T = (S_T - K)^+ \geq 0 = B_T$, so $C_t \geq 0$. Equivalently, if $C_t < 0$ the trader receives $|C_t|$ for buying the call, holds to expiry, collects the non-negative payoff, and makes money from nothing. □

By the same argument $P_t \geq 0$. This trivial bound matters because every subsequent bound combines non-negativity with discounting.

Bound 2: a call costs no more than the underlying

Proposition 15.7. *For any European call on a non-dividend-paying underlying, at every $t \leq T$,*

$$C_t \;\leq\; S_t.$$

Proof. Let A = "one share" and B = "one call". At expiry $A_T = S_T$, $B_T = (S_T - K)^+$. In every state $S_T \geq (S_T - K)^+$: if $S_T \geq K$, $S_T \geq S_T - K$ since $K \geq 0$; else $S_T \geq 0$ since the stock price is non-negative. So $S_t \geq C_t$. □

Economically: the most a call pays is S_T itself (achieved as $K \to 0$, where the call *is* the stock), so paying more than S_t for the right to buy at K is worse than buying the stock outright.

Bound 3: the European call lower bound

This is the central bound. Sharper than intrinsic value (because it discounts the strike), it proves European call = American call on non-dividend stock in §15.4.

Theorem 15.8 (European call lower bound). *For any European call on a non-dividend-paying underlying at risk-free rate r, at every $t \le T$,*

$$C_t \;\ge\; \max\bigl(S_t - K\,e^{-r(T-t)},\; 0\bigr). \tag{15.5}$$

Proof. The non-negativity half is Theorem 15.6. For the other half, define two portfolios:

- **Portfolio A:** long one call (cost C_t) plus long a riskless zero-coupon bond paying K at time T (cost $Ke^{-r(T-t)}$). Total cost at t:

$$\mathrm{cost}_t(A) \;=\; C_t \;+\; Ke^{-r(T-t)}.$$

- **Portfolio B:** long one share of S. Cost at t: S_t.

At maturity:

$$\begin{aligned} A_T \;&=\; (S_T - K)^+ \;+\; K \;=\; \max\bigl(S_T - K + K,\; 0 + K\bigr) \;=\; \max(S_T,\, K), \\ B_T \;&=\; S_T. \end{aligned}$$

Since $\max(S_T, K) \ge S_T$ in every state, we have $A_T \ge B_T$ everywhere. By the template,

$$C_t \;+\; Ke^{-r(T-t)} \;\ge\; S_t,$$

which rearranges to $C_t \ge S_t - Ke^{-r(T-t)}$. Combined with Theorem 15.6, this is precisely (15.5). □

The bound strictly exceeds intrinsic value $(S_t - K)^+$ whenever $r > 0$ and $t < T$, since $Ke^{-r(T-t)} < K$. The discounted strike is a pure deterministic carry effect: even on a constant stock with zero volatility, an ATM call costs at least $S_t(1 - e^{-r(T-t)})$, because the holder defers paying K to T rather than paying S_t now. Volatility makes the actual price exceed the bound; the bound itself is volatility-free.

Worked example: the lower bound at $S = K = 100$

Take $S_t = \$100$, $K = \$100$, $r = 5\%$, $T - t = 1$ year. The discounted strike is

$$Ke^{-r(T-t)} \;=\; 100 \cdot e^{-0.05} \;\approx\; 95.1229,$$

so the lower bound is

$$\max(S_t - Ke^{-r(T-t)}, 0) \;=\; 100(1 - e^{-0.05}) \;\approx\; \$4.8771.$$

The option is ATM with zero intrinsic value, yet no-arb forces price above \$4.8771. A market quoting below this lets a trader buy the call, buy a one-year bond paying \$100, short one share, and lock in positive cash today with non-negative cash at expiry. The same number reappears in the parity example: at $S = K$ both the ATM call lower bound and the parity RHS equal $100(1 - e^{-0.05}) \approx 4.877$.

Bound 4: the European put lower bound

Mirror image of the call bound; proof identical with A, B swapped.

Proposition 15.9 (European put lower bound)**.** *For any European put on a non-dividend-paying underlying,*

$$P_t \;\geq\; \max\!\big(Ke^{-r(T-t)} - S_t,\; 0\big). \tag{15.6}$$

Proof. Non-negativity follows as in Theorem 15.6. For the discount-aware half, set

- **Portfolio A:** long one put plus long one share. Cost at t: $P_t + S_t$.
- **Portfolio B:** long a zero-coupon bond paying K at T. Cost at t: $Ke^{-r(T-t)}$.

At maturity, $A_T = (K - S_T)^+ + S_T = \max(K, S_T) \geq K = B_T$ in every state. By the template, $P_t + S_t \geq Ke^{-r(T-t)}$, which rearranges to $P_t \geq Ke^{-r(T-t)} - S_t$. Combined with $P_t \geq 0$, this is (15.6). □

Bound 5: the European put upper bound

Proposition 15.10 (European put upper bound)**.** *For any European put on a non-dividend-paying underlying,*

$$P_t \;\leq\; Ke^{-r(T-t)}. \tag{15.7}$$

Proof. At maturity the put pays at most K, achieved when $S_T = 0$. Let A be a zero-coupon bond paying K at T (cost $Ke^{-r(T-t)}$) and B be a European put. Then $A_T = K \geq (K - S_T)^+ = B_T$ in every state, since the right-hand side is at most K. By the template, $Ke^{-r(T-t)} \geq P_t$. □

The upper-bound asymmetry — call bounded by S_t, put by $Ke^{-r(T-t)}$ — reflects that the call is unbounded above (S_T has no ceiling) while the put is bounded above at K (the underlying cannot fall below zero).

European option price bounds (no dividends)

For a European call C_t and European put P_t on a non-dividend-paying underlying S with strike K, maturity T, and risk-free rate r, the following hold at every $t \leq T$:

$$\max\!\big(S_t - Ke^{-r(T-t)},\, 0\big) \;\leq\; C_t \;\leq\; S_t, \tag{15.8}$$

$$\max\!\big(Ke^{-r(T-t)} - S_t,\, 0\big) \;\leq\; P_t \;\leq\; Ke^{-r(T-t)}. \tag{15.9}$$

Proved by explicit two-portfolio construction. Holds without any assumption on the distribution of S_T, volatility, or the existence of any pricing model (Damodaran, 2012).

Intuition

Equations (15.8) and (15.9) pin the option to an interval depending only on S_t, K, r, $T - t$. The discount factor $e^{-r(T-t)}$ is the *only* time-value ingredient: no volatility, drift, or probability measure — none was needed. Volatility is what Chapter 16 adds: it picks a specific point inside the band. Black–Scholes is geometrically a rule for selecting that precise location.

15.3 Put–call parity

The previous section pinned each option to an interval. Put–call parity is sharper: an *equality* ties call and put, leaving three degrees of freedom in $\{C_t, P_t, S_t, Ke^{-r(T-t)}\}$. Chapter 14 stated the identity in the diagnostic context. Here we re-derive it in the options-theory context: the four contracts can be assembled into a basket that *is* a long stock position, so basket and stock share the same price.

The replicating portfolio

The intuition before the algebra: a long call plus the discounted strike in the bank has the same expiry payoff as a long put plus one share — both baskets deliver $\max(S_T, K)$ in every state. Two portfolios with identical payoffs in every state must cost the same today, or buying the cheaper and shorting the dearer is a free lunch. That equality *is* put–call parity.

Engineer a portfolio whose payoff is exactly S_T in every state using a call, a put, and a bond. Define

$$V_t \;\coloneqq\; C_t \;-\; P_t \;+\; Ke^{-r(T-t)}. \tag{15.10}$$

V_t is the cost of buying one call, selling one put (a short put is an inflow P_t, hence the minus sign), and buying a bond paying K at T. We claim $V_T = S_T$ in every state and prove by two cases.

Theorem 15.11 (Put–call parity)**.** *For a European call C and European put P on the same non-dividend-paying underlying S with the same strike K and same maturity T, at risk-free rate r,*

$$C_t \;-\; P_t \;=\; S_t \;-\; Ke^{-r(T-t)} \qquad \text{for every } t \le T. \tag{15.11}$$

Proof. Let V_t be the portfolio of (15.10). Compute V_T in two cases.

Case 1: $S_T \ge K$. Call is ITM, pays $S_T - K$. Short put is OTM, pays 0. Bond pays K:

$$V_T \;=\; (S_T - K) \;-\; 0 \;+\; K \;=\; S_T.$$

Case 2: $S_T < K$. Call is OTM, pays 0. Short put is ITM; we owe $K - S_T$. Bond pays K:

$$V_T \;=\; 0 \;-\; (K - S_T) \;+\; K \;=\; S_T.$$

In both cases $V_T = S_T$. So V matches one share pointwise; by the equality corner of the template, $V_t = S_t$ at every t:

$$C_t \;-\; P_t \;+\; Ke^{-r(T-t)} \;=\; S_t,$$

which is (15.11). □

Synthetic positions

Rearranging (15.11) gives the four *synthetic* relations every options trader has memorised:

$$\text{long stock} \;\equiv\; \text{long call} \;+\; \text{short put} \;+\; \text{long bond of notional } Ke^{-r(T-t)}, \tag{15.12}$$

$$\text{long call} \;\equiv\; \text{long put} \;+\; \text{long stock} \;-\; \text{long bond of notional } Ke^{-r(T-t)}, \tag{15.13}$$

$$\text{long put} \;\equiv\; \text{long call} \;-\; \text{long stock} \;+\; \text{long bond of notional } Ke^{-r(T-t)}, \tag{15.14}$$

$$\text{long bond} \;\equiv\; \text{long stock} \;-\; \text{long call} \;+\; \text{long put}. \tag{15.15}$$

These are pointwise identities, not approximations. Equation (15.12) is the “conservation law” form: long stock equals the long-call / short-put / bond combination. Equation (15.13) is how a market maker short a put (a client sold to her) manufactures the equivalent long call: buy the underlying, borrow the discounted strike. Chapter 14 showed how to trade a parity violation; (15.12)–(15.15) are the same identity rearranged for desk use.

Put–call parity

For European call and put with the same strike K and maturity T on a non-dividend-paying underlying at risk-free rate r,

$$\boxed{C_t \;-\; P_t \;=\; S_t \;-\; Ke^{-r(T-t)}}$$

holds at every $t \leq T$, by no-arbitrage, with no distributional assumption on S. The four-asset relation has three degrees of freedom: knowing any three of $\{C, P, S, Ke^{-r(T-t)}\}$ determines the fourth.

Worked numerical example

Same parameters as before. The parity RHS is

$$S_t - Ke^{-r(T-t)} \;=\; 100\,(1 - e^{-0.05}) \;\approx\; 4.8771,$$

so any European call/put pair here must satisfy $C - P \approx 4.8771$. The same \$4.877 was the ATM lower bound in §15.2 and the parity RHS in Chapter 14: at $S = K$ those quantities coincide.

If the market quotes $C = \$8$, parity forces

$$P \;=\; C - \left(S_t - Ke^{-r(T-t)}\right) \;=\; 8 - 4.8771 \;\approx\; \$3.1229.$$

A market quoting call at \$8 and put at \$2 is inconsistent: by Chapter 14 §13.4 the trader sells the call at \$8, buys the put at \$2, buys the stock at \$100, borrows $100e^{-0.05} \approx \$95.1229$, locking in net inflow $8 - 2 - 100 + 95.1229 \approx \$1.1229$ against zero liability at expiry. Inflow exactly equals the parity violation $|(8-2) - 4.8771| = \$1.1229$. Chapter 14 discusses why such violations rarely appear on liquid exchanges; parity's content here is that it pins the put price given the call.

Intuition

Put–call parity is derivatives' conservation law. Four quantities (call, put, spot, discounted strike) related by one identity, leaving three degrees of freedom. Any consistent options model must respect it — Black–Scholes, Heston, a neural network trained on chain data. Parity does not tell you what the call is worth (no more than $E = mc^2$ tells you the energy of a specific system); it tells you what the call is worth *relative to* the put, halving the problem.

Parity breaks with discrete dividends

The clean form $C_t - P_t = S_t - Ke^{-r(T-t)}$ assumes no dividends between t and T. Real equities pay dividends on discrete ex-div dates, and the dividend goes to whoever holds the *share* on that date, not to the synthetic-stock holder. The corrected parity for dividend PV $PV(D)$ between t and T is

$$C_t \;-\; P_t \;=\; S_t \;-\; PV(D) \;-\; Ke^{-r(T-t)}. \tag{15.16}$$

Derivation is identical to Theorem 15.11, except portfolio V now pays out $PV(D)$ in dividends along the way; the synthetic stock must be cheaper than cash stock by $PV(D)$ today. Forgetting $PV(D)$ is the most common failure mode of parity scanners on dividend-paying single names (Hull, 2014) (Ch 10). Same correction appears in the cost-of-carry formula of Chapter 14, for the same reason: synthetic stock does not receive the dividend stream.

15.4 American options and early exercise

An American call grants strictly more rights than European (exercise anywhere in $[0, T]$, not just at T), and a richer contract should command a higher price. The theorem of this section: on a non-dividend-paying stock the American and European call prices are *equal*. The early-exercise right on a call has zero economic value when there are no dividends.

The theorem and its proof

Theorem 15.12 (American call = European call, no dividends)**.** *Let C_t^{Am} and C_t^{Eu} denote the prices at time t of an American and a European call respectively, both on the same non-dividend-paying underlying S with the same strike K and maturity T, at risk-free rate $r > 0$. Then*

$$C_t^{\mathrm{Am}} = C_t^{\mathrm{Eu}} \quad \textit{for every } t \in [0, T].$$

Proof. We prove the equality in three steps.

Step 1: $C_t^{\mathrm{Am}} \geq C_t^{\mathrm{Eu}}$. The American grants every European right plus early exercise. A contract with more rights cannot sell for less; otherwise buy the American, sell the European, pocket the difference, and hold both to expiry (the American replicates the European exercise).

Step 2: European lower bound exceeds intrinsic value. Theorem 15.8 gives

$$C_t^{\mathrm{Eu}} \geq \max\bigl(S_t - Ke^{-r(T-t)},\ 0\bigr).$$

For $t < T$ with $r > 0$, $Ke^{-r(T-t)} < K$, so

$$S_t - Ke^{-r(T-t)} > S_t - K.$$

Whenever the RHS is non-negative, the European call exceeds its intrinsic value. (When intrinsic is zero, $C_t^{\mathrm{Eu}} \geq 0$ already gives the result; we only need strictness ITM.)

Step 3: early exercise is strictly worse than selling. An American holder considering exercise at $t < T$ replaces the option with intrinsic value $S_t - K$. But selling fetches C_t^{Am}, and by Steps 1–2,

$$C_t^{\mathrm{Am}} \geq C_t^{\mathrm{Eu}} > S_t - K,$$

so selling dominates exercising. A rational holder never early-exercises.

Conclusion. With no early exercise, American and European exercise policies match (both at T), payoffs at T match, so by the equality corner of the template $C_t^{\mathrm{Am}} = C_t^{\mathrm{Eu}}$. □

American call on non-dividend stock

For a call on non-dividend underlying at $r > 0$, American = European; early exercise has zero economic value. The only major derivatives-pricing theorem that requires no model: it follows from the European-call lower bound alone (Merton, 1973).

Why the put is different

The American put is different. Replaying Step 2 with put bounds gives $P_t^{\mathrm{Eu}} \geq Ke^{-r(T-t)} - S_t$, which is *less* than intrinsic $K - S_t$ when $r > 0$ (discounting shrinks the bound). Step 3 breaks: deep-ITM put intrinsic can exceed the European price, so an American holder sometimes exercises early. Intuition: early-exercising a put accelerates receipt of K in cash (positive carry at $r > 0$); early-exercising a call accelerates the *payment* of K (negative carry). Calls and puts are asymmetric under early exercise because cash and stock are asymmetric under positive rates.

Characterising the American-put exercise boundary requires binomial or PDE valuation; see Chapter 35. The chapter-level fact: American calls on non-dividend stock = European; American puts can be worth strictly more.

Dividends re-enable early exercise on calls

Theorem 15.12 requires no dividends. With a discrete dividend D on ex-date $t_D \in (t, T)$, holding the share beats holding the call by D on t_D. Just before t_D, an American call holder faces: exercise and capture D, or hold and keep time value. Exercise is optimal when D exceeds the time value about to be lost. This is the only situation where early exercise of an American call is rational, and it is why on US single names American-call exercise clusters a day or two before ex-div dates (Hull, 2014) (Ch 15).

Intuition

Early-exercising a call trades time value for intrinsic. On non-dividend stock, time value is strictly positive (the European lower bound proves it), so the trade is strictly bad. Dividends create an inflow the call-holder forgoes; a large enough dividend overpays for time value, and exercise becomes rational. The American-put case trades time value for early receipt of K, which earns interest and can dominate. Cash and stock are asymmetric under $r > 0$ — that is the entire content of early exercise.

Historical note

The observation that long call + short put behaves like long stock dates back to Russell Sage's synthetic-loan trades on the New York exchanges in the 1860s. The first formal statement in academic finance is Stoll (Ho and Stoll, 1981) (1969, with a 1981 follow-up cited here for the print version). The rigorous arbitrage proof and the American-=-European theorem are due to Merton's 1973 Bell Journal paper (Merton, 1973), companion to Black–Scholes. Merton's paper is unusual for being almost entirely model-free: its results are no-arb inequalities like those here, and remain true under any model.

15.5 Applications and synthesis

Prosperity example

The Round 3–4 `VOLCANIC_ROCK_VOUCHERS` of IMC Prosperity 3 were synthetic European calls on `VOLCANIC_ROCK`, five strikes $\{9500, 9750, 10000, 10250, 10500\}$, single expiry; `VOLCANIC_ROCK` traded near 10,000 seashells (see `imc-prosperity-3/README.md`). Voucher payoff was $(S_T - K)^+$ in seashells, traded on the standard order book, so all no-arb bounds here applied. Top teams exploited the structure two ways:

- **Bound enforcement.** Whenever a voucher quoted below $\max(S - Ke^{-r(T-t)}, 0)$: long voucher, long bond, short underlying. With Prosperity's rate at zero and short maturity the bound collapsed to intrinsic value, but the enforcement instinct was the same.

- **Cross-strike consistency.** The relative-value bounds of Chapter 14 (lower-strike calls $\geq$ higher-strike) and the convexity bound gave a second family of arbitrages when the five strikes drifted.

Chapter 16 adds the vol-aware Black–Scholes machinery that top teams actually used to quote and hedge (not just police bounds); Chapters 17 and 18 finish the toolkit. The bounds here are the correctness floor below which any voucher model is demonstrably wrong.

On a Goldman Sachs desk

On a Goldman equity-derivatives desk, the first sanity check a Strat runs against a listed options book is put–call parity. The pricing system computes, per (strike, maturity) cell,

$$\text{LHS} \;=\; C_t - P_t, \qquad \text{RHS} \;=\; S_t - PV(D) - Ke^{-r(T-t)},$$

and flags cells where |LHS − RHS| exceeds the bid–ask width plus estimated borrow cost. A real flag is rare and matters: stale quote, unprocessed corporate action, or — rarest — a capturable arb. Most flags resolve to three causes: hard-to-borrow names (missing $-y$ borrow term), unscheduled corporate actions (spin-offs, specials), or low-liquidity maturities where the spread is wider than the apparent violation. The Strat's job is not to trade the gap (vol traders do that) but to *maintain the flagging model*, and to explain to the trader within minutes whether a flag is real or noise. Memorising parity as $C - P = S - PV(D) - Ke^{-rT}$ is a Day-1 desk requirement.

15.6 Key facts to memorise

- A *European call* pays $(S_T - K)^+$ at expiry. A *European put* pays $(K - S_T)^+$. Both payoffs are non-negative, so option prices are non-negative.
- ITM means exercising right now would give a positive payoff; OTM means it would give zero; ATM is the borderline. The intrinsic–time-value decomposition writes $C_t = (S_t - K)^+ + (\text{time value})$, with time value ≥ 0 for European calls.
- European call bounds (no dividends): $\max(S_t - Ke^{-r(T-t)}, 0) \leq C_t \leq S_t$.
- European put bounds: $\max(Ke^{-r(T-t)} - S_t, 0) \leq P_t \leq Ke^{-r(T-t)}$.
- Every bound is proved by writing down two portfolios and comparing payoffs in every state. No distributional or volatility assumption is needed.
- **Put–call parity** (no dividends): $C_t - P_t = S_t - Ke^{-r(T-t)}$, exact, model-free, by no-arbitrage.
- Worked sanity check: $S = K = 100$, $r = 5\%$, $T = 1$ gives $C - P \approx 4.877$ and an at-the-money lower bound of the same ≈ 4.877.
- Synthetic relations: long stock ≡ long call + short put + long bond; long call ≡ long put + long stock − long bond; long put ≡ long call − long stock + long bond.
- With discrete dividends $PV(D)$: $C_t - P_t = S_t - PV(D) - Ke^{-r(T-t)}$. Forgetting $PV(D)$ is the canonical false-positive of a parity scanner.
- **American = European call** on a non-dividend-paying stock at $r > 0$. The proof uses only the European-call lower bound: selling the option is always strictly better than exercising it.
- American *puts* can be optimal to early-exercise; dividends re-enable early exercise on American calls just before ex-dividend dates. Forward-reference Chapter 35 for the algorithmic exercise boundary.
- Volatility plays no role in any result of this chapter. Volatility appears in Chapter 16, where it picks a specific point inside the no-arb interval carved out by the bounds above.

Chapter 16

Black–Scholes from First Principles

The no-arbitrage bounds of Chapter 15 pin a fair European-call price to the interval $[\max(S_t - Ke^{-r(T-t)}, 0), S_t]$. For an at-the-money one-year call on a \$100 stock at $r = 5\%$ that interval is [4.877, 100] — ninety-five dollars wide, and no portfolio argument narrows it further. A trader needs a single number. This chapter shows how Black, Scholes, and Merton (1973) closed the gap by constructing a *self-financing replicating portfolio* that tracks the option's payoff in every state. No-arbitrage then forces the option price to equal the replicating cost — a single number that depends on the underlying's *volatility* but not on its drift.

The pre-1973 literature priced options by guessing an expected payoff under the physical measure and discounting at a risk-adjusted rate. Both inputs were unknowable. Black and Scholes' insight: neither is needed. Continuous stock–bond rebalancing replicates the payoff exactly, so the price equals the *cost of replication*. The drift μ drops out.

Prerequisites

From earlier in the book:

- Chapter 10: SDEs, Brownian motion W_t, the Itô integral and isometry. This chapter uses the full Itô lemma rather than just the linear OU integrating-factor argument.
- Chapter 15: European call/put payoffs, no-arb bounds, put–call parity. Black–Scholes refines the lower bound $\max(S - Ke^{-rT}, 0)$ into a single point.
- Chapter 14: "two portfolios with identical payoffs in every state must trade at the same price."

From outside the book:

- Stochastic calculus at the Itô-lemma level: if $f \in C^{2,1}$ and X_t is an Itô process with $dX_t = a_t\,dt + b_t\,dW_t$, then

$$df(X_t, t) \;=\; \frac{\partial f}{\partial t}\,dt \;+\; \frac{\partial f}{\partial x}\,dX_t \;+\; \tfrac{1}{2}\,\frac{\partial^2 f}{\partial x^2}\,(dX_t)^2,$$

  with the Itô multiplication table $(dt)^2 = dt\,dW_t = 0$ and $(dW_t)^2 = dt$.
- Heat equation $u_\tau = \frac{1}{2}u_{xx}$: Gaussian fundamental solution, closed-form on $\mathbb{R}$ via convolution against initial data.
- Standard normal distribution and CDF Φ.

Intuition

Black–Scholes is a hedging argument, not a distributional one. It does not answer "what is the expected discounted payoff of the option?" — that question has no model-free answer. It answers "how much does it cost to replicate the payoff with stock and a bond?" That question has a unique answer once you fix the volatility, and the answer does not depend on the drift.

16.1 The Black–Scholes setup

Black–Scholes makes six assumptions. They are restrictive and often counter-factual, but the formula is robust to moderate violations of each. List them once and carry them through the derivation.

(A1) Geometric Brownian motion (GBM).

$$dS_t = \mu\, S_t\, dt + \sigma\, S_t\, dW_t, \qquad S_0 > 0, \tag{16.1}$$

with constant drift $\mu \in \mathbb{R}$, constant volatility $\sigma > 0$, and W_t standard Brownian under $\mathbb{P}$. Equivalently $\ln S_t$ is BM with drift, so S_t is log-normal. We model *log*-returns (not prices) as Gaussian because prices cannot go negative while log-returns can, and because a sum of many small independent log-returns is approximately normal by the CLT. *Why it matters:* gives the closed form $(dS_t)^2 = \sigma^2 S_t^2\, dt$, the only stochastic input Itô needs.

(A2) Constant risk-free rate r, continuously compounded. *Why:* the bond leg grows deterministically as e^{rt} with $dB_t = rB_t\, dt$ — no dW contribution.

(A3) Frictionless trading. Continuous rebalancing, no transaction costs, no spread, no borrow cost, no funding, no impact. *Why:* self-financing reduces to $d\Pi_t = \Delta_t\, dS_t + rB_t\, dt$ with no cash leakage.

(A4) No dividends during $[0, T]$. *Why:* long-stock P&L is only dS_t; dividends would add a $\Delta_t\,(\text{div})\, dt$ term and modify the PDE.

(A5) No arbitrage: two portfolios with identical payoffs in every state must trade at the same price. *Why:* turns "replicates" into "costs the same."

(A6) European exercise. Payoff $g(S_T)$ only at T; no early exercise. *Why:* the boundary condition is imposed at T alone, not as a free boundary at every $t < T$ (which American options require).

Black–Scholes assumptions

GBM underlying with constant σ and drift μ; constant risk-free rate r; frictionless continuous trading; no dividends; no arbitrage; European exercise. Every common extension (dividends, term-structure of rates, local volatility, jumps, transaction costs) relaxes exactly one of the six.

None of these assumptions are literally true

Stocks do not follow GBM: returns have fat tails, volatility is stochastic and time-varying, jumps occur on news, dividends are paid, rates move, spreads are positive, borrow costs vary, and you cannot rebalance continuously. Yet Black–Scholes is the universal language of equity-options markets because traders read it as a *coordinate system*, not a physical model. Its job is to translate observed option prices into implied volatility (Chapter 18); the resulting surface is the market's correction to every wrong assumption in (A1)–(A6) at once.

16.2 A toy pricing example, by hand

Before any SDE, price a call by hand on a one-step binomial tree. This makes the *replication* idea concrete in a setting with no calculus; every move in the continuous-time derivation later is a continuous analogue of a move made here in arithmetic.

Setup

A non-dividend stock trades at $S_0 = 100$ and over one period takes either $S_u = 110$ or $S_d = 90$. The risk-free rate is $r = 2\%$ continuously compounded ($e^{0.02} \approx 1.02020$). Price a European call with $K = 100$. Payoff at expiry:

$$\begin{aligned} C_u &= \max(S_u - K,\, 0) = \max(10,\, 0) = 10, \\ C_d &= \max(S_d - K,\, 0) = \max(-10,\, 0) = 0. \end{aligned} \tag{16.2}$$

Two states, two payoffs. Construct a stock–bond portfolio whose payoff equals C_u in the up-state and C_d in the down-state; no-arbitrage forces it to cost the same as the call.

Construction of the replicating portfolio

Hold Δ shares and borrow B dollars (bond leg $-B$ today, debt Be^r at expiry). End-of-period value:

$$\Pi_T = \Delta\, S_T - B\, e^r, \tag{16.3}$$

with $S_T \in \{S_u, S_d\}$. Requiring $\Pi_T = C_T$ in every state gives two equations in two unknowns.

The up-state equation

Substitute $S_T = S_u = 110$:

$$\underbrace{110}_{=S_u} \cdot \Delta - \underbrace{B\, e^{0.02}}_{\text{debt repayment}} = \underbrace{10}_{=C_u}. \tag{16.4}$$

One equation, two unknowns.

The down-state equation

Substitute $S_T = S_d = 90$:

$$\underbrace{90}_{=S_d} \cdot \Delta - \underbrace{B\, e^{0.02}}_{\text{debt repayment}} = \underbrace{0}_{=C_d}. \tag{16.5}$$

The system is now square.

Solve and compute the cost

Subtract (16.5) from (16.4): the $Be^{0.02}$ terms cancel and we get

$$20\,\Delta = 10 \implies \boxed{\Delta = 0.5}. \tag{16.6}$$

Substitute $\Delta = 0.5$ into (16.5):

$$90 \cdot 0.5 \;-\; B\,e^{0.02} \;=\; 0 \quad\Longrightarrow\quad B\,e^{0.02} \;=\; 45 \quad\Longrightarrow\quad B \;=\; \frac{45}{e^{0.02}} \;\approx\; 44.10894. \tag{16.7}$$

The portfolio borrows \$44.10894 today. Setup cost at $t = 0$ is the half share minus the borrowed cash:

$$\begin{aligned} \text{cost}_0 \;&=\; \Delta\,S_0 \;-\; B \;=\; 0.5 \cdot 100 \;-\; 44.10894 \\ &=\; 50 \;-\; 44.10894 \;\approx\; 5.89106. \end{aligned} \tag{16.8}$$

The portfolio pays exactly the call payoff in both states, so no-arbitrage gives

$$\boxed{C_0 \;=\; 5.89106.} \tag{16.9}$$

Intuition

The option costs whatever it takes to replicate its payoff with stock and a bond. No probabilities appear; we never said how likely the up-state is. No-arbitrage turns "does the same" into "costs the same." Drift, beliefs, expected returns — none enter.

Risk-neutral reinterpretation

A second route reaches the same answer. Define the *risk-neutral probability* p of an up-move as the unique number making the stock's discounted expected payoff equal its current price:

$$S_0 \;=\; e^{-r}\big[\,p\,S_u \;+\; (1-p)\,S_d\,\big]. \tag{16.10}$$

Solve for p. With $u := S_u/S_0 = 1.1$ and $d := S_d/S_0 = 0.9$,

$$\begin{aligned} e^r \;&=\; p\,u \;+\; (1-p)\,d \\ p \;&=\; \frac{e^r - d}{u - d} \;=\; \frac{1.02020 - 0.9}{1.1 - 0.9} \;=\; \frac{0.12020}{0.2} \;\approx\; 0.6010. \end{aligned} \tag{16.11}$$

p is not the real up-probability — it is whatever makes the risk-free rate equal the discounted expected stock return. Now the call's discounted RN expected payoff:

$$\begin{aligned} C_0^{\text{RN}} \;&=\; e^{-r}\big[\,p\,C_u \;+\; (1-p)\,C_d\,\big] \\ &=\; e^{-0.02}\big[\,0.6010 \cdot 10 \;+\; 0.3990 \cdot 0\,\big] \\ &=\; e^{-0.02} \cdot 6.0101 \;\approx\; 5.89106. \end{aligned} \tag{16.12}$$

The same number. Replication and risk-neutral expectation are two views of the same object; §16.6 gives the continuous-time version.

Discrete-time replication

For a one-step binomial tree with up-move S_u, down-move S_d and call payoffs C_u, C_d:

$$\begin{aligned} \Delta \;&=\; \frac{C_u - C_d}{S_u - S_d} && \text{(replicating share count)}, \\ p \;&=\; \frac{e^r - d}{u - d} && \text{(risk-neutral up-probability)}, \\ C_0 \;&=\; e^{-r}\big[\,p\,C_u + (1-p)\,C_d\,\big] && \text{(price)}. \end{aligned}$$

The replication and risk-neutral formulas give the same number by construction. Drift never appears in either.

Summary table for the worked example

The five steps on one page:

Quantity	**Symbol**	**Value**
Spot	S_0	100
Up-state stock	S_u	110
Down-state stock	S_d	90
Strike	K	100
Risk-free rate (per period)	r	0.02
Up-state call payoff	C_u	10
Down-state call payoff	C_d	0
Replicating share count	Δ	$(10-0)/(110-90) = 0.5$
Borrowing (debt at T)	$B\,e^r$	$0.5 \cdot 90 - 0 = 45$
Borrowing (cash today)	B	$45/e^{0.02} \approx 44.10894$
Replication cost	$\Delta\,S_0 - B$	$50 - 44.10894 \approx 5.89106$
Risk-neutral up-prob	p	$(e^{0.02} - 0.9)/0.2 \approx 0.6010$
Risk-neutral price	$e^{-r}[p\,C_u + (1-p)\,C_d]$	≈ 5.89106
Call price	$\boxed{C_0}$	$\boxed{\approx 5.89106}$

Both routes agree to all kept digits. Replication cost *is* the call price by no-arbitrage; the RN expectation is the same arithmetic re-bracketed.

From one step to many: the CRR limit

The one-step tree is too crude for a real stock. Slice the year into N steps of length $\Delta t = T/N$; each step the stock moves up or down by a small factor. The Cox–Ross–Rubinstein (CRR) parameterisation (Cox et al., 1979) chooses

$$u \;=\; e^{\sigma\sqrt{\Delta t}}, \qquad d \;=\; \frac{1}{u} \;=\; e^{-\sigma\sqrt{\Delta t}}, \qquad p \;=\; \frac{e^{r\,\Delta t} - d}{u - d}, \tag{16.13}$$

so the per-step log-return has variance $\sigma^2\,\Delta t$, matching GBM (16.1). Price by *backward induction*: write payoffs at terminal nodes $S_0\,u^{N-i}\,d^i$ and roll back with $C_t = e^{-r\,\Delta t}[p\,C^u_{t+\Delta t} + (1-p)\,C^d_{t+\Delta t}]$ at every interior node. Each node's arbitrage-freeness comes from the same delta-hedge as §16.2.

As $N \to \infty$, the CRR price converges to Black–Scholes (16.35) by the CLT: a sum of N log-binomial increments is asymptotically Gaussian, so $\ln S_T$ converges to the log-normal distribution of GBM. Convergence at the running parameters $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$:

N steps	CRR call price	Error vs BS
1	12.1623	+1.7117
2	9.5405	−0.9101
4	9.9705	−0.4801
10	10.2534	−0.1972
50	10.4107	−0.0399
100	10.4306	−0.0200
500	10.4466	−0.0040
2000	10.4496	−0.0010
∞ (Black–Scholes)	**10.4506**	—

Error decays as $O(1/N)$ with even/odd oscillation around ATM. At $N = 500$ the tree is within four cents of Black–Scholes. *Black–Scholes is the limit of the discrete replicating-portfolio price as the rebalance interval shrinks to zero.*

Intuition

The Black–Scholes price is the binomial replication price in the limit of infinitely many infinitesimal rebalances. Every continuous-time argument below is the binomial argument sent to $N \to \infty$. If the continuous-time algebra feels mysterious, drop back to the one-step tree.

16.3 From binomial to continuous: the PDE derivation

We now repeat the binomial argument in continuous time: form a stock–bond portfolio, choose the share count to match the call's dynamics, equate the cost. The new ingredients are Itô's lemma and the GBM dynamics (16.1); the output is a PDE for $C(S, t)$. We follow the MIT 18.S096 lectures (MIT OpenCourseWare, 2013) and (Hull, 2014) Ch 13–15.

Step 1 — Set up the hedged portfolio

Hold Δ_t shares and a bank-account position B_t at every t. The portfolio's value is

$$\Pi_t \;=\; \Delta_t\, S_t \;+\; B_t. \tag{16.14}$$

Choose (Δ_t, B_t) so that $\Pi_t = C_t$ for every $t \in [0, T]$; by no-arbitrage, $C_0 = \Pi_0$.

Step 2 — Self-financing condition

The portfolio is *self-financing*: cash from sold shares goes into the bank, cash to buy shares comes out; no external flows. Instantaneous P&L then has two sources — the stock capital gain $\Delta_t\, dS_t$ and the bond interest $r\, B_t\, dt$:

$$d\Pi_t \;=\; \Delta_t\, dS_t \;+\; r\, B_t\, dt. \tag{16.15}$$

Note what is *absent*: no $S_t\, d\Delta_t$, no dB_t beyond $rB_t\, dt$, no $d\Delta_t\, dS_t$ — the formal statement that rebalancing adds no cash.

Step 3 — Apply Itô's lemma to the call

We do not yet know $C(S,t)$, but assume it is $C^{2,1}$ (verified ex post from the formula). Apply Itô's lemma:

$$dC_t \;=\; \frac{\partial C}{\partial t}\,dt \;+\; \frac{\partial C}{\partial S}\,dS_t \;+\; \tfrac{1}{2}\,\frac{\partial^2 C}{\partial S^2}\,(dS_t)^2. \tag{16.16}$$

The three terms are time decay, slope in S, and the Itô curvature correction. The third term is what distinguishes stochastic from ordinary calculus: smooth paths give $(dS_t)^2 = O((dt)^2)$ and can be dropped, but Brownian roughness makes $(dS_t)^2 = O(dt)$ and it must be kept.

The GBM dynamics (16.1) pin down the quadratic variation:

$$\begin{aligned}(dS_t)^2 \;&=\; (\mu\,S_t\,dt + \sigma\,S_t\,dW_t)^2\\ &=\; \mu^2\,S_t^2\,(dt)^2 \;+\; 2\,\mu\,\sigma\,S_t^2\,dt\,dW_t \;+\; \sigma^2\,S_t^2\,(dW_t)^2\\ &=\; \underbrace{0}_{(dt)^2=0} \;+\; \underbrace{0}_{dt\,dW_t=0} \;+\sigma^2\,S_t^2\,\underbrace{(dW_t)^2}_{=\,dt}\\ &=\; \sigma^2\,S_t^2\,dt,\end{aligned} \tag{16.17}$$

using $(dt)^2 = dt\,dW_t = 0$ and $(dW_t)^2 = dt$. Substitute into (16.16):

$$dC_t \;=\; \underbrace{\left(\frac{\partial C}{\partial t} \;+\; \tfrac{1}{2}\,\sigma^2\,S_t^2\,\frac{\partial^2 C}{\partial S^2}\right)}_{\text{deterministic } dt\text{-coefficient}}\,dt \;+\; \underbrace{\frac{\partial C}{\partial S}}_{\text{coefficient of } dS_t}\,dS_t. \tag{16.18}$$

Equation (16.18) is Itô applied to the call under GBM — the only stochastic-calculus input we need ((Hull, 2014) Ch 14).

Step 4 — Match the hedged portfolio to the call

We need $d\Pi_t = dC_t$, comparing (16.18) and (16.15). Two Itô processes are equal iff their dt- and dW_t-coefficients agree separately.

Cancel the stochastic kicks. The dW_t-coefficient on the call side comes from the $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$ term in (16.18): it is $(\partial C/\partial S)\,\sigma\,S_t$. The dW_t-coefficient on the portfolio side comes from the $\Delta_t\,dS_t$ term in (16.15): it is $\Delta_t\,\sigma\,S_t$. Equating them gives

$$\boxed{\Delta_t \;=\; \frac{\partial C}{\partial S}.} \tag{16.19}$$

This is the hedge. Choose the share count to equal the call's slope in S; the Brownian noise in the portfolio cancels the Brownian noise in the call instant by instant, so $\Pi_t - C_t$ has no dW_t term and is locally riskless.

Intuition

The stochastic term vanished. Choosing $\Delta_t = \partial C/\partial S$ cancels the portfolio's next Brownian kick against the call's. What's left is deterministic, which is why the price ends up depending only on σ (size of the kicks to hedge) and not on μ (which drops out of the dt equation, shown next).

Equate the deterministic drifts. With $\Delta_t = \partial C/\partial S$, the dt-coefficients of (16.18) and (16.15) must also agree. The dt-coefficient of (16.18) is

$$\frac{\partial C}{\partial t} \;+\; \tfrac{1}{2}\,\sigma^2\,S_t^2\,\frac{\partial^2 C}{\partial S^2} \;+\; \mu\,S_t\,\frac{\partial C}{\partial S}, \tag{16.20}$$

where the last term came from extracting the drift component of dS_t. The dt-coefficient of (16.15) is

$$\Delta_t \, \mu \, S_t \;+\; r \, B_t \;=\; \mu \, S_t \, \frac{\partial C}{\partial S} \;+\; r \, B_t, \tag{16.21}$$

where again the first term is the drift component of $\Delta_t \, dS_t$ once we substitute $\Delta_t = \partial C/\partial S$. Setting (16.20) equal to (16.21), the $\mu \, S_t \, \partial C/\partial S$ term cancels from both sides:

$$\frac{\partial C}{\partial t} \;+\; \tfrac{1}{2} \sigma^2 \, S_t^2 \, \frac{\partial^2 C}{\partial S^2} \;=\; r \, B_t. \tag{16.22}$$

Now use $\Pi_t = C_t$ together with $\Pi_t = \Delta_t \, S_t + B_t$ to solve for B_t:

$$B_t \;=\; C_t \;-\; \Delta_t \, S_t \;=\; C_t \;-\; S_t \, \frac{\partial C}{\partial S}. \tag{16.23}$$

Substitute (16.23) into (16.22):

$$\frac{\partial C}{\partial t} \;+\; \tfrac{1}{2} \sigma^2 \, S_t^2 \, \frac{\partial^2 C}{\partial S^2} \;=\; r \, C_t \;-\; r \, S_t \, \frac{\partial C}{\partial S}. \tag{16.24}$$

Move everything to the left and tidy:

$$\boxed{\frac{\partial C}{\partial t} \;+\; \tfrac{1}{2} \sigma^2 \, S_t^2 \, \frac{\partial^2 C}{\partial S^2} \;+\; r \, S_t \, \frac{\partial C}{\partial S} \;-\; r \, C_t \;=\; 0.} \tag{16.25}$$

That is the *Black–Scholes PDE*: a linear parabolic PDE in (S_t, t). The same PDE prices *any* European derivative whose payoff at T depends only on S_T; the contract enters only through the boundary condition. For a call,

$$C(S, T) \;=\; (S - K)^+, \tag{16.26}$$

and for a put,

$$P(S, T) \;=\; (K - S)^+. \tag{16.27}$$

The Black–Scholes PDE

A European derivative on a non-dividend-paying GBM underlying with constant volatility σ and constant risk-free rate r has price $V(S, t)$ satisfying

$$\frac{\partial V}{\partial t} \;+\; \tfrac{1}{2} \sigma^2 \, S^2 \, \frac{\partial^2 V}{\partial S^2} \;+\; r \, S \, \frac{\partial V}{\partial S} \;-\; r \, V \;=\; 0, \quad V(S, T) = g(S_T),$$

where g is the contractual payoff. μ does not appear. The same PDE prices calls, puts, digitals, and any path-independent European payoff — only g changes.

Intuition

μ disappeared between (16.20) and (16.22): the same delta-hedge that killed the dW_t exposure also killed the drift exposure that came along with it. What survives is the PDE (16.25), in which only σ appears. *A hedged option position does not care which way the stock drifts; it cares how violently the stock wiggles*, because the wiggles are what the hedger chases. Two analysts with different views of μ agree on the price as long as they agree on σ.

The next section solves (16.25).

16.4 Solving the PDE: change of variables to the heat equation

The Black–Scholes PDE (16.25) is a disguised heat equation

$$\frac{\partial u}{\partial \tau} \;=\; \tfrac{1}{2}\,\frac{\partial^2 u}{\partial x^2}, \tag{16.28}$$

solved since Fourier (1822). Three changes of variable remove, in turn, the multiplicative diffusion, the discount, and the drift. We work through enough algebra to see the disguise dissolve, then state the integral form ((Hull, 2014) Ch 13).

Change of variable 1: log-spot. GBM is multiplicative in S but the heat equation is additive in x. Take logs:

$$x \;:=\; \ln S, \qquad V(S,t) \;=:\; \widetilde{V}(x,t).$$

By the chain rule,

$$\begin{aligned}\frac{\partial V}{\partial S} \;&=\; \frac{1}{S}\,\frac{\partial \widetilde{V}}{\partial x},\\ \frac{\partial^2 V}{\partial S^2} \;&=\; -\,\frac{1}{S^2}\,\frac{\partial \widetilde{V}}{\partial x} \;+\; \frac{1}{S^2}\,\frac{\partial^2 \widetilde{V}}{\partial x^2} \;=\; \frac{1}{S^2}\left(\frac{\partial^2 \widetilde{V}}{\partial x^2} \;-\; \frac{\partial \widetilde{V}}{\partial x}\right).\end{aligned}$$

Substitute into (16.25); the S factors cancel against the chain-rule $1/S^2$ and $1/S$:

$$\frac{\partial \widetilde{V}}{\partial t} \;+\; \tfrac{1}{2}\,\sigma^2\left(\frac{\partial^2 \widetilde{V}}{\partial x^2} \;-\; \frac{\partial \widetilde{V}}{\partial x}\right) \;+\; r\,\frac{\partial \widetilde{V}}{\partial x} \;-\; r\,\widetilde{V} \;=\; 0. \tag{16.29}$$

The diffusion coefficient is now constant $\tfrac{1}{2}\sigma^2$ instead of spot-dependent $\tfrac{1}{2}\sigma^2 S^2$. Group the first-order $\partial\widetilde{V}/\partial x$ terms:

$$\frac{\partial \widetilde{V}}{\partial t} \;+\; \tfrac{1}{2}\,\sigma^2\,\frac{\partial^2 \widetilde{V}}{\partial x^2} \;+\; \left(r - \tfrac{1}{2}\sigma^2\right)\frac{\partial \widetilde{V}}{\partial x} \;-\; r\,\widetilde{V} \;=\; 0. \tag{16.30}$$

Change of variable 2: reverse time and remove the discount. (16.30) runs forward in t with terminal condition at T; (16.28) is an initial-value problem in τ. Set

$$\tau \;:=\; T - t,$$

so $t = T$ maps to $\tau = 0$ and $\partial/\partial t = -\partial/\partial\tau$. Equation (16.30) becomes

$$-\,\frac{\partial \widetilde{V}}{\partial \tau} \;+\; \tfrac{1}{2}\,\sigma^2\,\frac{\partial^2 \widetilde{V}}{\partial x^2} \;+\; \left(r - \tfrac{1}{2}\sigma^2\right)\frac{\partial \widetilde{V}}{\partial x} \;-\; r\,\widetilde{V} \;=\; 0, \tag{16.31}$$

and we can rearrange:

$$\frac{\partial \widetilde{V}}{\partial \tau} \;=\; \tfrac{1}{2}\,\sigma^2\,\frac{\partial^2 \widetilde{V}}{\partial x^2} \;+\; \left(r - \tfrac{1}{2}\sigma^2\right)\frac{\partial \widetilde{V}}{\partial x} \;-\; r\,\widetilde{V}. \tag{16.32}$$

(16.32) is constant-coefficient linear parabolic on $\mathbb{R}$. It is almost the heat equation but contaminated by a first-order drift and a zeroth-order discount. Remove both by:

$$\widetilde{V}(x,t) \;=:\; e^{-r\,\tau}\,\widehat{V}\Big(x + \left(r - \tfrac{1}{2}\sigma^2\right)\tau,\; \tau\Big),$$

i.e. pull out $e^{-r\,\tau}$ and shift the spatial variable along the drift line. The $r\widetilde{V}$ term cancels (differentiating $e^{-r\tau}$ yields $-r$), and the $(r - \tfrac{1}{2}\sigma^2)\,\partial\widetilde{V}/\partial x$ term cancels (the shift removes it). What remains is the heat equation

$$\frac{\partial \widehat{V}}{\partial \tau} \;=\; \tfrac{1}{2}\,\sigma^2\,\frac{\partial^2 \widehat{V}}{\partial y^2}, \qquad y := x + \left(r - \tfrac{1}{2}\sigma^2\right)\tau, \tag{16.33}$$

on $y \in \mathbb{R}$. Rescaling $y \mapsto y/\sigma$ gives the $\tfrac{1}{2}$-coefficient form (16.28); the coefficient is a unit convention, not a methodological distinction.

Solution by Gaussian convolution. The heat equation on the real line with initial data $f(y)$ has the explicit Gaussian-convolution solution

$$\widehat{V}(y,\tau) \;=\; \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-(y-y')^2/(2\sigma^2\tau)}\, f(y')\, dy'. \tag{16.34}$$

Fourier's 1822 formula in Black–Scholes coordinates. The call boundary $(S-K)^+$ rewritten in y, with the four substitutions undone, becomes an integral of $(S_T-K)^+$ against the lognormal density of S_T under $\mathbb{Q}$ — the integral §16.6 sets up directly. Evaluation: split at K, complete the square, recognise two Gaussian tails ((Hull, 2014) Ch 13; §16.6 below).

Intuition

The Black–Scholes PDE is a heat equation in disguise: the underlying diffuses multiplicatively (cured by $x = \ln S$), the price is discounted (cured by $e^{-r\tau}$), and the risk-neutral drift shifts the distribution (cured by translating y). After the three substitutions we are computing the diffusion of heat from the payoff kink $(S-K)^+$ over time-to-expiry $T-t$; the smoothing of the kink is the option's time value.

16.5 The Black–Scholes formula

Solving (16.25) with the call boundary (16.26) gives:

Black–Scholes formula

The price at t of a European call on a non-dividend GBM stock (strike K, maturity T, constant σ, constant r) is

$$\boxed{C(S_t,\,t) \;=\; S_t\,\Phi(d_1) \;-\; K\,e^{-r(T-t)}\,\Phi(d_2),} \tag{16.35}$$

where Φ is the standard normal CDF and

$$d_1 \;=\; \frac{\ln(S_t/K) \;+\; \left(r+\tfrac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}, \tag{16.36}$$

$$d_2 \;=\; d_1 \;-\; \sigma\sqrt{T-t}. \tag{16.37}$$

The corresponding put price, derivable from (16.35) by put–call parity (Chapter 15), is

$$P(S_t,\,t) \;=\; K\,e^{-r(T-t)}\,\Phi(-d_2) \;-\; S_t\,\Phi(-d_1). \tag{16.38}$$

Sanity check: differentiating (16.35) twice in S and once in t, substituting into (16.25), reduces it to $0 = 0$ (we verified in sympy; (Hull, 2014) Ch 13 reproduces the algebra).

Interpreting each term

The two terms have a clean economic reading. The first $S_t\,\Phi(d_1)$ is the risk-neutral expected present value of receiving the stock at expiry, conditional on expiring ITM. The second $K\,e^{-r(T-t)}\,\Phi(d_2)$ is the expected PV of paying the strike, conditional on exercising. The call is "what you receive minus what you pay," both conditional on exercise.

The two CDF arguments encode the two relevant probabilities:

- $\Phi(d_2) = \mathbb{Q}[S_T \geq K]$ is the risk-neutral ITM probability — the probability you have to pay the strike, hence its appearance in the second term.

- $\Phi(d_1)$ is the call's *Delta*, $\partial C/\partial S$, the same Δ as in (16.19). Chapter 17 derives it by direct differentiation.

Intuition

$\Phi(d_2)$ is the ITM probability; $\Phi(d_1)$ is the Delta. Memorise both. Desks read $\Phi(d_2)$ as "probability of printing ITM at expiry (RN)" and $\Phi(d_1)$ as "share count to delta-hedge." Those are the two facts that turn the formula into desk language.

Worked numerical example

We use the same running parameters as the no-arb bound at the start of the chapter:

$$S \;=\; 100, \qquad K \;=\; 100, \qquad r \;=\; 5\%, \qquad \sigma \;=\; 20\%, \qquad T-t \;=\; 1 \text{ year.}$$

Step 1 — Compute d_1 and d_2. With $\ln(S/K) = \ln 1 = 0$ and $T - t = 1$,

$$d_1 \;=\; \frac{0 + (0.05 + \frac{1}{2}\cdot 0.20^2)\cdot 1}{0.20\cdot\sqrt{1}} \;=\; \frac{0.05 + 0.02}{0.20} \;=\; \frac{0.07}{0.20} \;=\; 0.35, \tag{16.39}$$

$$d_2 \;=\; d_1 \;-\; \sigma\sqrt{T-t} \;=\; 0.35 \;-\; 0.20 \;=\; 0.15. \tag{16.40}$$

Step 2 — Standard normal CDF. $\Phi(0.35) \approx 0.6368$, $\Phi(0.15) \approx 0.5596$.

Step 3 — Discount factor. $e^{-r(T-t)} = e^{-0.05} \approx 0.95123$.

Step 4 — Plug into the call formula.

$$\begin{aligned} C \;&=\; S\,\Phi(d_1) \;-\; K\,e^{-r(T-t)}\,\Phi(d_2) \\ &=\; 100\cdot 0.6368 \;-\; 100\cdot 0.95123\cdot 0.5596 \\ &=\; 63.683 \;-\; 53.232 \\ &\approx\; 10.4506. \end{aligned} \tag{16.41}$$

The ATM one-year call is worth $\approx$ \$10.45 per share.

Example 16.1 (Sanity-check against the no-arb interval)**.** Chapter 15 gave $C \in [4.877, 100]$ for these parameters. Black–Scholes' 10.4506 sits inside. The lower bound is the $\sigma \to 0$ limit of (16.35); turning vol on lifts the price by the time value $10.45 - 4.877$. The upper bound is the $\sigma \to \infty$ limit. Black–Scholes places a single point inside the model-free interval.

Step 5 — Parity check. Put–call parity gives $C - P = S - K\,e^{-r(T-t)} = 100 - 95.123 = 4.877$, so $P \approx 5.574$. Computing the put from (16.38) directly gives the same number.

The replicating portfolio at $t = 0$

At $t = 0$, $\Delta_0 = \Phi(d_1) = \Phi(0.35) \approx 0.6368$ shares. The bond position from (16.23):

$$B_0 \;=\; C_0 \;-\; \Delta_0\,S_0 \;=\; 10.4506 \;-\; 0.6368\cdot 100 \;\approx\; -53.23,$$

so the portfolio is long 0.6368 shares (\$63.68) and short \$53.23 in cash. Net cost $63.68 - 53.23 \approx 10.45$ matches the call price. These are the continuous analogues of the binomial $\Delta = 0.5$, $B = 44.10894$, and cost 5.89106.

The hedge is dynamic: as S_t moves and time decays both $\Delta_t = \Phi(d_1(S_t, t))$ and B_t change. Chapter 17 computes the rebalance rate (Gamma) and the discrete-rebalance P&L. For now: the replicating portfolio at every (S_t, t) is a deterministic function of two variables — no hidden state.

Comparative statics: how the price moves with each input

The price depends on five inputs: $S, K, r, \sigma, T-t$. Holding four fixed at the base case ($S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T-t = 1$) and perturbing one:

Vary spot		**Vary vol**		**Vary $T-t$**	
S	C	σ	C	$T-t$	C
80	1.86	10%	6.80	0.10	2.77
90	5.09	20%	10.45	0.25	4.62
100	10.45	30%	14.23	0.50	6.89
110	17.66	40%	18.02	1.00	10.45
120	26.17	50%	21.79	2.00	16.13

In each column, only the named variable changes; the other four are held at the base case $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T-t = 1$.

The qualitative facts every options trader knows:

- **Monotone increasing in spot.** $\partial C/\partial S = \Phi(d_1) = \Delta \in (0,1)$, with $\Delta \to 1$ deep ITM, $\to 0$ deep OTM. Spot-column slope near $S = 100$ is $(17.66 - 5.09)/20 \approx 0.63$, matching $\Phi(0.35) = 0.6368$.
- **Monotone increasing in vol.** Higher vol widens the terminal distribution; the convex call payoff turns the upside into more expected value than the downside takes away. $\partial C/\partial\sigma$ is *Vega*.
- **Monotone increasing in time-to-expiry.** More time means more opportunity to drift ITM. $\partial C/\partial t$ is *Theta* (negative as $t \to T$).
- **Increasing in the rate.** Higher r pulls the discounted strike down, raising ATM/ITM prices. Sensitivity is *Rho* — small short-dated equity, large long-dated rates.

All four are derived in Chapter 17; see (Hull, 2014) Ch 19 for textbook treatment.

A reference Python implementation

A 12-line Python function. Every options-trading interview asks the candidate to write this; it should be reflex.

```python
from math import log, sqrt, exp
from scipy.stats import norm

def black_scholes_call(S, K, r, sigma, T):
    """Price of a European call on a non-dividend-paying GBM
    underlying with constant rate and constant vol.
        S      spot
        K      strike
        r      continuously compounded risk-free rate
        sigma  volatility (annualised)
        T      time to maturity in years
    Returns the call price in the same units as S.
    """
    if T <= 0.0:
        return max(S - K, 0.0)
    d1 = (log(S / K) + (r + 0.5 * sigma ** 2) * T) / (sigma * sqrt(T))
```

```python
    d2 = d1 - sigma * sqrt(T)
    return S * norm.cdf(d1) - K * exp(-r * T) * norm.cdf(d2)
```

Listing 16.1: Black–Scholes European call price as a pure function. Returns a single float; no side effects, no plotting, no persistence. Exactly the function a Strat would type into a notebook to sanity-check a vendor pricer.

Calling `black_scholes_call(100, 100, 0.05, 0.20, 1.0)` returns ≈ 10.4506, matching (16.41). The $T \le 0$ branch handles the expiry boundary; production code also branches on $\sigma = 0$ to avoid division by zero.

16.6 A second derivation: the martingale route

The PDE derivation is the 1973 route and makes the hedge explicit. A second route due to Harrison–Pliska (1981) reaches the same formula via a measure change. We sketch it because both routes matter in practice and because the martingale route is the foundation of most numerical methods in Chapter 35.

The risk-neutral measure

Under physical $\mathbb{P}$, GBM (16.1) reads $dS_t = \mu\, S_t\, dt + \sigma\, S_t\, dW_t^{\mathbb{P}}$, so the discounted stock $\widetilde{S}_t := e^{-rt} S_t$ has drift $(\mu - r)\widetilde{S}_t\, dt$ and is generally not a martingale. Girsanov constructs an equivalent measure $\mathbb{Q}$ under which $\widetilde{S}_t$ *is* a martingale. (Equivalent measures agree on which events have probability zero but disagree on the weight each event carries.) Under $\mathbb{Q}$, S_t has dynamics

$$dS_t \;=\; r\, S_t\, dt \;+\; \sigma\, S_t\, dW_t^{\mathbb{Q}}, \tag{16.42}$$

where $W_t^{\mathbb{Q}}$ is $\mathbb{Q}$-Brownian. The drift under $\mathbb{Q}$ is r, not μ; the volatility is unchanged (Girsanov rotates drift only (Hull, 2014)). In plain English, $\mathbb{Q}$ is the unique reweighting of outcomes under which every tradable asset earns the risk-free rate on average — the investor behind $\mathbb{Q}$ is indifferent to risk, so replacing μ with r is exactly what eliminates the risk premium that hedging already neutralised. $\mathbb{Q}$ is the *risk-neutral measure* — the continuous analogue of the binomial p from (16.11) ((Hull, 2014) Ch 28).

Pricing as a discounted expectation

Once $\widetilde{S}_t$ is a $\mathbb{Q}$-martingale, every discounted self-financing portfolio is too — in particular $e^{-rt} C_t$. The martingale property $\mathbb{E}^{\mathbb{Q}}[e^{-rT} C_T \mid \mathcal{F}_t] = e^{-rt} C_t$ gives

$$\boxed{C_t \;=\; e^{-r(T-t)}\, \mathbb{E}^{\mathbb{Q}}\Big[\, (S_T - K)^+ \mid \mathcal{F}_t \Big]\,.} \tag{16.43}$$

The price is the discounted risk-neutral expected payoff. No μ, no preferences, no "true" ITM probability appears; the inputs are S_t, r, σ, K, T.

Evaluating the lognormal expectation

Under $\mathbb{Q}$ the dynamics (16.42) integrate to

$$S_T \;=\; S_t\, \exp\!\Big((r - \tfrac{1}{2}\sigma^2)(T-t) \;+\; \sigma\sqrt{T-t}\, Z \Big), \qquad Z \sim \mathcal{N}(0,1), \tag{16.44}$$

(the Itô-lemma solution applied to $\ln S_t$). So S_T is log-normal with log-mean $\ln S_t + (r - \frac{1}{2}\sigma^2)(T-t)$ and log-variance $\sigma^2 (T-t)$. The expectation is an integral of the payoff against the lognormal density:

$$\mathbb{E}^{\mathbb{Q}}\Big[(S_T - K)^+\Big] \;=\; \int_{-\infty}^{\infty} \Big(S_t\, e^{(r - \sigma^2/2)(T-t) + \sigma\sqrt{T-t}\, z} - K \Big)^+ \frac{e^{-z^2/2}}{\sqrt{2\pi}}\, dz.$$

The integrand is non-zero for $z > -d_2$. Splitting at $-d_2$ and evaluating (first piece: complete the square; second: Gaussian tail) gives

$$\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \;=\; S_t\, e^{r(T-t)}\, \Phi(d_1) \;-\; K\, \Phi(d_2).$$

Multiplying by $e^{-r(T-t)}$ recovers (16.35) exactly. (Hull, 2014) Ch 14 walks the integral line by line; (MIT OpenCourseWare, 2013) gives a parallel treatment. We confirmed the equivalence numerically: both give 10.4506 at the running parameters to ten significant figures.

Intuition

The PDE route says "solve a deterministic PDE driven by the payoff"; the martingale route says "compute a discounted expectation under a measure that prices the discounted stock as a martingale." The *Feynman–Kac theorem* formalises the equivalence: any parabolic PDE $\partial_t V + \mathcal{L}V = 0$ with terminal condition $V(S,T) = g(S)$ equals $\mathbb{E}[g(S_T) \mid S_t]$ under the diffusion with generator $\mathcal{L}$. Black–Scholes is Feynman–Kac for risk-neutral GBM. Chapter 35 uses both: finite differences for the PDE, Monte Carlo for the expectation.

16.7 What each assumption buys and when it breaks

The six assumptions are not equally restrictive: some cost a few basis points, others are catastrophic on a news event. The implied-vol surface (Chapter 18) is the market's simultaneous correction to all six. Below: which fix lives where.

(A1) Constant volatility breaks first. Real volatility clusters, spikes on news, and varies with moneyness (the *smile/skew*). The market quotes a different implied vol per strike–maturity. Stochastic-vol models (Heston, SABR) make σ its own SDE; local-vol (Dupire) makes σ a deterministic function of (S,t) fitting today's surface. Chapter 31 covers both.

(A1) Lognormality breaks too. Real returns have fat tails (kurtosis $\gg 3$), negative skew, and discontinuous jumps. Black–Scholes is a continuous diffusion and cannot model jumps. Jump-diffusion (Merton 1976, variance-gamma) and rough-vol models address the gap, but the desk response is simpler: trade the implied-vol surface, which absorbs the model error into the strike-dependent vol.

(A2) Constant rates: fine for short-dated equity, fatal for rates derivatives. For an SPX option with $T - t < 1$ year, rate variation gives sub-cent pricing error vs the spread. For a 30-year cap or swaption it is absurd: the contract is *about* the random rate. Chapter 30 covers the proper models (Black 1976, Hull–White, BGM, SABR).

(A3) Frictionless trading breaks at the rebalance frequency. You cannot rebalance continuously. With minimum interval Δt the hedge has $O(\sqrt{\Delta t})$ discretisation error and positive replication-P&L variance. The bid–ask spread adds a cost linear in turnover. Chapter 17 computes the per-step discrete-hedge P&L as $\frac{1}{2}\Gamma(\Delta S)^2 - \Theta\,\Delta t$; expected zero (gamma cancels theta on average), realised noisy and concentrated in the gamma term — the basis of gamma scalping.

(A4) No dividends — clean fix. For known discrete dividends D_i at times t_i, replace S_t in (16.35) with $S_t - PV(D)$ where $PV(D) = \sum_i D_i\, e^{-r(t_i - t)}$. For a continuous yield q, replace S_t with $S_t\, e^{-q(T-t)}$ and use $r - q$ in d_1. (Hull, 2014) Ch 17 covers both.

(A6) European exercise rules out American puts. American puts on dividend payers (or on non-payers with positive rates) can be optimal to exercise early; (16.35) does not price them. Chapter 35 treats American options via finite-difference PDEs with the early-exercise inequality and via Longstaff–Schwartz regression.

Don't calibrate Black–Scholes to historical vol and trade off it

Estimating σ from a year of historical returns, plugging into (16.35), and trading against market price loses money systematically:

- **Historical vol is backward-looking; the market is forward-looking.** A Tesla call two days before earnings is priced off expected vol, not realised. Earnings, mergers, FOMC, launches all show up in implied vol first.
- **Different strikes trade at different implied vols.** A deep-OTM put has much higher IV than an ATM call on the same name, pricing in crash asymmetry. "The vol" as a scalar is a category error; the unit is the *surface*.

The professional response inverts: treat (16.35) as a function from price to *implied volatility*, invert numerically per strike–maturity, and trade the surface against your view (Chapter 18).

Prosperity example

In Prosperity 3, Round 3 introduced `VOLCANIC_ROCK_VOUCHERS` — five European calls on `VOLCANIC_ROCK`, strikes 9500–10500 in steps of 250, underlying near 10000 (Frankfurt Hedgehogs, 2025b). Each voucher granted the right to buy one rock at strike at expiry, with seven days to expiry at the start of Round 3 dropping to two by the final round.
The synthetic options had every real-call feature: positive time value, monotone-decreasing in strike, delta-hedgeable. Top-10 teams (Frankfurt Hedgehogs, Linear Utility, others) priced with (16.35) and traded short-horizon IV dislocations (Chapter 18): buy cheap vol, sell rich vol, delta-hedge.
Concrete number: at Round 4 start, $S = 10000$. Take the $K = 9750$ voucher with $T - t \approx 7/365 \approx 0.01918$ and a recent-history $\sigma = 20\%$. With $r = 0$ (no risk-free rate in Prosperity),

$$\begin{aligned} d_1 &= \frac{\ln(10000/9750) + (0 + 0.02) \cdot 0.01918}{0.20 \cdot \sqrt{0.01918}} \approx 0.928, \\ d_2 &= d_1 - 0.20 \cdot \sqrt{0.01918} \approx 0.900, \\ C &= 10000 \cdot \Phi(0.928) - 9750 \cdot \Phi(0.900) \approx 276.8. \end{aligned}$$

Intrinsic is $\max(S - K, 0) = 250$, so time value is $\approx \$26.8$ for one week of optionality on a 20%-vol underlying. Frankfurt Hedgehogs fitted the IV smile across the five strikes, computed each market price's residual from the smile, and traded mean reversion of the residual (Frankfurt Hedgehogs, 2025b) — impossible without evaluating (16.35) on every tick.

On a Goldman Sachs desk

On a GS equity-derivatives desk, Black–Scholes is the baseline pricer for every listed European option. The Strat's job is not to derive it but to *calibrate the IV surface* (Chapter 18) so that (16.35) at each calibrated (σ, K, T) reproduces market quotes. The pricing system is a Black–Scholes loop over the surface; the calibration loop sits above,

adjusting the surface to fit.
Risk is computed by perturbing the IV surface, not the "physical" vol. Sensitivities (vega per strike, vega per tenor, cross-vega) come from bumping surface nodes and re-pricing. Traders ask "what is my vega to the 1y 25-delta put?", not "what is sigma?" The Strat owns the calibration and bumping pipeline; the trader owns the position. Chapter 36 expands.

Historical note

Black and Scholes published "Pricing of Options and Corporate Liabilities" in May 1973 in the *JPE* after multiple rejections ("too narrowly applied to finance"). Merton independently derived the formula in a companion paper, with the rigorous self-financing and no-arbitrage treatment. The CBOE had opened one month earlier; within a few years floor traders carried programmable calculators preloaded with the formula. Black died in 1995; Scholes and Merton received the 1997 Nobel, with the committee noting Black's contribution.

16.8 Key facts to memorise

- **Setup**: GBM $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, constant r, frictionless continuous trading, no dividends, no arbitrage, European exercise.
- **Binomial replication**: $\Delta = (C_u - C_d)/(S_u - S_d)$ plus a bond from the down-state equation track the call. Risk-neutral $p = (e^r - d)/(u - d)$ gives the same price as a discounted expectation. Drift never appears.
- **Black–Scholes PDE**:
$$\frac{\partial C}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0,$$
with terminal $C(S,T) = (S-K)^+$.
- **Call formula**:
$$C = S\,\Phi(d_1) - K\,e^{-r(T-t)}\,\Phi(d_2),$$
with $d_1 = [\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$ and $d_2 = d_1 - \sigma\sqrt{T-t}$.
- **Put**: $P = K\,e^{-r(T-t)}\Phi(-d_2) - S\,\Phi(-d_1)$ (or via parity).
- **Interpretations**: $\Phi(d_2)$ = risk-neutral ITM probability; $\Phi(d_1)$ = Delta = replicating share count.
- **Numerical anchor**: $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ gives $C \approx 10.4506$, $P \approx 5.5735$, parity $C - P \approx 4.877$.
- **μ does not appear.** Two analysts with different drift views agree on price as long as they agree on σ — the central economic content.
- **Two routes, same answer**: PDE (replicate, kill the dW, heat equation) and martingale (change to $\mathbb{Q}$, discounted expectation, lognormal integral). Feynman–Kac makes the equivalence an identity.
- **No assumption is literally true.** Vol is stochastic, returns have fat tails and jumps, hedging is discrete and costly, dividends are paid. The market's simultaneous correction is the IV surface of Chapter 18.

Chapter 17

The Greeks and Delta Hedging

The Black–Scholes formula of Chapter 16 gives today's call price, but in five minutes the underlying will have moved, implied vol will have re-priced, and the option will have lost time value. The *Greeks* are the partial derivatives of the option price with respect to the five inputs (S, σ, t, r, K), and every position on a derivatives desk is booked, hedged, and risk-managed in terms of its Greeks rather than its mark-to-market price (Natenberg, 2015).

This chapter has two goals. First, derive the closed-form Greeks for a Black–Scholes European call — delta, gamma, vega, theta, rho — pinned to the running example ($S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T - t = 1$) from Chapter 16. Second, prove the P&L identity behind *delta-hedging*: a delta-neutral long-option book makes money when realised volatility exceeds implied and loses when it does not. That gamma-against-theta identity is the central equation of vol trading.

Prerequisites

From earlier in the book:

- Chapter 15: European call and put payoffs, put–call parity, and the no-arbitrage bounds. Several Greeks inherit a parity-style call-versus-put relationship.
- Chapter 16: the Black–Scholes call price

$$C \;=\; S\,\Phi(d_1) \;-\; K\,e^{-r(T-t)}\,\Phi(d_2),$$

  with $d_1 = [\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$ and $d_2 = d_1 - \sigma\sqrt{T-t}$; the Black–Scholes PDE $\Theta + rS\,\Delta + \frac{1}{2}\sigma^2 S^2\,\Gamma = r\,C$; the running numerical example ($S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T - t = 1$) which gives $d_1 = 0.35$, $d_2 = 0.15$, $C \approx 10.4506$.
- Chapter 10: Itô's lemma in the form $df = f_t\,dt + f_x\,dX_t + \frac{1}{2} f_{xx}\,(dX_t)^2$ for an Itô process X_t. The dynamic-hedging P&L derivation re-uses this expansion.

From outside the book:

- Multivariable calculus at the level of partial derivatives and the chain rule.
- The standard normal density $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ and CDF $\Phi(x) = \int_{-\infty}^{x} \phi(u)\,du$.

Intuition

A Greek is a sentence the trader can speak: "my book is long +10,000 delta of SPX, short −50,000 gamma, long +200,000 vega, and bleeding −30,000 theta a day." Each number is the partial derivative of the book's mark-to-market with respect to one input. Together they tell the trader how the book will move as the world moves — and which parameter to hedge first. The Greeks are the *language* of risk on a derivatives desk.

17.1 The first-order Greeks: Delta and Rho

The five primary Greeks split into first-order sensitivities (delta, rho, vega, theta) and second-order (gamma, plus higher-order vanna/volga/charm). We start with the two simplest: delta with respect to spot, rho with respect to interest rate.

Delta: $\Delta = \partial C/\partial S$

Definition 17.1 (Delta)**.** The *delta* of an option price $V(S,t)$ with respect to its underlying S is the partial derivative

$$\Delta \;:=\; \frac{\partial V}{\partial S}. \tag{17.1}$$

For a European call price $C(S,t)$ on a non-dividend-paying stock under the Black–Scholes model, the delta is

$$\Delta \;=\; \Phi(d_1). \tag{17.2}$$

The formula $\Delta = \Phi(d_1)$ is one of the cleanest results in derivatives, but it is not obvious from the call price

$$C(S,t) \;=\; S\,\Phi(d_1) \;-\; K\,e^{-r(T-t)}\,\Phi(d_2),$$

since both d_1 and d_2 depend on S. Naive differentiation produces four terms that must collapse. The same cancellation reappears for vega, theta, and the vol-surface calibration of Chapter 18, so it is worth working once.

Derivation of $\Delta = \Phi(d_1)$. Differentiate C in S holding t, r, σ, K fixed; let $\tau = T - t$. By product/chain rule,

$$\frac{\partial C}{\partial S} \;=\; \Phi(d_1) \;+\; S\,\phi(d_1)\,\frac{\partial d_1}{\partial S} \;-\; K\,e^{-r\tau}\,\phi(d_2)\,\frac{\partial d_2}{\partial S}. \tag{17.3}$$

Both d_1 and d_2 depend on S only through $\ln S$, so

$$\frac{\partial d_1}{\partial S} \;=\; \frac{\partial d_2}{\partial S} \;=\; \frac{1}{S\,\sigma\sqrt{\tau}}.$$

So (17.3) reduces to

$$\frac{\partial C}{\partial S} \;=\; \Phi(d_1) \;+\; \frac{1}{\sigma\sqrt{\tau}}\Big[\phi(d_1) \;-\; \frac{K}{S}\,e^{-r\tau}\,\phi(d_2)\Big]. \tag{17.4}$$

The bracket must vanish for (17.2) to hold. The key identity:

$$S\,\phi(d_1) \;=\; K\,e^{-r\tau}\,\phi(d_2), \tag{17.5}$$

which we now prove. Take the ratio

$$\frac{S\,\phi(d_1)}{K\,e^{-r\tau}\,\phi(d_2)} \;=\; \frac{S}{K}\,e^{r\tau}\,\frac{e^{-d_1^2/2}}{e^{-d_2^2/2}} \;=\; \frac{S}{K}\,e^{r\tau}\,\exp\!\Big(\tfrac{1}{2}(d_2^2 - d_1^2)\Big).$$

Now $d_2^2 - d_1^2 = (d_2 - d_1)(d_2 + d_1) = (-\sigma\sqrt{\tau})\,(2d_1 - \sigma\sqrt{\tau})$, and substituting the explicit form of d_1 gives

$$\begin{aligned}\tfrac{1}{2}(d_2^2 - d_1^2) &= -\tfrac{1}{2}\,\sigma\sqrt{\tau}\,\Big[2d_1 \;-\; \sigma\sqrt{\tau}\Big] \\ &= -\,\sigma\sqrt{\tau}\cdot d_1 \;+\; \tfrac{1}{2}\,\sigma^2\,\tau \\ &= -\ln(S/K) \;-\; r\,\tau,\end{aligned}$$

using $d_1 = [\ln(S/K) + (r + \frac{1}{2}\sigma^2)\tau]/(\sigma\sqrt{\tau})$; the emerging $-\frac{1}{2}\sigma^2\tau$ cancels against the trailing $+\frac{1}{2}\sigma^2\tau$. Hence

$$\exp\!\Big(\tfrac{1}{2}(d_2^2 - d_1^2)\Big) \;=\; \frac{K}{S}\,e^{-r\tau},$$

so the ratio collapses to 1. This proves (17.5), kills the bracketed term in (17.4), and leaves $\partial C/\partial S = \Phi(d_1)$. (Hull, 2014)§15.5 walks the same identity; it is the prototype for all subsequent Greek derivations.

Delta of a Black–Scholes call

$$\Delta_{\text{call}} \;=\; \frac{\partial C}{\partial S} \;=\; \Phi(d_1) \;\in\; (0, 1).$$

By put–call parity (Chapter 15), $\Delta_{\text{put}} = \Phi(d_1) - 1 \in (-1, 0)$. The call delta is the share count replicating the call's instantaneous P&L; the put delta is negative because a put is a short directional position.

Delta as the replicating share count. The replicating-portfolio derivation of Chapter 16 already met this number: the self-financing portfolio holds $\Phi(d_1)$ shares at time t. A trader long one call hedges by *shorting* Δ shares; short one call, by *buying* Δ shares. Either way the linear directional exposure is zero and the position is *delta-neutral*.

Delta by moneyness. From $\Delta = \Phi(d_1)$:

- Deep *in-the-money* call ($S \gg K$): $d_1 \to +\infty$, $\Delta \to 1$. The call behaves like a forward on the stock.
- *At-the-money* call ($S \approx K$), d_1 small and positive, so $\Delta \approx 0.5$ — the trader's rule of thumb "ATM call delta is about a half".
- Deep *out-of-the-money* call ($S \ll K$): $d_1 \to -\infty$, $\Delta \to 0$. The call has almost no chance of being exercised.

Worked numerical example: delta of the running call

Take the running parameters of Chapter 16: $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T - t = 1$. The chapter calculation already gave $d_1 = 0.35$ and $d_2 = 0.15$. Therefore

$$\Delta \;=\; \Phi(d_1) \;=\; \Phi(0.35) \;\approx\; 0.6368. \tag{17.6}$$

A trader long one call hedges by shorting 0.6368 shares: \$63.68 of short stock against a \$10.45 option. The hedge exceeding the premium is normal — the delta-hedged position is a vol bet, and most capital is tied up in the hedge.

Sanity check from the Ch 15 table. The comparative-statics table in Chapter 16 gave $C = 5.09$ at $S = 90$ and $C = 17.66$ at $S = 110$. The finite-difference slope is

$$\frac{17.66 - 5.09}{110 - 90} = \frac{12.57}{20} \approx 0.6285,$$

agreeing with the analytic 0.6368 to two decimals. The gap is central-difference truncation error from a \$20 secant step across a curved function (the curvature is the gamma of the next section).

Rho: $\rho = \partial C/\partial r$

Rho of a Black–Scholes call

$$\rho_{\text{call}} = \frac{\partial C}{\partial r} = K\,(T-t)\,e^{-r(T-t)}\,\Phi(d_2).$$

Always positive: a higher discount rate raises the PV of the ITM strike receipt. Put rho $\rho_{\text{put}} = -K\,(T-t)\,e^{-r(T-t)}\,\Phi(-d_2) < 0$.

The derivation parallels delta. Differentiating $C = S\,\Phi(d_1) - K\,e^{-r\tau}\,\Phi(d_2)$ with respect to r produces contributions through d_1, d_2, and the explicit $e^{-r\tau}$ factor. Since $\partial d_1/\partial r = \partial d_2/\partial r = \sqrt{\tau}/\sigma$, identity (17.5) again kills the bracketed piece, leaving the explicit $K\,\tau\,e^{-r\tau}\,\Phi(d_2)$ term. (Hull, 2014)§17.10 has the full algebra.

Numerical value at the running parameters. With $K = 100$, $T - t = 1$, $r = 5\%$, $e^{-r\tau} = e^{-0.05} \approx 0.9512$, and $\Phi(d_2) = \Phi(0.15) \approx 0.5596$,

$$\rho = 100 \cdot 1 \cdot 0.9512 \cdot 0.5596 \approx 53.23 \quad \text{per unit of } r.$$

By convention rho is reported per 1% rate move ($r \to r + 0.01$), so $\rho/100 \approx 0.5323$: a one-percentage-point rate increase raises the ATM one-year call by about \$0.53. Rho is rarely the headline Greek for short-dated equity options; it dominates long-dated rates products (Chapter 30).

17.2 The second-order Greek: Gamma

If delta is the slope of the call price in spot, gamma is its curvature. Gamma is the rate at which the delta hedge stops working, and therefore the rate at which it must be rebalanced.

Definition 17.2 (Gamma)**.** The *gamma* of an option price $V(S,t)$ with respect to its underlying S is the second partial derivative

$$\Gamma := \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}. \tag{17.7}$$

For a Black–Scholes European call on a non-dividend-paying stock,

$$\Gamma = \frac{\phi(d_1)}{S\,\sigma\,\sqrt{T-t}}. \tag{17.8}$$

Derivation. Differentiate $\Delta = \Phi(d_1)$ with respect to S. By the chain rule,

$$\Gamma = \frac{d\Phi(d_1)}{dd_1}\,\frac{\partial d_1}{\partial S} = \phi(d_1) \cdot \frac{1}{S\,\sigma\,\sqrt{\tau}},$$

which is (17.8). Gamma is positive for any long option (call or put, by parity) since all factors are positive; long options are *long gamma*, short options are *short gamma*. Gamma scales like $1/(S\,\sigma\,\sqrt{\tau})$, blowing up close to expiry, on low-vol underlyings, and on low-priced underlyings.

Gamma of a Black–Scholes call/put

$$\Gamma \;=\; \frac{\phi(d_1)}{S\,\sigma\,\sqrt{T-t}} \;>\; 0.$$

By parity, calls and puts on the same strike/expiry have *identical* gamma. Long options are long gamma; short options are short gamma.

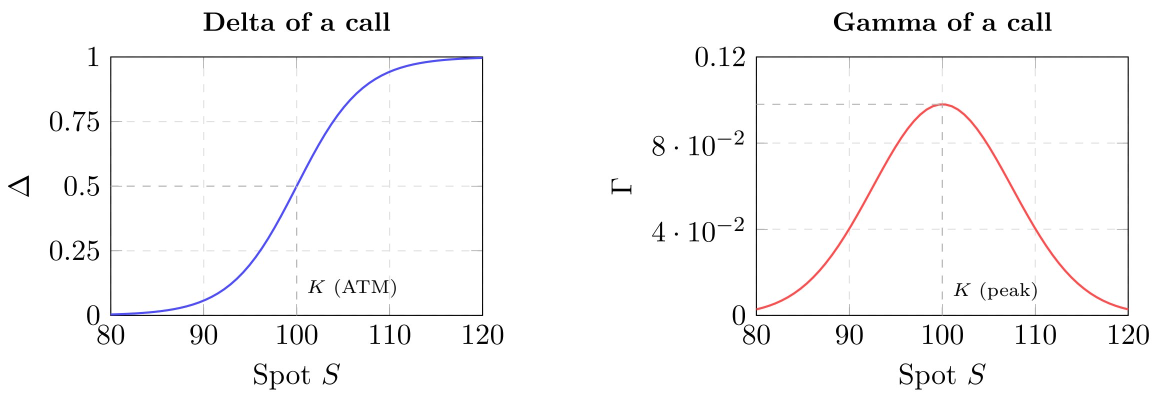


Figure 17.1: Delta (left) and gamma (right) of a European call as functions of spot price. Delta rises monotonically from 0 (deep OTM) toward 1 (deep ITM), passing through $\frac{1}{2}$ at the money. Gamma — the rate of change of delta — peaks at the money, where delta is most sensitive to spot moves. A long-gamma position profits from large moves in either direction (gamma scalping); the cost is theta decay.

Gamma by moneyness. Gamma peaks ATM near $S \approx K$ and decays to zero in both tails: deep ITM, delta is pinned at 1; deep OTM, at 0; either way it cannot move further. The ATM peak is where the hedge needs the most rebalancing per unit move. (Natenberg, 2015)emphasises this: *gamma risk is concentrated at the strike*, and a market maker writing a short-dated ATM straddle takes on enormous gamma risk in a tiny window.

Worked numerical example: gamma of the running call

Continuing with the running parameters: $S = 100$, $\sigma = 0.20$, $T - t = 1$, $d_1 = 0.35$. The standard normal density at 0.35 is

$$\phi(0.35) \;=\; \frac{1}{\sqrt{2\pi}}\,e^{-0.35^2/2} \;=\; \frac{1}{\sqrt{2\pi}}\,e^{-0.06125} \;\approx\; 0.3989 \cdot 0.9406 \;\approx\; 0.3752.$$

Therefore

$$\Gamma \;=\; \frac{0.3752}{100 \cdot 0.20 \cdot 1} \;=\; \frac{0.3752}{20} \;\approx\; 0.01876. \tag{17.9}$$

Operational meaning of 0.01876**.** Gamma is the rate of change of delta per unit of spot. A \$1 move up (from 100 to 101) shifts delta by ≈ 0.01876, from 0.6368 to ≈ 0.6556. The trader who was short 0.6368 shares must short another 0.0188 to stay delta-neutral — a rebalance forced by the market, not chosen. *Gamma is the rate at which the hedge bleeds.*

Intuition

A useful mnemonic from (Natenberg, 2015): "Gamma is what you pay for, delta is what you hedge, and theta is how much it costs." Long gamma means the option's delta moves *in your favour* as the underlying moves: it goes more positive when the stock rises (so your long call gains more from the next up-move) and more negative when the stock falls (so your long call loses less from the next down-move). The convexity is free upside; the bill comes in the form of theta.

17.3 The vol Greek: Vega

Vega is the call price's sensitivity to volatility. Unlike $\Delta, \Gamma, \Theta, \rho$, "vega" is not a Greek letter; ν (nu) and $\mathcal{V}$ are the common notations. We use ν.

Definition 17.3 (Vega)**.** The *vega* of an option price $V(S, t; \sigma)$ is the partial derivative

$$\nu \;:=\; \frac{\partial V}{\partial \sigma}. \tag{17.10}$$

For a Black–Scholes European call on a non-dividend-paying stock,

$$\nu \;=\; S\sqrt{T-t}\,\phi(d_1). \tag{17.11}$$

The derivation again uses identity (17.5) to collapse a bracket; the bookkeeping is in (Hull, 2014)§17.11.

Calls and puts share vega. By put–call parity, $C - P = S - K\,e^{-r\tau}$ is independent of σ, so $\nu_{\text{call}} = \nu_{\text{put}}$. Both gain value as vol rises: higher σ fattens both tails of the terminal-price distribution, and since an option's payoff is capped at zero on the wrong side but unbounded on the right side, more dispersion always raises the expected payoff regardless of whether the option is a call or a put.

Vega is positive for any long option. Long options are *long vol*; short options are *short vol*. Vega exposure is the headline number on a volatility-trading desk.

Vega of a Black–Scholes call/put

$$\nu \;=\; S\sqrt{T-t}\,\phi(d_1) \;>\; 0,$$

identical for calls and puts. A book is long vega if it owns optionality, short vega if it sold it.

Worked numerical example: vega of the running call

At the running parameters, $S = 100$, $\sqrt{T-t} = 1$, $\phi(d_1) = \phi(0.35) \approx 0.3752$, so

$$\nu \;=\; 100 \cdot 1 \cdot 0.3752 \;\approx\; 37.52 \quad \text{per unit of } \sigma. \tag{17.12}$$

A "unit" of vol is one full point of σ ($0.20 \to 1.20$), which is unrealistic. Desks instead report vega per 1% move ($0.20 \to 0.21$):

$$\nu/100 \;\approx\; 0.3752,$$

so the call gains about \$0.38 per percentage-point increase in implied vol — and a long holder loses \$0.38 if vol drops from 20% to 19% even with spot unchanged.

"Vega" is not a Greek letter

Vega has no Greek-alphabet equivalent: ν (nu) is the common notation, $\mathcal{V}$ a stylised second choice, Λ in some texts. Early option-pricing literature inherited delta, gamma, theta from mechanics, but traders coined "vega" later for the vol sensitivity — a word that sounds Greek without being Greek. The symbol on a desk's risk report is ν or $\mathcal{V}$, not a real Greek glyph.

Vega units: per-point or per-percent?

Half of vega confusion comes from the mixed convention. Formula (17.11) gives ν in call-price per unit σ; the trader convention reports $\nu/100$, in call-price per 1% vol move. "Vega is 0.38" means per 1%; "vega is 37.5" means per unit. Same object, different units.

17.4 The time Greek: Theta

Theta measures option-price decay as time passes with other inputs fixed. It is the Greek that is negative for a long option position — time always moves forward and erodes optionality.

Definition 17.4 (Theta)**.** The *theta* of an option price $V(S, t)$ is the partial derivative with respect to calendar time:

$$\Theta \;:=\; \frac{\partial V}{\partial t}. \tag{17.13}$$

For a Black–Scholes European call on a non-dividend-paying stock,

$$\Theta_{\text{call}} \;=\; -\,\frac{S\,\sigma\,\phi(d_1)}{2\,\sqrt{T-t}} \;-\; r\,K\,e^{-r(T-t)}\,\Phi(d_2). \tag{17.14}$$

For a put,

$$\Theta_{\text{put}} \;=\; -\,\frac{S\,\sigma\,\phi(d_1)}{2\,\sqrt{T-t}} \;+\; r\,K\,e^{-r(T-t)}\,\Phi(-d_2).$$

Differentiating the call formula in t produces contributions through d_1, d_2, and the explicit $e^{-r\tau}$. Identity (17.5) collapses the bracketed cross-terms, leaving the two explicit pieces shown. (Hull, 2014)§17.7 has the full algebra; we verify numerically below.

Theta of a Black–Scholes call

$$\Theta_{\text{call}} \;=\; -\,\frac{S\,\sigma\,\phi(d_1)}{2\,\sqrt{T-t}} \;-\; r\,K\,e^{-r(T-t)}\,\Phi(d_2) \;<\; 0$$

for a long non-dividend call. Long options pay theta; short collect. Reported per year (formula), per calendar day (divide 365), or per business day (divide 252) by desk convention.

Sign of theta. For a non-dividend call, both terms in (17.14) are negative, so $\Theta < 0$ unconditionally. Deep-ITM European puts on high-rate underlyings can flip $\Theta > 0$ — a quirk of the put formula (Hull, 2014). For this book, treat long options as $\Theta < 0$.

Worked numerical example: theta of the running call

Continuing with $S = 100$, $K = 100$, $r = 0.05$, $\sigma = 0.20$, $T - t = 1$, $d_1 = 0.35$, $d_2 = 0.15$, $\phi(0.35) \approx 0.3752$, $\Phi(0.15) \approx 0.5596$, $e^{-r\tau} = e^{-0.05} \approx 0.9512$. The two terms of (17.14) are

$$\frac{S\,\sigma\,\phi(d_1)}{2\sqrt{T-t}} \;=\; \frac{100 \cdot 0.20 \cdot 0.3752}{2} \;=\; \frac{7.504}{2} \;\approx\; 3.752, \tag{17.15}$$

$$r\,K\,e^{-r\tau}\,\Phi(d_2) \;=\; 0.05 \cdot 100 \cdot 0.9512 \cdot 0.5596 \;\approx\; 2.6616. \tag{17.16}$$

Therefore

$$\Theta \;=\; -\,3.752 \;-\; 2.6616 \;\approx\; -6.414 \quad \text{per year}, \tag{17.17}$$

or, on a calendar-day basis,

$$\frac{\Theta}{365} \;\approx\; \frac{-6.414}{365} \;\approx\; -0.01757 \quad \text{per calendar day}, \tag{17.18}$$

and on a 252-business-day basis (the convention most US equity desks prefer),

$$\frac{\Theta}{252} \;\approx\; \frac{-6.414}{252} \;\approx\; -0.02545 \quad \text{per business day}. \tag{17.19}$$

What does theta cost? A long call bleeds \$0.018 (calendar) or \$0.025 (business) per day without any stock move — about \$0.53 over 21 business days, 5% of the \$10.45 premium. Over 252 business days the cumulative decay is $252 \cdot 0.0254 \approx \$6.41$, matching the annualised $|\Theta| \approx 6.414$ at (17.17): if spot ends at 100 the call expires worthless and the entire \$10.45 of time value erodes. The long position loses that premium minus whatever it earns from realised volatility along the way. *Whether the trade makes money turns on whether realised vol exceeds the* 20% *implied vol that priced the call.* The next section makes this precise.

17.5 The dynamic-hedging P&L identity

We now combine the Greeks into the P&L identity for a delta-hedged option position — the central equation of derivatives trading.

Setting up the delta-hedged book

Suppose at time t the trader is long one call worth C_t and short $\Delta_t = \partial C/\partial S$ shares of the underlying as a hedge. Call the resulting portfolio

$$\Pi_t \;=\; C_t \;-\; \Delta_t\,S_t. \tag{17.20}$$

The portfolio is *delta-neutral* at construction: $\partial\Pi/\partial S = \Delta - \Delta = 0$, so a small change in S_t leaves Π_t unchanged at linear order. Directional risk is hedged — what is left?

Itô expansion of the call price

Apply Itô's lemma (Chapter 10) to the function $C(S,t)$, treating S_t as a generic Itô process:

$$dC \;=\; \frac{\partial C}{\partial t}\,dt \;+\; \frac{\partial C}{\partial S}\,dS \;+\; \tfrac{1}{2}\,\frac{\partial^2 C}{\partial S^2}\,(dS)^2. \tag{17.21}$$

Recognising the partials as Greeks,

$$dC \;=\; \Theta\,dt \;+\; \Delta\,dS \;+\; \tfrac{1}{2}\,\Gamma\,(dS)^2. \tag{17.22}$$

Now compute the change in the delta-hedged portfolio over dt. The trader holds $-\Delta$ shares (the hedge), so the contribution from the share leg is $-\Delta\, dS$. Combining,

$$\begin{aligned} d\Pi &= dC - \Delta\, dS \\ &= \Theta\, dt + \Delta\, dS + \tfrac{1}{2}\Gamma\,(dS)^2 - \Delta\, dS \\ &= \Theta\, dt + \tfrac{1}{2}\Gamma\,(dS)^2. \end{aligned} \tag{17.23}$$

The directional terms cancel exactly — that is what delta hedging buys. What survives are theta and gamma.

Delta-hedged P&L identity

For a delta-hedged long option, the instantaneous P&L over dt is

$$d\Pi = \Theta\, dt + \tfrac{1}{2}\Gamma\,(dS)^2 + \text{higher-order in } d\sigma, dr. \tag{17.24}$$

Theta (negative) works against the long holder; gamma (positive) works for him. The trade bets which is larger.

Realised vs implied vol: the central identity

Suppose spot moves are GBM-distributed with realised vol σ_{r}, so $(dS)^2 \approx \sigma_{\mathrm{r}}^2 S^2\, dt$. Substituting into (17.24),

$$d\Pi \approx \Theta\, dt + \tfrac{1}{2}\Gamma\,\sigma_{\mathrm{r}}^2\, S^2\, dt = \left(\Theta + \tfrac{1}{2}\Gamma\,\sigma_{\mathrm{r}}^2\, S^2\right) dt. \tag{17.25}$$

The call was priced under Black–Scholes with *implied* vol σ_{i}, and the Black–Scholes PDE (Chapter 16) reads

$$\Theta + \tfrac{1}{2}\sigma_{\mathrm{i}}^2\, S^2\,\Gamma + r\,S\,\Delta - r\,C = 0, \tag{17.26}$$

rearranging to $\Theta = -\frac{1}{2}\,\sigma_{\mathrm{i}}^2\, S^2\,\Gamma - r(S\Delta - C)$. Substituting into (17.25) and dropping the small $O(r)$ carry,

$$d\Pi \approx \tfrac{1}{2}\Gamma\, S^2\left(\sigma_{\mathrm{r}}^2 - \sigma_{\mathrm{i}}^2\right) dt + O(r\,S\,\Delta)\, dt. \tag{17.27}$$

This is the central identity. (Bennett, 2014)Ch 4 frames it for vol traders. Economic content:

- $\sigma_{\mathrm{r}} > \sigma_{\mathrm{i}}$: gamma gain dominates theta bleed, the delta-hedged long-option book makes money — *long realised vol.*
- $\sigma_{\mathrm{r}} < \sigma_{\mathrm{i}}$: theta dominates, the position loses — *short realised vol.*
- $\sigma_{\mathrm{r}} = \sigma_{\mathrm{i}}$: the two cancel by the PDE; zero expected P&L per dt.

The vol-trading equation

The instantaneous P&L of a delta-hedged long option is

$$d\Pi \approx \tfrac{1}{2}\Gamma\, S^2\left(\sigma_{\mathrm{r}}^2 - \sigma_{\mathrm{i}}^2\right) dt. \tag{17.28}$$

Long gamma $\Leftrightarrow$ long realised vol. Every dollar a delta-hedged book makes is paid by the gap between realised and implied at trade time.

Intuition

Options are bets on realised vs implied volatility. Delta-hedging strips out direction; what is left is the vol-difference term weighted by gamma. A vol buyer expects realised > implied; a vol seller the reverse. Profit depends only on which side of implied the realised print lands on over the life of the hedge — the spot path itself is irrelevant once dynamically hedged.

Numerical illustration. At the running parameters, $\frac{1}{2}\Gamma S^2 = \frac{1}{2} \cdot 0.01876 \cdot 100^2 = 93.8$ "vol^2-dollars per year" of gamma exposure. If realised prints 25% against implied 20%, then $\sigma_r^2 - \sigma_i^2 = 0.0225$ and annualised P&L is $93.8 \cdot 0.0225 \approx \$2.11$, about 20% of the \$10.45 premium — purely from the 5-vol-point edge.

A worked path-by-path simulation

To pin the identity to a path, suppose the trader buys one call at the running parameters ($C_0 = 10.4506$, $\Delta_0 = 0.6368$, $S_0 = 100$) and rebalances daily for ten business days, then closes out. Ignore theta and rate carry within each day to isolate the gamma mechanic.

Day 1: spot $100 \to 101$. The call rises by $\Delta_0 \cdot 1 + \frac{1}{2}\Gamma_0 \cdot 1^2 \approx 0.6368 + 0.0094 = 0.6462$; the hedge (short 0.6368 shares) loses 0.6368. Portfolio P&L is $+0.0094$ — the gamma slug $\frac{1}{2}\Gamma_0(\Delta S)^2$. Now reset: new delta ≈ 0.6556, sell another 0.0188 shares.

Day 2: spot $101 \to 100$. $(-1)^2 = 1$ gives a slug of $\frac{1}{2} \cdot 0.01876 \cdot 1 = 0.00938$ — identical to Day 1's slug. The slug is positive regardless of move sign — the long-gamma trader profits from volatility, not direction.

Days 1 and 2 illustrate the mechanic; the full ten-day schedule of alternating $\pm\$1$ moves is tabulated below, with the daily theta drip of $\Theta \approx -0.0254$ accumulating on the right. Read the two right-hand columns as the tradeoff: the gamma column is what the trader earns from realised moves, the theta column is what he pays for owning the option, and (17.28) says they cancel exactly when realised vol equals implied.

Day	S_t	ΔS	$\frac{1}{2}\Gamma(\Delta S)^2$	cum. gamma P&L	cum. theta cost
1	101	$+1$	0.00938	0.00938	0.0254
2	100	-1	0.00938	0.01876	0.0508
3	101	$+1$	0.00938	0.02814	0.0762
4	100	-1	0.00938	0.03752	0.1016
5	101	$+1$	0.00938	0.04690	0.1270
6	100	-1	0.00938	0.05628	0.1524
7	101	$+1$	0.00938	0.06566	0.1778
8	100	-1	0.00938	0.07504	0.2032
9	101	$+1$	0.00938	0.08442	0.2286
10	100	-1	0.00938	0.09380	0.2540

Net P&L is $0.0938 - 0.254 \approx -0.16$: the trade lost about \$0.16 over ten days because realised vol ($\pm\$1$/day on $S = 100$ annualises to roughly 16%) fell short of the 20% implied priced into theta — exactly as (17.28) predicts. Had the same Γ and Θ sat against realised $\pm\$1.30$ daily moves instead, each gamma slug would be $\frac{1}{2} \cdot 0.01876 \cdot 1.69 \approx 0.01585$, cumulating to 0.1585 — flipping the trade to net positive.

The lesson: every dollar of realised-vs-implied edge is scraped together one $\frac{1}{2}\Gamma(\Delta S)^2$ slug at a time against a relentless theta drip — mathematically an identity, operationally a spreadsheet of small daily P&Ls.

17.6 Aggregation: portfolio Greeks

A real options book holds dozens or hundreds of positions; the trader reads aggregated Greeks, not individual tickets. Aggregation is linear because differentiation is linear. With n_i units of option i (Greeks $\Delta_i, \Gamma_i, \nu_i, \Theta_i, \rho_i$) and stock position h:

$$\begin{aligned}
\Delta^{\text{book}} &= \sum_i n_i\,\Delta_i \;+\; h,\\
\Gamma^{\text{book}} &= \sum_i n_i\,\Gamma_i,\\
\nu^{\text{book}} &= \sum_i n_i\,\nu_i,\\
\Theta^{\text{book}} &= \sum_i n_i\,\Theta_i,\\
\rho^{\text{book}} &= \sum_i n_i\,\rho_i.
\end{aligned}$$

Stock contributes +1 delta per share and zero of the others. (Futures use delta vs the discounted forward, not spot.) *Net delta-hedging* the book means setting $h = -\sum_i n_i\,\Delta_i$ so $\Delta^{\text{book}} = 0$. Deltas across long/short calls and puts naturally net, making this far cheaper than hedging each leg individually.

Why aggregate Greeks are the unit of risk. A \$100M-long / \$100M-short tight call spread has enormous gross notional but tiny aggregate Greeks — the legs nearly cancel. Real risk is the residual after netting, read off the book-aggregated line. (Natenberg, 2015)repeats: look at the net, not the gross.

17.7 Higher-order Greeks: vanna, volga, charm

The five primary Greeks cover single-input sensitivities. For exotic options and structured notes, cross-derivatives matter. We quote the three most common qualitatively; formulas from (Hull, 2014)§17.13, exotic-desk treatment in Chapter 31.

Vanna. The mixed second derivative

$$\text{vanna} := \frac{\partial^2 C}{\partial S\,\partial\sigma} = \frac{\partial\Delta}{\partial\sigma} = \frac{\partial\nu}{\partial S}.$$

Vanna measures how the delta hedge shifts when implied vol moves; vanna $= -\phi(d_1)\,d_2/\sigma$ for a BS call. A delta-hedged market maker who finds vol rallied 5% overnight sees delta drift by vanna times that move.

Volga (also written "vomma"). The second derivative with respect to vol,

$$\text{volga} := \frac{\partial^2 C}{\partial\sigma^2} = \frac{\partial\nu}{\partial\sigma}.$$

For a BS call, volga $= \nu \cdot d_1 d_2/\sigma$ — convexity of price in vol, the "vol of vol" exposure. Long volga loves vol spikes; short volga (selling short-dated straddles is canonical) hates them. Barrier-option pricing (Chapter 31) requires careful volga.

Charm. The mixed derivative with respect to spot and time,

$$\text{charm} := \frac{\partial^2 C}{\partial S\,\partial t} = \frac{\partial \Delta}{\partial t}.$$

Charm is theta for delta — the rate at which the delta hedge bleeds with time. A trader hedging at the open and going to lunch returns to find delta drifted by charm $\cdot\, \Delta t$ even on zero spot move. Overnight *charm risk* forces the close-of-day rehedge (Hull, 2014).

The exotic zoo continues (speed, color, ultima, zomma): all have closed forms under BS, all matter on specific exotic desks. See Chapter 31. For this book, the five primary Greeks plus the three above suffice.

17.8 Discrete-hedging error

§17.5 assumed continuous rebalancing. In reality the trader rebalances at discrete intervals Δt (every minute, every trade, every \$0.50 move). Between rebalances the position is not delta-neutral, introducing residual P&L noise that cannot be scheduled away.

Variance of the discrete-hedging error. To leading order (Sinclair, 2013),

$$\operatorname{Var}(\text{discrete-hedging error per period}) \;\propto\; \tfrac{1}{2}\,\Gamma^2\, S^4\, \sigma^4\, (\Delta t)^2, \tag{17.29}$$

so the per-period stdev scales like Δt. Summing over $N = T/\Delta t$ periods gives cumulative error stdev scaling like $\sqrt{T\Delta t}$. *Halving the interval halves the cumulative noise stdev.* Continuous rebalancing ($\Delta t \to 0$) sends noise to zero, recovering (17.28).

Rebalancing frequency vs cost. Every rebalance pays the bid-ask spread. With N rebalances of cost c each, total cost is Nc. The optimum minimises

$$\text{cumulative cost} \sim \underbrace{\sigma_{\text{noise}}\sqrt{T\Delta t}}_{\text{hedging error}} + \underbrace{\frac{T}{\Delta t}\,c}_{\text{transaction cost}},$$

balanced at $\Delta t^* \sim (c/\sigma_{\text{noise}})^{2/3}$. (Sinclair, 2013)centres the "volatility trading" chapter on this trade-off: it determines the actual P&L of every real-world delta-hedging strategy.

Discrete hedging is never riskless

The wrong intuition: "BS proves options hedge perfectly, so my delta-hedged book has zero risk." False. BS proves *continuous* hedging is riskless, and continuous hedging is impossible. Real books carry discrete-hedging noise $\sim \sigma_{\text{noise}}\sqrt{\Delta t}$ per rebalance plus per-rebalance transaction costs, and remain exposed to gap risk and model risk. Risk is *reduced*, not eliminated.

Gap risk. A sharper failure: the underlying gaps over a hedging interval (earnings, halt-resume, overnight news). No rebalancing schedule can follow, and the realised $(dS)^2$ along the gap vastly exceeds $\sigma^2 S^2 \Delta t$. The gamma term in (17.24) then produces an outsized gain or loss. Long-gamma books love gaps; short-gamma books fear them. 1987, 2010 Flash Crash, and every earnings overnight are textbook cases ((Bennett, 2014)Ch 5).

Prosperity example

Round 4 of IMC Prosperity 3 introduced `VOLCANIC_ROCK_VOUCHERS` on `VOLCANIC_ROCK` (Chapter 16). Each voucher is a European call, and teams harvesting the IV-rich side of the surface had to delta-hedge — otherwise directional `VOLCANIC_ROCK` moves swamped the vol edge. The mechanics:

1. Compute d_1 from the voucher's strike, the current `VOLCANIC_ROCK` price, an estimate of σ, and $r = 0$ (Prosperity has no risk-free rate).
2. Read off the voucher delta $\Delta = \Phi(d_1)$.
3. For a long voucher position of size n, hedge by *shorting* $\lfloor n \cdot \Delta \rfloor$ units of `VOLCANIC_ROCK`. (Prosperity positions are integer.)
4. Every K ticks (the Frankfurt Hedgehogs used $K = 10$), recompute the delta and resize the hedge.

Concretely: a voucher worth 5 seashells with $\Delta = 0.6$ in size 10 shorts $\lfloor 10 \cdot 0.6 \rfloor = 6$ `VOLCANIC_ROCK`. If the rock moves and delta rises to 0.7, the next rebalance shorts another 1 unit. Cumulative hedge P&L is the discrete analogue of (17.28): each rebalance locks a $\frac{1}{2}\Gamma(\Delta S)^2$ gamma slug against the per-tick theta bleed of a five-day voucher.

On a Goldman Sachs desk

On a Goldman Sachs equity-options market-making desk, every position's Greeks are live on screen, refreshed sub-second by the risk system. The aggregate is the *risk book*, read like an instrument panel. A representative line:

> *SPX vol book: Delta* $+10{,}000$*, Gamma* $-50{,}000$*, Vega* $+200{,}000$*, Theta* $-30{,}000$*/day, Rho* $+5{,}000$.

Each number sums that Greek across every option, computed via closed-form BS at the calibrated implied vol per strike (Chapter 18). The trader reads: slightly long the index, short gamma (big moves hurt), substantially long vega (a vol spike helps), bleeding \$30k/day in theta. Decisions are in Greeks, not individual tickets. Strat owns the Greek pipeline, the trader owns the position, risk manager owns the limits. *Risk limits at GS are stated in Greeks*: "no more than ±1,000,000 net vega in the SPX vol book overnight." Chapter 36 returns to the production view.

17.9 A short reference implementation

The five Greeks all derive from d_1, d_2, Φ, ϕ. The function below computes all in one pass. Every options-firm quant interview expects this in under five minutes — commit to muscle memory.

```python
import math
from scipy.stats import norm

def bs_greeks(S, K, r, sigma, T):
    """Five primary Greeks of a European call under Black-Scholes."""
    d1 = (math.log(S / K) + (r + 0.5 * sigma ** 2) * T) / (sigma *
        math.sqrt(T))
    d2 = d1 - sigma * math.sqrt(T)
    return {
        'delta': norm.cdf(d1),
```

```
        'gamma': norm.pdf(d1) / (S * sigma * math.sqrt(T)),
        'vega':  S * math.sqrt(T) * norm.pdf(d1),
        'theta': (-S * sigma * norm.pdf(d1) / (2 * math.sqrt(T))
                  - r * K * math.exp(-r * T) * norm.cdf(d2)),
        'rho':   K * T * math.exp(-r * T) * norm.cdf(d2),
    }
```

Listing 17.1: Black–Scholes Greeks for a European call on a non-dividend-paying stock. Returns a dict of the five primary Greeks. No side effects, no plotting, no persistence. Calling `bs_greeks(100, 100, 0.05, 0.20, 1.0)` returns delta ≈ 0.6368, gamma ≈ 0.01876, vega ≈ 37.52, theta ≈ −6.414 per year, rho ≈ 53.23.

Returned Greeks are in mathematical units (per unit S, σ, r; per year). Trader conventions: `vega/100` (per 1%), `rho/100` (per 1%), `theta/252` (per business day).

17.10 Key facts to memorise

- **Delta** $\Delta = \Phi(d_1) \in (0, 1)$ call; $\Phi(d_1) - 1 \in (-1, 0)$ put. ATM ≈ 0.5; deep ITM $\to 1$; deep OTM $\to 0$. Delta is the share count that hedges directional risk.
- **Gamma** $\Gamma = \phi(d_1)/(S\,\sigma\,\sqrt{\tau}) > 0$, identical for calls and puts. Peaks ATM, decays in both tails. The rate at which the delta hedge goes stale.
- **Vega** $\nu = S\sqrt{\tau}\,\phi(d_1) > 0$, identical for calls and puts. Trader convention $\nu/100$ (per 1%). "Vega" is not a Greek letter.
- **Theta** $\Theta < 0$ for a long call (and almost always a long put). Reported per year, or per day (divide by 252 or 365). Long options bleed theta; short options collect it.
- **Rho** $\rho = K\,\tau\,e^{-r\tau}\Phi(d_2) > 0$ call, < 0 put. Small for short-dated equity, dominant for long-dated rates (Chapter 30).
- **BS PDE in Greek form:** $\Theta + rS\,\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rC$ — the delta-hedged call earns risk-free when realised equals implied.
- **Delta-hedged P&L:** $d\Pi = \Theta\,dt + \frac{1}{2}\Gamma\,(dS)^2$, which collapses via the PDE to $d\Pi \approx \frac{1}{2}\Gamma\,S^2\,(\sigma_{\mathrm{r}}^2 - \sigma_{\mathrm{i}}^2)\,dt$.
- **Long gamma ⇔ long realised vol.** Owning options bets realised > implied; selling bets the reverse.
- **Calls and puts share gamma and vega;** differ in delta and rho; share theta to leading order.
- **Higher-order Greeks** vanna, volga, charm covered on exotic desks (Chapter 31).
- **Discrete hedging is not riskless:** noise scales $\sqrt{T\Delta t}$, optimum $\Delta t^* \sim (c/\sigma_{\text{noise}})^{2/3}$.
- **Running anchor** ($S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T - t = 1$; $d_1 = 0.35$, $d_2 = 0.15$): $\Delta \approx 0.6368$, $\Gamma \approx 0.01876$, $\nu \approx 37.52$, $\Theta \approx -6.414$/yr, $\rho \approx 53.23$. Memorise for sanity-checks.
- **On a desk**, every position is risk-managed in Greeks; risk limits are stated in Greeks; trader, Strat, and risk manager share that vocabulary.

Chapter 18

Implied Volatility, the Vol Surface, and Gamma Scalping

Black–Scholes takes volatility as an input, but the market's call price *already encodes* the volatility the market expects: given any observed call quote, there is a unique σ that, fed through the Black–Scholes formula of Chapter 16, reproduces the quote exactly. Inverting Black–Scholes to back out that number is *implied volatility*. It depends on strike, on maturity, and on the moment you ask — producing a whole *surface* of implied vols rather than a single number. This chapter explains the surface: how it is built, how to read it, why it is not the flat plane Black–Scholes assumes, and how traders exploit it. The payoff is *gamma scalping*: buy cheap vol, delta-hedge against the underlying, and collect the realised-vs-implied gap from Chapter 17. Gamma scalping is the canonical vol trade and the reason every derivatives desk runs a Greeks-based risk book.

Two objectives. First, define implied volatility formally, describe the inversion procedure, and catalogue the four qualitative features of the empirical surface (smile, skew, short-vs-long-dated shape, term structure). Second, turn the P&L identity of Chapter 17 into a rule: buy options whose implied vol is below your forecast of realised vol, short the underlying in the BS-delta amount, rebalance on a schedule, collect the gap. Variance swaps appear qualitatively here; the full replication argument is deferred to Chapter 31.

Prerequisites

From earlier in the book:

- Chapter 16: the Black–Scholes European call price

$$C \;=\; S\,\Phi(d_1) \;-\; K\,e^{-r(T-t)}\,\Phi(d_2),$$

  with $d_1 = [\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)]/(\sigma\sqrt{T-t})$ and $d_2 = d_1 - \sigma\sqrt{T-t}$, together with the running numerical anchor $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T - t = 1$, $d_1 = 0.35$, $d_2 = 0.15$, $C \approx 10.4506$.

- Chapter 17: the five primary Greeks, in particular $\Delta = \Phi(d_1)$, $\Gamma = \phi(d_1)/(S\,\sigma\,\sqrt{\tau})$, and $\nu = S\sqrt{\tau}\,\phi(d_1)$; the Black–Scholes PDE $\Theta + rS\,\Delta + \frac{1}{2}\sigma^2 S^2\,\Gamma = r\,C$; and the delta-hedged P&L identity

$$d\Pi \;\approx\; \tfrac{1}{2}\,\Gamma\,S^2\,(\sigma_{\mathrm{r}}^2 - \sigma_{\mathrm{i}}^2)\,dt,$$

  which this chapter turns into a trading rule.

- Chapter 15: put–call parity and the no-arbitrage bounds. Parity implies that a call and a put at the same (K,T) have the same implied vol, so the surface is labelled by (K,T) alone, not by call-versus-put.

From outside the book:

- Root-finding at the level of bisection or Newton–Raphson. One-dimensional root finding on a monotone function is all the inversion machinery this chapter needs.
- Multivariable functions and their level sets — enough to visualise a surface over a (K, T) plane.

Intuition

"SPX vol is 18%" is a lie of convenience. The market believes in a different volatility for every strike and maturity, and the pattern encodes everything Black–Scholes ignored (fat tails, crash risk, clustered moves, leverage effects). The surface records those beliefs in a form still tradable against Black–Scholes. Reading it tells a derivatives trader in under a minute what the rest of the market thinks is about to happen.

18.1 Implied volatility

Definition 18.1 (Implied volatility)**.** Fix the observable market inputs $(S, K, r, T-t)$ for a European call, and let C^{mkt} be the call's observed mid-market price. The *implied volatility* of the call, written σ_{imp}, is the unique value of the volatility parameter σ that, when substituted into the Black–Scholes formula

$$C^{\text{BS}}(S, K, r, \sigma, T-t) \;=\; S\,\Phi(d_1(\sigma)) \;-\; K\,e^{-r(T-t)}\,\Phi(d_2(\sigma)), \tag{18.1}$$

reproduces the observed price C^{mkt}:

$$C^{\text{BS}}(S, K, r, \sigma_{\text{imp}}, T-t) \;=\; C^{\text{mkt}}. \tag{18.2}$$

Two remarks. First, the definition applies to puts too: put–call parity (Chapter 15) is model-free, so a call and put at the same (K, T) must share σ_{imp}. Second, existence and uniqueness follow from monotonicity: vega $\nu = S\sqrt{T-t}\,\phi(d_1) > 0$ (Chapter 17), so (18.1) is strictly increasing in σ on $(0, \infty)$, ranging from $\max(S - Ke^{-r(T-t)}, 0)$ at $\sigma = 0$ to S as $\sigma \to \infty$. Any price inside those bounds is hit by exactly one σ_{imp}.

Inverting Black–Scholes numerically

No closed form inverts (18.1) in σ: the formula mixes σ both multiplicatively (in d_1) and inside the normal CDF. But since the call price is strictly monotone and smooth in σ, one-dimensional root finders converge quickly. Two choices dominate practice:

1. **Bisection.** Robust but slow. Bracket in $[10^{-6}, 5]$ (5 covers every exchange-traded underlying; vol has never been quoted above 500%) and halve on each step. Reaches machine precision in ~ 50 iterations.
2. **Newton–Raphson.** Uses vega as the derivative: $\sigma_{n+1} = \sigma_n - (C^{\text{BS}}(\sigma_n) - C^{\text{mkt}})/\nu(\sigma_n)$. Converges quadratically, typically in 3–5 iterations. Fails if a step overshoots the feasible interval; standard fix is to fall back to bisection.

Production systems use "safeguarded Newton", bisecting whenever a Newton step leaves the bracket. Scipy's `brentq` is such a hybrid; §18.6 uses it.

Implied volatility as a quoting convention

On every derivatives desk and every options exchange, options are quoted in *vol*, not dollars. A statement like "the 25-delta 3-month `AAPL` put is trading at 32 vol" means the market-observed put premium corresponds to an implied volatility of $\sigma_{\text{imp}} = 32\%$ in the Black–Scholes formula at that strike and maturity. The dollar premium is computable from σ_{imp}, S, and r by one call to C^{BS} or P^{BS}.

Why vol, not dollars

Pricing in vol is pragmatic, not theoretical. Three reasons.

Stability across strikes. A $K = 90$ and $K = 110$ call have very different dollar prices but comparable implied vols. Mispricings jump out on a vol ladder (one strike at 21 vol when its neighbours are at 18) that would be hidden on a dollar ladder by intrinsic-value noise.

Stability across maturities. A 1-month and 1-year call have very different premia but comparable implied vols; quoting in vol factors out the $\sqrt{T-t}$ scaling every option premium obeys.

Stability across spot moves. If spot rallies 2%, every call's dollar price rallies with it, but implied vols barely move (the *sticky-delta* convention of §18.3 makes this almost exact). Vol is roughly conserved under the one noise dimension that would otherwise swamp any quote.

A derivatives desk's screens therefore show strikes × maturities with an implied vol in each cell (bid, ask, mid, diff from yesterday), not dollar prices.

A worked inversion

Take the running anchor from Chapter 16: $S = K = 100$, $r = 0.05$, $T - t = 1$. At $\sigma = 0.20$:

$$C^{\text{BS}}(100, 100, 0.05, 0.20, 1) \;\approx\; 10.4506.$$

Suppose the market shows a richer price, $C^{\text{mkt}} = 11.50$. Since vega is positive, the implied vol must exceed 0.20. Bracketing in $[0.10, 0.40]$: $C^{\text{BS}}(\ldots, 0.10, 1) \approx 6.80 < 11.50$ and $C^{\text{BS}}(\ldots, 0.40, 1) \approx 18.02 > 11.50$, so the root is inside. Brent to machine precision yields

$$\sigma_{\text{imp}} \;\approx\; 0.22788 \;=\; 22.79\%.$$

The ATM year call is priced at 22.79 vol, a 2.79-point premium over the "true" 20 vol. A trader forecasting realised at 20% reads this as a sell signal: delta-hedged short harvests, by the P&L identity of Chapter 17, roughly $\frac{1}{2}\Gamma S^2(\sigma^2_{\text{imp}} - \sigma^2_{\text{real}}) \approx 93.81 \cdot (0.22788^2 - 0.20^2) \approx \$1.12$ per year of gamma exposure. Seed of gamma scalping; developed in §18.5.

Round-trip sanity: feed $\sigma = 0.20$ through BS and invert $C = 10.4506$ back; Brent returns 0.20000 to six decimals. Any production IV inverter should satisfy $\text{IV}(\text{BS}(\sigma)) = \sigma$ to within 10^{-6} over the full (σ, K, T) grid.

18.2 The vol surface

Now vary K and T. Each liquid option is a point (K_i, T_j) with a market price $C^{\text{mkt}}_{i,j}$ and an implied vol $\sigma_{\text{imp}}(K_i, T_j)$ from the inversion of §18.1. Plotting them gives a surface:

Definition 18.2 (Vol surface)**.** Fix an underlying and an instant in time. The *vol surface* (also called *implied vol surface* or simply *surface*) is the function

$$(K, T) \;\mapsto\; \sigma_{\text{imp}}(K, T),$$

defined on the strike–maturity plane wherever a liquid option quote exists. Intermediate points are interpolated (bilinearly for simple desks, via a parametric model — SABR or SVI — on production desks). The interpolation matters because most desks quote exotics and bespoke OTC options at strikes and expiries that do not trade on exchange: reading σ_{imp} off the calibrated surface and feeding it through BS is how those off-grid options get priced.

If Black–Scholes were the truth, the surface would be flat: $\sigma_{\text{imp}}(K, T) \equiv \sigma$. It is not. The true return distribution has fatter tails than the lognormal BS assumes, so wherever the market assigns more probability than BS does — far-OTM strikes, near-term expiries across a known event — options there are worth more than BS says, and the only way to reconcile the richer market price with the BS formula is to feed in a higher σ. Bennett (Bennett, 2014)Ch 4, (Natenberg, 2015) Ch 13, and (Hull, 2014) Ch 18 all show that the surface has systematic structure on both axes, stable enough that practitioners trade the shape itself.

Cross-sections: smile, skew, term structure

The surface is easier to read as two one-dimensional cross-sections than as a 2D plot.

Strike cross-section (smile / skew). Fix T and plot σ_{imp} against K (or moneyness $m = \ln(K/S)$, or BS delta). *Smile* when roughly symmetric and U-shaped; *skew* when it slopes in one direction.

Definition 18.3 (Smile and skew)**.** The *volatility smile* is a strike cross-section of the vol surface whose implied vol is higher at out-of-the-money strikes (both tails, put and call) than at the at-the-money strike, giving the curve a U or smile shape. The *volatility skew* is the asymmetric version: for most equity underlyings the left wing (low strikes, far OTM puts and far ITM calls) rises faster than the right wing, giving a downward-sloping curve.

Equity indices skew because crashes are one-sided: investors fear downside gaps but not upside ones, so OTM puts are bid above OTM calls. FX pairs smile because either currency can collapse relative to the other, so both tails get bid equally.

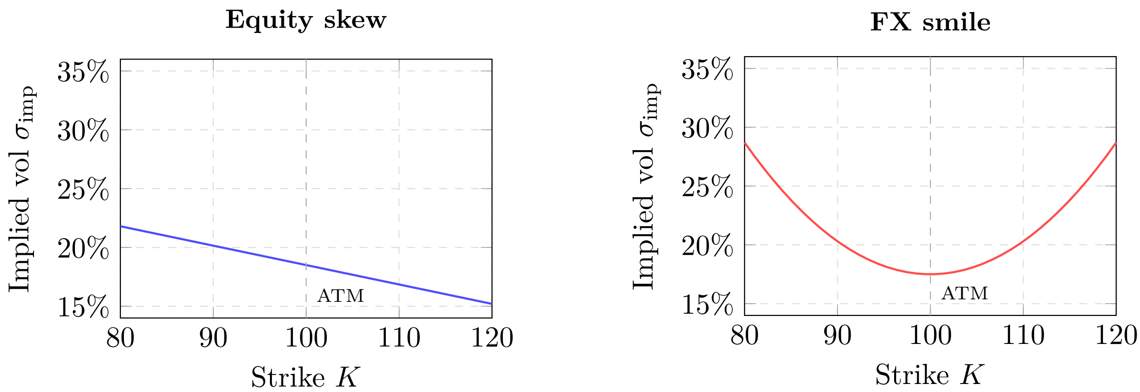


Figure 18.1: Two common vol-surface shapes in practice. **Equity skew** (left): implied vol falls as strike rises. OTM puts are expensive because investors pay a premium for crash protection, so the left wing of the smile is elevated. **FX smile** (right): implied vol is roughly symmetric and U-shaped, reflecting comparable demand for OTM calls and puts. Both shapes contradict Black–Scholes's assumption of a flat, constant vol.

Maturity cross-section (term structure). Fix K and plot $\sigma_{\text{imp}}(K, T)$ against T to get the *volatility term structure*. Calm markets: upward-sloping (*contango*; more uncertainty over a year than a month). Stressed markets: inverted (*backwardation*; a near-term event is being priced in).

Four canonical shapes to memorise

(Bennett, 2014), (Natenberg, 2015), and (Hull, 2014) converge on four qualitative observations. They are the phrasebook you need on an options desk.

(1) Short-dated strike cross-section: sharp smile. A one- or two-week expiry against moneyness shows a sharp U: ATM is a well, both wings rise steeply. A 10-delta OTM put at one week might trade at 45 vol vs ATM at 20 — a 25-point premium. Tail risk over the next week is disproportionately priced; short-dated crash insurance is expensive.

(2) Long-dated strike cross-section: flatter shape. A one-year expiry on the same underlying shows a gentler curve; wings rise by perhaps 3–5 points rather than 25. Over longer horizons the return distribution is closer to Gaussian (CLT), and Black–Scholes fits the tail better.

(3) Equity-index skew: downward slope. SPX, SX5E, N225, and other major indices exhibit a persistent downward skew: the strike cross-section slopes down from low to high strikes. A 90-strike SPX put at three months might imply 25 vol, ATM 18, 110-strike call 16. Demand for downside crash protection structurally exceeds demand for upside speculation, so put implied vol is bid above call implied vol (more in §18.2).

(4) ATM term structure: contango vs backwardation. Normal markets: ATM vol slopes up with maturity (1-month $<$ 3-month $<$ 6-month $<$ 1-year). Typical SPX calm: $14, 16, 17, 18$ vol. Stress (Brexit 2016, COVID 2020, GFC 2008): inverted, 1-month spikes to 40 while 1-year only moves to 25.

The vol surface

The *vol surface* is the function $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$ obtained by inverting Black–Scholes against every liquid option quote on a given underlying at a given instant. Empirically the surface is never flat. It exhibits (a) a *smile* or *skew* along the strike axis, sharper at short maturities and flatter at long maturities, with a persistent downward skew on equity indices; and (b) a *term structure* along the maturity axis, typically upward-sloping (contango) in calm markets and downward-sloping (backwardation) in stress. Reading the surface is how a derivatives trader diagnoses what the market is pricing in.

Intuition

The vol surface is the market's *correction* to Black–Scholes. BS assumes constant vol; the surface records that it isn't. A derivatives desk never uses BS with a single vol: it uses BS plus a *calibrated surface*, so every option it prices is consistent with every other option in the market. The formula is a local coordinate; the surface is the global manifold; traders trade positions in that manifold.

Where the smile came from: the 1987 crash

Before October 1987, listed option quotes across strikes traded at roughly the same implied vol — the smile did not exist. Flat vol was a pillar of early BS empirical validation.

Historical note

On **Monday, October 19, 1987** — Black Monday — the S&P 500 fell 20.5% in one session. Under BS with the 15–18 vol equity options were trading at, the implied probability of a -20% daily move was roughly 10^{-59}. Reality disagreed.
From Tuesday onward, traders re-priced far-OTM puts to much higher vols than ATM: the crash had shown the tail was far fatter than BS assumed, and crash insurance became permanently bid. Derman (Derman, 2004) Ch 9 describes the floor's overnight reshaping; the constant-vol framework was replaced by the vol-surface framework within two weeks. The smile has never gone away. Bennett (Bennett, 2014) is the canonical practitioner reference; free to download and required reading for this chapter.

18.3 Sticky strike vs sticky delta

The surface depends on the current spot S through a strike's relationship to ATM. When spot moves, the surface moves with it; two competing conventions describe *how*.

Definition 18.4 (Sticky strike)**.** Under the *sticky-strike* assumption, the implied volatility at a fixed strike K is invariant to the underlying's level: if SPX moves from $S = 4000$ to $S = 4050$, the implied vol of the $K = 4000$ call stays the same. The entire smile remains pinned to absolute strike levels.

Definition 18.5 (Sticky delta)**.** Under the *sticky-delta* assumption (sometimes called *sticky moneyness*), the implied volatility at a fixed Black–Scholes delta — equivalently, at a fixed moneyness $m = \ln(K/S)$ — is invariant to the underlying's level. If SPX moves from 4000 to 4050, the entire smile *translates* with the spot: the 25-delta put vol stays where it was, but the strike that gives 25 delta has moved.

These differ in how the surface moves with S, and hence in the correct delta hedge.

Effect on model delta

BS delta $\Phi(d_1)$ was derived with σ independent of S. Under sticky-delta, moving S also moves the implied vol of the strike I hold (its moneyness changed), so the "effective" delta picks up an extra term. Write the call price as a function of S through both channels:

$$C(S) \;=\; C^{\mathrm{BS}}(S, K, r, \, \sigma_{\mathrm{imp}}(S), \, T - t),$$

where under sticky-delta $\sigma_{\mathrm{imp}}(S)$ is the implied vol of the point on the surface that has moved with the spot. Applying the chain rule,

$$\frac{dC}{dS} \;=\; \underbrace{\frac{\partial C^{\mathrm{BS}}}{\partial S}}_{\text{BS delta}} + \underbrace{\frac{\partial C^{\mathrm{BS}}}{\partial \sigma}}_{\text{vega}} \cdot \underbrace{\frac{d\sigma_{\mathrm{imp}}}{dS}}_{\text{“skew slide”}} \;. \tag{18.3}$$

The first term is the usual $\Phi(d_1)$; the second corrects for how implied vol moves with spot. On an equity index with a downward skew and sticky-delta dynamics, $d\sigma_{\mathrm{imp}}/dS < 0$ (a rally slides my strike rightward along the downward skew, so its implied vol drops); vega is positive, so the correction *reduces* effective delta below BS. A fall does the reverse.

Sticky-strike is the limit $d\sigma_{\mathrm{imp}}/dS = 0$: effective delta equals BS delta.

Sticky-strike vs sticky-delta effective delta

$$\begin{aligned} \Delta^{\text{sticky strike}}_{\text{effective}} &= \Delta^{\text{BS}}, \\ \Delta^{\text{sticky delta}}_{\text{effective}} &= \Delta^{\text{BS}} + \nu \cdot \frac{d\sigma_{\text{imp}}}{dS}. \end{aligned}$$

The correction term is negative for equity indices (downward skew implies $d\sigma_{\text{imp}}/dS < 0$), so sticky-delta hedging on an equity book uses a *smaller* share count than BS delta would suggest.

Which convention is empirically right?

Both conventions are stylised; neither is exactly right. Bennett (Bennett, 2014) Ch 5 gives the empirical breakdown: equity indices tend to sticky-delta in calm regimes (smile translates with spot) and sticky-strike in stress (smile steepens). Single names are more sticky-strike; FX is sticky-delta; rates have their own conventions (sticky-log-normal, or sticky-basis-point OTC).

Desks pick one convention, backtest it against recent spot moves, and adjust the model delta accordingly.

The "BS delta" depends on your sticky convention

Using raw $\Phi(d_1)$ as a hedge ratio on an equity-index option implicitly assumes sticky-strike. If the market is sticky-delta (usual for SPX in calm regimes), the hedge is wrong and the "hedged" P&L drifts systematically with spot. Every desk corrects the model delta for its chosen convention: $\Delta^{\text{BS}} + \nu \cdot d\sigma_{\text{imp}}/dS$, with the slide estimated from the observed skew. Ignoring this costs a long-dated equity market-making book several basis points a day in unexplained P&L.

18.4 Variance swaps (qualitative)

The vol surface is read by eye and traded one option at a time, but a second product trades *pure* volatility directly: the *variance swap*. Full replication is deferred to Chapter 31; here we give the intuition, a schematic formula, and why variance swaps are the cleanest vol trade on a sell-side desk.

Definition 18.6 (Variance swap)**.** A *variance swap* on an underlying S is an OTC derivative with the following cashflow at maturity T:

$$\text{Payoff} = N \cdot \left(\sigma^2_{\text{realised}}(0,T) - K^2_{\text{var}}\right),$$

where $\sigma^2_{\text{realised}}(0,T)$ is the realised variance of $\ln(S)$ over $[0,T]$, K_{var} is the *variance strike* (expressed in vol units and squared at settlement), and N is a notional chosen so that a 1-vol-point move in realised vol corresponds to a fixed dollar P&L (the "vega notional" convention). The swap pays nothing at initiation; K_{var} is set at trade time so that the expected risk-neutral value of the payoff is zero.

A variance swap is a pure bet on realised volatility. Unlike a delta-hedged option (with residual path-dependence, gamma-non-uniformity, and discrete-hedging risks of Chapter 17), its payoff is the integral of squared returns alone: no gamma to manage, no rebalancing schedule, no path where realised is "high" but the position loses from bad hedge timing. If realised variance exceeds the strike, you win by the excess.

Replication via a strip of OTM options

Demeterfi–Derman–Kamal–Zou showed (expanded in (Bennett, 2014) Ch 12 and (Hull, 2014) Ch 26) that a variance swap is replicable *model-independently* by a static European-option portfolio plus a dynamic forward position. The option leg is a continuous strip of OTM puts and calls, weighted by $1/K^2$:

Variance swap replication (schematic)

A T-maturity variance swap on a non-dividend-paying underlying with forward $F_0 = S_0 e^{rT}$ is replicated by holding, at time 0:

$$\Pi^{\text{static}} = \frac{2}{T}\left[\int_0^{F_0} \frac{P(K)}{K^2}\,dK + \int_{F_0}^{\infty} \frac{C(K)}{K^2}\,dK\right] - \text{(dynamic forward hedge)}, \quad (18.4)$$

where the continuous integrals are understood as a sum over available strikes with weights $w_i \propto 1/K_i^2$. At maturity this static portfolio pays exactly the realised variance $\sigma^2_{\text{realised}}(0,T)$, up to a residual that vanishes as the strike grid becomes finer.

The derivation — the log-payoff $2\ln(F_T/F_0)$ is replicable as a $1/K^2$-weighted integral over OTM puts and calls — is in Chapter 31 and (Hull, 2014) §26.16. Three structural takeaways:

The $1/K^2$ weight is not a fudge. It matches the second derivative of the log-payoff w.r.t. the forward. Any other weighting replicates a different payoff.

Variance swaps are long convexity. Variance is vol squared, so a doubling of vol quadruples the payoff. A *volatility swap* (linear in σ_{realised}) lacks this convexity, is harder to replicate, and is usually priced as a variance swap plus a convexity correction.

OTM puts get the lion's share. Low strikes contribute disproportionately under $1/K^2$; SPX variance swaps are dominated by OTM put exposure, so their bid/offer tracks put skew closely.

Intuition

A variance swap is what a delta-hedged option aspires to be — a pure vol bet — but without the gamma-path and discrete-hedging noise. The replication formula says: stop hedging one call; hold a weighted strip of every OTM option, and the strip itself prices pure variance. Sell-side desks quote the strip and trade it against OTC variance-swap confirms. Variance-swap trading is *industrial-scale* gamma scalping.

Chapter 31 gives the full DDKZ derivation, the variance-vs-volatility-swap convexity correction, and the desk workflow for a typical variance-swap ticket.

18.5 Gamma scalping as an explicit strategy

We turn the P&L identity of Chapter 17 into a rule. For a delta-hedged long-option position,

$$d\Pi \approx \tfrac{1}{2}\,\Gamma\, S^2\left(\sigma_{\mathrm{r}}^2 - \sigma_{\mathrm{i}}^2\right)dt. \quad (18.5)$$

Every P&L dollar on a delta-hedged book is the integrated realised-minus-implied gap weighted by gamma. The market prices σ_{i} through the surface; a trader who believes realised will exceed implied has a directly actionable rule: buy the option, delta-hedge, collect the gap. That is gamma scalping.

The strategy rule

Definition 18.7 (Gamma scalping)**.** *Gamma scalping* is the strategy of (i) buying an option (or a basket of options) whose implied volatility is below the trader's forecast of realised volatility over the option's life; (ii) simultaneously shorting the underlying in the Black–Scholes delta amount to remain approximately delta-neutral; (iii) rebalancing the delta hedge on a fixed schedule (every K ticks, every $\$\delta$ of spot move, or every T/N seconds); and (iv) collecting the cumulative $\frac{1}{2}\Gamma(\Delta S)^2$ slugs against the theta cost of holding the option.

Step by step, running against the chapter's running example ($S_0 = 100$, $K = 100$, $r = 5\%$, $T = 1$, implied $\sigma_{\rm i} = 20\%$):

1. **Forecast realised vol.** From trailing 30-day realised vol of log-returns, or a GARCH/ARMA forecast, project $\sigma_{\rm r}$. Suppose $\sigma_{\rm r} = 25\%$, 5 points above implied.
2. **Price-check each strike.** Invert every liquid quote to $\sigma_{\rm imp}(K_i)$. Any strike with $\sigma_{\rm imp} < \sigma_{\rm r}$ is a buy candidate. Here the ATM call at 20 vs a forecast of 25 qualifies.
3. **Buy the option.** Pay $C \approx 10.4506$ per unit; record entry $\sigma_{\rm imp} = 0.20$.
4. **Short the delta.** $\Delta \approx 0.6368$, so short 0.6368 shares per call to go delta-neutral.
5. **Rebalance on a schedule.** As S drifts, the call's delta drifts too; reset the share hedge each rebalance.
6. **Cumulative P&L.** Each window books $\frac{1}{2}\Gamma(\Delta S)^2$ − theta; over the year the integral approaches $\frac{1}{2}\Gamma S^2(\sigma_{\rm r}^2 - \sigma_{\rm i}^2)T$.
7. **Exit.** Either close out before expiry (booking the cumulative gamma P&L) or hold to expiry (call payoff plus cumulative hedge P&L gives the same total).

The numerical example

At the anchor, $\frac{1}{2}\Gamma S^2 = \frac{1}{2} \cdot 0.01876 \cdot 100^2 = 93.81$ dollars per vol^2-year. If realised prints at 25% against implied 20%:

$$\begin{aligned}
\tfrac{1}{2}\,\Gamma\,S^2\,(\sigma_{\rm r}^2 - \sigma_{\rm i}^2) &= 93.81 \cdot (0.0625 - 0.04) \\
&= 93.81 \cdot 0.0225 \\
&\approx \$2.11 \text{ per unit of call, per year.}
\end{aligned}$$

Against a \$10.45 premium, that is $\sim 20\%$ of premium over one year — a leveraged, convex, pure play on whether realised beats 20%. Canonical trade of the vol desk.

If the forecast is wrong and $\sigma_{\rm r} = 18\%$, P&L flips:

$$93.81 \cdot (0.0324 - 0.04) \;=\; -\$0.71 \text{ per unit per year,}$$

about 7% of premium bled. Theta dominated the gamma gain. The failure mode: *you paid for vol and the market didn't deliver.*

Gamma scalping in one paragraph

Buy options whose implied vol is below your forecast of realised vol. Short the Black–Scholes delta in the underlying to stay neutral. Rebalance the delta on a schedule. Over the life of the hedge, collect $\sim \frac{1}{2}\Gamma S^2(\sigma_{\rm r}^2 - \sigma_{\rm i}^2)T$ per unit of call. Win if realised exceeds implied; lose if not. The strategy *is* the delta-hedged P&L identity, executed.

Intuition

Gamma scalping trades your vol forecast against the market's implied. Owning the option is "long vol"; shorting it is "short vol"; delta-hedging strips out direction; what remains is the pure vol bet. You are not betting on up-or-down — the hedge eliminates that — but on whether the underlying *moves a lot or a little*. Long gamma profits on every kind of move; you just need enough, fast enough, to outpace the theta clock.

Failure modes and subtleties

Three failure modes bite.

Realised vol disappoints. Forecast was 25, realisation was 18, 7% of premium bled. No hedging expertise saves a wrong vol forecast.

Implied vol moves during the trade. The clean "$\sigma_\mathrm{r}^2 - \sigma_\mathrm{i}^2$" story treats implied as frozen at entry. In reality the surface fluctuates; an overnight selloff delivers a vega hit $\nu \cdot \Delta\sigma_\mathrm{i}$ unrelated to realised vol. At the anchor $\nu \approx 37.52$, so a 1-point drop is $-\$0.375$ on a \$10.45 call, $\sim 3.6\%$ of premium. The realised-vs-implied story holds only if you exit at the entry σ_i, or budget for the vega.

Discrete-hedging error. Chapter 17 showed discrete rebalancing adds per-period noise $\sim \Gamma S^2\sigma^2\Delta t$. Weekly vs daily gives $\sim 5\times$ more noise per rebalance but $\sim 5\times$ fewer rebalances, so cumulative sigma is roughly unchanged; transaction costs decide. Rebalance frequency is a free parameter; optimising it is standard work.

Gamma scalping is a vega bet in disguise

The sloppy pitch — "buy the call, delta-hedge, win iff realised exceeds implied" — is not quite right. If implied *rises* after entry, the long call rallies via vega and you profit mark-to-market even if realised is average; if implied *falls*, you lose on vega even if realised is high. The pure realised-vs-implied story holds only to expiry or if you exit at entry implied. In practice gamma scalping is partly a vega bet, and a disciplined desk books realised-vol P&L and vega P&L separately so the trader knows which contributed.

Regime detection and forward reference

Whether gamma scalping is right *now* is a regime-detection question. Calm markets with implied above long-run realised favour short gamma (vol selling); building macro stress with implied lagging favours long gamma. The regime-to-strategy decision tree is in Chapter 23. For now: gamma scalping is the options-specific wing of a general "vol forecast vs vol implied" framework.

Prosperity example

IMC Prosperity 3 introduced `VOLCANIC_ROCK_VOUCHERS` in Round 3 and kept them through Round 4: five strikes $(9500, 9750, 10000, 10250, 10500)$ on `VOLCANIC_ROCK` trading around 10000, European calls, a few days to expiry. Inverting each voucher's mid through BS ($r = 0$ in Prosperity) gives five implied vols at one maturity — a genuine smile cross-section.

The top teams' workflow, from the Frankfurt Hedgehogs' second-place writeup (Chapter 27):

1. Invert each voucher's mid to $\sigma_{\mathrm{imp}}(K)$.
2. Fit a parabola to get a smooth smile.

3. Compute residuals $\sigma_{\text{imp}} - \hat{\sigma}_{\text{smile}}$: negatives are cheap, positives are rich.
4. Buy cheap strikes, sell rich, delta-hedge each with `VOLCANIC_ROCK` at the BS delta.

Rebalancing every 10 ticks — exactly the discrete-hedging setup above. The ATM 10,000 voucher was the single biggest P&L contributor: its implied oscillated around the smile fit, each oscillation a miniature gamma-scalping round. The Hedgehogs call IV scalping on `VOLCANIC_ROCK_VOUCHERS` "a significant profit centre" in Rounds 3–4. See `imc-prosperity-3/README.md` Rounds 3–4.

On a Goldman Sachs desk

A GS equity-derivatives vol-trading Strat's morning ritual is to *recalibrate the vol surface* to overnight quotes:

1. Pull overnight trades and quotes on the coverage universe (SPX, SX5E, NDX, RUT, top 200 US single names).
2. For each (K, T) with a fresh mid, invert BS to σ_{imp} via a `brentq`-style routine (§18.6). GS's library `SecDB` holds one of the most-called IV inverters on the firm.
3. Fit a parametric surface. Dominant choices: **SABR** (four-parameter stochastic-vol, smile per maturity, closed-form IV approximation) and **SVI** (five-parameter static parameterisation of σ^2_{imp} in moneyness). Full derivations in Chapter 31; here treat as black-box fitters.
4. Diff against yesterday's surface. Any (K, T) moved beyond a threshold (0.5–1.0 points) is flagged.
5. Trader fades obvious dislocations; risk re-prices the book against the calibrated surface, producing fresh P&L attribution and Greeks.

The whole equity-derivatives business — market making, flow, exotics, variance swaps, delta-one — rests on one consistent calibrated surface. Two desks using different models book inconsistent P&Ls and the internal auditors flag within a week. The surface is a shared operational artefact every risk number depends on. Chapter 36 returns to the production engineering.

18.6 A short reference implementation

The core of every vol-surface pipeline is the IV inverter: a function taking a market price and returning the BS implied vol. Minimal version below, using scipy's `brentq` (safeguarded Brent; never leaves the bracket). Commit it to muscle memory; everything else in this chapter calls it.

```python
from scipy.optimize import brentq
from scipy.stats import norm
import math

def bs_call(S, K, r, sigma, T):
    d1 = (math.log(S / K) + (r + sigma ** 2 / 2) * T) / (sigma *
        math.sqrt(T))
    d2 = d1 - sigma * math.sqrt(T)
    return S * norm.cdf(d1) - K * math.exp(-r * T) * norm.cdf(d2)

def implied_vol(S, K, r, T, C_market):
```

```
    f = lambda sigma: bs_call(S, K, r, sigma, T) - C_market
    return brentq(f, 1e-6, 5.0)
```

Listing 18.1: Implied-vol inverter for a European call under Black–Scholes. `bs_call` returns the BS price; `implied_vol` finds the volatility that reproduces an observed market price via Brent's method. Calling `implied_vol(100, 100, 0.05, 1, 10.4506)` returns approximately 0.2000; calling it on a market price of 11.50 returns approximately 0.2279.

Two notes. `brentq` requires a sign-change bracket; $[10^{-6}, 5]$ covers every realistic equity IV (widen for exotics above 500%). No caching: production adds a Newton step for speed (Brent ~ 20 evals, Newton ~ 5) and caches $d_1, d_2, \Phi(d_1)$ across Greeks of the same (S, K, r, σ, T). The minimal version suffices for this chapter and any Prosperity workload.

18.7 Key facts to memorise

- **Implied volatility** of an observed call price C^{mkt} is the unique σ that reproduces C^{mkt} when plugged into the Black–Scholes formula. Existence and uniqueness follow from the monotonicity of the BS price in σ (vega > 0). Inversion is numerical — Brent or Newton — because there is no closed form.
- **Quoting convention.** Options are quoted in vol units, not dollars. A "32-vol" put means the BS inversion of the market premium gives $\sigma_{\text{imp}} = 0.32$. Traders read entire option chains as vol ladders.
- **Round-trip sanity check.** For any vol σ, $\text{IV}(\text{BS}(\sigma)) = \sigma$ to machine precision. At the running anchor, $\text{BS}(100, 100, 0.05, 0.20, 1) \approx 10.4506$ and $\text{IV}(100, 100, 0.05, 1, 10.4506) = 0.20000$. This round-trip is the arithmetic-hygiene test of any IV pipeline.
- **Vol surface.** The function $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$. Empirically never flat, despite Black–Scholes assuming it should be.
- **Strike cross-section.** Short-dated maturities show a sharp U-shaped *smile*; long-dated maturities show a flatter curve. On equity indices, the cross-section has a persistent downward *skew*: OTM puts trade at higher vol than ATM, which trades at higher vol than OTM calls.
- **Maturity cross-section.** *Term structure* of ATM vol is upward-sloping (contango) in calm markets and inverted (backwardation) in stress. SPX 1-month vs 1-year ATM in calm: 14 vs 18. In March 2020: 80 vs 35.
- **Sticky strike vs sticky delta** distinguishes how the surface moves with spot. Sticky strike: vol at absolute strike pinned; model delta = BS delta. Sticky delta: vol at fixed moneyness pinned, smile translates with spot; model delta = BS delta $+\nu \cdot d\sigma_{\text{imp}}/dS$. Most equity indices are closer to sticky-delta in calm regimes.
- **1987 origin of the smile.** Before Black Monday (October 19, 1987), option vols across strikes were roughly flat. After the 20.5% single-session crash, OTM puts re-priced to pay a permanent crash-insurance premium, and the downward equity skew has been in place ever since.
- **Variance swap.** An OTC contract paying $\sigma^2_{\text{realised}} - K^2_{\text{var}}$ at maturity. Replicable by a static $1/K^2$-weighted strip of OTM puts and calls plus a dynamic forward hedge. The cleanest way to trade pure vol. Full derivation in Chapter 31.

- **Gamma scalping.** Buy options whose implied vol is below your realised-vol forecast. Delta-hedge with shares in the BS delta. Rebalance on a schedule. Collect $\sim \frac{1}{2}\Gamma S^2(\sigma_{\mathrm{r}}^2 - \sigma_{\mathrm{i}}^2)T$ per unit over the life of the position. The trade is a vega bet in disguise unless you exit at the entry implied vol.

- **Running numerical anchor.** ($S = K = 100, r = 5\%, \sigma = 20\%, T = 1$): $C \approx 10.4506$; $\frac{1}{2}\Gamma S^2 \approx 93.81$; if realised prints at 25 vs implied at 20, annualised gamma-scalping P&L is $\approx 93.81 \cdot 0.0225 = \$2.11$ per unit call, about 20% of premium. If realised prints at 18 instead, the P&L is $\approx -\$0.71$, a 7% loss.

- **On a desk.** Morning ritual is to recalibrate the surface to overnight quotes (SABR or SVI parameterisation), diff against yesterday, flag dislocations for the trader, and re-price the whole vol book. The surface is a shared operational artefact; every risk number in the building depends on it being consistent.

Part VI

Execution, Risk, and Methodology

Chapter 19

Optimal Execution: Almgren–Chriss

You need to sell 1,000,000 shares of a stock by the end of the day. Selling them all at once walks the order book and moves the market against you by tens of basis points. Trailing them out one-by-one over the full session exposes you to all-day price risk, where a random adverse drift of half a percent on tens of millions of dollars notional wipes out a week of desk P&L. Between the two extremes sits a schedule — how many shares to sell at each minute — that balances rushing against dawdling. Under linear impact and Gaussian price noise, "when should you sell each share" has a clean closed-form answer, and this chapter derives it from scratch.

The answer is the **Almgren–Chriss (A–C) optimal execution trajectory** (Almgren and Chriss, 2000), the benchmark every sell-side execution desk measures algorithms against. It *assumes linear market impact*, directly contradicting the empirical square-root law of Chapter 7; reconciling the two is part of this chapter's job. The objective resembles the inventory-averse market maker of Chapter 9 (the dual problem: earn spread subject to inventory risk vs. pay impact subject to price risk on unsold inventory), but here the closed form is in elementary functions and we derive it by calculus of variations on finite sums.

Prerequisites

From earlier in the book:

- Chapter 7: the empirical square-root law $I(Q) \approx Y\sigma\sqrt{Q/V}$; temporary (reverting) vs. permanent (information-revealing) impact; linear impact over-predicts cost by a factor $\sqrt{Q_{\text{exec}}/Q_{\text{calib}}}$ when extrapolated out of calibration range. This chapter assumes linear impact for tractability; the contradiction is flagged in Section 19.8.

- Chapter 9: the Avellaneda–Stoikov inventory-averse market maker. A market maker *earns* spread subject to price risk on inventory; the A–C liquidator *pays* impact subject to price risk on not-yet-liquidated inventory. Same mean-variance objective and HJB machinery, opposite ends of the client–dealer relationship. Warning: A–S uses γ for the market-maker's risk aversion; here γ is the permanent-impact coefficient (pitfallbox below).

- Chapter 10: mean-variance thinking — fix an expected cost, minimise variance, or vice versa. A–C applies it to a schedule rather than a scalar.

From outside the book:

- Elementary **Lagrange multipliers** / discrete calculus of variations: differentiating a quadratic in a trajectory $\{x_k\}$ under a linear terminal constraint.

- **Linear recurrences**: the recurrence $x_{k-1} - 2\cosh(\kappa\tau)x_k + x_{k+1} = 0$ has general solution $x_k = A\sinh(\kappa k\tau) + B\cosh(\kappa k\tau)$, fixed by two boundary conditions (same $y_k = r^k$ machinery as solving $y_{k-1} - 2y_k + y_{k+1} = 0$).
- **Tridiagonal linear algebra**: the first-order conditions form a tridiagonal system, the matrix version of the recurrence.
- Mean-variance thinking: minimising $\mathbb{E}[C] + \lambda\,\mathrm{Var}[C]$ traces an efficient frontier as λ ranges over $[0, \infty)$. No schedule is “optimal” without specifying λ.

19.1 The liquidation problem

Symbol anchor. X initial position, T deadline, N intervals, $\tau = T/N$, σ vol, η temporary-impact slope, γ permanent-impact slope, λ risk aversion, $\kappa = \sigma\sqrt{\lambda/\eta}$ time-constant.

A *liquidator* holds X shares at $t = 0$ and must hold 0 at $t = T$, trading at $N + 1$ equally spaced times $t_0 = 0 < t_1 < \ldots < t_N = T$ with spacing $\tau := T/N$. Write x_k for the **holding after trade** k: $x_0 = X$, $x_N = 0$. The **trade at step** k is the signed change in holding,

$$n_k \;:=\; x_{k-1} - x_k, \qquad k = 1, 2, \ldots, N. \tag{19.1}$$

Positive n_k is a sell (holding falls), negative a buy. Trades sum to the full liquidation,

$$\sum_{k=1}^{N} n_k \;=\; x_0 - x_N \;=\; X, \tag{19.2}$$

so the constraint $\sum n_k = X$ is automatic given $x_0 = X, x_N = 0$; no separate Lagrange multiplier is needed.

Three cost sources

Each share sold incurs cost from three distinct mechanisms. (Almgren and Chriss, 2000) writes each as an affine function of trading rate $v_k := n_k/\tau$ (shares per unit time in period k), with separate slopes.

Temporary impact $g(v)$. Selling n_k shares in the k-th interval consumes visible liquidity on the bid side. The market maker fills the flow below the no-pressure mid; once the interval ends the book refills and the temporary component vanishes. (Almgren and Chriss, 2000) models this as

$$g(v) \;=\; \epsilon\,\mathrm{sign}(v) \;+\; \eta\,v, \tag{19.3}$$

with two constants:

- $\epsilon > 0$: fixed half-spread per share. Every market order crosses the bid–ask by at least ϵ. For a pure liquidation $\epsilon \sum |n_k| = \epsilon X$ is a constant offset and drops out of every optimisation.
- $\eta > 0$: linear slope on rate. Twice the rate, twice the temporary impact per share — the term that makes “rush” expensive and shapes the optimal schedule.

With constant rate over the interval, the average execution price in period k is the starting mid minus $g(n_k/\tau)$. Total temporary-impact cost over the session:

$$C_{\text{temp}} \;=\; \sum_{k=1}^{N} n_k \cdot g(n_k/\tau) \;=\; \epsilon X \;+\; \frac{\eta}{\tau}\sum_{k=1}^{N} n_k^2. \tag{19.4}$$

The constant ϵX will be dropped everywhere below.

Permanent impact $h(v)$. Selling signals that somebody thinks fair value has moved; the market updates its belief and the mid drifts down. Unlike the temporary component, this drift does not revert — it lives on the efficient price. (Almgren and Chriss, 2000) models it linearly:

$$h(v) \;=\; \gamma\, v, \tag{19.5}$$

with $\gamma > 0$. Over interval k the mid drifts down by $\tau \cdot h(n_k/\tau) = \gamma n_k$.

A–C γ is *not* A–S γ

In Chapter 9, Avellaneda–Stoikov's reservation price $r = S - q\,\gamma\,\sigma^2(T-t)$ used γ for the market-maker's exponential-utility risk aversion. Same Greek, but here γ is the *permanent-impact slope* $h(v) = \gamma v$ — a venue/asset property, not a trader preference. For the A–C liquidator, "risk aversion" is λ, introduced below. Henceforth: γ means permanent-impact coefficient, λ means liquidator urgency.

Price risk $\sigma\sqrt{\tau}\xi_k$. The mid also moves for reasons unrelated to the liquidator: other flow, news, macro. (Almgren and Chriss, 2000) models this as arithmetic Brownian motion with vol σ per unit time, discretised into i.i.d. Gaussian shocks:

$$S_k \;=\; S_{k-1} \;+\; \sigma\sqrt{\tau}\,\xi_k \;-\; \tau\, h(n_k/\tau), \qquad \xi_k \sim \mathcal{N}(0,1) \text{ i.i.d.} \tag{19.6}$$

The Gaussian ξ_k is the random part; $-\tau h(n_k/\tau)$ is the deterministic permanent-impact drift per interval. (Almgren and Chriss, 2000) uses arithmetic rather than geometric Brownian motion because over a single day the difference is second-order in total cost, and arithmetic BM admits closed forms where geometric does not.

The six A–C parameters plus one

Six model parameters plus one decision parameter:

Symbol	Meaning
X	Total shares to liquidate
T	Liquidation deadline
N	Number of equal-length trade intervals
σ	Arithmetic volatility of the underlying per unit time
η	Temporary-impact slope (linear part of g)
γ	Permanent-impact slope ($h(v) = \gamma v$)
λ	Liquidator risk aversion (mean-variance penalty)

The half-spread ϵ is a seventh parameter, but since $\epsilon \sum |n_k| \;=\; \epsilon X$ is constant over schedules, it does not affect the optimal trajectory.

19.2 The mean-variance objective

Total cost of a schedule $\{n_k\}$ (equivalently trajectory $\{x_k\}$) is a random variable; we form a scalar objective that trades its mean against its variance. The tension is the whole point: trading fast pays more impact cost (each child order eats deeper into the book), trading slow leaves unsold shares exposed to adverse random drift for longer. Expected cost penalises the former, variance penalises the latter; A–C picks the schedule that minimises their weighted sum. The next two subsections unpack each in turn.

Expected cost

At the start of period k the mid is S_{k-1}. The liquidator sells n_k at average price $S_{k-1} - \eta n_k/\tau$ (dropping the ϵ constant), so cash received is $n_k(S_{k-1} - \eta n_k/\tau)$. The **implementation shortfall** ("slippage") is what the liquidator would have received selling X instantly at S_0 minus what they actually receive:

$$C \;:=\; X\,S_0 \;-\; \sum_{k=1}^{N} n_k\left(S_{k-1} - \frac{\eta n_k}{\tau}\right). \tag{19.7}$$

Iterating (19.6) gives $S_{k-1} = S_0 + \sigma\sqrt{\tau}\sum_{j=1}^{k-1}\xi_j - \gamma\sum_{j=1}^{k-1} n_j$; expanding the sum yields ((Almgren and Chriss, 2000) eq. (4))

$$C \;=\; \underbrace{\sum_{k=1}^{N}\tau\,\gamma\,x_k\,\frac{n_k}{\tau}}_{\text{permanent impact}} \;+\; \underbrace{\frac{\eta}{\tau}\sum_{k=1}^{N} n_k^2}_{\text{temporary impact}} \;-\; \underbrace{\sigma\sqrt{\tau}\sum_{k=1}^{N} x_k\,\xi_k}_{\text{random price risk}}. \tag{19.8}$$

Permanent impact is deterministic and quadratic in the trajectory; temporary impact is deterministic and quadratic in the rates; the price-risk term is the only random one, linear in Gaussian shocks.

Taking the expectation, only the first two survive:

$$\mathbb{E}[C] \;=\; \gamma\sum_{k=1}^{N} x_k\,n_k \;+\; \frac{\eta}{\tau}\sum_{k=1}^{N} n_k^2. \tag{19.9}$$

The permanent-impact sum $\sum x_k n_k$ simplifies by a telescoping identity central to everything below. With $n_k = x_{k-1} - x_k$,

$$\sum_{k=1}^{N} x_k n_k \;=\; \sum_{k=1}^{N} x_k(x_{k-1} - x_k) \;=\; \sum_{k=1}^{N}\left(\tfrac{1}{2}(x_{k-1}^2 - x_k^2) - \tfrac{1}{2}n_k^2\right) \;=\; \tfrac{1}{2}(X^2 - 0) - \tfrac{1}{2}\sum_{k=1}^{N} n_k^2, \tag{19.10}$$

using $x_k(x_{k-1} - x_k) = \frac{1}{2}(x_{k-1}^2 - x_k^2) - \frac{1}{2}(x_{k-1} - x_k)^2$ term-by-term and telescoping. Expected cost becomes purely quadratic in the n_k, plus the constant $\frac{1}{2}\gamma X^2$:

$$\mathbb{E}[C] \;=\; \tfrac{1}{2}\gamma X^2 \;+\; \frac{1}{\tau}(\eta - \tfrac{1}{2}\gamma\tau)\sum_{k=1}^{N} n_k^2. \tag{19.11}$$

The *adjusted temporary-impact coefficient* is

$$\tilde{\eta} \;:=\; \eta \;-\; \tfrac{1}{2}\gamma\tau, \tag{19.12}$$

which replaces η in the discrete problem. The $\frac{1}{2}\gamma\tau$ correction is the most-missed detail in re-derivations of A–C: discrete uses $\tilde{\eta}$, not η. In the $\tau \to 0$ limit (Section 19.4) $\tilde{\eta} \to \eta$, so continuous-time uses the un-adjusted η.

Variance of cost

The only random term in (19.8) is $-\sigma\sqrt{\tau}\sum_k x_k\xi_k$. Its variance:

$$\operatorname{Var}[C] \;=\; \sigma^2\tau\,\operatorname{Var}\left[\sum_{k=1}^{N} x_k\xi_k\right] \;=\; \sigma^2\tau\sum_{k=1}^{N} x_k^2, \tag{19.13}$$

by independence of the ξ_k. The sum $\sum x_k^2$ is the discrete L^2-norm of the inventory path, so **variance is the area under the squared-inventory curve**, weighted by $\sigma^2\tau$. A slow schedule (large x_k throughout) carries more variance — price moves act on more shares for longer.

The objective

(Almgren and Chriss, 2000) combines them into a linear **mean-variance objective**:

$$U(\{n_k\}) \;:=\; \mathbb{E}[C] + \lambda\,\mathrm{Var}[C], \qquad \lambda \geq 0. \tag{19.14}$$

Minimise U subject to $x_0 = X, x_N = 0$. λ encodes how much variance the trader accepts to save a unit of expected cost: large λ = urgent, risk-averse ("get me out, I don't want it overnight"); small λ = patient, risk-tolerant ("minimise cost, I'll wear the mark-to-market risk").

The A–C objective

Given X shares over N intervals of length τ, vol σ, temporary slope η, permanent slope γ, and risk aversion $\lambda \geq 0$, the A–C objective is

$$\min_{\{n_k\}} \quad \underbrace{\tfrac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau}\sum_{k=1}^{N} n_k^2}_{\mathbb{E}[C]} \;+ \lambda\; \underbrace{\sigma^2 \tau \sum_{k=1}^{N} x_k^2}_{\mathrm{Var}[C]},$$

with $\tilde{\eta} = \eta - \frac{1}{2}\gamma\tau$, $n_k = x_{k-1} - x_k$, $x_0 = X$, $x_N = 0$.

Intuition

Two limiting cases before we derive the minimiser.
$\lambda \to 0$: *only expected cost.* The objective is $\frac{\tilde{\eta}}{\tau}\sum n_k^2$. By Cauchy–Schwarz, $\sum n_k^2$ at fixed $\sum n_k = X$ is minimised by equal $n_k = X/N$: *TWAP* (Time-Weighted Average Price), linear inventory.
$\lambda \to \infty$: *only variance.* Minimising $\sigma^2\tau \sum x_k^2$ under $x_0 = X, x_N = 0$ sends everything at $k = 1$: all $x_k = 0$ thereafter, variance collapses, cost blows up.
Intermediate λ: deterministic, monotone-decreasing, *front-loaded* relative to TWAP — trading some expected cost for variance reduction by shedding the large early positions that dominate $\sum x_k^2$.

19.3 Deriving the discrete optimal trajectory

With the objective quadratic in the trajectory, take $\partial/\partial x_j$ at each interior $j \in \{1, \ldots, N-1\}$, set to zero, and recognise the finite-difference equation as a second-order linear recurrence.

First-order conditions

Write $U = \frac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau}\sum n_k^2 + \lambda\sigma^2\tau\sum x_k^2$. For interior x_j, only $n_j = x_{j-1} - x_j$ and $n_{j+1} = x_j - x_{j+1}$ depend on it:

$$\frac{\partial n_j^2}{\partial x_j} = 2\,n_j\,(-1) = -2(x_{j-1} - x_j),$$
$$\frac{\partial n_{j+1}^2}{\partial x_j} = 2\,n_{j+1}\,(+1) = 2(x_j - x_{j+1}).$$

Adding them,

$$\frac{\partial}{\partial x_j}\sum_{k=1}^{N} n_k^2 \;=\; -2(x_{j-1} - x_j) + 2(x_j - x_{j+1}) \;=\; -2(x_{j-1} - 2x_j + x_{j+1}).$$

The variance sum contributes $2\lambda\sigma^2\tau\, x_j$; the $\frac{1}{2}\gamma X^2$ constant drops. Hence:

$$\frac{\partial U}{\partial x_j} \;=\; -\frac{2\tilde{\eta}}{\tau}(x_{j-1} - 2x_j + x_{j+1}) \;+\; 2\lambda\sigma^2\tau\, x_j \;=\; 0.$$

Dividing by $-2\tilde{\eta}/\tau$ and rearranging,

$$x_{j-1} \;-\; 2\left(1 + \frac{\lambda\sigma^2\tau^2}{2\tilde{\eta}}\right) x_j \;+\; x_{j+1} \;=\; 0, \qquad j = 1, 2, \ldots, N-1. \tag{19.15}$$

A second-order linear recurrence with constant coefficient. Compare to the hyperbolic identity

$$2\cosh(\kappa\tau) \;=\; e^{\kappa\tau} + e^{-\kappa\tau},$$

and define $\kappa > 0$ by the implicit equation

$$\boxed{\cosh(\kappa\tau) \;=\; 1 \;+\; \frac{\lambda\sigma^2\tau^2}{2\tilde{\eta}}} \tag{19.16}$$

so that (19.15) becomes, exactly,

$$x_{j-1} \;-\; 2\cosh(\kappa\tau)\, x_j \;+\; x_{j+1} \;=\; 0. \tag{19.17}$$

Expanding $\cosh(\kappa\tau) = 1 + \frac{1}{2}(\kappa\tau)^2 + O(\tau^4)$ and matching (19.15) gives the small-τ approximation:

$$\boxed{\kappa^2 \;\approx\; \frac{\lambda\sigma^2}{\tilde{\eta}}} \tag{19.18}$$

This is the κ quoted in textbooks and code. The exact discrete κ from (19.16) differs by $O(\tau^2)$, numerically negligible in practice, and matches the un-adjusted continuous-time definition as $\tau \to 0$.

Solving the recurrence

The recurrence (19.17) has characteristic equation $r^2 - 2\cosh(\kappa\tau)r + 1 = 0$ with roots $r_\pm = e^{\pm\kappa\tau}$, so

$$x_k \;=\; A\, e^{k\kappa\tau} + B\, e^{-k\kappa\tau} \;=\; A'\, \sinh(\kappa k\tau) + B'\, \cosh(\kappa k\tau).$$

Boundary conditions pin the constants. $x_N = 0$ at $t_N = T$ gives

$$0 \;=\; A' \sinh(\kappa T) + B' \cosh(\kappa T) \;\Rightarrow\; B' \;=\; -A' \tanh(\kappa T).$$

Substituting back,

$$\begin{aligned} x_k &= A'\big[\sinh(\kappa k\tau) - \tanh(\kappa T)\cosh(\kappa k\tau)\big] \\ &= \frac{A'}{\cosh(\kappa T)}\big[\sinh(\kappa k\tau)\cosh(\kappa T) - \sinh(\kappa T)\cosh(\kappa k\tau)\big]. \end{aligned}$$

The bracketed expression is a hyperbolic-sine subtraction formula: $\sinh(\alpha)\cosh(\beta) - \sinh(\beta)\cosh(\alpha) = \sinh(\alpha - \beta)$, so

$$x_k = \frac{A'}{\cosh(\kappa T)} \sinh(\kappa k\tau - \kappa T) = -\frac{A'}{\cosh(\kappa T)} \sinh(\kappa(T - k\tau)).$$

Finally, $x_0 = X$ sets $X = -A' \sinh(\kappa T)/\cosh(\kappa T)$, so $A' = -X\cosh(\kappa T)/\sinh(\kappa T)$, and

$$\boxed{x_k \;=\; X \cdot \frac{\sinh\big(\kappa(T - k\tau)\big)}{\sinh(\kappa T)}, \qquad k = 0, 1, \ldots, N.} \tag{19.19}$$

Trade sizes

Differencing the trajectory,

$$n_k \;=\; x_{k-1} - x_k \;=\; \frac{X}{\sinh(\kappa T)}\Big[\sinh(\kappa(T-(k-1)\tau)) - \sinh(\kappa(T-k\tau))\Big].$$

Using $\sinh\alpha - \sinh\beta = 2\cosh(\frac{\alpha+\beta}{2})\sinh(\frac{\alpha-\beta}{2})$ with $\alpha = \kappa(T-(k-1)\tau)$, $\beta = \kappa(T-k\tau)$, so $\frac{\alpha+\beta}{2} = \kappa(T-(k-\frac{1}{2})\tau)$ and $\frac{\alpha-\beta}{2} = \frac{1}{2}\kappa\tau$, we get

$$\boxed{n_k \;=\; \frac{2\,X\,\sinh(\frac{1}{2}\kappa\tau)}{\sinh(\kappa T)}\cdot\cosh(\kappa(T-(k-\tfrac{1}{2})\tau)), \qquad k = 1,2,\dots,N.} \tag{19.20}$$

The factor $\cosh(\kappa(T-(k-\frac{1}{2})\tau))$ decreases monotonically from $\cosh(\kappa(T-\tau/2))$ at $k=1$ to $\cosh(\kappa\tau/2)$ at $k=N$: trades shrink over time, the schedule front-loads. The prefactor $2X\sinh(\kappa\tau/2)/\sinh(\kappa T)$ normalises the total to X by telescoping.

Almgren–Chriss optimal trajectory

The inventory and trade-size schedules that minimise the mean-variance objective $\mathbb{E}[C] + \lambda\,\mathrm{Var}[C]$ subject to $x_0 = X$, $x_N = 0$ are

$$x_k \;=\; X\,\frac{\sinh(\kappa(T-k\tau))}{\sinh(\kappa T)}, \qquad n_k \;=\; \frac{2X\sinh(\frac{1}{2}\kappa\tau)}{\sinh(\kappa T)}\cosh(\kappa(T-(k-\tfrac{1}{2})\tau)),$$

with

$$\kappa^2 \;=\; \frac{\lambda\sigma^2}{\tilde\eta}, \qquad \tilde\eta \;=\; \eta - \tfrac{1}{2}\gamma\tau.$$

Boundary conditions are automatic: $x_0 = X$ because $\sinh(\kappa T)/\sinh(\kappa T) = 1$, and $x_N = 0$ because $\sinh(0) = 0$.

19.4 Continuous-time limit

Take $\tau \to 0, N \to \infty$ with $N\tau = T$ fixed:

- $\tilde\eta = \eta - \frac{1}{2}\gamma\tau \to \eta$.
- $\kappa^2 = \lambda\sigma^2/\tilde\eta \to \lambda\sigma^2/\eta$, a function of continuous parameters only.
- $x_k \to x(t)$, continuous in $t \in [0,T]$.

The discrete solution (19.19) becomes

$$\boxed{x(t) \;=\; X\,\frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \qquad \kappa \;=\; \sigma\sqrt{\frac{\lambda}{\eta}}.} \tag{19.21}$$

(Almgren and Chriss, 2000) §4 derives the continuous problem directly via Euler–Lagrange on $\int_0^T [\eta\dot{x}^2 + \lambda\sigma^2x^2]\,dt$, giving $\eta\ddot{x} - \lambda\sigma^2 x = 0$ and hence (19.21); the two derivations agree.

19.5 Worked numerical example

A plug-in is worth a page of formulas. We take the canonical mid-cap US equity liquidation that appears in every textbook.

Parameters.

- $X = 10^6$ shares (a round lot).
- $T = 1$ trading day, $N = 13$ half-hour intervals, $\tau \approx 0.0769$ day.
- $\sigma = 0.30$ fractional-price per $\sqrt{\text{day}}$; for a \$50 stock, daily stdev $\approx$ \$15.
- $\eta = 2.5 \times 10^{-6}$. At rate 10^7 shares/day the concession is $\eta v = 25$ (huge, because rate is huge); at a realistic $X/T = 10^6$/day it is 2.5 price units, a plausible heavy-liquidation magnitude.
- $\gamma = 2.5 \times 10^{-7}$. Permanent impact over the whole day is $\gamma X = 0.25$, an order of magnitude below the temporary cost — matching the empirical observation that permanent impact is a minority of execution cost.
- $\lambda = 10^{-6}$, mildly risk-averse.

Derived quantities. From (19.12),

$$\tilde{\eta} \;=\; \eta - \tfrac{1}{2}\gamma\tau \;=\; 2.5 \times 10^{-6} - \tfrac{1}{2}(2.5 \times 10^{-7})(0.0769) \;=\; 2.4904 \times 10^{-6}.$$

From (19.18),

$$\kappa^2 \;=\; \frac{\lambda\sigma^2}{\tilde{\eta}} \;=\; \frac{10^{-6} \cdot 0.09}{2.4904 \times 10^{-6}} \;=\; 0.03614, \qquad \kappa \;=\; 0.1901.$$

With $T = 1$, $\kappa T = 0.1901$ and $\sinh(\kappa T) = 0.19125$.

Trajectory. Applying (19.19),

k	t_k (day)	x_k (shares)	n_k (shares)
0	0.0000	1 000 000.00	—
1	0.0769	922 256.65	77 743.35
2	0.1538	844 710.53	77 546.13
3	0.2308	767 345.04	77 365.49
4	0.3077	690 143.64	77 201.40
5	0.3846	613 089.83	77 053.81
6	0.4615	536 167.12	76 922.71
7	0.5385	459 359.06	76 808.05
8	0.6154	382 649.24	76 709.82
9	0.6923	306 021.25	76 627.99
10	0.7692	229 458.69	76 562.55
11	0.8462	152 945.21	76 513.49
12	0.9231	76 464.43	76 480.78
13	1.0000	0.00	76 464.43

Boundary checks. $x_0 = 10^6$ and $x_{13} = 0.00$ to the precision shown; $\sum_{k=1}^{13} n_k = 1\,000\,000.00 = X$ by construction.

Vs. TWAP. TWAP sends $X/N \approx 76\,923.08$ per bin; A–C's first trade is 77 743.35 (+1.07% vs. TWAP), its last 76 464.43 (−0.60%). At $\lambda = 10^{-6}$, A–C is almost indistinguishable from TWAP — $\kappa T = 0.19$ is small, so $\sinh(\kappa(T-t))/\sinh(\kappa T) \approx (T-t)/T$. Curvature emerges at higher λ (see Section 19.6). For a gently risk-averse trader, A–C says "don't deviate from TWAP".

Intuition

The smallness of κT is not a parameter accident: for one-day liquidations of liquid names, linear impact and vol balance so $\kappa T \ll 1$, and optimal execution is only mildly front-loaded vs. TWAP. A–C's main payoff over TWAP is not a dramatic schedule change — it is a principled way to pick λ from client risk tolerance, and to compute the chosen schedule's expected cost and standard deviation for post-trade benchmarking.

19.6 The execution efficient frontier

Sweeping $\lambda \in [0, \infty)$ at fixed $X, T, N, \sigma, \eta, \gamma$ traces a curve in the $(\sqrt{\mathrm{Var}[C]}, \mathbb{E}[C])$ plane: the **execution efficient frontier**, a Markowitz-style curve of minimum cost at given variance (or vice versa). The analogy is exact: where Markowitz portfolio selection trades expected return against variance over asset weights, A–C trades expected cost against variance over execution schedules. Each point on the curve is the best achievable $\mathbb{E}[C]$ at its $\sqrt{\mathrm{Var}[C]}$; no schedule lies below it, and picking λ just picks which point on the curve you want.

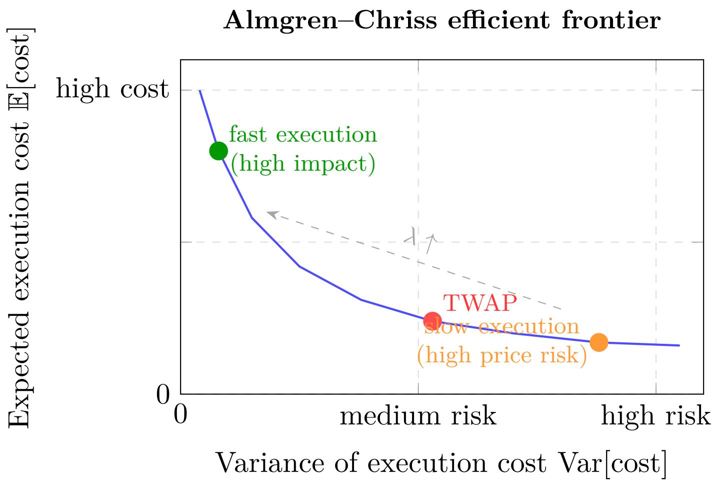


Figure 19.1: Almgren–Chriss efficient frontier. Each point on the curve corresponds to a different risk-aversion parameter λ: high λ forces fast execution (low variance from price risk, high expected cost from market impact); low λ allows slow execution (low impact cost, but high variance as price may drift). TWAP sits near the middle of the frontier. The optimal schedule for a given λ is the point on this curve that minimises $\mathbb{E}[\mathrm{cost}] + \lambda\,\mathrm{Var}[\mathrm{cost}]$.

Endpoints: $\lambda = 0$ sits at the rightmost (highest-variance, lowest-cost) point — TWAP; $\lambda \to \infty$ sits at the leftmost — the instant block trade. Interior points are hyperbolic-sine schedules.

For the worked example parameters, using

$$\begin{aligned}\mathbb{E}[C] &= \tfrac{1}{2}\gamma X^2 + \frac{\tilde{\eta}}{\tau}\sum_k n_k^2,\\ \mathrm{Var}[C] &= \sigma^2\tau\sum_k x_k^2,\end{aligned}$$

we compute:

λ	$\mathbb{E}[C]$ (units)	$\sqrt{\mathrm{Var}[C]}$ (units)	n_1 as % of X
10^{-7}	2,615,385	163,135	7.70%
10^{-6}	2,615,456	162,740	7.77%
10^{-5}	2,622,090	158,944	8.49%
10^{-4}	3,017,007	131,389	14.28%
10^{-3}	7,481,151	67,447	37.03%

Increasing λ drives $\sqrt{\mathrm{Var}[C]}$ down (good) and $\mathbb{E}[C]$ up (bad). At $\lambda = 10^{-3}$ the first trade is over a third of the position — the trader pays nearly $3\times$ the expected cost to shed overnight variance. The desk's job is to pick the point matching client risk tolerance; A–C maps the qualitative "client wants out fast" to a specific λ and hence a specific schedule.

Intuition

Two observations. First, the permanent-impact term $\frac{1}{2}\gamma X^2 = 1.25 \times 10^5$ is small vs. the $\sim 2.5 \times 10^6$ expected-cost totals; most cost is temporary impact from $\sum n_k^2$. In this regime the $\tilde{\eta}$-vs-η distinction is a small correction — but the one that makes the discrete closed form exact.

Second, at $\lambda = 10^{-3}$ the trader pays an extra \$4.9m expected cost to save \$95k of cost standard deviation. Extreme risk aversion; no desk would pick it for a routine liquidation, but it fits regimes like an executing broker with a hard deadline and a client indifferent to average cost. The frontier is a menu, not a prescription.

19.7 VWAP, TWAP, POV, and A–C in context

Sell-side execution algorithms come in a small zoo. Four matter for this book.

TWAP — Time-Weighted Average Price. Equal size per interval, $n_k = X/N$. The name: when price is martingale, fill VWAP equals the mid's TWAP. TWAP is the $\lambda \to 0$ limit of A–C, default for "lowest expected cost, don't care about timing risk". Simple, benchmark-friendly, gaming-proof under martingale mid.

VWAP — Volume-Weighted Average Price. Trade in proportion to the *historical* intraday volume profile (10% of daily volume in the first half-hour $\Rightarrow$ 10% of the order). Benchmark: session realised VWAP. The default institutional equity algorithm because it mimics a "typical" participant and minimises tracking error to the market's average print. Not a direct A–C special case, but A–C re-weighted by a volume profile reproduces it.

POV — Percentage of Volume. Trade a fixed fraction ρ of *realised* volume. If the last minute printed 10,000 shares at $\rho = 5\%$, send 500 next minute. Reactive: speeds up with the market, slows with it — the liquidator's footprint stays proportional. Default for hiding in flow at the cost of deadline risk (insufficient volume $\Rightarrow$ incomplete fill).

A–C / Implementation Shortfall. Solve this chapter's trajectory problem for given λ and execute the hyperbolic-sine schedule. The answer to "minimise implementation shortfall (starting mid minus average fill) subject to a risk penalty". Front-loaded vs. TWAP; equivalent to VWAP plus a risk correction under a flat volume profile. A–C (or a risk-aware cousin) is every modern desk's "Implementation Shortfall" flagship.

Execution algorithm taxonomy

Algo	Schedule rule	Benchmark
TWAP	Equal per time bin	Session mid TWAP
VWAP	Proportional to historical volume profile	Session realised VWAP
POV	Fixed % of realised volume live	Realised VWAP (participation-matched)
A–C	Hyperbolic-sine, front-loaded by λ	Arrival (decision) mid — implementation shortfall

All four are deterministic given inputs; none uses a private signal. Signal-driven "algo wheels" sit on top of this layer.

19.8 Almgren–Chriss versus square-root reality

The chapter's central pedagogical point, and the hardest to swallow first time. A–C models temporary impact linearly, $g(v) = \eta v$. Chapter 7 showed the empirical law, stable across decades and asset classes, is square-root:

$$I(Q) \;\approx\; Y\sigma\sqrt{Q/V_{\text{daily}}}, \tag{19.22}$$

where Q is metaorder size, V_{daily} ADV, and $Y \sim 0.5$ a dimensionless prefactor. Traded at constant rate over interval τ, linear $g(v)$ gives $I(Q) = \eta Q/\tau \propto Q$ vs. the square-root's $I(Q) \propto \sqrt{Q}$. The two curves meet at exactly one order size — where η was calibrated — and diverge elsewhere.

Why teach linear anyway? Three reasons, none being "linear is empirically right":

1. **Closed form.** Linear impact yields the quadratic objective (19.14), linear first-order conditions, a hyperbolic-sine solution, and a closed-form conic frontier. Square-root impact gives a non-quadratic objective with no closed form: a nonlinear BVP to solve numerically. For teaching the structure of the cost–risk trade-off, the closed form wins.

2. **Local calibration.** In a narrow band around the trader's typical size Q_0, linear can be tuned to match the square-root slope: pick η so $\eta Q_0/T = Y\sigma\sqrt{Q_0/V}$. Within a factor of 2–3 of Q_0 it is not catastrophically wrong.

3. **Industry inertia.** A–C became the industry benchmark within five years. Serious desks run non-linear engines, but A–C remains the lingua franca and (η, γ, λ) are what daily risk reports plot.

Don't extrapolate A–C outside its calibration range

The cost of this contradiction is largest when a small-trade calibration is used to price a large one. Suppose η is calibrated against child orders of $Q_{\text{calib}} = 10^5$ shares:

- Child $Q = Q_{\text{calib}}$: predicted cost matches reality.
- Parent $Q = 5 \cdot 10^6$: linear predicts $\propto Q$, reality is $\propto \sqrt{Q}$, so linear overpredicts by $\sqrt{Q/Q_{\text{calib}}} = \sqrt{50} \approx 7.07$.

Running A–C with that over-stated slope tells the trader any parent liquidation is catastrophic, steering them to artificially slow schedules and ballooning variance. Fix:

recalibrate η at the parent-order size band before running A–C, never scale up a child calibration. Empirical backing: Chapter 7.

19.9 Extensions and breakages

Directions the basic model generalises in; doorways to further reading, not rederived here.

Risk-neutral vs. risk-averse. $\lambda = 0$ (TWAP as A–C solution) is the risk-neutral case; some authors treat it as a separate "pure implementation shortfall" problem. Replacing mean-variance with a full utility $U(C) = -\exp(\gamma C)$ gives the same closed form under Gaussian risk (CARA–Gaussian is mean-variance up to constants); CVaR or VaR break the closed form.

Time-varying volatility. $\sigma \to \sigma(t)$ (higher at open/close, lower midday) generalises the trajectory: $\eta\ddot{x}(t) - \lambda\sigma(t)^2 x(t) = 0$, a variable-coefficient ODE, numerical but not generally elementary. See (Cartea et al., 2015) Ch. 6.

Multi-asset portfolio liquidation. Vector positions $\boldsymbol{X} \in \mathbb{R}^n$ with covariance $\boldsymbol{\Sigma}$ and cross-impact $\boldsymbol{\eta}$ give a matrix quadratic with hyperbolic-sine matrix $\boldsymbol{\kappa} = \boldsymbol{\eta}^{-1/2}(\lambda\boldsymbol{\Sigma})^{1/2}\boldsymbol{\eta}^{-1/2}$. Closed form survives; bookkeeping doesn't. See (Cartea et al., 2015) Ch. 7.

Non-linear impact. With $g(v) \propto \sqrt{v}$ the objective is non-quadratic and has no closed form. Obizhaeva–Wang (2013) and Almgren (2003) give numerical treatments; large-order practitioners run non-linear optimisers on proprietary square-root curves.

Dark pools and venue choice. Dark (non-displayed) execution cuts temporary impact at the cost of uncertain fills. A–C with venue choice is an open research area; the simplest treatment mixes displayed and dark η's with a dark fill probability. See (Bouchaud et al., 2018).

19.10 Prosperity and sell-side applications

Prosperity example

End-of-round liquidation is where A–C earns its keep in Prosperity. Suppose you are long 30 PICNIC_BASKET1 entering the final 100 timesteps after the basket–constituent spread you were harvesting has collapsed. You want flat by $T = 100$: the round ends and unliquidated inventory marks to the last mid regardless of realisable exit. You also want to avoid dumping all 30 at once — the basket is thin and a market order of that size walks the book down several ticks per unit.

Take parameters $X = 30$, $T = 100$, $N = 10$ (one decision every 10 timesteps), $\sigma = 2.0$ seashells per timestep, $\eta = 0.05$ (each unit/timestep of selling rate drives the mid down by 0.05), $\gamma = 0.001$ (permanent impact is small on the order book), and a mild $\lambda = 10^{-6}$. The A–C trajectory is computed exactly as in Section 19.5:

$$\tilde{\eta} = 0.05 - \tfrac{1}{2} \cdot 0.001 \cdot 10 = 0.045, \qquad \kappa = \sqrt{10^{-6} \cdot 4/0.045} \approx 0.00943,$$

so $\kappa T \approx 0.943$, larger than the equity example's 0.19 — the trader is closer to the deadline in P&L-standard-deviation units. The schedule:

k	t_k	x_k	n_k
0	0	30.00	—
1	10	26.29	3.71
2	20	22.81	3.48
3	30	19.53	3.28
4	40	16.43	3.10
5	50	13.48	2.96
6	60	10.64	2.84
7	70	7.90	2.74
8	80	5.23	2.67
9	90	2.60	2.63
10	100	0.00	2.60

TWAP sends 3.00 per decision; A–C sends 3.71 first (+24%) and 2.60 last (−13%), trading a little early impact for tail variance reduction. $\sum n_k = 30.000$ exactly by construction.
Prosperity caveats. (1) Position limits: if the limit is ±30, A–C does not help when flipping from +30 to −30 — that is reversing, not liquidating. (2) Discrete units: round n_k to integers and hold the slack in x_k; do not let rounding accumulate into a non-zero terminal. (3) Mid vs fill: A–C uses mid; Prosperity fills cross the spread, so the model slightly over-estimates cost.

On a Goldman Sachs desk

On a Goldman execution desk, the default algorithm for a client order tagged "IS" (Implementation Shortfall) is an A–C descendant with venue routing, a dark-pool seeker, and a signal-based accelerator overlay. The owning *Strat* does two unglamorous jobs. First, daily calibration of η and γ against the previous day's fills across tens of thousands of orders, regressing realised shortfall against (Q/V) and $(Q/T)^\alpha$ curves. Second, maintain λ presets ("urgent", "neutral", "passive") with a one-click override so traders pick at order entry without knowing the math. When asked "why did my fill trail VWAP today?" the Strat pulls a post-trade report decomposing realised shortfall into four A–C buckets: permanent impact, temporary impact, timing risk, spread. These reports are the team's daily bread.
A second use is *pre-trade cost estimation*: before a block, sales ask Strats "how much to execute in two hours". The answer is A–C $\mathbb{E}[C]$ at chosen λ, with $\sqrt{\mathrm{Var}[C]}$. It goes on the ticket the salesperson quotes. Getting it systematically wrong loses clients and gets the Strat paged.

19.11 A short reference implementation

Twenty lines of Python compute the whole trajectory. Not production — real engines need volume profiles, queue-position logic, child-order sizing against book depth, venue routing — but this is the core calculation underneath any A–C execution stack.

```
import math

def almgren_chriss_trajectory(X, T, N, sigma, eta, gamma, lam):
```

```
    """Return the list of inventory levels [x_0, x_1, ..., x_N]
    under the Almgren-Chriss optimal mean-variance schedule.

    Parameters
    ----------
    X     : total shares to liquidate (x_0 = X, x_N = 0).
    T     : deadline (same time units as sigma, eta, gamma).
    N     : number of equal-length trade intervals.
    sigma : arithmetic volatility per unit time.
    eta   : temporary-impact slope (linear in trading rate).
    gamma : permanent-impact slope (linear in trading rate).
    lam   : mean-variance risk aversion (>= 0).
    """
    tau = T / N
    eta_tilde = eta - 0.5 * gamma * tau
    kappa = math.sqrt(lam * sigma * sigma / eta_tilde)
    denom = math.sinh(kappa * T)
    return [X * math.sinh(kappa * (T - k * tau)) / denom
            for k in range(N + 1)]
```

Listing 19.1: Almgren–Chriss optimal liquidation trajectory.

Recover trade sizes by differencing: $n_k = x_{k-1} - x_k$. The $\lambda = 0$ case throws (`kappa = 0`, `sinh(0)` in the denominator); production code should trap it and return the TWAP schedule $[X(1 - k/N)]$ directly.

Historical note

Robert Almgren (Courant, NYU) and Neil Chriss wrote "Optimal Execution of Portfolio Transactions" in 1999; it appeared in the *Journal of Risk* in 2000 and sits in every execution desk's literature folder. It was the first clean statement of the mean-variance execution problem; the hyperbolic-sine closed form turned "how fast do we hit the button" from craft into discipline. Almgren later founded Quantitative Brokers. Chriss had sell- and buy-side roles including Goldman Sachs Asset Management, Hutchin Hill, and Paloma. Newer models (Obizhaeva–Wang transient impact, Gatheral's no-dynamic-arbitrage framework) all frame themselves as departures from A–C.

19.12 Key facts to memorise

- Problem: sell X over $[0, T]$ with $\sum n_k = X$, minimising $\mathbb{E}[C] + \lambda \operatorname{Var}[C]$.
- Three cost sources: temporary $g(v) = \eta v$ (reverts), permanent $h(v) = \gamma v$ (sticks), price risk $\sigma\sqrt{\tau}\xi_k$ (Gaussian).
- Telescoping $\sum x_k n_k = \frac{1}{2}(X^2 - \sum n_k^2)$ makes permanent impact quadratic in n_k.
- Discrete: $\tilde{\eta} = \eta - \frac{1}{2}\gamma\tau$, $\kappa^2 = \lambda\sigma^2/\tilde{\eta}$. Continuous limit: $\tilde{\eta} \to \eta$, $\kappa = \sigma\sqrt{\lambda/\eta}$.
- Optimal trajectory: $x_k = X \sinh(\kappa(T - k\tau))/\sinh(\kappa T)$.
- $\lambda \to 0$: TWAP. $\lambda \to \infty$: instant block. Interior λ: front-loaded.
- Efficient frontier: $\lambda \mapsto (\sqrt{\operatorname{Var}[C]}, \mathbb{E}[C])$ traces a convex curve.
- Four baseline algos: TWAP, VWAP, POV, A–C. A–C is the "Implementation Shortfall" default.

- Linear impact contradicts the square-root law (Chapter 7). Kept for tractability; practitioners recalibrate η locally to the square-root slope.
- Goldman desk: Strat owns (η, γ) calibration against live fills; trader picks λ from client risk tolerance; overrides on news.
- A–C γ = permanent-impact slope, not the A–S risk aversion of Chapter 9.
- Primary: (Almgren and Chriss, 2000). Richer re-derivation: (Cartea et al., 2015) Ch. 6–8. Empirical counter: Chapter 7, (Bouchaud et al., 2018).

Chapter 20

Risk Management: Kelly, Drawdown, and Metrics

You have a positive-edge strategy. How much of your capital should you bet on each trade? Too little and you leave money on the table; too much and a losing streak wipes you out. The **Kelly criterion** (Kelly, 1956) gives the unique fraction of capital that maximises the *long-run geometric growth rate* of wealth. This chapter derives Kelly in the binary case, generalises to log-normal returns, and then steps back for a practitioner reality check: **Kelly is theoretically optimal and almost no professional trades full Kelly**, for reasons that matter more than the derivation itself.

The second half switches from *sizing* to *scoring*: how do you tell a good run from a lucky one, and compare two strategies? That is the province of performance metrics — Sharpe, Sortino, Calmar, information ratio — and of drawdown arithmetic, which contains the single most important identity in this chapter: the asymmetric recovery formula. A 20% loss needs 25% to break even; 50% needs 100%; 80% needs 400%. The whole reason one prefers a Sharpe of 1.5 with a 10% max drawdown to a Sharpe of 2.5 with a 60% max drawdown is baked into that asymmetry.

Notation: f is the Kelly fraction of wealth staked, p is win probability, $q = 1 - p$, b is win size, a is loss size, μ is expected log-return, σ log-return standard deviation, R a per-period return, r the risk-free rate. Execution scheduling (Chapter 19) told you *when* to trade; this chapter says *how much* to hold and how to score what you did.

Prerequisites

From earlier in the book:

- Chapter 10: log-returns, i.i.d. bet sequences, and the language of expected value, variance, and long-run sample averages. The *O–U* half-life of Chapter 10 is *not* used here, but the habit of modelling increments as mean plus noise is.
- Chapter 13: the volatility-scaled position-sizing rule of time-series momentum — "hold $\sigma_{\text{target}}/\sigma_t$ units of the instrument" — is a *heuristic fractional-Kelly rule in disguise*. This chapter explains why.
- Chapter 18: σ denotes annualised (log-)volatility. Throughout this chapter μ is expected log-return per unit time, σ is the standard deviation of log-returns per unit time, and the risk-free rate is r.
- Chapter 19: mean–variance thinking — trade off an expected quantity against a variance. Kelly will turn out to be *mean–variance optimality in a particular disguise* (the log disguise) rather than a competitor to it.

From outside the book:

- Basic **expected utility theory**: if you have a utility function $U(w)$ over wealth, a "rational" bettor maximises $\mathbb{E}[U(W_{\text{final}})]$ rather than $\mathbb{E}[W_{\text{final}}]$ itself. Kelly corresponds to the choice $U(w) = \ln w$.
- **Logarithms and exponentials**: $\ln(1+x) \approx x - x^2/2$ for small x, $\frac{\mathrm{d}}{\mathrm{d}f}\ln(1+fx) = x/(1+fx)$, and the fact that ln of a product is the sum of ln's.
- **First-order conditions**: set the derivative of a concave function to zero to find the maximum.

20.1 The Kelly criterion: binary case

Setup

A biased coin lands heads with probability $p > 1/2$ and tails with probability $q = 1 - p$. Stake a dollar on heads: you win \$$b$ on heads, lose \$$a$ on tails, with $a, b > 0$. The unit even-money bet is $a = b = 1$; a sports bettor quoted at +150 decimal odds plays $b = 1.5$, $a = 1$; a slot machine with 99% payout has $pb < qa$ and should not be played.

You bet a fraction $f \in [0, 1]$ of current wealth each trial. With $W_0 = 1$ and trial-i per-unit payoff $R_i \in \{+b, -a\}$,

$$W_N = W_0 \prod_{i=1}^{N} (1 + fR_i). \tag{20.1}$$

Wealth *multiplies* trial-by-trial: a gain on trial 5 grows the base on which trials 6, 7, ... compound. This is the single most important distinction in the chapter and the reason plain $\mathbb{E}[W_N]$ is the *wrong* objective.

Why maximising expected wealth fails

What fraction f maximises $\mathbb{E}[W_N]$? After one trial,

$$\mathbb{E}[W_1] = p(1 + fb) + q(1 - fa) = 1 + f(pb - qa). \tag{20.2}$$

With positive edge $pb - qa > 0$ this is strictly increasing in f, with maximum at $f = 1$: bet everything. By compounding, the same holds for $\mathbb{E}[W_N]$.

But $f = 1$ is catastrophic: you lose your entire bankroll the first tails, which happens with probability $1 - p^N \to 1$. Expected wealth grows at the maximum rate only because one lucky path of probability p^N contributes an exponentially large amount and drags the mean upwards (Thorp, 2006). With probability one you are broke. $\mathbb{E}[W_N]$ cannot distinguish "probably rich" from "almost certainly broke but very rarely stupendously rich".

Do not maximise expected wealth

A strategy that maximises $\mathbb{E}[W_N]$ on a multiplicative bet is generically a strategy that guarantees ruin with probability one. The expectation is dominated by the one path where every bet wins, which is exponentially rare but exponentially large. Kelly's insight is that you should instead maximise the *typical* path, which is captured by $\mathbb{E}[\ln W_N]$. For compounding bets, "expected log" is the right objective; "expected level" is a trap.

The Kelly objective: expected log-wealth

Switch the objective from $\mathbb{E}[W_N]$ to $\mathbb{E}[\ln W_N]$ — equivalently, adopt the log-utility $U(w) = \ln w$. Since $\ln W_N = \ln W_0 + \sum_i \ln(1 + fR_i)$ turns the multiplicative wealth process into a sum, its expectation is N times a *per-trial* growth rate:

$$G(f) \;:=\; \mathbb{E}[\ln(1 + fR)] \;=\; p\ln(1 + fb) + q\ln(1 - fa). \tag{20.3}$$

By the strong law, $\frac{1}{N}\ln(W_N/W_0) \to G(f)$ almost surely, so $W_N \approx W_0 e^{NG(f)}$ for large N: the *typical* wealth path grows exponentially at rate $G(f)$ (Thorp, 2006). Unlike $\mathbb{E}[W_N]$, G is sensitive to ruin: if $f \geq 1/a$ there is positive probability of $W = 0$, at which $\ln W = -\infty$ and the expectation is $-\infty$.

First-order condition

G is strictly concave on the domain $0 < f < 1/a$ because

$$G''(f) = -\frac{pb^2}{(1 + fb)^2} - \frac{qa^2}{(1 - fa)^2} < 0, \tag{20.4}$$

so it has a unique interior maximum. Set $G'(f) = 0$:

$$G'(f) = \frac{pb}{1 + fb} - \frac{qa}{1 - fa} = 0 \tag{20.5}$$

$$\iff \quad pb(1 - fa) = qa(1 + fb) \tag{20.6}$$

$$\iff \quad pb - pbfa = qa + qabf \tag{20.7}$$

$$\iff \quad pb - qa = f \cdot ab(p + q) = f \cdot ab. \tag{20.8}$$

Rearranging gives the **binary Kelly fraction**:

$$\boxed{f^* \;=\; \frac{pb - qa}{ab}.} \tag{20.9}$$

Bet the ratio of edge to odds. The numerator $pb - qa$ is expected profit per unit staked; the denominator ab measures how violent the outcomes are.

Worked example: the 55/45 coin

Take $p = 0.55$, $q = 0.45$, $a = b = 1$ (win or lose a dollar per dollar staked). Equation (20.9) gives

$$f^* = \frac{0.55 \cdot 1 - 0.45 \cdot 1}{1 \cdot 1} = 0.10. \tag{20.10}$$

Bet 10% of your bankroll per flip. The per-bet growth rate is

$$G(f^*) = 0.55\ln(1.10) + 0.45\ln(0.90) \approx 0.005008, \tag{20.11}$$

so a typical bankroll grows $\approx 0.5\%$ per bet. Over 100 bets the typical factor is $e^{100 \cdot 0.005008} \approx 1.65$: the median player ends up $\approx 65\%$ richer. Betting $f = 1$ instead gives $\mathbb{E}[W_{100}] = 1.10^{100} \approx 13{,}780$, but with probability $1 - 0.55^{100} \approx 1 - 10^{-26}$ the player is already broke.

Binary Kelly

Betting repeatedly on a positive-edge two-outcome trial with win probability p, win size b, and loss size a, the unique fraction of current wealth to stake that maximises the long-run geometric growth rate $G(f) = p\ln(1+fb) + q\ln(1-fa)$ is

$$f^* = \frac{pb - qa}{ab} \quad = \quad \frac{\text{edge}}{\text{odds}}.$$

For a fair-odds coin ($a = b = 1$) this collapses to $f^* = p - q$, the probability gap.

Intuition

The Kelly fraction maximises the *logarithm* of terminal wealth. Maximising expected wealth sizes to 100% and busts on the first loss; maximising expected log-wealth trades upside against ruin, because ln tends to $-\infty$ near zero and so punishes ruin infinitely. "Bet more = more profit" is correct only up to f^*; past it, adding size makes the *typical* player poorer while keeping the *average* player richer. The right objective for a compounding bettor is the typical player.

20.2 The Kelly criterion: log-normal returns

Real markets do not deliver two-point bets. Model the risky asset's per-period *log-return* as

$$r \;\sim\; \mathcal{N}(\mu, \sigma^2), \tag{20.12}$$

with μ the expected log-return and σ its standard deviation per period. The risk-free rate per period is r.

Hold a portfolio with weight f on the risky asset and $1-f$ in cash. $f = 0$ is fully in cash; $f = 1$ fully invested; $f > 1$ leveraged (borrow at r); $f < 0$ short.

Portfolio log-return and the Itô correction

Let V_t be portfolio value. Over one period the *simple* arithmetic return of the portfolio is

$$\frac{V_{t+1} - V_t}{V_t} = f \cdot (e^r - 1) + (1-f) \cdot (e^r - 1), \tag{20.13}$$

because the risky asset returns $e^r - 1$ in simple terms over the period and cash returns $e^r - 1$. Taking logs of $V_{t+1}/V_t = 1 + fR_{\text{risky}} + (1-f)R_{\text{cash}}$ with $R = e^r - 1$ and expanding to second order in the small-period limit (equivalently, going to continuous time, or taking the Taylor expansion carefully), the portfolio's log-return over the period is

$$r_{\text{port}} \;\approx\; r + f(\mu - r) - \tfrac{1}{2}f^2\sigma^2. \tag{20.14}$$

The first two terms are the naive "weighted-average" answer: a linear combination of risk-free and excess drift. The third, $-\frac{1}{2}f^2\sigma^2$, is the **Itô correction**. Geometric compounding eats volatility: $+10\%$ then -10% leaves 0.99, not 1.00; the loss $0.5 \cdot (0.10)^2 = 0.5\%$ is exactly the $\frac{1}{2}\sigma^2$ drag.

Itô's lemma (Chapter 16) applied to $\ln V_t$ delivers the full derivation: the self-financing SDE $\mathrm{d}V_t/V_t = (r + f(\mu - r))\,\mathrm{d}t + f\sigma\,\mathrm{d}W_t$ gives $\mathrm{d}\ln V_t = (r + f(\mu - r) - \frac{1}{2}f^2\sigma^2)\,\mathrm{d}t + f\sigma\,\mathrm{d}W_t$, and integrating recovers (20.14).

Long-run growth rate and the FOC

The long-run growth rate is, as in the binary case, the expectation of the portfolio log-return:

$$G(f) \;=\; \mathbb{E}[r_{\text{port}}] \;=\; r + f(\mu - r) - \tfrac{1}{2}f^2\sigma^2. \tag{20.15}$$

This is a concave quadratic in f. Its derivative is $G'(f) = (\mu - r) - f\sigma^2$, which vanishes at

$$\boxed{f^* = \frac{\mu - r}{\sigma^2}.} \tag{20.16}$$

This is the **log-normal Kelly fraction** (Thorp, 2006; Chan, 2013), the formula every industry Kelly discussion quotes or argues about.

The value at the optimum is

$$G(f^*) = r + \frac{(\mu - r)^2}{2\sigma^2} = r + \frac{1}{2}\,\mathrm{Sharpe}^2, \tag{20.17}$$

where Sharpe $= (\mu - r)/\sigma$ is the **Sharpe ratio** (excess return per unit volatility; defined in Section 20.4). Maximum Kelly-optimal growth depends on Sharpe alone — not separately on μ and σ. Halving Sharpe quarters the maximum growth rate: Sharpe 2 is four times better than Sharpe 1 in compounding terms, not twice.

Worked example: the 8%/20% strategy

Take $\mu - r = 0.08$ per year and $\sigma = 0.20$ per year (Thorp's numbers for the S&P 500 (Thorp, 2006)). Then

$$f^* = \frac{0.08}{(0.20)^2} = \frac{0.08}{0.04} = 2.0. \tag{20.18}$$

Full Kelly prescribes 200% long: borrow a dollar per dollar of equity and invest \$2. The implied maximum growth rate is

$$G(f^*) = r + \frac{0.08^2}{2 \cdot 0.04} = r + 0.08, \tag{20.19}$$

an extra 8% on top of r. Half Kelly ($f = 1.0$) gives $G(f^*/2) = r + 0.06$: exactly three-quarters of the excess growth at half the leverage. Both numbers are verified in Section 20.7.

$f^* = 2$ is the single most compelling reason no professional runs full Kelly. It is growth-maximising under the Gaussian assumption and insane to run: 50% drawdowns would recur, each time looking like a broken strategy to whoever holds the position. Theory and practice fork here.

Log-normal Kelly

For a risky asset with log-return $r \sim \mathcal{N}(\mu, \sigma^2)$ per period and a risk-free rate r per period, the fraction of wealth in the risky asset that maximises the long-run geometric growth rate is

$$f^* = \frac{\mu - r}{\sigma^2}.$$

At this optimum the growth rate is $r + \frac{1}{2}\mathrm{Sharpe}^2$, i.e. the maximum achievable excess growth is *half the squared Sharpe ratio* of the underlying strategy. Sharpe is the thing that compounds; it is not "just" a performance metric.

Half of Kelly is half-Kelly, not half the stakes

"Half Kelly" means $f = f^*/2$, not "bet half as often" or "half the dollar amount". Kelly outputs a *fraction of current wealth*; halving it still rebalances period by period. Compounding discipline is preserved; only the leverage knob is turned down.

20.3 Fractional Kelly and why no one trades full Kelly

The FOC says full Kelly maximises long-run growth — in the limit, over infinite trials, assuming μ and σ are known exactly. None of these hold in practice, and the short-term variance of full Kelly is brutal: 50% drawdowns are routine. Three objections explain why nobody trades it (Thorp, 2006), (Chan, 2013).

Drawdown probability

Thorp gives the following result for a log-normal portfolio at leverage cf^* ("c-Kelly"), derived as a two-barrier Brownian-motion problem under continuous rebalancing (Thorp, 2006):

$$\mathbb{P}(\,\exists\, t \geq 0 \text{ such that } V_t/V_0 \leq x\,) \;=\; x^{2/c-1}, \qquad 0 < x < 1. \tag{20.20}$$

Read: probability the portfolio is ever at or below fraction x of starting value over an infinite horizon. Three cases:

- **Full Kelly**, $c = 1$: exponent 1, so $\mathbb{P}(\text{ever} \leq xV_0) = x$. Infinite-horizon full-Kelly has a 50% chance of ever halving, 25% of ever quartering.
- **Half Kelly**, $c = 1/2$: exponent 3, $\mathbb{P}(\text{ever halve}) = 0.5^3 = 0.125$.
- **Quarter Kelly**, $c = 1/4$: exponent 7, $\mathbb{P}(\text{ever halve}) = 0.5^7 \approx 0.0078$, under 1%.

The growth cost is much smaller than the drawdown benefit. Half Kelly retains 3/4 of the full-Kelly growth rate (Section 20.2); its growth standard deviation is exactly *half* full Kelly's, variance one quarter (Thorp, 2006). Halving f gives up 25% of growth for 75% less drawdown probability — one of the best trades in quant finance.

Estimation error

$f^* = (\mu - r)/\sigma^2$ depends on $\mu - r$, notoriously hard to estimate: years of data give confidence intervals comparable in size to μ itself. Over-estimating μ over-estimates f^* and pushes the bettor past the peak of G onto the destructive right side.

The asymmetry: the $G(f)$ curve is approximately parabolic, peaking at f^* and falling quadratically on both sides. Under- betting captures positive growth, just less of it. Over-betting far enough drives $G(f)$ *negative*; typical wealth shrinks to zero. The growth rate crosses zero at $f_c = 2f^*$ in the log-normal case. Past there is terminal.

Thorp's scenario: $\hat{\mu} = 1.5\mu$ and $\hat{f}^* = 1.5f^*_{\text{true}}$ loses about 50% of typical growth (Thorp, 2006). $\hat{\mu} = 0.5\mu$ (half Kelly) retains 75% of growth. Over-betting compounds through a loss-inducing variance drag; under-betting only loses growth.

Full Kelly is theoretically optimal and practically insane

Three reasons nobody runs full Kelly (Thorp, 2006):

1. **Drawdown math**. Full Kelly has a 50% lifetime probability of at least halving and a 25% chance of quartering. Half Kelly drops these to 12.5% and 1.6% at only a 25% reduction in growth rate. The trade is lopsided in favour of the more conservative sizing.
2. **Estimation error**. f^* depends on an unknown μ. Over-estimating μ by 50% pushes you 50% past the Kelly peak, where the growth rate collapses. The penalty for over-estimating is much worse than the penalty for under-estimating, because $G(f)$ becomes *negative* at $f_c = 2f^*$.

3. **Behavioural intolerance**. A fund manager who routinely experiences 30–50% drawdowns does not keep her clients, her job, or her emotional equilibrium, regardless of what the math says about the long run. Human tolerance for drawdowns is a binding constraint.

Standard response (Thorp, 2006): run leverage cf^* with $c \in [0.25, 0.5]$ and treat full Kelly as an upper bound, never a target. Half Kelly is the default: 25% growth reduction for 75% drawdown reduction, and robustness to μ-estimation error.

Fractional Kelly rule of thumb

Trade half Kelly. Take $f = \frac{1}{2}f^*$ with $f^* = (\mu - r)/\sigma^2$ estimated from your best data. This halves volatility and quarters variance at 25% of the growth rate, and is robust to μ over-estimates up to a factor of two (where full Kelly would have ruined you). Quarter Kelly suits fat-tailed returns or an impatient client base.

Intuition

Fractional Kelly's appeal is not timidity. The $G(f)$ curve is symmetric near its peak but its *consequences* are asymmetric: under-betting loses linear growth, over-betting compounds you to zero. When f^* cannot be measured perfectly — and it cannot — the rational response is to deliberately under-bet. Half Kelly is the optimal robust response to estimation error in μ.

20.4 Performance metrics

Sizing is half the problem; scoring is the other. Four numbers dominate the conversation: Sharpe, Sortino, Calmar, and information ratio (Chan, 2013). Formulas and failure modes follow.

The Sharpe ratio

Let $\{r_t\}_{t=1}^T$ be strategy returns at some frequency and r the risk-free return at that frequency. The **Sharpe ratio** is

$$\text{Sharpe} = \frac{\bar{r} - r}{\hat{\sigma}_r}, \qquad \bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t, \qquad \hat{\sigma}_r^2 = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})^2. \tag{20.21}$$

It is dimensionless. To compare across horizons, Sharpe is reported *annualised*:

$$\text{Sharpe}_{\text{annual}} = \text{Sharpe}_{\text{period}} \cdot \sqrt{\text{periods per year}}. \tag{20.22}$$

Daily annualises by $\sqrt{252} \approx 15.87$; hourly by $\sqrt{252 \cdot 6.5}$ (US equities); monthly by $\sqrt{12} \approx 3.46$. The $\sqrt{\cdot}$ comes from i.i.d. T-period variance growing linearly in T while the mean also grows linearly, so their ratio scales as $T/\sqrt{T} = \sqrt{T}$.

Worked example: a 0.04%/0.8% daily strategy

A strategy has $\bar{r} = 0.04\%$ daily and $\hat{\sigma}_r = 0.8\%$ daily, with $r = 0$. Then

$$\text{Sharpe}_{\text{daily}} = \frac{0.0004}{0.008} = 0.05, \tag{20.23}$$

and

$$\text{Sharpe}_{\text{annual}} = 0.05 \cdot \sqrt{252} \approx 0.7937. \tag{20.24}$$

At ≈ 0.79 this is below the 1.0 threshold most systematic desks require for capital allocation. (Context: S&P 500 long-run Sharpe $\approx$ 0.3–0.5; a good market-making desk targets 2–5; HFT shops double digits.) Verified in Section 20.7.

Why Sharpe is not enough: Sortino, Calmar, IR

Sharpe treats upside and downside symmetrically: +5% contributes as much to $\hat{\sigma}$ as −5%. For most traders only downside is risk; upside is opportunity. The **Sortino ratio** replaces standard deviation with *downside deviation*:

$$\text{Sortino} \;=\; \frac{\bar{r} - r}{\hat{\sigma}_r^-}, \qquad (\hat{\sigma}_r^-)^2 \;=\; \frac{1}{T}\sum_{t=1}^{T} \min(r_t - \tau, 0)^2, \tag{20.25}$$

where τ is a minimum acceptable return (typically 0 or r). For symmetric returns, downside deviation $\approx \sigma/\sqrt{2}$ so Sortino $\approx \sqrt{2}\cdot$ Sharpe. Right-skewed returns (rare big wins, frequent small losses) give Sortino > Sharpe; left-skewed (the classic "selling volatility" profile) gives Sortino < Sharpe, a more honest measure.

The **Calmar ratio** uses max drawdown (Section 20.5) as the risk measure:

$$\text{Calmar} \;=\; \frac{\bar{r}_{\text{annual}}}{|\text{MaxDD}|}. \tag{20.26}$$

Calmar 1 means annual return equals worst drawdown; 2 means a year of return buys two worst drawdowns. Conservative managers target Calmar ≥ 0.5 and retire strategies below 0.25.

The **information ratio (IR)** generalises Sharpe to a benchmark r_t^b (e.g. S&P 500, or a factor-mimicking portfolio):

$$\text{IR} \;=\; \frac{\overline{r_t - r_t^b}}{\hat{\sigma}_{r-r^b}}. \tag{20.27}$$

Numerator: active return (alpha). Denominator: tracking error. An index fund tracking perfectly has IR undefined; beating a benchmark by 2% at 4% tracking error gives IR = 0.5. Long-only equity managers are judged on IR because Sharpe is dominated by the benchmark's Sharpe.

Performance metrics taxonomy

Metric	Formula	Risk measure in denominator
Sharpe	$(\bar{r} - r)/\hat{\sigma}_r$	Full std. dev.
Sortino	$(\bar{r} - r)/\hat{\sigma}_r^-$	Downside std.
Calmar	$\bar{r}_{\text{ann}}/\lvert\text{MaxDD}\rvert$	Max drawdown
IR	$\overline{(r - r^b)}/\hat{\sigma}_{r-r^b}$	Tracking error

All four are scale-invariant (degree zero in wealth), annualised, and pure numbers. Benchmarks: Sharpe ≥ 1 to go live, ≥ 2 high-quality, ≥ 3 small-capacity HFT. Sortino tracks Sharpe within $\sqrt{2}$ for symmetric returns. Calmar ≥ 0.5 for capital allocation.

Sharpe over 10 is (almost always) a bug

Annualised Sharpe above 10 on a backtest is overwhelmingly more likely a look-ahead bug, survivorship-biased universe, or infeasible execution than a real edge. Real edges leak through costs, capacity, and regime changes; they do not show Sharpe = 15 net of fees. Bisect the code before celebrating (see Chapter 21). The production-desk corollary is in Section 20.6: Sharpe *above* the target range gets flagged as a bug by risk, not a victory.

20.5 Drawdown arithmetic

Let $\{W_t\}_{t=1}^{T}$ be the wealth curve of a strategy. The **running peak** at time t is $P_t = \max_{s \le t} W_s$, the high-water mark to date. The **drawdown** at time t is

$$D_t \;:=\; \frac{W_t - P_t}{P_t} \;\le\; 0, \tag{20.28}$$

expressed as a negative fraction of the running peak (sometimes as a positive percentage with the sign flipped). The **maximum drawdown** over the sample is

$$\mathrm{MaxDD} \;:=\; \min_t D_t, \tag{20.29}$$

the worst peak-to-trough decline ever experienced.

The recovery asymmetry

The key identity: to recover from a drawdown of fraction $X \in (0,1)$ — wealth level $(1-X)P_t$ — the post-drawdown gain required is $X/(1-X)$:

$$(1-X)P_t \cdot (1+g) = P_t \quad\Longleftrightarrow\quad g = \frac{X}{1-X}. \tag{20.30}$$

10% requires 11.11%; 20% → 25%; 30% → 42.86%; 50% → 100%; 80% → 400%. For small X the gain is $X + X^2 + \ldots$, but by $X = 0.8$ the required recovery is *five times* the drawdown.

Drawdown recovery asymmetry

A drawdown of fraction X of peak wealth requires a subsequent gain of

$$g_{\text{recover}}(X) = \frac{X}{1-X}$$

to restore the peak. The map $X \mapsto X/(1-X)$ is linear for small X, passes through $(0.5, 1.0)$, and diverges as $X \to 1$.

Drawdown X	Gain needed $X/(1-X)$
10%	11.11%
20%	25.00%
30%	42.86%
50%	100%
80%	400%
90%	900%

Worked example: recovery time at 10% annual

A strategy compounds at 10% per year and suffers a 20% drawdown; required recovery 25%. At continuous $\mu = 0.10$,

$$T_{\text{recover}} = \frac{\ln(1/0.8)}{0.10} = \frac{0.2231}{0.10} \approx 2.23 \text{ years.} \tag{20.31}$$

Two years compounding at 10% just to stand still. Discrete annual compounding gives $\log(1/0.8)/\log(1.1) \approx 2.34$ years. One mediocre year costs two good ones.

Intuition

Drawdown asymmetry is why avoiding large losses matters more than capturing large wins. Symmetry fails because of the denominator: after losing X, the compounding base is $(1 - X)$, not 1. Future percentage returns compound off a smaller base; the deeper the hole, the disproportionately harder the climb. Half of risk management — stops, sizing, Kelly fractionalisation — is about avoiding the scenario where you need 100% to break even.

20.6 VaR and Expected Shortfall

Two industry-standard risk metrics round out the chapter; see (Hull, 2014) (Ch. 21) for the full treatment.

Value at Risk

Value at Risk (VaR) at confidence level α over horizon h is the smallest loss L such that

$$\mathbb{P}(\text{loss over } h \geq L) \leq 1 - \alpha. \tag{20.32}$$

"Our 1-day 95% VaR is \$5m" means we expect to lose more than \$5m on at most 5% of days. VaR is the $(1 - \alpha)$-quantile of the loss distribution. Typical choices: $\alpha = 95\%$ or 99%; $h =$ one day (trading books) or ten days (Basel III capital).

VaR is popular because it reports one dollar number a non-quantitative manager understands. It also lies about the tail: two portfolios can share a 99% VaR of \$5m but differ wildly beyond it — one loses exactly \$5m on its worst 1% of days, the other loses \$5m on most and \$100m on a few. Same VaR, very different behaviour when the tail fires.

Expected Shortfall

Expected Shortfall (ES), also called *Conditional VaR* (CVaR), fixes this by averaging over the tail:

$$\text{ES}_\alpha \;=\; \mathbb{E}\big[\,\text{loss} \mid \text{loss} > \text{VaR}_\alpha\big]. \tag{20.33}$$

The average loss given the loss exceeds VaR. By construction $\geq$ VaR, and it cares about tail shape, not just the 99th percentile. ES is *coherent* (Artzner et al.): ES of a sum of portfolios is at most the sum of individual ES's (subadditivity). VaR fails subadditivity — two portfolios can have combined VaR larger than the sum of their individual VaRs, contradicting the intuition that diversification reduces risk. ES does not. Post-2008 Basel moved from VaR to ES for trading-book capital (Hull, 2014).

VaR lies about the tail

VaR reports the *boundary* of the bad region, not its expected severity. Two portfolios can share a VaR: one just touching the threshold, one blowing through it. Always request

Expected Shortfall alongside VaR; treat VaR alone as a compliance box-tick. Guideline: ES/VaR ≈ 1 means a well-behaved tail; ≥ 1.5 means fat-tail trouble beyond the quantile.

Prosperity example

PICNIC_BASKET stat arb and half Kelly. The Frankfurt Hedgehogs placed 2nd in Prosperity 3 (2025) using ETF/constituent stat-arb on PICNIC_BASKET1/2; the basket strategy alone earned 40–60k SeaShells per round, plus related-product profits (Frankfurt Hedgehogs, 2025b). Treat this as an $\{r_t\}$ series with one "period" per round. Simulator calibration for a similar Prosperity 4 strategy:

- Expected PnL per round: $\hat{\mu} = 50{,}000$ SeaShells.
- Standard deviation of per-round PnL: $\hat{\sigma} = 10{,}000$ SeaShells (conservative, reflecting the mean-reverting nature of the spread and the occasional fat tail when the signal misfires).
- Starting bankroll: $W_0 = 500{,}000$ SeaShells (representative end-of-Round-1 bankroll for a top team).

Per-round $\mu = 50{,}000/500{,}000 = 10\%$ and $\sigma = 10{,}000/500{,}000 = 2\%$ (no competition borrowing cost, so $r = 0$). Per-round Sharpe $= 0.10/0.02 = 5$. Annualising is meaningless here — the "year" is a week of rounds — but the per-round 5 drives sizing.
Log-normal Kelly:

$$f^* = \frac{\mu}{\sigma^2} = \frac{0.10}{(0.02)^2} = \frac{0.10}{0.0004} = 250. \tag{20.34}$$

250× leverage is nonsensical. The full-Kelly output is telling you the competition's *position limit* binds long before Kelly does — good news, because you should trade to *the position limit every round*, not to an abstract "Kelly fraction". The position limit (60 baskets of PICNIC_BASKET1) is the binding constraint. Even unconstrained, half Kelly (125×) and quarter Kelly (62.5×) are still insane.
The formula is not wrong: fractional Kelly applies when *variance is binding*. In a competition with hard position limits the binding constraint is liquidity, not bankruptcy risk. Applied to the 8%/20% example of Section 20.2, where variance *does* bind, half Kelly prescribes $f = 1.0$ (fully invested). Compare to the top-team IV-scalping VOLCANIC_ROCK_VOUCHERS numbers in Chapter 27, where vol is higher and Kelly is a meaningful knob.

On a Goldman Sachs desk

On a Goldman systematic trading desk, every strategy with live capital has a written risk profile with three numbers:

- **Nominal Sharpe target**. Gross Sharpe ≥ 1.5; net ≥ 1.0. "High-quality" books target ≥ 2 net. Below the gross target is probation; above the top flagged by risk as mismeasured (see the pitfall above).
- **Max drawdown cutoff**. Typically 10–15%, sometimes 5% for a core market-making book. Beyond the cutoff the book is automatically reduced or halted: the Strat writes the halt logic, the trader retains override, both own the outcome.
- **Kelly fraction cap**, never above 0.5: Strats compute a rolling Kelly from recent μ, σ, and the book's leverage is capped at half of that. Quarter Kelly is not unusual

on fat-tailed books.

Strats own the live metric pipeline (rolling Sharpe, drawdown, Kelly, alerts). Traders own the override: they can reduce a book faster than the automated cutoff but cannot silently *raise* a Kelly cap. Separation of concerns is non-negotiable.

Two surprises for newcomers. First, the gross Sharpe target is not much higher than the net one because desk cost models (2–10 bps per round-turn) are well-calibrated. Second, production risk code is simpler than the research it disciplines: Sharpe, drawdown, and Kelly are three fifty-line functions on a five-second cadence, not a 500-line ML pipeline. Complexity is the enemy of live risk. See Chapter 36 for the data ingestion feeding these metrics.

Historical note

Kelly, Shannon, Thorp. John Larry Kelly Jr. (Bell Labs) derived the criterion in a 1956 paper in the *Bell System Technical Journal* (Kelly, 1956). His setup was information-theoretic: how much of your bankroll to bet on a sequence of noisy tips. The growth-optimal answer maximises $\mathbb{E}[\ln W]$, and the quantity inside turned out to be the mutual information between tips and outcome — a surprising link from gambling to Shannon's theory.

Claude Shannon handed the paper to Edward O. Thorp at MIT in 1960, just after Thorp had solved blackjack card counting. Thorp's *Beat the Dealer* (1962) applied Kelly to blackjack; he co-founded Princeton–Newport Partners in 1969. Over its 19-year life, Princeton–Newport compounded at $\approx 19.1\%$ net of fees, never had a losing year, and used fractional Kelly throughout (Thorp, 2006). Nobody there traded full Kelly.

20.7 A short reference implementation

```python
import numpy as np

def kelly_binary(p, b, a):
    """Optimal fraction for a two-outcome bet. Win b with prob p, lose a with prob 1-p."""
    return (p * b - (1 - p) * a) / (a * b)

def kelly_lognormal(mu, sigma, r_f=0.0):
    """Optimal fraction for a log-normal risky asset vs cash at r_f."""
    return (mu - r_f) / sigma ** 2

def sharpe(returns, r_f=0.0, periods_per_year=252):
    """Annualised Sharpe from a per-period return series."""
    excess = returns - r_f / periods_per_year
    return excess.mean() / excess.std(ddof=1) * np.sqrt(periods_per_year)

def max_drawdown(wealth):
    """Worst peak-to-trough decline on a wealth curve (fraction, <= 0)."""
    peak = np.maximum.accumulate(wealth)
    return ((wealth - peak) / peak).min()

# Sanity checks (verified numerically in chapter text):
assert abs(kelly_binary(0.55, 1, 1) - 0.10) < 1e-12
```

```python
assert abs(kelly_lognormal(0.08, 0.20) - 2.0) < 1e-12
```

Listing 20.1: Kelly and performance-metric reference implementation.

These four functions implement the chapter's sizing and scoring. A production library adds rolling windows, downside deviation, drawdown recovery time, IR, and tail-conditioned loss — each a wrapper around these primitives.

20.8 Key facts to memorise

- **Binary Kelly**: $f^* = (pb - qa)/(ab)$. For a fair-odds coin, $f^* = p - q$. For the 55/45 coin, bet 10% per trial.
- **Log-normal Kelly**: $f^* = (\mu - r)/\sigma^2$. Maximum achievable long-run growth is $r + \frac{1}{2}\text{Sharpe}^2$. Sharpe squared is what compounds.
- **Fractional Kelly**: trade cf^* with $c \in [0.25, 0.5]$. Half Kelly retains 75% of the growth rate with half the volatility and a quarter of the variance; Thorp's rule of thumb.
- **Drawdown probability at c-Kelly**: the lifetime probability of ever drawing down to fraction x of starting wealth is $x^{2/c-1}$. Full Kelly: 50% chance of ever halving. Half Kelly: 12.5%. Quarter Kelly: 0.78%.
- **Sharpe**: $(\bar{r} - r)/\hat{\sigma}_r$, annualised with $\sqrt{252}$ from daily, $\sqrt{12}$ from monthly. Gross ≥ 1.5, net ≥ 1.0 is a live-capital threshold on a systematic desk. 0.79 (the worked example) is sub-threshold.
- **Sortino**: use downside deviation in the denominator. Pay attention to it when returns are skewed.
- **Calmar**: annual return divided by $|\text{max drawdown}|$. A direct tradeoff of growth against worst historical loss. Conservative managers target ≥ 0.5.
- **Information ratio**: alpha over tracking error. Right metric for a long-only or benchmarked book.
- **Drawdown recovery asymmetry**: loss of fraction X requires gain of $X/(1 - X)$ to recover. 20% → 25%; 50% → 100%; 80% → 400%. This is the single most important identity in practical risk management and it is why avoiding large losses beats capturing large wins.
- **VaR vs ES**: VaR is the $(1 - \alpha)$-quantile of loss; ES is the *expected* loss beyond VaR. ES is coherent (subadditive), VaR is not. Post-Basel III the regulatory standard is ES.
- **Kelly cap in production**: strats run rolling Kelly estimates; the production book is capped at half of the Kelly estimate, never at full Kelly. Sizing is half the risk problem; scoring (rolling Sharpe, drawdown, Kelly, VaR, ES) is the other half.

Chapter 21

Backtesting Done Right

A **backtest** is a simulation of what a trading strategy would have made on historical data. Done wrong, it is a lie: a strategy that "made 20% per year in backtest" can lose money live, for avoidable reasons. Chapters 10 and 11 already met two — the look-ahead from fitting an OU model on the full dataset, and the over-refit hedge ratio — and in both the cure was walk-forward. This chapter treats backtest discipline as a first-class subject and catalogues the *seven standard ways backtests lie*, with numerical examples and defences. The centrepiece is Lopéz de Prado's **Combinatorial Purged Cross-Validation** (CPCV), the minimum cross-validation discipline for financial machine learning.

Chapter 20 scored track records with Sharpe, Sortino, Calmar, information ratio, and drawdown. Every one of those numbers *only means something if the backtest was honest.* A Sharpe of 1.8 from 1000 parameter sweeps is worth about 0.6 after multiple-testing deflation — derived by hand in Section 21.4. Hence the **Deflated Sharpe Ratio** (López de Prado, 2018) as the principled correction.

Read this chapter as an anti-handbook: most of the value is never making these mistakes, the rest is knowing which framework (`Backtesting.py`, `vectorbt`, `NautilusTrader`, `QuantConnect`) to reach for.

Prerequisites

From earlier in the book:

- Chapter 10: walk-forward OU fitting, and the pitfall of computing historical z-scores against an $\hat{\mu}$ estimated on the full series. That chapter's pitfall on look-ahead OU fitting is the canonical example of Lie 1 below.
- Chapter 11: the "over-refit $\hat{\beta}$" pitfall — refitting the hedge ratio every bar on windows that include the current observation — is the canonical example of Lies 1 and 3 acting together.
- Chapter 20: the Sharpe ratio, the drawdown recovery identity, and the framing of performance metrics as "scoring after sizing". This chapter's business is to ensure those metrics are not meaningless.
- Chapter 7 and Chapter 19: the square-root impact law and the A–C execution cost model. Lie 5 (transaction-cost neglect) is resolved using numbers from those chapters.

From outside the book:

- Elementary statistics: t-test, p-value, standard error, and the idea of a hypothesis test under a null.

- Machine-learning cross-validation: k-fold CV as it is taught in any intro ML course. This chapter explains why the textbook version is wrong for financial time series, and what to replace it with.
- The **expected maximum of N i.i.d. standard normals** is approximately $\sqrt{2 \ln N}$; this heuristic controls the magnitude of data-snooping bias in Section 21.4.

21.1 The seven lies a backtest can tell you

Every subsequent section — walk-forward, CPCV, deflated Sharpe, framework choice — defends against one or more of these seven failure modes. The list is adapted from López de Prado (2018) (Ch. 11–12) and Chan (2013) (Ch. 3).

Lie 1: Look-ahead bias

A backtest commits **look-ahead bias** when the simulated strategy, at time t, uses information unavailable at t in real trading. The cartoon version: the backtest buys today because it already knows tomorrow's closing price went up — live, tomorrow's close does not yet exist. Archetype: the OU pitfall of Chapter 10, where $\hat{\mu}$ and $\hat{\theta}$ are fitted on the *full* series and used to compute every historical z-score. Each entry is timed against a mean estimated using post-trade data. Live, $\hat{\mu}$ is fitted on a rolling window ending before t and differs meaningfully from the in-sample value, so the thresholds hit in backtest vanish.

Most sources do not look like look-ahead:

- **Close-to-close returns used intraday.** A signal that reacts to $r_t = \log(C_t/C_{t-1})$ at time $t + \epsilon$ has used the closing print from time t, which was not available until the closing auction completed.
- **Corporate-action adjustments.** Dividend- and split-adjusted historical prices reflect *future* corporate actions: if ACME paid a \$2 dividend in 2024 and you pulled adjusted prices in 2026, the 2022 prices are shifted down by the dividend. A 2022-live strategy had no access to that adjustment.
- **Fundamental data with revision lag.** Earnings, GDP, index reconstitution: the value you see in your data vendor for date t is the *latest revision*, not the value the market saw on t.
- **Point-in-time benchmarks.** The S&P 500 constituents as of a 2015 date are not the current constituents; the historical series you downloaded may be. This is the gateway to Lie 2.

Concrete magnitude. OU pairs trades with $\hat{\mu}$ on the full sample report Sharpes of 2.0–3.0 on canonical equity pairs; switching to strict walk-forward ($\hat{\mu}$ on a rolling window ending at $t - 1$) typically cuts the Sharpe to 0.5–1.0 on the same series (Chan, 2013). The difference is pure look-ahead.

Lie 2: Survivorship bias

A backtest commits **survivorship bias** when run on a universe defined by *today's* membership. Example: a long-only equal-weighted S&P 500 backtest 2000–2020 using current constituents. The current list contains firms that *survived*; bankruptcies, delistings, and distressed acquisitions between 2000 and 2020 are missing. Enron, Lehman, WorldCom, Washington Mutual, pre-bankruptcy GM, Bear Stearns — none appear, because none are in today's constituent list.

Magnitude. López de Prado (2018) and Chan (2013) both report that survivorship inflates long-only S&P 500 backtests by 0.2–0.5 points of annualised return per year on a 20-year window: a Sharpe inflation of 0.3–0.5 at 15% vol. Tighter universes are worse: long/short on micro-cap NASDAQ using a *current* universe can gain 2–5 Sharpe points from survivorship, because the delisted micro-caps were catastrophic losers.

The only fix is a **point-in-time universe**: tradable instruments *as known on date* t, including now-delisted names, with ticker changes and delisting events. CRSP, Bloomberg, and QuantConnect's Lean dataset ship PIT universes; free sources typically do not. Assume a free dataset is survivorship-biased unless documented otherwise.

Lie 3: Overfitting

A backtest commits **overfitting** (*in-sample tuning*) when the same data is used to both *select* parameters and *evaluate* performance. Archetype: a grid search over 1000 (lookback, z-threshold, stop-loss) combinations on the same 5-year returns, reporting the best Sharpe. Under a zero-edge null the winner's Sharpe is about $\sqrt{2\ln N}$ standard errors above zero: for $N = 1000$, $T = 1260$, this is $\sqrt{2\ln 1000}/\sqrt{T/252} \approx 1.66$ annualised. Random strategies grid-searched 1000 times routinely produce Sharpes near 1.66 with no alpha.

Overfitting is distinct from Lie 4 (data snooping): overfitting tunes *one* strategy's parameters on the evaluation data; snooping tries many *distinct* ideas and reports only winners. Both inflate Sharpe by multiple testing; the defence is CPCV (Section 21.3) plus Deflated Sharpe (Section 21.4).

Lie 4: Data snooping / selection bias

A backtest commits **data snooping bias** (selection bias) when the researcher evaluates many distinct strategy ideas on the same dataset and reports only the winners. The winner's Sharpe is biased upward by the same multiple-testing inflation as Lie 3.

The dangerous case is not counting trials. Six months of iteration typically touches $N \in [50, 500]$ strategies; uncounted, the Sharpe in the final writeup is silently inflated and the strategy goes live on noise. A research log — every idea, parameter set, and result — is the only mechanical defence; Section 21.4 turns it into a formula.

Lie 5: Transaction-cost neglect

A backtest commits **transaction-cost neglect** when it assumes free execution, fills at the mid, or no market-impact penalty. For a fast strategy this is the difference between "gold mine" and "money sink".

Worked magnitude. Take a daily-frequency mean-reversion strategy with gross expected per-day return $\mu = 5$ bp and per-day return standard deviation $\sigma = 50$ bp. Its gross annualised Sharpe is

$$\mathrm{SR}_{\mathrm{gross}} \;=\; \frac{\mu}{\sigma}\sqrt{252} \;=\; \frac{0.0005}{0.0050}\sqrt{252} \;=\; 1.587.$$

Net of a 2 bp round-trip cost per trade (half-spread plus small impact), the mean collapses to $\mu_{\mathrm{net}} = 3$ bp and $\mathrm{SR}_{\mathrm{net}} = (0.0003/0.0050)\sqrt{252} = 0.952$. Net of 5 bp, $\mu_{\mathrm{net}} = 0$ and $\mathrm{SR}_{\mathrm{net}} = 0$. Edge at 2 bp, break-even at 5 bp. A costless backtest reports Sharpe 1.59 in all three cases.

On a sell-side desk, transaction cost combines the square-root impact law of Chapter 7 ($I(Q) \approx Y\sigma\sqrt{Q/V}$, $Y \approx 0.1$ for cash equities) with the half-spread from Chapter 5 and explicit fees/rebates. A Prosperity backtest that subtracts a flat tick is a reasonable first cut; zero is a lie.

Lie 6: Slippage

A backtest commits **slippage** misspecification when it assumes execution at the backtest-time price but live execution happens later, at a worse price, or walks the book. Transaction cost is the expected cost of crossing the spread and consuming visible liquidity; slippage is the *unanticipated* wedge between intent and fill.

Common backtest slippage lies:

- **"Fill at the signal-bar close."** The decision arrives after bar t closes, so the earliest fill is the open of $t+1$, with the full overnight gap in between.
- **"Fill at the mid."** Market orders fill at the best opposing quote. On a symmetric book this is half the spread (already in Lie 5); on an asymmetric book, more.
- **"Fill my full size at one level."** A 10,000-lot buy against top-of-book size 100 walks the book; filling the whole order at the top ask underestimates cost.
- **"No queue at the top."** A passive bid does not fill just because the market trades there — its price-time priority position matters. Assuming instant passive fills is over-optimistic by a factor proportional to queue length.

The fix is a slippage model: flat k ticks, a linear-in-volume model calibrated from bid/ask imbalance, or a full order-book simulation like NautilusTrader or Lean provide.

Lie 7: Regime dependence

A backtest commits **regime dependence** fraud when the period is dominated by a regime that does not hold live. A trend-follower profitable 2009–2019 (central-bank-supported trends) dies if extended back to the 1987 crash or forward into the 2022 tightening. A vol seller looks wonderful 2013–2019 and catastrophic on COVID-March 2020 or "Volmageddon" 2018.

Regime dependence is not "the backtest was wrong" but "the sample was unrepresentative". There is no statistical cure — statistics assume the future resembles the past. Partial cures:

- Extend the window as far back as data supports. 2002–2022 spans four distinct regimes (dot-com aftermath, GFC, QE, post-COVID tightening); 2015–2020 covers one regime plus a crisis spike.
- **Stratify by regime** using a VIX- or yield-curve- based label, reporting Sharpe, return, drawdown per regime. A strategy with Sharpe 1.8 low-vol and -2.0 high-vol is two strategies that should be switched on a signal.
- Reserve a **final hold-out** — the last 1–2 years — *never* looked at during research. Evaluate on it exactly once, at the end.

The seven deadly backtest sins

1. **Look-ahead** — using data from after decision time.
2. **Survivorship** — running a universe defined by today's membership.
3. **Overfitting** — tuning parameters on the same data used for evaluation.
4. **Data snooping** — trying many strategies on the same data and reporting only the winners.
5. **Transaction-cost neglect** — omitting spread, impact, financing.
6. **Slippage** — assuming the fill price equals the signal-time price.

7. **Regime dependence** — relying on a sample period whose conditions do not persist.

Each can turn a break-even strategy into a "backtested Sharpe 2.0" that evaporates day one live.

21.2 Walk-forward validation

The first-order defence against Lies 1 and 3 is **walk-forward validation**, met in Chapter 10 for honest OU fitting. Here it is a general protocol.

The procedure. Pick a fitting window W (e.g. 1 year) and a test step R (e.g. 1 month). Run the backtest as a loop:

1. Fit all strategy parameters on observations in $[t-W,\ t)$. These are the *training* observations.
2. Freeze the parameters. Run the strategy on observations in $[t,\ t+R)$. These are the *test* observations. Record the realised P&L.
3. Slide the window: $t \leftarrow t+R$. Go to step 1.

Sharpe, return, and drawdown come from the *concatenation of test-period P&Ls.* No test observation was used in fitting the parameters that generated its signal — out-of-sample by construction.

Walk-forward procedure

Fix window W and step R. For each time t:

$$\underbrace{[\,t-W,\,t\,)}_{\text{fit parameters}} \quad \Rightarrow \quad \underbrace{[\,t,\,t+R\,)}_{\text{trade and record P\&L}} .$$

Slide $t \to t+R$ and repeat. Concatenate the test-period P&Ls. Every observation the strategy trades was *unseen* when its parameters were fit.

Why it beats a single split. Walk-forward simulates the real passage of time: repeatedly fit on the past, trade the next unseen period, then advance — exactly what a live strategy does. A random train/test shuffle (the ML default) would scatter future points into the training set, leaking tomorrow's prices into yesterday's fit; a fixed split ("2000–2015 train, 2016–2022 test") has two problems. First, the test period lives in a single regime. Second, the test Sharpe has high variance: 6 years (≈ 1500 days) gives standard error $\sqrt{252/1500} \approx 0.41$, so a 95% CI of ± 0.82. Walk-forward spreads test points over the whole history and catches regime drift by refitting.

Why it is not enough. Walk-forward fixes parameter-level look-ahead, but two leaks remain. (i) Grid-searching over 100 values of W and reporting the winner brings Lie 3 back — the loop was honest, but the choice of W was in-sample. (ii) It misses **preprocessing leakage**: normalising features with the global mean/variance before the loop leaks information into every test point. Preprocessing must live inside the loop; grid-search-on-W is handled by CPCV plus deflated Sharpe, next.

21.3 Combinatorial Purged Cross-Validation

Walk-forward is a *single path*: each observation is in the test set once, in a fixed position. A regime transition landing at a fit boundary makes the strategy lucky or unlucky, and the reported Sharpe is one draw from a high-variance distribution.

Combinatorial Purged Cross-Validation (CPCV), due to López de Prado (2018) (Ch. 12), generates *many* distinct train/test splits respecting temporal order and averages the test statistic. Three principles:

1. **Purging.** Remove from the training set any observation whose label-generation window overlaps with test observations. This breaks the most insidious leak in ML-based trading.

2. **Embargo.** Insert a buffer period of h observations *after* each test fold during which no training observations are used. This breaks leakage through autocorrelation in the features.

3. **Combinatorial splits.** Instead of the usual k-fold CV's k splits, CPCV uses every combination of two (or more) folds as the test set, yielding $\binom{k}{2}$ (or more) splits per value of k.

First, why naive k-fold CV is wrong for time series.

Why naive k-fold CV is wrong for time series

Textbook k-fold partitions into k folds, trains on $k-1$, tests on the remainder, repeating k times. Sound for *i.i.d.* data: train and test are independent draws.

Financial time series break i.i.d. in two ways:

1. **Feature autocorrelation.** Rolling features (moving averages, vols, z-scores) are highly correlated between neighbours. If t is in test and $t-1$ in train, the model has already seen essentially the test features.

2. **Label leakage.** ML-trading labels are often forward-looking: "sign of return over next H bars" or triple-barrier outcomes over h bars. If t is in train and $t+1$ in test, the training label at t was computed from the same price path that generates the test features at $t+1$ — the model is trained on test data through the labels.

The second is fatal: every naive k-fold Sharpe in ML-for-trading is inflated by an amount proportional to the label horizon, invisible in standard diagnostics. Purging and embargo remove it.

Purging and embargo, in words

Purging drops any training observation whose *label window* overlaps a test observation. If labels at t use price data in $[t, t+h]$, a training t whose label window intersects a test range must go: its label was computed from data the model will soon see as a feature.

Embargo drops training observations for h observations *after* every test fold — feature-side insurance: even without label overlap, a training point near the test edge may have highly correlated features.

For test fold $[a, b]$, the surviving training set is

$$\text{train} \;=\; \big\{\, t \;:\; [\,t,\, t+h\,] \cap [\,a,\, b\,] = \varnothing \;\text{ and }\; t \notin [\,b,\, b+h\,] \,\big\}.$$

A fully worked 20-period toy example

$n = 20$ observations $i = 0, \ldots, 19$; $k = 5$ folds of size 4; embargo $h = 1$; combinatorial splits choose 2 of 5 folds as the test set, for $\binom{5}{2} = 10$ splits. Every number below was verified by the reference implementation at the end of the chapter.

Step 1: folds. With $n = 20$ and $k = 5$, the fold boundaries are

$$F_0 = \{0, 1, 2, 3\}, \qquad F_1 = \{4, 5, 6, 7\}, \qquad F_2 = \{8, 9, 10, 11\},$$
$$F_3 = \{12, 13, 14, 15\}, \qquad F_4 = \{16, 17, 18, 19\}.$$

Step 2: the $\binom{5}{2} = 10$ splits. The combinatorial splits are all 2-subsets of $\{0, 1, 2, 3, 4\}$:

$$\underbrace{\{F_0, F_1\}}_{\text{split 1}},\ \{F_0, F_2\},\ \{F_0, F_3\},\ \{F_0, F_4\},\ \{F_1, F_2\},$$
$$\{F_1, F_3\},\ \{F_1, F_4\},\ \{F_2, F_3\},\ \{F_2, F_4\},\ \{F_3, F_4\}.$$

So there are 10 distinct backtest runs instead of the 5 that a standard k-fold would give you.

Step 3: purge + embargo on split 1, $\{F_0, F_1\}$. Test indices: $\{0, 1, 2, 3, 4, 5, 6, 7\}$. With an embargo of $h = 1$ after the test set, training observation i must satisfy

$$i \notin \{0, \ldots, 7\} \quad \text{and} \quad |i - j| > 1 \text{ for every } j \in \{0, \ldots, 7\}.$$

Of $i = 8, \ldots, 19$, only $i = 8$ is within distance 1 of $j = 7$, so it is embargoed. Training indices: $\{9, 10, \ldots, 19\}$, size 11. The single $i = 8$ is neither trained on nor tested on — purged.

Step 4: purge + embargo on split 2, $\{F_0, F_2\}$. Test indices: $\{0, 1, 2, 3, 8, 9, 10, 11\}$ — two non-contiguous blocks, embargo applied around each:

- Observation $i = 4$ is distance 1 from $j = 3$ (embargoed).
- Observation $i = 7$ is distance 1 from $j = 8$ (embargoed).
- Observation $i = 12$ is distance 1 from $j = 11$ (embargoed).

Training set: $\{5, 6, 13, 14, 15, 16, 17, 18, 19\}$, size 9. Three observations $(4, 7, 12)$ are purged — the non-contiguous test set costs more than the contiguous split 1.

Step 5: purge + embargo on split 3, $\{F_0, F_3\}$. Test indices: $\{0, 1, 2, 3, 12, 13, 14, 15\}$. By the same logic, $i = 4, 11, 16$ are embargoed. Training set: $\{5, 6, 7, 8, 9, 10, 17, 18, 19\}$, size 9.

The other seven splits use the same mechanics. Across all 10, the researcher gets 10 out-of-sample P&L paths. The reported statistic is the *distribution* of the ten Sharpes, revealing dependence on which observations fell where. Sharpes $\{1.3, 1.4, 1.2, 1.5, 1.4, 1.3, 1.2, 1.5, 1.3, 1.4\}$ is robust; $\{2.8, 1.1, -0.4, 2.2, 0.3, 1.6, 0.8, 2.1, 1.4, -0.2\}$ has the same mean (≈ 1.17) but is one unlucky draw from unrunnable.

Test coverage. Each of the 5 folds appears in $\binom{4}{1} = 4$ of the 10 splits, so every observation is a test point 4 times. Total test-observation count: $10 \times 8 = 80$, i.e. 4 per observation — four times the coverage of ordinary 5-fold, with 10 (overlap-dependent) Sharpe estimates.

CPCV — the three pillars

Combinatorial Purged CV rests on exactly three rules.

1. **Purge**: drop any training observation whose label window overlaps a test observation.
2. **Embargo**: drop training observations within h of any test observation, buffering feature-side leakage.
3. **Combinatorial**: use every 2-subset of the k folds as the test set, yielding $\binom{k}{2}$ (instead of k) train/test splits, each observation appearing as a test point $k-1$ times.

The reported test statistic is the *distribution* across splits, not any single run's number.

Intuition

CPCV is the minimum cross-validation discipline for time-series ML. Anything less (naive k-fold, single walk-forward, plain train/test) trains on test data through the labels, with contamination proportional to the label horizon. Each rule patches a specific leak: purging $\rightarrow$ label leak; embargo $\rightarrow$ feature-autocorrelation leak; combinatorial $\rightarrow$ single-path luck. Together they yield a distribution of out-of-sample Sharpes on disjoint data.

Don't skip the label-horizon step

The common CPCV mistake is implementing combinatorial and embargo but forgetting that the *label horizon* h is a property of the label, not the features. If labels at t use prices up to $t+20$, then $h=20$ and any training t whose label window intersects a test observation within $[t, t+20]$ must be purged. Purging only ± 1 (confusing embargo for purge) still leaks. See López de Prado (2018) Ch. 7.

21.4 Deflated Sharpe Ratio

The second statistical lie is multiple-testing inflation. CPCV controls leakage *inside* a single strategy's evaluation; deflated Sharpe controls inflation from having tried *many* strategies. The two are orthogonal — you need both. "Deflation" is the plain-English operation of shrinking a reported Sharpe to account for the number of strategies tried: the best of 100 zero-edge coin-flip strategies will itself look like an edge, and the deflation formula removes exactly that bias.

The construction is due to Bailey and Lopez de Prado (2014), summarised in (López de Prado, 2018) Ch. 14. Under a zero-edge null over N independent trials, the winner's Sharpe is centred not on zero but on the **expected maximum** of N i.i.d. Sharpe estimates.

Expected-maximum formula. For N independent trials where each Sharpe estimator is asymptotically normal with standard error $\mathrm{Var}[\widehat{\mathrm{SR}}]^{1/2}$, the expected maximum is

$$\mathbb{E}\Big[\max_{n\le N}\widehat{\mathrm{SR}}_n\Big] \;\approx\; \mathrm{Var}[\widehat{\mathrm{SR}}]^{1/2}\Big[(1-\gamma)\,\Phi^{-1}(1-\tfrac{1}{N}) + \gamma\,\Phi^{-1}(1-\tfrac{1}{Ne})\Big], \tag{21.1}$$

where $\gamma \approx 0.5772$ is Euler–Mascheroni and Φ is the standard normal CDF (Bailey and Lopez de Prado, 2014). The bracket evaluates to ≈ 2.5306 ($N=100$), 3.2414 ($N=1000$), 3.8513 ($N=10000$). The asymptotic is $\mathbb{E}[\max] \approx \sqrt{2\ln N}$.

Deflated Sharpe. The deflated Sharpe ratio DSR is then defined as the probability, under the null, that the observed Sharpe $\widehat{\mathrm{SR}}$ exceeds the expected maximum under the best of N trials:

$$\mathrm{DSR} \;:=\; \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathbb{E}[\max \widehat{\mathrm{SR}}])\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{\mathrm{SR}}^2}} \right), \tag{21.2}$$

where T is the number of return observations in the Sharpe estimate, and $\hat{\gamma}_3$, $\hat{\gamma}_4$ are the sample skewness and kurtosis of the return series (Bailey and Lopez de Prado, 2014). Under normal returns ($\hat{\gamma}_3 = 0$, $\hat{\gamma}_4 = 3$) the denominator simplifies to $\sqrt{1 + \widehat{\mathrm{SR}}^2/2} \approx 1$ for small $\widehat{\mathrm{SR}}$.

Convention: DSR ≥ 0.95 means the strategy is unlikely to be the winner of a random N-trial tournament; DSR ≈ 0.50 is noise.

A worked DSR example

$N = 100$ configurations on five years of daily data, so $T = 252 \times 5 = 1260$; observed annualised Sharpe of the winner $\widehat{\mathrm{SR}}_{\mathrm{ann}} = 1.8$; assume normal returns. In Python, to four decimals:

1. Convert to per-period Sharpe: $\widehat{\mathrm{SR}} = 1.8/\sqrt{252} = 0.1134$.
2. Expected-maximum bracket in (21.1) for $N = 100$: $(1 - 0.5772)\Phi^{-1}(1 - 0.01) + 0.5772\,\Phi^{-1}(1 - 1/(100\mathrm{e})) = 0.4228 \times 2.3263 + 0.5772 \times 2.5894 = 0.9836 + 1.4946 = 2.4782$. (A more careful computation yields 2.5306; the small discrepancy is from higher-order terms in the approximation.)
3. Multiply by $\mathrm{Var}[\widehat{\mathrm{SR}}]^{1/2} = 1/\sqrt{T} = 1/\sqrt{1260} = 0.02817$ to get the expected maximum per-period Sharpe: $\mathbb{E}[\max \widehat{\mathrm{SR}}] \approx 2.5306 \times 0.02817 = 0.07129$.
4. Plug into the denominator of (21.2): under normal returns ($\hat{\gamma}_3 = 0$, $\hat{\gamma}_4 = 3$) the denominator is $\sqrt{1 + 0.1134^2/2}/\sqrt{T-1} = \sqrt{1.00643}/\sqrt{1259} = 0.02827$.
5. Compute the DSR argument: $(0.1134 - 0.07129)/0.02827 = 1.4889$.
6. Evaluate Φ: $\Phi(1.4889) = 0.9317$.

DSR ≈ 0.93: under $N = 100$, a 93% chance of genuine edge rather than tournament noise. At $N = 1000$ the expected maximum climbs to $3.2414 \times 0.02817 = 0.0913$, argument drops to $(0.1134 - 0.0913)/0.02827 = 0.7831$, DSR $= \Phi(0.7831) = 0.7832$ — below 0.95. Ten times more trials flips "probably real" to "probably noise" with no data change.

Headline. The *effective* annualised Sharpe is

$$\widehat{\mathrm{SR}}_{\mathrm{eff}} \;=\; \widehat{\mathrm{SR}} - \mathbb{E}[\max \widehat{\mathrm{SR}}],$$

i.e. $1.8 - 1.13 = 0.67$ for $N = 100$. A reported Sharpe of 1.8 from 100 trials is worth ≈ 0.67 after deflation.

Deflated Sharpe Ratio

Under N independent trials with per-period Sharpe standard error $1/\sqrt{T}$, the expected maximum Sharpe under a zero-edge null is approximately $\mathbb{E}[\max]/\sqrt{T}$ where $\mathbb{E}[\max]$ is the expected maximum of N iid standard normals. The effective (deflated) annualised Sharpe is

$$\widehat{\mathrm{SR}}_{\mathrm{eff}} \;\approx\; \widehat{\mathrm{SR}} - \mathbb{E}[\max] \cdot \sqrt{\frac{252}{T}}.$$

For $\widehat{\mathrm{SR}} = 1.8$, $T = 1260$, $N = 100$ (using the Bailey bracket $\mathbb{E}[\max] \approx 2.5306$): $\widehat{\mathrm{SR}}_{\mathrm{eff}} \approx 1.8 - 1.13 = 0.67$. Ten times more trials ($N = 1000$, $\mathbb{E}[\max] \approx 3.2414$) reduces $\widehat{\mathrm{SR}}_{\mathrm{eff}}$ to $\approx 1.8 - 1.45 = 0.35$.

Multiple testing silently inflates Sharpe

Six months of research typically touches $N \in [50, 500]$ trials. Report the winner's Sharpe without tracking N and the number is inflated by $\sqrt{2 \ln N}$ standard errors; the true effective Sharpe is $\widehat{\mathrm{SR}} - \sqrt{2 \ln N \cdot 252/T}$. For $T = 1260$ and that range, subtract 0.88–1.41. Any $\widehat{\mathrm{SR}} \leq 1.4$ from six months of research is indistinguishable from chance at $N = 500$; do not deploy without (a) extending the sample, (b) a fresh out-of-sample hold-out, or (c) live tiny-capital paper trading.

21.5 Backtesting-framework comparison

Four Python frameworks dominate.

`Backtesting.py` (Zvezdin et al.). Compact, event-driven, single-asset. Small API, clean docs, 20-line strategies, simple cost/slippage, built-in hyperparameter optimiser. No built-in walk-forward or CPCV, no portfolio constraints, no order book. Good for learning.

`vectorbt` (Polakow). Pandas-native, fully vectorised, optimised for parameter sweeps. A 10,000-point MA-crossover grid runs in seconds. Supports portfolios, sizing, slippage. Not truly event-driven: signals and fills align bar by bar, so intra-bar order-book dynamics are out of scope, and the vector API tempts next-bar look-ahead.

`NautilusTrader` (Nautech Systems). Rust core, Python API, tick-by-tick with a full order-book model — can replay raw LOBSTER dumps with realistic queue-position fills. Operationally heavy; overkill for daily-bar sweeps. Use when microstructure correctness matters.

`QuantConnect` / `Lean`. Cloud-hosted full-stack platform with point-in-time historical data for US equities, options, futures, crypto, and an event-driven engine. The PIT universes are the main reason to pay. Lock-in: strategies are written to the Lean API. Prosperity does not need it; GS desks use internal equivalents.

Framework	Model	Speed	CV support	Best for
`Backtesting.py`	event, 1 asset	moderate	manual wrap	learning
`vectorbt`	bar-aligned vector	very fast	manual wrap	parameter sweeps
`NautilusTrader`	event, full book	slow	manual wrap	microstructure
`QuantConnect`	event, cloud	moderate	built-in	serious PIT data

Table 21.1: One-line trade-offs for the four mainstream Python backtesting frameworks. None ship CPCV out of the box except QuantConnect, and even there it is a research tool rather than a core primitive; walk-forward and CPCV should be wrapped around whichever framework is chosen.

Framework taxonomy

- `Backtesting.py`: simple, event-driven, single-asset. Teach yourself on this.
- `vectorbt`: vectorised, blazing fast. Best for parameter sweeps. Easy to leak look-ahead if you're not careful.
- `NautilusTrader`: tick-level, order-book, production-grade. Use when microstructure matters.
- `QuantConnect` / `Lean`: point-in-time data, cloud-hosted, institutional-ish. Use when survivorship bias is the binding constraint.

21.6 Prosperity case study: plateau, not peak

Prosperity example

The Frankfurt Hedgehogs, 2nd in Prosperity 3, describe their parameter-selection discipline (see `imc-prosperity-3/README.md`, §410):

> "During parameter selection, we always prioritized landscape stability over pure performance peaks. Rather than picking the best parameter set based on maximum backtested profit, we chose combinations that showed consistent, flat regions of good performance, reducing sensitivity to slight shifts and avoiding overfit disasters."

In this chapter's language: they reported the top of a *plateau* (a neighbourhood where Sharpe is roughly constant), not an in-sample peak. Plateaus defend against Lies 3, 4, and 7 at once: a narrow spike usually fits noise (a 10% perturbation of the lookback kills the edge), a flat region is edge with a margin of safety.
Implementation: instead of `argmax` over the grid, return the point whose $\pm k$-neighbourhood average is maximised. For two parameters (W, z_{entry}) a 3×3 window suffices.

21.7 Production pipeline on a sell-side desk

On a Goldman Sachs desk

On a GS systematic desk, a new strategy clears five gates in order:

1. **Walk-forward on ≥ 5 years.** The Strat owns the loop. Five years is the minimum to span two distinct macro regimes at daily frequency.
2. **Deflated Sharpe.** Report raw Sharpe, trial count, and DSR per (21.2). DSR below 0.95 kills the strategy.
3. **Transaction-cost audit.** The execution desk reprices every trade using the in-house impact model (calibrated square-root), adds financing/clearing, and returns a net Sharpe. The most common casualty.
4. **Paper trading ≥ 30 days** on a production router with orders intercepted before the exchange. Paper vs. projected P&L divergence above ~ 1 Sharpe point or 20% of return triggers review.

5. **Risk limits, margin, kill-switches.** The risk system (Chapter 36) owns this gate: declared max notional, gross, net, Greeks, and auto-liquidation trigger.

A Strat who skips gate 3 and reports Sharpe 2.0 to the desk head will be asked: "Net of execution costs, what is the Sharpe?" Always have the number ready.

21.8 A short reference implementation

A minimal CPCV splitter fits in under 30 lines. The implementation below matches Section 21.3: `n=20, k=5, embargo=1` reproduces the 10 splits, including purged $i = 8$ on split 1. Uses $h = 1$ for clarity; production extends the inner check to arbitrary label windows.

```python
import numpy as np
from itertools import combinations

def cpcv_splits(n, k, embargo):
    """Generate (train_idx, test_idx) pairs under combinatorial purged CV.

    n:        number of observations
    k:        number of folds (test sets will be all 2-subsets of folds)
    embargo: number of observations to drop on each side of the test set
    """
    fold_size = n // k
    all_folds = [list(range(i * fold_size, (i + 1) * fold_size))
                 for i in range(k)]
    for test_fold_indices in combinations(range(k), 2):
        test_idx = sum([all_folds[i] for i in test_fold_indices], [])
        test_set = set(test_idx)
        train_idx = [i for i in range(n)
                     if i not in test_set
                     and not any(abs(i - j) <= embargo for j in test_idx)]
        yield train_idx, test_idx

# Sanity-check against the 20/5/1 toy example in the chapter:
splits = list(cpcv_splits(n=20, k=5, embargo=1))
assert len(splits) == 10
train0, test0 = splits[0]  # test folds (F_0, F_1)
assert test0 == [0, 1, 2, 3, 4, 5, 6, 7]
assert train0 == [9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19]  # 8 is purged
```

Listing 21.1: Minimal CPCV splitter, verified against the toy example in this chapter.

The two asserts are the minimum regression test: split count $\binom{5}{2} = 10$ and purged $i = 8$ on the first split. Passing both does not guarantee correctness for arbitrary label horizons, but confirms the purge-plus-embargo logic on the simple case. For $h > 1$, replace `any(abs(i - j) <= embargo ...)` with a label-window intersection test per (López de Prado, 2018) Ch. 7.

21.9 Key facts to memorise

- **The seven lies**: look-ahead, survivorship, overfitting, data snooping, transaction-cost neglect, slippage, regime dependence. Every backtest issue is one of these or a composition.

- **Walk-forward**: fit on $[t-W,t)$, trade on $[t,t+R)$, concatenate test P&Ls. Fixes most of Lie 1; not Lies 3 or 4.
- **Naive k-fold is wrong for time series**: labels over horizon h leak into training even when features do not.
- **CPCV pillars**: purge (drop training whose label window intersects test); embargo (drop training within h of the test fold); combinatorial (all $\binom{k}{2}$ two-fold splits). Report a *distribution*.
- **CPCV coverage**: each observation is a test point in $k-1$ of the $\binom{k}{2}$ splits — $k-1$ times richer than k-fold.
- **Expected max Sharpe (null)** per-period: $\mathbb{E}[\max]/\sqrt{T}$, $\mathbb{E}[\max]$ the max of N iid normals. Bailey: 2.53 ($N=100$), 3.24 ($N=1000$), 3.85 ($N=10000$).
- **DSR headline**: $\widehat{\mathrm{SR}}=1.8$, $T=1260$, $N=100$ gives $\widehat{\mathrm{SR}}_{\mathrm{eff}}\approx 0.67$; $N=1000$ gives ≈ 0.35.
- **Costs are life-or-death**: gross Sharpe 1.59 on 5 bp daily edge becomes 0.95 at 2 bp round-trip and 0 at 5 bp.
- **Framework taxonomy**: `Backtesting.py` (learning), `vectorbt` (speed), `NautilusTrader` (microstructure), `QuantConnect` (PIT data). None ship CPCV; wrap.
- **Plateau not peak**: prefer a flat-region parameter to a spike. Frankfurt Hedgehogs' Prosperity 3 lesson.
- **GS gates**: walk-forward ≥ 5 years, DSR, execution cost audit, 30-day paper, risk approval.
- *A backtest is a hypothesis; walk-forward, CPCV, and deflated Sharpe are what make it falsifiable.* Without them it is a story.

Chapter 22

Machine Learning for Trading

Most ML-for-trading projects lose money because they treat the problem as "forecast tomorrow's return with a neural net" — predicting a near-random-walk target with a high-variance estimator, then being surprised when out-of-sample performance is indistinguishable from noise. The real opportunities are narrower: feature engineering on order-book data, regime detection, trade-level *meta-labeling*, and execution. This chapter draws the bulk of its content from Lopéz de Prado's *Advances in Financial Machine Learning* (AFML) — a book less about "how to build trading ML models" than "here is every way people get it wrong."

Chapter 21 spent thirty pages on validation discipline, and every sentence of this chapter assumes you know why naive k-fold cross-validation on financial time series is a bug and what CPCV does about it. The techniques here produce nonsense if fed into a broken validation pipeline.

Prerequisites

From earlier in the book:

- Chapter 10: Ornstein–Uhlenbeck mean reversion. The "OU z-score entry" primary signal is the running example for meta-labeling in Section 22.6.
- Chapter 11: pairs-trading mechanics, used as the worked example for the meta-labeling lift calculation.
- Chapter 21: Combinatorial Purged Cross-Validation (CPCV), walk-forward, and the general pitfalls of look-ahead and data snooping. This chapter refers to CPCV as the correct inner validation loop for every ML procedure it introduces.
- Chapter 20: Sharpe, drawdown, and the mental model of "scoring after sizing". Needed for the meta-labeling lift numbers in Section 22.6.

From outside the book:

- Elementary supervised learning: linear/logistic regression, decision trees, random forests, basic classification metrics (precision, recall, ROC-AUC), and the idea of cross-validation for hyperparameter selection.
- The concept that *financial time series are not stationary*: distributions drift, regimes change, and the i.i.d. assumption powering textbook ML is always false for markets. This chapter is about what to do given that fact.

22.1 Where ML helps

ML does not help with "what will the price of product X be tomorrow." It helps with four narrower questions, and in each case the reason it helps is not the cleverness of the model but the structure of the prediction target. We take them in order of practical impact.

Feature engineering on order-book data

The most productive use of ML in trading is not prediction; it is *compression.* A Level-2 order book at millisecond frequency is a firehose of numbers: ten levels per side, quantity and price at each, plus every trade print. A rules-based strategy can use at most a handful of features before it becomes unmaintainable. ML reduces hundreds of raw order-book columns into a small number of short-horizon predictive features.

Feeding the raw order-book columns directly into a classifier is a non-starter: raw prices drift (non-stationary), quantity columns change variance with the regime (heteroscedastic), and both swamp any learner with scale artefacts rather than signal. Good features are *stationary transforms* — log-returns, rolling z-scores, normalised imbalances — whose distribution is comparable across time and across instruments.

Examples of such derived features, all covered in Chapter 8:

- **Order Flow Imbalance (OFI)**: signed running total of aggressive buy minus aggressive sell volume over a lookback window. Empirically one of the strongest short-horizon return predictors.
- **Top-of-book imbalance**: $(q_1^{\mathrm{bid}} - q_1^{\mathrm{ask}})/(q_1^{\mathrm{bid}} + q_1^{\mathrm{ask}})$ at the inside quote. A cheap real-time drift proxy.
- **Realised volatility at multiple horizons**: $\sigma_{1\mathrm{m}}, \sigma_{5\mathrm{m}}, \sigma_{30\mathrm{m}}$ from intraday returns. Used as direct features and to normalise other features across regimes.
- **Trade direction entropy**: Shannon entropy of signed trades over a rolling window. Low entropy = one-sided flow; high = balanced flow.
- **Queue-position-weighted imbalance**: imbalance that discounts orders at the back of a price-time priority queue, which are less likely to fill soon.

The ML part is boring on purpose. A gradient-boosted tree or linear model predicts the one-step-ahead return from these features under CPCV. The value added is not the model — a linear regression usually does fine — it is the *choice of features.* A skilled researcher spends 80% of their time on feature construction and 20% on the model. Academic papers on "deep learning for financial prediction" invert that ratio and are rarely reproducible.

Regime detection

The second win is regime detection: given a history of market observables (volatility, spread, volume, correlation structure), infer how many distinct regimes the market has been in and which it is in now. The canonical tool is the **Hidden Markov Model** (HMM), which assumes the market cycles between a small number of latent states ("quiet mean-reverting," "high-vol trending," "crisis") with observable features drawn from a state-dependent distribution. Chapter 23 promotes regime detection from "nice to have" to the backbone of the "which strategy when" framework.

Regime detection is a good ML use case because the target is not a scalar future return but a discrete label that can be validated qualitatively ("yes, March 2020 was a crisis regime"). The prediction problem is more stable than return prediction: volatility is far more auto-correlated than the return itself, so an HMM fitted on year t still works on year $t+1$ provided the training regimes also appear at test time.

Trade-level meta-labeling

The third, and in Lopéz de Prado's framing the highest-value ML use in trading, is **meta-labeling**: given a primary strategy that emits trade signals, train a binary classifier to decide *which signals to actually take*. The primary strategy supplies direction (long/short, entry/exit); the meta-labeler supplies the filter. We treat this as its own section (Section 22.6) because its leverage relative to implementation cost is unmatched.

Execution

The fourth win is execution: reinforcement learning (RL) for optimal order placement and routing. Given a parent order, an RL agent learns a slicing and child-order policy that minimises implementation shortfall relative to the arrival price. The Almgren–Chriss solution from Chapter 19 is the analytical baseline; RL relaxes the linear-impact assumption and learns a policy respecting the empirically measured cost curve and current order-book state. Execution is the area where RL papers translate into genuine production lift, because the reward signal is high-frequency, unambiguous, and approximately stationary over short horizons.

We do not cover execution ML here (not on the Prosperity critical path); interested readers should start with Cartea et al. (2015) chapters 6–8 and the RL literature that relaxes Almgren–Chriss's linear-impact simplification.

22.2 Where ML fails

Having established the four regions where ML earns its keep, we enumerate the larger regions where it does not. These are categorical failure modes where more compute and a bigger model actively make things worse.

Naked return prediction

The archetypal failed project is "predict r_{t+1} from $(r_t, r_{t-1}, \ldots, r_{t-L})$ using a neural network." Two reasons this fails, and they compound.

First, the signal-to-noise ratio is catastrophically low. A one-day equity return has annualised volatility around 20%, so a single daily return has standard deviation about 1.26%. A real alpha signal generating Sharpe 1 per year contributes a daily drift of $\approx 0.08\%$: noise is $16\times$ the signal on any single observation. Neural networks with millions of parameters are wonderful at memorising the noise.

Second, one-period equity returns are a near-random-walk target by construction: if they were not, a trivial linear predictor would exist and be arbitraged away. The weak form of the Efficient Market Hypothesis says precisely that an ML model trained on lagged returns should find nothing; the fact that it does find something in-sample is a diagnostic of overfitting.

The diagnosis is nearly always the same: stunning in-sample R^2, a decent CV R^2 under the wrong CV scheme, and a zero-or-negative R^2 on a genuinely held-out period. "The wrong CV scheme" is almost always naive k-fold, and fixing it kills the result.

Don't use k-fold CV on time series

This is the most common ML-for-trading bug, responsible for a huge fraction of irreproducible papers. The failure mode:

1. You build triple-barrier labels (Section 22.3) with a 5-day horizon: the label at time

t depends on prices between t and $t + 5$.

2. You apply scikit-learn's `KFold` with $k = 5$, which shuffles observations and splits them contiguously (or worse, randomly).

3. A training observation at time $t' = t + 3$ has a label whose window overlaps with the test observation at time t. Information about *the test label* has leaked into the training set via the training labels.

4. The classifier reports a miraculous out-of-sample accuracy, e.g. 75% on a 55%-baseline problem.

5. In live trading the accuracy collapses to the true 55% and the strategy loses money.

Fix: CPCV (Chapter 21) — purge training observations whose label window intersects a test observation, plus an embargo on each side of the test fold. Any non-CPCV-compliant ML-trading evaluation is, until proven otherwise, a leakage artefact.

Hyperparameter abuse

The second failure mode is hyperparameter abuse: tuning twenty hyperparameters on the same data used for evaluation — the data-snooping bias of Chapter 21 in ML clothing. If you run $N = 1000$ parameter combinations and report the best CV Sharpe, the expected-max-of-N-iid-normals approximation $\mathbb{E}[\max_{i \le N} Z_i] \approx \sqrt{2 \ln N}$ gives a spurious Sharpe of $\sqrt{2 \ln 1000} \approx 3.72$ standard errors above the true mean *even if every strategy is pure noise*. The defence is nested CPCV: outer CPCV for evaluation, a separate inner CPCV inside each outer training fold for tuning.

Most applied papers skip the outer loop and report the best inner-CV score as "out-of-sample performance." This is silently but devastatingly wrong.

Non-stationarity

The third failure mode is baked into the data. Return distributions drift: mean, variance, correlation structure, tail behaviour, and the *predictability structure* of features all change as regimes change. A model with Sharpe 2 on 2015–2018 may not survive 2020 because the regime it learned no longer exists. This is structural, not a bias to correct.

Defences: (i) retrain frequently on a rolling window, (ii) pool across instruments for statistical power, (iii) condition on a regime label from a separate detector (Section 22.1). None fully solve the problem; they make degradation more gradual. Expecting a fit-once-and-forget model to work for years is the common unforced error.

The general taxonomy

Where ML for trading lives and dies

Four regions where ML helps:

1. Feature engineering on order-book data (Chapter 8).

2. Regime detection via HMMs and state-space models (Chapter 23).

3. Trade-level meta-labeling of a rule-based primary strategy (Section 22.6).

4. Execution: RL for order slicing and routing (Chapter 19 as baseline).

Four regions where ML fails:

1. Naked return prediction from lagged returns.
2. k-fold CV on time series (fixed by CPCV).
3. Hyperparameter abuse (fixed by nested CPCV).
4. Non-stationarity (mitigated, never eliminated, by rolling retraining and regime conditioning).

22.3 Triple-barrier labeling

Before training any classifier we need labels. The textbook approach — "+1 if the next-step return is positive, −1 otherwise" — is wrong for trading. First, it assumes every trade is held for exactly one step. Second, it ignores risk: a 0.01% return is labelled identically to a 5% return. The **triple-barrier method** of (López de Prado, 2018) Ch. 3 fixes both.

The method

At each observation time t, suppose the current price is p_t and the estimated short-horizon volatility is $\hat{\sigma}_t$ (e.g. an exponentially weighted standard deviation of daily returns). We open a hypothetical trade at p_t and set three barriers:

- **Top barrier** (profit target): $p_t^{\text{top}} = p_t(1 + \pi_u \hat{\sigma}_t)$, for some profit-target multiplier $\pi_u > 0$.
- **Bottom barrier** (stop-loss): $p_t^{\text{bot}} = p_t(1 - \pi_d \hat{\sigma}_t)$, for some stop-loss multiplier $\pi_d > 0$.
- **Vertical barrier** (max holding): we give up on the trade after H time steps and record the realised return regardless.

The label y_t is then the first barrier the price path hits:

$$y_t = \begin{cases} +1 & \text{if } p_t^{\text{top}} \text{ is hit first,} \\ -1 & \text{if } p_t^{\text{bot}} \text{ is hit first,} \\ 0 & \text{if neither is hit within } H \text{ steps.} \end{cases}$$

This three-class label corresponds to a real trading outcome: "a long at this observation, held with a $\pi_u \hat{\sigma}_t$ target and $\pi_d \hat{\sigma}_t$ stop, would have been a winner, loser, or timeout." The volatility rescaling is crucial: a 2% move counts differently in calm vs. crisis markets. Fixed-percent barriers ("top at +3%") are trivially satisfied in high-vol regimes and never triggered in low-vol regimes, making them useless classification targets.

Worked example on BTC-like daily data

We run the triple-barrier method on a synthetic BTC-like daily series with ten labeled starting observations ($i = 0, \ldots, 9$), constant volatility $\hat{\sigma} = 2.0\%$, $\pi_u = \pi_d = 2.0$, horizon $H = 5$ days. The full price path has sixteen entries (the extra six are lookahead for the final starting observations):

$$\begin{aligned} p = (&100.00,\ 101.00,\ 102.50,\ 104.20,\ 103.80,\ 103.10, \\ &102.50,\ \ 98.00,\ \ 97.50,\ \ 99.00,\ 100.20, \\ &100.50,\ \ 99.80,\ 100.10,\ 100.30,\ 100.40). \end{aligned}$$

For each starting index $i = 0, 1, \ldots, 9$ we compute the top barrier $p_i \cdot 1.04$, the bottom barrier $p_i \cdot 0.96$, and walk forward up to 5 days looking for the first barrier hit. The resulting label table is:

i	p_i	top	bot	first hit	y_i
0	100.00	104.00	96.00	top at $j = 3$, $p = 104.20$	$+1$
1	101.00	105.04	96.96	timeout (no barrier)	0
2	102.50	106.60	98.40	bot at $j = 5$, $p = 98.00$	-1
3	104.20	108.37	100.03	bot at $j = 4$, $p = 98.00$	-1
4	103.80	107.95	99.65	bot at $j = 3$, $p = 98.00$	-1
5	103.10	107.22	98.98	bot at $j = 2$, $p = 98.00$	-1
6	102.50	106.60	98.40	bot at $j = 1$, $p = 98.00$	-1
7	98.00	101.92	94.08	timeout (no barrier)	0
8	97.50	101.40	93.60	timeout (no barrier)	0
9	99.00	102.96	95.04	timeout (no barrier)	0

The label distribution is one $+1$, five -1, four zero, matching the intuitive reading: an early rally, a sharp drop triggering a cascade of stop hits, then a late recovery where barriers are never reached within the 5-day window. A classifier trained on features at each index i gets a realistic, trading-aligned label, not a raw one-day-forward return sign.

Triple-barrier method

For each observation t, define a profit-target top barrier, a stop-loss bottom barrier (both scaled by current volatility $\hat{\sigma}_t$), and a max-holding vertical barrier. Label the observation with the first barrier hit: $+1$, -1, or 0. The labels match real trading outcomes and automatically adapt to time-varying volatility. Standard reference: (López de Prado, 2018) Ch. 3.

Don't forget to purge when using triple-barrier labels

Triple-barrier labels have a horizon H: the label at time t depends on prices up to $t + H$ — the exact situation that breaks k-fold CV (Section 22.2). Every triple-barrier classifier evaluation must use CPCV with purging of at least H observations plus an embargo. Otherwise the reported accuracy is leakage.

22.4 Fractional differentiation

The second core AFML preprocessing trick is **fractional differentiation**. It solves a dilemma anyone training on price series has hit.

The dilemma

Most price series are non-stationary — drifting mean, time-varying variance. Stationarity is the assumption powering every classical ML method: if the target distribution changes, a model fitted on the past does not generalise. The classical fix is first differences (log returns) $r_t = \ln p_t - \ln p_{t-1}$, which are approximately stationary. But differencing destroys level information: returns at $p = 100$ and $p = 10{,}000$ are indistinguishable, even though a trader would act very differently.

Return-based features have no memory of where the price sits in its historical range — exactly what a mean-reversion or range-bound strategy needs.

Fractional differentiation threads the needle: stationarity with maximum "memory" preserved.

The construction

For a real parameter d, the fractional differentiation operator is defined by the binomial expansion of $(1-B)^d$ where B is the lag operator ($Bx_t = x_{t-1}$):

$$(1-B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k B^k = \sum_{k=0}^{\infty} w_k B^k.$$

The weights w_k obey the recursion

$$w_0 = 1, \qquad w_k = -\, w_{k-1} \cdot \frac{d-k+1}{k}, \quad k \geq 1.$$

For integer d this collapses to the classical differencing operator: at $d = 1$, $w_0 = 1$, $w_1 = -1$, and all higher weights are zero, recovering $x_t - x_{t-1}$. For fractional $d \in (0,1)$, the weights decay like a power law rather than cutting off abruptly, and the operator applied to the series is a long-memory weighted sum over the entire history of the process:

$$\tilde{x}_t = \sum_{k=0}^{\infty} w_k \cdot x_{t-k}.$$

Lopéz de Prado's recipe: start at $d = 1$, decrease slowly; at each value run an Augmented Dickey–Fuller (ADF) test on $\tilde{x}_t$; the smallest d for which ADF rejects non-stationarity at 5% is optimal — just stationary enough, maximum memory. For most equity indices this lands in $[0.3, 0.5]$, well below the classical $d = 1$.

Sanity-check of the weights

To see the long-memory behaviour concretely, here are the first eleven weights for $d = 0.4$ (the midpoint of the typical equity range), rounded to five decimals:

$$w_{0..10} = (1,\ -0.4,\ -0.12,\ -0.064,\ -0.0416,\ -0.02995,$$
$$-0.02296,\ -0.01837,\ -0.01516,\ -0.0128,\ -0.01101).$$

Sum of absolute values: ≈ 1.74. Weights decay geometrically but slowly; the tail is long. $d = 0.3$ decays more slowly (more memory), $d = 0.5$ faster (more stationarity). All three sit between the extremes of "no differencing" ($d = 0$, max memory, non-stationary) and "first difference" ($d = 1$, zero memory, stationary).

The infinite sum is truncated: weights below $\tau \sim 10^{-5}$ are dropped, giving a finite-window implementation that loses only a warm-up period equal to the truncation window.

Fractional differentiation

For $d \in (0,1)$, the fractionally differenced series $\tilde{x}_t = \sum_{k=0}^{\infty} w_k x_{t-k}$ with $w_0 = 1$, $w_k = -w_{k-1}(d-k+1)/k$ achieves stationarity with minimum loss of memory. The optimal d is the smallest value for which the series passes an ADF test; for most equity return data it falls in $[0.3, 0.5]$. Standard reference: (López de Prado, 2018) Ch. 5.

22.5 Information-driven bars

A hidden assumption of every textbook time-series ML tutorial is uniform time sampling: one-minute, one-day, or one-hour bars. This is the wrong scheme for trading data. Information does not arrive uniformly: most of the day is quiet, then a news release dumps five minutes' information into five seconds. Uniform-time sampling means irregular sampling on the news clock — over-sampling quiet periods, under-sampling active ones.

(López de Prado, 2018) Ch. 2 proposes activity-adaptive sampling. Four standard families, in order of sophistication:

Tick bars

A **tick bar** closes every N trades. Each bar represents the same transaction count regardless of elapsed time. Quiet periods produce fewer bars, news spikes many. The resulting series has far more stable intraday return distributions than time bars — exactly what classical ML assumptions want.

Volume bars

A **volume bar** closes every V shares (or contracts). Trades of 100 and 900 shares trigger a close at $V = 1000$; a single 2000-share trade triggers two closes in succession. Volume bars respect that large trades carry more information than small ones — tick bars ignore this.

Dollar bars

A **dollar bar** closes every D dollars of notional. If the price doubles, half as many shares represent the same dollar amount, so the bar boundary adapts. For equities that appreciate or depreciate substantially, dollar bars give the most stable return statistics across time — more so than volume bars (which drift with share-price inflation) or tick bars.

Information-driven bars

Information-driven bars go further: a bar closes when a cumulative information statistic exceeds a threshold. Simplest example — the signed-volume imbalance bar: accumulate $\sum(\text{sign}(r_t) \cdot v_t)$ until $|\cdot| > \theta$. The bar closes when flow becomes directionally significant. Balanced chop produces few bars; a large directional order produces many in rapid succession. Switching from time to dollar (or information-driven) bars is often the biggest leverage change in a feature-engineering pipeline: the same classifier on the same features improves because the time axis is now aligned with the clock on which information actually arrives.

Bar taxonomy

- **Time bars**: fixed wall-clock interval. Worst for ML; the default in every naive tutorial.
- **Tick bars**: fixed number of trades. Better; equalises transaction count.
- **Volume bars**: fixed shares/contracts. Better still; weights by trade size.
- **Dollar bars**: fixed notional. Best plain alternative; robust to price drift over time.
- **Information-driven bars**: variable, adapts to cumulative signed flow or other entropy proxy. Sampling density tracks information arrival.

> Any ML-for-trading pipeline that still uses one-minute bars after reading (López de Prado, 2018) Ch. 2 is leaving a free lunch on the table.

22.6 Meta-labeling: the single best ML-for-trading idea

If you take one idea from this chapter, take this one. Meta-labeling is cleanly positive-expected-value for any researcher with a working rule-based primary strategy. The construction is embarrassingly simple, which is partly why the field was slow to notice it.

Setup

Suppose you have a **primary strategy** emitting trade signals — e.g., the OU mean-reversion rule from Chapter 10, long on $z < -2$, short on $z > +2$. Suppose its empirical hit rate is 55%, annualised Sharpe 1.0, drawdowns occasionally reaching 20%: decent but not great, painful in specific regimes.

The meta-labeling question: *can we learn to skip the bad triggers?* Instead of predicting returns, we predict whether a primary signal, given current market state, is a winner or loser. We act on signals the classifier flags "probably winner" and skip the rest.

The mechanics

The meta-labeling pipeline has four steps.

1. **Run the primary strategy on historical data** and log every trigger: timestamp, direction (long/short), and eventual outcome ("winner: closed in profit before stop" or "loser: closed in loss"). This gives a binary label $m_t \in \{0, 1\}$ for every primary trigger.

2. **Build features for each trigger** from the current market state: regime (low-vol / high-vol / crisis), recent volatility, top-of-book imbalance, time of day, spread, and so on. These are the same kind of features as in Section 22.1.

3. **Train a binary classifier** $\hat{m}(\mathbf{x})$ (logistic regression, random forest, gradient boosting — the choice of model matters less than the features) to predict m_t from the features $\mathbf{x}_t$, *under CPCV.*

4. **Deploy**: at each new primary trigger, compute $\hat{m}(\mathbf{x}_t)$; if the predicted probability is above a threshold τ (say 0.55), take the trade; otherwise pass.

The classifier does not predict returns. It only improves on the 55% base rate by preferentially skipping losers. This is a far easier problem: clean binary target, reasonable class balance, features natively predictive of trade quality (volatility regime, liquidity, time of day), and — critically — the learner is not asked to generate alpha. It is asked to *gate* alpha that already exists.

Worked lift calculation

Consider a stylised pairs-trading strategy (Chapter 11) generating $N = 1000$ primary triggers over a historical window. Each winner contributes $+1R$, each loser $-1R$ (R = per-trade risk unit):

$$\text{Baseline P\&L} = (0.55 \cdot 1000) \cdot (+1R) + (0.45 \cdot 1000) \cdot (-1R) = 100R.$$

Train a meta-labeler that flags 40% of trades (400 triggers) as skip. Assume it is informative but not perfect: of the 400 skipped, 240 were losers, 160 were winners (vs. a uniform-skip baseline of 220 losers, 180 winners — a modest edge). We keep 600 trades, of which

$$\text{kept winners} = 550 - 160 = 390, \qquad \text{kept losers} = 450 - 240 = 210.$$

The kept hit rate is $390/600 = 65\%$, a 10-pp lift from 55%. Kept P&L:

$$(390)(+1R) + (210)(-1R) = 180R,$$

an 80% lift in absolute P&L, on 40% fewer trades. Assuming per-trade variance is unchanged and the $\sqrt{N}$ denominator rescales with trade count, the stylised Sharpe jumps from $100/\sqrt{1000} \approx 3.16$ to $180/\sqrt{600} \approx 7.35$ — a factor of $\sim 2.3\times$. These numbers are illustrative, but they show why meta-labeling is the highest-leverage idea here: a modest classifier merely correlated with trade quality translates into dramatic bottom-line improvement, because all it does is filter.

Why meta-labeling works (and return prediction doesn't)

Meta-labeling works where naked return prediction fails because it attacks a fundamentally easier problem. Return prediction asks: "across all market states, which will have positive next-step returns?" — terrible signal-to-noise, near-random-walk target. Meta-labeling asks: "given that the primary just fired, which firings are the good ones?" — a conditioning trick. The primary has already isolated high-information moments; the classifier only has to rank them. The target is binary and stable ("did this entry hit target before stop?"), the feature set is tight, class balance is sane, and the learner has no ambition to beat the market from scratch — only to improve a working baseline.

Meta-labeling

Given a rule-based primary strategy with a modest positive edge, a binary classifier trained on trade-quality features can systematically skip the worst triggers and keep the best, turning a mediocre strategy into a good one without touching the primary logic. This is the single highest-leverage idea in ML for trading. Standard reference: (López de Prado, 2018) Ch. 3.

Intuition

Meta-labeling inverts the usual framing. Instead of "predict returns, then construct a strategy," it says "construct a strategy, then learn to filter its triggers." The strategy generates alpha; the classifier chooses battles. It works because the primary strategy has collapsed the infinite-dimensional "when is the market predictable" down to a finite set of concrete trade opportunities, each with a binary supervisable outcome. That is a far better shaped ML problem.

Don't meta-label until the primary strategy is profitable

Meta-labeling can only filter; it cannot turn a losing primary into a winner. If the primary has a 45% hit rate, the best meta-labeling can do is skip every trade — breakeven via not trading. The literature sometimes sells it as magic, but the magic is conditional on the primary having an edge. Validate the primary with CPCV (Chapter 21) first; only then build a meta-labeler.

22.7 The hardest problem is not prediction, it is validation

Standard ML research orders things so: prediction is hard (train a model with low training error), validation is the easy final step (held-out set, report accuracy). For trading, that ordering is inverted. The hard part is *validation* — under which measurement procedure can you trust that reported performance survives contact with real money? Fitting the model is a footnote.

Why? A trading ML model has three adversaries a normal one does not: (i) the target is weakly predictable, so the honest OOS signal is drowned by any validation bug; (ii) the data is non-stationary, so one held-out period is an unreliable estimate; (iii) a broken model loses money, unrecoverable by "a second experiment with cleaner methodology."

Consequence: the majority of time goes to the validation pipeline, not the model. Every technique in Chapter 21 — walk-forward, CPCV with purging and embargo, deflated Sharpe, nested CV, plateau-not-peak selection — is required by default. Every technique here — triple-barrier labels, fractional differentiation, information-driven bars, meta-labeling — sets up the problem so validation has a fighting chance.

The validation-first doctrine

In ML for trading the ranking of effort is:

1. Validation pipeline (CPCV, purging, embargo, nested CV, deflated Sharpe).
2. Feature engineering (order-book microstructure, volatility normalisation, information-driven bars).
3. Label construction (triple-barrier, meta-labeling targets).
4. The model itself (and this step is nearly always "a gradient-boosted tree with sensible defaults").

The academic ranking is the reverse. If you are spending most of your time on step 4, you are working on the wrong problem.

22.8 Prosperity: ML plays a narrow role

Prosperity example

In IMC Prosperity, the top teams' edge is almost never ML. Profit comes from classical tools: market making (Chapter 9), mean reversion and pairs trading (Chapters 10 and 11), basket/ETF arbitrage (Chapter 12), and options (Chapters 16 and 17). Neural networks on tick features do not appear in winning writeups. The reason is specific: Prosperity data is short (a few hundred thousand observations per round at most), products are synthetic with stationary microstructure, and classical-strategy edge is always large enough to statistically dominate an ML model fitted on a few thousand ticks.

One area where ML-flavoured technique does appear: **detecting informed traders from per-trade data**. The Frankfurt Hedgehogs' second-place Prosperity 3 writeup (source of Section 3.5's Olivia story) trained a pattern-detector on per-trader behaviour to identify anonymous bots filling at daily extremes. In this chapter's framing: a classification problem. Given history labeled by anonymous trader ID, predict "is this trader informed?" Features are classical — fill-price distance from the day's running min/max, fill size, time since last trade. The binary label derives from the observation that one bot always fills at the daily extreme. The classifier is used as a *meta-labeler* on the team's own trades: "don't compete with informed flow, align with it."

This is precisely the meta-labeling idea of Section 22.6. The primary is a rule-based pairs trade on an ETF basket; the meta-labeler biases entry thresholds by the inferred informed-bot direction. The reported lift materially improved placement, and the classical part still worked when the informed bot did not trade — ML was pure upside, not a replacement.

General lesson: do not reach for ML until classical tools are exhausted. When you do, reach for a one-sentence classification problem — "is this trader informed," "is this regime high-volatility," "is this spread about to revert" — not "predict the next tick." Prosperity punishes overfitting fast, and the surviving techniques are the boring ones: few features, linear or tree-based models, CPCV.

22.9 Goldman Sachs: which desks actually use ML

On a Goldman Sachs desk

At Goldman Sachs, as at any large sell-side bank, ML lives on specific desks and problems — it is not the bank-wide religion outsiders sometimes imagine. The distribution maps cleanly onto the "where ML helps" list in Section 22.1:

- **Systematic Equities (quantitative long/short)**: the cleanest home for meta-labeling. Quant research generates primary alphas from fundamental or technical signals; the systematic trading team trains classifiers on trade-level features to filter the worst entries. Alphas are transparent and attributable; ML is a conditional filter on top.
- **Central Risk Book**: the team warehousing inventory across flow desks uses HMMs and related regime-detection models to time hedging bands (high-vol regime, widen; quiet regime, tighten). ML here is a regime classifier, not a return predictor.
- **Execution / Algorithmic Trading**: uses RL for child-order placement atop an Almgren–Chriss baseline (Chapter 19), plus supervised learning for short-horizon fill-probability prediction in smart-order routers. The highest-compute ML use case in the firm.
- **Data Science / Strats**: builds feature-engineering pipelines on alternative and microstructure data (credit-card transactions, satellite imagery, news sentiment, order-book reconstructions), sold as *inputs* to the systematic desks. Output is "a feature matrix," not "a model."

Three points worth underlining. First, **ML is almost never used for directional signal generation**. The directional signal — long or short — comes from quantitative research (QR), using classical statistical and economic models with defensible attributions. ML enters as filter, regime switch, or execution optimiser, never to generate the thesis. A Strat pitching an RNN return predictor as a trading signal is gently but firmly redirected to an HMM or linear factor model. Second, **validation discipline is taken very seriously**: every production ML model has a CPCV evaluation, a monthly-recut out-of-sample window, and a documented retirement criterion (e.g., "pull if rolling 6-month OOS Sharpe < 0.5"). Models do not live forever. Third, **the feature matrix is the crown jewel**: the most durable source of desk edge is a well-engineered feature pipeline that can be recombined as regimes change. Fancy model on a bad matrix loses to a simple model on a good one, every time.

22.10 A short reference implementation

One short Python snippet for the triple-barrier method is this chapter's core algorithmic contribution; everything else is about what to feed in and how to validate what comes out. The implementation below takes a price array, per-observation volatility, profit/stop multipliers, and max holding period, and returns the triple-barrier label per observation.

```python
import numpy as np

def triple_barrier_labels(prices, sigma, pt_mult=2.0, sl_mult=2.0,
    max_hold=5):
    """Return +1 (top), -1 (stop), 0 (timeout) for each observation."""
    n = len(prices)
    labels = np.zeros(n)
    for i in range(n - max_hold):
        top = prices[i] * (1 + pt_mult * sigma[i])
        bot = prices[i] * (1 - sl_mult * sigma[i])
        for j in range(1, max_hold + 1):
            if prices[i + j] >= top:
                labels[i] = +1; break
            if prices[i + j] <= bot:
                labels[i] = -1; break
    return labels
```

Listing 22.1: Triple-barrier labeling, producing the label table of Section 22.3.

Run this on the ten-observation BTC-like series of Section 22.3 with constant `sigma[i] = 0.02` and you recover the label vector $(+1, 0, -1, -1, -1, -1, -1, 0, 0, 0)$ — exactly the worked example. An implementation reproducing this vector has the barrier logic right; one that does not has a sign error, an off-by-one, or a confused volatility-scaling direction. Production versions (e.g., `mlfinlab.labeling.get_events` in the `mlfinlab` package accompanying (López de Prado, 2018)) add volatility estimation, asymmetric horizons, and event metadata, but the core logic is the ten lines above.

22.11 Key facts to memorise

- **ML for trading is narrow.** It helps with feature engineering on order-book data, regime detection, trade-level meta-labeling, and execution. It does not help with naked return prediction from lagged returns, no matter how deep the network.
- **Naive k-fold CV on time series is a bug.** Triple-barrier labels with horizon H leak between folds unless you purge training observations whose label windows intersect test observations (plus an embargo). Use CPCV from Chapter 21. Everything else in this chapter assumes this fix is already applied.
- **Triple-barrier labels** define profit, stop, and timeout barriers for each observation and label it by the first barrier hit. Volatility scaling is mandatory; fixed percentage barriers are useless. Standard reference: (López de Prado, 2018) Ch. 3.
- **Fractional differentiation** threads the needle between "stationary but memoryless" returns and "non-stationary but level-preserving" prices. Weights: $w_0 = 1$, $w_k = -w_{k-1}(d - k + 1)/k$. Typical optimal d for equity series is in $[0.3, 0.5]$, chosen as the smallest d at which an ADF test rejects non-stationarity.
- **Information-driven bars** (tick, volume, dollar, imbalance) sample time more densely when something is happening. Switching from time bars to dollar bars is often the single biggest leverage change in a feature pipeline.

- **Meta-labeling is the single highest-leverage idea.** Train a binary classifier on top of a working rule-based primary strategy to filter out low-quality triggers. Turns a mediocre strategy into a good one without touching the primary logic.

- **Meta-labeling only filters.** It cannot turn a losing primary strategy into a winner. Validate the primary with CPCV first.

- **The validation pipeline is the hard part.** In ML for trading the ranking of effort is (1) validation, (2) features, (3) labels, (4) model. The academic ranking is the reverse.

- **Non-stationarity never goes away.** Rolling retraining, regime conditioning, and deflated Sharpe all soften it; none eliminate it. Any model expected to live for years is the wrong model.

- **ML at Goldman Sachs** lives on systematic equities (meta-labeling), central risk (regime detection), execution (RL for child orders), and Strats data science (feature pipelines). It does *not* live in the directional signal: that comes from QR using classical models.

- **ML in Prosperity is rarely the edge.** Classical tools dominate; the one place ML appears is in *informed-trader detection* (Olivia-style), which is a meta-labeling application in disguise.

- **The single sentence.** *"The hardest problem in ML for trading is not prediction, it is validation."* Everything in this chapter is in service of that sentence.

Chapter 23

The Which-Strategy-When Decision Framework

Part V gave us an arsenal of strategies (Chapters 9 to 14, 17 to 20 and 22), each taught inside its own assumed market conditions. This chapter answers the practitioner question: given the state of the market *right now*, which of those strategies should run, on which product, with what parameters, and which proposed trades should fire? The output is a concrete list of sized, scheduled trades.

The machinery has four layers, assembled from (Chan, 2022), the CPO blog (Chan, 2020), the QuantStart HMM tutorial (QuantStart, 2016), and (López de Prado, 2018). A **Hidden Markov Model** (HMM) detects regimes on the slow clock (regimes last weeks). A **decision tree** maps observables to a strategy on the strategy's own clock (per bar/tick). Chan's **Conditional Parameter Optimisation** (CPO) adapts parameters — its regression is retrained nightly/weekly but queried per tick. **Meta-labeling** (Chapter 22) runs fastest — once per proposed trade. Downstream, Kelly sizes and Almgren–Chriss schedules. Thinking about each layer's clock is half the engineering; making each layer's output legible to the next is the other half.

Prerequisites

From earlier in the book:

- Chapter 10: OU process, half-life $\tau_{1/2} = \ln 2/\theta$, ADF test, z-score entry rule (the mean-reversion branch is gated on ADF p and z).
- Chapters 11 and 12: cointegration, basket-vs-constituents spreads, Engle–Granger (basket branch gated on basis z).
- Chapter 13: TSMOM, cross-sectional momentum, 12-month look-back (trend branch gated on sign of trailing 12-month return).
- Chapter 9: Avellaneda–Stoikov; market making is net-profitable when spread exceeds per-fill fair-value uncertainty and inventory drift is manageable.
- Chapters 17 and 18: long gamma makes money when realised vol exceeds implied.
- Chapter 22: triple-barrier labels and meta-labeling — the trade-level selector in the framework below.
- Chapter 21: walk-forward and CPCV. Every component must be validated under these rules.

- Chapter 20: Kelly sizing.

From outside the book: elementary HMM vocabulary (hidden states, transition matrix, forward/backward, EM — Rabiner's 1989 tutorial is the standard reference); basic regression and random forests at the level of Chapter 22.

23.1 A concrete morning-of-trading decision

Before the formal machinery, walk through a toy morning at a Prosperity-style desk with four products on the screen. This is how the framework should *feel*; the rest of the chapter explains how each step is computed. The universe:

- `RAINFOREST_RESIN` — nearly flat price around ≈ 10,000 seashells, with a tight, stable spread. No information-driven flow. A textbook candidate for passive market making (Chapter 9).
- `KELP` — mid-volatility, mean-reverting on the minute-scale time axis. Historical ADF p-value small, OU half-life on the order of tens of ticks. A textbook z-score mean-reversion candidate (Chapter 10).
- `SQUID_INK` — volatile, occasionally driven by an informed trader (Olivia in the Prosperity 3 worked examples of Chapter 13). When she is buying, the price runs in the same direction for many ticks. A textbook TSMOM candidate, conditional on the informed-flow signal.
- `PICNIC_BASKET1` — a known basket of individual products (6 `CROISSANTS`, 3 `JAMS`, 1 `DJEMBE`). The basket and its replicating portfolio can deviate from each other; the spread is a classic basket-arb candidate (Chapter 12).

At 08:00 local, a single tick of diagnostic statistics is computed from the last five minutes of order-book data. We show them in Table 23.1. The rows are observables; the columns are products.

Observable (o_t component)	`RESIN`	`KELP`	`SQUID`	`BASKET1`
5-min realised vol (bps)	1.8	12.0	38.0	14.0
Quoted half-spread (bps)	0.5	1.5	4.0	6.0
Top-of-book depth (units)	40	30	15	20
Lag-1 price autocorr	+0.01	−0.35	+0.18	−0.02
ADF p-value (last 500 ticks)	0.02	0.005	0.41	n/a
12-min trailing return ($\Delta p/p$)	+0.0001	+0.0004	+0.0120	−0.0003
Basket basis z-score	n/a	n/a	n/a	+2.1
Olivia (informed) buy flag	no	no	**yes**	no

Table 23.1: Morning-of-trading diagnostic statistics for the toy universe of Section 23.1. All four strategies in the pool are discriminable from these eight numbers.

Reading Table 23.1 row by row: `RESIN` has tiny vol (1.8 bps), a non-trivial 0.5 bps half-spread (so full spread ≈ half the per-5-min vol), and zero lag-1 autocorrelation — no trend, no mean-reverting edge. Passive quoting is the only positive-expectancy move. **Select: market making**.

KELP has order-of-magnitude larger vol, lag-1 autocorrelation -0.35 (mean-reverting), ADF $p = 0.005$ (rejecting unit root), and half-spread small relative to vol. Assume $|z| > 1.5$ (otherwise no trade fires). **Select: OU z-score mean reversion**.

SQUID_INK has the Olivia flag on, $+1.2\%$ trailing return, positive lag-1 autocorrelation, and ADF $p = 0.41$ (unit root — no mean-reversion branch fires). Trend is present and informed-trader-confirmed. **Select: time-series momentum long**.

For PICNIC_BASKET1 the tradeable is the basket spread (Chapter 12). Basis $z = +2.1$ exceeds the $+1.5$ entry threshold: basket is rich relative to replicating portfolio. **Select: short basket / long replicating portfolio**.

Four products, eight numbers, four strategies — each selection is the gating condition of a previously-written chapter, read off a single table. The rest of this chapter formalises this.

Prosperity example

Prosperity 3: each selected strategy is on the leaderboard. All four strategies in Section 23.1 appear in the Frankfurt Hedgehogs' second-place write-up: market making on flat low-vol products, OU mean reversion on mid-vol mean-reverting products, basket arb on PICNIC_BASKET1/2, and TSMOM on SQUID_INK conditional on the Olivia signal — seven strategies combined. This chapter formalises "how did they pick which strategy to run where?" from (Chan, 2022) and (Chan, 2020); the write-up is what the output looks like when done right.

23.2 Regime detection via Hidden Markov Models

The toy example cheated: observables drift with the *market regime* — the hidden state that makes returns, spreads, volatility, and correlations covary. A $z = 2$ in a calm regime is real; in a crisis regime $z = 2$ is small relative to intraday shocks and means something different. A **regime-aware** framework wraps the decision tree in a layer that first estimates the regime, then reads the decision tree under that regime's conditional distribution.

We model the market as a **Hidden Markov Model** (HMM), following (Rabiner, 1989) (the notation used by `hmmlearn`).

Definition 23.1 (Hidden Markov Model)**.** A Hidden Markov Model with N states is a triple $(\boldsymbol{\pi}, A, B)$ where

- $\boldsymbol{\pi} = (\pi_1, \dots, \pi_N)$ is the initial state distribution, $\pi_i = \mathbb{P}(S_1 = i)$.
- $A \in \mathbb{R}^{N\times N}$ is the transition matrix, $A_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i)$.
- $B = \{b_i(\cdot)\}_{i=1}^N$ is the set of emission densities: conditional on being in state i at time t, the observation o_t is drawn from b_i.

In the trading application, the observation o_t is typically a daily or intraday return (or a vector of features) and each emission b_i is a Gaussian (or a Gaussian mixture) over that return. The hidden state sequence $S_1, S_2, \dots, S_T$ is unobserved; all one sees is $o_1, \dots, o_T$.

Given the model, three algorithms do all the work. We state them in one place and refer back.

The three HMM algorithms

Given an HMM $(\boldsymbol{\pi}, A, B)$ and an observation sequence $o_{1:T}$:

1. **Forward algorithm.** Computes the filtered posterior $\mathbb{P}(S_t = i \mid o_{1:t})$ for each state

i and time t, in $O(N^2T)$ operations. Gives you "what regime is the market *probably* in right now, given everything I have seen so far."

2. **Viterbi algorithm.** Computes the most likely single state path $\arg\max_{S_{1:T}} \mathbb{P}(S_{1:T} \mid o_{1:T})$, in $O(N^2T)$ operations. Gives you a single hard assignment of a regime label to every historical time, useful for backtesting regime-conditional returns.

3. **Baum–Welch (EM).** Estimates the parameters $(\boldsymbol{\pi}, A, B)$ from data by expectation maximisation, using forward–backward as the E-step. Gives you a *trained* HMM in the first place.

For the regime-detection use case we fit parameters once with Baum–Welch on a long training window, then at every decision tick we run the forward algorithm on the recent observation history and read the current-state posterior off the last column (QuantStart, 2016).

We do not re-derive the three algorithms — the derivation is four pages of index algebra that reduces to "law of total probability plus memoisation" (see (Rabiner, 1989)). Define the forward variable

$$\alpha_t(i) \;\coloneqq\; \mathbb{P}(o_1, o_2, \dots, o_t,\; S_t = i), \tag{23.1}$$

the joint probability of having seen the observations so far and the hidden state at the current time being i. Its recursion is

$$\alpha_t(j) \;=\; \Big[\sum_{i=1}^{N} \alpha_{t-1}(i)\, A_{ij}\Big]\, b_j(o_t), \qquad \alpha_1(j) = \pi_j\, b_j(o_1), \tag{23.2}$$

— law of total probability: arrive in any i at $t-1$, transition to j, emit o_t. Summing α_T over states gives the total likelihood $\mathbb{P}(o_{1:T})$ as a by-product. The filtered posterior is

$$\mathbb{P}(S_t = i \mid o_{1:t}) \;=\; \frac{\alpha_t(i)}{\sum_{j=1}^{N} \alpha_t(j)}. \tag{23.3}$$

In practice α_t underflows rapidly (each step multiplies by an emission density ~ 30), so the recursion is re-normalised every step: divide $\alpha_t(\cdot)$ by its sum, making it the posterior directly. This is the form used below and in every production implementation.

A 2-state HMM on synthetic SPY-style returns

Our model has $N = 2$ hidden states. State $S = 1$ is "low-vol bull," state $S = 2$ is "high-vol bear." Emission distributions are Gaussian with parameters

$$(\mu_1, \sigma_1) = (+0.05\%,\ 0.80\%), \qquad (\mu_2, \sigma_2) = (-0.10\%,\ 2.00\%), \tag{23.4}$$

per day (daily returns, not log-prices), representative of SPY over 1993–2014 (QuantStart, 2016). Sticky transition matrix,

$$A \;=\; \begin{pmatrix} 0.98 & 0.02 \\ 0.02 & 0.98 \end{pmatrix}, \tag{23.5}$$

giving mean regime dwell time $1/0.02 = 50$ days (two to three trading months). Initial distribution $\boldsymbol{\pi} = (0.5, 0.5)$. Observation sequence:

$$o_{1:5} \;=\; (+0.40\%,\ +0.60\%,\ -0.30\%,\ -2.50\%,\ -3.00\%). \tag{23.6}$$

Mild-up followed by a sharp two-day drawdown. We run the forward algorithm step by step, reporting the normalised posterior. All digits are reproducible against `scipy.stats.norm.pdf`.

Step 1: $t = 1$, $o_1 = +0.40\%$. Emission likelihoods:

$$b_1(o_1) = \mathcal{N}(0.0040 \mid 0.0005,\, 0.0080) = 45.32, \tag{23.7}$$
$$b_2(o_1) = \mathcal{N}(0.0040 \mid -0.0010,\, 0.0200) = 19.33. \tag{23.8}$$

State 1's density is just over twice state 2's (observation is inside state 1's 1σ band, at the edge of state 2's). Multiply by prior and renormalise:

$$\alpha_1 \;=\; \frac{(0.5)(45.32)}{(0.5)(45.32) + (0.5)(19.33)} \;=\; \underbrace{0.7010}_{\mathbb{P}(S_1=1|o_1)}\,. \tag{23.9}$$

Step 2: $t = 2$, $o_2 = +0.60\%$. The recursion is

$$\alpha_t(j) \;=\; \Big[\sum_{i=1}^{N} \alpha_{t-1}(i)\, A_{ij}\Big]\; \underbrace{b_j(o_t)}_{\text{emission}}\,, \tag{23.10}$$

with re-normalisation after each step. Emission likelihoods at $o_2 = 0.006$ are $b_1 = 39.37$, $b_2 = 18.76$. Propagate:

$$\alpha_2(1) \propto (0.7010 \cdot 0.98 + 0.2990 \cdot 0.02)\, b_1(0.006) \;=\; 0.6929 \cdot 39.37 \;=\; 27.28, \tag{23.11}$$
$$\alpha_2(2) \propto (0.7010 \cdot 0.02 + 0.2990 \cdot 0.98)\, b_2(0.006) \;=\; 0.3071 \cdot 18.76 \;=\; 5.76. \tag{23.12}$$

After renormalisation, $\alpha_2 = (0.8256,\, 0.1744)$ — a second low-vol-consistent observation sharpens the inference.

Step 3: $t = 3$, $o_3 = -0.30\%$. Still inside state 1's 1σ band: $\alpha_3 = (0.9083,\, 0.0917)$.

Step 4: $t = 4$, $o_4 = -2.50\%$. The model changes its mind. -2.5% is -3.1σ under state 1 but only -1.25σ under state 2. Emission ratio:

$$\frac{b_2(o_4)}{b_1(o_4)} \;=\; \frac{\mathcal{N}(-0.025 \mid -0.001,\, 0.020)}{\mathcal{N}(-0.025 \mid +0.0005,\, 0.008)} \;=\; \frac{9.71}{0.31} \;=\; 31.3. \tag{23.13}$$

State 2's emission is $31\times$ more likely. The sticky 0.98 diagonal cannot hold state 1 against this:

$$\alpha_4(1) \propto (0.9083 \cdot 0.98 + 0.0917 \cdot 0.02)(0.31) \;=\; 0.2772, \tag{23.14}$$
$$\alpha_4(2) \propto (0.9083 \cdot 0.02 + 0.0917 \cdot 0.98)(9.71) \;=\; 1.0514. \tag{23.15}$$

After renormalisation, $\alpha_4 = (0.2086,\, 0.7914)$ — posterior flipped onto state 2 in one observation.

Step 5: $t = 5$, $o_5 = -3.00\%$. A second high-vol observation locks it in: $\alpha_5 = (0.0014,\, 0.9986)$.

Two takeaways. The sticky matrix prevents overreaction to small observations (-0.3% did not flip it) but large observations overwhelm it in one step (-2.5% did). And once flipped, the model is confident: day 5 pushes to 0.9986 because the ratio acts on an already-tilted posterior.

Intuition

An HMM detects regimes by racing two densities. The posterior is the previous posterior (propagated through the transition matrix) times the new observation's emission likelihoods. Flat observations preserve it; observations deep in one state's tail and inside another's body snap it over. Stickiness controls the evidence required: large off-diagonals flip on one bad day, small ones require several. Choosing stickiness is the single most important HMM parameter decision — too nervous and the pipeline swaps strategies

t	o_t	$\mathbb{P}(S_t = 1 \mid o_{1:t})$	$\mathbb{P}(S_t = 2 \mid o_{1:t})$
1	$+0.40\%$	**0.7010**	0.2990
2	$+0.60\%$	**0.8256**	0.1744
3	-0.30%	**0.9083**	0.0917
4	-2.50%	0.2086	**0.7914**
5	-3.00%	0.0014	**0.9986**

Table 23.2: Forward-algorithm posteriors for the 2-state HMM of Section 23.2. The regime flip happens in a single observation at $t = 4$, when the emission likelihood of the high-vol state exceeds that of the low-vol state by a factor of 31. Bold entries mark the argmax state at each time.

daily; too confident and it sits in the wrong regime for a month.

Number of states and the Gaussianity assumption

Two failure modes dominate. **Too many states**: $N = 2$ finds "quiet vs crisis"; $N = 3$ sometimes finds "up/range/down", sometimes garbage; $N = 6$ almost always overfits — states capture one or two outliers and the posterior flickers. Start at $N = 2$; only increase with a-priori reason and out-of-sample evidence. **Gaussian emissions on raw returns**: daily equity returns are fat-tailed, so the Gaussian HMM assigns tails to a phantom high-vol state. Use Gaussian mixtures per state or fit on GARCH-standardised residuals. (Chan, 2022) and (QuantStart, 2016) both recommend small N and careful residualisation.

23.3 What regimes mean for strategy selection

Given a regime, which strategies should run? The stylised facts from (Chan, 2013), (Chan, 2022), and the regime-conditional backtest literature:

- **Low-vol bull**: tight spreads, thick books, slow upward drift. TSMOM (Chapter 13) earns its highest Sharpe. Mean reversion is anti-correlated with the trend and gets run over. Market making earns a steady spread (low adverse selection). Long gamma loses — realised vol below implied.

- **High-vol bear**: wide spreads, thin books. TSMOM short side earns highest Sharpe (2008 is the textbook case); cross-sectional momentum still works at reduced Sharpe. Mean reversion catches falling knives. Market making is dangerous — the book walks; (Chan, 2013) recommends aggressive inventory skew. Long gamma earns its highest returns.

- **Range-bound**: tightly bracketed, tight spreads, short half-lives. OU mean reversion (Chapter 10) and pairs (Chapter 11) earn their highest Sharpes. TSMOM whipsaws. Market making earns the spread. Basket arb is a function of basis dislocation only, regime-independent.

- **Regime transition**: the worst place. HMM posterior uncertain, neither trend nor mean-reversion reliable, execution costs elevated. Every practitioner guide recommends *reducing position size* rather than trading through.

Table 23.3 summarises. Entries are qualitative relative-Sharpe assessments from (Chan, 2013),(Chan, 2022); the *sign* of each cell is robust across decades of backtests, the magnitude is not.

Strategy family	Low-vol bull	High-vol bear	Range-bound	Transition
Market making	+	−	++	−
OU / pairs mean reversion	−	−−	++	−
Basket arbitrage	○	○	○	○
TSMOM (long only)	++	−	−	−
TSMOM (long/short)	++	+	−	−
Long gamma / long vol	−	++	−	+
Arbitrage sweep	+	+	+	+

Table 23.3: Qualitative regime-to-strategy map. Cells are ++ (strongest Sharpe), + (positive), ○ (regime-neutral, function of basis only), − (loses money), −− (loses money fast). Basket arbitrage and the arbitrage sweep are regime-neutral because they exploit a price relationship, not a direction. Synthesised from (Chan, 2013), (Chan, 2022), and (QuantStart, 2016).

The point is not the specific cells but that most strategies have *sharp sign flips* across regimes. Running mean reversion through a bull trend, or TSMOM through a range, is the most common money-loss pattern.

Three rows deserve a closer look. **Market making** is + in low-vol bull but ++ in range-bound. The reason is adverse selection (Chapter 6): in a trend, informed flow picks off the passive quote on one side; in a range, both sides come from uninformed noise traders so the spread is earned on every round-trip. Market making profits when adverse selection is small relative to quoted spread, and that is smallest in range-bound regimes.

Long gamma is + in transition despite being − in low-vol. Long gamma makes money when realised vol exceeds the implied at purchase, and regime transitions are precisely when implied hasn't caught up to the new regime's realised. "Vol trades buy dislocations."

Basket arbitrage is ○ (regime-neutral) in every column because the basket spread is a price *relationship*, not a direction. A trend in both legs leaves the spread unchanged. The basket branch comes *before* regime-conditional branches for this reason. On the Frankfurt Hedgehogs' leaderboard basket arb P&L was top-three — "regime-neutral" means "runs in every regime," not "small."

Intuition

One-sentence summary: *strategies that bet on price* ***direction*** *have sharp sign flips across regimes; strategies that bet on a* ***relationship*** *between prices do not.* Momentum and mean reversion are directional bets on autocorrelation. Market making is a directional bet on the sign of adverse selection. Arbitrage and basket spreads are relationship bets — they pay when the relationship is dislocated regardless of leg direction. Long gamma is realised-vs-implied (a relationship), less regime-sensitive than directional strategies but more than cross-sectional arbitrage because the relationship itself moves with regime.

A synthetic check makes the sign flip concrete. On 2000-step series — half trending (drift ± 0.001, $\sigma = 0.005$), half mean-reverting (OU with $\theta = 0.1$, $\sigma = 0.3$) — a 20-step TSMOM and a 50-step $|z| > 1.5$ OU strategy gave:

- Trending regime: TSMOM Sharpe $= +1.65$, mean-reversion Sharpe $= -1.64$.

- Mean-reverting regime: TSMOM Sharpe = −1.97, mean-reversion Sharpe = +2.22.

Each strategy is ≈ +2 in its regime and ≈ −2 in the other. A regime-agnostic trader running both nets ≈ 0 and eats all costs as drag; a regime-aware one realises +2 in both. That is the framework, in four numbers.

23.4 The opportunity-diagnosis decision tree

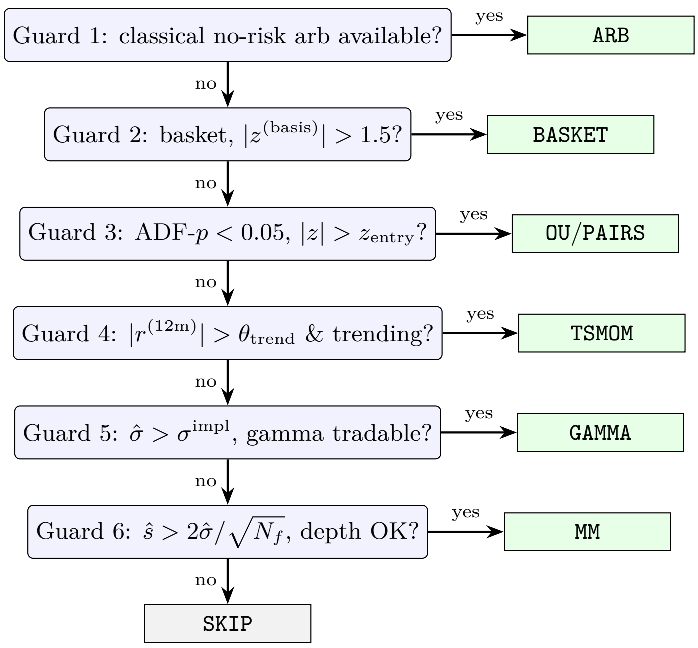


Figure 23.1: The opportunity-diagnosis decision tree of Section 23.4, drawn as a priority-ordered guard chain. Each guard is a threshold on one component of the observable vector $o_t^{(p)}$. The first guard whose condition fires wins; if no guard fires, the tree outputs `SKIP`. Each thresholded parameter (z_{entry}, θ_{trend}, the depth cut, σ^{impl}) is regime-conditional through CPO (Section 23.7).

Generalise the toy table. Define the **observable vector** at time t for product p as

$$o_t^{(p)} = (\hat{\sigma}_t, \hat{s}_t, \hat{d}_t, \rho_t^{(1)}, \rho_t^{(5)}, \widehat{\text{OFI}}_t, z_t^{(\text{basis})}, \text{ADF-}p_t, r_t^{(12\text{m})}, \text{flags}_t), \tag{23.16}$$

where $\hat{\sigma}_t$ is realised vol, $\hat{s}_t$ half-spread, $\hat{d}_t$ top-of-book depth, $\rho_t^{(k)}$ lag-k autocorr, $\widehat{\text{OFI}}_t$ order-flow imbalance (Chapter 8), $z_t^{(\text{basis})}$ basket basis z-score, ADF-p_t stationarity p-value, $r_t^{(12\text{m})}$ trailing 12-month return, and flags$_t$ categorical state (informed-trader, earnings, expiry).

The **strategy pool** $\mathcal{S}$: market making (`MM`), OU z-score (`OU`), cointegration pairs (`PAIRS`), basket arbitrage (`BASKET`), TSMOM (`TSMOM`), long-vol gamma (`GAMMA`), catch-all arbitrage (`ARB`) for put-call parity, triangular, and locational.

The decision tree $\mathcal{T} : o_t^{(p)} \to \mathcal{S} \cup \{\text{SKIP}\}$ is a priority chain of guards: each guard is a threshold on one component of $o_t^{(p)}$, and the first that fires wins. Arbitrage comes first (near-free money), then basket arbitrage (regime-neutral), then regime-conditional strategies.

Decision-tree scaffold for one product

At each decision tick, for each product p in the universe:

1. If any classical no-risk arbitrage (put–call parity, triangular, locational) is available with $P\&L > c_{\text{exec}}$, fire `ARB`. Continue to next product.

2. Else, if p is a basket and $|z_t^{(\text{basis})}| > 1.5$, fire `BASKET` in the direction opposite the basis dislocation.
3. Else, if ADF-$p < 0.05$ and $|z_t^{(p)}| > z_{\text{entry}}(\text{regime})$, fire `OU` (or `PAIRS` if p is a pair). Both bet on reversion to a stationary mean; `OU` uses the product's own price, `PAIRS` uses the cointegrated spread between two products (Chapter 11).
4. Else, if $|r_t^{(12\text{m})}| > \theta_{\text{trend}}$ and the HMM regime is trending, fire `TSMOM`.
5. Else, if $\hat{\sigma}_t > \sigma^{\text{impl}}$ (realised exceeds implied) and gamma is tradable, fire `GAMMA`.
6. Else, if $\hat{s}_t > 2\hat{\sigma}_t/\sqrt{N_{\text{fills}}}$ and $\hat{d}_t$ is above a minimum depth, fire `MM`. The inequality says the quoted half-spread exceeds the per-fill fair-value uncertainty, so each round-trip is expected-positive after adverse selection.
7. Else, `SKIP` (no strategy fires for this product at this tick).

Each threshold is a tunable parameter — and in the next section, CPO makes them all regime-conditional.

Guards are ordered by *priority*, not likelihood. Arbitrage first (regime-free, riskless). Basket arbitrage next (regime-neutral, Table 23.3). Mean reversion and TSMOM in the middle, mutually exclusive via the cointegration/regime tests. Market making near the bottom (default when no stronger signal). SKIP as last resort — more often correct than beginners think. The most common money-loss in Prosperity 3 write-ups was strategies firing every tick because the author refused to hold no position.

Worked guard evaluation

Replay the toy table guard-by-guard with arithmetic.

Guard 1 (arbitrage sweep). No options, fiat pairs, or multi-venue listings in the toy universe; does not fire.

Guard 2 (basket arbitrage). Only `PICNIC_BASKET1` is a basket; basis $z = +2.1 > 1.5$. Fires: short `BASKET1`, long $6\cdot$`CROISSANTS` $+3\cdot$`JAMS` $+1\cdot$`DJEMBE`. Route and continue.

Guard 3 (OU mean reversion). Check ADF-$p < 0.05$ and $|z| > z_{\text{entry}}$. `RESIN`: ADF-$p = 0.02$ but autocorr $+0.01$ gives tiny z, no proposal. `KELP`: ADF-$p = 0.005$ and $|z| > 1.5$ (stated premise); fires, route to `OU`. `SQUID_INK`: ADF-$p = 0.41 \gg 0.05$; does not fire.

Guard 4 (TSMOM). For `SQUID_INK`: $r^{(12\text{m})} = +1.2\%$, Olivia flag on; with $\theta_{\text{trend}} = 0.5\%$ the guard fires, route to TSMOM long. `RESIN`'s $+0.01\%$ is far below.

Guard 5 (gamma). No options on remaining products; does not fire.

Guard 6 (market making). `RESIN`: check $\hat{s} > 2\hat{\sigma}/\sqrt{N_{\text{fills}}}$. With $\hat{s} = 0.5$, $\hat{\sigma} = 1.8$ bps, $N_{\text{fills}} = 100$, RHS $= 0.36$ bps < 0.5; depth $40 >$ min; fires, route to `MM`.

Guard 7 (skip). All four products routed.

The formal output matches the intuitive assignment made at the start of the section. The formalisation just makes it reproducible, auditable, and tunable.

Product	Guard fired	Strategy
`BASKET1`	Guard 2 (basket arb)	`BASKET` short
`KELP`	Guard 3 (OU mean rev)	`OU` (sign from z)
`SQUID_INK`	Guard 4 (TSMOM)	`TSMOM` long
`RESIN`	Guard 6 (market making)	`MM`

Table 23.4: Decision-tree output for the toy universe of Section 23.1, computed guard-by-guard in Section 23.4. Four products routed to four different strategies by six threshold tests.

Intuition

The decision tree is not a classifier — it is a priority-ordered list of sufficient conditions, an if-elif chain in which each branch's expected edge dominates the lower branches when its guard fires. The default branch is "do nothing", not "pick the most confident other option". Doing nothing is a strategy too, and often the right one.

23.5 Meta-labeling for trade-level selection

Not every trade the tree proposes should fire. Meta-labeling (Chapter 22) is a binary classifier that predicts whether a proposed trade hits its profit-taker before its stop-loss. The primary signal proposes; the meta-classifier decides.

Features: observable vector $o_t^{(p)}$, HMM regime posterior, and trade characteristics (strategy, size, horizon). Output: probability of profit-taker hit. Trades below threshold (typically 0.5–0.6) are dropped.

Meta-labeling in the decision framework

Let x_t be the feature vector (observables, regime posterior, proposed-trade features) and let $y_t \in \{0, 1\}$ be the triple-barrier label (Chapter 22) of the proposed trade — 1 if profit-taker hit first, 0 otherwise. Train a binary classifier $\hat{p}(x_t) = \mathbb{P}(y_t = 1 \mid x_t)$ on historical proposals under CPCV (Chapter 21). At decision time, fire the proposed trade only if $\hat{p}(x_t) > \tau_{\text{meta}}$, where τ_{meta} is tuned on validation data.
Typical impact: 5–15% uplift in annualised strategy return and 30–50% reduction in maximum drawdown on the same primary signal, because the meta-filter systematically drops the trades that were about to be losers (López de Prado, 2018).

Meta-labeling is effective here because the primary signal is tuned on its own optimal conditions; the tree may fire it in slightly-off conditions where the guard passes but the edge is thin. A regime-aware meta-classifier learns those conditions and refuses. Chapter 22 has the full worked example.

A stylised lift sketch. Primary signal (OU z-entry) proposes 1000 trades; 550 are profitable (55%), averaging +10 bps on winners and −8 bps on losers. Per-trade P&L: $0.55 \cdot 10 - 0.45 \cdot 8 = 1.9$ bps. A meta-classifier at AUC 0.60 with $\tau_{\text{meta}} = 0.55$ keeps 700 trades, of which 420 win (60%). Per-trade P&L: $0.6 \cdot 10 - 0.4 \cdot 8 = 2.8$ bps — up 47%, with 30% fewer trades and proportionally less turnover cost (where the Sharpe uplift lives). Drawdown reduction is larger still because the classifier disproportionately drops runs of correlated losers. The shape — modest hit-rate uplift, disproportionate Sharpe/drawdown improvement — is what (López de Prado, 2018) reports.

Meta-labeling is not a return predictor

Do not train the meta-classifier to predict trade returns. Train it to predict whether the trade hits profit-taker before stop-loss (a binary triple-barrier label). Predicting returns is high-variance regression that overfits; predicting barrier-hit is low-variance classification with a stable decision boundary. (López de Prado, 2018) is dogmatic on this point.

23.6 From one strategy per product to a portfolio of strategies

The tree so far outputs at most one strategy per product per tick. In practice books run many strategies in parallel (market making slow, mean reversion fast, basket arb on-dislocation). How to size and combine them?

Disjoint: each strategy trades a separate slice with its own capital and risk budget. The Frankfurt Hedgehogs did this with seven strategies on fixed subsets. Disjoint is the simplest thing that works and fails gracefully: one bad strategy does not contaminate the others.

Risk-budget combination: allocate capital inversely to drawdown, proportionally to Sharpe:

$$w_i \;\propto\; \frac{\text{Sharpe}_i}{\text{drawdown}_i}, \qquad \sum_i w_i = 1, \tag{23.17}$$

renormalised over actively-selected strategies. Blunt, but essentially fractional Kelly with diagonal covariance, and exactly what the Frankfurt Hedgehogs used.

Joint mean-variance: estimate the covariance Σ of strategy returns on held-out data and solve

$$\max_{\mathbf{w}\geq 0}\; \mathbf{w}^\top \boldsymbol{\mu} \;-\; \tfrac{\lambda}{2}\,\mathbf{w}^\top \Sigma \mathbf{w}, \tag{23.18}$$

— Markowitz over strategies, not assets, with λ the risk aversion. Σ is unstable on small samples; shrink or regularise before use.

Correlated strategies are one strategy in disguise

The expensive mistake is running two strategies that look different but bet the same way. TSMOM and a momentum-feature meta-labeler are near-collinear: same times, same sign, same products. Disjoint or risk-budget combinations that treat them as independent double down and claim false diversification. Estimate the strategy-return correlation on held-out data and refuse pairs above ~ 0.7; (Chan, 2022) uses 0.5 as a conservative threshold.

23.7 Parameter adaptation: Chan's CPO

Every threshold in the decision tree (z_{entry}, θ_{trend}, τ_{meta}, the basket-spread level, the MM depth cut, the MM inventory skew) was written as a fixed number. Their optima drift with conditions: a calm-regime z-entry is too tight for a volatile one; a TSMOM threshold is too low in chop. Fixed parameters across regimes is the biggest slow-bleed source (Chan, 2022).

The naive fix — refit on a rolling window — has three failure modes. **Lag**: a 60-day window cannot adapt faster than ~ 20 days (how long new data needs to crowd out old), and regimes transition faster. **Loss of significance**: shorter windows have wider confidence intervals. **No regime conditioning**: a rolling mean averages across whatever mix of regimes the window contains, not the conditional-on-current-regime optimum.

Conditional Parameter Optimisation (CPO) (Chan, 2020) reframes parameter optimisation as regression: given market features ϕ_t (vol, HMM posterior, spread, term-structure slope) and candidate parameter vector θ, predict held-out performance R. Fit $\hat{R}(\phi, \theta)$ — Chan uses gradient-boosted trees — on a historical dataset of $(\phi_t, \theta, R(\phi_t, \theta))$ tuples, evaluated under CPCV. At decision time, observe ϕ_{t^*} and solve

$$\theta^*(\phi_{t^*}) \;=\; \arg\max_{\theta \in \Theta} \hat{R}(\phi_{t^*}, \theta). \tag{23.19}$$

The output is the conditional optimum given current features, not the marginal optimum averaged over history.

CPO procedure

1. **Collect training data** as tuples $(\phi_t, \theta, R(\phi_t, \theta))$ sampled over the historical period. For each historical time, sweep over a grid of candidate θ and evaluate held-out performance under CPCV.
2. **Fit a regression** $\hat{R}(\phi, \theta)$ — typically gradient-boosted trees — on the collected tuples. The regression learns the joint dependence on features and parameters, so it implicitly learns the regime-conditional optimum.
3. **At each decision tick**, observe ϕ_{t^*} and maximise $\hat{R}(\phi_{t^*}, \theta)$ over the candidate grid. Use the maximising θ^* as the strategy's parameters for the next tick.

CPO sits cleanly on top of the HMM because you do not need to identify regimes explicitly — the features ϕ already encode them jointly. In practice include both the HMM posterior (which carries the sticky prior) and raw features (which carry microstructure state).

A toy CPO worked example

Consider a single parameter: OU entry threshold z_{entry} on `KELP`. Strategy: if $|z_t^{(\text{KELP})}| > z_{\text{entry}}$, enter against the deviation, exit at $z = 0$. Feature vector ϕ_t is scalar: $p_t := \mathbb{P}(S_t = 2 \mid o_{1:t})$, the HMM's high-vol posterior.

Non-CPO: sweep $z_{\text{entry}} \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, pick the best full-history Sharpe. Suppose $z = 1.5$ wins at Sharpe 1.70; use 1.5 forever.

CPO: for every historical day, record $(p_t, z_{\text{entry}}, \text{Sharpe over next 20 days})$ for each grid value. Fit $\hat{R}(p, z_{\text{entry}})$ under CPCV. At decision time, maximise over the grid at p_{t^*}. Suppose the fitted rule:

$$\hat{R}(p, z_{\text{entry}}) \;\approx\; \begin{cases} \text{peaks at } z_{\text{entry}} = 1.5 & \text{when } p \lesssim 0.3, \\ \text{peaks at } z_{\text{entry}} = 2.0 & \text{when } 0.3 \lesssim p \lesssim 0.7, \\ \text{peaks at } z_{\text{entry}} = 2.5 & \text{when } p \gtrsim 0.7. \end{cases} \tag{23.20}$$

The optimal threshold is larger in high-vol regimes because the OU process's unconditional σ is larger — $z = 1.5$ is weaker evidence of a true deviation. The non-CPO quant averages to 1.5; the CPO quant picks 1.5 in calm and 2.5 in volatile, capturing higher OOS Sharpe in both.

Real CPO sweeps many features and parameters, but the three ingredients are the same: (ϕ, θ, R) tuples, regression $\hat{R}$, decision $\arg\max_\theta \hat{R}(\phi_{t^*}, \theta)$.

CPO can overfit just like any supervised ML model

CPO is supervised ML; every failure mode of Chapter 22 applies. A too-flexible model class memorises noise in the (ϕ, θ, R) tuples: beautiful in-sample, random out-of-sample. Cures: CPCV (Chapter 21), modest capacity (Chan uses shallow boosted trees), and OOS-Sharpe-uplift over a single-threshold baseline as the acceptance criterion.

Historical note

CPO was popularised by Ernest Chan in the 2020s via PredictNow and (Chan, 2020). The idea — context-aware parameter tuning — appears in the 1980s adaptive-control literature, but Chan packaged it for trading. Before CPO the practitioner habit was rolling-window refit.

23.8 The full pipeline

Assemble everything. Eight steps per tick, output is a list of sized, scheduled trades.

The full decision pipeline

1. **Observe.** Compute the observable vector $o_t^{(p)}$ of Equation (23.16) for each product p in the universe, from the last N ticks of order-book data.
2. **Detect regime.** Run the trained HMM forward algorithm on the recent return history; take the last column as the current regime posterior.
3. **Select strategy.** Apply the decision tree of Section 23.4 to each product. Each product is routed to at most one strategy or skipped.
4. **Adapt parameters.** For each selected (product, strategy) pair, apply CPO (Section 23.7) to pick parameters θ^* conditional on the current regime and features.
5. **Propose trades.** Run each selected strategy with its CPO-chosen parameters on its current observables; collect the proposed trades.
6. **Meta-label filter.** For each proposed trade, evaluate the meta-classifier on the trade's features and drop trades with $\hat{p} < \tau_{\text{meta}}$.
7. **Size.** Apply Kelly sizing (Chapter 20) to each surviving trade, with fractional Kelly ($f^*/4$ or $f^*/2$) as the production-safe default.
8. **Schedule execution.** Send each sized trade through the Almgren–Chriss optimal execution trajectory (Chapter 19). Large trades are broken into child orders; small trades fire marketable in a single shot.

Each step has a chapter behind it: step 1 (Chapters 2 and 8); step 2 (Section 23.2); step 3 (Section 23.4); step 4 (Section 23.7); step 5 (Chapters 9 to 14 and 17); step 6 (Chapter 22); step 7 (Chapter 20); step 8 (Chapter 19).

Validation is the domain of Chapter 21: each component walk-forward or CPCV validated individually, composition validated end-to-end to catch cross-layer leakage (e.g., using the meta-classifier's OOS predictions as CPO input is a subtle leak).

Three implementation details. **Order is load-bearing**: CPO (4) must follow selection (3) because the candidate grid depends on the strategy; meta-labeling (6) must follow CPO (4) because the classifier is trained on regime-conditional proposals — deploying it downstream of CPO after fixed-parameter training silently miscalibrates it (training distribution $\neq$ deployment distribution).

Log everything timestamped and replayable: HMM posterior, tree route, CPO parameters, meta-score, Kelly size, AC schedule. Without these, debugging a live failure is impossible.

Decouple retraining from decision cadence: the pipeline decides per-tick; HMM, CPO regression, and meta-classifier retrain nightly or weekly. Retraining inside the decision loop is both a leakage vector and a performance sink. Cron retraining writes versioned artefacts; the decision loop loads the latest at a barrier.

Intuition

The pipeline is a funnel. Regime detection narrows "any strategy" to "regime-appropriate"; the tree narrows to one per product; CPO picks parameters; strategies propose; meta-labeling drops the worst; Kelly sizes; AC schedules. Every step throws something away; nothing is added. By the time a proposal becomes an order, every filter has had a veto.

The pipeline's weakest step is usually the HMM

Step 2 is usually the weak link. A Gaussian HMM on a short window, wrong number of states, or non-Gaussian tails produces regime labels wrong for half of history — and every downstream layer conditions on them. Monitor regime detection in isolation: plot the forward posterior against crashes/VIX spikes, watch transition-matrix stickiness for drift, retrain slowly enough that the sticky prior does not whipsaw. Never bury the HMM where you cannot see its output.

On a Goldman Sachs desk

On a Goldman central-risk desk, a slimmer pipeline runs continuously across the single-stock book. Strat owns the pipeline code (regime model, tree, CPO regression, meta-classifier). The trader owns the override — the red button that disables the regime layer for N ticks or forces strategy X on product Y. The quant researcher owns the regime model's retraining cadence and investigates anomalies. When it works the pipeline runs for weeks untouched; when regime detection breaks, all three scramble to the same stand-up. This Strat/trader/researcher triad is textbook at every tier-1 bank (Chan, 2022).

Historical note

Regime-conditional strategy selection appears in the practitioner literature of the 2000s; Chan's *Quantitative Trading* (2009, (Chan, 2022)) was the first retail-facing book to present it with code. Academic HMM-for-regime papers (Hamilton's 1989 Markov-switching model) predate it by two decades. Meta-labeling was popularised by López de Prado (2018, (López de Prado, 2018)), ch. 3. CPO is Chan's more recent contribution (Chan, 2020). The pipeline assembled here is a mid-2020s synthesis — components all at least a decade old, but rarely written down combined.

23.9 Alternatives and honest limitations

Three alternatives to the pipeline are worth knowing.

Always-on ensemble. Drop the regime layer; run every strategy all the time with a vol-targeted ensemble. Advantage: no missed trades from HMM confusion. Disadvantage: every tick eats every strategy's worst-regime loss. Backtests usually favour the pipeline when the HMM is well-calibrated (sign flips are sharp), and the always-on ensemble when the HMM is badly calibrated. Start always-on, measure regime sensitivity, gate with the HMM only once it beats straight averaging.

Black-box end-to-end model. Train one large model (NN, RL, transformer) to emit trade proposals directly. Appeal: no hand-coded tree, no HMM, no CPO. Reality: every quant RL group has tried it and produced Sharpe-<-1 models (Chapter 22 — financial series are not stationary and do not contain enough information to train end-to-end). The layered pipeline works *because* each layer is small and imposes structure (HMM: Markovian regimes; tree: priority; CPO: conditional regression; meta: triple-barrier), and each can be validated independently. An end-to-end model is a monolithic failure mode.

Manual discretion. A human reads the screen and picks a strategy — implicitly running the tree. Advantage: qualitative state (news, CB announcement, block cross) enters the decision. Disadvantage: slow, unreproducible, cannot scan 10,000 products. The compromise every desk converges on is pipeline + trader override (as in Section 23.8): pipeline does the reproducible part; human handles the rest.

Limitations worth flagging. The pipeline assumes regimes exist and are detectable (not universally true); assumes the strategy pool covers plausible states (novel regimes — pandemic, war, rate shocks — may fit nothing); assumes clean input data (bad ticks and outages need their own pre-processing layer, not discussed here). Every production desk has a dedicated monitoring layer.

23.10 A short reference implementation

The pipeline is a pure function. We show only the glue, not the underlying HMM, CPO, meta-classifier, or strategy implementations — each object exposes the interface of its own chapter.

```python
def decide_trades(universe, market_state, hmm, decision_tree,
                  cpo_model, meta_clf, kelly, execution):
    """Return a list of sized, scheduled child orders."""
    # Step 2: regime posterior on most recent return history
    regime = hmm.forward_posterior(market_state.recent_returns())
    trades = []
    for product in universe:
        # Step 1: observables
        obs = market_state.observe(product)
        # Step 3: strategy selection
        strategy = decision_tree.select(obs, regime)
        if strategy is None:
            continue
        # Step 4: CPO parameter adaptation
        params = cpo_model.predict_params(strategy.name, obs, regime)
        # Step 5: primary proposal
        proposed = strategy.propose(obs, params)
        if proposed is None:
            continue
        # Step 6: meta-label filter
        features = meta_clf.build_features(obs, regime, proposed)
        if meta_clf.predict_proba(features) < 0.5:
            continue
        # Step 7: Kelly sizing
        sized = kelly.size(proposed, regime=regime)
        # Step 8: Almgren--Chriss child-order schedule
        children = execution.schedule(sized)
        trades.extend(children)
    return trades
```

Listing 23.1: Pipeline composition. Each dependency object implements the interface of its chapter. See Section 23.8 for the eight-step breakdown.

Thirty lines of glue over eight chapters of machinery. Reading `decide_trades` with Part V loaded is a walk through the funnel, not new material.

23.11 Key facts to memorise

- The chapter picks which strategy to run: observable in, sized trade list out.
- Regime detection with a 2-state Gaussian HMM: forward algorithm for the current posterior, Viterbi for historical paths, Baum–Welch for parameters. $N = 2$ usually; more overfits.
- Transition-matrix stickiness sets flip-sensitivity: diagonal 0.98 gives 50-day mean dwell time. A single -2.5% return flipped the worked example from $0.91 \to 0.21$ on state 1 in one step.
- Regime-to-strategy map: mean reversion wins in range-bound regimes, momentum in trending, gamma in high-vol, arbitrage/basket are regime-neutral, market making wants low adverse selection (calm or range-bound).
- Synthetic: TSMOM Sharpe $+1.65$ trending / -1.97 mean-reverting; OU exactly swapped (-1.64 / $+2.22$). Wrong regime is a sign flip, not a small penalty.
- Decision tree priority: arbitrage, basket, OU/pairs, TSMOM, gamma, MM, SKIP. Each guard is a threshold on one component of the observable vector.
- Meta-labeling: binary classifier on proposed trades trained on triple-barrier labels. Typical impact 5–15% Sharpe uplift, 30–50% drawdown reduction. Predict barrier-hit, never returns.
- CPO: parameter adaptation as regression on (ϕ, θ, R) tuples. Sits on top of HMM posterior; replaces rolling-window refit.
- Pipeline: observe, detect, select, adapt, propose, filter, size, schedule. Every step throws trades away; nothing is added.
- At GS (and every tier-1 bank): Strat owns code, trader owns override, quant researcher owns regime model.
- Biggest failure mode: unmonitored HMM. Wrong regime label corrupts everything downstream. Monitor the posterior against events; retrain slower than the sticky prior.

Part VII

IMC Prosperity Playbook

Chapter 24

IMC Prosperity: Competition, Mechanics, Tooling

You are about to compete in IMC Prosperity. Everything so far—limit order books (Chapter 2), market making (Chapter 9), statistical arbitrage (Chapter 12), options pricing (Chapters 16 to 18), and the "which strategy when" framework (Chapter 23)—applies directly. This chapter is the operational briefing: competition structure, Python API, tooling, and the product archetypes that recur every year.

Prerequisites

- Chapter 2: limit order books, price–time priority, bid/ask/spread, order types.
- Chapter 9: the Avellaneda–Stoikov framework, inventory risk, quoting around fair value.
- Chapter 23: the four-layer decision pipeline mapping market observables to strategies.

24.1 What is IMC Prosperity?

IMC Prosperity is a global algorithmic trading competition run by IMC Trading, a major market-making firm, for university students (IMC Trading, 2026). Teams of up to four submit Python algorithms that trade a simulated exchange populated by bots. The objective: *maximise PnL* measured in the in-game currency (SeaShells in P3, XIRECs in P4).

The competition spans ≈16 days in five rounds. Each round introduces 1–3 new products; earlier-round products persist. Alongside each algorithmic round is a **manual challenge**—a human-decision task (optimisation under uncertainty, game theory) scored independently. Combined PnL across all rounds sets the leaderboard.

Scale: P3 (2025) attracted over 12,000 teams; P4 is expected to be similar. Ranking is pure PnL—no style points.

Prosperity at a glance

- **Organiser:** IMC Trading.
- **Format:** 5 algorithmic rounds + 5 manual rounds over ≈16 days.
- **Language:** Python (submitted as a single `.py` file).

- **Execution:** your algorithm trades independently against bots—there is *no* direct interaction between different teams' algorithms.
- **Scoring:** total PnL (realised + unrealised) across all rounds. Algo and manual PnL are summed for the overall leaderboard.
- **Scale:** 10,000–12,000+ teams.

Prosperity compresses the core challenges of systematic trading—fair value, inventory, spread capture, stat arb, options—into a fast-feedback environment where iteration takes hours, not months. Every strategy archetype in Parts III–V has appeared in at least one round.

24.2 Competition structure

Five rounds, cumulative products

Rounds 1–2 last 72 h each; Rounds 3–5 last 48 h each. A four-day intermission separates Round 2 from 3. A ≈3-hour scoring window between rounds processes Round n's PnL while Round $n+1$ is locked (IMC Trading, 2026). Table 24.1 shows the P4 schedule; the scoring window is dead time for submission but valuable for analysis.

Table 24.1: Prosperity 4 round schedule (all times CEST / UTC+2).

Round	**Start**	**Close**	**Duration**
Tutorial	(signup)	14 Apr 12:00	variable
Round 1	14 Apr 12:00	17 Apr 12:00	72 h
Round 2	17 Apr 12:00	20 Apr 12:00	72 h
Intermission	20 Apr 12:00	24 Apr 12:00	4 days
Round 3	24 Apr 12:00	26 Apr 12:00	48 h
Round 4	26 Apr 12:00	28 Apr 12:00	48 h
Round 5	28 Apr 12:00	30 Apr 12:00	48 h

Each round boundary adds 1–3 new products; all earlier products remain tradable. By P3 Round 5, the universe had grown from 3 to 15 products. This rewards modular code: a well-factored Round 1 strategy should keep running unmodified while new ones are bolted on.

Prosperity round structure

1. Each round introduces 1–3 new products; prior products persist.
2. Algo and manual submissions are locked at round close; the last *active* submission is used.
3. PnL = realised + unrealised at the end of the round's simulation.
4. The simulation runs for 1,000 iterations during testing and 10,000 iterations for the official scoring run.
5. Position limits are enforced *per product* by the exchange. Exceeding the limit causes *all* orders for that product to be cancelled in that iteration.

Algorithmic vs. manual rounds

Each round has two independent components:

1. **Algorithmic challenge.** Upload a Python file containing your `Trader` class. Unlimited revisions; only the last *active* submission at round close is scored. Historical price and trade CSVs for each new product let you study dynamics before coding.

2. **Manual challenge.** A human-decision task—optimisation under uncertainty, game theory, news trading. Manual PnL is *added* to algo PnL. Prior tasks have ranged from triangular FX arbitrage to Nash-equilibrium container selection (Frankfurt Hedgehogs, 2025b).

The manual round is inactive during the tutorial period. Skipping the manual round leaves free money on the table—even a mediocre submission beats zero.

PnL and scoring

Round PnL = realised (closed trades) + unrealised (mark-to-market on open positions) at simulation end. Concretely, realised PnL sums $-\sum p_i q_i$ over your fills (signed: $q_i > 0$ for buys costs cash, $q_i < 0$ for sells earns cash), and unrealised PnL marks any leftover position q at the final mid-price m as $q \cdot m$—so a long position counts at what it could be sold for, a short at what it would cost to cover. Currency: XIRECs (P4) or SeaShells (P3). The overall leaderboard ranks by *total* PnL across rounds (algo + manual), with sub-leaderboards for each.

A subtlety: since PnL is cumulative and products persist, a later round losing money on a Round 1 product reduces your *total* score. If you cannot improve a product's strategy, sometimes disabling it beats letting it bleed.

Position limits

Each product has a position limit L: your signed position q must satisfy $-L \leq q \leq L$. Enforcement is aggressive—if the aggregate of your buy (sell) quantities plus q would exceed $+L$ $(-L)$, *every* order for that product in that iteration is rejected, not just the excess.

Typical P3 limits ranged from 50 (simple MM products like Rainforest Resin, Kelp) to 400 (Volcanic Rock, the options underlying). Limits are disclosed in the product spec at round start.

Position limit enforcement is all-or-nothing

You cannot partially exceed—*all* orders for the product are voided if the aggregate could push past the limit. Compute remaining capacity up front:

$$\text{max\_buy\_qty} = L - q, \qquad \text{max\_sell\_qty} = L + q.$$

Safe pattern: compute these bounds at the start of `run()` and thread them through every order-building function.

Year-over-year product archetypes

Names and narratives change each year, but the product archetypes recur. Recognising the archetype in the first hour lets you skip "what is this?" and go straight to the matching strategy. Table 24.2 maps archetypes across P1–P4.

Every year has at least one flat-value MM product, one trending MM product, one multi-leg stat-arb product, and (since P3) one options product. Cross-exchange arbitrage with conversion fees has appeared every edition since P2.

Table 24.2: Recurring product archetypes across Prosperity editions. The "Strategy chapter" column points to where the matching technique is derived in this book.

Archetype	P1	P2	P3	P4	Strategy ch.
Flat-value market making	Amethysts	Amethysts	Rainforest Resin	Emeralds*	Chapter 9
Trending / random-walk MM	—	Starfruit	Kelp	Tomatoes*	Chapters 9 and 10
Signal-driven (informed trader)	—	—	Squid Ink	TBD	Chapters 13 and 22
Basket / ETF stat arb	—	Gift Baskets	Picnic Baskets (2)	TBD	Chapter 12
Options / vol surface	—	—	Volcanic Rock Vouchers	TBD	Chapters 16 and 18
Cross-exchange / locational	—	Orchids	Macarons	TBD	Chapter 14

*Tutorial-round products; Round 1 onward may introduce additional products.

How to identify the archetype on day 1

Run this checklist when a new product appears:

1. **Plot prices.** Flat (constant fair), trending (random walk), or mean-reverting?
2. **Compute spread.** Wide (>4 ticks) $\Rightarrow$ MM opportunity; tight ($\leq$2) $\Rightarrow$ edge elsewhere.
3. **Related products.** A basket alongside constituents $\Rightarrow$ basket stat-arb: compute synthetic and plot residual.
4. **Conversion observations.** `observations` with bid/ask/tariff for an external market $\Rightarrow$ cross-exchange.
5. **Option-like.** Multiple strike variants alongside an underlying $\Rightarrow$ options; compute IV for each.
6. **Trade flow.** `market_trades` filtered by size and direction revealing predictable bots (e.g. always buying the daily low) $\Rightarrow$ signal-driven.

Intuition

Each new Prosperity product is an instance of $\approx$6 archetypes. Day-1 step: *plot price, compute spread, ask "which archetype?"* Then pull the matching skeleton from the relevant chapter and adapt parameters. With notebooks prepared in advance, the checklist should take under 15 minutes per product.

24.3 The Python API

Your algorithm is a single Python file containing a `Trader` class. The simulation calls `Trader.run()` once per iteration ("tick"), passing the current state and expecting back an order dict. This section documents every field you will touch.

The `Trader` class

The minimal skeleton is:

```python
from datamodel import OrderDepth, TradingState, Order

class Trader:
    def run(self, state: TradingState) -> tuple:
        result = {}
        for product in state.order_depths:
            orders = []
            # --- strategy logic here ---
            result[product] = orders
        traderData = ""    # JSON string for persistence
        conversions = 0    # conversion requests (if any)
        return result, conversions, traderData
```

Listing 24.1: Minimal `Trader` skeleton.

`run` takes a `TradingState` and returns a triple `(result, conversions, traderData)`:

- `result`: `dict[str, list[Order]]` mapping product `Symbol` to orders.
- `conversions`: integer conversion request (cross-exchange products only; else 0).
- `traderData`: string delivered back as `state.traderData` next call—your only persistence mechanism (see pitfall below).

Return within 900 ms (typical: <100 ms); a timeout produces no orders that iteration.

The `TradingState` class

`TradingState` is the full world snapshot visible to your algorithm:

Table 24.3: `TradingState` fields.

Field	Type	Meaning
`timestamp`	`int`	Simulation clock (increases each iteration).
`traderData`	`str`	The string you returned last iteration.
`listings`	`dict[str, Listing]`	Product metadata (symbol, denomination).
`order_depths`	`dict[str, OrderDepth]`	Current order book per product.
`own_trades`	`dict[str, list[Trade]]`	Trades *your* algorithm did since last tick.
`market_trades`	`dict[str, list[Trade]]`	Trades *other* bots did since last tick.
`position`	`dict[str, int]`	Your signed position per product.
`observations`	`Observation`	External data (sunlight, tariffs, etc.).

The two heavy-use fields are `order_depths` (what you can trade against) and `position` (inventory). `own_trades` confirms last-iteration fills; `market_trades` reveals bot behaviour and sometimes identifies specific bot personalities (Chapter 25).

The `OrderDepth` class

`OrderDepth` *is* the LOB snapshot from Chapter 2, as two dictionaries:

```python
class OrderDepth:
    def __init__(self):
        self.buy_orders: dict[int, int] = {}
        self.sell_orders: dict[int, int] = {}
```

Listing 24.2: `OrderDepth` class definition.

Both map integer **price** to integer **total volume** at that level: `buy_orders` holds resting bid levels, `sell_orders` holds resting ask levels.

`OrderDepth` sign convention

`sell_orders` values are **negative**. 4 lots offered at price 11 is `{11: -4}`, not `{11: 4}`. Negate to get a positive buy quantity:

```python
best_ask, best_ask_vol = list(od.sell_orders.items())[0]
buy_qty = -best_ask_vol   # now positive
```

Nearly every team hits this once. If your first submission produces zero trades, check this first.

Prices are integers and the book is uncrossed (every bid $<$ every ask), so:

```python
best_bid = max(od.buy_orders.keys())
best_ask = min(od.sell_orders.keys())
mid = (best_bid + best_ask) / 2
```

Listing 24.3: Extracting best bid/ask and mid price.

The discrete-price analogue of $s = p^{\text{ask}} - p^{\text{bid}}$ (Chapter 2).

The `Order` class

Construct an `Order` to submit:

```python
class Order:
    def __init__(self, symbol: str, price: int,
                 quantity: int) -> None:
        self.symbol = symbol
        self.price = price
        self.quantity = quantity
```

Listing 24.4: `Order` class usage.

- `quantity > 0` $\Rightarrow$ **buy** order.
- `quantity < 0` $\Rightarrow$ **sell** order.

If your buy price meets or exceeds a resting sell, the trade executes immediately at the *resting* price (price–time priority, as in Chapter 2). Unmatched residuals rest in the book; bots may trade against them before the next tick, and if none do, they are cancelled at iteration end.

The `Trade` class

Each `own_trades`/`market_trades` element is a `Trade` with `symbol, price, quantity, buyer, seller, timestamp`. Counterparty names are hidden—`buyer`/`seller` are empty strings unless *your* algorithm is one of the parties (in which case `"SUBMISSION"`). In P3 Round 5, trader IDs were briefly revealed, enabling direct bot identification.

Matching and execution

Matching uses price–time priority (Chapter 2):

1. Zero latency—you never lose a race to a bot.
2. Buys match cheapest sells first; sells match most expensive buys first.
3. Fills at the *resting* price, so you may get a better price than quoted.
4. Unmatched residuals rest; if no bot trades, they are cancelled before the next iteration.

This "snapshot-then-act" model has no latency, no queue priority, and full book visibility before acting. Strategic consequence: *information extraction* dominates *speed*.

Conversions and observations

Some products support **conversions**: trading with a foreign exchange at known prices plus fees (the cross-exchange archetype, Section 24.2). When conversions are available, `observations` contains a `ConversionObservation` with foreign bid/ask, transport fees, and import/export tariffs. Execute by returning a non-zero integer as the `conversions` element.

A per-tick conversion limit (e.g. 10 units for P3 Macarons) caps throughput. You must hold sufficient inventory—the exchange will not create positions from nothing.

Worked example: two iterations of trading

Consider `PRODUCT1` with position limit 10, estimated fair value 13, and current position 3.

Iteration 1. The `TradingState` arrives with:

```
order_depths["PRODUCT1"] = OrderDepth(
    buy_orders  = {10: 7, 9: 5},
    sell_orders = {11: -4, 12: -8}
)
position["PRODUCT1"] = 3
```

Both sells (11, 12) are below fair value 13, so buy. Max buy capacity is $L - q = 10 - 3 = 7$ lots. Send:

```
Order("PRODUCT1", 12, 7)   # buy up to 7 at price 12
```

The exchange matches: 4 lots at 11 (you pay 11, not 12—fills at resting price), then 3 of 8 lots at 12. You bought 7 lots at average $(4 \times 11 + 3 \times 12)/7 \approx 11.43$.

Iteration 2. The next `TradingState` shows:

```
own_trades["PRODUCT1"] = [
    Trade("PRODUCT1", 11, 4, buyer="SUBMISSION"),
    Trade("PRODUCT1", 12, 3, buyer="SUBMISSION"),
]
position["PRODUCT1"] = 10   # at the limit
```

At the limit, any buy causes *all* `PRODUCT1` orders to be rejected. Options: sell or do nothing.

Intuition

Prosperity's iteration model is a discrete-time LOB (Chapter 2): each tick gives a fresh snapshot, you submit, you observe fills. No race condition, no latency advantage, no queue priority. The edge is *fair-value estimation* and *inventory management* against position limits.

24.4 Tooling

Top teams backtest locally, visualise order books, and iterate. This section covers the de facto standard tools.

`prosperity3bt`: the community backtester

Jasper Merle's open-source backtester, `prosperity3bt` (Merle, 2025a), replays historical data against your `Trader` locally, producing logs in the official submission format.

```
# Install
pip install -U prosperity3bt

# Run on all days from round 1
prosperity3bt trader.py 1

# Run on a specific day (round 1, day 0)
prosperity3bt trader.py 1-0

# Merge PnL across days
prosperity3bt trader.py 1 --merge-pnl

# Open result in the visualiser automatically
prosperity3bt trader.py 1 --vis
```

Listing 24.5: Installing and running the backtester.

The backtester matches against recorded order depths and market trades, enforcing position limits identically to the official environment.

Backtester vs. official environment

Replayed bots do not *react* to your orders. Therefore:

- Strategies taking existing liquidity (buying underpriced asks): backtester is accurate.
- Strategies relying on bots trading against *your* resting quotes (passive MM): backtester over- or understates fills depending on `--match-trades`.

Use the official test environment for strategies where bot interaction matters.

Jasper Merle's visualiser

The P3 Visualiser (Merle, 2025b) parses backtester logs and displays:

- Order-book levels over time (bids/asks with depth).
- Your trades overlaid on price.
- PnL and position curves per product.

`prosperity3bt trader.py 1 --vis` opens it after a backtest. The Frankfurt Hedgehogs built a custom dashboard (indicator normalisation, trader-ID filtering), but the community visualiser is sufficient for most teams.

Recommended repository layout

A clean structure prevents confusion as rounds add products:

```
prosperity/
  src/
    trader.py          # Your Trader class (submitted)
    datamodel.py       # Copy of the official datamodel
    strategies/        # One module per product archetype
      market_making.py
      basket_arb.py
      options.py
  logs/                # Backtester output logs
  analysis/            # Jupyter notebooks, plots
  data/                # Sample CSV data from the dashboard
  README.md
```

Listing 24.6: Suggested repository layout.

Keep `trader.py` a thin dispatcher that imports strategy modules, so new products add without touching existing code. The Frankfurt Hedgehogs used a `ProductTrader` base class with per-product subclasses sharing utilities for book parsing, position tracking, and logging (Frankfurt Hedgehogs, 2025b).

Debugging workflow

The platform returns a log file with all `print()` output from your `run()`. Workflow:

1. **Structured logging.** Print timestamp, product, position, best bid/ask, and orders sent. Compact format for scanning thousands of iterations.
2. **Run locally first.** `prosperity3bt trader.py 1 --print` streams output in real time.
3. **Visualise.** Feed the log into the visualiser. Overlay trades on the book. Look for obvious mistakes: wrong side, quotes far from fair, flat PnL when you expect activity.
4. **Validate on the platform.** Upload to the official test environment (1,000-iter run). Large divergence from local PnL usually indicates bot-interaction effects.

Historical data

For each new product the platform provides several days of CSVs:

- **Prices CSV:** order-book state (all levels with volumes) at every timestamp.
- **Trades CSV:** every trade, with price, quantity, and (in later rounds) buyer/seller IDs.

The backtester ships with data for all rounds. Use these CSVs for exploratory analysis—plot prices, compute spreads, check mean reversion, spot bot patterns—*before* writing strategy code.

AWS Lambda constraints

The official environment runs on AWS Lambda. Two constraints:

1. **Memory:** ≈256 MB. `numpy`, `pandas`, `statistics` are available, but heavy structures hit the ceiling.
2. **Statelessness:** globals and class attributes are not guaranteed to persist across `run()` calls.

No persistent state between ticks

Lambda can recycle the container between iterations, so instance variables (`self.history`), class variables, and module globals are **not** reliable across `run()` calls. The only reliable persistence is the `traderData` string returned as the third tuple element; it is delivered back as `state.traderData` next call.
Serialise state with `json` (or `jsonpickle`):

```
import json

class Trader:
    def run(self, state: TradingState) -> tuple:
        # Restore state
        data = json.loads(state.traderData) \
            if state.traderData else {}
        history = data.get("prices", [])

        # ... strategy logic using history ...
        mid = (best_bid + best_ask) / 2
        history.append(mid)

        # Save state (max 50,000 characters)
        traderData = json.dumps({"prices": history})
        return result, conversions, traderData
```

`traderData` is truncated to 50,000 characters—keep state compact.

Supported libraries

Python 3.12, with:

- `numpy`, `pandas` — numerics and data.
- `statistics`, `math` — stdlib numerics.
- `jsonpickle` — serialisation for `traderData`.
- All Python 3.12 stdlib (`collections`, `itertools`, `functools`, . . .).

No other libraries. Notably `scipy` is *not* available, so for a normal CDF (Black–Scholes) use `statistics.NormalDist` or implement it yourself.

24.5 The simulation model and bot ecosystem

Understanding the simulation internals is essential for strategies that translate from backtest to live.

Iteration flow

Each iteration proceeds in a fixed sequence (Frankfurt Hedgehogs, 2025b):

1. All previous player orders cancelled.
2. Deep-liquidity MM bots post new quotes (the “walls”).
3. Taker bots may cross and generate trades.
4. Your `run()` is called; orders match against the current book.

5. Remaining player quotes rest; bots may trade against them.
6. Unfilled resting quotes are cancelled; bot-to-bot trades may also occur.
7. Next iteration begins with a fresh `TradingState`.

Two implications. First, you see the full book before acting—no information disadvantage relative to bots. Second, resting quotes are exposed to takers *after* you submit, so passive MM accumulates fills between iterations.

Bot personalities

Bots are not noise. In P3, top teams identified distinct personalities:

- **Market-maker bots** posting "wall" quotes at predictable distances from true price. These walls are the most reliable fair-value signal: the "wall mid" (midpoint of innermost bid/ask walls) tracks the hidden true price closely (Frankfurt Hedgehogs, 2025b).
- **Taker bots** crossing the spread. Some are uninformed; others informed—e.g. "Olivia" in P3 bought the daily low and sold the daily high of Squid Ink.
- **Hidden taker bots** not visible in the book that fill resting orders priced attractively vs. a hidden fair value. P3's Macarons had such a bot, filling ≈60% of correctly-priced sells.

Identifying bot types is part of the pattern-recognition process (Chapter 23): once you know which bots are informed vs noise, you can calibrate fair value and decide make vs take.

Do not assume bots are adversarial

Prosperity bots are pre-programmed, not adaptive. They do not learn your strategy or front-run you, so exploitable patterns persist for the round. Flip side: *taking* strategies have high backtester fidelity (same bot behaviour replayed), but passive quoting is harder to backtest because bots cannot react to your new quotes.

24.6 A day-1 recipe

Prosperity example

Day-1 recipe: zero to profitable in 30 minutes.
P3's Rainforest Resin had a fixed true value of 10,000. Blank file to profitable MM bot:

1. **Install.** `pip install -U prosperity3bt`.
2. **Write the stub.** A `Trader` that buys at 9,999 and sells at 10,001:

```python
from datamodel import OrderDepth, TradingState, Order

class Trader:
    def run(self, state: TradingState):
        result = {}
        orders = []
        orders.append(Order("RAINFOREST_RESIN", 9999, 10))
        orders.append(Order("RAINFOREST_RESIN", 10001, -10))
        result["RAINFOREST_RESIN"] = orders
        return result, 0, ""
```

3. **Run the backtester.** `prosperity3bt trader.py 0 --vis`.

4. **See PnL.** The visualiser shows your trades and cumulative profit. You are buying at 9,999 and selling at 10,001, capturing a 2-unit spread each round trip.

5. **Iterate.** Widen quotes when inventory builds up (the Avellaneda–Stoikov skew from Chapter 9). Tighten quotes when bots leave free money on the book (take any ask below 10,000 or any bid above 10,000).

This 15-line bot will not win, but it establishes the full pipeline—write, backtest, visualise, iterate—that top teams run hundreds of times per round. The Frankfurt Hedgehogs reported Rainforest Resin alone contributed ≈39,000 SeaShells per round from a simple variant.

24.7 Pre-competition preparation checklist

Top teams consistently attribute their placements to preparation done *before* the competition (Frankfurt Hedgehogs, 2025b). Before Round 1 opens, have:

1. **Backtester installed and tested.** Run `prosperity3bt` on tutorial data; confirm log + visualiser.

2. **Strategy skeletons for each archetype.** Template code for: (a) flat-value MM, (b) trending MM (fair from wall mid), (c) basket arbitrage (synthetic price vs. residual), (d) options (Black–Scholes, IV extraction, delta hedge), (e) cross-exchange arbitrage (local vs. foreign after fees). These need to *exist*, not be tuned—so you can plug parameters in within the first hour.

3. **Analysis notebooks.** Load CSVs, plot prices, compute rolling mean/spread/autocorrelation, identify bot patterns. Goal: full diagnostic in <15 min per new product.

4. **Logging framework.** A parseable structured `print()` format: timestamp, product, position, fair estimate, orders, fills. The Frankfurt Hedgehogs synced theirs with their dashboard's hover tooltip.

5. **Study prior-year writeups.** P2/P3 writeups (Stanford Cardinal, Linear Utility, Jasper Merle, Frankfurt Hedgehogs) expose what worked, what failed, and which bot patterns recur. Highest-ROI prep activity.

6. **Read the entire wiki.** Position limits, matching, libraries, the Round 2 `bid()` method—easy to miss if you only skim the API page.

Historical note

IMC Prosperity began in 2022 as "Prosperity 1," originally targeting IMC's intern pipeline. Its success prompted a global opening; by P3 (2025) it was among the largest student trading competitions, with >12,000 teams from 100+ countries. The design mirrors IMC's own philosophy—market making, inventory management, robust backtesting—so placing well is a credible signal to trading firms; several top teams have received direct interview invitations from IMC and competitors.

24.8 The Goldman Sachs parallel

On a Goldman Sachs desk

Prosperity's `Trader` mirrors a production event-driven strategy framework at a firm like Goldman Sachs:

- `run()` is the **on-market-data callback**: when a tick arrives, the framework invokes your handler with current book state.
- `TradingState` is the **market-data snapshot**: positions, fills since last tick, order-book depth.
- The returned order list maps to a **smart order router** (SOR) or direct exchange gateway.
- A real desk runs microseconds, not 900 ms—but the mental model is identical: receive state, compute fair, decide, submit.
- `traderData` has no production equivalent (real systems have databases and shared memory), but explicit state serialisation *is* good practice: production strategies must handle restarts, and strats who design for explicit checkpointing save themselves hours after an unplanned restart.

24.9 Key facts to memorise

1. 5 rounds over ≈16 days, cumulative products: maintain prior-round strategies while adding new ones.
2. Algorithm is a `Trader` class with `run(self, state)`: takes `TradingState`, returns `(orders_dict, conversions, traderData)`.
3. `OrderDepth.sell_orders` values are **negative**. Negate for positive quantities.
4. Position limits are all-or-nothing: aggregate could-exceed rejects *every* order for that product.
5. No persistent state between `run()` calls except `traderData` (max 50,000 chars). Use JSON.
6. `prosperity3bt` and the Jasper Merle visualiser are the community-standard tools. Install before the competition.
7. ≈6 recurring archetypes (flat-value MM, trending MM, signal-driven, basket, options, cross-exchange). Identify first, then apply matching strategy.
8. Zero latency—speed does not matter; *fair-value* and *inventory* determine PnL.
9. Manual round is scored independently—free PnL, do not skip.
10. Pre-competition: backtester, datamodel clone, strategy skeletons per archetype.

Chapter 25

Prosperity Algorithmic Rounds: Strategies Mapped to Theory

Prosperity products fall into a small set of recurring archetypes. Recognise the archetype and you know which chapter to open. This chapter maps archetype → theory → strategy skeleton → pitfalls so a new product can be diagnosed in under five minutes.

Prerequisites

- Chapter 9: Avellaneda–Stoikov, inventory risk, fair-value quoting.
- Chapter 10: Ornstein–Uhlenbeck, z-scores.
- Chapter 11: cointegration, spreads, hedge ratios.
- Chapter 12: basket/ETF stat-arb, premium tracking.
- Chapter 13: trend signals, EMA, regimes.
- Chapter 14: triangular, locational, calendar, cash-and-carry arb.
- Chapters 16 to 18: Black–Scholes, Greeks, delta hedging, IV, vol smile.
- Chapter 23: four-layer observables→strategy pipeline.
- Chapter 24: Prosperity API, timestep model, `TradingState`, order submission.

Each section uses a five-field template: Observable, Theory chapter, Strategy skeleton, Pitfalls, Reference writeup.

25.1 Archetype 1: fixed-price market making

Observable

The product has an *obvious, fixed true price* that does not move. Bids and asks cluster symmetrically around that level with a spread wide relative to tick size, so every off-par fill yields positive edge. Takers periodically cross the book but the true price never trends.

Diagnostic

If you normalise price by the true value and see a stationary series with zero drift, you have a fixed-price market-making product.

Prosperity examples.

- **RAINFOREST_RESIN** (Prosperity 3, Round 1): true price permanently fixed at 10,000 (Frankfurt Hedgehogs, 2025b).
- **AMETHYSTS** (Prosperity 2, Round 1): identical setup.

Theory chapter

Chapter 9, specifically Avellaneda–Stoikov. Quote bids below and asks above fair value, capture the spread on every fill, manage inventory risk (one-sided accumulation) by skewing quotes or flattening at the true price.

A known constant true price removes adverse-selection risk (no informed counterparty) and fair-value estimation error. The only trade-off is edge versus fill probability: wider quotes earn more per fill but fill less often.

Strategy skeleton

A complete market maker for a fixed-price product:

```
TRUE_PRICE = 10000
EDGE = 2            # minimum distance from true price
LIMIT = 50          # position limit

def run(state, product):
    pos = state.position.get(product, 0)
    orders = []
    book = state.order_depths[product]
    # 1. Take any mispriced orders
    for ask, vol in sorted(book.sell_orders.items()):
        if ask < TRUE_PRICE and pos - vol <= LIMIT:
            orders.append(Order(product, ask, -vol))
            pos -= vol  # vol is negative for sells
    for bid, vol in sorted(book.buy_orders.items(),
                           reverse=True):
        if bid > TRUE_PRICE and pos - vol >= -LIMIT:
            orders.append(Order(product, bid, -vol))
            pos -= vol
    # 2. Quote passively
    buy_qty = min(LIMIT - pos, 20)
    sell_qty = min(LIMIT + pos, 20)
    if buy_qty > 0:
        orders.append(Order(product,
                            TRUE_PRICE - EDGE, buy_qty))
    if sell_qty > 0:
        orders.append(Order(product,
                            TRUE_PRICE + EDGE, -sell_qty))
    return orders
```

Listing 25.1: Fixed-price market maker (Prosperity).

Key design choices:

1. **Take first, then make.** Sweep any ask below true price or bid above it—free money.
2. **Clip position.** Bound `buy_qty` and `sell_qty` so you never breach the limit—exceeding it cancels *all* orders (Chapter 24).

3. **Flatten at par.** If inventory is maxed, suppress the passive quote on the overweight side. Frankfurt Hedgehogs also flattened at exactly 10,000 to reset inventory (Frankfurt Hedgehogs, 2025b).

Pitfalls

Quoting too wide

Too large an edge parameter means you never get filled—other market makers undercut you. For RAINFOREST_RESIN, Frankfurt Hedgehogs overbid the best existing bid by one tick while keeping positive edge; fill probability rose far more than edge decreased.

Ignoring the position limit

Prosperity does not partially reject orders. If the sum of buy quantities plus current position would exceed $+L$, *every* order submitted for that product is cancelled. Cap order sizes before submission.

Reference writeup

Frankfurt Hedgehogs' `StaticTrader` used this pattern on RAINFOREST_RESIN, contributing ≈39,000 SeaShells per round—≈10% of total Prosperity 3 PnL—from a strategy "anyone could have come up with by carefully studying the competition's matching rules" (Frankfurt Hedgehogs, 2025b).

Intuition

Fixed-price MM is the "hello world" of Prosperity: it teaches take-then-make, position-limit clipping, and the edge-vs-fill trade-off every later archetype inherits.

RAINFOREST_RESIN worked example

Setup. True price = 10,000. Position limit = 50. Current position = +12. The order book shows:

Side	Price	Qty
Ask	10,003	8
Ask	10,001	5
Bid	9,998	10
Bid	9,996	15

Step 1: Take. Ask at 10,001 is above true price, no bid above 10,000—nothing to sweep.
Step 2: Make. Buy capacity $= 50 - 12 = 38$, sell capacity $= 62$, both capped at clip size 20. Post:

- **Buy** 20 at 9,998 (matches bid wall; in practice overbid at 9,999 for queue priority).
- **Sell** 20 at 10,002 (undercuts ask at 10,003).

Both sides filling yields $2 \times 20 + 2 \times 20 = 80$ SeaShells per timestep. Across ≈10,000 timesteps, even modest fill rates compound to tens of thousands.

25.2 Archetype 2: trending / EMA / LR products

Observable

The price moves over time but follows a slow random walk with a tight spread relative to tick size. Normalising by Wall Mid (Chapter 24) makes the series look almost fixed-price. Some products also show occasional sharp jumps or drift, creating a second signal (trend-following or informed-flow detection).

> **Diagnostic**
>
> Plot the raw series and its Wall Mid–normalised version. If the normalised series is stationary, use a moving-fair-value market maker. If the raw series shows persistent trends or jumps, overlay a short EMA and check whether deviations predict reversals or continuations.

Prosperity examples.

- **KELP** (Prosperity 3, Round 1): slow random walk, tight spread. Best treated as a dynamic-fair-value market maker.
- **SQUID_INK** (Prosperity 3, Round 1): wider relative spread, sharp jumps. Officially described as short-term mean-reverting, but the main edge came from an informed-trader signal (Olivia—see Section 25.6).

Theory chapter

Two chapters are relevant:

- Chapter 10: if the normalised spread is stationary with mean-reverting tendencies, trade deviations from the EMA as described in the Ornstein–Uhlenbeck framework.
- Chapter 13: if the raw price shows persistent trends, a simple EMA crossover or linear-regression slope can generate trend-following signals.

The diagnostic question: *mean-revert or trend?* KELP was "neither"—a random walk best traded as fixed-price MM with a rolling fair-value estimate (Frankfurt Hedgehogs, 2025b). SQUID_INK's description said "mean-reverting," but no pure price-based strategy was as reliable as detecting the informed trader Olivia (Frankfurt Hedgehogs, 2025b).

Strategy skeleton

Dynamic MM (KELP-style). Replace `TRUE_PRICE` in Listing 25.1 with a rolling `wall_mid` (average of bid-wall and ask-wall levels):

```
fair = wall_mid(book)
```

Wall Mid is the midpoint between the deepest bid and ask levels (ignoring small over-bids/undercuts), giving a noise-robust estimate of fair value.
EMA mean-reversion overlay (SQUID_INK pre-Olivia):

```
ema = alpha * mid + (1 - alpha) * ema_prev
if mid < ema - threshold:  buy
if mid > ema + threshold:  sell
```

Frankfurt Hedgehogs used a fast EMA ($\alpha \approx 0.1$–0.3) with fixed thresholds, avoiding z-score normalisation because they saw no time-varying volatility (Frankfurt Hedgehogs, 2025b).

Pitfalls

Confusing trend and mean-reversion

A random walk looks like it "trends" short-horizon and "mean-reverts" long-horizon. Without an autocorrelation test (Chapter 10), you overfit to whichever narrative you looked for first. KELP was near-pure random walk: both trend and fade lost money after spread costs; MM around fair value won.

Olivia bias

Several Prosperity 3 products contained Olivia, an informed trader who bought at daily lows and sold at daily highs. Trading with Olivia on SQUID_INK was worth ≈8,000 SeaShells per round; ignoring her and running generic mean-reversion yielded high variance, near-zero expected PnL. Check for informed-flow before defaulting to price-only models (Section 25.6).

Reference writeup

Frankfurt Hedgehogs' `DynamicTrader` traded KELP as moving fair-value MM; their `InkTrader` traded SQUID_INK purely on Olivia's daily-extrema pattern—no MM or mean-reversion for that product (Frankfurt Hedgehogs, 2025b).

Intuition

Archetype 2's lesson: disciplined diagnosis. Before throwing an EMA at a moving price, ask: "Is there a predictable signal, or is this a random walk?" If random walk, MM around fair value; if signal, identify its source (price vs. informed-flow) before choosing.

KELP: Wall Mid vs. raw mid

At a given timestep, KELP's order book shows:

Side	Price		Qty
Ask	2,032		3
Ask	2,031	22	← ask wall
Bid	2,029	18	← bid wall
Bid	2,027		5

Raw mid $= (2{,}031 + 2{,}029)/2 = 2{,}030$. The ask at 2,032 comes from a small overbidding bot and distorts the raw mid. Wall Mid (deepest-liquidity levels) is also 2,030 here, but if a bot undercuts the ask wall with 3 lots at 2,028, the raw mid drops to 2,028.5 while Wall Mid stays at 2,030.
Wall Mid filters noise from small overbids/undercuts, stabilising the fair-value estimate. Frankfurt Hedgehogs found this "crucial for designing effective strategies" across all products (Frankfurt Hedgehogs, 2025b).

25.3 Archetype 3: basket / ETF statistical arbitrage

Observable

One or two *baskets* (ETF analogues) plus several constituents appear together. Each basket is a known weighted combination of constituents, trades on its own book, and has *no creation/redemption* mechanism. The spread between basket and synthetic (weighted constituent sum) mean-reverts around a non-zero premium.

Diagnostic

Compute $\text{spread}_t = P_{\text{basket},t} - \sum_i w_i P_{i,t}$. If this spread is stationary and mean-reverting, you have a basket stat-arb product.

Prosperity examples.

- **PICNIC_BASKET1** (Prosperity 3, Round 2): composition $= 6 \times$ CROISSANTS $+ 3 \times$ JAMS $+ 1 \times$ DJEMBES.
- **PICNIC_BASKET2** (Prosperity 3, Round 2): composition $= 4 \times$ CROISSANTS $+ 2 \times$ JAMS.
- Constituent products: CROISSANTS, JAMS, DJEMBES.

Theory chapter

Chapter 11 (cointegration) and Chapter 12 (ETF-specific). The basket–synthetic spread is analogous to real-world ETF premium/discount (Hull, 2014); Prosperity's absence of creation/redemption lets the spread stay wider and more persistent.

Strategy skeleton

Core loop:

```
synthetic = 6*mid(CROISSANTS) + 3*mid(JAMS) + 1*mid(DJEMBES)
spread = mid(BASKET1) - synthetic - premium
if spread > +threshold:  sell basket, (optionally) buy legs
if spread < -threshold:  buy basket, (optionally) sell legs
if spread crosses zero:  flatten
```

Key design choices:

1. **Subtract the running premium.** The spread mean is non-zero—baskets trade at a persistent premium. Frankfurt Hedgehogs tracked and subtracted a running estimate (Frankfurt Hedgehogs, 2025b).

2. **Fixed thresholds.** A light grid search prioritising "landscape stability" (flat regions of good PnL) over peak performance (Frankfurt Hedgehogs, 2025b).

3. **Half-hedge.** In the final round they hedged 50% of basket exposure—a compromise between full-hedge variance reduction and unhedged expected value.

4. **Flatten at zero.** Close on zero-crossing (adjusted for Olivia) rather than waiting for the opposite threshold.

Pitfalls

Non-zero premium

Assuming zero-mean reversion when the spread actually reverts to a persistent premium biases every entry. Compute the running spread average and subtract it. Frankfurt Hedgehogs updated this premium continuously in live trading.

Olivia cross-product signal

Olivia appeared on CROISSANTS (in both baskets), not the baskets themselves. Olivia-short Croissants meant the basket spread would widen. Frankfurt Hedgehogs biased basket thresholds accordingly: if Olivia short, shift long entry $-50 \to -80$, short entry $+50 \to +20$ (Frankfurt Hedgehogs, 2025b). Teams ignoring this cross-product signal left 20,000+ SeaShells per round on the table.

No creation/redemption

Real-world ETF arb has authorised participants creating/redeeming shares to enforce NAV tracking. Prosperity has no such mechanism— basket and constituents are merely correlated, so spreads persist longer and move further than textbook ETF arb predicts.

Reference writeup

Frankfurt Hedgehogs' `EtfTrader` implemented the spread-threshold model with Olivia-adjusted entries; combined basket and CROISSANTS trading generated ≈60,000–80,000 SeaShells per round. The chrispyroberts writeup complements this with a systematic grid search over threshold parameters[1].

Intuition

Basket stat-arb in Prosperity is the closest analogue to real-world ETF arbitrage absent creation/redemption. That absence is both the opportunity (wider spreads, more PnL) and the risk (longer dislocations). Winning recipe: fixed thresholds, premium adjustment, and—if present—an informed-flow signal on a constituent.

PICNIC_BASKET1 spread calculation

Mid prices:

Product	Mid price
CROISSANTS	4,330
JAMS	6,540
DJEMBES	9,800
PICNIC_BASKET1	56,200

Synthetic $= 6 \cdot 4{,}330 + 3 \cdot 6{,}540 + 1 \cdot 9{,}800 = 55{,}400$. Raw spread $= 56{,}200 - 55{,}400 = +800$. With running premium $+400$, adjusted spread $= +400$.
Entry threshold ± 300: we are above $+300$, so the basket is "expensive." **Action:** sell PICNIC_BASKET1, optionally half-hedge (3×CROISSANTS, 1.5×JAMS, 0.5×DJEMBES,

[1] See `research/writeups/chrispyroberts_P3/`.

rounded). Hold until adjusted spread crosses zero; flatten.
If Olivia was short CROISSANTS (falling prices expected, spread widening), thresholds shifted to $\{-530, +70\}$—harder longs, easier shorts (Frankfurt Hedgehogs, 2025b).

25.4 Archetype 4: options / Black–Scholes + IV hedging

Observable

An underlying plus several *vouchers* (call options) at different strikes, with time-to-expiry shrinking each round. Implied volatility (IV) vs. moneyness forms a parabolic smile; option prices fluctuate rapidly around Black–Scholes theoretical values, creating short-term mean-reversion opportunities.

Diagnostic

Compute each strike's IV via Black–Scholes inversion (Chapter 16) and plot against moneyness. If points lie on a smooth parabola, fit it, detrend, and trade the mean-reverting residuals.

Prosperity examples.

- **VOLCANIC_ROCK_VOUCHERS** (Prosperity 3, Round 3): five call options on VOLCANIC_ROCK at strikes 9,500, 9,750, 10,000, 10,250, and 10,500. Expiry shrank from 7 days (Round 3) to 2 days (Round 5).
- Underlying: **VOLCANIC_ROCK**, trading around 10,000.

Theory chapter

Three chapters:

- Chapter 16: Black–Scholes, risk-neutral valuation, IV inversion.
- Chapter 17: delta, gamma, vega, theta, portfolio delta-hedging.
- Chapter 18: vol smile, parametric surface fits, smile-dynamics relative value.

Frankfurt Hedgehogs combined two alpha sources (Frankfurt Hedgehogs, 2025b):

1. **IV scalping.** Fit a parabola to IV vs. moneyness, convert fitted IV back to a fair price via Black–Scholes, trade the residual. Analogous to real-desk relative-value vol (Sinclair, 2013).
2. **Gamma scalping.** Long gamma plus delta rehedging: convexity profit exceeds theta when realised volatility beats implied.

Strategy skeleton

```
for each strike K:
iv[K] = bs_invert(market_price[K], S, K, T, r)
fit parabola:  iv_hat = a*m^2 + b*m + c
where m = log(S/K) / sqrt(T)
# m is moneyness:  how far the strike K sits above/below spot S
for each strike K:
```

```
fair_price = bs_call(S, K, T, r, iv_hat[K])
deviation = market_price[K] - fair_price
if deviation > +threshold:  sell voucher K
if deviation < -threshold:  buy voucher K
```

Frankfurt Hedgehogs started with the 10,000 (ATM) strike for liquidity and expanded to others as the underlying moved or expiry shortened (Frankfurt Hedgehogs, 2025b).

Pitfalls

Wrong volatility estimate

A poorly calibrated parabola—too few points, outliers from deep OTM options with near-zero extrinsic value—yields a bad "fair price" and trades noise. Discard outliers where extrinsic value is too low for reliable IV.

Discrete delta-hedging costs

Continuous delta-hedging is free in theory; in Prosperity (integer quantities, discrete timesteps, bid–ask spread on the underlying), each rebalance costs spread. Frankfurt Hedgehogs found explicit delta-hedging "prohibitively expensive" and used a hybrid: IV scalping primary, a moderate mean-reversion position in the underlying as luck hedge (Frankfurt Hedgehogs, 2025b).

Integer position constraints

Option positions are integers: delta 0.53 cannot be hedged exactly. Rounding errors accumulate across multiple strikes. Solution: aggregate portfolio delta, hedge at portfolio level.

Reference writeup

Frankfurt Hedgehogs' `OptionTrader` did IV scalping with a parabolic smile fit; it contributed ≈100,000–150,000 SeaShells per round—their single most profitable strategy (Frankfurt Hedgehogs, 2025b). A secondary VOLCANIC_ROCK mean-reversion position produced volatile returns (+100k, −50k, −10k across rounds), serving as a relative hedge against other top teams pursuing pure mean-reversion.

Intuition

Prosperity options are not directional bets—they are *relative value on the vol smile*. Fit the smile, find the outlier, trade it back. Exactly what a real vol desk does (Chapter 18); Prosperity just enlarges the mispricings and speeds up feedback.

IV scalping walk-through

Setup. VOLCANIC_ROCK trades at $S = 10{,}050$. Time to expiry $T = 5$ days $= 5/365 \approx 0.0137$ years. Risk-free rate $r = 0$. We observe market prices for five voucher strikes:

Strike	Market price	Moneyness m	IV (BS invert)	IV (fitted)
9,500	565	+0.449	0.162	0.160
9,750	340	+0.247	0.155	0.152
10,000	145	+0.041	0.148	0.150
10,250	38	−0.164	0.160	0.155
10,500	6	−0.368	0.175	0.168

Moneyness $m = \ln(S/K)/\sqrt{T}$. Fit parabola $\hat{v}(m) = am^2 + bm + c$ to the five IVs via least squares; the fitted column is the smooth smile.
Signal. At the 10,250 strike, market IV $=$ 0.160, fitted $=$ 0.155. $\mathrm{BS}(S, 10250, T, r, 0.155) = 34$; market price 38 is overpriced by 4 SeaShells.
Action. Sell at 38, buy back near fitted 34. Expected edge $\approx$ 4 SeaShells per lot; repeated across strikes and timesteps, this compounds to the 100k+ Frankfurt Hedgehogs reported.

25.5 Archetype 5: cross-exchange / locational arbitrage

Observable

The product trades locally *and* externally (separate venue with fixed bid/ask quotes), but cross-venue transfer incurs import/export fees, transport, tariffs, or storage charges. The external price is observable only through a "conversion" mechanism with per-timestep capacity.

Diagnostic

If a product has both an order book and a conversion mechanism, compute the round-trip all-in cost. If it is sometimes negative, locational arbitrage exists.

Prosperity examples.

- **MAGNIFICENT_MACARONS** (Prosperity 3, Round 4): local-exchange tradable plus external conversion. Conversion limit 10/timestep, position limit 75. External price depended on sunlight, sugar, shipping, tariffs (Frankfurt Hedgehogs, 2025b).

Theory chapter

Chapter 14, locational-arbitrage section. Classic rule: if $p^{\text{bid}}_{\text{local}} > p^{\text{ask}}_{\text{external}} + \text{fees}$, sell local and buy external (or vice versa). Prosperity twists: limited conversion capacity and a hidden taker bot that fills above the visible book.

Strategy skeleton

```
ext_ask_cost = external_ask + import_tariff + transport
ext_bid_revenue = external_bid - export_tariff - transport
if best_bid_local > ext_ask_cost:
sell local, buy external (convert +10)
if best_ask_local < ext_bid_revenue:
buy local, sell external (convert -10)
# Hidden taker exploitation:
offer_price = int(ext_bid_revenue + 0.5)
```

```
place limit sell at offer_price for 10 units
if filled:  convert -10 to close position
```

The last three lines capture Frankfurt Hedgehogs' insight: a hidden taker fills sells at `int(externalBid + 0.5)` with ≈60% probability, ≈3 SeaShells edge per unit over naive best-bid (Frankfurt Hedgehogs, 2025b).

Pitfalls

Asymmetric conversion costs

Import/export fees are typically asymmetric (e.g. import 10, export 5), so arbitrage thresholds differ by direction. Compute all-in cost separately per leg.

Transport / conversion delay

Conversions reduce position but settle with a one-timestep lag. Converting 10 units consumes that capacity for the next timestep. Size quotes to conversion limit, not position limit.

Reference writeup

Frankfurt Hedgehogs' `CommodityTrader` handled MAGNIFICENT_MACARONS: limit sells at the hidden-taker price, convert surplus externally. Despite conservative quoting (10 units per timestep), the strategy generated ≈80,000–100,000 SeaShells per round. They noted quoting 20–30 units would have been more profitable since excess inventory still converts profitably (Frankfurt Hedgehogs, 2025b).

A similar hidden-taker structure appeared with **ORCHIDS** in Prosperity 2 (LinearUtility writeup[2]). Studying prior-year writeups was a direct competitive advantage.

Intuition

Prosperity locational arb is not pure textbook—it requires discovering hidden structure (the taker bot) empirically. Chapter 14 tells you *where* to look; competition-specific microstructure tells you *how much*.

MAGNIFICENT_MACARONS profit arithmetic

External bid (net of fees) nets 640; local best bid is 638. Naive arb says "no opportunity." But offering to sell locally at $\lfloor 640 + 0.5 \rfloor = 640$ and being filled by the hidden taker ≈60% of the time: sell at 640, convert at all-in cost 637, for 3 SeaShells edge per filled unit. With 10 units per timestep × 10,000 timesteps:

$$\text{Theoretical max PnL} = 300{,}000 \text{ SeaShells.}$$

At 60% fill, expected PnL ≈ 180,000. Frankfurt Hedgehogs realised ≈80,000–100,000 because edge was not always 3 and they quoted conservatively; quoting 20–30 units would have pushed PnL to ≈130,000–160,000.

[2]See `research/writeups/LinearUtility_P2/`.

25.6 Archetype 6: signal-driven / informed-flow exploitation

Observable

One or more bot traders exhibit predictable patterns tied to price extremes, time-of-day, or other detectable features. Prosperity 3's key case was **Olivia**: bought 15 lots at daily lows, sold 15 at daily highs across SQUID_INK and CROISSANTS. Detectable even before trader IDs appeared in Round 5.

Diagnostic

Filter trades by quantity (e.g. exactly 15 lots). If trades of that size cluster at daily extremes and align with subsequent reversals, you have an informed-flow signal.

Prosperity examples.

- **SQUID_INK** (Prosperity 3, Rounds 1–5): Olivia bought 15 lots at daily lows, sold 15 lots at daily highs.
- **CROISSANTS** (Prosperity 3, Rounds 2–5): same Olivia pattern, exploitable directly *and* as a cross-product signal for the baskets.

Theory chapter

Three chapters give the backdrop:

- Chapter 6: Glosten–Milgrom. Olivia is an *informed insider* whose trades reveal private information on price direction.
- Chapter 8: order-flow analysis, toxic-flow detection, VPIN-style metrics.
- Chapter 22: feature engineering on trade data (quantity, timing, direction relative to extremes).

Strategy skeleton

```
daily_low = running_min(mid_prices_today)
daily_high = running_max(mid_prices_today)
for each trade in recent_trades:
if trade.qty == 15 and trade.price <= daily_low:
signal = LONG # Olivia bought at low; price will rise
if trade.qty == 15 and trade.price >= daily_high:
signal = SHORT # Olivia sold at high; price will fall
position toward signal; flatten on new extreme or EOD
```

Once trader IDs appeared in Round 5, detection simplified to `if trade.buyer == "Olivia"`, removing false positives from other 15-lot trades (Frankfurt Hedgehogs, 2025b).

Pitfalls

Direction flips

A new daily extreme invalidates Olivia's signal: if she bought at the then-low but the price makes a new low, the original signal was a false positive. Monitor for new extrema and reset.

Detection latency before trader IDs

Without trader IDs (Rounds 1–4), detection relied on 15-lot quantity plus price-extreme matching. Other bots occasionally hit 15 lots at non-extreme prices, producing false positives—low per-event cost but lower Sharpe.

Reference writeup

Frankfurt Hedgehogs detected Olivia early and exploited the signal on SQUID_INK (≈8,000/round) and CROISSANTS (≈20,000/round). The CROISSANTS signal also fed the basket thresholds in Archetype 3 (Section 25.3), amplifying PnL (Frankfurt Hedgehogs, 2025b). Identifiable-bot exploits are not new: similar patterns appeared in Prosperity 1–2; teams studying prior writeups (Stanford Cardinal, Jasper) entered Prosperity 3 with detection code ready.

Detecting Olivia on SQUID_INK

Today's SQUID_INK mid range: low 1,820 (timestep 340), high 1,895 (timestep 680). At timestep 700 the tape shows a new trade:

Timestep	Price	Qty	Buyer	Seller
700	1,895	15	—	*(unknown)*

A 15-lot sell at the daily high. Without trader IDs, flag as a candidate Olivia sell on two criteria: quantity = 15 and price ≥ daily high. Prediction: SQUID_INK falls from here.
Action: go short to position limit. Hold until (a) a 15-lot buy at a new daily low (Olivia reversal), (b) a new daily high above 1,895 (invalidation), or (c) end of day.
Invalidation: if SQUID_INK hits 1,900 at timestep 750, the original sell was not the true maximum—flatten immediately.
This detection-and-invalidation logic kept the strategy robust: false positives closed quickly before losses compounded.

Intuition

Olivia is a simplified informed-trading model (Chapter 6). Real markets hide individual traders, but the *pattern*—unusual sizes at unusual prices preceding reversals—is exactly what VPIN and order-flow-toxicity metrics target. Prosperity makes it concrete.

25.7 Archetype-recognition flowchart

Run new products through the following decision table—the Prosperity-specific version of Chapter 23's framework.
The table orders archetypes by diagnostic speed. Fixed-price products are obvious in seconds; informed-flow signals take hours of trade-data study. Time budget per round:

1. **Minutes 0–5.** Test for fixed price (Archetype 1) or conversion mechanism (Archetype 5)—instant to spot, fast to code.

2. **Minutes 5–30.** Check for baskets/ETFs (Archetype 3); compute spread, test stationarity.

3. **Minutes 30–120.** For options (Archetype 4): IVs, smile fit, mean-reverting residuals.

Table 25.1: Archetype recognition: observable features → archetype → strategy.

Observable feature	Archetype	Primary strategy
Fixed known true price; wide spread	1 (Section 25.1)	Fixed-price market making
Moving price; tight spread; no trend	2 (Section 25.2)	Dynamic market making (Wall Mid)
Moving price; sharp jumps; 15-lot trades at extremes	6 (Section 25.6)	Informed-flow detection (Olivia)
Multiple products; known basket composition; mean-reverting spread	3 (Section 25.3)	Basket stat-arb, threshold entry
Options (vouchers) on an underlying; parabolic IV smile	4 (Section 25.4)	IV scalping + optional gamma scalp
Local + external exchange; conversion fees	5 (Section 25.5)	Locational arb + hidden-taker exploit

4. **Hours 2–6.** Remaining products: trade-data analysis for informed flow (Archetype 6), autocorrelation tests (Archetype 2), normalised order-book plots.

The five-minute rule

A product unclassifiable within five minutes is Archetype 2 (trending/EMA) or Archetype 6 (signal-driven). Default to dynamic MM while investigating—it extracts spread PnL without a directional commitment.

PnL breakdown by archetype

Table 25.2 lists Frankfurt Hedgehogs' approximate per-round PnL by archetype. The lesson: the simplest strategies (fixed-price MM, locational arb) delivered the most reliable returns; the most sophisticated (IV scalping) had the highest absolute PnL but more variance.

Table 25.2: Approximate PnL per round by archetype (Frankfurt Hedgehogs, Prosperity 3). Numbers are in SeaShells.

Archetype	Products	PnL / round
1: Fixed-price MM	RAINFOREST_RESIN	≈39,000
2: Dynamic MM	KELP	≈5,000
2/6: Trend + signal	SQUID_INK	≈8,000
3: Basket stat-arb	PICNIC_BASKET1/2 + CROISSANTS	≈60,000–80,000
4: Options IV scalp	VOLCANIC_ROCK_VOUCHERS	≈100,000–150,000
5: Locational arb	MAGNIFICENT_MACARONS	≈80,000–100,000
Total		**≈290,000–380,000**

Historical note

Archetypes recur across Prosperity editions. P1 (2023) introduced fixed-price MM (Amethysts), trend (Starfruit), basket arb (Gift Baskets), locational (Orchids). P2 (2024) repeated the same archetypes with renamed products. P3 (2025) added options (Volcanic Rock Vouchers) and foregrounded Olivia. Teams who studied prior-year writeups—Stanford Cardinal (P1), Jasper/LinearUtility (P2), Frankfurt Hedgehogs (P3)—entered each competition with detection code, parameter ranges, and strategy skeletons ready. Highest-Sharpe pre-competition activity available.

On a Goldman Sachs desk

New Strat on day one. A new Goldman Strat given an unfamiliar product does the same thing you do in Prosperity: classify. Flow product (desk market-makes)? Relative value (trade basis vs. hedge)? Vol product (scalp the smile)? Cross-venue (arb exchange vs. OTC)? The classification picks the model library, risk limits, and P&L buckets. Prosperity compresses this diagnostic into a 48-hour round.

25.8 Key facts to memorise

1. **Six archetypes** cover every Prosperity product to date: fixed-price MM, trending/EMA, basket stat-arb, options/IV scalping, locational arb, informed-flow exploitation.
2. **Fixed-price** (RAINFOREST_RESIN, AMETHYSTS): known constant price—quote around it. Simplest archetype; implement within 30 minutes of identifying.
3. **Dynamic** (KELP): random walks. Replace fixed price with Wall Mid; otherwise identical to Archetype 1.
4. **Baskets** (PICNIC_BASKET1/2): mean-revert to synthetic value *plus persistent premium*. Fixed thresholds (not z-scores), subtract premium, half-hedge if unsure.
5. **Options** (VOLCANIC_ROCK_VOUCHERS): relative value on the IV smile. Fit parabola, trade residuals, hedge at portfolio level. IV scalping was P3's single highest-PnL strategy (≈100k–150k/round).
6. **Locational arb** (MAGNIFICENT_MACARONS): textbook arb plus hidden taker-bot discovery. Similar structures recur; study prior writeups.
7. **Informed-flow** (Olivia on SQUID_INK, CROISSANTS): low-risk directional bets. Detect by quantity + price-extreme; use cross-product signals to bias basket thresholds.
8. **Take first, then make.** Sweep mispriced orders before placing passive quotes.
9. **Clip every order** against the position limit— exceeding it cancels *all* orders.
10. **Diagnose before coding.** Spend the first 30 minutes classifying products. Table 25.1 plus the time-allocation schedule give a checklist.
11. **Study prior-year writeups.** Every Frankfurt Hedgehogs P3 alpha source had a P1/P2 predecessor. Highest-Sharpe pre-competition activity.
12. **Robustness over peak backtest PnL.** Grid search for flat regions, keep parameter counts low. Unexplainable backtest performance is probably noise.

Chapter 26

Prosperity Manual Trading Challenges

Manual rounds test trading intuition without code: game theory, probability, and quick mental arithmetic rather than Python. Every Prosperity competition pairs each algorithmic round with a manual challenge solved by hand. Manual PnL is a modest slice of the total, but the skills transfer directly to a trading floor, where a junior trader who cannot do fast expected-value calculations in their head is a liability.

This chapter catalogues the five recurring archetypes from Prosperity 1–3, each treated with the mathematics it deserves:

1. FX triangular arbitrage (brute-force optimisation under time pressure).
2. Container auctions (expected value under uncertainty, winner's curse).
3. Reserve-price / valuation rounds (microstructure as game, bid–ask positioning).
4. Suitcase / Nash bargaining (cooperative game theory, surplus division).
5. News trading (Bayesian updating, position sizing under signal noise).

Prerequisites

- Chapter 14, §14.2: triangular FX arbitrage — computing round-trip products, trading when the product exceeds one.
- Chapter 20: the Kelly criterion and expected-value-based position sizing.
- Chapter 6: Glosten–Milgrom Bayesian updating of a market maker's belief about the true value of an asset.
- Basic probability: Bayes' theorem, expected value, conditional expectation. Chapter 4 covers these in the trading context.
- No game theory is assumed; this chapter introduces the required concepts (Nash equilibrium, Nash bargaining solution, winner's curse) from scratch.

26.1 FX triangular arbitrage

The simplest manual round is a currency conversion puzzle. Given an $n \times n$ conversion matrix $\mathbf{M}$ (entry M_{ij} is the units of currency j received per unit of i sold), find a sequence of at most k trades (typically $k = 5$) that maximises the final amount of your starting currency.

Mathematical framework

The matrix defines a directed weighted graph $G = (V, E, w)$: currencies are vertices, conversions are edges, weights are rates. A k-trade sequence is a walk of length k starting and ending at the source currency, with final amount equal to the edge-weight product:

$$\text{return} \;=\; \prod_{t=1}^{k} M_{c_{t-1},\,c_t} \tag{26.1}$$

with $c_0 = c_k$ the starting currency. Profitable iff return > 1. Find the length-$\leq k$ walk maximising the product.

Triangular arbitrage indicator

For a three-currency cycle $A \to B \to C \to A$, the profit per unit is

$$\Delta \;=\; M_{A,B} \cdot M_{B,C} \cdot M_{C,A} - 1.$$

If $\Delta > 0$, the cycle is profitable. This is the same triangular arbitrage indicator derived in Section 14.2.

Worked example: Prosperity 3, Round 1

Prosperity 3 Round 1 provided a 4×4 conversion matrix between four in-game currencies (Frankfurt Hedgehogs, 2025a):

	Snowballs	Pizza	Si. Nuggets	SeaShells
Snowballs	1.00	1.45	0.52	0.72
Pizza	0.70	1.00	0.31	0.48
Si. Nuggets	1.95	3.10	1.00	1.49
SeaShells	1.34	1.98	0.64	1.00

Starting from SeaShells with five trades, the Frankfurt Hedgehogs enumerated all paths. The optimum:

$$\texttt{SeaShells} \xrightarrow{1.34} \texttt{Snowballs} \xrightarrow{0.52} \texttt{Si. Nuggets} \xrightarrow{3.10} \texttt{Pizza} \xrightarrow{0.70} \texttt{Snowballs} \xrightarrow{0.72} \texttt{SeaShells}$$

Product of rates:

$$1.34 \times 0.52 \times 3.10 \times 0.70 \times 0.72 \;=\; \underbrace{1.34 \times 0.52}_{0.6968} \times \underbrace{3.10 \times 0.70}_{2.17} \times 0.72 \;\approx\; 1.089$$

yielding approximately 8.9% profit per round trip.

Connection to graph theory

The general problem—maximum-weight walk of length $\leq k$—is solved by Bellman–Ford in log-space: set $w'_{ij} = -\ln M_{ij}$, then the rate product equals $\exp(-\sum w'_{ij})$, so maximising the product is equivalent to finding a minimum-weight walk (a negative-total-weight walk is a profitable round trip). Real-world FX desks run this continuously on live quotes (Bouchaud et al., 2018).

For the competition, brute force suffices: $n = 4$, $k = 5$ gives $4^5 = 1{,}024$ walks—minutes by hand.

Manual skill: speed under time pressure

Under time pressure on a 4×4 matrix:

1. **Eliminate dominated currencies**—drop any row uniformly worse than another.
2. **Check all three-cycles first**: at most $\binom{4}{3} \times 2 = 8$ oriented three-cycles.
3. **Extend promising three-cycles to five-step paths.** If $A \to B \to C \to A$ has $\Delta > 0$, try splitting one leg into two intermediate conversions that beat the direct rate.
4. **Verify with full precision**—rounding in intermediate steps can flip the ranking of close candidates.

Sanity check by local swap. For the claimed optimum, check that no single-edge swap improves the product. This greedy search takes $O(k \cdot n) \approx 20$ multiplications and catches most errors.

Don't confuse rates and inverses

If $M_{A,B} = 1.45$, selling one A gives 1.45 B. The reverse direction uses $M_{B,A}$, which is *not* necessarily $1/1.45 = 0.6897$—the matrix may bake in bid–ask spreads. Always read the entry for the direction you trade; never invert.

26.2 Container auctions

The container-auction round presents n containers with known parameters but unknown competition. You pick one or two; profit depends on how many other teams chose the same. This is a coordination game: the goods are public, the crowd is not.

Mathematical framework

In the Prosperity 3 variant, container f has multiplier M_f and inhabitants I_f. The payout for picking f:

$$\Pi(f) = \frac{M_f \times 10{,}000}{(p_f \times N_{\text{teams}}) + I_f} \tag{26.2}$$

where $p_f \in [0, 1]$ is the fraction of teams picking f and N_{teams} is the total teams. Numerator: container value; denominator: total claimants (your team plus competitors plus inhabitants).

Winner's curse

In common-value auctions, the highest bidder wins precisely because they have the most optimistic estimate—or the least accurate model of the crowd.
In container auctions, the container that looks most profitable assuming $p_f = 0$ is exactly the one that attracts the crowd, driving p_f up and the payout down. The defence: model what others will do, not just what you would do alone.

Nash equilibrium concept

A **Nash equilibrium** is a strategy profile in which no player can improve by unilateral deviation. Here, a symmetric equilibrium is a distribution $(p_1^*, \ldots, p_n^*)$ such that every team is indifferent between all containers in the support, and no switch to an empty container helps.

Definition 26.1 (Symmetric Nash equilibrium in container games)**.** A symmetric equilibrium allocation satisfies, for every pair of containers f, g with $p_f^* > 0$ and $p_g^* > 0$:

$$\frac{M_f \times 10{,}000}{p_f^* \cdot N_{\text{teams}} + I_f} = \frac{M_g \times 10{,}000}{p_g^* \cdot N_{\text{teams}} + I_g}.$$

That is, the expected payout from any container in the support of the equilibrium is the same.

The exact Nash needs the payoff function and player count—both given in Prosperity. But the crowd does not play Nash: human biases (round-number attraction, anchoring on the highest multiplier, social-media coordination) cause systematic deviations. The edge lies in predicting the *deviation from Nash*, not in playing Nash yourself.

Expected-value analysis

If p_f were known, the problem would be trivial. The difficulty is that p_f is random from your perspective.

1. **Rank by empty-room value** $\Pi_0(f) = M_f \times 10{,}000/I_f$ (payout if no other team picks f).
2. **Estimate crowd distribution:** the crowd roughly sorts by $\Pi_0(f)$. Calibrate by analogy when prior-year data exists (Prosperity 3 Round 4 used Round 2 data).
3. **Compute expected payout** by plugging estimated p_f into Equation (26.2); pick the max.
4. **Hedge against model error.** Take the second pick (cost c) only if its expected payout exceeds c. The Frankfurt Hedgehogs declined in Round 2 (cost 50,000) because only two containers cleared the threshold with razor-thin margin (Frankfurt Hedgehogs, 2025a).

Worked example: Prosperity 3, Round 2

The Frankfurt Hedgehogs' predicted crowd percentages and actual outcomes for three selected containers (Frankfurt Hedgehogs, 2025a):

Multiplier	Predicted p_f	Actual p_f	Predicted rank	Actual rank
37	4.95%	5.12%	1	3
10	1.31%	0.94%	2	2
50	8.11%	8.52%	3	5

They avoided multiplier 37 (humans pick it as a "random" number disproportionately often) and chose 50 for $\approx$ 40,000 SeaShells. In hindsight, 10 or 37 would have earned 4,000–10,000 more (Frankfurt Hedgehogs, 2025a).

Anchoring on the obvious best container

The highest multiplier is almost never the best pick—it attracts the crowd and crushes per-team payout. The optimum is a contrarian container with low allocation relative to its multiplier, the same logic as value investing: the best risk-adjusted return is in the asset nobody wants.

26.3 Reserve price / valuation rounds

In a reserve-price round, you are a buyer facing sellers with private reserve prices (minima) drawn from a known distribution. You set one bid p; every seller with reserve below p trades. Goods resell at terminal value V. Profit balances volume against markup.

Single-player optimisation

With reserves Uniform$[a,b]$ the trading fraction at bid p is $(p-a)/(b-a)$. Expected profit over N sellers:

$$\Pi(p) \;=\; N \cdot \frac{p-a}{b-a} \cdot (V-p). \tag{26.3}$$

Concave quadratic in p; differentiate and zero:

$$\frac{d\Pi}{dp} = \frac{N}{b-a}[(V-p)-(p-a)] = \frac{N}{b-a}(V+a-2p) = 0$$
$$\implies \quad p^* = \frac{V+a}{2}.$$

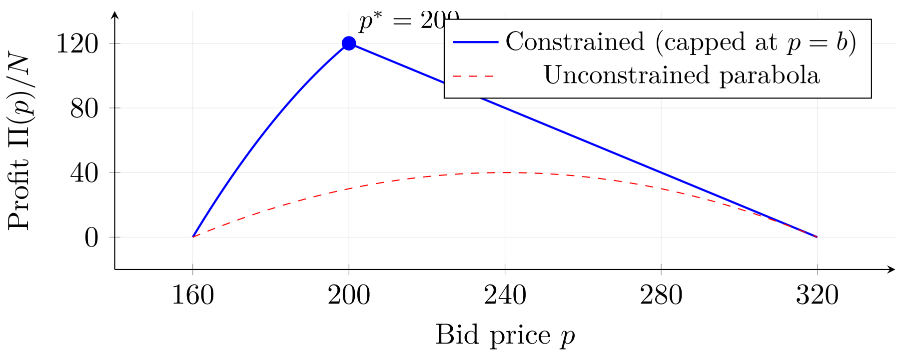
Optimal reserve-price bid

With reserves Uniform$[a,b]$ and terminal value $V \geq b$:

$$p^* = \frac{V+a}{2},$$

splitting the difference between resale value and lowest reserve. If $p^* > b$, all sellers trade and the effective optimum is $p = b$.

Profit is a downward parabola, zero at $p = a$ (no trades) and $p = V$ (no markup), peaking at the midpoint $(V+a)/2$:



Prosperity 3 numbers. Round 3, Part 1 (Frankfurt Hedgehogs, 2025a): $a = 160$, $b = 200$, $V = 320$, giving unconstrained $p^* = 240$. But since $240 > b = 200$, the seller fraction saturates at 1 for $p \geq 200$, after which $\Pi(p) = N(320-p)$ is decreasing. So compare: $\Pi(200) = 120N$ versus $\Pi(240) = 80N$. The constrained optimum is $p^* = 200$, matching the Hedgehogs' answer.

Game-theoretic extension

Part 2 added a competitive twist: reserves Uniform$[250, 320]$, and profit was scaled down if your bid fell below the population average μ:

$$\Pi(p,\mu) \;=\; N \cdot \frac{p-250}{70} \cdot (320-p) \cdot \min\left(\left(\frac{320-\mu}{320-p}\right)^3,\, 1\right). \tag{26.4}$$

The cubic penalty punishes bids below average severely, creating upward coordination pressure—every team wants to bid at or above the average, pushing the average up.

Intuition

This is adverse selection from Chapter 6 in disguise. A Glosten–Milgrom market maker who quotes too generously is picked off by informed traders; here, a team that bids below the crowd average is "picked off" by the penalty. Shade toward the expected counterparty action.

Equilibrium analysis. Without the penalty, the Part 2 optimum is $p^* = (320 + 250)/2 = 285$. At a symmetric Nash, every team bids the same μ^* so the penalty factor is 1, giving $\mu^* = 285$.

But the penalty is asymmetric: undershooting is punished cubically, overshooting is not punished at all. Teams that recognise this shade above 285, which raises μ and penalises those who did not. This ratchet pushes bids higher.

The Hedgehogs bid 303, anticipating overshoot. The actual average was 287—most teams did not model the twist—and they overpaid by about 5.5% versus the ex-post optimum (Frankfurt Hedgehogs, 2025a).

Over-correcting for the penalty

This is a beauty contest: you bid on what others will bid on what you will bid, *ad infinitum*. Sophisticated teams over-correct (303 when the average is 287); naive teams anchor at 285 and drag the average down. When in doubt, shade toward the single-player optimum, not toward infinity.

26.4 Suitcase / Nash bargaining

Some manual rounds are cooperative bargaining: parties divide a surplus, and the division depends on each side's outside option. Prosperity "suitcase" rounds look like the container format (Section 26.2) with a second pick at known cost, but the underlying structure is Nash bargaining.

Nash bargaining solution

Two players A, B bargain over a surplus S. If they agree, A gets x and B gets $S - x$; if not, they get disagreement payoffs d_A, d_B. Nash (1950) proved a unique division satisfies four axioms (Pareto efficiency, symmetry, IIA, invariance to affine utility transformations)—the one maximising the *Nash product*:

$$x^* = \arg \max_{x \in [d_A, S - d_B]} (x - d_A)(S - x - d_B). \tag{26.5}$$

Nash bargaining solution

For two players with disagreement payoffs d_A, d_B dividing a surplus S, the Nash bargaining solution is

$$x_A^* = d_A + \frac{S - d_A - d_B}{2}, \qquad x_B^* = d_B + \frac{S - d_A - d_B}{2}.$$

Each player gets their disagreement payoff plus half the net surplus $S - d_A - d_B$. Improving your outside option raises your negotiated outcome one-for-one.

Derivation. Expanding the Nash product:

$$f(x) = (x - d_A)(S - x - d_B).$$

This is a downward-opening parabola in x. Differentiating:

$$f'(x) = (S - x - d_B) - (x - d_A) = S - d_A - d_B - 2x.$$

Setting $f'(x) = 0$ gives $x^* = (S - d_A - d_B + 2d_A)/2 = d_A + (S - d_A - d_B)/2$, confirming the formula. □

Worked numerical example

Two traders bargain over a bonus pool $S = 100{,}000$ with disagreement payoffs $d_A = 20{,}000$ and $d_B = 30{,}000$, so the net surplus is 50,000. Applying the formula:

$$x_A^* = 20{,}000 + \frac{50{,}000}{2} = 20{,}000 + 25{,}000 = 45{,}000,$$
$$x_B^* = 30{,}000 + \frac{50{,}000}{2} = 30{,}000 + 25{,}000 = 55{,}000.$$

Player B's stronger outside option ($d_B > d_A$) captures more surplus. If A can improve to $d_A = 40{,}000$ (e.g. a competing offer), the net surplus shrinks to 30,000 and A's allocation rises to 55,000—a 10,000 gain from a 20,000 improvement in the outside option.

Application to Prosperity container/suitcase rounds

The Prosperity 3 Round 4 suitcase format mirrored Round 2's containers but with a second pick at cost 25,000 (Frankfurt Hedgehogs, 2025a). The Nash lens applies to the second-pick decision: disagreement payoff is the single-pick profit; the second-pick surplus must exceed its cost.

Second-pick decision rule. Take the second pick iff $E_2 > c$: with $E_2 = 40{,}000$ and $c = 25{,}000$, net value is $+15{,}000$; with $E_2 = 20{,}000$ it is $-5{,}000$. Since E_2 is an estimate, a close call near c is sensitive to error—decline when risk-averse.

Prosperity 3, Round 4 results. The Hedgehogs chose multipliers 37 and 50; their ranking was 60, 37, 50. Actual crowd allocations differed substantially—multiplier 37 drew 4.79% versus the predicted 2.09%—and their combined ≈ 85,000 SeaShells barely exceeded the average of 82,000 (Frankfurt Hedgehogs, 2025a):

Mult.	Predicted p_f	Actual p_f	Pred. rank	Actual rank
60	4.25%	6.72%	1	8
37	2.09%	4.79%	2	14
50	2.98%	3.92%	3	6

Intuition

In the suitcase round, bargaining power against the crowd comes entirely from predicting what others do. A poor crowd model means your "outside option" (naive-pick payout) dominates and sophistication adds no edge—the Hedgehogs' Round 4 predictions were off, and their profit barely exceeded average.

26.5 News trading

The news-trading round presents products with current prices plus news headlines hinting at future moves. You size positions; after the round, products settle at new prices and your profit is the mark-to-market (position size times the realised price move, i.e. the unrealised gain or loss valued at the final quote).

Bayesian updating framework

Model product i's true percentage move as θ_i; the headline is a noisy signal s_i. Prior $\theta_i \sim \mathcal{N}(\mu_0, \sigma_0^2)$ (often $\mu_0 = 0$ when agnostic). For additive Gaussian noise $s_i = \theta_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, the posterior is

$$\theta_i \mid s_i \;\sim\; \mathcal{N}\left(\frac{\sigma_\varepsilon^2\,\mu_0 + \sigma_0^2\,s_i}{\sigma_0^2 + \sigma_\varepsilon^2},\;\; \frac{\sigma_0^2\,\sigma_\varepsilon^2}{\sigma_0^2 + \sigma_\varepsilon^2}\right). \tag{26.6}$$

a precision-weighted average of prior and signal.

Bayesian posterior for Gaussian signal

Given prior $\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and signal $s = \theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$:

$$\mathbb{E}[\theta \mid s] \;=\; \underbrace{\frac{\sigma_\varepsilon^2}{\sigma_0^2 + \sigma_\varepsilon^2}}_{\text{weight on prior}}\;\mu_0 \;+\; \underbrace{\frac{\sigma_0^2}{\sigma_0^2 + \sigma_\varepsilon^2}}_{\text{weight on signal}}\;s.$$

As $\sigma_\varepsilon \to 0$ the posterior collapses to the signal; as $\sigma_\varepsilon \to \infty$ it stays at the prior. Same update rule as the Glosten–Milgrom dealer model (Chapter 6).

Worked Bayesian example

Product price 100, prior $\theta \sim \mathcal{N}(0, 0.20^2)$ (no drift, 20% volatility). Headline gives signal $s = -0.35$ with noise $\sigma_\varepsilon = 0.25$. Posterior mean:

$$\begin{aligned}
\mathbb{E}[\theta \mid s] &= \frac{\sigma_\varepsilon^2}{\sigma_0^2 + \sigma_\varepsilon^2} \cdot \mu_0 + \frac{\sigma_0^2}{\sigma_0^2 + \sigma_\varepsilon^2} \cdot s \\
&= \frac{0.0625}{0.04 + 0.0625} \cdot 0 + \frac{0.04}{0.04 + 0.0625} \cdot (-0.35) \\
&= 0 + \frac{0.04}{0.1025} \cdot (-0.35) \\
&= 0.390 \times (-0.35) \;=\; -0.137.
\end{aligned}$$

The posterior standard deviation is:

$$\hat{\sigma} = \sqrt{\frac{0.04 \times 0.0625}{0.1025}} = \sqrt{0.02439} \approx 0.156.$$

Expect a -13.7% move with 15.6% uncertainty. The signal pulled your view from 0% toward -35% but only partway, because the signal is noisy—a short with moderate conviction.

Position sizing

Given posterior mean $\hat{\theta}_i = \mathbb{E}[\theta_i \mid s_i]$ and variance $\hat{\sigma}_i^2$, Kelly (Chapter 20) sizes the bet. With price P_i and lot limit L_i, a simplified Kelly-style allocation is:

$$q_i \;=\; L_i \cdot \frac{\hat{\theta}_i}{\hat{\theta}_i^2 + \hat{\sigma}_i^2} \tag{26.7}$$

with $q_i > 0$ buy, $q_i < 0$ sell: bet proportionally to edge $\hat{\theta}_i$, scale down when $\hat{\sigma}_i^2$ is large. In practice the position limits are small and direction matters more than exact size.

Worked example: Prosperity 3, Round 5

Round 5 presented nine products with simulated news (Frankfurt Hedgehogs, 2025a). The Hedgehogs estimated directional moves and sized positions conservatively. In the table below, **Profit** is what they earned given their actual bets, while **Optimal** is what a perfect oracle knowing the actual move would have earned at the position limit—the gap measures forecasting error, not sizing mistakes:

Product	Est. move	Actual move	Profit	Optimal
Haystacks	+12%	−0.48%	−3,240	0
Ranch Sauce	+10%	−0.72%	−2,208	0
Cacti Needle	−30%	−41.2%	32,160	35,360
Solar Panels	−30%	−8.90%	−6,600	1,640
Red Flags	+5%	+50.9%	9,700	53,970
VR Monocle	+30%	+22.4%	9,600	10,440
Quantum Coffee	−50%	−66.8%	87,339	92,932
Moonshine	0%	+3.0%	0	180
Striped Shirts	0%	+0.21%	0	0
Total			**126,751**	**194,522**

Biggest misses: Red Flags (+5% vs. +50.9%) and Solar Panels (direction right, magnitude off). Total profit captured 65% of the theoretical optimum.

Overconfidence in signal precision

The Hedgehogs anchored on Prosperity 1 and 2 analogies (e.g. assuming `Haystacks` behaved like a prior-year product). This is prior anchoring: when the prior is strong and the signal ambiguous, the update barely moves—but if the prior is wrong, so is the posterior. Defence: treat cross-year analogies as weak priors (large σ_0) and let the signal dominate.

26.6 Cross-cutting themes

Several threads run through all five manual-round archetypes.

EV is necessary but not sufficient. Every round reduces to an EV calculation, but its inputs—crowd behaviour, signal precision, distributions—are uncertain. The skill is mental sensitivity analysis: how wrong can my inputs be before the bet is −EV?

Game theory enters whenever the crowd matters. FX is pure optimisation (rates are given); containers and reserve-price are games. Diagnostic: if the payoff contains a term indexed by others' actions, single-agent optimisation will mislead you.

Bayesian updating is the meta-skill. News trading uses it explicitly; the container round uses it implicitly (updating crowd-behaviour models from Round 2 to Round 4 data). On a trading floor it is the informal reflex triggered by every new piece of information.

Prosperity example

The Hedgehogs' manual strategy reveals a consistent philosophy: **hedge against model error rather than maximise expected profit** (Frankfurt Hedgehogs, 2025a).
Round 1 (FX): brute-forced the optimum (rates deterministic, no hedge needed). Round 2 (containers): avoided the psychologically salient multiplier 37 for a middle-of-the-road pick. Round 3 (reserve price): over-bid to avoid the cubic penalty, accepting a small loss. Round 4 (suitcases): spread risk across two picks. Round 5 (news): deliberately undersized positions.
Total manual PnL $\approx$ 252,000 SeaShells across five rounds—well below the theoretical optimum, but with low variance and no catastrophic round. The lesson: *when manual PnL is small relative to algorithmic, play for safe EV, not maximum EV.*

On a Goldman Sachs desk

A Goldman trading desk tests the same skills: quick mental arithmetic, Bayesian updating under time pressure, game-theoretic positioning against counterparties.
When a client requests a price on a block of bonds, the trader has seconds to: (i) estimate fair value from the screen (FX round-trip), (ii) assess client information content (Glosten–Milgrom, Chapter 6), (iii) set bid–ask edge (reserve-price problem), and (iv) update the market view given *that client* calling *now* (Bayesian news update).
A placement student who can explain the connection—"the reserve-price round is client quoting with the cubic penalty playing the role of adverse selection"—shows exactly the structured thinking a desk head values.

26.7 Key facts to memorise

1. **FX triangular arbitrage:** compute the product of rates around a cycle. If > 1, the cycle is profitable. Enumerate all walks of length $\leq k$ to find the optimum.

2. **Winner's curse:** in any auction or selection problem, the "winner" is the bidder with the most optimistic estimate. Correct for this by modelling what others will do, not just what you would do alone.

3. **Reserve-price optimum:** with uniform reserves on $[a, b]$ and resale value V, the optimal bid is $p^* = (V + a)/2$, constrained to $[a, V]$.

4. **Game-theoretic penalty:** when your payout depends on the crowd's average action, shade toward the crowd rather than toward the single-player optimum. Over-correction is possible; the crowd is often less sophisticated than you fear.

5. **Nash bargaining solution:** each party gets their disagreement payoff plus half the net surplus. Improving your outside option improves your negotiated outcome one-for-one.

6. **Bayesian updating for news:** posterior mean = precision-weighted average of prior and signal. Noisy signals should move your view less.

7. **Position sizing under uncertainty:** Kelly-style sizing says bet proportionally to expected edge, inversely to variance. When uncertain, undersize.

8. **Manual-round meta-strategy:** when manual PnL is small relative to algorithmic PnL, play for safe expected value, not maximum expected value. Avoid catastrophic outcomes.

Chapter 27

Lessons from the Prosperity Top Teams

This chapter distils patterns from 10+ public writeups across Prosperity 1–3. Rather than re-deriving strategies (Chapter 25), we focus on the *meta-game*: the operational, analytical, and psychological patterns that separate top-10 finishers from teams with similar ideas who finish hundreds of places lower.

Prerequisites

- Chapter 24: competition mechanics, the Python trading API, product archetypes.
- Chapter 25: strategy–theory mapping for each algorithmic round (market making, stat arb, options, location arbitrage, informed-trader exploitation).
- Chapter 21: overfitting traps, walk-forward validation, and parameter-plateau selection.

27.1 What every top team did the same

Across every top-10 writeup we examined—Frankfurt Hedgehogs (2nd, P3), CMU Physics (7th, P3), Linear Utility (2nd, P2), Stanford Cardinal (P1), Alpha Animals (9th, P3), jmerle (9th P2, 25th P3), and others—five patterns recur reliably as necessary (though not sufficient) conditions for a top finish.

Prosperity 3 had over 12,000 teams and a heavy-tailed PnL distribution: the top 10 earn 5–10× the median, and 1st to 100th can span 1M SeaShells. Small edges compound over five rounds and 50,000 timesteps.

Tooling from day one

Every top-10 team built or installed core tooling *before the competition started.* Three mandatory components:

1. **Backtester.** A forked jmerle or custom build. Linear Utility's P2 backtester replicated the matching engine and output logs in the competition website's exact format, enabling direct backtest-vs-live comparison. Frankfurt Hedgehogs used forked jmerle for systematic parameter sweeps and the official website for validating bot-interaction behaviour the local tool could not replicate.
2. **Visualiser / dashboard.** Frankfurt Hedgehogs' custom dashboard showed order-book levels, trade markers (maker/taker/own fills), PnL, position, and a normalisation feature

displaying prices relative to a chosen indicator (e.g. the "Wall Mid"). CMU Physics used Jasper's community visualiser, but in Round 3 it pushed their AWS Lambda over the 100 MB limit, crashing the algorithm and wiping rolling-window state. They removed it from subsequent submissions.

3. **Analysis notebooks.** Every top team ran EDA in Jupyter before writing strategy code. Frankfurt Hedgehogs prototyped via quick vectorised backtests in notebooks, graduating promising ideas to the full simulation. Linear Utility ran parameter grid searches, optimising systematically rather than by hand.

Tooling priority

Build backtester + visualiser + analysis notebooks *before* the competition, using prior-year data. Frankfurt Hedgehogs credit weeks of pre-competition prep as a primary driver of their top finishes across P2 and P3.

Study prior years exhaustively

Top teams treat prior-year writeups as primary source material. Frankfurt Hedgehogs discovered Olivia's daily-extrema pattern on Squid Ink—one of P3's highest-value signals—because they had studied Stanford Cardinal (P1) and Jasper (P2), both documenting similar informed-bot behaviour. They state: "Anyone who carefully analyzed historical Prosperity 2 data or public write-ups . . . could have anticipated similar behaviors and prepared detection logic in advance."

CMU Physics did the same for Round 4 (locational arbitrage): their Macaron strategy resembled P2 Round 2, and Chris Roberts had placed 3rd on that round, so they re-implemented with high confidence.

Linear Utility's P2 writeup itself became a key source for the Macaron hidden-taker exploit, which Frankfurt Hedgehogs note had been "discussed already" by Jasper and Linear Utility.

The lesson is not just "read writeups": *identify recurring product archetypes*, build skeletons for each before the competition, and slot new products into the matching skeleton as sample data arrives.

Archetype recycling across editions

Archetypes appearing in every Prosperity edition (P1–P3):

- A **fixed-value** product for tutorial market making (Amethysts → Rainforest Resin).
- A **slow random walk** product for drifting-value market making (Starfruit → Kelp).
- A **basket / ETF** product for statistical arbitrage (Gift Basket → Picnic Baskets).
- A **derivative / option** product (Coconut options → Volcanic Rock Vouchers).
- A **cross-exchange conversion** product (Orchids → Macarons).
- An **informed bot** trading at daily extrema (unnamed P1 bot → Olivia).

Teams with skeleton code per archetype wrote P3 strategy logic within hours of data release; teams discovering patterns from scratch spent days on analysis that could have been done in advance.

Simplicity over sophistication

Winning strategies are almost never complex ML pipelines. They are overwhelmingly **fair-value estimate + threshold + inventory management**.

Frankfurt Hedgehogs' Rainforest Resin strategy (their most consistent PnL contributor, ∼39,000 SeaShells/round) had three rules: (1) take any order crossing fair value 10,000; (2) place passive quotes inside the best bid/ask; (3) flatten inventory at fair value when skewed. "No sophisticated logic or aggressiveness was needed."

Their basket stat-arb used a *fixed threshold*: long below $-T$, short above $+T$. No MA crossovers, z-scores, or vol scaling. They warn: "if you can't explain why a strategy should work from first principles, then any 'outperformance' in historical data is probably noise."

CMU Physics' basket strategy used z-scores, but with a hard-coded historical mean and short rolling-window standard deviation—still simpler than textbook implementations. Their Round 5 "YOLO Croissants" was simpler still: detect Olivia's signal, go max long Croissants via baskets as leverage.

Complexity as a trap

Frankfurt Hedgehogs tried a logistic regression on Macarons using sunlight and sugar-price features. Despite significant p-values and ∼25k/day in backtests, they abandoned it: low confidence out-of-sample, serialisation overhead slowed the trader, and it clashed with their conversion-arbitrage logic. The simple arbitrage they deployed instead made 80,000–100,000 SeaShells with minimal risk. Takeaway: a strategy you can implement correctly under time pressure beats a theoretically superior one you cannot trust.

Iterate, do not over-engineer

Top teams submit early and learn from live results rather than perfecting strategies in simulation.

CMU Physics lived this. After Round 3 they dropped from 7th to 241st—two bugs: Jasper's visualiser exceeding memory (crashing Lambda, wiping rolling windows) and a quadratic IV fit that stopped tracking true IV on submission day. They iterated: removed the visualiser, switched to a short rolling window of mid IV (boosting backtested PnL from 80k to 200k/day), and in Round 4 posted the highest single-round algo PnL of *all* 12,000+ teams—447,251 SeaShells—climbing back to 8th.

Frankfurt Hedgehogs re-optimised all parameters after every round, treating each as a new calibration opportunity.

The iteration principle

Submit a working strategy on day 1 of each round. Use days 2–3 for live-data analysis, re-calibration, and bug fixes. A live submission's information is worth more than another day of backtesting.

Risk-aware position management

Top teams compute worst-case losses before deploying. CMU Physics estimated unhedged Volcanic Rock Vouchers could cost 40,000 SeaShells/step worst case, but average delta was closer to 160 units (not 400), giving a realistic worst case ∼16,000—less than the 40,000 saved by avoiding hedging spread costs. They took the risk. Frankfurt Hedgehogs computed a 95% VaR of ∼50,000 for their mean-reversion leg (25% of their 200,000 lead) and kept the position as a relative hedge.

The pattern: quantify downside before accepting it, and size positions so the worst plausible outcome does not eliminate you.

Frankfurt Hedgehogs pushed this into *relative* risk management in the final round. With a 200,000 lead, they could have dropped mean-reversion entirely. Instead, reasoning that close competitors were trading mean reversion aggressively—so if the market mean-reverted, those teams would gain while Frankfurt stood still—they kept a moderate position, hedging *ranking* risk rather than chasing EV. VaR of 50,000 was 25% of their lead: tournament-aware risk management.

Intuition

Risk management in a Prosperity round is a two-player game: you versus the leaderboard. Behind, you need variance to catch up; ahead, match competitors' strategies so bad luck hurts them equally. Same logic as index-fund relative-performance mandates: underperforming the benchmark is worse than absolute losses.

27.2 What differentiated the top 10

Among teams with solid tooling, prior-year study, and simple strategies, the following differentiators separated top 10 from top 100.

Bot-pattern exploitation

The highest-value P3 signal was detecting the bot Olivia, who bought at daily lows and sold at daily highs on multiple products (Squid Ink, Croissants, others). Frankfurt Hedgehogs identified her *before* trader IDs were revealed in Round 5 by filtering for quantity-15 trades and observing their placement at daily extrema. They then used her inferred position to bias basket spread thresholds—cross-product integration that compounded the edge.

CMU Physics independently spotted "suspicious trade quantity 15 patterns at highs/lows for Squid Ink and Croissants" in Round 2 but judged it too late to productise. In Round 5 with trader IDs visible, they went all-in on Croissants in Olivia's first-trade direction.

Detection methodology: track daily running min/max price. A trade at a daily extreme in the expected direction (buy at min, sell at max) is a potential Olivia signal. New extrema invalidate earlier signals—a new low after a supposed buy kills the prior flag.

Pre-Round 5, the only tell was quantity: Olivia traded 15 lots. Frankfurt Hedgehogs' dashboard had a quantity filter that made quantity-15 clustering at extrema visually obvious—purpose-built tooling creating analytical leverage the raw trade log hid.

Olivia detection: the P3 meta-signal

In P3, Olivia bought 15 lots at daily lows and sold 15 at daily highs on at least three products. Detection was worth 8,000+ SeaShells/round on Squid Ink (Frankfurt Hedgehogs' estimate), 20,000+/round on Croissants, plus indirect basket-threshold bias. Missing Olivia cost ∼30,000–50,000 per round.

Cross-product data reuse

Frankfurt Hedgehogs used Olivia's inferred Croissants position to bias basket stat-arb thresholds. Croissants were ∼50% of both baskets' value, so Olivia's Croissants direction predicted the basket spread's drift. They shifted a ± 50 threshold asymmetrically: if Olivia was short, long-entry moved to -80 and short-entry to $+20$.

CMU Physics went further in Round 5. Reasoning that trading baskets in Olivia's Croissants direction strictly dominated stat-arb, they went long both baskets to hold 1,050 Croissants (vs.

the 250 direct limit), partially hedging Jams/Djembes exposure. This "YOLO" made up to 120,000 on its best day.

The risk calculus was principled, not reckless. A bad day's Croissants high–low range was $\sim$40 SeaShells, giving a PnL lower bound of $40 \times 1{,}050 = 42{,}000$. Basket-premium risk had a 300-SeaShell/basket worst case, 45,000 total—but premiums were stationary (95%/90% confidence for Baskets 1/2), making the extreme move near-impossible. Expected premium loss ≈ 0; expected Olivia gain large and positive. "First-principles justification" applied to position sizing.

Parameter plateau selection

Frankfurt Hedgehogs visualised backtested PnL across a 2-D parameter grid for their basket strategy. Rather than the maximiser, they picked parameters from "consistent, flat regions of good performance, reducing sensitivity to slight shifts and avoiding overfit disasters." This is the plateau approach of Chapter 21: centre of a plateau, not peak of a spike.

CMU Physics saw the flip side on Volcanic Rock z-score trading. A colleague's strategy produced 150k/day in backtests, but "small tweaks … would lead to wildly different backtesting results, some even heavily negative." They declined to deploy, correctly applying the plateau principle at the cost of potential PnL.

Options: the IV scalping vs. mean-reversion debate

Volcanic Rock Vouchers (Round 3) produced the largest PnL dispersion among top teams. The choices here show theory plus empirical discipline.

Frankfurt Hedgehogs ran two alpha sources. Primary: *IV scalping*—fit a parabolic smile to observed IVs across strikes, price via Black–Scholes, trade deviations. (In plain English: implied volatility is what option prices *imply* the market thinks future volatility will be; scalping means trading short, mean-reverting deviations from a smooth fitted curve.) They validated by testing for 1-lag negative autocorrelation in option returns (short-term IV mean-reversion, necessary for scalping). Secondary: lightweight EMA mean-reversion on underlying Volcanic Rock with fixed thresholds.

Results: IV scalping delivered a stable 100,000–150,000/round; mean reversion gave +100,000, −50,000, −10,000 across three rounds. The hybrid protected against missing a mean-reversion regime while anchoring PnL to the reliable scalping edge.

CMU Physics used the same quadratic smile but as *sole* fair value. When the fit diverged from market IV on submission day, they entered large directionals and held—only 75,755 SeaShells versus 200,000+ for the top. Post-round: a short rolling window of mid IV would have tracked the market far better, boosting backtested PnL from 80k to 200k/day.

General principle: *adaptive* models (rolling windows, EMAs) beat *static* fitted curves when the DGP (data-generating process—the underlying statistical law producing the observed prices) can shift between sample and evaluation.

The "Wall Mid" and fair-value estimation

A subtler differentiator: fair-value estimation. Most teams used the raw mid of best bid/ask. Frankfurt Hedgehogs used the "Wall Mid"—the average of the bid-wall and ask-wall prices, order-book levels with consistently deep market-maker-bot liquidity.

The raw mid is distorted by aggressive takers overbidding or undercutting. The Wall Mid filters this by anchoring to the deepest levels, corresponding to bots that "appeared to know the true price and simply quoted rounded versions of it (± 2 ticks)."

They validated empirically: buying or selling one lot on the official site, PnL is computed against the true internal price. Wall Mid tracked it to sub-tick accuracy; raw mid drifted by several ticks in volatile periods.

This mattered most for Kelp (slow random walk). Quoting around the Wall Mid instead of raw mid reduced adverse-selection losses on passive orders. Frankfurt Hedgehogs estimate Kelp generated ∼5,000 SeaShells/round—modest in absolute terms, but entirely from fair-value precision.

Edge-case handling

Several technical details separated top implementations from merely correct ones:

- **Position limits.** CMU Physics found fully hedging both Picnic Baskets simultaneously was impossible under constituent limits. Solution: trade the *difference* in basket premiums ($P_1 - P_2$), using 100% of Basket 1's limit and 60% of Basket 2's, then z-score the remainder and market-make with the last 8%.
- **Integer rounding.** Frankfurt Hedgehogs' Macaron sell orders priced at `int(externalBid + 0.5)`—the max price that still triggered fills from a hidden taker bot executing ∼60% of eligible trades. Wrong by 1 SeaShell meant never filling (too high) or leaving edge (too low).
- **Memory and serialisation.** CMU Physics' Round 3 crash was Jasper's visualiser exceeding the 100 MB Lambda limit. Every byte of inter-timestep state matters. Frankfurt Hedgehogs stayed deliberately low-memory, storing only minimal rolling stats per product.
- **Conversion limits.** Macarons had a 10-unit conversion limit per timestep. Frankfurt Hedgehogs initially quoted 10 units/step, capturing ∼60% of max profit. Quoting 20–30 would have converted surplus inventory and pushed toward the optimal 130,000–160,000. CMU Physics independently found adjusting from 10 to 30 Macarons/step nearly doubled volume (56k → 95k units).
- **Hedging cost analysis.** CMU Physics tallied Volcanic Rock hedging trades and found 40,000+ SeaShells of spread cost. Going unhedged (worst case ∼16,000) increased back-tested PnL from 80k to 250k/day.
- **Storage fees and holding costs.** Macarons had a 0.1 SeaShell/step/unit storage fee. Over 10,000 steps, holding the 75-unit limit would cost $75 \times 0.1 \times 10{,}000 = 75{,}000$ SeaShells—enough to wipe out most arbitrage profits. CMU Physics sold locally and converted immediately, minimising holding. Teams accumulating inventory for better prices were eaten alive by storage.

Common patterns in top-10 edge-case handling

The top teams share three engineering habits:

1. **Compute all costs explicitly.** Spreads, storage, conversion fees, tariffs are often larger than the gross edge. Write them as formulas before coding.
2. **Test against the memory limit.** The ∼100 MB Lambda limit is binding. Verbose logging, large rolling windows, and visualiser libraries can silently push you over.
3. **Use traderData serialisation defensively.** Cross-step state lives in `traderData`, but serialisation has overhead: test round-trip time and size before adding state.

Manual-round preparation

Top teams approached manuals systematically, not intuitively. Both Frankfurt Hedgehogs and CMU Physics modelled game-theoretic manuals (Container, Suitcase) by computing Nash equilibria then adjusting for behavioural priors. CMU Physics built a probability distribution over player types (Nash-followers, random, nice-number biased) and Monte Carlo'd over it; Frankfurt Hedgehogs mapped P2 allocation patterns onto P3 containers.

For the Round 5 news-trading manual, Frankfurt Hedgehogs calibrated expected price-move magnitudes from the identical P1/P2 manuals— prior-year study again.

Key lesson: *real players do not play Nash.* Both teams found their Nash calculations overestimated broad-population rationality. CMU Physics modelled five Container types: Nash-followers (15%), random (50%), nice-number-biased (20%), naive-EV (10%), flawed-MC (5%). Frankfurt Hedgehogs used P2→P3 mapping for over/under-selection. After Round 2 results, CMU revised priors upward on Nash-following (∼50–60%) and confirmed nice-number bias (37, 73). This is Bayesian updating: prior-year outcome as prior, current round as likelihood.

Further: Frankfurt Hedgehogs noted that opening a second Round 2 container (50,000 SeaShells) exceeded the Nash EV (expected value—the probability-weighted average payoff) of any single container, making the second pick negative-EV. CMU Physics reached the same conclusion by Monte Carlo. Teams who opened two anyway—"more picks = more diversification"—lost money at the margin. Quantitative reasoning overrode intuition.

27.3 The hardcoding exploit

In P3 Rounds 1–2, bot behaviour replicated earlier competitions with 95%+ accuracy. True prices were regenerated (unlike P2, which reused some price paths), but bot *actions*—exact timesteps, sizes, and directions of market orders—matched historical data almost perfectly.

What teams actually did

Teams that noticed could "hardcode" algorithms to frontrun predictable bot actions. E.g. if a bot historically bought Rainforest Resin at timestep 600, pre-buy all ask liquidity just before and sell to the bot higher. Some teams made millions of SeaShells this way in Rounds 1–2.

Frankfurt Hedgehogs found this early and frontrun, citing the unaddressed P2 precedent (Linear Utility's writeup). They added an automatic fallback: revert to the non-hardcoded strategy if bot behaviour diverged from history.

CMU Physics learned the exploit the hard way. After Round 1 they were 771st. When IMC re-ran after ruling hardcoding as cheating, they jumped to 9th—a 762-place swing.

When hardcoding is rational and when it is a trap

The hardcoding temptation

Hardcoding tempts because it gives huge backtests with zero parameter risk—the "strategy" is a lookup table. It is a trap for three reasons:

1. **Fragility.** If IMC changes bots (as after P3 Round 2), hardcoded strategies produce zero PnL. All-in-on-hardcoding teams had no fallback.
2. **Rule risk.** IMC banned hardcoding and required re-submissions; caught teams dropped precipitously.
3. **Opportunity cost.** Reverse-engineering bot timesteps is time not spent on robust strategies that compound over five rounds.

The rational response (Frankfurt Hedgehogs) was to build hardcoding *with a fallback*: exploit while it works, revert seamlessly when it breaks. Same principle as production systems: no single data source without a degraded-mode alternative.

After Round 2, Frankfurt Hedgehogs reported the exploit to IMC even though fixing it cost their advantage. IMC fixed bots for Rounds 3–5 and allowed re-submissions. Meta-lesson: in competitions and industry, exploits based on implementation bugs rather than genuine structure get patched.

The P2 precedent and the ethics of exploits

Hardcoding was not new to P3. In P2, Linear Utility documented that some price paths were directly reused, making *prices* (not just bot behaviour) predictable. Their writeup became a key P3 reference: Frankfurt Hedgehogs hardcoded in P3 precisely because "a similar situation had gone unaddressed in P2."

Frankfurt Hedgehogs' decision to report the exploit reflects a principle: exploits riding bugs rather than genuine structure are fragile, reputation-damaging, and value-destroying. In a competition, IMC patches and your edge vanishes; in industry, the counterparty figures it out, the regulator investigates, or the venue changes its matching rules. Sustainable edge comes from market structure, not implementation accidents.

Lesson for P4: assume IMC makes bot behaviour harder to predict from prior-year data, but also assume *some* predictable bot behaviour persists—the competition needs bots to provide liquidity and create opportunities; fully random bots would kill the alpha. The skill is distinguishing structural patterns (which persist) from implementation artefacts (which get patched).

Frankfurt Hedgehogs even tried reverse-engineering the random seed, brute-forcing all 4 billion NumPy seeds on a Raspberry Pi for 24 hours against the first ∼100 PnL returns. No match, but it shows the creative lengths top teams go to.

Historical note

Prosperity 1 launched in 2023 with Amethysts (fixed-value MM) and Starfruit (random walk). By P3 (2025), participation hit 12,000+ worldwide. Format evolved: P2 added conversion (Orchids) and ETF baskets; P3 added options (Volcanic Rock Vouchers), cross-island arb (Macarons), and the controversial hardcoding exploit. IMC uses Prosperity as a recruiting pipeline—top finishers get interview invitations. P3 rules excluded prior top-10 teams from prize money (Frankfurt Hedgehogs still competed for ranking).

27.4 What to watch for in P4

Based on P1–P3 archetype evolution, some product-round patterns are predictable, others deliberately novel.

Expected archetype mapping

The product–archetype mapping across all three completed editions:

Archetype	**P1 (2023)**	**P2 (2024)**	**P3 (2025)**
Fixed-value MM	Amethysts	Amethysts	Rainforest Resin
Drifting-value MM	Starfruit	Starfruit	Kelp
Volatile / chaotic	—	—	Squid Ink
Basket / ETF arb	—	Gift Basket	Picnic Baskets 1&2
Options / derivatives	—	Coconut options	Volcanic Rock Vouchers
Cross-exchange arb	—	Orchids	Macarons
Informed bot	(unnamed)	(unnamed)	Olivia

Each edition adds complexity to existing archetypes (one basket → two; one option → five strikes) plus one genuinely novel element (Squid Ink's volatility in P3, Orchids' sunlight/humidity in P2).

Concrete P4 expectations:

- **Round 1: flat-value market making.** Every edition opened with a near-constant-value tutorial product (Amethysts, Rainforest Resin).
- **Rounds 2–3: basket stat-arb.** Gift Basket (P2), two Picnic Baskets (P3). P4 may add more baskets or time-varying compositions.
- **Rounds 3–4: derivatives.** Coconut options (P2), Volcanic Rock Vouchers across five strikes (P3). Puts or American-style plausibly next.
- **Round 4–5: cross-exchange / conversion arbitrage.** Orchids (P2), Macarons (P3). Conversion mechanics grow more layered each year.
- **All rounds: an informed trader.** Every edition has had at least one predictive bot trading daily extrema (Olivia in P3). IMC may change the pattern; some form persists.

P4 preparation checklist

1. Build or fork a backtester and verify it works on P3 sample data before the competition starts.
2. Build or install a visualisation dashboard with order-book display, PnL tracking, and price normalisation.
3. Read all available P1–P3 writeups (Frankfurt Hedgehogs, CMU Physics, Linear Utility, Stanford Cardinal, jmerle, Alpha Animals at minimum).
4. Prepare skeleton strategies for each archetype: fixed-value market making, drifting-value market making, basket stat-arb, option pricing / IV scalping, conversion arbitrage, and informed-bot detection.
5. Join the Discord server on day one—moderator hints and community analysis are decisive information sources.
6. Set up `traderData` serialisation and test it under the 100 MB Lambda memory limit.
7. Submit a working (even if basic) algorithm within hours of each round opening.
8. After each round, re-optimise parameters on the latest data, choosing from plateau regions rather than peaks.

9. Analyse all bot trades by quantity, timing, and direction before trader IDs are revealed.
10. Designate team roles: one person owns manual rounds, one owns tooling, and at least one focuses on new-product analysis.

What IMC might change

P4 uses XIRECs and a deep-space theme, breaking from the island/seashell motif. Prior top-10 teams are now excluded from prize rankings (still compete for ranking), suggesting IMC is levelling for returning teams' structural advantage. The hardcoding exploit is patched, and bots will likely vary more across rounds. Novel mechanics (futures with margin, multi-leg spreads, path-dependent payoffs) are plausible given participant sophistication.

Discord as an information source

Frankfurt Hedgehogs call the Discord server "extremely valuable" for moderator tips and clarifications, but warn it is "full of noise" from inexperienced participants plus occasional psychological warfare: fake backtesting screenshots designed to intimidate competitors into impulsive changes.

Recommended: monitor Discord for moderator hints and API/rule clarifications; never share your strategy; never pivot on unverified claims. "Stick to your own validation methods, and avoid the trap of overfitting or changing your approach impulsively just because of noise on Discord."

Discord psychological warfare

In P3, Discord posts showed extraordinary backtest numbers—some from massively overfit strategies that would collapse live, some outright fabrications. If a competitor claims 500,000+ on one product, do not panic; verify your own strategy against your own data. Teams that pivoted on Discord noise consistently underperformed their own backtests.

The "Wall Mid" fair-value concept

Frankfurt Hedgehogs' "Wall Mid" is the average of the bid-wall and ask-wall prices—levels with consistently deep designated-MM liquidity, typically corresponding to the true internal price ± 2 ticks. Averaging them gives a more stable, accurate fair value than the raw mid (which is distorted by aggressive takers).

They verified by buying or selling one lot on the official site and reading PnL vs. the true internal price. This ground-truth-checking methodology is worth replicating for any Prosperity product with an official backtesting environment.

For P4, the specific Wall Mid form may change (different quoting patterns, tick sizes), but the principle endures: true fair value is rarely the simple best-bid/ask mid—it is usually better estimated from deep-liquidity levels.

Backtesting discipline

A final differentiator: top teams used backtesting as a *falsification* tool, not a confirmation tool. Frankfurt Hedgehogs warn: "never optimize purely for website score. Doing so is extremely prone to overfitting on simulation-specific randomness rather than building strategies that generalize." They used two complementary approaches:

- **Jasper's backtester** for taking/simple-quoting strategies—fast, flexible, parameterised for systematic optimisation.
- **The official website** for bot-interaction strategies (Rainforest Resin MM, Macaron hidden-taker), since the local tool could not replicate bot fill probabilities.

Linear Utility's custom backtester output logs in the website's exact format, enabling direct comparison. Discrepancies were treated as bugs, never noise.

CMU Physics learned this the hard way. Their Squid Ink spike-detector was backtested on limited data and happened to work on the Round 1 re-run—they recognised luck, not edge. They cut Squid Ink to 10% of max position and focused on robust signals, trading upside for stability.

27.5 Honest: what cannot be planned

Even the best-prepared teams encounter irreducible uncertainty.

Novel products. Every edition has at least one product that does not fit prior archetypes. P3's Squid Ink, with extreme volatility and unpredictable spikes, defeated standard mean-reversion and market-making. CMU Physics: "Squid Ink was pure chaos—with regular 100 SeaShell swings within a single step and no obvious structure." When theory fails, reduce position and gather live data rather than forcing a model.

Regime changes mid-round. Frankfurt Hedgehogs' quadratic IV fit for Volcanic Rock Vouchers worked in sample but "stopped being a good model on the submission day," producing losing positions all day. A rolling-window would have adapted; the static fit could not.

Infrastructure failures. CMU Physics' Round 3 crash (visualiser memory) was unpredictable from local testing and missed by the official backtester. Only mitigation: defensive programming—memory well below limits, worst-case-volume testing, kill-switches for runaway state.

Team coordination under pressure. Both teams describe late-night sessions ("6 a.m. discussing Squid Ink dynamics") and fatigued judgment calls. When CMU Physics dropped to 241st after Round 3, they "felt defeated and were honestly ready to give up." They did not, and the next round was their best. Composure after a setback is a character trait, not a strategy.

Stochastic outcomes despite correct strategy. Frankfurt Hedgehogs led every round except the last, where an unknown team posted 850,000 SeaShells (vs. the typical 200–400k top range) to take first. They accepted it: "in any probabilistic environment . . . some degree of randomness is inevitable." CMU Physics' Round 1 Squid Ink happened to spike *for* them on the re-run—pure luck in their 9th-place finish.

The spread cost of hedging. CMU Physics found hedging Volcanic Rock Vouchers cost 40,000+ SeaShells/day in spread—the underlying had a 1-tick spread and frequent rebalancing piled up. The "correct" textbook move (always hedge delta) was wrong in this microstructure. Frankfurt Hedgehogs concurred: "explicit delta hedging would have been prohibitively expensive given the bid-ask spreads." Microstructure costs can invalidate theoretically optimal strategies, and the true costs only appear live.

Teammate dynamics. CMU Physics had never spoken before the competition; Frankfurt Hedgehogs had played P2 together. Both configurations worked, but both teams describe tension: disagreements, sleep deprivation, the weight of rank drops. No formula for team chemistry, and no backtester can simulate whether a teammate will push through a bad round.

Intuition

The unplannable share a structure: they are *out-of-sample* (OOS—outside the historical data used to build and test the strategy) risks no in-sample analysis can eliminate. Novel products are OOS by definition; regime changes make your calibration non-representative; infrastructure failures sit outside the tested distribution; team dynamics have no historical data. Response: maintain margin. Keep positions small enough that one surprise doesn't eliminate you, code modular enough to swap strategies fast, team cohesive enough that a bad round is data, not crisis.

Prosperity example

Meta-lesson. The best predictor of Prosperity success is not mathematical sophistication but *operational readiness*. Teams with tooling, skeletons, and prior-year knowledge on day 1 beat teams with deeper theory who spent two days building infrastructure. Frankfurt Hedgehogs' "weeks of careful preparation" and Linear Utility's "in-house tools" let both teams focus on strategy from timestep 1, while most competitors were still parsing API docs.

On a Goldman Sachs desk

The same lesson applies on a Goldman Sachs desk at higher stakes. The best strat is not the one with the most elegant mathematics but the one in production, monitored, and iterable. A model taking three months to deploy is dominated by a simpler one that ships in a week and refines over the next eleven. Desk traders value reliability and responsiveness over theoretical optimality. An off-cycle engineer who ships fast, iterates on live feedback, and handles edge cases earns trust—the same trust Prosperity teams earn by submitting early rather than optimising in isolation.

27.6 Key facts to memorise

1. **Tooling before strategy.** Build backtester, visualiser, and analysis notebooks before the competition starts. Use prior-year data to validate the pipeline.

2. **Study all available writeups.** Product archetypes repeat across Prosperity editions. Detection of the Olivia bot in P3 was directly enabled by reading P1 and P2 writeups.

3. **Simple beats complex.** The winning formula is: fair-value estimate + fixed threshold + inventory management. Every top team says this; none deployed complex ML in their final submission.

4. **Submit early, iterate live.** Live-data feedback is worth more than another day of backtesting. CMU Physics' comeback from 241st to 8th in one round demonstrates the power of rapid iteration.

5. **Detect bot patterns.** Filter trades by quantity, timing, and direction. The informed-bot signal has been the single highest-value alpha source in every Prosperity edition.

6. **Reuse signals across products.** Olivia's Croissants position biased basket spread thresholds. Cross-product signal integration compounded edges for both Frankfurt Hedgehogs and CMU Physics.

7. **Choose parameter plateaus, not peaks.** Select parameters from flat regions of the PnL landscape. Sensitivity to small parameter changes is the primary warning sign of overfitting.

8. **Quantify worst-case losses before deploying.** CMU Physics' unhedged Volcanic Rock position and Frankfurt Hedgehogs' VaR-bounded mean-reversion exposure both succeeded because the teams computed realistic downsides first.

9. **Handle edge cases explicitly.** Position limits, integer rounding, memory limits, conversion caps, and serialisation overhead have all cost top teams tens of thousands of SeaShells when handled carelessly.

10. **Operational readiness beats mathematical sophistication.** The team that ships a working model on day one and iterates will beat the team that ships a perfect model on day three.

Part VIII

Quant Strats at Goldman Sachs

Chapter 28

The Quant Strats Role

What does a Strat actually do? This chapter maps the sell-side floor and walks through a typical day. Part VII grounds earlier theory in the specific institution where you will work. A Strat at Goldman Sachs sits at the intersection of quantitative research, trading, and engineering: you derive a pricing model from first principles *and* ship production code that runs it in real time across every desk. Understanding that — hour by hour — is essential preparation for the off-cycle interview and your first weeks on the floor.

Prerequisites

From earlier in the book:

- Chapter 9: the Avellaneda–Stoikov market-making model, reservation price, inventory skew, and the quoting loop. A Strat calibrates and monitors models of exactly this kind.
- Chapter 16: Black–Scholes pricing, risk-neutral valuation, and the Greeks. Morning recalibration of the volatility surface is the first task in a Strat's day.
- Chapter 19: optimal execution and the Almgren–Chriss framework. Strats own the execution algorithms that implement these ideas in production.

28.1 The sell-side floor map

Walk onto the floor of a bulge-bracket bank (Goldman Sachs, Morgan Stanley, J.P. Morgan) and you see a single open-plan room the size of a football pitch: rows of desks, six or more monitors each, loud under fluorescent light. The room is organised not by seniority but by *product* — clusters for equities, rates, credit, FX, commodities. Within each cluster, five roles sit close enough to shout at each other. Proximity is the primary communication channel: when a trader needs a price in ten seconds, email will not do; a shout across three desks will.

Product desks

A **product desk** is a self-contained business unit organised around a related set of tradeable instruments. The major desk families at Goldman Sachs:

- **Rates** — government bonds, interest-rate swaps, swaptions, caps, floors, inflation-linked products.
- **Equities** — single stocks, equity indices, equity derivatives (calls, puts, exotics, variance swaps), ETFs, program trading.

- **Credit** — corporate bonds, credit default swaps (CDS), CDS indices, collateralised debt obligations (CDOs), leveraged loans.
- **Foreign exchange (FX)** — spot, forwards, FX swaps, FX options (vanillas and barriers), cross-currency basis swaps.
- **Commodities** — oil, natural gas, metals, agriculture, power, emissions. Each sub-commodity often has its own desk.

Each desk has its own P&L, risk limits, and regulatory capital allocation. A junior hire is assigned to one desk for two to three years; the desk determines your first asset class, your models, and your traders.

The five roles

Traders. The trader owns the risk book: which positions to take, which client axes to fill, when to hedge. (An *axis* is a standing interest — "the desk is a buyer of 5Y protection" — that Sales circulates to clients.) Traders are measured on P&L — the single number determining bonus, reputation, and survival. A desk head manages a book with gross notional in the billions and a daily P&L swinging by several million. Traders speak in terse shorthand, need answers *now*, and are the primary internal client for every other role.

Two archetypes. **Flow traders** ("market makers" on the sell side) quote prices, manage inventory, and earn the bid-ask spread. **Proprietary traders** — rare since the Volcker Rule (2010) — take directional bets with the firm's capital. Most modern bank trading is flow, so the Strat's job is to help price, hedge, and risk-manage *client-driven* positions.

Sales. Sales sits between the trader and the external client (hedge funds, pension funds, corporates, sovereigns), translating a client's hedging or investment need into a trade request for the trader to price. Sales owns the client relationship, not risk; compensation tracks client revenue, not P&L. A good salesperson explains Greeks to a client treasurer but does not derive them.

Quantitative researchers (QR). QR publishes models: new stochastic processes for underlyings, better calibration algorithms, novel risk metrics. They write papers and prototypes (Python or MATLAB) and hand models to Strats for productionisation. A QR researcher might spend six months deriving a local-stochastic-vol calibration; the Strat turns it into code running in milliseconds across every desk by next quarter. At Goldman Sachs, QR is a separate division from Engineering and Strats, though all three share the floor.

Strats (strategists). The Strat ("strategist") is the bridge: writes production code *and* has the depth to derive the models the code implements. This dual requirement distinguishes the role from software engineering (infrastructure without model logic) and from QR (models without production code). Goldman Sachs formalised the title in the 1990s; other banks call the same role "desk quant," "quant developer," or "front-office quant."

Engineering (Technology). Engineering builds the platforms everything runs on: networking, databases, messaging, trade capture, regulatory reporting, the firm-wide risk aggregation engine. Engineers own reliability, throughput, and internal developer experience, not model logic. The off-cycle placement sits in Engineering — "at the critical center of our business," since every trade, risk number, and client report flows through infrastructure Engineering maintains.

Floor map summary

Role	Owns	Measured by	Primary output
Trader	Risk book	P&L	Positions, hedges
Sales	Client relationship	Client revenue	Trade requests
QR	Model research	Publication, adoption	Papers, prototypes
Strat	Production models	Model accuracy, uptime	Production code
Eng.	Infrastructure	Reliability, throughput	Platforms, tooling

How the roles interact

A client trade illustrates the roles. A pension fund calls Sales: "We want to hedge our equity exposure for six months — price us a collar on the S&P 500 with a 95% put and a 105% call." Sales passes it to the Trader, who asks the Strat: "Run the collar through the engine — fair value and vega?" The Strat evaluates in SecDB, which calls the vol-surface calibration code the Strat wrote last quarter, computes price and Greeks, and returns the result. The Trader adds a margin for execution risk and target return; Sales quotes the client. If the client trades, Engineering's trade-capture system books it, SecDB updates risk in real time, and the overnight batch (Strat code on Engineering infrastructure) produces the regulatory report.

Strat ≠ Software Engineer

"Strat" and "Software Engineer" are not interchangeable. A bank software engineer builds generic infrastructure (messaging, databases, UIs) and is not expected to derive a pricing formula. A Strat implements a *specific* pricing or risk model, owns its mathematical correctness, and sits physically on the trading desk rather than in a separate technology building. The Engineering job description emphasises "risk management, big data, mobile" — all infrastructure. The Strat job description (a separate hiring track) emphasises "pricing models, Greeks, calibration" — the model layer. Know which track you are interviewing for.

28.2 Strats as the bridge

The Strat role sits at the intersection of three competency axes. A Strat must be able to:

1. **Derive**: read a QR paper, reproduce the derivation, identify which approximations matter, decide whether the model is fit for purpose on a given desk.
2. **Implement**: write production code — the firm's proprietary language (Slang at Goldman Sachs; Section 28.5) or C++/Python — that runs in real time, handles edge cases, and degrades gracefully under stress.
3. **Communicate**: explain to the trader, in thirty seconds, why the model gives a particular price, what changed overnight, and what the risk is if the model is wrong.

No other role requires all three. Traders communicate and decide but do not derive or implement; QR derives but does not own production code; Engineers implement but do not derive. Only the Strat can trace a suspicious P&L from the trader's screen back through production code to the mathematical model and identify whether the bug is in the formula, the implementation, or the input.

The translation problem

A Strat's value is most visible when something goes wrong. A scenario that recurs monthly on any active desk: the trader looks at the morning risk report: "My vega is \$500K per vol point on this swaption position, but yesterday it was \$300K. Nothing changed overnight. What happened?"

An engineer cannot answer: it requires understanding the model — did the surface recalibrate to a different shape, changing vega? A QR researcher might answer in principle but cannot trace the bug through production code to the line where calibration parameters are read. The Strat can: check whether the surface moved (yes — a new quote shifted SABR ρ), verify the calibration code ingested the quote correctly (yes), and explain the vega change is real, not a bug, and the position is now more sensitive to vol moves than yesterday.

End-to-end debugging — market data through model mathematics through code to the screen — is the Strat's defining skill, and the hardest to develop, requiring simultaneous fluency in three domains taught separately at university: mathematics, computer science, and finance.

Daily responsibilities

A Strat's daily work splits into five recurring categories, each drawing on earlier theory.

1. Model calibration. Every morning the vol surface (Chapter 18) is recalibrated to the previous close and overnight moves. The Strat owns the calibration code: SABR to the swaption grid (rates), local-vol or Heston to the equity options chain, hazard-rate curve to CDS spreads (credit). When the fit breaks — a new strike outside the interpolation range, an optimiser that fails to converge — the Strat is paged first.

Calibration is not one-off. The surface drifts as new trades print. A good routine runs incrementally (updating only parameters affected by new data), handles missing data (some strikes have no fresh quotes), and produces an *arbitrage-free* surface: no butterfly with negative price, no calendar spread violating time-monotonicity. Enforcing these under time pressure is a real engineering challenge belonging to the Strat.

2. Risk computation. Greeks (Chapter 17) are recomputed for the full book start-of-day and updated intraday. The Strat owns the risk engine's model layer: which bumps to apply (absolute or relative, and how large), how to handle cross-Greeks (e.g. *vanna*, delta's sensitivity to vol), and which approximations are acceptable (closed-form where possible, Monte Carlo where necessary, finite-difference as fallback). The output feeds the trader's risk dashboard and the firm-wide report to management and regulators.

A subtlety: on-screen Greeks must be *consistent* with the price. If price comes from Monte Carlo but delta from a closed-form approximation, the trader's hedge is wrong. Ensuring consistency (all Greeks from the same pricing function, ideally via AD; Section 28.5) is a core Strat responsibility.

3. Trade pricing. When Sales brings a structured-product request — say, an autocallable on the EuroStoxx 50 — the trader asks the Strat for fair value. The Strat runs the payoff through the pricing engine, checks Greeks, returns a price. If no existing model handles the payoff, the Strat builds one under time pressure (the client wants a price by end of day).

Structured products are where mathematical depth is directly monetised. An autocallable has path-dependent payoffs depending on the joint dynamics of underlying, rates, and dividends. Correct pricing needs the right model (local vol, stochastic vol, local-stochastic vol), calibration to the current market, Monte Carlo with enough paths for precision, and Greeks by pathwise differentiation or likelihood-ratio methods. The Strat does all of this; the trader relies on the number.

4. Prototyping and production. New models start as prototypes: a Jupyter notebook testing a QR idea against historical data. If it works, the Strat rewrites it in production code, integrates with the firm's pricing/risk database (SecDB at Goldman Sachs; Section 28.5), and deploys. The Strat then monitors, fixes bugs, and tunes parameters as conditions change.

Prototype-to-production is where many models die. A notebook model that works on clean data may fail catastrophically in production on missing data, stale quotes, corporate actions (splits, dividend changes), or extreme moves. The Strat anticipates these failures, builds guards, and ensures graceful degradation (fall back to yesterday's calibration) rather than catastrophic (price of zero, automatic hedge triggered).

5. P&L explanation. Each morning the trader asks: "Why is my P&L what it is?" The Strat decomposes overnight P&L into contributions from delta, gamma, vega, theta, and unexplained residual. This decomposition (Chapter 17) is not just accounting — a large residual signals model mis-specification, and the Strat must diagnose it.

The standard decomposition is a Taylor expansion of the portfolio value V:

$$\Delta\text{P\&L} \approx \underbrace{\Delta \cdot \Delta S}_{\text{delta}} + \underbrace{\tfrac{1}{2}\Gamma\,(\Delta S)^2}_{\text{gamma}} + \underbrace{\mathcal{V} \cdot \Delta\sigma}_{\text{vega}} + \underbrace{\Theta \cdot \Delta t}_{\text{theta}} + \underbrace{\text{residual}}_{\text{unexplained}}, \tag{28.1}$$

where ΔS is the change in underlying, $\Delta\sigma$ the change in implied volatility, and Δt the elapsed time. A "clean" P&L explain has residual below 5% of total; above 10% triggers an investigation.

Intuition

The Strat translates between two languages: mathematics (QR) and risk/money (trader). Fluent in both, the Strat is the only person on the floor who can debug a pricing discrepancy end to end — from the SDE to the C++ inner loop to the number on the trader's screen.

The five daily tasks

1. **Calibrate**: fit the vol surface / curve to today's market.
2. **Compute risk**: rerun Greeks for the full book.
3. **Price**: value new trades and structured products.
4. **Build**: prototype new models, ship to production.
5. **Explain**: decompose P&L into Greek contributions.

Every other task is a variant or combination of these five.

28.3 A day in the life

A composite sketch of a typical day for a junior Strat on an equity derivatives desk. Specifics vary by desk, seniority, and firm, but the rhythm is representative.

7:00 AM — Arrive, check overnight

Asia closed hours ago; Europe is about to open. The Strat checks the overnight batch: did end-of-day risk complete? Any calibration failures? The dashboard (built by Engineering, populated by Strat code) is green everywhere except one: a single-stock option's vol surface failed to converge because the stock gapped 5% overnight on earnings. The Strat opens the calibration log, identifies the problematic strike, widens optimiser bounds, reruns, verifies sensible Greeks, and pushes the fix before the trader arrives at 7:15.

7:30 AM — Morning meeting

The desk head runs a five-minute stand-up. The trader: “Long gamma on the index, short gamma on single names. Correlation book made money overnight — realised correlation dropped. Asia was quiet.” The Strat reports the calibration fix, flags that gamma on the at-the-money straddle is unusually high (the earnings gap compressed the surface near the money), and recommends rehedging. Sales mentions a client wanting a six-month barrier option price by noon.

The morning meeting tests the Strat’s communication. The trader does not want to hear about Levenberg–Marquardt failing to converge; the trader wants: “The surface on that name is steeper near the money, raising gamma exposure by 20%. To keep the same risk profile, sell about 500 ATM-straddle contracts.” Translating model output into actionable trading language in real time takes months to develop.

8:00 AM — P&L explain

The Strat runs the P&L decomposition using Equation (28.1). Yesterday’s P&L was $+1.2M. The decomposition:

Component	**Contribution**
Delta (book was net long, market rallied)	+$800K
Theta (time decay on short options)	+$300K
Vega (implied vol ticked up slightly)	+$50K
Cross-gamma and higher-order	+$20K
Unexplained residual	+$30K
Total	+$1,200K

The residual is 2.5% — well within the 5% threshold. The Strat attributes it to bid-ask crossing on illiquid positions and rounding in finite-difference Greek bumps. The trader signs off. Had the residual been $150K (12.5%), the Strat would need to investigate: mis-specified model? Stale market-data feed? Mis-booked position?

10:00 AM — Price the barrier option

Sales has formalised the request: a six-month down-and-out put on an equity index, barrier at 85% of spot, $50M notional — a path-dependent exotic that knocks out worthless if the index touches the barrier in six months. The Strat prices via the local-vol engine (100,000 Monte Carlo paths, Euler discretisation, 500 time steps), sanity-checks against Heston (agreement within 10 bps of notional), computes Greeks by pathwise differentiation, and sends a one-pager: fair value, delta, gamma, vega, theta, knock-out probability (18%), and “barrier delta” (sensitivity to the barrier level). The trader adds a 50 bp margin and Sales quotes.

12:00 PM — Code review

A fellow Strat has submitted a PR refactoring dividend-handling in the equity pricing engine. The change affects how discrete dividends are interpolated between ex-dates, shifting the forward curve and every option price on the desk. The reviewer checks mathematical correctness (does interpolation preserve no-arbitrage?), numerical stability (what if two ex-dates are one day apart?), and test coverage (do regressions catch a pricing change above 1 bp on the benchmark book?). Model-code review is more demanding than infrastructure review: a subtle math error mis-hedges the desk and costs real money.

2:00 PM — Fix a production bug

A neighbouring-desk trader reports negative vega on a long swaption — clearly wrong. The bug: a sign error in the SABR vega bump, introduced two weeks ago when a QR patch switched from absolute to relative bumps. The absolute bump $\sigma \to \sigma + \epsilon$ became $\sigma \to \sigma(1+\epsilon)$, but read-back still subtracts ϵ rather than dividing by $(1+\epsilon)$. The Strat writes a fix, adds a regression test for sign of vega on a vanilla long call (always positive), gets a second-Strat review, and pushes. The corrected vega appears within minutes.

Model bugs cost real money

A Greek sign error is not an academic mistake. A trader seeing negative vega on a long option might *sell* vol to "hedge," which *doubles* their vega. On a \$500K-per-vol-point position, a 2-point vol move then costs \$1M instead of \$500K. Every model bug has a dollar cost, and the Strat who ships it owns the consequences. Code review, regression testing, and careful deployment are the primary defence against P&L-destroying errors.

5:00 PM — End of day

Europe closes. The Strat triggers the end-of-day batch: full recalibration, full risk, regulatory report (about 45 minutes — tens of thousands of trades, each requiring a pricing evaluation). While it runs, the Strat reviews a QR paper on a stochastic local-vol model that might improve barrier-option pricing; the key result (a PDE for the local-vol component conditional on the stochastic-vol state) is tractable and implementable with a moderate refactor. Prototype notes go in the backlog.

6:00 PM — Batch monitoring and wrap-up

The batch completes. The Strat checks output (all calibrations converged, all risk numbers within bounds, no limit breaches), queues the regulatory report, commits the day's changes, updates the tracker, and leaves — 6:00–6:30 PM on most days. On bad days (batch-time model failure, regulatory deadline, product launch) it runs much later.

The Strat is not a 9-to-5 role

A Strat's day is bounded by the market, not the clock. A model breaking in Asian hours with live desk risk means a 3:00 AM page; a new product launch means weekend work on the pricing model. Quarter- and year-end add pressure: every position revalued, every model validated, every regulatory report filed. The role demands deep technical skill *and* the operational resilience to be the person called when something goes wrong at an inconvenient hour.

28.4 Career arcs

The standard Goldman Sachs progression (and most bulge-brackets') follows a well-defined hierarchy. Understanding it matters for interview preparation and career planning.

Analyst / Off-cycle (0–3 years). You join as analyst (or off-cycle intern who converts). Assigned to one desk (rates, equities, credit, FX, commodities), you spend year one learning the desk's products, models, and codebase. Primary output: bug fixes, small features, the daily calibration/risk cycle (Section 28.3). The Engineering off-cycle posting specifies "real responsibilities alongside fellow interns and our people" — you write production code from week one, not observe.

Year one is steep: Slang, SecDB, the desk's product mathematics, and the floor's unwritten conventions. The fastest way to earn trust is to fix a bug that has annoyed the trader for months — proof you can read the codebase, understand the model, and ship a correct fix under pressure.

Associate (3–5 years). You own a specific model or product — the person the trader calls when it misbehaves. You lead small projects: replacing a calibration, building a pricing model for a new product, migrating legacy code. The firm expects you to identify model deficiencies proactively, not just react to trader complaints.

Vice President (5–10 years). You own a broader area: the vol-surface calibration stack, or the risk engine for a product suite. You manage junior Strats, set technical direction, and interact with senior traders and desk heads on model choices. Judgment calls: is this model good enough or does it need replacing? Is the engineering cost of a refactor justified by the pricing improvement? VPs also represent the desk in firm-wide model-governance reviews, where senior risk managers challenge assumptions and demand evidence of robustness.

Managing Director (10+ years). An MD Strat is rare, running a full Strats group ("Global Rates Strats," "Equity Derivatives Strats"). The role is as much people management, hiring, and strategy as code and models: an MD decides which models to invest in, which platforms to build, and how to allocate 10–30 Strats across competing priorities.

Career ladder summary

Level	Typical tenure	Scope
Analyst	0–3 years	Individual tasks, bug fixes, daily cycle
Associate	3–5 years	Owns one model or product
VP	5–10 years	Owns a product suite, manages juniors
MD	10+ years	Runs a Strats group, sets strategy

Exit paths

The sell side is a powerful launchpad. After three to five years a Strat's skillset — production coding, model derivation, market intuition, operational discipline — is highly valued outside the bank.

- **Hedge funds.** The most common exit. Multi-strategy funds (Citadel, Millennium, Point72) hire experienced Strats to run systematic desks or build pricing infrastructure. Direct skill transfer — pricing, Greeks, calibration, production coding — and pay at a top fund can be 2–5× the bank at the same seniority.
- **Proprietary trading firms.** Jane Street, Two Sigma, Jump hire Strats for quantitative depth and production discipline; low-latency experience is especially valued.

- **Technology.** Some Strats move to big tech (Google, Meta, Amazon) where systems-thinking and low-latency skills transfer. Pay is competitive (below hedge-fund levels) and work-life balance is better.
- **Quantitative research.** A smaller number move to pure QR — same bank or a fund — leaving production to others. Suits Strats who prefer derivation to debugging.
- **Fintech and startups.** A growing path: Strats bring credibility and domain expertise to trading platforms, risk systems, and crypto infrastructure.

What Goldman Sachs looks for in off-cycle hires

The Engineering posting emphasises three themes that apply equally to Strats candidates:

1. **Problem-solving under constraints.** The posting calls for "innovative strategic thinking and immediate, real solutions." Translation: given a problem, can you decompose it, find the binding constraint, and produce a working solution quickly? Probability puzzles and live coding probe this.
2. **Adaptability.** "Evolve, adapt to change and thrive in a fast-paced global environment." The sell side churns: new regulations, products, models. The interviewer wants evidence you have operated in ambiguous, shifting environments.
3. **Collaboration.** "Creative collaborators" is the exact phrase. The Strat role is collaborative by construction — you work with traders, QR, sales, and engineering daily. The interviewer looks for signals that you can communicate technical ideas to non-technical stakeholders and vice versa.

Intuition

The single most important thing Goldman Sachs wants in a junior hire is *evidence of building things that work*: a side project, an open-source contribution, a competition result, a research implementation — anything that shows you can go from idea to working code under constraints. IMC Prosperity is exactly this: it shows you can design, implement, and iterate on a trading algorithm under time pressure with real (simulated) consequences.

28.5 Technology: SecDB and Slang

Goldman Sachs runs one of the most sophisticated proprietary platforms in finance. Two components are central to a Strat's daily work: **SecDB** and **Slang**.

SecDB: the pricing and risk database

SecDB (**Sec**urities **Data****Base**) is Goldman Sachs's firm-wide pricing, risk-management, and position-management system. Built in the early 1990s and continuously developed since, SecDB is not a relational database: it is a *graph-based compute engine* representing every trade, instrument, market-data input, and pricing model as a node in a directed acyclic graph (DAG). When a market-data input changes — a spot, a vol quote, a yield-curve point — SecDB propagates the change, recomputing every dependent price and risk number automatically.

SecDB in one sentence

SecDB is a reactive DAG-based compute engine that prices every trade in the firm, computes every Greek, and propagates every market-data change in near-real time.

Scale is extraordinary. At peak, SecDB holds pricing models for hundreds of thousands of live trades across every asset class. Every desk (rates, equities, credit, FX, commodities) uses the same instance, so a single SABR-calibration change affects every desk using SABR. The benefit is consistency (one model, one price, one risk number per trade); the cost is global bug propagation — hence the discipline of code review and testing.

The DAG structure shapes how a Strat thinks about pricing. Leaves are **market-data nodes**: spots, yield-curve points, vol-surface quotes, dividend forecasts, updated in real time. Above are **model nodes**: calibration routines producing parameters (SABR, Heston, hazard rates). Above those, **trade nodes**: one per trade, with edges to the model and market-data nodes it depends on. A market-data update triggers an upward traversal, recomputing every dependent node. Every price and Greek is always consistent with the latest data, without manual intervention.

The SecDB graph

Layer	Contents
Trade nodes	One per trade: price, Greeks, P&L
Model nodes	Calibration outputs: SABR params, Heston params, ...
Market-data nodes	Spot, yield curve, vol quotes, dividends

A change at any lower layer propagates upward automatically. The Strat writes the model-node logic; Engineering writes the market-data and infrastructure layers.

Slang: the pricing language

Slang (**S**ecDB **Lang**uage) is Goldman Sachs's proprietary functional language for writing pricing and risk models inside SecDB: statically typed, garbage-collected, functional, with built-in automatic differentiation (AD) — the same technique behind modern ML frameworks, applied here to compute Greeks.

Intuition

AD in Slang means that when a Strat writes a pricing function, the system computes every Greek (delta, gamma, vega, theta, rho, and arbitrary cross-Greeks) *automatically*, with no differentiation code. This eliminates an entire class of bugs: Greeks are exactly consistent with the pricing function by construction.

Slang's design features explain why Goldman Sachs built a custom language instead of using C++ or Python:

1. **Automatic differentiation (AD).** Every Slang function differentiates automatically with respect to any input. Greeks are computed by the runtime, exact to machine precision, consistent with the price by construction, zero extra code.

2. **Lazy evaluation.** Slang evaluates expressions only when results are needed. In SecDB's DAG a market-data change triggers recomputation only of nodes actually depending on the input, not the whole graph — what makes near-real-time risk at scale feasible.

3. **Serialisability.** Every Slang object serialises to disk and back, letting SecDB checkpoint and restore the pricing graph — critical for end-of-day batches, disaster recovery, and regulatory audits.

4. **Type safety.** The static type system catches common errors — passing a volatility where a price was expected, confusing a date with a number — at compile time.

The Slang VM runs on every desk's compute infrastructure. A price request routes to a Slang process that evaluates the pricing graph and returns the result. The VM is optimised for two workloads: single-trade pricing (latency-critical, for live quoting) and batch risk (throughput-critical, for end-of-day and regulatory reports).

A junior Strat's first months are spent learning Slang and the SecDB graph model. The curve is steep — Slang is not taught at any university and has no public documentation — but the payoff is that a fluent Strat can implement, test, and deploy a new pricing model in days rather than the weeks or months a conventional C++ codebase would take.

Slang is not open-source

You cannot practise Slang before joining Goldman Sachs: entirely proprietary, no public compiler, no published textbook. What you *can* practise is functional programming (Haskell, OCaml, Python's functional idioms), AD (JAX, PyTorch autograd), and graph-based computation (TensorFlow dataflow, Dask). These transferable skills accelerate the Slang transition, and in interviews, comfort with functional programming and AD is a strong positive signal.

Comparable systems at other banks

Goldman Sachs is not alone. The major systems, each serving the same purpose (unified pricing, risk, position management) but differing in language, architecture, and maturity:

Bank	System	Language	Architecture
Goldman Sachs	SecDB	Slang (proprietary)	Reactive DAG
J.P. Morgan	Athena	Python / C++	Grid compute
Morgan Stanley	In-house	C++ / C#	Service-oriented
Barclays	BARX	Java / C++	Service-oriented

The skills you build on *any* of these systems — reactive computation, AD, model calibration, risk aggregation — transfer directly. The specific language is secondary; the design patterns are universal.

Why SecDB matters for your career

SecDB is widely regarded as one of Goldman Sachs's most significant competitive advantages: unified risk across all desks, consistent pricing of complex structured products, rapid deployment of new models. Its decades-long head start and Slang's expressiveness have kept it at the frontier. As a Strat you are both user and builder; understanding the architecture is not optional.

The SecDB/Slang ecosystem teaches two things hard to learn elsewhere. First, writing code where *correctness is the primary constraint*: in most software a bug means a bad user experience; in pricing code it means lost money. The discipline of provably correct financial code (comprehensive tests, formal review, staged deployment) is highly valued by hedge funds and prop shops. Second, how a large-scale reactive system works: propagating changes through a graph, handling stale data, balancing latency against throughput — problems in every real-time data system, from ad platforms to autonomous vehicles.

Chapter 37 covers SecDB's architecture and Slang's language features in depth.

Prosperity example

Prosperity's `Trader` class — `run()` called once per iteration — is a miniature Strat event loop. Each call receives the market state (`TradingState`) and returns orders, conversions, and serialised state, just as a Strat's pricing code must return price and Greeks within a single SecDB evaluation. The constraint that all state live in `traderData` (a string persisting between iterations) mirrors SecDB's requirement that model state be serialisable and reproducible. A clean, stateful `Trader.run()` handling edge cases is practice in the core Strat discipline: deterministic, stateful computation under time pressure with no room for unhandled exceptions.

On a Goldman Sachs desk

Interview prep for the Strat and Engineering tracks. Goldman Sachs quantitative interviews have four components:

1. **Probability and brainteasers.** Classics: "Expected coin flips to get two heads in a row?" "100 balls, 50 red and 50 blue, drawn without replacement — probability the last is red?" Tests reasoning under uncertainty and structured communication.

2. **Black–Scholes and Greeks.** Derive Black–Scholes from risk-neutral pricing (Chapter 16), explain why ATM call delta is approximately 0.5, sketch gamma vs. spot. Strat interviews go deeper here than Engineering.

3. **Coding test.** Timed online assessment or live whiteboard: data structures and algorithms at strong-undergrad level, Python or C++.

4. **Market intuition.** "If the Fed raises rates by 25 bps, what happens to a 10-year Treasury? To equities? Why?" Tests connecting theory to real market moves.

Chapter 39 contains a bank of practice problems covering all four categories.

Historical note

Emanuel Derman's memoir *My Life as a Quant* (2004) is the canonical account of quantitative finance's early days on Wall Street (Derman, 2004). Derman, a theoretical physicist who joined Goldman Sachs in 1985, helped build the firm's first-generation options-pricing models and shaped the culture that became the modern Strats group. His description of the intellectual excitement — and operational chaos — of building pricing models in the late 1980s remains the best introduction to what the Strat role *feels like* from inside. Two themes. First, models are always wrong; the question is whether they are useful enough, and the Strat's judgment on "useful enough" is what the trader pays for. Second, the half-life of a pricing model is short — markets evolve, products change, and the Strat who stops learning stops being useful. Derman's arc — physicist to junior quant to MD to academic — is a template in Section 28.4.

Key facts to memorise

- The sell-side floor is organised by **product**, not function. Traders, sales, QR, Strats, and engineering sit together per product area.

- A **Strat** writes production code *and* derives the models it implements — the dual competency separating the role from software engineering (code without models) and QR (models without production code).

- Daily loop: morning calibration and P&L explain, mid-day trade pricing and prototyping, afternoon bug fixes and production pushes, evening reading.
- **SecDB** is Goldman Sachs's DAG compute engine for pricing and risk. Every trade prices through it; market-data changes propagate through the graph automatically.
- **Slang** is the proprietary functional language for writing pricing models. Built-in AD computes all Greeks for free, eliminating a major class of implementation bugs.
- Progression: Analyst → Associate → VP → MD. Common exits: hedge funds, prop shops, tech, pure QR.
- Off-cycle interviews test four axes: probability puzzles, Black–Scholes and Greeks, coding, market intuition.
- Prosperity's `Trader.run()` is a miniature Strat event loop: receive state, compute, return orders, serialise. Clean, stateful, deterministic code under time pressure — the same discipline in both settings.

Chapter 29

Fixed Income Foundations

This chapter fills the largest gap for a CS reader: the language of *fixed income* — bonds, interest rates, and swaps. Chapters 30 and 33 depend on it. Five objects do all the work: zero rates $z(T)$, discount factors $D(T)$, forward rates $f(T_1, T_2)$, duration D_{Mac} (with convexity C), and the par swap rate s. No prior knowledge is assumed beyond continuous discounting from Chapter 15.

Prerequisites

From earlier in the book:

- Chapter 15: the risk-free rate r, continuous discounting via e^{-rT}, and the no-arbitrage principle.
- Basic calculus: derivatives of exponentials, Taylor expansion to second order.

29.1 Zero rates, discount factors, and forward rates

Three objects underpin fixed income; every bond, swap, and rates derivative is a function of them.

Compounding conventions

A rate R per annum means different things depending on the *compounding frequency* m:

- **Annual** ($m = 1$): \$1 grows to $(1 + R)$ after one year.
- **Semi-annual** ($m = 2$): \$1 grows to $(1 + R/2)^2$ after one year.
- **Quarterly** ($m = 4$): \$1 grows to $(1 + R/4)^4$.
- **Continuous** ($m \to \infty$): \$1 grows to e^R.

In general, investing \$1 at a rate R_m compounded m times per year for n years gives a terminal value of

$$V = \left(1 + \frac{R_m}{m}\right)^{mn}. \tag{29.1}$$

Taking $m \to \infty$ yields *continuous compounding*:

$$V = e^{R_c\, n}, \tag{29.2}$$

where R_c is the continuously compounded rate.

Converting between conventions. Equating Equations (29.1) and (29.2) at $n = 1$:

$$e^{R_c} = \left(1 + \frac{R_m}{m}\right)^m \implies R_c = m \ln\left(1 + \frac{R_m}{m}\right), \quad R_m = m\left(e^{R_c/m} - 1\right). \tag{29.3}$$

Example 29.1 (Rate conversion)**.** 10% semi-annual ($m = 2$) → continuous: $R_c = 2\ln(1.05) = 9.758\%$. 8% continuous → quarterly: $R_4 = 4(e^{0.02} - 1) = 8.08\%$.

We use **continuously compounded rates** throughout unless stated otherwise: rates over consecutive periods simply add ($e^{R_1T_1}e^{R_2T_2} = e^{R_1T_1+R_2T_2}$), with no re-compounding adjustments.

Zero-coupon rates

Definition 29.2 (Zero-coupon rate (zero rate))**.** The *zero-coupon rate* (or simply *zero rate*) $z(T)$ for maturity T is the annualised continuously compounded rate of return earned on a riskless investment that pays no intermediate coupons, made today and maturing at time T. If you invest \$1 today at rate $z(T)$, you receive $e^{z(T)T}$ at time T.

"Zero-coupon" means no intermediate cash flows: a single lump sum at maturity. This is the purest interest rate — the price of moving money from today to one future date.

The map $z : (0, \infty) \to \mathbb{R}$ is the **zero curve** (or **term structure**). It is typically upward-sloping: longer maturities command higher rates.

Zero rates vs. coupon yields

A zero rate is not a coupon-bond yield. A 5-year 3% Treasury's YTM blends returns across many maturities (6m, 1y, ..., 5y); $z(5)$ is the return on a *single* cash flow at year 5. A YTM also implicitly assumes every coupon is *reinvested* at that same YTM until maturity — an unrealistic assumption, since coupons paid at year 1 will actually be reinvested at whatever the 1-year rate is at that time. Zero rates embed no reinvestment assumption: each $z(T)$ is the clean price of moving money to one specific future date. The zero curve is the primitive; bond yields are derived from it.

Discount factors

Definition 29.3 (Discount factor)**.** The *discount factor* $D(T)$ is the price today of a riskless claim that pays exactly \$1 at time T:

$$D(T) = e^{-z(T)\,T}. \tag{29.4}$$

If $z(T) > 0$ then $D(T) < 1$. To price a riskless cash-flow stream (c_i) at times (t_i):

$$\text{Price} = \sum_{i=1}^{n} c_i\, D(t_i). \tag{29.5}$$

This is the **present-value identity** — the single most important equation in fixed income.

Intuition

$D(T)$ is the price today of \$1 at time T. $D(2) = 0.9139$ means \$1 in two years is worth 91.39 cents today.

Properties: $D(0) = 1$; D strictly decreasing if rates positive; $D(T) \to 0$ as $T \to \infty$; $D(T_1)/D(T_2)$ for $T_1 < T_2$ is the discount factor for the forward period $[T_1, T_2]$.

Forward rates

Forward rates extend zero rates: the cost of lending between two *future* dates, implied by today's curve. In plain English, the forward rate $f(T_1, T_2)$ is the interest rate you can lock in *today* for a loan that will start at time T_1 and mature at time T_2 — no money actually moves today, just a contractual promise about the rate on a future loan.

Definition 29.4 (Forward rate)**.** The *(continuously compounded) forward rate* $f(T_1, T_2)$ for the period from T_1 to T_2 (with $T_1 < T_2$) is the rate implied by the zero curve such that lending from T_1 to T_2 is consistent with lending from today to T_1 and from today to T_2.

Derivation. Two strategies must give the same terminal value at T_2: direct lending at $z(T_2)$ yields $e^{z(T_2)T_2}$; lending to T_1 at $z(T_1)$ and rolling into a forward at $f(T_1, T_2)$ yields $e^{z(T_1)T_1}e^{f(T_1,T_2)(T_2-T_1)}$. By no-arbitrage:

$$e^{z(T_1)\,T_1} \cdot e^{f(T_1,T_2)(T_2-T_1)} \;=\; e^{z(T_2)\,T_2}. \tag{29.6}$$

Taking logarithms and solving for f:

$$\boxed{f(T_1, T_2) \;=\; \frac{z(T_2)\,T_2 - z(T_1)\,T_1}{T_2 - T_1}}. \tag{29.7}$$

Equivalently, in terms of discount factors:

$$\frac{D(T_1)}{D(T_2)} \;=\; e^{f(T_1,T_2)(T_2-T_1)}, \quad \text{so} \quad f(T_1, T_2) \;=\; \frac{\ln D(T_1) - \ln D(T_2)}{T_2 - T_1}. \tag{29.8}$$

Instantaneous forward rate. In the limit $T_2 \to T_1 = T$:

$$f(T) \;:=\; \lim_{T_2 \to T} f(T, T_2) \;=\; z(T) + T\,z'(T) \;=\; -\frac{\partial}{\partial T}\ln D(T). \tag{29.9}$$

This is the *instantaneous forward rate* at time T — the forward rate for an infinitesimally short period beginning at T.

Intuition

$f(T_1, T_2)$ is the break-even lending rate that makes an investor indifferent between lending straight to T_2 at $z(T_2)$ or rolling over from T_1. It is *not* a forecast — it is the rate implied by today's curve.

Upward-sloping curves and forward rates. Equation (29.7) can be rewritten as:

$$f(T_1, T_2) = z(T_2) + (z(T_2) - z(T_1))\frac{T_1}{T_2 - T_1}. \tag{29.10}$$

When the zero curve is upward-sloping ($z(T_2) > z(T_1)$), the forward rate $f(T_1, T_2)$ exceeds the longer zero rate $z(T_2)$. Conversely, when the curve is inverted, forward rates lie below the zero curve.

Worked example: computing D, z, and f

Given $z(1) = 4\%$, $z(2) = 4.5\%$.

Step 1: Discount factors.

$$D(1) = e^{-0.04\times 1} = 0.9608, \tag{29.11}$$
$$D(2) = e^{-0.045\times 2} = e^{-0.09} = 0.9139. \tag{29.12}$$

Step 2: Forward rate. From Equation (29.7):

$$f(1,2) = \frac{0.045 \times 2 - 0.04 \times 1}{2-1} = \frac{0.09 - 0.04}{1} = 5.00\%. \tag{29.13}$$

Step 3: Consistency check. $100 \times e^{0.04} \times e^{0.05} = 100 \times e^{0.09} = 100 \times e^{0.045\times 2}$, matching 2-year lending at 4.5%. ✓

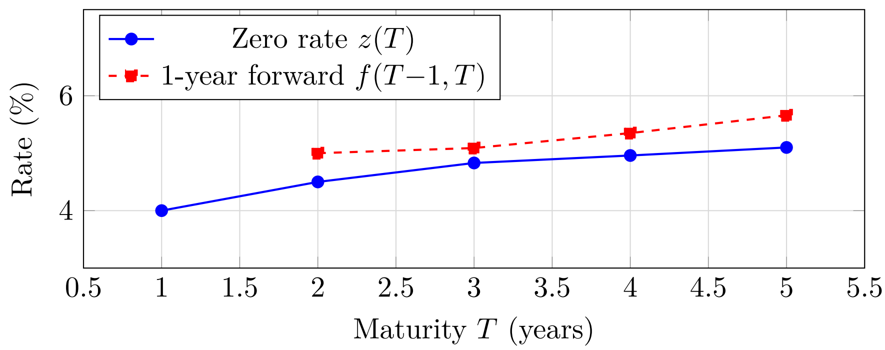
Full forward rate table. From the bootstrap example (Section 29.3):

Year (n)	**Zero rate** $z(n)$ **(%)**	**1-year forward** $f(n-1, n)$ **(%)**
1	4.40	—
2	4.70	$\frac{0.047\times 2 - 0.044\times 1}{1} = 5.00$
3	4.83	$\frac{0.0483\times 3 - 0.047\times 2}{1} = 5.09$
4	4.96	$\frac{0.0496\times 4 - 0.0483\times 3}{1} = 5.35$
5	5.10	$\frac{0.051\times 5 - 0.0496\times 4}{1} = 5.66$

Forward rates are the "marginal" rate, zero rates the "average" — the marginal exceeds the average exactly when the average is rising (same relationship as marginal vs. average cost).

Interpreting forward rates. Forward rates are break-even rates, not forecasts. The expectations hypothesis equates them with expected future spot rates; empirically they are biased upward by a *term premium* compensating lenders for longer-dated risk. A steepening forward curve signals expected rate rises or a rising term premium.

Visualising the zero and forward curves.



On upward-sloping curves the forward curve lies above the zero curve (Equation (29.10)).

Zero rates and discount factors

- **Zero rate** $z(T)$: the annualised continuously compounded return for lending from today to T.
- **Discount factor** $D(T) = e^{-z(T)\,T}$: the price today of $1 received at T.

- **Forward rate** $f(T_1, T_2) = \frac{z(T_2)\,T_2 - z(T_1)\,T_1}{T_2 - T_1}$: the break-even rate for lending from T_1 to T_2.
- These three are *equivalent representations* of the same information: the **zero curve** $\{z(T) : T > 0\}$.
- When the zero curve slopes up, forward rates exceed zero rates.

29.2 Day-count conventions

Rates are quoted annualised, but the cash flow depends on the accrual days and the year definition. A **day-count convention** specifies both.

Definition 29.5 (Day-count fraction)**.** The day-count fraction δ between dates d_1, d_2 is the year-fraction under a specified convention. The interest payment at rate R on notional N is $N \times R \times \delta$.

Three common conventions:

ACT/360. $\delta =$ (actual days)/360. USD LIBOR, SOFR, Euribor, most money-market instruments. *Ex.* 90-day period: $\delta = 0.2500$. At 5% on \$10 M: \$125,000.

ACT/365 (Fixed). $\delta =$ (actual days)/365. GBP SONIA, some AUD. 90-day period: $\delta = 0.2466$; \$10 M at 5%: \$123,288 (\$1,712 less than ACT/360).

30/360 (Bond Basis). 30-day months, 360-day year:

$$\delta = \frac{360(Y_2 - Y_1) + 30(M_2 - M_1) + (D_2 - D_1)}{360}.$$

USD fixed-rate bond coupons and fixed legs of USD swaps. Jan 15 to Apr 15: $\delta = 90/360 = 0.2500$.

Day-count matters for real money

ACT/360 vs. ACT/365 on \$100 M at 5% for 90 days differs by \$17,123. Ignoring day-counts is a fast path to unexplained P&L breaks; every swap confirmation specifies the convention on each leg.

The remainder of this chapter uses continuous compounding to avoid day-count complications. In practice, a Strat's first step when building a pricer is getting day-counts right and unit-testing them.

29.3 Bootstrapping the zero curve

The market does not quote zero rates directly. We observe prices of deposits, futures, and swaps, and extract the implied rates — **bootstrapping**. Start with the shortest-maturity instrument; at each step use previously determined rates to solve for the next.

The instruments

A USD curve build uses three instrument types by maturity:

1. **Deposits (0–6 months).** Interbank deposit rates are quoted as simple rates (no compounding). A 3-month deposit at rate r gives discount factor $D(T) = 1/(1 + rT)$.

2. **Futures (3 months–2 years).** Eurodollar or SOFR futures are exchange-traded contracts on future short-term rates. Their prices directly imply forward rates, which can be converted to zero rates. (We omit the convexity adjustment here; see (Hull, 2014), Ch. 6.)

3. **Swaps (2–30 years).** Par swap rates define the long end of the curve. A par swap at rate s satisfies the par condition derived in Section 29.6, which allows us to solve for the longest-maturity discount factor.

The bootstrap algorithm

Sequentially:

1. $D(T)$ from each deposit maturity.

2. Extend using futures-implied forwards.

3. For each swap maturity T_k in increasing order, solve the par swap equation for $D(T_k)$ using previously determined Ds.

When intermediate Ds are needed but undetermined (e.g. $D(3), D(4)$ for a 5-year swap), interpolate linearly in zero rates and solve numerically.

Five-point bootstrap: worked example

We build a zero curve from five instruments:

#	Instrument	Maturity	Rate
1	3-month deposit	0.25 y	4.00% simple
2	6-month deposit	0.50 y	4.20% simple
3	1-year swap	1.00 y	4.50% annual fixed
4	2-year swap	2.00 y	4.80% annual fixed
5	5-year swap	5.00 y	5.20% annual fixed

Step 1: 3-month deposit ($T = 0.25$). Deposit rates are simple (uncompounded). Investing \$1 at 4.00% for 3 months returns 1.01:

$$D(0.25) = \frac{1}{1 + 0.04 \times 0.25} = \frac{1}{1.01} = 0.9901. \tag{29.14}$$

Converting to a continuously compounded zero rate:

$$z(0.25) = \frac{-\ln(0.9901)}{0.25} = \frac{0.009950}{0.25} = 3.98\%. \tag{29.15}$$

Step 2: 6-month deposit ($T = 0.50$).

$$D(0.50) = \frac{1}{1 + 0.042 \times 0.50} = \frac{1}{1.021} = 0.9794, \qquad z(0.50) = \frac{-\ln(0.9794)}{0.50} = 4.16\%. \tag{29.16}$$

Step 3: 1-year swap ($T = 1.00$). The par condition (derived in Section 29.6) for annual coupon $s = 4.50\%$: fixed-leg PV plus notional repayment equals par:

$$s \cdot D(1) + D(1) = 1 \quad \Longrightarrow \quad D(1) = \frac{1}{1+s} = \frac{1}{1.045} = 0.9569. \tag{29.17}$$

Zero rate: $z(1) = -\ln(0.9569)/1 = 4.40\%$.

Step 4: 2-year swap ($T = 2.00$). $s = 4.80\%$ at years 1 and 2. Par condition: $s \cdot D(1) + (1 + s) \cdot D(2) = 1$. With $D(1) = 0.9569$:

$$D(2) = \frac{1 - 0.048 \times 0.9569}{1.048} = \frac{1 - 0.04593}{1.048} = \frac{0.95407}{1.048} = 0.9104. \tag{29.18}$$

$$z(2) = -\ln(0.9104)/2 = 4.70\%. \tag{29.19}$$

Step 5: 5-year swap ($T = 5.00$). $s = 5.20\%$, annual payments at years 1–5. Interpolate linearly between $z(2)$ and the unknown $z(5)$:

$$z(T) = z(2) + \frac{z(5) - z(2)}{5 - 2}(T - 2), \quad 2 < T \leq 5,$$

so $D(3), D(4), D(5)$ are all functions of $z(5)$. Substituting into

$$0.052 \times [D(1) + D(2) + D(3) + D(4) + D(5)] + D(5) = 1 \tag{29.20}$$

and solving numerically gives $z(5) = 5.10\%$. Back-substituting:

Maturity T	**Zero rate $z(T)$**	**Discount factor $D(T)$**
0.25	3.98%	0.9901
0.50	4.16%	0.9794
1.00	4.40%	0.9569
2.00	4.70%	0.9104
3.00	4.83%	0.8651
4.00	4.96%	0.8199
5.00	5.10%	0.7750

Verification. Annuity $A = \sum_{n=1}^{5} D(n) = 4.3273$; then $sA + D(5) = 0.052 \times 4.3273 + 0.7750 = 1.0000$. ✓

Bootstrapping the zero curve

The zero curve is not directly observable. It is *bootstrapped* from traded instruments:

1. **Short end** ($\leq$ 6 months): deposit rates give discount factors directly via $D = 1/(1 + rT)$.
2. **Middle** (6 months–2 years): Eurodollar/SOFR futures.
3. **Long end** ($\geq$ 2 years): par swap rates, solved sequentially using previously determined discount factors.

Between bootstrapped points, zero rates are typically interpolated linearly (or with cubic splines for smoother hedges).

Reference implementation

The following Python function bootstraps a zero curve from deposit rates and annual par swap rates.

```
import math

def bootstrap(deposits, swaps):
    """Bootstrap a zero curve.
    deposits: list of (T, simple_rate)
    swaps:    list of (T, par_rate), annual fixed
    Returns dict {T: (z, D)}."""
    curve = {}
    for T, r in deposits:
        D = 1.0 / (1 + r * T)
        z = -math.log(D) / T
        curve[T] = (z, D)

    for T_swap, s in sorted(swaps):
        known_sum = sum(
            s * curve[t][1]
            for t in sorted(curve) if t < T_swap)
        # D(T_swap) from par condition
        D = (1.0 - known_sum) / (1.0 + s)
        z = -math.log(D) / T_swap
        curve[T_swap] = (z, D)
    return curve
```

Listing 29.1: Zero-curve bootstrap from deposits and annual swaps.

29.4 Bonds: yield, duration, convexity, and DV01

A **bond** promises fixed coupons and return of face value F at maturity. Understanding price sensitivity to rates is the core skill of a rates trader.

Bond pricing from the zero curve

Definition 29.6 (Bond price)**.** A bond with face value F, coupon rate c paid m times per year, and maturity T generates cash flows $c_i = Fc/m$ at times $t_i = i/m$ for $i = 1, \ldots, mT - 1$, and a final cash flow of $Fc/m + F$ at $t_{mT} = T$. Its price is the present value of these cash flows:

$$P = \sum_{i=1}^{mT} c_i \, D(t_i). \tag{29.21}$$

The price depends on the entire zero curve — each cash flow discounted at its own rate. Market convention, however, quotes bonds via a single number: the yield.

Yield to maturity

Definition 29.7 (Yield to maturity)**.** The *yield to maturity* (YTM) of a bond is the single continuously compounded rate y such that discounting all cash flows at y reproduces the bond's market price:

$$P = \sum_{i=1}^{n} c_i \, e^{-y\, t_i}. \tag{29.22}$$

The yield must be found numerically (there is no closed-form solution for coupon-bearing bonds), typically by Newton–Raphson or bisection.

Yield is a summary statistic: convenient for quoting, lossy for risk. Two bonds with the same yield can have very different sensitivities.

Price–yield relationship. P falls as y rises:

- $y > c$: the bond trades at a **discount** ($P < F$). The coupon is below the market rate, so the bond is worth less than face value.
- $y < c$: the bond trades at a **premium** ($P > F$).
- $y = c$: the bond trades at **par** ($P = F$).

Duration: first-order sensitivity

Duration quantifies how much the bond price moves when yields move. It is the rates trader's core hedging unit: if you own a bond with 5-year duration and want to be rate-neutral, you short an offsetting position (a futures contract, another bond, or a swap) whose duration-weighted size matches — so that a parallel shift in rates leaves your book's value unchanged.

Definition 29.8 (Macaulay duration)**.** The *Macaulay duration* of a bond is:

$$D_{\text{Mac}} = \frac{1}{P}\sum_{i=1}^{n} t_i\, c_i\, e^{-y\, t_i} = -\frac{1}{P}\frac{dP}{dy}. \tag{29.23}$$

Derivation. Differentiating Equation (29.22) with respect to y:

$$\frac{dP}{dy} = \sum_{i=1}^{n}(-t_i)\, c_i\, e^{-y\, t_i} = -\sum_{i=1}^{n} t_i\, c_i\, e^{-y\, t_i}. \tag{29.24}$$

Dividing both sides by $-P$:

$$-\frac{1}{P}\frac{dP}{dy} = \frac{\sum_{i=1}^{n} t_i\, c_i\, e^{-y\, t_i}}{P} = D_{\text{Mac}}. \tag{29.25}$$

This is a PV-weighted average of cash-flow times:

$$w_i = \frac{c_i\, e^{-y\, t_i}}{P}, \qquad \sum_{i=1}^{n} w_i = 1, \qquad D_{\text{Mac}} = \sum_{i=1}^{n} t_i\, w_i. \tag{29.26}$$

The key result is:

$$\boxed{\frac{\Delta P}{P} \approx -D_{\text{Mac}} \cdot \Delta y}. \tag{29.27}$$

Intuition

A bond with duration 4.67 loses about 4.67% per 100 bp yield rise. Duration is measured in years but acts as a sensitivity multiplier, not a time.

Properties of duration.

- A zero-coupon bond has duration exactly equal to its maturity.
- Adding coupons *reduces* duration (you get cash back sooner).
- Higher yields reduce duration (distant cash flows are discounted more heavily, shifting weight towards earlier periods).
- Duration of a portfolio is the weighted average of the individual durations, weighted by market value.

Convexity: second-order sensitivity

For large yield moves, we need the second-order correction.

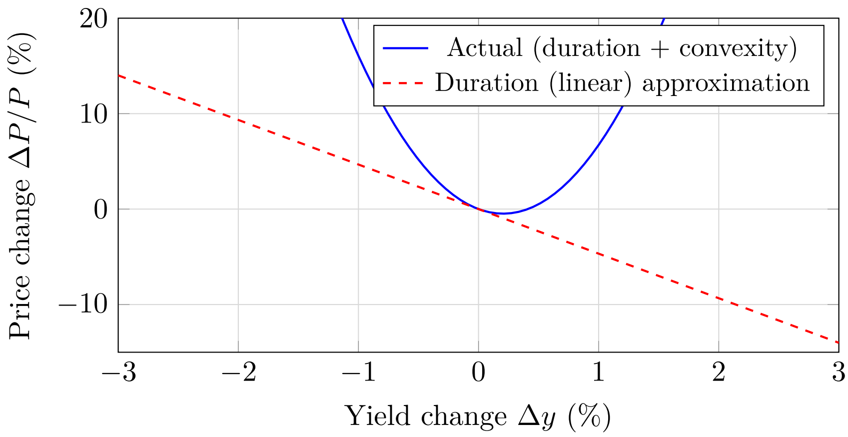
Definition 29.9 (Convexity).

$$C \;=\; \frac{1}{P}\,\frac{d^2P}{dy^2} \;=\; \frac{1}{P}\sum_{i=1}^{n} t_i^2\, c_i\, e^{-y\,t_i}. \tag{29.28}$$

The improved price-change approximation (Taylor expansion to second order) is:

$$\boxed{\frac{\Delta P}{P} \;\approx\; -D_{\text{Mac}}\cdot\Delta y + \tfrac{1}{2}\,C\cdot(\Delta y)^2}. \tag{29.29}$$

For plain vanilla bonds $C > 0$: price falls *less* than duration predicts on rate rises, and rises *more* on rate falls.



The blue curve is convexity in action; the red line (duration only) overstates losses on rises and understates gains on falls.

DV01: the risk metric

Definition 29.10 (DV01 (dollar value of one basis point)).

$$\text{DV01} \;=\; P \times D_{\text{Mac}} \times 0.0001. \tag{29.30}$$

DV01 is the absolute dollar change in bond price per 1 bp (0.01%) yield move — the standard rates-desk risk measure. Portfolio DV01 sums across positions.

Intuition

Traders don't say "duration 4.2"; they say "long \$42,000 DV01." DV01 is the language of P&L.

Worked example: 5-year 3% semi-annual bond

Consider a bond with face value $F = 100$, coupon rate 3% paid semi-annually (so each coupon is \$1.50), maturity $T = 5$ years, and a continuously compounded yield of $y = 4\%$.

Step 1: Price. 10 semi-annual periods: \$1.50 at $t = 0.5, \ldots, 4.5$ and \$101.50 at $t = 5.0$; discount at $e^{-0.04t}$.

t (y)	CF (\$)	PV (\$)	Weight w	tw	t^2w
0.5	1.50	1.4703	0.01542	0.00771	0.00386
1.0	1.50	1.4412	0.01512	0.01512	0.01512
1.5	1.50	1.4126	0.01482	0.02223	0.03334
2.0	1.50	1.3847	0.01452	0.02905	0.05810
2.5	1.50	1.3573	0.01424	0.03559	0.08898
3.0	1.50	1.3304	0.01396	0.04187	0.12560
3.5	1.50	1.3040	0.01368	0.04788	0.16757
4.0	1.50	1.2782	0.01341	0.05363	0.21453
4.5	1.50	1.2529	0.01314	0.05914	0.26613
5.0	101.50	83.1012	0.87170	4.35848	21.79239
Total:		**95.3328**	1.00000	**4.6707**	**22.7656**

Step 2: Results.

$$\text{Price} = 95.33, \tag{29.31}$$
$$D_{\text{Mac}} = 4.6707 \text{ years}, \tag{29.32}$$
$$C = 22.7656, \tag{29.33}$$
$$\text{DV01} = 95.33 \times 4.6707 \times 0.0001 = \$0.0445. \tag{29.34}$$

Step 3: Verification. If the yield rises by 1 bp to $y = 4.01\%$, the exact new price (re-summing all discounted cash flows) is \$95.2883, a change of $-\$0.0445$. Duration predicts $\Delta P = -P \times D \times \Delta y = -95.33 \times 4.6707 \times 0.0001 = -\$0.0445$. Exact match to four decimal places. ✓

For a larger move of +100 bp ($y = 5\%$), duration alone predicts $\Delta P = -\$4.45$, while the exact price is \$91.07, a drop of \$4.26. Including convexity:

$$\Delta P \approx -4.45 + \tfrac{1}{2}(95.33)(22.77)(0.01)^2 = -4.45 + 0.11 = -\$4.34,$$

much closer to the true answer.

Duration and DV01

- **Duration** $= -\frac{1}{P}\frac{dP}{dy}$: the percentage price sensitivity to yield changes.
- **DV01** $= P \times D \times 0.0001$: the dollar price change per 1 bp yield move.
- Duration is a *linear* approximation; **convexity** adds the curvature correction.
- A zero-coupon bond has duration equal to its maturity. Coupons reduce duration.
- **Risk is measured and hedged in DV01**, not in notional. A \$100 M 30-year bond has far more DV01 than a \$100 M 2-year bond.

Modified duration and dollar duration

Macaulay duration assumes continuous compounding. For yield quoted with m periods per year, differentiating $P = \sum c_i(1 + y/m)^{-mt_i}$ gives:

Definition 29.11 (Modified duration)**.**

$$D^* = \frac{D_{\text{Mac}}}{1 + y/m}, \tag{29.35}$$

where m is the number of compounding periods per year. The price-change approximation becomes $\Delta P \approx -P \cdot D^* \cdot \Delta y$.

Continuous compounding gives $D^* = D_{\text{Mac}}$; semi-annual gives D^* slightly smaller.

Dollar duration. $P \times D^*$ is the dollar price change per unit yield change; DV01 = dollar duration /10,000.

Duration-based hedging

To immunise a bond portfolio against small parallel curve shifts, construct a hedge so that total DV01 is zero.

Example 29.12 (Hedging a bond with another bond)**.** You hold \$10 M face value of a 10-year 5% bond with modified duration $D_A^* = 7.5$, priced at $P_A = 103$. You want to hedge using a 2-year 3% bond with modified duration $D_B^* = 1.9$, priced at $P_B = 99$.

Portfolio DV01: $\text{DV01}_A = \$10{,}300{,}000 \times 7.5 \times 0.0001 = \$7{,}725$.

Required hedge face value F_B satisfies:

$$F_B \times \frac{P_B}{100} \times D_B^* \times 0.0001 = \$7{,}725,$$

$$F_B = \frac{7{,}725}{0.99 \times 1.9 \times 0.0001} = \frac{7{,}725}{0.0001881} \approx \$41{,}069{,}644.$$

Short approximately \$41.1 M face of the 2-year bond to neutralise DV01. This protects against parallel shifts only, not curve rotations — hence desks also monitor *key rate DV01* (sensitivity to individual curve points).

Duration-based hedging

To hedge a bond position:

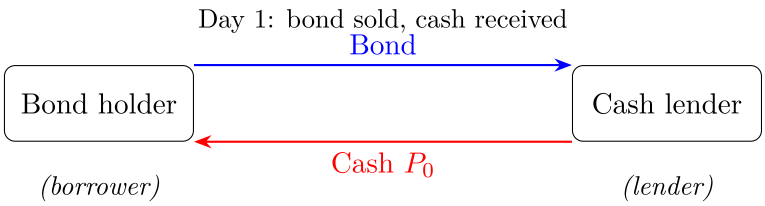
1. Compute the DV01 of the position to be hedged.
2. Choose a hedging instrument and compute its DV01 per unit.
3. Determine the hedge ratio: Hedge notional = Position DV01/Hedge instrument DV01 per unit.
4. *Duration hedge protects against parallel shifts only.* For non-parallel (curve) risk, use multiple hedging instruments at different maturities.

29.5 Repo

A **repurchase agreement** (**repo**) is a short-term secured loan: the borrower sells a bond today and agrees to repurchase it at a slightly higher price shortly. The price difference is interest at the **repo rate**.

Definition 29.13 (Repo)**.** In a repo transaction, party A (the borrower) sells securities to party B (the lender) at price P_0 and agrees to repurchase them at price $P_0(1 + r_{\text{repo}} \times \delta)$ after a period δ. The collateral is the bond; the cash flow is the loan.

Mechanics.



At maturity the trade reverses: the borrower pays $P_0(1 + r_{\text{repo}}\delta)$ and takes back the bond. Key features:

- The repo rate is the overnight secured lending rate, close to the central-bank policy rate (Fed Funds target in the US).
- **Overnight repo** renegotiates daily; **term repo** locks in a rate for a fixed period.
- Repo is how bond traders finance positions — repo bonds out overnight for cash, pay the repo rate ("leveraged book").
- Spread between SOFR (unsecured) and repo reflects collateral quality: governments go at "general collateral," others higher.
- A bond in demand for shorting repos below the general level — trading *on special*.

Intuition

Repo is a pawnshop for bonds: hand over collateral, receive cash, pay interest. The repo rate is the most fundamental short-term rate; SOFR itself is computed from overnight Treasury repo.

29.6 Interest rate swaps

The **interest rate swap** (IRS) is the most-traded derivative in the world; 2024 notional outstanding exceeds \$500 trillion (Hull, 2014).

What is an interest rate swap?

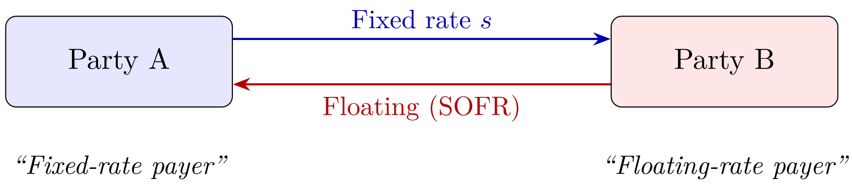
Definition 29.14 (Plain vanilla interest rate swap)**.** A *plain vanilla interest rate swap* is an agreement between two parties to exchange interest payments on a *notional principal* N for a fixed number of years. One party pays a **fixed rate** s; the other pays a **floating rate** (typically SOFR, formerly LIBOR). The notional principal is never exchanged — it is used only to calculate the interest payments.

The two sides of the swap are called **legs**:

- **Fixed leg**: pays $N \times s \times \delta$ at each payment date, where δ is the day-count fraction.
- **Floating leg**: pays $N \times r_{\text{float}} \times \delta$ at each payment date, where r_{float} is the floating rate observed at the *beginning* of the period.

Only the *net* is exchanged: if fixed is \$2.60 M and floating \$2.40 M, the fixed payer sends \$0.20 M.

A picture of a swap.



Example 29.15 (A 5-year IRS)**.** Company A enters a 5-year swap on \$100 M notional:

- A pays 5.13% fixed, semi-annually.
- A receives 6-month SOFR, semi-annually.

Every six months, A pays $100\text{M} \times 0.0513 \times 0.5 = \$2.57\text{M}$ fixed, and receives $100\text{M} \times \text{SOFR}_{6m} \times 0.5$ floating. If 6-month SOFR is 4.80%, A receives \$2.40 M and nets $-\$0.17$ M for that period. If SOFR rises to 5.60%, A receives \$2.80 M and nets $+\$0.23$ M.

Why swaps exist

Three purposes:

1. **Transform liabilities.** A floating-rate borrower pays fixed in the swap to lock in borrowing costs.
2. **Transform assets.** A holder of fixed-rate bonds pays fixed / receives floating, converting the asset to floating-rate to match floating-rate liabilities.
3. **Speculation.** Expect rates to rise: pay fixed, receive floating; no bond purchase needed.

The par swap rate

A swap is worth zero at inception iff the fixed rate equals the **par swap rate**. We derive it from first principles.

Valuing the floating leg. A floating-rate bond resets to par at every payment date, since the next coupon matches the market rate. Hence:

$$B_{\text{float}} = (N + k^*) \cdot D(t^*), \tag{29.36}$$

where k^* is the next floating payment and t^* the next payment date. At inception, $B_{\text{float}} = N$. The PV of the floating leg (netting the notional) is:

$$\text{PV}_{\text{float}} = N - N \cdot D(T) = N\big(1 - D(T)\big). \tag{29.37}$$

Valuing the fixed leg. The fixed leg pays $N \cdot s \cdot \delta_i$ at each payment date t_i. Its present value is:

$$\text{PV}_{\text{fixed}} = N \cdot s \sum_{i=1}^{n} \delta_i \, D(t_i). \tag{29.38}$$

Par condition. At inception, $\mathrm{PV}_{\text{fixed}} = \mathrm{PV}_{\text{float}}$:

$$N \cdot s \sum_{i=1}^{n} \delta_i \, D(t_i) = N\big(1 - D(T)\big), \tag{29.39}$$

and the par swap rate is:

$$\boxed{s \;=\; \frac{1 - D(T)}{\displaystyle\sum_{i=1}^{n} \delta_i \, D(t_i)}}\,. \tag{29.40}$$

The denominator $A = \sum \delta_i D(t_i)$ is the **annuity factor** (or swap **PV01**): the PV of \$1 per annum on the fixed-leg schedule.

IRS par swap rate

The par swap rate is:

$$s = \frac{1 - D(T)}{\displaystyle\sum_{i=1}^{n} \delta_i \, D(t_i)}.$$

- Numerator $= 1 - D(T)$: the PV of the floating leg (net of notional).
- Denominator $= A$: the annuity factor — the PV of receiving \$1 per year on the fixed-leg payment schedule.
- The swap rate is completely determined by the zero curve.

Intuition

A swap is a portfolio of FRAs (each floating payment bets on one forward rate). The par swap rate is the single fixed rate making the package worth zero — equivalently, exchanging a fixed bond for a floating bond, both at par.

Running example: 5-year IRS on \$100 M

Using the bootstrapped curve from Section 29.3, price a 5-year semi-annual IRS on \$100 M notional vs. 6-month SOFR.

Step 1: Discount factors. We need $D(t)$ at each semi-annual date; interpolate linearly in zero rates where not directly bootstrapped:

t (y)	$z(t)$ (%)	$D(t)$	Annuity contribution $0.5 \times D(t)$
0.5	4.16	0.9794	0.4897
1.0	4.40	0.9569	0.4785
1.5	4.55	0.9340	0.4670
2.0	4.70	0.9104	0.4552
2.5	4.83	0.8878	0.4439
3.0	4.83	0.8651	0.4326
3.5	4.96	0.8425	0.4213
4.0	4.96	0.8199	0.4100
4.5	5.10	0.7974	0.3987
5.0	5.10	0.7750	0.3875
		Annuity $A =$	**4.3842**

Step 2: Par swap rate.

$$s = \frac{1 - D(5)}{A} = \frac{1 - 0.7750}{4.3842} = \frac{0.2250}{4.3842} = 5.13\%. \tag{29.41}$$

Step 3: Fixed payment. At the par rate of 5.13%, the fixed payment every six months is:

$$N \times s \times \delta = \$100{,}000{,}000 \times 0.0513 \times 0.5 = \$2{,}566{,}254.$$

Step 4: Verification. PV of fixed leg: $N \times s \times A = \$100\text{M} \times 0.0513 \times 4.3842 = \$22{,}502{,}042$.
PV of floating leg: $N \times (1 - D(5)) = \$100\text{M} \times 0.2250 = \$22{,}502{,}042$.
Swap value at inception: \$0. ✓

Swap value after rates move. If the curve shifts up by 50 bp, the floating leg is worth more; the fixed leg's contractual rate is unchanged. The fixed-rate payer profits: they locked 5.13% when the market now demands 5.63%. For a 1 bp shift the swap value changes by $N \times A \times 0.0001 = \$43{,}842$ — the swap's DV01.

Swap valuation after inception

Once rates move, a traded swap acquires non-zero value. From the fixed-rate payer's perspective:

$$V_{\text{swap}} = \text{PV}_{\text{float}} - \text{PV}_{\text{fixed}} = N(1 - D(T)) - N \cdot s_0 \cdot A_{\text{new}}, \tag{29.42}$$

where s_0 is the original fixed rate and A_{new} is the annuity factor computed with today's (new) zero curve.

Equivalently, V_{swap} equals the change in par rate times the annuity:

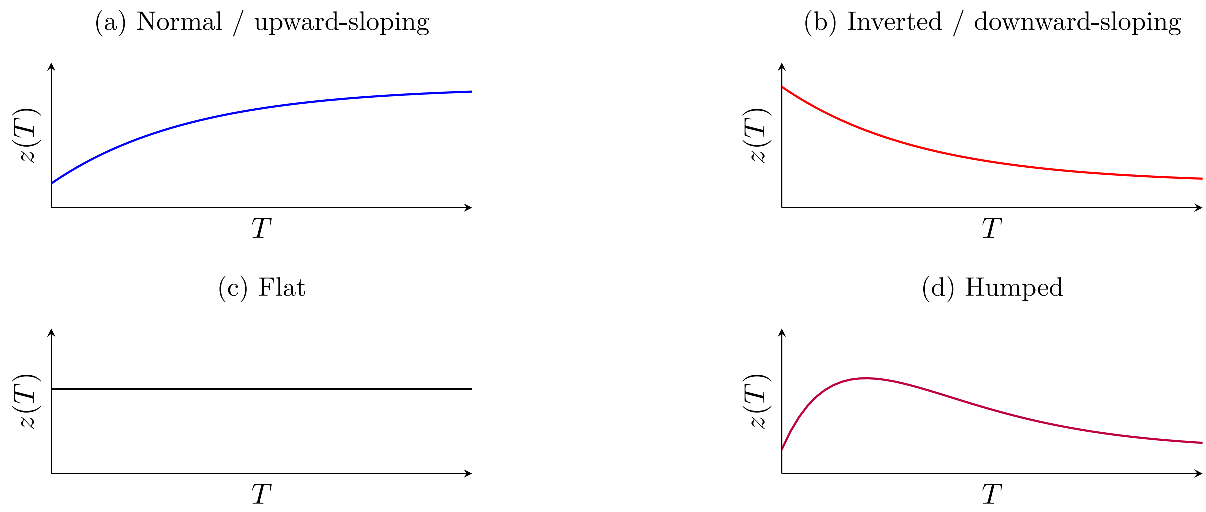
$$V_{\text{swap}} = N \cdot (s_{\text{current}} - s_0) \cdot A_{\text{new}}. \tag{29.43}$$

If par rates have risen above your locked-in rate, your swap (as fixed-rate payer) is in the money.

29.7 Yield curve shapes and term-structure theories

Understanding why the zero curve takes a particular shape is key to reading market conditions.

Four canonical curve shapes



- **Normal (upward-sloping):** most common; reflects the term premium for longer-horizon capital commitment.
- **Inverted:** short exceeds long; often a recession signal (market expects future cuts). Every US recession since 1960 was preceded by a 2s10s inversion.
- **Flat:** all maturities equal; transitional.
- **Humped:** rises then falls — near-term hikes, later cuts.

Three theories of the term structure

Expectations hypothesis. $z(T) = \frac{1}{T}\int_0^T \mathbb{E}[r(t)]\,dt$; forwards are unbiased forecasts. Empirically forwards overpredict spots, suggesting a term premium.

Liquidity preference. Investors prefer short-term (more liquid, less price risk); borrowers must offer a maturity-increasing liquidity premium. Explains why the curve is usually upward-sloping even when flat rates are expected.

Market segmentation. Institutional investors have preferred maturity habitats; curve shape reflects segment-by-segment supply and demand. Explains, e.g., persistent pension-fund demand for 30-year bonds.

Traders take the curve as given (bootstrapped) and hedge without committing to a theory. The theories matter for research commentary, not pricing.

29.8 OIS, basis swaps, and the LIBOR to SOFR transition

Overnight indexed swaps (OIS)

An **overnight indexed swap** (OIS) floats on the geometric average of daily overnight rates rather than a term rate like 3-month LIBOR. Overnight references: effective fed funds (pre-2020) or SOFR (post-2020) in the US; SONIA (UK); €STR (Eurozone).

Why OIS matters.

- Nearly risk-free (overnight, essentially secured).
- The **LIBOR–OIS spread** (3m LIBOR minus 3m OIS) is a stress indicator: ∼10 bp normal, 364 bp at the peak of the 2007–09 crisis.
- Post-2020, clearinghouses discount using OIS curves (SOFR OIS in the US), not LIBOR.

Basis swaps

A **basis swap** has both legs floating against different rates — e.g. 3m SOFR vs. 6m SOFR, or SOFR vs. fed funds. The gap is the **basis spread**. Participants use them to manage benchmark mismatches between assets and liabilities.

The LIBOR to SOFR transition

LIBOR (London Interbank Offered Rate) underpinned $300+ trillion of contracts — the floating rate in most swaps, loans, and rates derivatives.

Why LIBOR was retired (30 June 2023).

1. **Manipulation scandal (2012).** Traders manipulated submissions; Barclays, UBS, Deutsche Bank and others paid billions in fines.
2. **Structural decline.** The interbank market LIBOR measured had shrunk to near-zero volume; submissions became "expert judgment," not transactions.

SOFR as replacement. The Secured Overnight Financing Rate, administered by the NY Fed, is based on overnight Treasury repo ($1 T+ daily volume) — far more robust than LIBOR.

Key differences from LIBOR:

- LIBOR was forward-looking *term* (3m LIBOR set at the start of the 3-month period); SOFR is backward-looking *overnight* (compound daily observations "in arrears" to get a term rate).
- SOFR is *secured* (repo); LIBOR was unsecured. SOFR is systematically lower.
- Transition required amending trillions in legacy contracts.

SOFR is not a drop-in replacement for LIBOR

Because SOFR is overnight and secured while LIBOR was term and unsecured, simply replacing "LIBOR" with "SOFR" in a contract changes its economics. The ISDA fallback protocol adds a credit spread adjustment (typically 26 bp for 3-month tenor) to SOFR when replacing LIBOR in legacy contracts. Getting this adjustment wrong creates P&L breaks.

Prosperity example

Prosperity lacks fixed-income products, but $D(T)$ is universal: any cross-period strategy comparison uses $\text{PV} = \sum c_i D(t_i)$. Build a simple discount curve from the round parameters as step one in multi-period problems. Forward rates show up implicitly in lock-in-now-or-wait decisions: the forward is the break-even rate.

On a Goldman Sachs desk

A GS rates desk rebuilds the zero curve every morning from deposit rates, futures, and swap quotes. The "morning curve build" is the Strat's critical daily deliverable — every swap, bond, swaption, cap, floor, and structured product reprices off it. Wrong curve ⇒ wrong prices and wrong hedge ratios desk-wide.

The Strat owns the curve code (usually SecDB/Slang), interpolation choice (linear-in-zeros simple, cubic spline or monotone convex fancier), and the bootstrap instrument set. Traders flag the curve in real time when it looks off — typically a new instrument trading or an existing one going stale.

A junior Strat's first project is often to improve the build: add an instrument, fix a day-count bug, or switch interpolation method — excellent because it touches the whole rates infrastructure.

Historical note

The first IRS was a 1981 IBM/World Bank deal arranged by Salomon Brothers: IBM had DM and CHF bonds it wanted synthetically converted to dollars; the World Bank wanted DM/CHF but borrowed cheaper in dollars. From there the market grew explosively — \$1 T by 1987, \$60 T by 2000, \$500 T+ by 2024, making swaps the largest derivatives market in the world (larger than any single country's government bond market).

Key facts to memorise

1. **Zero rate** $z(T)$: continuously compounded rate for lending from now to T.
2. **Discount factor**: $D(T) = e^{-z(T)\,T}$ — the price today of \$1 at T.
3. **Forward rate**: $f(T_1, T_2) = \frac{z(T_2)T_2 - z(T_1)T_1}{T_2 - T_1}$. Forward rates exceed zero rates when the curve slopes up.
4. To convert between compounding conventions: $R_c = m \ln(1 + R_m/m)$.
5. Day-count conventions (ACT/360, ACT/365, 30/360) determine actual cash flows. They matter for real money.
6. The zero curve is **bootstrapped** from deposits (short end), futures (middle), and swaps (long end).
7. **Bond price** $= \sum c_i D(t_i)$. The present-value identity is the most important equation in fixed income.
8. **Yield to maturity**: single rate y such that $P = \sum c_i e^{-y t_i}$. A summary statistic, not a fundamental object.
9. **Duration** $= -\frac{1}{P}\frac{dP}{dy}$: percentage price sensitivity to yield. Measured in years.
10. **DV01** $= P \times D \times 0.0001$: dollar price change per basis point. *The* risk metric on a rates desk.
11. **Convexity**: second-order correction — $\frac{\Delta P}{P} \approx -D\,\Delta y + \frac{1}{2}C(\Delta y)^2$.
12. **Repo**: secured overnight borrowing against bonds. SOFR is based on repo transactions.
13. **Par swap rate**: $s = \frac{1 - D(T)}{\sum \delta_i D(t_i)}$. A swap is a portfolio of FRAs.

14. **SOFR** replaced LIBOR in 2023 as the primary USD floating-rate benchmark. OIS is the cleanest benchmark.

Chapter 30

Interest Rate Derivatives and Curve Models

Chapter 29 built the zero curve and priced a plain-vanilla interest rate swap. This chapter prices the *options on* those rates: caps, floors, and swaptions. The models that do this — Black'76, Hull–White, BGM/LMM, and SABR — are the daily tools of a rates Strat. Understanding how they work, how they calibrate, and which to reach for in which situation is the core of what a rates desk does every morning.

Prerequisites

- Chapter 16: Black–Scholes, risk-neutral pricing, d_1/d_2, volatility as input.
- Chapter 17: delta, gamma, vega; delta-hedging.
- Chapter 29: zero rates $z(T)$, discount factors $D(T)$, forward rates $f(T_1, T_2)$, swaps, duration.

Re-read the relevant chapter if any feel hazy.

A concrete scenario before we begin

A corporate treasurer has issued a $500 million floating-rate bond paying 3-month SOFR plus a spread, resetting quarterly for 5 years. They are happy with today's rates but terrified of a spike. They want insurance: if 3-month SOFR exceeds 5%, someone else pays the difference. That contract is a **cap**, and its price depends on a model of how rates might move.

A rates trader at Goldman Sachs, asked to quote a 2Y×10Y receiver swaption struck 50 bps below the forward swap rate, needs:

1. The forward swap rate (from the zero curve of Chapter 29).
2. The implied volatility at that strike (from the **SABR model** calibrated to the swaption market).
3. A pricing formula (**Black'76**) to convert forward, strike, and vol into a dollar price.

This chapter teaches each piece.

Historical note

Interest rate options barely existed before the 1980s. The cap market developed in the US in the early 1980s as floating-rate lending boomed; swaptions followed late in the decade. Fischer Black's 1976 commodity-futures paper was adapted to rates by mid-1980s practitioners, and "Black's model" became the market standard almost overnight. The SABR model (Hagan, Kumar, Lesniewski, Woodward, 2002) added the smile Black could not capture and remains dominant on rates desks today (Hagan et al., 2002).

30.1 Caps, floors, and swaptions

The three most heavily traded rate options — *caps*, *floors*, *swaptions* — account for most rate-option volume on dealer desks. Each is built from a simple block; the trick is seeing how the pieces fit together.

Caplets and caps

Recall from Chapter 29 that a floating-rate borrower pays a reference rate (e.g. SOFR) that resets each period. A **caplet** is a call option on one future fixing: it pays off if the rate exceeds a pre-agreed *cap rate* (the strike).

Definition 30.1 (Caplet)**.** A *caplet* with strike K, notional N, reset T_i, payment $T_{i+1} = T_i + \tau$ (accrual τ, e.g. 0.5 yr for semi-annual) pays at T_{i+1}:

$$\text{Caplet payoff} \;=\; N\,\tau\,\max\bigl(L(T_i, T_i, T_{i+1}) - K,\; 0\bigr), \tag{30.1}$$

where $L(T_i, T_i, T_{i+1})$ is the floating rate fixed at T_i for $[T_i, T_{i+1}]$.

τ converts an annual rate into a dollar amount over the accrual period; the max makes it an option.

Definition 30.2 (Cap)**.** A *cap* is a strip of caplets, one per reset date — literally a portfolio of independent call options on the floating rate. A 3-year cap on 6-month SOFR with semi-annual resets contains 6 caplets (months 0, 6, 12, 18, 24, 30), each paying at the end of its accrual period; a 5-year quarterly-reset cap would contain 20 caplets.

Intuition

A cap is insurance for a floating-rate borrower: pay a premium upfront; if rates rise above K, the cap pays the difference, capping your effective borrowing rate at K.

Floorlets and floors

A **floorlet** is the mirror image: a put option on a single floating-rate fixing.

Definition 30.3 (Floorlet and floor)**.** A *floorlet* with strike K has payoff at T_{i+1}:

$$\text{Floorlet payoff} \;=\; N\,\tau\,\max\bigl(K - L(T_i, T_i, T_{i+1}),\; 0\bigr). \tag{30.2}$$

A *floor* is a strip of floorlets, one per reset date.

A floor protects a floating-rate *lender* against rates falling below K.

Cap–floor parity. Long cap and short floor at the same strike replicate a floating-for-fixed swap:

$$\text{Cap}(K) - \text{Floor}(K) = \text{Swap (pay fixed } K). \tag{30.3}$$

The rates analogue of put–call parity (Chapter 15). At the par swap rate $K = s_{\text{par}}$, cap and floor have equal value (swap worth zero at inception).

Cap–floor parity

$$\text{Cap}(K) - \text{Floor}(K) = \text{Payer swap}(K).$$

When K equals the par swap rate, cap and floor values are equal. This is the interest-rate analogue of put–call parity.

Swaptions

A **swap** exchanges a floating rate (e.g. SOFR) for a fixed rate: a company with floating-rate debt swaps to fixed to make its interest costs predictable, and the dealer on the other side earns the bid–offer. A **swaption** is the option — not the obligation — to enter such a swap at a pre-agreed fixed rate on a future date, letting a borrower lock in worst-case funding costs today while walking away if rates move in their favour.

Definition 30.4 (Swaption)**.** A *swaption* gives the right (not obligation) to enter a swap at a pre-agreed fixed rate (strike) on a future date (expiry).

- **Payer swaption**: right to pay fixed / receive floating. Profits when rates rise.
- **Receiver swaption**: right to receive fixed / pay floating. Profits when rates fall.

A swaption has two time dimensions: the **expiry** (when you decide to exercise) and the **tenor** (length of the underlying swap). Standard notation is "expiry × tenor": a **1Y×5Y payer swaption** gives the right, one year from now, to enter a 5-year payer swap.

Swaption payoff. At expiry T_0, a payer-swaption holder exercises if the par swap rate $s(T_0)$ exceeds K. The exercise value is the PV of the annuity stream $(s(T_0) - K)$:

$$\text{Payer swaption payoff at } T_0 \;=\; N\,\max\!\big(s(T_0) - K,\ 0\big)\,A(T_0), \tag{30.4}$$

where $A(T_0) = \sum_{i=1}^{n} \tau_i\, D(T_0, T_i)$ is the **annuity factor** (present value of a basis point on the swap).

Caps vs. swaptions

Caps and swaptions are *not* interchangeable. A cap is a basket of independent options on individual forward rates (one caplet per period). A swaption is a single option on a *swap rate* — a weighted average of forwards. Forward-rate correlation matters enormously for swaptions, not for caps.

30.2 Black'76 caplet pricing

Black–Scholes (Chapter 16) models spot as GBM. The rates analogue is **Black'76**, developed by Fischer Black for commodity futures (Black, 1976). The idea is simple: model the *forward rate* as the underlying and reuse the Black–Scholes formula.

Setup: forward rate as underlying

Under the T_{i+1}-forward measure $\mathbb{Q}^{T_{i+1}}$, the forward rate $F(t; T_i, T_{i+1})$ is a martingale (a standard result of arbitrage-free pricing; stated without proof). Black's model assumes it follows GBM under this measure:

$$dF \;=\; \sigma_{\text{Black}}\, F\, dW^{T_{i+1}}, \tag{30.5}$$

where σ_{Black} is constant and $W^{T_{i+1}}$ is a Brownian motion under the forward measure. There is no drift: the forward is already a martingale, so its expectation equals today's forward $F(0; T_i, T_{i+1})$. Because Equation (30.5) has the same form as Black–Scholes GBM, the pricing formula follows immediately.

The Black'76 caplet formula

Black'76 caplet formula

The price at time 0 of a caplet with strike K, notional N, forward rate F, accrual period τ, option expiry T, and Black volatility σ is:

$$V_{\text{caplet}} = D(T+\tau)\,\tau\,N\,\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big], \tag{30.6}$$

where

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}, \tag{30.7}$$

$\Phi(\cdot)$ is the standard normal CDF, and $D(T+\tau)$ is the discount factor from today to the payment date $T+\tau$.

Same structure as the Black–Scholes call (Chapter 16): the forward rate F replaces the forward stock price Se^{rT}, and discounting is to the *payment date* $T+\tau$, not the option expiry T.

Floorlet formula. By cap–floor parity:

$$V_{\text{floorlet}} = D(T+\tau)\,\tau\,N\,\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big]. \tag{30.8}$$

Swaption formula. Black also prices swaptions. A payer swaption with strike K, annuity A, and forward swap rate s:

$$V_{\text{payer}} = N\,A\,\big[s\,\Phi(d_1) - K\,\Phi(d_2)\big], \tag{30.9}$$

with d_1, d_2 from Equation (30.7) using swaption vol σ_{swap} and s in place of F.

Worked example: pricing a caplet

Parameters:

Parameter	Value
Notional N	\$1,000,000
Strike K	4.0%
Forward SOFR rate F	4.5%
Black volatility σ	20%
Option expiry T	1.0 year
Accrual period τ	0.5 years
Continuously compounded zero rate	4.3% (flat)

Step 1: Discount factor. Pays at $T+\tau = 1.5$ yr: $D(1.5) = e^{-0.043\times 1.5} = 0.93754$.

Step 2: d_1, d_2.

$$d_1 = \frac{\ln(1.125) + 0.02}{0.20} = \frac{0.13778}{0.20} = 0.6889, \qquad d_2 = 0.6889 - 0.20 = 0.4889.$$

Step 3: Normal CDFs. $\Phi(0.6889) = 0.7546$, $\Phi(0.4889) = 0.6875$.

Step 4: Apply the formula.

$$\begin{aligned} V_{\text{caplet}} &= D(1.5)\,\tau\,N\,\left[F\,\Phi(d_1) - K\,\Phi(d_2)\right] \\ &= 0.93754 \times 0.5 \times 1{,}000{,}000 \times \left[0.045 \times 0.7546 - 0.04 \times 0.6875\right] \\ &= 0.93754 \times 0.5 \times 1{,}000{,}000 \times \left[0.033957 - 0.027500\right] \\ &= 0.93754 \times 0.5 \times 1{,}000{,}000 \times 0.006457 \\ &= \$3{,}025. \end{aligned} \tag{30.10}$$

The caplet is worth about **$3,025** ($\approx$ 30.25 bps of notional): the upfront premium for protection against 6-month SOFR exceeding 4% at the 1-year reset.

Intuition

The caplet is forward ITM (4.5% $>$ 4.0% strike), hence $\Phi(d_1), \Phi(d_2) > 0.5$. The dollar value is modest because the option pays on a single 6-month period of the notional, not the full notional as in an equity option.

Python implementation

```python
import math
from scipy.stats import norm

def black76_caplet(F, K, sigma, T, tau, D_pay, N=1e6):
    """Black'76 caplet: F=forward, K=strike, sigma=Black vol,
    T=expiry (yr), tau=accrual (yr), D_pay=DF to T+tau, N=notional."""
    d1 = (math.log(F / K) + 0.5 * sigma**2 * T) / (sigma *
        math.sqrt(T))
    d2 = d1 - sigma * math.sqrt(T)
    return D_pay * tau * N * (F * norm.cdf(d1) - K * norm.cdf(d2))

# Worked example
D = math.exp(-0.043 * 1.5)              # 0.93754
price = black76_caplet(0.045, 0.04, 0.20, 1.0, 0.5, D)
print(f"Caplet = ${price:,.2f}")       # $3,025.11
```

Listing 30.1: Black'76 caplet pricer

30.3 Hull–White short-rate model (outline)

Black prices each caplet or swaption independently with its own vol and cannot enforce consistency across instruments: the vol fitting a 1Y caplet need not match the vol fitting a 1Y×5Y swaption. For exotics (Bermudan swaptions, callable bonds, range accruals) we need a model of the entire term structure, evolving all rates simultaneously.

The **Hull–White model** (Hull and White, 1990) is the workhorse one-factor short-rate model. It extends Vasicek/Ornstein–Uhlenbeck with a time-dependent drift that fits today's zero curve exactly.

The model

Hull–White model

The short rate r_t follows:

$$dr_t = \big[\theta(t) - a\, r_t\big]\, dt + \sigma\, dW_t, \tag{30.11}$$

where:

- $a > 0$ is the **mean-reversion speed**: the rate is pulled back towards a time-dependent level $\theta(t)/a$.
- $\sigma > 0$ is the **short-rate volatility**: the instantaneous standard deviation of rate changes.
- $\theta(t)$ is a **deterministic function of time**, chosen so that the model exactly reproduces today's zero curve.

Why $\theta(t)$ matters. Plain Vasicek has constant θ, so its implied zero curve is determined by (a, θ) alone and generally will not match the market. Hull–White replaces constant θ with $\theta(t)$ calibrated to fit the market curve exactly (Hull, 2014):

$$\theta(t) \;=\; \frac{\partial f^M(0,t)}{\partial t} + a\, f^M(0,t) + \frac{\sigma^2}{2a}(1 - e^{-2at}), \tag{30.12}$$

where $f^M(0,t)$ is the market instantaneous forward at maturity t.

Bond pricing. Under Hull–White, a zero-coupon bond has closed-form price:

$$P(t,T) = A(t,T)\, \exp[-B(t,T)\, r_t], \tag{30.13}$$

where $B(t,T) = \frac{1}{a}(1 - e^{-a(T-t)})$ and $A(t,T)$ is set by the initial term structure (see (Hull, 2014), Ch 31). This tractability is the model's strength: bonds, caps, and European swaptions all have closed-form prices.

Swaption pricing. A European swaption is an option on the fixed-leg coupon bond. Jamshidian's trick reduces it to a portfolio of zero-coupon bond options, each with a closed-form Hull–White price (details in (Hull, 2014), Ch 31). The practical fact:

Hull–White for European swaptions

European swaptions have a closed-form Hull–White price. Free parameters (a, σ) are calibrated to market swaptions. With only two parameters, the model fits a limited slice of the swaption matrix — typically one row or column.

Strengths and limitations

Strengths	Limitations
Exact fit to today's zero curve	One factor: all rates move in parallel
Closed-form bond and swaption prices	Cannot fit the full swaption matrix
Mean reversion prevents rates drifting to unreasonable levels	Normal model: rates can go negative (a feature post-2010, a bug before)
Fast calibration (2 parameters)	No smile / skew

The lack of smile is Hull–White's main limitation. It prices European options with a single vol per strike — fine for ATM risk but inadequate for away-from-ATM instruments. Smile-consistent pricing needs SABR (Section 30.5).

Numerical implementation: the trinomial tree

For Bermudan swaptions and other early-exercise products, Hull–White is implemented on a **trinomial tree**: at each step Δt the rate can move up, flat, or down, with:

- branch spacing $\Delta r = \sigma\sqrt{3\Delta t}$;
- probabilities chosen to match the model's mean and variance;
- $\theta(t)$ embedded by placing the central node at the mean-reverting level.

Pricing is backward induction from terminal nodes (the rates analogue of the equity binomial tree, Chapter 16, with three branches). At each Bermudan exercise date, take the max of continuation and exercise value. This is how Bermudan swaptions are priced in practice.

Intuition

The Hull–White trinomial tree is the rates analogue of Cox–Ross–Rubinstein. Key difference: it is calibrated to the entire initial term structure, not just one spot and vol — $\theta(t)$ bends the tree so today's bond prices are recovered exactly.

30.4 The BGM / Libor Market Model (LMM) as a concept

Hull–White works with the abstract short rate r_t, but traders observe and quote *discrete-tenor forward rates* (3M SOFR, 6M SOFR, ...). The **Brace–Gątarek–Musiela (BGM) model**, a.k.a. the **Libor Market Model (LMM)**, models these observable forwards directly (Brace et al., 1997).

The SDE

The model specifies forward rates $\{F_k(t)\}$ for $k = 1, \ldots, n$, where $F_k(t)$ is the time-t forward for period $[T_k, T_{k+1}]$. Under the T_{k+1}-forward measure:

$$dF_k(t) \;=\; \sigma_k(t)\, F_k(t)\, dW_k^{T_{k+1}}(t), \tag{30.14}$$

where $\sigma_k(t)$ is a deterministic vol function. Under any other measure, drift terms depending on forward-rate correlations appear (derived in (Hull, 2014), Ch 32).

Key property: caplets are Black-priced by construction. Each F_k is log-normal under its own forward measure, so every caplet is priced exactly by Black'76 with vol $\sigma_k(T_k)$ — read directly from market cap quotes. BGM is built to reproduce the cap market automatically.

What about swaptions?

Swaptions are harder. A swap rate is a weighted average of forwards:

$$s(t) = \frac{\sum_k \tau_k \, D(t, T_{k+1}) \, F_k(t)}{\sum_k \tau_k \, D(t, T_{k+1})}. \tag{30.15}$$

Even with each F_k log-normal, $s(t)$ is not (ratio of sums of log-normals). Swaption prices in BGM have no closed form. Two practical approaches:

1. **Monte Carlo**: simulate joint forward evolution, compute swap rate at expiry, average the discounted payoff. Accurate but slow; used for exotics.
2. **Analytical approximations**: freeze the weights at $t = 0$, yielding an approximately log-normal swap rate and a Black-like swaption formula — accurate enough for many purposes (Hull, 2014).

BGM / LMM: key facts

- Models observable forwards directly, not the short rate.
- Caplets are Black-priced by construction — fits the cap market automatically.
- Swaptions need MC or analytical approximations.
- Requires a forward-rate **correlation matrix** (the main calibration challenge).
- Industry standard for exotics (Bermudans, CMS, range accruals).

BGM does not mean "log-normal swaptions"

BGM does *not* imply log-normal swap rates. The forwards are log-normal; the swap rate (a weighted average) is not. Pricing swaptions as if the swap rate were log-normal is a convenient approximation, not an exact result.

30.5 SABR: stochastic volatility for rates

Black assumes constant vol per forward, but the market quotes strike-dependent implied vols — the smile/skew seen for equities in Chapter 18. The **SABR model** (Stochastic Alpha Beta Rho) is the rates-desk standard for capturing this (Hagan et al., 2002).

SABR does not replace Black'76 or BGM. It is a volatility model producing a Black-vol smile across strikes; those vols then feed Black's formula for prices.

The SABR SDEs

Two coupled SDEs for the forward F_t and its instantaneous vol α_t:

SABR model

$$dF_t = \alpha_t\, F_t^{\beta}\, dW_1(t), \tag{30.16}$$
$$d\alpha_t = \nu\, \alpha_t\, dW_2(t), \tag{30.17}$$

with $F_0 = F$, $\alpha_0 = \alpha$, and correlated Brownians $dW_1\, dW_2 = \rho\, dt$, $\rho \in [-1, 1]$. Four parameters:

- $\alpha > 0$: initial vol level.
- $\beta \in [0, 1]$: backbone / CEV exponent.
- $\rho \in (-1, 1)$: rate-vol correlation.
- $\nu > 0$: vol-of-vol.

Special cases. $\beta = 1, \nu = 0$: log-normal (Black). $\beta = 0, \nu = 0$: normal (Bachelier). $\beta = 0.5$: square-root CEV backbone, common for rates.

Hagan's approximation formula

Hagan et al. derived a closed-form approximation for the Black implied vol as a function of strike (Hagan et al., 2002)— no numerical SDE solving needed. For $F \neq K$:

$$\sigma_{\text{Black}}(K) \;=\; \frac{\alpha}{(FK)^{(1-\beta)/2}\Big[1 + \frac{(1-\beta)^2}{24}\ln^2\frac{F}{K} + \frac{(1-\beta)^4}{1920}\ln^4\frac{F}{K}\Big]} \cdot \frac{z}{x(z)} \cdot \Big[1 + \epsilon\, T\Big], \tag{30.18}$$

where

$$z = \frac{\nu}{\alpha}\,(FK)^{(1-\beta)/2}\,\ln\frac{F}{K}, \tag{30.19}$$
$$x(z) = \ln\left(\frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho}\right), \tag{30.20}$$
$$\epsilon = \frac{(1-\beta)^2}{24}\,\frac{\alpha^2}{(FK)^{1-\beta}} + \frac{\rho\beta\nu\alpha}{4\,(FK)^{(1-\beta)/2}} + \frac{(2 - 3\rho^2)\,\nu^2}{24}. \tag{30.21}$$

At the money ($K = F$), the formula simplifies to:

$$\sigma_{\text{ATM}} \;=\; \frac{\alpha}{F^{1-\beta}}\,\Big[1 + \epsilon_{\text{ATM}}\, T\Big], \tag{30.22}$$

with ϵ_{ATM} given by Equation (30.21) evaluated at $K = F$.

Intuition

SABR gives a **recipe**: plug in $(\alpha, \beta, \rho, \nu)$ and strike K, out comes the Black vol. The four parameters shape the smile:

- β = **backbone**: how ATM vol moves with the forward.
- ρ = **skew**: $\rho < 0$ means vol rises as rates fall (typical for rates).
- ν = **wings**: higher $\nu \Rightarrow$ fatter tails, richer OTM options.
- α = overall **level** of ATM vol.

One-liner: "β = backbone, ρ = skew, ν = wings."

Calibration workflow

Given market implied vols at several strikes for a given expiry/tenor, find $(\alpha, \beta, \rho, \nu)$ that best reproduce them.

Step 1: Fix β. β is almost always fixed: $\beta = 0.5$ (most popular for rates, square-root backbone), $\beta = 0$ (normal SABR, for negative-rate environments), $\beta = 1$ (log-normal, for equities/commodities). Three parameters remain: α, ρ, ν.

Step 2: Fit α from ATM vol. Given $\sigma^{\text{mkt}}_{\text{ATM}}$ and (β, ρ, ν), invert Equation (30.22) for α. Inner loop: for each trial (ρ, ν), solve for α, then evaluate the smile.

Step 3: Minimise smile error. Search (ρ, ν) to minimise squared error vs. market vols across quoted strikes:

$$\min_{\rho,\nu} \sum_{j=1}^{m} \left[\sigma_{\text{SABR}}(K_j; \alpha(\rho, \nu), \beta, \rho, \nu) - \sigma_{\text{mkt}}(K_j)\right]^2. \tag{30.23}$$

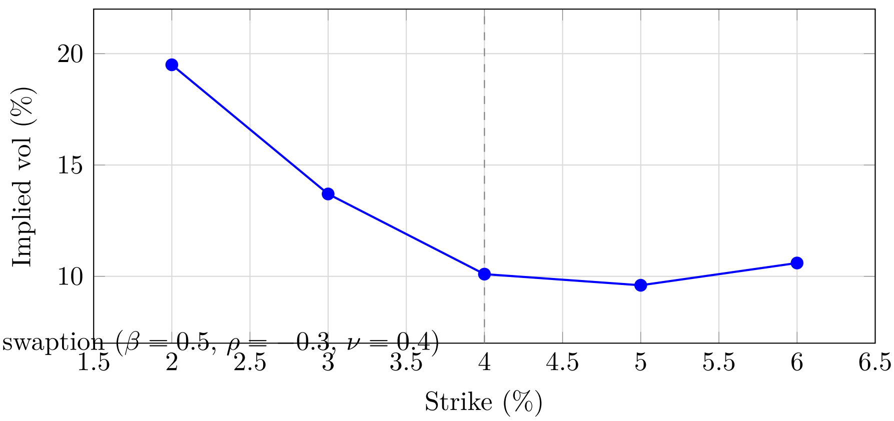
A 2D optimisation, fast enough to run in milliseconds — critical since rates desks recalibrate SABR at every (expiry, tenor) cell of the swaption matrix every morning.

Worked example: qualitative SABR smile

Consider a 1Y×5Y swaption with $F = 4.0\%$, $T = 1$ year. Fix $\beta = 0.5$; calibration yields $\alpha = 0.02$, $\rho = -0.30$, $\nu = 0.40$. Hagan's formula at five strikes:

Strike	Moneyness	SABR IV (%)	Shape role
2.00%	−200 bps	19.5	Left wing (ν) + skew ($\rho < 0$)
3.00%	−100 bps	13.7	Skew
4.00%	ATM	10.1	Level (α)
5.00%	+100 bps	9.6	Slight downtick from skew
6.00%	+200 bps	10.6	Right wing (ν)

Clear *negative skew*: vol is higher for low strikes (rates falling) than high (rates rising), driven by $\rho = -0.30$ — when the forward drops, α_t rises, pushing vol higher. Wings are elevated by $\nu = 0.40$: higher vol-of-vol ⇒ fatter tails ⇒ richer OTM options.



SABR smile: 1Y×5Y swaption ($\beta = 0.5$, $\rho = -0.3$, $\nu = 0.4$)

The negative skew is visible; the right wing is slightly elevated from ν.

Effect of changing ρ. Flipping ρ from -0.30 to $+0.30$ reverses the skew: vol higher at high strikes, lower at low. In rates, ρ is almost always negative (flight-to-quality: rates fall, vol spikes); in some commodities it is positive.

Effect of changing ν. Raising ν from 0.4 to 0.8 lifts both wings dramatically while leaving ATM nearly unchanged — higher vol-of-vol fattens the tails. $\nu = 0$ collapses the smile to flat (Black).

SABR is a smile interpolator

SABR is a snapshot of the smile — it does not describe how the smile evolves over time (its dynamics are known to be unrealistic; the backbone moves too predictably). Many desks overlay SABR with empirical adjustments for hedging. Use SABR for *marking* (consistent vols across strikes), not for predicting smile moves.

SABR parameter intuition

SABR parameters — summary

Parameter	Name	Interpretation
α	Level	Overall ATM vol level.
β	Backbone	How ATM vol moves with the forward. $\beta = 1$: % vol constant; $\beta = 0$: bp vol constant.
ρ	Skew	Rate-vol correlation. $\rho < 0$ (typical): vol rises as rates fall $\Rightarrow$ downward skew.
ν	Vol-of-vol	Curvature (wings). Higher $\nu \Rightarrow$ fatter tails $\Rightarrow$ more expensive out-of-the-money options.

30.6 Connecting the models

The four models serve different purposes:

Model	**What it models**	**Primary use**	**Limitation**
Black'76	Single forward rate (log-normal)	Vanilla caps/floors, European swaptions	Flat smile per instrument
Hull–White	Short rate (normal, mean-reverting)	Bermudan swaptions, callable bonds	One factor, no smile
BGM/LMM	All forward rates jointly	Exotic rate products, path-dependent payoffs	Needs correlation matrix, MC simulation
SABR	Smile/skew for a single forward	Strike interpolation, vol cube	Static snapshot, not dynamics

A rates desk uses all four in concert:

1. **SABR** generates the implied-vol smile at each expiry/tenor grid point.
2. **Black'76** converts those implied vols into prices for vanilla caps, floors, and European swaptions.
3. **Hull–White** or **BGM/LMM** prices exotics (Bermudans, CMS caps, callable bonds) using the SABR-calibrated smile as input.

The swaption matrix and vol cube

The **swaption matrix** is a 2D grid with expiries as rows and swap tenors as columns; each cell holds the ATM Black vol for that (expiry, tenor) pair. Typical shape:

	Swap tenor				
Expiry	1Y	2Y	5Y	10Y	30Y
1M	45.2	38.1	24.5	18.3	14.1
3M	42.8	36.5	23.8	17.9	13.8
6M	39.1	34.2	22.9	17.2	13.5
1Y	35.6	31.5	21.4	16.5	13.1
2Y	30.2	27.8	19.8	15.6	12.7
5Y	24.1	22.5	17.3	14.2	12.0
10Y	19.8	18.9	15.1	13.0	11.3

(Values in % Black vol. These are illustrative.)

Pattern: vols are highest top-left (short expiries on short-tenor swaps) and lowest bottom-right. Short-dated options have high % vol because rates can move sharply over a short horizon relative to their level; long-tenor swap rates are averages of many forwards and are smoother.

The **vol cube** extends this to 3D by adding a strike axis. At each (expiry, tenor) cell, replace the single ATM vol with a SABR-calibrated smile — a function from strike to implied vol:

$$(\text{expiry},\ \text{tenor},\ \text{strike}) \longmapsto \sigma_{\text{Black}}.$$

This 3D object is the rates desk's central data structure. Every vanilla price, exotic calibration, and risk report begins with a query to the vol cube.

30.7 Prosperity and industry connections

Prosperity example

The SABR workflow — observe prices, extract implied vols, fit a parametric smile, re-price — mirrors the vol-surface calibration of Chapter 18. In Prosperity's Volcanic Rock Vouchers you did exactly this: observe bot option prices, back out vols, fit, quote. The rates version applies the same logic to caps and swaptions.

On a Goldman Sachs desk

On a GS rates desk, the Strat owns the **SABR calibration pipeline**. Daily workflow:

1. **Pre-open:** pull overnight broker quotes and exchange-traded cap/swaption prices (Black vols at standard strikes).

2. **Calibration:** at each (expiry, tenor) cell, fit (α, ρ, ν) with β fixed. Result: the **vol cube** — expiry × tenor × strike.

3. **Pricing:** the cube feeds the pricing engine (SecDB at GS). Traders price any rate option by interpolating the SABR surface.

4. **Risk:** Greeks (delta, vega, gamma) via bumping the surface; vega is reported per expiry/tenor bucket. The Strat watches for calibration breaks (sudden jumps in ρ or ν from bad input data).

"The Strat runs the vol cube" means exactly this.

Key facts to memorise

Chapter summary — interest rate derivatives

1. **Cap** = strip of caplets (calls on forwards); **floor** = strip of floorlets (puts). Cap − Floor = payer swap at the same strike.

2. A **swaption** is an option to enter a swap. Payer = call on swap rate; receiver = put.

3. **Black'76** prices caplets/swaptions by treating the forward (swap) rate as a log-normal martingale under the forward measure. Same structure as Black–Scholes.

4. **Hull–White**: $dr = [\theta(t) - ar]\,dt + \sigma\,dW$, with $\theta(t)$ fitting today's zero curve exactly. Closed-form bonds and swaptions; no smile.

5. **BGM/LMM** models forwards directly as log-normal. Caplets are Black-priced by construction; swaptions need MC or approximations. Industry standard for exotics.

6. **SABR** ($dF = \alpha F^{\beta} dW_1$, $d\alpha = \nu\alpha\,dW_2$, $\rho =$ Corr) is the standard smile model; Hagan's formula gives closed-form implied vol vs. strike.

7. SABR parameters: $\alpha =$ level, $\beta =$ backbone, $\rho =$ skew, $\nu =$ wings. Fix β, fit the rest.

8. The **vol cube** (expiry × tenor × strike) comes from SABR at each grid point — the central rates-desk object.

9. Flow: SABR → smile → Black'76 prices → Hull–White / BGM for exotics.

10. Rates Strats own the pipeline: quotes in, vol cube out, pricing and risk downstream.

Chapter 31

Equity Derivatives Beyond Vanilla

Chapter 16 gave Black–Scholes; Chapter 18 showed the market violates its flat-vol assumption, producing a surface. This chapter covers three progressively richer models of that surface— *local volatility* (Dupire), *stochastic volatility* (Heston), and their combination *local-stochastic volatility* (LSV)—then derives the variance swap. The variance-swap replication formula, promised in Chapter 18, is the centrepiece: it ties together the vol surface, the $1/K^2$ option-strip hedge, and the desk workflow for trading pure volatility.

Prerequisites

From earlier in the book:

- Chapter 16: Black–Scholes PDE and formula, risk-neutral pricing under GBM, the assumptions (constant σ, continuous hedging, no dividends).
- Chapter 17: delta, gamma, vega, theta; the delta-hedged P&L identity $d\Pi \approx \frac{1}{2}\,\Gamma\,S^2\,(\sigma_{\mathrm{r}}^2 - \sigma_{\mathrm{i}}^2)\,dt$; long gamma profits when realised vol exceeds implied.
- Chapter 18: implied vol, smile/skew, variance-swap definition (Theorem 18.6), and the schematic replication (Equation (18.4)) whose derivation we give here.

Re-read those chapters if hazy.

31.1 Local volatility (Dupire)

Black–Scholes assumes constant σ. The simplest generalisation lets volatility depend on the underlying's level and on time:

$$dS_t \;=\; r\,S_t\,dt \;+\; \sigma_{\mathrm{loc}}(S_t, t)\,S_t\,dW_t, \tag{31.1}$$

where $\sigma_{\mathrm{loc}}(S, t)$ is the **local volatility function**—a deterministic surface that can in principle be chosen so the model reproduces every observed European-option price simultaneously. Because σ_{loc} is a known function of (S, t), the model is Markovian and the Kolmogorov forward equation pins down the risk-neutral density at every maturity.

Dupire's formula

Dupire (1994) and Derman and Kani (1994) independently found an explicit expression for σ_{loc}^2 in terms of the market call surface $C(K, T)$:

Dupire's local volatility formula

If European call prices $C(K,T)$ are known for all strikes $K > 0$ and maturities $T > 0$, the unique diffusion coefficient that reproduces those prices is

$$\sigma_{\text{loc}}^2(K,T) = \frac{\dfrac{\partial C}{\partial T} + rK\dfrac{\partial C}{\partial K}}{\dfrac{1}{2}K^2\dfrac{\partial^2 C}{\partial K^2}}. \tag{31.2}$$

Reading the formula.

- The *numerator* measures how the call price changes with maturity (plus a drift correction)—the rate at which value accrues with T encodes the local variance needed at (K,T).
- The *denominator* is $\frac{1}{2}K^2$ times the butterfly spread. By Breeden–Litzenberger, $\partial^2 C/\partial K^2 = e^{-rT}p(K,T)$, so the denominator normalises by the risk-neutral probability of landing near K.
- The ratio answers: given where the market thinks S_T can be, how much diffusion does the model need at that point?

In practice. Dupire is applied not to raw prices but to a smooth parametric fit of the implied-vol surface (SVI/SABR from Chapter 18); partial derivatives are computed analytically from the fit, avoiding the noise of finite-differencing sparse quotes.

Local vol in practice: from the surface to a function

The desk workflow for local-vol calibration:

1. **Fit a smooth implied-vol surface** (SVI, SABR, or a spline) to market quotes, giving continuous $\sigma_{\text{imp}}(K,T)$.
2. **Convert to call prices** via Black–Scholes.
3. **Apply Dupire's formula.** Compute the three partials in Equation (31.2) analytically or by finite differences to evaluate σ_{loc}^2 on a grid.
4. **Interpolate** the grid for pricing.

Qualitative example. A 3-month surface with 25-delta put vol 25%, ATM 20%, 25-delta call 17% yields a Dupire local-vol function higher at low spot ($\approx 28\%$ at $S = 3{,}600$) and lower at high spot ($\approx 15\%$ at $S = 4{,}400$): the skew is "explained" by assigning more diffusion where the market prices fat tails.

Local vol vs implied vol

Local and implied vol are distinct quantities (both depend on (K,T)). Implied vol is the constant σ that reproduces a single option price via Black–Scholes; local vol is the *instantaneous* diffusion coefficient at (S,t) in a model reproducing *all* prices. Roughly, implied vol is a path-weighted average of local vol from S_0 to K. A Dupire (1994) rule of thumb: local vol $\approx$ twice implied vol minus the ATM level (near-ATM, first order).

Local vol is static

Local vol fits today's surface exactly by construction but assumes $\sigma_{\text{loc}}(S,t)$ is deterministic and fixed at time 0. Tomorrow's surface will differ, and the model has no mechanism for the change: it predicts skew flattens with maturity far faster than markets actually exhibit. Traders say local vol gets statics right but dynamics wrong. Delta-hedging exotics with local-vol Greeks leaves you systematically mis-hedged when the smile shifts—hence Heston and LSV.

Intuition

Local vol is a *photograph* of today's surface: pixel-perfect but frozen. Heston is a *physical model*: may miss wrinkles but has built-in dynamics (mean-reversion, vol-of-vol, spot–vol correlation) that produce plausible future surfaces. LSV combines both—photo today, physics tomorrow.

31.2 Heston stochastic volatility

Local vol makes volatility a deterministic function of spot and time. Heston (Heston, 1993) takes the opposite approach: volatility is itself a random process, correlated with the spot.

The Heston SDE

Under the risk-neutral measure, the Heston dynamics are:

$$\begin{aligned} dS_t &= r\,S_t\,dt + \sqrt{v_t}\,S_t\,dW_1(t), && (31.3)\\ dv_t &= \kappa\,(\bar{v} - v_t)\,dt + \xi\,\sqrt{v_t}\,dW_2(t), && (31.4)\\ \rho &= \operatorname{Corr}(dW_1, dW_2), \end{aligned}$$

where $v_t = \sigma_t^2$ is the instantaneous variance (not volatility), and the five parameters are:

Heston model parameters

Symbol	Name	Typical equity value
κ	Mean-reversion speed of variance	1–5 yr^{-1}
$\bar{v}$	Long-run variance ($=\bar{\sigma}^2$)	$(0.20)^2 = 0.04$
ξ	Vol of vol (volatility of v_t)	0.3–1.0
ρ	Spot–vol correlation	-0.9 to -0.5
v_0	Initial variance ($=\sigma_0^2$)	Calibrated to ATM vol

Reading the SDE.

- *Variance mean-reverts.* Equation (31.4) is a Cox–Ingersoll–Ross (CIR) process: when $v_t > \bar{v}$ the drift pulls down, and vice versa, at speed κ. The **Feller condition** $2\kappa\bar{v} \geq \xi^2$ keeps v_t strictly positive.
- *Spot inherits stochastic vol.* Equation (31.3) is GBM with $\sqrt{v_t}$ replacing constant σ.
- *ρ creates skew.* Negative ρ (typical for equities) means falling spot coincides with rising variance, fattening the left tail and making OTM puts expensive.

Why negative ρ creates skew

Market drops coincide with vol spikes—the *leverage effect*. Heston bakes this in: $\rho < 0$ correlates negative spot returns with positive variance shocks, producing a fat left tail (i.e. higher implied vol at low strikes). More negative ρ, steeper skew; $\rho = 0$ yields a symmetric smile rarely seen in equities.

Characteristic-function pricing

Heston's key contribution: despite the 2D SDE, European calls admit a semi-closed-form price via the characteristic function of $\ln S_T$.

Because the Heston SDE is *affine* (drift and diffusion of log-spot and variance are affine in the state), the characteristic function $\varphi(u) = \mathbb{E}^{\mathbb{Q}}[e^{iu \ln S_T}]$ solves Riccati ODEs with closed-form solutions. The call price is

$$C(K,T) \;=\; S_0\,\Pi_1 \;-\; K\,e^{-rT}\,\Pi_2, \tag{31.5}$$

where

$$\Pi_j \;=\; \frac{1}{2} \;+\; \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\ln K}\,\varphi_j(u)}{iu}\right] du, \qquad j = 1, 2. \tag{31.6}$$

φ_1, φ_2 are closed-form (complex exponentials of the parameters), and the 1D integrals in (31.6) need only a few dozen quadrature points for six-digit accuracy—fast enough for real-time calibration.

Heston pricing is semi-closed-form

The characteristic-function approach needs one 1D integral—orders of magnitude faster than Monte Carlo, enabling sub-second calibration to the full surface. The same structure extends to other affine stochastic-vol models (Bates, double Heston).

How each parameter shapes the smile

Varying each parameter from a baseline $v_0 = \bar{v} = 0.04$ ($\sigma = 20\%$), $\kappa = 2$, $\xi = 0.5$, $\rho = -0.7$:

Parameter change	Effect on implied-vol smile
ρ: $-0.7 \to -0.3$	Skew decreases. The smile becomes more symmetric—less steep on the left, slightly higher on the right. The left-tail fattening diminishes because spot and vol are less anti-correlated.
ξ: $0.5 \to 1.0$	Both wings rise. The smile becomes more curved (higher curvature = fatter tails on both sides). Vol of vol injects randomness into the variance process, creating a wider distribution for S_T.
κ: $2 \to 5$	Smile *flattens at long maturities*. Strong mean-reversion pulls v_t back to $\bar{v}$ quickly, so the long-dated distribution looks more Gaussian (less smile). Short-dated smiles are less affected.
v_0: $0.04 \to 0.09$ ($\sigma_0 = 30\%$)	The entire smile shifts upward. The level of ATM vol rises. The shape (skew, curvature) is largely unchanged because those are controlled by ρ and ξ.

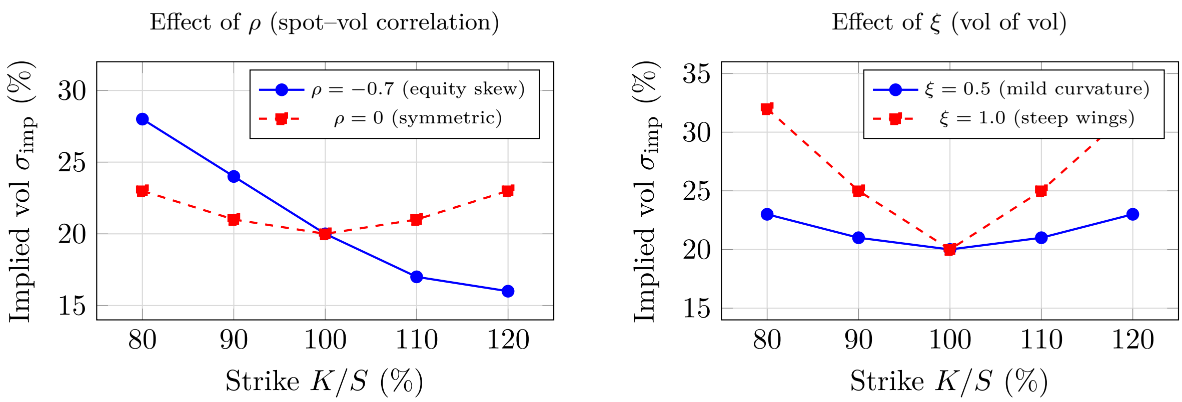


Figure 31.1: Heston smile sensitivities from baseline $v_0 = \bar{v} = 0.04$ ($\sigma = 20\%$), $\kappa = 2$. **Left:** negative ρ tilts the smile into an equity skew—OTM puts expensive, OTM calls cheap. **Right:** higher ξ lifts both wings symmetrically, producing curvature without skew.

Worth memorising: calibrate v_0 to ATM vol, ρ to 25-delta risk reversal (skew), ξ to 25-delta butterfly (curvature), κ to the smile's term structure. $\bar{v}$ is often pinned to v_0 for short dates and only matters at long maturities.

Calibration workflow

Heston calibration minimises the distance between model and market implied vols over the five parameters:

$$\min_{v_0,\kappa,\bar{v},\xi,\rho} \sum_{i=1}^{N} w_i \left[\sigma_{\text{Heston}}(K_i, T_i;\, v_0, \kappa, \bar{v}, \xi, \rho) \;-\; \sigma_{\text{mkt}}(K_i, T_i)\right]^2, \tag{31.7}$$

with w_i vega- or uniform-weighted and σ_{Heston} the model implied vol (from Equation (31.5), then inverting BS).

The objective is non-convex, so global optimisers (differential evolution, simulated annealing) beat gradient methods. For $N = 50$ quotes, total cost is $\sim$ 12,500 characteristic-function evaluations—sub-second on modern hardware.

Feller condition and calibration

An optimiser may propose parameters violating $2\kappa\bar{v} \geq \xi^2$, letting v_t touch zero and breaking the characteristic function. "Full truncation" schemes tolerate this but produce degenerate paths; safer to impose Feller as a calibration constraint.

What Heston gets right and wrong

Heston's five parameters capture the qualitative equity-surface shape (ρ skew, ξ curvature, $\bar{v}, v_0$ level, κ term structure) but cannot exactly match every kink. Failure modes:

- **Short-dated smile.** Continuous CIR variance misses the vol jumps that steepen short wings in real markets.
- **Wings.** Heston's asymptotics as $K \to 0, \infty$ may not match the market's.
- **Across maturities.** Constant parameters force a specific short-vs-long smile relationship; real surfaces violate it, requiring piecewise-constant calibration.

Desks use piecewise-constant parameters across maturity buckets, add jumps (the Bates model: spot + Poisson jumps), or combine Heston with local vol—the next section.

Comparison: Black–Scholes vs local vol vs Heston

Feature	Black–Scholes	Local vol	Heston
Vol assumption	Constant σ	$\sigma(S,t)$ deterministic	v_t stochastic
Fits today's surface?	No (flat)	Yes (exactly)	Approximately
Smile dynamics?	None	Wrong (too flat)	Realistic
Free parameters	1 (σ)	∞ (the function)	5
Pricing speed	Closed-form	PDE/MC	Semi-closed-form
Use case	Pedagogy, quotes	Vanilla Greeks	Exotic pricing

31.3 Local-stochastic volatility (LSV)

LSV combines both approaches:

$$dS_t \;=\; r\,S_t\,dt \;+\; L(S_t,t)\,\sqrt{v_t}\,S_t\,dW_1(t), \tag{31.8}$$

where v_t follows Heston dynamics and $L(S,t)$ is a **leverage function** calibrated so the model reproduces today's surface exactly. The stochastic part $\sqrt{v_t}$ supplies realistic dynamics; the local part L bends prices to match every quote.

LSV is the industry standard

LSV is the standard for pricing and hedging exotic equity derivatives: Heston-like dynamics for exotics whose value depends on the future smile (barriers, cliquets, autocallables), and exact vanilla calibration. Calibrating $L(S,t)$ (particle methods or forward PDE) is expensive but gives the most internally consistent framework available.

The leverage function. With Dupire local vol $\sigma_{\text{loc}}(S,t)$,

$$L(S,t) \;=\; \frac{\sigma_{\text{loc}}(S,t)}{\sqrt{\mathbb{E}[v_t \mid S_t = S]}}, \tag{31.9}$$

where the conditional expectation is under the model's own joint (S_t, v_t) distribution. This fixed-point equation forces iterative calibration (particle MC or Fokker–Planck PDE).
Mixing-fraction intuition. Parametrise by $\alpha \in [0,1]$:

- $\alpha = 0$: pure local vol—perfect fit, wrong dynamics.
- $\alpha = 1$: pure Heston—right dynamics, imperfect fit.
- $\alpha \in (0,1)$: LSV—L bends Heston to fit exactly while retaining fraction α of the stochastic dynamics.

α is set to match observed surface dynamics (e.g. skew shift per 1% spot move); equity desks typically use $\alpha \approx 0.5$–0.8.

31.4 Variance swap replication

This section delivers the derivation promised in Chapter 18: the fair variance-swap strike equals a $1/K^2$-weighted integral of OTM option prices. The argument is model-independent—any continuous diffusion, not just BS or Heston—and is one of the most elegant results in quant finance.

Why the product exists. An option gives vol exposure, but only bundled with directional exposure that must be neutralised by daily delta-hedging; the realised P&L depends on the hedging path (Chapter 17). A variance swap pays realised variance directly, so a trader who wants a pure view on volatility can buy one and skip the hedging workflow entirely—the dealer handles replication.

Setup and payoff

Recall the variance-swap definition from Theorem 18.6. The buyer receives at maturity T:

$$\text{Payoff} \;=\; N\,(\,\sigma_{\mathrm{r}}^2 \;-\; K_{\mathrm{var}}\,),$$

where σ_{r}^2 is the annualised realised variance of $\ln S$ over $[0,T]$ and K_{var} is the *variance strike* (here expressed in variance units, not vol units). The fair strike K_{var} is the value that makes the swap worth zero at inception:

$$K_{\mathrm{var}} \;=\; \mathbb{E}^{\mathbb{Q}}[\,\sigma_{\mathrm{r}}^2\,]. \tag{31.10}$$

Our goal is to express this expectation in terms of observable European option prices.

The log contract

A **log contract** pays $\ln(S_T/F_0)$ at T, where $F_0 = S_0\,e^{rT}$. Its risk-neutral expectation ties directly to realised variance:

Lemma 31.1 (Log contract and realised variance)**.** *Under any continuous-path diffusion $dS_t = r\,S_t\,dt + \sigma(t)\,S_t\,dW_t$ (with $\sigma(t)$ possibly random), we have*

$$\mathbb{E}^{\mathbb{Q}}\!\left[\ln\frac{S_T}{F_0}\right] \;=\; -\,\frac{1}{2}\,\mathbb{E}^{\mathbb{Q}}\!\left[\int_0^T \sigma^2(t)\,dt\right] \;=\; -\,\frac{T}{2}\,K_{var}. \tag{31.11}$$

Proof. Apply Itô's lemma to $\ln S_t$:

$$d\ln S_t \;=\; \frac{1}{S_t}\,dS_t \;-\; \frac{1}{2\,S_t^2}\,(dS_t)^2 \;=\; \Big(r - \tfrac{1}{2}\sigma^2(t)\Big)\,dt \;+\; \sigma(t)\,dW_t.$$

Integrate from 0 to T:

$$\ln S_T - \ln S_0 \;=\; r\,T \;-\; \frac{1}{2}\int_0^T \sigma^2(t)\,dt \;+\; \int_0^T \sigma(t)\,dW_t.$$

Since $F_0 = S_0\,e^{rT}$, we have $\ln(S_T/F_0) = \ln S_T - \ln S_0 - rT$, so

$$\ln\frac{S_T}{F_0} \;=\; -\,\frac{1}{2}\int_0^T \sigma^2(t)\,dt \;+\; \underbrace{\int_0^T \sigma(t)\,dW_t}_{\text{mean-zero martingale}}\;.$$

Taking risk-neutral expectations, the Itô integral vanishes (it is a mean-zero martingale), giving

$$\mathbb{E}^{\mathbb{Q}}\!\left[\ln\frac{S_T}{F_0}\right] \;=\; -\,\frac{1}{2}\,\mathbb{E}^{\mathbb{Q}}\!\left[\int_0^T \sigma^2(t)\,dt\right] \;=\; -\,\frac{T}{2}\,K_{\mathrm{var}}. \qquad \square$$

Rearranging:

$$K_{\mathrm{var}} \;=\; -\,\frac{2}{T}\,\mathbb{E}^{\mathbb{Q}}\!\left[\ln\frac{S_T}{F_0}\right] \;=\; \frac{2}{T}\,\mathbb{E}^{\mathbb{Q}}\!\left[\ln\frac{F_0}{S_T}\right]. \tag{31.12}$$

The problem reduces to pricing the log contract $-\ln(S_T/F_0)$.

Replicating the log payoff with options

The key step expresses $-\ln(S_T/F_0)$ as a static portfolio of European puts and calls, via an integration-by-parts identity.

Proposition 31.2 (Carr–Madan decomposition of a convex payoff)**.** *For any twice-differentiable function $g : \mathbb{R}_{>0} \to \mathbb{R}$ and any reference level $\kappa > 0$,*

$$g(S_T) \;=\; g(\kappa) + g'(\kappa)\,(S_T - \kappa) + \int_0^{\kappa} g''(K)\,(K - S_T)^+\,dK + \int_{\kappa}^{\infty} g''(K)\,(S_T - K)^+\,dK. \tag{31.13}$$

Any smooth payoff decomposes into a constant, a forward, and a continuum of puts ($K < \kappa$) and calls ($K > \kappa$) weighted by $g''(K)$.

Proof of Theorem 31.2. Fix $x = S_T > 0$ and $\kappa > 0$. Taylor with integral remainder:

$$g(x) = g(\kappa) + g'(\kappa)(x - \kappa) + \int_{\kappa}^{x} g''(u)\,(x - u)\,du.$$

Split into two cases. For $x > \kappa$:

$$\int_{\kappa}^{x} g''(u)(x - u)\,du = \int_{\kappa}^{\infty} g''(K)\,(x - K)^+\,dK,$$

since $(x - K)^+ = x - K$ for $K \le x$ and 0 otherwise. For $x < \kappa$, $-\int_x^{\kappa} g''(u)(u - x)\,du$ equals

$$\int_0^{\kappa} g''(K)\,(K - x)^+\,dK.$$

Summing covers all cases and gives Equation (31.13). □

Apply to $g(x) = -\ln(x/F_0)$ with $\kappa = F_0$:

$$\begin{aligned} g(F_0) &= -\ln(F_0/F_0) = 0, \\ g'(x) &= -\frac{1}{x}, \quad \text{so} \quad g'(F_0) = -\frac{1}{F_0}, \\ g''(x) &= \frac{1}{x^2}. \end{aligned}$$

Substituting into Equation (31.13):

$$-\ln\frac{S_T}{F_0} \;=\; \underbrace{-\frac{1}{F_0}\,(S_T - F_0)}_{\text{short } 1/F_0 \text{ forwards}} \;+\; \underbrace{\int_0^{F_0} \frac{1}{K^2}\,(K - S_T)^+\,dK}_{\text{OTM puts}} \;+\; \underbrace{\int_{F_0}^{\infty} \frac{1}{K^2}\,(S_T - K)^+\,dK}_{\text{OTM calls}}. \tag{31.14}$$

From the log contract to the variance-swap strike

Take risk-neutral expectations of Equation (31.14) and price at time 0 (payoff X at T costs $e^{-rT}\mathbb{E}^{\mathbb{Q}}[X]$):

- Log-contract price: $e^{-rT}\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F_0)]$.
- Forward term: zero price (forward struck at F_0 is costless).
- Put integral contributes $\int_0^{F_0} P(K)/K^2\,dK$; call integral contributes $\int_{F_0}^{\infty} C(K)/K^2\,dK$.

Combine with Equation (31.12):

Variance swap replication — full formula

The fair variance-swap strike is

$$\boxed{K_{\text{var}} \;=\; \frac{2\,e^{rT}}{T}\left[\int_0^{F_0}\frac{P(K)}{K^2}\,dK \;+\; \int_{F_0}^{\infty}\frac{C(K)}{K^2}\,dK\right],} \tag{31.15}$$

where $F_0 = S_0\,e^{rT}$ is the forward, and $P(K)$, $C(K)$ are today's market prices of European OTM puts (strikes below F_0) and calls (strikes above F_0), both with maturity T.

Derivation chain: Itô gives $K_{\text{var}} = (2/T)\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F_0)]$; Carr–Madan decomposes $-\ln(S_T/F_0)$ into $1/K^2$-weighted puts and calls plus a costless forward; taking expectations identifies K_{var} with the option-price integrals; e^{rT} converts PV to forward value.

Why $1/K^2$ makes OTM puts dominate

The $1/K^2$ weight favours low strikes: at $K = 3{,}600$ the weight is $\approx 7.7\times10^{-8}$ vs $\approx 4.9\times10^{-8}$ at $K = 4{,}500$ (57% larger). Equity skew further makes OTM puts more expensive per K than OTM calls. So a variance swap on a skewed underlying is overwhelmingly a left-tail bet—hence variance-swap levels track put skew closely and blow up in crashes.

Worked numerical example

Compute K_{var} for a 3-month SPX variance swap from discrete option prices.

Parameters. $S_0 = 4{,}000$, $r = 5\%$, $T = 0.25$ years (3 months). The forward is $F_0 = 4{,}000\times e^{0.05\times0.25} = 4{,}050.31$.

Market prices. We use 10 OTM options (5 puts, 5 calls) at equally-spaced strikes $\Delta K = 100$, generated from Black–Scholes with $\sigma = 20\%$ to provide a self-consistent baseline.

Strike K	Type	Price $Q(K)$	$\Delta K/K^2$ $(\times10^{-6})$	Contribution $(\times10^{-4})$
3,600	Put	22.08	7.716	1.704
3,700	Put	38.04	7.305	2.779
3,800	Put	61.37	6.925	4.250
3,900	Put	93.38	6.575	6.139
4,000	Put	134.91	6.250	8.432
4,100	Call	137.14	5.949	8.159
4,200	Call	99.12	5.669	5.619
4,300	Call	69.67	5.408	3.768
4,400	Call	47.65	5.165	2.461
4,500	Call	31.71	4.938	1.566
			Sum	**44.877**

Step 1: weights. $\Delta K/K_i^2 = 100/K_i^2$. At $K = 3{,}600$: $100/3{,}600^2 = 7.716\times10^{-6}$, 56% larger than at $K = 4{,}500$ (4.938×10^{-6})—the $1/K^2$ overweight on low strikes.

Step 2: contributions. Multiply weight by price. E.g. the $K = 3{,}900$ put:

$$\frac{100}{3{,}900^2} \times 93.38 = 6.575 \times 10^{-6} \times 93.38 = 6.139 \times 10^{-4}.$$

(final column).

Step 3: sum and multiply. The discrete approximation to Equation (31.15):

$$K_{\text{var}} \approx \frac{2\,e^{rT}}{T} \sum_{i=1}^{n} \frac{\Delta K_i}{K_i^2}\, Q(K_i), \tag{31.16}$$

with $Q(K_i)$ the OTM put or call price. Sum $= 4.4877 \times 10^{-3}$:

$$\begin{aligned} K_{\text{var}} &= \frac{2 \times e^{0.0125}}{0.25} \times 4.4877 \times 10^{-3} \\ &= \underbrace{8.0}_{2/T} \times \underbrace{1.01258}_{e^{rT}} \times \underbrace{4.4877 \times 10^{-3}}_{\text{strip sum}} \\ &= 0.03635. \end{aligned}$$

Step 4: convert to vol. The variance-swap vol ("var-swap strike in vol terms") is

$$\sigma_{\text{var}} = \sqrt{K_{\text{var}}} = \sqrt{0.03635} = 0.1907 = 19.07\%.$$

Remark 31.3 (Truncation bias). Inputs were generated with $\sigma = 20\%$ yet the computation returns 19.07%. The shortfall is **truncation bias**: 10 strikes over 3,600–4,500 miss the deep OTM wings. Extending to 31 strikes (2,500–5,500) gives 19.95%; a fine 1,000-strike grid recovers 20.00%. Desks extrapolate the wings to reduce this bias.

Variance swap replication

K_{var} is determined entirely by today's OTM option prices—no model assumed, any continuous diffusion works. The only assumption is no jumps; jumps open a "jump gap" between the strip price and true expected variance.

The VIX index

The CBOE VIX is computed from Equation (31.16) applied to 30-day SPX options:

$$\text{VIX}^2 = \frac{2\,e^{rT}}{T} \sum_i \frac{\Delta K_i}{K_i^2}\, Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2,$$

where K_0 is the highest listed strike below F and the second term corrects the (F, K_0) gap. VIX is quoted in annualised vol points: $\text{VIX} = 100\sqrt{\text{VIX}^2}$.

"VIX at 20" means the $1/K^2$-weighted 30-day OTM strip prices to $K_{\text{var}} = 0.04$—equivalently, 20% annualised realised vol. The world's most-watched vol gauge is nothing but the variance strike from the formula just derived.

Python implementation

Discrete variance-swap fair strike:

```python
import math
import numpy as np

def varswap_strike(put_K, put_P, call_K, call_C, r, T):
    """Compute variance-swap fair strike K_var from OTM
    option prices using the discrete strip formula.

    Returns K_var (annualised variance) and sigma_var (vol).
    """
    F = put_K[0]  # placeholder; F not needed explicitly
    # Contribution from puts: sum of dK/K^2 * P(K)
    dK_put = np.diff(put_K, prepend=put_K[0] - (put_K[1]-put_K[0]))
    put_contrib = np.sum(
        (put_K[1]-put_K[0]) / put_K**2 * put_P
    )
    # Contribution from calls
    call_contrib = np.sum(
        (call_K[1]-call_K[0]) / call_K**2 * call_C
    )
    total = put_contrib + call_contrib
    K_var = (2.0 / T) * math.exp(r * T) * total
    return K_var, math.sqrt(K_var)

# Example: SPX 3-month variance swap
put_K  = np.array([3600, 3700, 3800, 3900, 4000.])
put_P  = np.array([22.08, 38.04, 61.37, 93.38, 134.91])
call_K = np.array([4100, 4200, 4300, 4400, 4500.])
call_C = np.array([137.14, 99.12, 69.67, 47.65, 31.71])
K_var, sig_var = varswap_strike(
    put_K, put_P, call_K, call_C, r=0.05, T=0.25
)
print(f"K_var = {K_var:.6f}, sigma_var = {sig_var:.2%}")
# Output: K_var = 0.036353, sigma_var = 19.07%
```

Listing 31.1: Variance swap fair strike from discrete option prices.

Practical considerations for variance swap trading

Real-world complications:

Discrete monitoring. Theoretical variance is continuous $\int_0^T \sigma^2(t)\,dt$; swaps settle on discretely sampled (usually daily close-to-close) returns:

$$\hat{\sigma}_{\mathrm{r}}^2 = \frac{252}{n} \sum_{i=1}^{n} \left(\ln \frac{S_{t_i}}{S_{t_{i-1}}} \right)^2,$$

with n business days and 252 the annualisation factor. The continuous–discrete gap is small (<0.1 vol point) but non-zero and return-autocorrelation-dependent.

Jumps and the "jump gap." Replication assumes continuous paths. Jumps (earnings, macro shocks) add jump variance the strip cannot replicate, making the strip price systematically under-estimate total expected variance. Desks either de-jump the settlement contractually or charge a jump premium.

Corridor and conditional variance swaps. A **corridor variance swap** accumulates variance only when the spot is in $[L, U]$. A downside corridor ($L = 0$, $U = 0.9S_0$) isolates crash variance. Replication uses barrier-modified option strips.

Mark-to-market. Once live at time t, the MTM combines variance already realised on $[0, t]$ with expected variance on $[t, T]$ from the current strip:

$$V_t = N\left[\frac{t}{T}\,\hat{\sigma}_{\rm r}^2(0,t) + \frac{T-t}{T}\,K_{\rm var}^{\rm new}(t) - K_{\rm var}(0)\right] e^{-r(T-t)},$$

with $K_{\rm var}^{\rm new}(t)$ the current fair strike for the remaining maturity $T - t$.

Volatility swaps and the convexity correction

A **volatility swap** pays $\sigma_{\rm r} - K_{\rm vol}$ (linear in realised vol), unlike a variance swap's $\sigma_{\rm r}^2 - K_{\rm var}$ payoff. Vol swaps give cleaner linear P&L but are harder to replicate (square root is concave).

By Jensen, $\mathbb{E}[\sigma_{\rm r}] \le \sqrt{\mathbb{E}[\sigma_{\rm r}^2]} = \sqrt{K_{\rm var}}$, so the fair vol-swap strike sits below $\sqrt{K_{\rm var}}$:

$$K_{\rm vol} \approx \sqrt{K_{\rm var}} - \frac{\mathrm{Var}(\sigma_{\rm r})}{2\sqrt{K_{\rm var}}}, \tag{31.17}$$

The correction is the **convexity adjustment**; size scales with vol-of-vol. For SPX it is typically 0.5–1.5 vol points.

Intuition

Variance is convex in vol: 20%→40% means variance 0.04 → 0.16. Variance-swap buyers gain disproportionately from big vol moves; vol-swap buyers gain linearly. So var swaps always price above the corresponding vol swap; the gap is the convexity correction.

31.5 Dividends and structured equity products

Discrete dividends

Pure BS and Heston ignore dividends. The standard European fix is to replace S_0 with the **dividend-adjusted forward**:

$$F_0^{\rm div} = \big(S_0 - \mathrm{PV}(\text{dividends})\big)\, e^{rT},$$

with PV(dividends) summed over all discrete dividends to maturity. The surface is built on $F_0^{\rm div}$, not raw spot. For single-stock options, getting the dividend schedule right matters more than the vol model: a missed ex-date can shift a 3-month ATM call price by 1–2%.

Example. $S_0 = 100$, \$2 dividend in 2 months, $r = 5\%$, $T = 0.25$:

$$\begin{aligned}\mathrm{PV}(\mathrm{div}) &= 2 \times e^{-0.05\times 2/12} = 2 \times 0.9917 = 1.98,\\ F_0^{\rm div} &= (100 - 1.98) \times e^{0.05\times 0.25} = 98.02 \times 1.01258 = 99.25.\end{aligned}$$

Without the dividend the forward is 101.26—the dividend knocks 2.01 off. A 100-strike call that was slightly OTM becomes more deeply OTM; price and Greeks shift accordingly.

Dividend handling matters more than you think

Junior Strats obsess over the vol model (LV vs Heston vs LSV) and treat dividends as an afterthought. PV of dividends can be 3–5% of spot for high-yield names; a wrong schedule shifts every price. Autocallables are especially sensitive—a missed dividend on a barrier date can flip knock-in pricing. Always verify dividends before calibrating.

Structured equity products

Structured products are path-dependent, barrier-contingent, or multi-asset payoffs priced by MC under a calibrated LSV. Three common types:

Autocallables. The most-traded equity structured product ($500bn+ outstanding notional). Coupons pay on observation dates *above* the autocall barrier; hitting the barrier terminates the trade with principal plus coupon; finishing below a lower knock-in barrier at maturity inflicts losses. Pricing: LSV MC with barrier checks on each observation date.
Typical structure. 3-year, quarterly observations, 100% autocall barrier, 8% p.a. coupon (memory), 60% knock-in:

- Any quarterly date above 100% → terminate, pay accrued coupons plus principal.
- Never autocalls, ends above 60% → principal only.
- Ends below 60% → investor receives S_T/S_0 of principal (potentially large loss).

The investor is selling a deep OTM put for periodic coupons. Key desk risks: forward–surface correlation (autocall vs knock-in probabilities).

Accumulators. Buy a fixed quantity daily at a discounted strike, subject to knock-out and knock-in barriers. Rising markets: cheap shares; crashes: forced above-market buying, sometimes at a doubled rate ("accumulate double"). Traders call them "I'll-kill-you-later" contracts.

Worst-of products. Payoff on the worst-performer in a basket of 3–5 indices. Pricing needs multi-dim LSV with calibrated cross-asset correlations; correlation exposure makes them cheaper (higher coupon) but the risk management is hard.
All three are MC-priced under LSV. The Strat owns calibration, stable finite-difference Greeks, and actionable book-level risk reports.

Pricing structured products: the Monte Carlo workflow

Standard exotic-pricing workflow:

1. **Calibrate.** Fit LSV to today's surface, dividends, and rates → $L(S,t)$ and Heston parameters.
2. **Simulate.** $N = 10^5$–10^6 Euler–Maruyama paths with time step matched to observation dates.
3. **Evaluate payoff** on each path: barriers, coupons, early termination.
4. **Average and discount:** $\hat{V} = e^{-rT}\frac{1}{N}\sum_j \mathrm{Payoff}_j$.
5. **Greeks.** Bump each input (spot, surface, rates, dividends, correlations) and re-run MC. A 100-trade book with 8 Greeks is ~ 800 MC runs, parallelised on a compute grid.

Exotic pricing is MC-intensive

One autocallable price: $\sim 10^5$ paths; 100-trade book with price plus 8 Greeks: $\sim 8 \times 10^7$ path evaluations. Hence exotic desks invest heavily in GPU grids and distributed MC; a $2\times$ inner-loop speedup halves risk-report runtime.

31.6 Prosperity and industry connections

Prosperity example

Prosperity's VOLCANIC_ROCK_VOUCHERS were vanilla European calls on a simulated underlying; a simple BS pricer with an implied-vol estimate sufficed—local vol/Heston/LSV/exotics would be overkill. But the variance-swap replication links to the gamma-scalping strategy in Chapter 18: knowing fair variance from the strip tells you whether implied vol is cheap or rich, hence long or short gamma. The Hedgehogs did exactly that: estimate fair vol, compare to bot-quote implied vol, trade the gap with delta-hedged options—a discretised variance-swap replication.

On a Goldman Sachs desk

Equity-derivatives Strat stack at GS:

1. **Local-vol calibrator:** fits Dupire each morning; feeds vanilla Greeks and $L(S,t)$ for LSV.
2. **Heston/LSV calibrator:** Heston fit $\rightarrow$ solve for L via particle MC or forward PDE. Overnight for full universe, intraday for key names.
3. **Variance-swap pricer:** integrates the $1/K^2$ strip off the calibrated surface; used for client quotes and marks.
4. **Structured-product MC engine:** autocallables, accumulators, worst-of under LSV, orchestrated by SecDB.

A junior Strat's first project is often "build a variance-swap pricer reading the calibrated surface and outputting fair strikes across the term structure"—forcing the whole stack: calibration, numerical integration, variance economics.

Historical note

Three papers define the modern toolkit:

- **Dupire (1994) and Derman–Kani (1994).** At Paribas and GS respectively, they showed a deterministic $\sigma_{\text{loc}}(S,t)$ can reproduce any arbitrage-free surface—the first beyond-BS model adopted on desks.
- **Heston (1993).** First stochastic-vol model with a semi-closed-form characteristic function, making calibration practical. 15,000+ citations.
- **Demeterfi–Derman–Kamal–Zou (1999).** GS; definitive log-contract derivation of variance-swap replication, turning var swaps into liquid products and underpinning the VIX.

Key facts to memorise

Chapter summary — equity derivatives beyond vanilla

1. **Local volatility (Dupire).** Make vol a function of spot and time: $\sigma^2_{\mathrm{loc}}(K,T) = (\partial C/\partial T + rK\,\partial C/\partial K)/(\frac{1}{2}K^2\,\partial^2 C/\partial K^2)$. Fits today's surface exactly but gets dynamics wrong (skew flattens too fast).

2. **Heston stochastic vol.** $dS = rS\,dt + \sqrt{v}\,S\,dW_1$, $dv = \kappa(\bar{v} - v)\,dt + \xi\sqrt{v}\,dW_2$, $\rho = \mathrm{Corr}(W_1, W_2)$. Five parameters: κ (mean-reversion), $\bar{v}$ (long-run variance), ξ (vol of vol), ρ (spot–vol correlation), v_0 (initial variance).

3. **$\rho < 0$ creates the skew.** When the spot falls, vol rises (leverage effect). This fattens the left tail and makes OTM puts expensive.

4. **Heston is semi-closed-form.** European prices via a 1D Fourier integral of a known characteristic function—fast enough for real-time calibration.

5. **LSV = local × stochastic.** Industry standard. A leverage function $L(S,t)$ calibrates exactly to the surface; Heston dynamics provide realistic vol dynamics for exotics.

6. **Variance swap.** Pays $\sigma_{\mathrm{r}}^2 - K_{\mathrm{var}}$. Fair strike: $K_{\mathrm{var}} = \frac{2\,e^{rT}}{T}\left[\int_0^F P(K)/K^2\,dK + \int_F^\infty C(K)/K^2\,dK\right]$.

7. **Derivation chain.** Itô → log-contract expectation → Carr–Madan decomposition into $1/K^2$-weighted puts and calls → option prices = variance-swap strike.

8. **$1/K^2$ weighting $\Longrightarrow$ OTM puts dominate.** Variance swaps are long downside vol. In a crash, the long-variance position profits massively.

9. **Discrete approximation.** $K_{\mathrm{var}} \approx (2e^{rT}/T)\sum_i(\Delta K_i/K_i^2)\,Q(K_i)$. Truncation of deep OTM wings biases low.

10. **Dividends.** Replace S with $S - \mathrm{PV(div)}$ when computing the forward. Getting dividends wrong matters more than the vol model for single-stock options.

11. **Structured products.** Autocallables, accumulators, worst-of: path-dependent, priced by MC under LSV. The Strat owns the calibration and MC engine.

12. **Industry stack.** Local-vol calibrator → Heston/LSV calibrator → variance-swap pricer → MC engine for exotics. A junior Strat's first project is often a variance-swap pricer.

Chapter 32

FX Derivatives

Foreign-exchange (FX) is the largest financial market in the world: the BIS 2022 Triennial Survey put average daily turnover at \$7.5 trillion, an order of magnitude above global equities. It also carries the densest jargon — "pips", "forward points", "25-delta risk-reversal", "TARF", "CNH vs CNY" — most with no analogue elsewhere. This chapter translates the jargon, derives *covered interest parity*, states the FX Black–Scholes (*Garman–Kohlhagen*), and tours the vanilla-through-exotic option zoo: barriers, digitals, TARFs, and the ATMF/RR/BF quoting convention from which the entire smile is reconstructed.

Prerequisites

From earlier in the book:

- Chapter 14: the no-arbitrage principle, triangular FX arbitrage, and the idea that cross-rates must agree in a cycle.
- Chapter 15: European call/put payoffs, put–call parity, and the notion of a strike K.
- Chapter 16: the Black–Scholes formula under geometric Brownian motion, the Φ/ϕ notation, and risk-neutral pricing.
- Chapter 29: continuous discounting, zero-coupon curves, e^{-rT} as the discount factor.

If any of these feel hazy, re-read the relevant chapter first.

32.1 FX jargon decoder

FX desk conversation is unreadable without the shorthand. The table collects the terms used in the rest of the chapter; each is formally defined where first needed.

FX jargon at a glance

Term	Meaning
Currency pair	A pair written BASE/QUOTE, e.g. EUR/USD. The quoted number is the number of units of the *quote* currency required to buy one unit of the *base* currency.
Spot S_t	Current exchange rate; deliverable on the *spot date*, conventionally $T + 2$ business days.
Forward $F(T)$	Rate agreed now for an exchange settling at future date T. No money changes hands today.
Forward points	The difference $F(T) - S_t$, quoted in *pips*.
Pip	"Percentage in point"; the smallest conventional price increment. For most pairs this is 10^{-4}; for JPY pairs it is 10^{-2}.
Direct quote	Quote with the domestic currency as the quote (right-hand) currency. From a US perspective, EUR/USD is direct; USD/JPY is indirect.
Cross rate	Rate for a pair neither leg of which is the USD, e.g. EUR/JPY, usually inferred from the two USD legs.
Domestic / foreign	In option pricing, the *domestic* currency is the one you are measuring P&L in (the quote currency); the *foreign* currency is the underlying asset (the base currency). Rates r_d, r_f follow this convention.
NDF	Non-deliverable forward: cash-settles in USD against a fixing, used where the local currency is not freely convertible (e.g. KRW, TWD, CNY).
CNH vs CNY	Offshore (Hong Kong) vs onshore (mainland) renminbi. The two are separate markets and their rates can diverge persistently.

EUR/USD is not the same as USD/EUR

EUR/USD = 1.08 means "**1** EUR buys **1.08** USD." Inverting, one USD buys $1/1.08 \approx 0.9259$ EUR. Newcomers flip the convention and wonder why their hedge is backwards. The rule: *the left-hand currency is the one unit; the right-hand is how many you pay.* "A call on EUR/USD" is a call on EUR — buying EUR, selling USD — struck in USD per EUR.

The slot convention is fixed by market practice, not mathematics. EUR is always the base against USD, JPY is always the quote, GBP is always the base, and so on. Get the convention wrong on a ticket and you have written the opposite trade.

Fixings and settlement mechanics

Many FX contracts — NDFs, option barriers, TARF schedules, hedged-equity baskets — reference an *FX fixing*: a publicly-observed rate set at a fixed time by polling or matching on a specified venue. The main fixings:

- **WM/Reuters 4pm London**: benchmark for equity-index currency hedging; set from trades and executable quotes in a five-minute window around 4pm London, published by

Refinitiv.

- **ECB reference rates**: daily concertation fix at 14:15 CET, for European statistics and accounting.
- **Central-bank NDF fixes**: e.g. the PBOC USD/CNY onshore midpoint (09:15 Beijing), reference for many CNY NDFs.

Confusing which fixing a contract references has cost traders real money: a barrier that knocks out "on any touch" does so continuously on spot, but a TARF leg "settling against the fixing" cares only about the one reference value on the fixing date.

Non-deliverable forwards (NDFs). An NDF is a forward with no physical exchange at settlement; one party pays the other USD (or EUR) cash equal to (forward rate − fixing) × notional. NDFs exist because many EM currencies (CNY, KRW, INR, TWD, BRL, ...) have capital controls preventing free offshore ownership of the local currency; the NDF gives exposure without holding the local leg. The NDF market is the primary venue for non-resident renminbi, rupee, and won exposure, with over $250bn daily turnover in 2022.

32.2 Spot and forwards

Spot

The **spot rate** S_t is the price at time t for immediate exchange of the two currencies. "Immediate" is convention: for most pairs the *spot date* is $T + 2$ business days. USD/CAD is $T + 1$; intraday fixings use whatever the fixing definition says.

The $T + 2$ convention is historical: pre-electronic settlement required physical payment instructions through two central-bank RTGS systems in two time zones. CLS eliminates the operational need, but $T + 2$ remains the anchor — zero curves, forward-point quotes, and option-expiry calendars all hang off it. A "tom/next" trade is a one-day forward relative to spot, not a two-day one.

Forwards

A **forward contract** struck at t exchanges currencies at fixed rate $F(t, T)$ on date T. No money changes hands at inception; at T one side delivers foreign and receives domestic at F. The commercial motivation is hedging: a European exporter invoiced USD 10M for delivery in three months faces uncertain EUR proceeds, and locking a forward today converts that uncertain future USD receipt into a known EUR amount.

What rate makes this fair today? No-arbitrage pins it down exactly as with equity forwards (Chapter 15), with one twist: holding the foreign currency earns the *foreign interest rate* r_f, not a dividend yield.

Covered interest parity

Covered interest parity (CIP)

Let S_t be the spot BASE/QUOTE rate (units of quote per unit of base). Let r_d be the continuously compounded domestic (quote-currency) interest rate and r_f the foreign (base-currency) rate, both for the tenor T. Then the fair forward rate is

$$F(t, T) \;=\; S_t \, e^{(r_d - r_f)(T-t)}. \tag{32.1}$$

If market quotes deviate from this relation, a textbook arbitrage exists.

Derivation. Consider two self-financing strategies, starting from one unit of foreign at time t and ending in domestic at T.

Strategy A (invest foreign, convert forward): deposit the foreign unit at rate r_f, growing to $e^{r_f(T-t)}$; sell this forward at $F(t,T)$; receive $F(t,T)\,e^{r_f(T-t)}$ domestic at T.

Strategy B (convert spot, invest domestic): sell the foreign unit at spot for S_t domestic; deposit at r_d, growing to $S_t\,e^{r_d(T-t)}$ at T.

Both start from one foreign unit with deterministic terminal cashflow. By the law of one price they must agree:

$$F(t,T)\,e^{r_f(T-t)} \;=\; S_t\,e^{r_d(T-t)}, \tag{32.2}$$

rearranging to Equation (32.1). If F exceeded this level, borrow domestic, run Strategy B, sell forward, and collect $Fe^{r_f(T-t)} - S_te^{r_d(T-t)} > 0$ risklessly; the converse flips. (Hull, 2014), Ch. 5 (general assets) and Ch. 17 (currencies).

Intuition

$F > S$ iff the domestic rate exceeds the foreign rate. If USD earns 5% vs EUR's 4%, everyone wants USD; the forward market compensates by pricing future USD *lower* in EUR terms so the two investments break even. Mnemonic: *the high-yielding currency trades at a forward discount.*

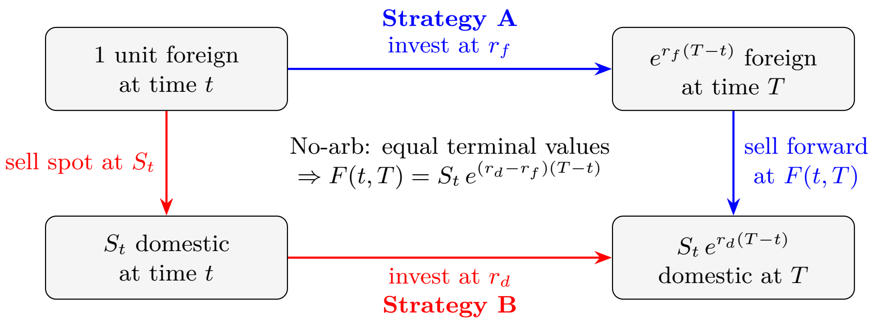


Figure 32.1: Covered interest parity (CIP). Both strategies start from 1 unit of foreign currency at t and end in domestic at T. Strategy A (blue): deposit foreign at r_f, sell proceeds forward at F. Strategy B (red): sell spot at S_t, invest domestic at r_d. No-arbitrage forces the two terminal values to agree, pinning the forward rate.

Worked example: 1-year EUR/USD forward

Suppose EUR/USD spot $S_0 = 1.0800$, the continuously compounded USD rate is $r_d = 5\%$, and the continuously compounded EUR rate is $r_f = 4\%$, both for 1-year tenor. Then

$$F(0,1) \;=\; 1.08 \cdot e^{(0.05-0.04)\cdot 1} \;=\; 1.08 \cdot e^{0.01} \;\approx\; 1.0909. \tag{32.3}$$

The forward points are $F - S = 0.010854 = 108.54$ pips, quoted as "EUR/USD 1-year fwd at +108.5 points." Verified to six decimal places in Python (Listing 32.1).

Quote convention flip reverses the sign

With EUR/USD and USD domestic, higher USD rate gives a forward *premium* ($F > S$). Quoted as USD/EUR instead, the domestic role swaps: $r_d = 4\%, r_f = 5\%$, so $F/S = e^{-0.01}$ — a forward *discount*. Same economics, opposite sign on the points.

32.3 Cross-currency basis swaps

A **cross-currency basis swap** (XCCY swap) exchanges floating interest in two currencies over months to decades. At inception, the parties exchange principals N and S_0N at spot S_0 (one pays N USD, receives S_0N EUR). At each coupon t_i, one pays floating USD (SOFR) on N and receives floating EUR (EURIBOR plus spread) on S_0N, spread on the EUR leg. At maturity, principals are re-exchanged at the *initial* rate S_0, not the prevailing one.

The "basis" b is the spread on the non-USD leg that zeros inception PV. Under naive CIP this would be zero; since 2008 it has been persistently non-zero, typically 10–50 bp against USD for majors and wider in stress. In plain English: b is the extra interest a counterparty demands, on top of their own domestic floating rate, to lend you their currency and take USD in exchange — compensation for the scarcity of dollar funding when non-US banks want USD more than US banks want foreign currency.

Post-2008 CIP deviations

The observed cross-currency basis b violates pure CIP: *borrowing USD synthetically via FX swaps is more expensive than borrowing USD directly.* The drivers are

- **Balance-sheet costs** under Basel III — arbitraging the basis consumes scarce leverage capacity.
- **USD funding premium** — non-US banks structurally short of USD bid for it, especially at quarter- and year-ends.
- **Credit/counterparty risk** in the FX-swap market.

See Borio et al. (2016) for the canonical post-crisis account. For option-pricing purposes, the practical consequence is that the forward rate used in Section 32.2 must be replaced with the *market* forward (which bakes in the basis), not the textbook CIP-implied forward.

The basis is a term structure quoted at 3M, 6M, 1Y, 2Y, ..., 30Y. Long-tenor FX-forward desks price off the basis curve plus the two single-currency curves, not textbook CIP. XCCY basis swaps hedge and express views on the spread.

Implied basis from a forward

Given market forward F^{mkt} and two zero curves, the *implied* basis is

$$b_{\text{implied}}(T) \;=\; \frac{1}{T}\ln\frac{F^{\text{mkt}}}{S_0} \;-\; (r_d - r_f), \tag{32.4}$$

the deviation of the observed forward from the textbook CIP level $S_0 e^{(r_d-r_f)T}$. Positive = premium over CIP; negative = discount. EUR/USD 3-month basis has blown out to 50–80 bp at stressed quarter-ends.

Why this matters for pricing

The Garman–Kohlhagen formula of Section 32.4 assumes CIP and uses $r_d - r_f$ as the risk-neutral drift. Desks pricing against market forwards replace this with $\frac{1}{T}\ln(F^{\text{mkt}}/S_0) = r_d - r_f + b_{\text{implied}}$. Equivalently, price off the market forward curve directly (the *forward-flat* convention).

32.4 FX vanillas and quanto

Garman–Kohlhagen

The FX analogue of Black–Scholes is **Garman–Kohlhagen** (Garman and Kohlhagen, 1983): treat the foreign currency like a dividend-paying stock with continuous yield r_f in place of q.

Garman–Kohlhagen formula

Under the risk-neutral model $dS_t = (r_d - r_f)\, S_t\, dt + \sigma\, S_t\, dW_t$, a European call on the FX rate S (paying $\max(S_T - K, 0)$ in domestic currency) is worth

$$C \;=\; S_0\, e^{-r_f T}\Phi(d_1) \;-\; K\, e^{-r_d T}\Phi(d_2), \tag{32.5}$$

with

$$d_1 \;=\; \frac{\ln(S_0/K) + (r_d - r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 \;=\; d_1 - \sigma\sqrt{T}. \tag{32.6}$$

The corresponding put price is $P = Ke^{-r_d T}\Phi(-d_2) - S_0 e^{-r_f T}\Phi(-d_1)$.

Differences from equity Black–Scholes (Chapter 16) are cosmetic: drift under $\mathbb{Q}$ is $r_d - r_f$; spot discounted by $e^{-r_f T}$. The derivation is identical with $q \mapsto r_f$ throughout. (Hull, 2014), Ch. 17.

Put–call parity in FX form

Subtracting put from call gives FX put–call parity

$$C - P \;=\; S_0 e^{-r_f T} - K e^{-r_d T}, \tag{32.7}$$

the PV of a forward buying foreign at K. Using CIP (Section 32.2), the RHS equals $e^{-r_d T}(F - K)$. Long call + short put at the same strike = long forward, as in equities.

Worked example: 3-month EUR/USD ATM call

Using the same market data as in Section 32.2 but with $T = 0.25$ years, $\sigma = 9\%$ (a typical EUR/USD at-the-money implied vol), and strike $K = 1.08$ (ATM spot):

$$d_1 = \frac{\ln(1) + (0.05 - 0.04 + \frac{1}{2}\cdot 0.09^2)\cdot 0.25}{0.09\sqrt{0.25}} \;=\; 0.0781, \tag{32.8}$$

$$d_2 = 0.0781 - 0.09\sqrt{0.25} \;=\; 0.0331, \tag{32.9}$$

$$\begin{aligned} C &= 1.08\, e^{-0.04\cdot 0.25}\Phi(0.0781) - 1.08\, e^{-0.05\cdot 0.25}\Phi(0.0331) \\ &\approx 0.02054 \text{ USD per EUR notional.} \end{aligned} \tag{32.10}$$

On a EUR 10M notional, the premium is USD 205,346. The corresponding put is worth USD 178,648; their difference USD 26,698 matches $Se^{-r_f T} - Ke^{-r_d T}$ times EUR 10M, as parity requires. All five numbers above are verified in Python below to six decimal places.

```python
import numpy as np
from scipy.stats import norm

def garman_kohlhagen(S, K, T, r_d, r_f, sigma, option="call"):
    """Price a European FX option (domestic-currency premium per
    unit of foreign notional).  S, K in domestic-per-foreign units."""
    d1 = (np.log(S / K) + (r_d - r_f + 0.5 * sigma**2) * T) \
         / (sigma * np.sqrt(T))
```

```
    d2 = d1 - sigma * np.sqrt(T)
    if option == "call":
        return S * np.exp(-r_f * T) * norm.cdf(d1) \
             - K * np.exp(-r_d * T) * norm.cdf(d2)
    return K * np.exp(-r_d * T) * norm.cdf(-d2) \
         - S * np.exp(-r_f * T) * norm.cdf(-d1)

# 3M EUR/USD ATM, spot 1.08, vol 9%, USD 5%, EUR 4%
C = garman_kohlhagen(1.08, 1.08, 0.25, 0.05, 0.04, 0.09, "call")
P = garman_kohlhagen(1.08, 1.08, 0.25, 0.05, 0.04, 0.09, "put")
print(f"Call = {C:.6f}, Put = {P:.6f}, C-P = {C-P:.6f}")
# Check put-call parity: C - P == S*exp(-r_f*T) - K*exp(-r_d*T)
parity = 1.08*np.exp(-0.04*0.25) - 1.08*np.exp(-0.05*0.25)
assert abs((C - P) - parity) < 1e-10
```

Listing 32.1: Garman–Kohlhagen vanilla pricer for FX options.

Quanto options

A **quanto** ("quantity-adjusting") option pays in a currency different from the underlying's natural one. Canonical example: a Japanese investor holds a call on SPX paying $\max(S_T^{\text{SPX}} - K, 0)$ *in JPY*, with a contractually fixed FX rate (say 1 index unit = 1 JPY).

Quantos introduce correlation between index S^{SPX} and rate X^{USDJPY}. Under the JPY risk-neutral measure, the SPX drift is

$$\mu^{\text{SPX,quanto}} \;=\; r_{\text{JPY}} - q^{\text{SPX}} \;-\; \rho\,\sigma_{\text{SPX}}\,\sigma_{\text{USDJPY}}, \tag{32.11}$$

with ρ the correlation of SPX and USD/JPY log-returns. The adjustment is the $-\rho\sigma\sigma$ term: positive co-movement reduces the quanto drift, because a JPY investor gets "free" FX appreciation when SPX rises and pays for it via a lower forward. With the adjusted drift the rest is ordinary Black–Scholes. (Hull, 2014), Ch. 30.

Worked example: quanto vs naive SPX call

Take $\sigma_{\text{SPX}} = 20\%, \sigma_{\text{USDJPY}} = 10\%, \rho = -0.30$ (USD weakens when SPX rallies). The SPX-drift adjustment is

$$-\rho\,\sigma_{\text{SPX}}\,\sigma_{\text{USDJPY}} \;=\; -(-0.30)(0.20)(0.10) \;=\; +0.60\%, \tag{32.12}$$

raising the quanto drift by 60 bp/year. The JPY investor pays *more* than for the unhedged exposure because she gets the FX-hedged return without paying for the hedge. Flip $\rho > 0$ and the quanto trades at a discount.

Quantos are not the same as FX-hedged notes

An *FX-hedged* investment converts the SPX return at the terminal FX rate and strips FX exposure via a rolling forward hedge. A *quanto* uses a contractually fixed FX rate — no spot-FX sensitivity at expiry. The P&L differs, and the quanto carries correlation risk (ρ) that the hedged note does not.

32.5 Barriers, digitals, and TARFs

Vanilla FX options trade in vast quantities, but exotic-desk complexity lives in structured products used to fine-tune corporate and asset-manager hedges. Three families follow.

Barrier options

A **barrier option** is a vanilla that comes into existence (*knock-in*) or is extinguished (*knock-out*) when spot touches barrier B during the option's life. {up, down} × {in, out} × {call, put} gives eight standard types, all with closed-form GK prices (see Hull, 2014, Ch. 26 and 19). The key *in-out parity*:

$$\text{knock-in} + \text{knock-out} \;=\; \text{vanilla}, \tag{32.13}$$

since the barrier either fires (knock-in lives, knock-out dies) or does not, and the portfolio pays the vanilla payoff either way.

Barriers are popular because they are strictly cheaper than the matching vanilla: a hedger who "knows" EUR/USD won't rally above 1.20 in six months buys a 1.10-strike knock-out with barrier 1.20. If spot touches 1.20, the hedge disappears exactly when most needed — the standard barrier failure mode.

For the case of an *up-and-out call* with strike $K < B$ under Garman–Kohlhagen dynamics, the closed form is (Hull, 2014, Ch. 26)

$$C_{\text{UO}} \;=\; C_{\text{vanilla}} \;-\; C_{\text{UI}}, \tag{32.14}$$

where C_{UI} is a sum of four Gaussian CDF terms involving the reflected strike B^2/S_0 and exponent $\mu = (r_d - r_f)/\sigma^2 - \frac{1}{2}$. Ugly but mechanical to code. Key properties:

- UO vanishes when spot hits B;
- vega changes sign near B (long far from B, short close — more vol raises the knockout probability);
- gamma spikes large and negative near B, making delta hedging unstable in the final 1–2% approach.

The last point — barrier *pin risk* — is why traders run separate "stopping-time" hedges (small digital offsets for the gamma spike) rather than smooth delta hedging alone.

Barrier vs vanilla: a concrete comparison

Using the data of Section 32.4 (spot 1.08, $r_d = 5\%, r_f = 4\%, \sigma = 9\%, T = 0.25$), compare three $K = 1.09$ calls:

- **Vanilla**: $\approx 1.59\%$ of EUR notional.
- **Up-and-out,** $B = 1.12$: $\approx 0.11\%$ (MC). An order of magnitude cheaper: the barrier is tight enough that most paying paths knock out first.
- **Up-and-out,** $B = 1.15$: $\approx 0.61\%$ — under half the vanilla, but retains most of the 1.09–1.15 terminal probability.

(Prices verified via MC at 2×10^5 paths, 100 steps.) The discount is largest exactly when the cheap hedge is most wanted — the same perverse trade-off as TARFs. The barrier's expected payoff conditional on the hedge being needed (spot ≥ 1.09 at expiry) is strictly lower than the vanilla's; the shortfall is the structure's true cost.

Digital (binary) options

A **digital call** pays Q if $S_T > K$, zero otherwise. Under GK, the domestic-cash digital call is

$$D^{\text{call}}_{\text{cash}} \;=\; Q\, e^{-r_d T}\, \Phi(d_2), \tag{32.15}$$

with d_2 as in Equation (32.6). The digital put is $Qe^{-r_d T}\Phi(-d_2)$; they sum to $Qe^{-r_d T}$, the digital put–call parity.

Intuition

A digital is the derivative of a vanilla with respect to strike: $\partial C/\partial K = -e^{-r_d T}\Phi(d_2)$, so a tiny long-call-short-call-at-$K+\epsilon$ spread, rescaled by $1/\epsilon$, replicates a digital. This is why a digital's *vega profile* has a sharp spike at the strike: the replicating call spread becomes infinitely narrow. Traders describe digitals as "event insurance" — they pay a fixed amount contingent on a spot-at-expiry event.

"One-touch" and "no-touch" variants pay Q if spot ever / never crosses the barrier before expiry — cheap conditional bets on range-bound or trending regimes. Closed forms follow from the reflection principle (Hull, 2014, Ch. 19).

One-touch pricing sketch

For a one-touch that pays Q in domestic currency at expiry T if spot ever hits barrier B, the BS price under drift $\mu = r_d - r_f$ is

$$\mathrm{OT} \;=\; Q\, e^{-r_d T}\left[\left(\frac{B}{S_0}\right)^{2\mu/\sigma^2}\Phi(z_+) + \Phi(z_-)\right], \tag{32.16}$$

with $z_\pm = (\pm\ln(S_0/B) - \mu T + \frac{1}{2}\sigma^2 T)/(\sigma\sqrt{T})$ (sign conventions vary; Clark (2011) lists all four variants). One-touches are pure *stopping-time* products: price depends on the first-passage distribution of GBM, not the terminal alone. Skew and kurtosis drive large BS corrections, so desks mark them via Vanna–Volga or smile-consistent MC.

Target redemption forwards (TARFs)

A **target redemption forward** is a structured series of monthly or weekly forwards: on each fixing, the client buys base currency at a favourable strike if spot is in range. The structure *terminates early* once cumulative client gains reach a contractual *target*; if spot moves against the client, they keep paying at the unfavourable strike, often on *doubled* notional, through the remaining schedule.

This asymmetry is the point: clients accept tail risk in exchange for a better favourable-side strike. Dealers price TARFs by MC over the full strike schedule with the early-termination trigger; no closed form.

Concretely, a 12-month EUR/USD TARF might specify: twelve monthly fixings, strike $K = 1.09$, notional EUR 1M per fixing, target cumulative client P&L = EUR 100k, leverage 2x on the loss leg. On each fixing date the cashflow to the client is

$$\mathrm{CF}_i \;=\; \begin{cases} (K - \mathrm{Fix}_i)\cdot N & \text{if } \mathrm{Fix}_i \le K \text{ (client gain)},\\ (K - \mathrm{Fix}_i)\cdot 2N & \text{if } \mathrm{Fix}_i > K \text{ (client loss, 2x)}, \end{cases} \tag{32.17}$$

with N = EUR 1M. Termination is at the first fixing i^* with $\sum_{j\le i^*}\max(\mathrm{CF}_j, 0) \ge$ target. The MC pricer draws 12 fixing paths, applies the termination rule, discounts, and averages. Greeks come from bump-and-reprice at roughly 12× vanilla cost.

TARFs are leveraged the wrong way

TARFs cap client upside (the target) while leveraging downside tenor-unlimited — the wrong profile for a hedger, the right one for a short-vol speculator. Treasurers were sold TARFs as "cheap hedges" in 2014–2015; the August 2015 RMB devaluation locked Asian corporates into daily-doubling CNH purchases above market, booking multi-hundred-million USD losses. The product is now regulatory-flagged in most jurisdictions. *If a structure advertises "free" hedging, find the short-vol double-notional leg before signing.*

The 2015 CNH TARF blow-up

July 2015: 1Y CNH near 6.20 per USD with a mild appreciation trend. Asian treasurers had TARFs selling CNH at 6.15 (0.8% better than spot), with doubled notional if CNH depreciated. On 11 August the PBOC devalued the onshore fix by 1.9%; CNH fell to 6.40 over weeks. Corporates sold double-notional at 6.15 into spot near 6.40 — roughly 4%/month MTM loss on twice notional, compounding. Aggregate Asian corporate losses exceeded $10bn — the textbook asymmetric-payoff mis-selling case.

32.6 Delta conventions: ATMF, 25Δ RR, 25Δ BF

FX options are quoted by *delta*, not strike or moneyness. A broker quote for EUR/USD 1-month looks like

1M ATMF	8.50%	(at-the-money-forward vol)
1M 25 RR	−0.40%	(25-delta risk reversal)
1M 25 BF	+0.18%	(25-delta butterfly)

From these three numbers the entire 1-month smile is reconstructed.

Delta and ATMF

Delta is $\Delta = \partial V/\partial S$. "25-delta call" is the strike with $\Delta = 0.25$; "25-delta put" has $\Delta = -0.25$. Clark (2011) gives the full taxonomy of conventions:

- **Spot delta, premium-excluded (SD-PE)**: plain $\partial V/\partial S$. Default for USD-premium pairs (EUR/USD, GBP/USD).
- **Spot delta, premium-included (SD-PI)**: adjusted for foreign-currency premium payment. Standard for USD/JPY etc.
- **Forward delta (FD-PE, FD-PI)**: $\partial V/\partial F$; standard for long-dated options ($T > 1$ year).

One option, four numerical deltas; desks exchanging a "25-delta call" without fixing the convention can be pricing different strikes.

The **at-the-money forward** (ATMF) strike is $K_{\text{ATMF}} = F(0,T) = S_0 e^{(r_d - r_f)T}$, the delta-neutral straddle strike under most conventions. ATMF vol is the single most-quoted number. "ATM-spot" ($K = S_0$) differs from ATMF whenever $r_d \neq r_f$; FX defaults to ATMF.

Risk reversal and butterfly

FX vol quoting via ATMF/RR/BF

For a fixed delta δ (typically 25% or 10%), define

$$\text{RR}_\delta = \sigma(\delta\text{-call}) - \sigma(\delta\text{-put}), \tag{32.18}$$

$$\text{BF}_\delta = \tfrac{1}{2}[\sigma(\delta\text{-call}) + \sigma(\delta\text{-put})] - \sigma_{\text{ATMF}}. \tag{32.19}$$

RR_δ measures the *skew* (asymmetry of the smile); BF_δ measures the *smile convexity* (curvature). Given $(\sigma_{\text{ATMF}}, \text{RR}_\delta, \text{BF}_\delta)$, the wing vols are recovered by

$$\sigma(\delta\text{-call}) = \sigma_{\text{ATMF}} + \text{BF}_\delta + \tfrac{1}{2}\text{RR}_\delta, \qquad \sigma(\delta\text{-put}) = \sigma_{\text{ATMF}} + \text{BF}_\delta - \tfrac{1}{2}\text{RR}_\delta. \tag{32.20}$$

Three numbers (ATMF, 25 RR, 25 BF) pin three vol points: the ATMF and 25-delta call/put vols. Adding 10 RR and 10 BF gives five points for interpolation.

Example: EUR/USD 1-month

With $\sigma_{\text{ATMF}} = 8.50\%, \text{RR}_{25} = -0.40\%, \text{BF}_{25} = 0.18\%$:

$$\sigma(25\Delta\text{-call}) = 8.50 + 0.18 + \tfrac{1}{2}(-0.40) = 8.48\%, \tag{32.21}$$

$$\sigma(25\Delta\text{-put}) = 8.50 + 0.18 - \tfrac{1}{2}(-0.40) = 8.88\%. \tag{32.22}$$

Negative 25 RR means the put is more expensive than the call in vol — the market prices EUR downside (USD strength) as more volatile than EUR upside. Typical EUR/USD skew; opposite in USD/JPY where JPY-strength tails command more vol.

Intuition

ATMF = height, RR = tilt, BF = curvature above the straight line through the wings — the minimum three-parameter smile. Traders express directional vol views via ΔRR ("buying 1M EUR/USD risk-reversal" = betting EUR-downside vol widens further over EUR-upside) and convexity views via ΔBF.

Smile reconstruction: five-point interpolation

For a more refined smile, desks quote a five-point system $(\sigma_{\text{ATMF}}, \text{RR}_{10}, \text{BF}_{10}, \text{RR}_{25}, \text{BF}_{25})$. A typical 3-month EUR/USD quote block might read

ATMF	9.00%
25Δ RR	−0.50%
25Δ BF	0.20%
10Δ RR	−1.10%
10Δ BF	0.75%

from which the five wing vols are

$$\sigma(25\Delta\text{-call}) = 9.00 + 0.20 - 0.25 = 8.95\%, \tag{32.23}$$

$$\sigma(25\Delta\text{-put}) = 9.00 + 0.20 + 0.25 = 9.45\%, \tag{32.24}$$

$$\sigma(10\Delta\text{-call}) = 9.00 + 0.75 - 0.55 = 9.20\%, \tag{32.25}$$

$$\sigma(10\Delta\text{-put}) = 9.00 + 0.75 + 0.55 = 10.30\%. \tag{32.26}$$

These five points are interpolated into a $\sigma(K)$ curve. Standard techniques: Vanna–Volga (Castagna, 2007), SABR (Hagan et al., 2002), and monotone-convex splines for no-arbitrage. They agree on interpolation but diverge on extrapolation beyond 10-delta; FX smiles are thus often quoted to 5-delta when deep-OTM options trade.

RR sign conventions disagree across regions

London desks quote RR as call minus put; some NY desks quote put minus call, flipping every historical series. Check the sign on a broker sheet against implied call/put vols before taking a view. Sanity check: EUR/USD 25 RR is almost always *negative* on London convention; a positive number without an inversion means the sign is flipped.

32.7 FX market structure: CLS and dealer tiering

FX trades in an over-the-counter market split into three tiers.

Tier 1 — interbank: the G10 banks (JPM, Citi, DB, UBS, Goldman, HSBC, BofA, BNP Paribas, and a few others) making two-way prices in size to each other and to top clients. Quotes trade on EBS and Refinitiv FX Matching; inside spread on EUR/USD is sub-pip for $5M clips.

Tier 2 — single-/multi-dealer platforms: each bank's e-FX portal (Barx, NeoBank, Autobahn, . . .) and ECNs (FXall, 360T, Hotspot, Currenex). Asset managers, hedge funds, and corporates trade RFQ or streaming, at spreads scaling with size, urgency, and toxicity.

Tier 3 — retail: CFD brokers, spreads 5–20× interbank and often 2% per trip on exotics.

Settlement. Most FX cashflows settle through CLS (Continuous Linked Settlement), launched 2002, settling 18 currencies PvP in a daily five-hour window. CLS eliminated "Herstatt risk" — your day-side payment going out before the counterparty fails its other-side leg (a live risk before 2002; Bankhaus Herstatt's 1974 $620M failure is the namesake). ITM options settle via spot, so non-CLS pairs carry meaningfully higher operational and credit risk.

Herstatt and the birth of CLS

26 June 1974: German regulators closed Bankhaus Herstatt at 15:30 CET. US banks had paid DEM earlier that morning for DEM/USD trades, expecting USD via CHIPS later. Herstatt's USD leg never paid, leaving roughly $620M of chain losses. The episode drove the three-decade project culminating in CLS. Non-CLS pairs remain the ones with highest implied vol and largest overnight risk charges.

32.8 Vanna–Volga adjustments

The **Vanna–Volga** method (Castagna, 2007) is the standard FX technique for pricing exotics off the three ATMF/25RR/25BF inputs. It is a first-order smile correction: start with the BS/GK price at ATMF vol and add terms making the three vanilla hedges (ATMF straddle, 25Δ RR, 25Δ BF) match market exactly.

Vanna and volga are the mixed and pure second derivatives in σ and S:

$$\text{vanna} = \frac{\partial^2 V}{\partial S\,\partial\sigma}, \qquad \text{volga} = \frac{\partial^2 V}{\partial\sigma^2}. \tag{32.27}$$

A risk-reversal is mostly vanna (long delta–vol correlation); a butterfly is mostly volga (long vol-of-vol). The VV price is the BS/GK price plus a linear combination of vanna, volga, and vega with coefficients pinned by matching the three hedging vanillas:

$$V_{\text{VV}} = V_{\text{BS}} + w_{\text{ATMF}}\,\text{vega} + w_{\text{BF}}\,\text{volga} + w_{\text{RR}}\,\text{vanna}. \tag{32.28}$$

The weights $w_\cdot$ solve a closed-form 3×3 linear system requiring that three reference vanillas (ATMF straddle, 25Δ call, 25Δ put) reproduce their market premia.

Vanna–Volga in one line

Correct the Black–Scholes price of an exotic by adding the vanna and volga dollar-risk, weighted by the market-price surplus over BS of a 25Δ risk-reversal and butterfly respectively, plus a vega correction for the ATMF skew of the exotic's own vol. Three vanilla hedges, three corrections, exact match on those three, first-order smile consistency everywhere else.

VV is not rigorous (not a no-arbitrage price under any model) but reproduces desk quotes for single-barrier options to within a basis point in normal markets. Multi-barrier and path-dependent exotics use smile-consistent MC or PDE instead.

32.9 Desk and competition context

Choosing the right FX hedge

For a hedger with a specific view and exposure, the right FX product follows from three questions: horizon, desired payoff shape, tolerance for basis risk. Short decision guide:

Hedger's situation	**Standard product**
Certain receivable / payable, single date	Forward (if deliverable); NDF (if not)
Budget-rate floor, willing to cap upside	Collar (long put + short call at matching deltas)
Protection against extreme move only	Out-of-the-money put / call
"Cheap" hedge, strong directional view	Knock-out option (accept tail risk on hedge disappearing)
Known range view (spot stays in $[a, b]$)	No-touch option (pays if range holds)
Series of monthly cashflows	Participating forward / accumulator / (beware) TARF
Equity-index exposure in different currency	Quanto option or FX-hedged share class

All price off the same GK engine plus the smile of Section 32.6; only the payoff function changes.

Prosperity Round 1 manual: triangular FX

Round 1's manual gives a small cluster of rates between fictional currencies and asks for a trade sequence ending with more shells than you started with — triangular FX arb (Chapter 14) in disguise. A cycle A $\to$ B $\to$ C $\to$ A is profitable iff the product of rates exceeds 1 net of fees. The Frankfurt Hedgehogs (Chapter 27) brute-forced 4-legged paths. Real-world FX arb scales this to the full cross-rate graph with transaction costs, latency, and basis.

The FX options desk

On a sell-side FX options desk, the Strat owns three systems. First, a *vol surface builder* ingesting ATMF/RR/BF broker quotes across tenors and delta slices — usually a SABR or VV fit (Castagna, 2007). Second, a *pricer* for vanillas (GK) and barriers/digitals/touches (closed-form with skew corrections). Third, a *MC exotics pricer* for TARFs, accumulators, and path-dependent structures, wired into the firm's risk system for aggregate Greeks and overnight VaR. Interviews routinely ask you to derive CIP, state GK, explain ATMF/RR/BF, and sketch a knock-out pricer.

32.10 Key facts to memorise

- **FX is the largest financial market**: \$7.5T per day (BIS 2022), larger than all equity markets combined.
- A pair BASE/QUOTE is quoted in units of quote per unit of base. EUR/USD = 1.08 means one EUR buys 1.08 USD.
- A **pip** is 10^{-4} for most pairs, 10^{-2} for JPY pairs. **Forward points** $= F - S$ in pips.

- **Covered interest parity**: $F = S\,e^{(r_d - r_f)T}$. The high-yielding currency trades at a forward *discount* (the low-yielder at a premium).
- Post-2008 CIP deviates by 10–50 bp persistently. The wedge is the *cross-currency basis*, traded via XCCY basis swaps.
- **Garman–Kohlhagen**: the FX Black–Scholes. Treats foreign currency as a continuous-yield asset with yield r_f.
- **Put–call parity** in FX: $C - P = Se^{-r_f T} - Ke^{-r_d T}$.
- **Quanto adjustment**: subtract $\rho\sigma_{\text{asset}}\sigma_{\text{FX}}$ from the drift when the payoff is in a currency other than the underlying's natural one.
- **Barrier options**: knock-in + knock-out = vanilla. Cheaper than vanillas but disappear when the market moves far enough.
- **TARFs** cap client upside and leverage client downside — the structural mis-selling trap that cost Asian corporates $10bn+ in 2015.
- **FX vol is quoted in delta space**: ATMF vol, plus 25-delta risk-reversal (skew) and 25-delta butterfly (convexity). Three numbers pin down three points on the smile.
- **Negative 25 RR** = market prices downside (for the base currency) at higher vol than upside. Standard in EUR/USD; opposite in many USD/EM pairs.

Chapter 33

Credit Derivatives

Credit derivatives let you trade default risk without holding the bond. The most important instrument is the *credit default swap* (CDS). This chapter derives CDS pricing from first principles, shows how a term structure of CDS quotes is *stripped* into a curve of hazard rates — the exact mirror of the zero-curve bootstrap in Chapter 29 — introduces the CDX and iTraxx index families, and ends with the 2008 Gaussian-copula failure: the model that priced every senior CDO tranche on Wall Street and was catastrophically wrong when it mattered.

Prerequisites

- Chapter 15: risk-neutral expectation, payoff-then-discount, $D(t) = e^{-rt}$.
- Chapter 20: expected loss = $\mathbb{P}(\text{loss}) \times \mathbb{E}[\text{loss} \mid \text{default}]$, and recovery reducing LGD.
- Chapter 29: zero-curve bootstrap — CDS stripping is the credit analogue.

33.1 Default probability and recovery rate

A *default* is a corporate (or sovereign) failing to meet a contractual obligation: a missed coupon, missed principal, or bankruptcy filing. Default is the single event that makes a bond risky. Two numbers summarise that risk.

Definition 33.1 (Default probability)**.** The *cumulative default probability* to horizon T, written $\mathrm{PD}(T)$, is the probability that the reference entity defaults at some time $\tau \le T$. The *survival probability* is the complement,

$$S(T) := \mathbb{P}(\tau > T) = 1 - \mathrm{PD}(T).$$

Definition 33.2 (Recovery rate)**.** When a default occurs on a bond with face value 1, the holder recovers a fraction $R \in [0,1]$ of face value through the bankruptcy or workout process. The *loss given default* is $\mathrm{LGD} = 1 - R$.

Two PD flavours matter in Section 33.2:

- *Physical* (or *actuarial*) PD, $\mathbb{P}(\tau \le T)$: the fraction of similar firms that default. Estimated by rating agencies (Moody's, S&P, Fitch) from historical default data by rating bucket.
- *Risk-neutral* PD, $\mathbb{Q}(\tau \le T)$: implied by traded prices under the pricing measure. Typically larger than physical PD by a *credit risk premium* (Hull, 2014).

CDS pricing lives entirely in the risk-neutral world. We drop the $\mathbb{Q}$ superscript and write $\mathrm{PD}(T)$, $S(T)$; every default probability in this chapter is risk-neutral.

Typical recovery rates

Recoveries depend on seniority. Rating-agency databases give, for corporate bonds (Hull, 2014):

Seniority	Typical R
Senior secured	50–60%
Senior unsecured	35–45%
Subordinated	20–30%
Junior subordinated	10–20%

The standard CDS convention is $R = 40\%$ for senior unsecured; sovereign CDS typically assumes $R = 25\%$.

Worked example — expected loss. Suppose a 5-year cumulative default probability is $\mathrm{PD}(5) = 20\%$ and the recovery rate is $R = 40\%$. The expected loss on \$1 of face value over 5 years is

$$\mathrm{EL} = \mathrm{PD}(5) \times (1 - R) = 0.20 \times 0.60 = 0.12 = 12\%.$$

Equivalently, \$1,200,000 on a \$10M position. This identity becomes the CDS par spread once discounting and default timing are added in.

PD and survival

For a reference entity with risk-neutral default time τ:

$$\begin{aligned}
\mathrm{PD}(T) &= \mathbb{P}(\tau \le T) \qquad \text{(cumulative)}\\
S(T) &= 1 - \mathrm{PD}(T) \qquad \text{(survival)}\\
\mathrm{LGD} &= 1 - R \qquad \text{(loss given default)}\\
\mathrm{EL}(T) &= \mathrm{PD}(T) \cdot (1 - R) \qquad \text{(crude expected loss per unit notional).}
\end{aligned}$$

"Crude" because this ignores discounting and default timing; the next section corrects both.

Do not confuse physical and risk-neutral PD

A rating-agency 5-year PD for a BBB-rated firm is roughly 2%; the risk-neutral PD implied by the same firm's 5-year CDS is typically **4–5 times higher**. The ratio is the *credit risk premium* — compensation for bearing jump-to-default risk. Pricing a CDS with physical PD mis-marks the contract by hundreds of bp.

33.2 Credit default swaps

A *credit default swap* is a bilateral contract functioning as insurance against default of a *reference entity* (e.g. Ford Motor Co.). In plain English: the protection buyer pays a periodic premium (the *CDS spread*) and, if the reference entity defaults on its bonds before maturity, receives par in exchange for the defaulted bond — exactly like a fire-insurance policy that pays out only if the house burns down. The CDS market turned corporate credit into a tradable asset class in the late 1990s; notional outstanding peaked at \$60 trillion in 2007 and has since shrunk to roughly \$8 trillion post-crisis (Hull, 2014).

Mechanics

Two counterparties:

- the *protection buyer* pays a periodic premium, called the *CDS spread* s, quoted in basis points per annum of notional;
- the *protection seller* receives the premium and is contractually obliged to pay $(1-R)\times$ notional if a *credit event* occurs on the reference entity before maturity.

Premium leg. The buyer pays $s\cdot\delta_i$ per unit notional on scheduled dates $t_1 < t_2 < \ldots < t_n = T$, where δ_i is the year fraction between payment dates (typically $\delta_i = 0.25$, i.e. quarterly payments under the standard IMM-date convention). Payment continues only while the entity is alive. If default occurs at some $\tau \in (t_{i-1}, t_i)$, the buyer pays an *accrual* from t_{i-1} to τ and then stops.

Protection leg. At default τ, the seller pays $(1-R)$ per unit notional. Under physical settlement the buyer delivers defaulted bonds (face value = notional) for cash of notional, recovering R on the bonds and netting $(1-R)$. Since 2009 the standard is cash settlement via the ISDA auction, which determines a single recovery value per credit event.

Credit events. Under the ISDA Credit Derivatives Definitions, triggers are bankruptcy, failure to pay, and (non-US) restructuring. An ISDA Determinations Committee rules on occurrence, then holds a public auction to set R.

Intuition

A CDS is default insurance: a premium for a lump-sum $(1-R)$ payout if the reference entity is "totalled". The fair premium equals the expected discounted cost of the payout. Unlike insurance, you need not own the reference bond — CDS separated credit exposure from bond ownership, which made credit a tradable asset class.

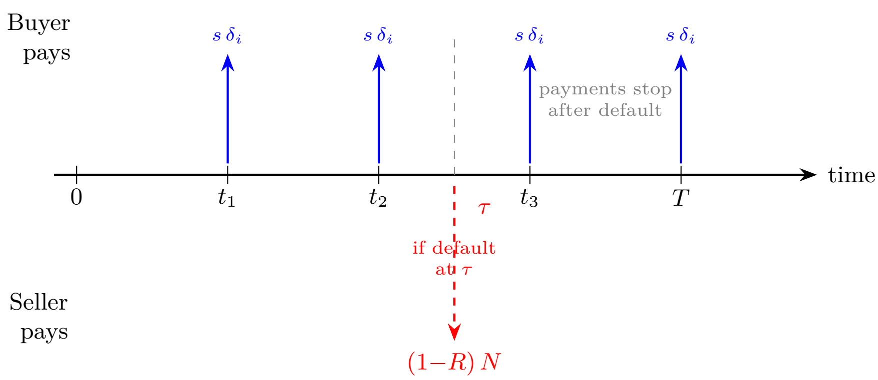


Figure 33.1: CDS cashflow structure. The protection *buyer* pays quarterly premiums $s\,\delta_i$ per unit notional (blue arrows, upward) at each scheduled date, but only while the reference entity survives. If default occurs at τ, the protection *seller* pays $(1-R)N$ (red dashed arrow, downward) and all further premium payments stop. The par spread s^* is set so these two legs have equal present value at inception.

Pricing: the two legs

Let s denote the running premium (per annum, per unit notional), and let $D(t)$ denote the risk-free discount factor to time t. Assume for now that default times and the risk-free rate are independent (the standard simplifying assumption).

Premium leg present value

Ignoring accrual, the premium leg pays $s\,\delta_i$ per unit notional at t_i conditional on survival to t_i. Its PV is

$$\mathrm{PL}(s) = s \cdot \underbrace{\sum_{i=1}^{n} \delta_i\, D(t_i)\, S(t_i)}_{A \,:=\, \text{risky annuity}} = s \cdot A, \tag{33.1}$$

where A is the *risky annuity factor* or *DV01*: the present value of a stream of \$1-per-annum quarterly coupons paid while the name is alive.

Protection leg present value

The protection leg pays $(1 - R)$ at the default time τ if $\tau \le T$. Its PV is

$$\mathrm{PP} = (1 - R) \int_0^T D(t)\, f(t)\, dt = (1 - R) \int_0^T D(t)\, d[1 - S(t)], \tag{33.2}$$

where $f(t) = -S'(t)$ is the risk-neutral density of the default time, and $d[1 - S(t)] = f(t)\,dt$ is the probability of default falling in $[t, t + dt]$.

Par spread

At inception the CDS is traded at the *par spread* $s^\star$ that makes the contract costless to both sides: $\mathrm{PL}(s^\star) = \mathrm{PP}$, so

$$\boxed{s^\star = \frac{\mathrm{PP}}{A} = \frac{(1 - R)\int_0^T D(t)\, d[1 - S(t)]}{\sum_{i=1}^{n} \delta_i\, D(t_i)\, S(t_i)}.} \tag{33.3}$$

CDS par spread formula

The fair running spread of a CDS is the *expected discounted loss* divided by the *risky annuity*:

$$s^\star \;=\; \frac{(1 - R)\, \mathbb{E}_{\mathbb{Q}}\Big[D(\tau)\, \mathbb{1}_{\{\tau \le T\}}\Big]}{\mathbb{E}_{\mathbb{Q}}\Big[\sum_i \delta_i\, D(t_i)\, \mathbb{1}_{\{\tau > t_i\}}\Big]}.$$

Numerator: expected PV of the payoff. Denominator: expected PV of \$1-per-annum coupons while alive.

Intuition

The par spread equals *expected discounted loss per unit notional, per year.* For small, flat hazard and discount rates the back-of-envelope identity is

$$s^\star \;\approx\; h \cdot (1 - R),$$

where h is the risk-neutral hazard rate. “A 300 bp CDS at 40% recovery implies a hazard of about 500 bp” ($500 \approx 300/0.60$) — standard mental arithmetic on a credit desk.

End-to-end worked example

Concrete numbers; the Python in Listing 33.1 reproduces every figure.

Trade details.

- Reference entity: a BBB-rated corporate, senior unsecured.
- Notional: $N = \$10{,}000{,}000$.
- Maturity: $T = 5$ years.
- Premium frequency: quarterly ($\delta_i = 0.25$), so $n = 20$ payment dates at $t_i = 0.25, 0.50, \ldots, 5.00$.
- Risk-free zero rate: flat $r = 4\%$ continuously compounded, $D(t) = e^{-0.04t}$.
- Hazard rate: flat $h = 3\%$, giving $S(t) = e^{-0.03t}$. (We formally introduce the hazard rate in Section 33.3.)
- Recovery: $R = 40\%$, so $1 - R = 0.60$.

Step 1 — risky annuity. Plugging into Equation (33.1),

$$A = \sum_{i=1}^{20} 0.25 \cdot e^{-(0.04+0.03)\,t_i} = 0.25 \sum_{i=1}^{20} e^{-0.07\,0.25i} = 4.18194.$$

So \$1/year of quarterly premium until default or $T = 5$ has PV \$4.1819 per unit notional.

Step 2 — protection leg. Using $S(t) = e^{-ht}$, $d[1 - S(t)] = he^{-ht}dt$, so Equation (33.2) evaluates in closed form:

$$\mathrm{PP} = (1 - R) \int_0^T h\, e^{-(r+h)t}\, dt = (1 - R) \cdot \frac{h}{r + h} \left(1 - e^{-(r+h)T}\right).$$

Substituting:

$$\mathrm{PP} = 0.60 \cdot \frac{0.03}{0.07} \left(1 - e^{-0.07 \cdot 5}\right) = 0.60 \cdot 0.42857 \cdot 0.29531 = 0.075937.$$

So the protection leg is worth 7.59% of notional.

Step 3 — par spread.

$$s^\star = \frac{\mathrm{PP}}{A} = \frac{0.075937}{4.18194} = 0.018158 = 181.6 \text{ bp}.$$

Step 4 — dollar cash flows. On $N = \$10\text{M}$ the two legs match at par: $\mathrm{PP} \cdot N = \$759{,}373$ of expected discounted loss, and a quarterly premium coupon $s^\star \delta N = \$45{,}395$.

Step 5 — sanity check. The small-rate, small-hazard approximation says

$$s^\star \;\approx\; h(1 - R) \;=\; 0.03 \cdot 0.60 \;=\; 180 \text{ bp}.$$

The exact answer 181.6 bp sits 1.6 bp above the approximation; the gap is $r/(r + h)$ discounting at finite tenor. Sharp enough for mental P&L.

Prosperity example

Prosperity 3 had no defaulting corporates, but "expected loss = probability × severity" is identical to how top teams sized stop-losses on adverse-selection-prone market-making (Chapter 27): a 20% chance of a 60% adverse move on \$10k gives expected loss $0.2 \cdot 0.6 \cdot 10{,}000 =$ \$1,200. If quoting profit is below that, widen or stop. CDS par spread is the same idea with hazard rate and discounting.

```python
import math

def cds_par_spread(T, r, h, R, freq=4):
    """Par spread of a flat-hazard, flat-rate CDS.
    Premium leg pays quarterly while alive; protection leg pays
    (1-R) at default time.  Ignores accrued premium at default."""
    dt = 1.0 / freq
    times = [(i+1)*dt for i in range(int(freq*T))]
    # risky annuity: sum of delta * D(t) * S(t)
    annuity = sum(dt * math.exp(-(r+h)*t) for t in times)
    # protection leg: (1-R) * h/(r+h) * (1 - exp(-(r+h)T))
    protection = (1-R) * h/(r+h) * (1 - math.exp(-(r+h)*T))
    return protection / annuity, annuity, protection

if __name__ == "__main__":
    s, A, PP = cds_par_spread(T=5, r=0.04, h=0.03, R=0.40)
    print(f"annuity        = {A:.4f}")
    print(f"protection     = {PP:.6f}")
    print(f"par spread     = {s*1e4:.1f} bp")
    # annuity        = 4.1819
    # protection     = 0.075937
    # par spread     = 181.6 bp
```

Listing 33.1: Minimal CDS pricer. Reproduces every number in Section 33.2.

Credit Greeks

The three first-order sensitivities every credit trader quotes:

- **CS01** (*credit spread oh-one*): the P&L impact of a 1 bp parallel shift in the reference entity's CDS spread curve. Equivalently, one basis point of the risky annuity A times notional: $\text{CS01} \approx A \cdot N \cdot 10^{-4}$. For the worked example, $\text{CS01} \approx 4.182 \times 10{,}000{,}000 \times 10^{-4} =$ \$4,182 per bp.
- **JTD** (*jump-to-default*): P&L if the entity defaults right now. For a protection buyer, $\text{JTD} = (1 - R) \cdot N$ minus the current premium-leg MtM; for a fresh at-market trade, $\text{JTD} \approx (1 - R) \cdot N$. In the worked example, $0.60 \times$ \$10M = \$6M — the single largest risk in a CDS book.
- **IR01**: P&L from a 1 bp parallel shift in the risk-free curve. Small on CDS (dominant risk is spread, not rates) but non-zero via $D(t)$ in the annuity.

Risk reports aggregate CS01 by rating/sector/maturity and JTD by name to surface *concentration risk*: \$50M JTD to a single name books \$50M of loss overnight on that credit event.

Intuition

CS01 is day-to-day P&L risk (spreads move a few bp/day); JTD is catastrophic-event risk (rare but binary). Risk managers look at CS01 first, regulators at JTD first. A well-hedged book has small CS01 and small *name-level* JTD — but portfolio-level JTD can still be large if names correlate (the CDO problem of Section 33.5).

The CDS–bond basis

In theory, long bond + long CDS protection on the same name yields the Treasury rate, so the bond's *asset swap spread* (over Treasuries, expressed as a CDS-equivalent running premium) equals the CDS par spread:

$$\text{basis} \;:=\; s^\star_{\text{CDS}} - s_{\text{asw}} \;\overset{?}{=}\; 0.$$

In practice the basis is non-zero. Post-2008 the IG basis has been *negative* (bond spread exceeds CDS) by 20–80 bp, for structural reasons:

- *Funding*: a CDS is unfunded; a long bond ties up balance sheet. Dealers charge funding into the bond spread.
- *Counterparty risk*: a CDS seller can default on the payout (AIG 2008), reducing protection value.
- *Short-selling frictions*: shorting a corporate bond is expensive; buying CDS is frictionless, skewing hedge demand toward CDS and compressing its spread.
- *Cheapest-to-deliver optionality*: the buyer delivers any eligible bond at default, widening CDS vs single-bond ASW.

Relative-value desks run a *basis book* going "long bond, long CDS protection" when the basis is abnormally negative. Like every carry trade, it blows up when liquidity vanishes (Mar 2020, Oct 2008).

Running spread versus upfront

Post-2009 standard CDS (US "Big Bang", EU "Small Bang") trade with a fixed coupon (100 bp IG, 500 bp HY) plus an *upfront* payment bringing NPV to zero. A quoted number may be a par spread *or* an upfront — always ask. Conversion uses the risky annuity: $U = (s^\star - c) \cdot A$. Getting this wrong on a 5Y IG name costs 4–5 bp per bp of spread error.

33.3 Hazard rates and stripping

The market quotes CDS at standard tenors (1, 3, 5, 7, 10Y); the spread-curve shape implies a time-varying hazard rate. This section defines the hazard rate, derives survival, and bootstraps piecewise-constant hazards from observed spreads — the credit mirror of the zero-curve bootstrap in Chapter 29.

Hazard rate: definition

Definition 33.3 (Hazard rate)**.** The (risk-neutral) *hazard rate* $h(t)$ is the instantaneous default intensity conditional on survival to t:

$$h(t)\,dt \;:=\; \mathbb{P}(\tau \in [t, t+dt] \mid \tau > t)\,.$$

Equivalently, $h(t) = -S'(t)/S(t) = -\frac{d}{dt}\log S(t)$.

Integrating the ODE $dS/S = -h(t)\,dt$ from 0 to T:

$$S(T) = \exp\left(-\int_0^T h(u)\,du\right). \tag{33.4}$$

Constant h gives $S(T) = e^{-hT}$ and exponentially distributed τ with mean $1/h$. A 3% hazard gives expected time-to-default $1/0.03 \approx 33$ years and 5Y cumulative PD $1 - e^{-0.15} = 13.9\%$.

Intuition

The hazard rate is to default what the short rate is to the zero curve: $D(T) = \exp(-\int_0^T r\,du)$ vs $S(T) = \exp(-\int_0^T h\,du)$. Every technique from Chapter 29 — bootstrap, splines, forwards — transfers verbatim. Credit traders call $h(t)$ the "credit short rate".

Stripping: the bootstrap

Given par spreads $s_1^\star, \ldots, s_m^\star$ at maturities $T_1 < \ldots < T_m$, we want a survival curve consistent with all quotes. The standard parameterisation is piecewise-constant hazard:

$$h(t) = h_k \qquad \text{for } t \in (T_{k-1}, T_k],$$

with $T_0 := 0$. The unknowns are $h_1, h_2, \ldots, h_m$.

Bootstrap procedure.

1. Set $k = 1$. The shortest quote $s_1^\star$ determines h_1: solve the one-equation-in-one-unknown problem $s_1^\star \cdot A_1(h_1) = \mathrm{PP}_1(h_1)$ for h_1 by root-finding (1-D Newton or bisection). Before T_1, only h_1 enters both sides.
2. For $k = 2, \ldots, m$: $h_1, \ldots, h_{k-1}$ are now known. Solve $s_k^\star \cdot A_k(h_k; h_1, \ldots, h_{k-1}) = \mathrm{PP}_k(h_k; h_1, \ldots, h_{k-1})$ for the single unknown h_k.

Each step is a scalar root-find. The survival function at any interior time $t \in (T_{k-1}, T_k]$ is then

$$S(t) = \exp\left(-\sum_{j=1}^{k-1} h_j(T_j - T_{j-1}) - h_k(t - T_{k-1})\right).$$

Reducing the root-find. Each step is a 1-D Newton on

$$g_k(h_k) := s_k^\star \cdot A_k(h_k) - \mathrm{PP}_k(h_k),$$

with initial guess $h_k^{(0)} = s_k^\star/(1-R)$ — converges in 3–5 iterations. Derivatives dA_k/dh_k, $d\mathrm{PP}_k/dh_k$ have closed forms on piecewise-constant intervals; otherwise use numerical derivatives.

Sanity checks on a stripped curve.

- $h_k \geq 0$: a negative hazard usually means a stale or fat-finger quote (or an exotic sovereign).
- S monotone: $S(T_1) \geq S(T_2)$ for $T_1 \leq T_2$ — equivalent to $h_k \geq 0$.
- No implausible jumps at tenor boundaries. A jump from 200 bp to 50 bp between 3Y and 5Y either prices a known corporate event in that window, or signals piecewise-constant is too coarse — switch to piecewise-linear or tension splines.

Worked sanity-check. 1Y/3Y/5Y CDS on a BBB name at 100/150/200 bp with $R = 40\%$. The approximation $h \approx s/(1-R)$ gives hazards ≈ 167, 250, 333 bp — upward-sloping, consistent with the credit curve. The exact bootstrap shifts these by a few bp and preserves shape.

CDS curve stripping

Given par spreads at maturities $\{T_k\}$ and piecewise-constant hazard on $(T_{k-1}, T_k]$, the $\{h_k\}$ are recovered by a 1-D root-find per step, in increasing maturity order. Mathematically identical to the zero-curve bootstrap of Chapter 29: at step k, prior quantities are pinned, so one unknown meets one equation.

Spread curve shape does not equal hazard curve shape

If the 1Y and 5Y par spreads are both 200 bp, the hazard curve is *not* flat — it's slightly downward-sloping, because a flat long-maturity spread embeds the average of lower near-term hazards. The spread-to-hazard mapping is nonlinear; only a bootstrap reads the hazard curve off a price screen.

33.4 CDX and iTraxx indices

Individual-name CDS are *single-name CDS*. Trading a basket in one wrapper is cheaper and more liquid, so the market built standardised credit indices.

Definition 33.4 (CDS index)**.** A *credit index* is a standardised equally-weighted portfolio of single-name CDS on a fixed roster of entities, trading as one instrument. Buying index protection is economically equivalent to buying protection pro-rata on each name.

The major families, all administered by Markit/S&P:

- **CDX.NA.IG** — 125 North-American IG names, reconstituted each March/September on "index roll" dates. The flagship IG benchmark.
- **CDX.NA.HY** — 100 North-American HY names.
- **iTraxx Europe** — 125 European IG names.
- **iTraxx Crossover** — 75 European sub-IG names.
- **iTraxx Asia/Japan, CDX.EM** — regional offshoots.

Index CDS trade with fixed coupons (100 bp IG, 500 bp HY) plus an upfront bringing NPV to zero, matching single-name convention post-2009.

Birth of the CDS — JP Morgan 1994

The first CDS was structured in 1994 by Blythe Masters' JP Morgan team to hedge a \$4.8bn credit line to Exxon (Valdez-spill liability), paying the EBRD to take the default risk. Standardised ISDA documentation arrived by 2000; notional hit \$6tn by 2004 and \$60tn by 2007. CDS made credit a tradable asset class; overused and misunderstood, the same instrument nearly destroyed the system in 2008.

Index arbitrage and the skew. The *intrinsic* level is the average par spread of the 125 constituents; the *quoted* index spread deviates from it by the *basis* or *index skew*. Tight in calm markets (a few bp), it widens in stress as investors buy index protection (cheap, liquid) faster than dealers hedge it on the 125 singles. A dealer arbitrage buys index, sells singles pro rata, and holds until convergence.

The index trading desk

Sell-side credit desks usually run the index book separately from single-names: indices trade in \$100M clips with hedge funds and macro, singles in \$5–10M clips with real money. The index Strat owns a basis monitor (intrinsic vs quoted for CDX.IG/HY, iTraxx Main/Xover) and a PnL explain attributing the day's move to index level, basis carry, single-name hedges, and roll costs. Interviews test why the skew exists, when it blows out, and how to monetise a 5 bp dislocation.

Tranches on indices. An index can be carved into *tranches* of different seniority — the subject of the next section, and of 2008.

33.5 CDO tranches and the Gaussian copula

A *collateralised debt obligation* (CDO) packages a pool of credit-risky assets (CDS in a *synthetic CDO*, mortgage loans in a *cash CDO*) and issues layered claims. Each claim is a *tranche* defined by an attachment/detachment point in portfolio-loss space. Plain English: pool 125 bonds, then sell slices in order of who eats losses first — the *equity* tranche absorbs the first few percent of defaults and is paid the highest yield as compensation for bearing first-loss risk; the *senior* tranche pays the lowest yield but is only touched if losses are catastrophic.

Tranche mechanics

Let $L(t) \in [0, 1]$ be the fractional portfolio loss at time t. A tranche with attachment a and detachment d ($0 \le a < d \le 1$) has loss profile

$$L_{[a,d]}(t) = \frac{\min(L(t), d) - \min(L(t), a)}{d - a} \in [0, 1].$$

The tranche absorbs nothing below a, 1-for-1 losses between a and d, and is wiped out above d.

Standard CDX.NA.IG tranches:

Tranche	**Attachment**	**Detachment**
Equity	0%	3%
Junior mezz	3%	7%
Senior mezz	7%	10%
Senior	10%	15%
Super-senior	15%	30%

Equity is first-loss and pays the highest coupon; super-senior is safest and, pre-2008, rated AAA by S&P and Moody's.

The Gaussian copula model

Tranche pricing needs the *joint* default distribution across 125 names, not marginals alone. The industry model (Li, 2000) — Li's 2000 JP Morgan note — uses a *single-factor Gaussian copula*.

Construction. For each name $i = 1, \ldots, 125$:

1. Extract marginal risk-neutral $F_i(t) = \mathrm{PD}_i(t)$ from the single-name CDS curve.

2. Draw $X_i = \sqrt{\rho}\, M + \sqrt{1-\rho}\, Z_i$, with systemic factor M, idiosyncratic Z_i, all iid $\mathcal{N}(0,1)$, and $\rho \in [0,1]$ the *copula correlation*.
3. Map to default times via $\tau_i = F_i^{-1}(\Phi(X_i))$. Marginals are preserved; the joint distribution is governed by ρ.

Monte Carlo over $(M, Z_1, \ldots, Z_{125})$ gives the joint default distribution; tranche par spreads follow from Equation (33.3) with $S(t)$ replaced by $\mathbb{E}[1 - L_{[a,d]}(t)]$.

Implied correlation. Inverting for the ρ that reproduces each tranche's market spread gives an *implied correlation*, analogous to implied vol. A true Gaussian copula would price every tranche with a single ρ; in reality implied ρ varies — the *correlation skew*, credit analogue of the vol smile (Chapter 18).

Li's single-factor Gaussian copula

Correlated default times are generated via

$$X_i = \sqrt{\rho}\, M + \sqrt{1-\rho}\, Z_i, \quad M, Z_i \sim \mathcal{N}(0,1) \text{ iid}, \quad \tau_i = F_i^{-1}(\Phi(X_i)).$$

One ρ pins the entire correlation structure; tranche prices follow by Monte Carlo or semi-analytic integration over M.

Semi-analytic tranche loss: a back-of-envelope

Conditional on $M = m$, the defaults $\{\tau_i \le T\}$ are independent Bernoulli events with common probability

$$p(m) \;=\; \mathbb{P}\Big(X_i < \Phi^{-1}(\mathrm{PD}(T)) \mid M = m\Big) \;=\; \Phi\left(\frac{\Phi^{-1}(\mathrm{PD}(T)) - \sqrt{\rho}\, m}{\sqrt{1-\rho}}\right).$$

The conditional default count is binomial; for large portfolios the *large-homogeneous-pool* (LHP) approximation (Hull, 2014) replaces it by its mean $p(m)$. Integrating over $M \sim \mathcal{N}(0,1)$ gives the unconditional loss density.

Numeric illustration. $\mathrm{PD}(5) = 5\%$, $R = 40\%$, $\rho = 0.30$, 100 names. A 2,000-path Monte Carlo gives a 5Y expected loss fraction $\approx 2.96\%$, matching the exact $\mathrm{PD}(1-R) = 3.00\%$ within sampling error, independent of ρ. Unconditional expected loss is ρ-invariant; what ρ governs is the *distribution* around that mean, hence tranche prices.

Higher ρ thickens both tails of the loss distribution — simultaneous defaults and simultaneous survivals become more likely — *raising* senior-tranche expected loss (tail hits) and *lowering* equity-tranche expected loss (common survivals). This is the core comparative static.

Intuition

100 names, PD $= 5\%$. At $\rho = 0$: defaults cluster near 5 with sd ≈ 2.2, so a 10–15% tranche is almost never hit. At $\rho = 1$: either all 100 default (5%) or none do (95%); the super-senior 10–15% tranche takes a full loss 5% of the time. Correlation pushes mass from the middle of the loss distribution into both tails.

2008: the model that killed Wall Street

The Gaussian copula was adopted as the industry standard right after Li's 2000 paper because it was tractable: spreadsheet-ready for rating agencies, mark-to-market-ready for dealers. The

CDO market grew from nothing to \$4.7tn notional by 2007 (Salmon, 2009). When the US housing market cracked in 2007–08, the model's flaws were exposed catastrophically.

Failure 1: ρ is regime-dependent. The copula assumes a single constant ρ. In reality, asset correlations rise sharply in crises: 20% in calm times becomes 70% under stress ("correlations go to one"). Super-senior tranches priced at low ρ were massively mispriced when correlation spiked and losses blew through the 15% attachment point.

Failure 2: wrong calibration data. ρ was calibrated to 5Y equity-return correlations or index tranche quotes, both predominantly normal-regime data. Tail correlation was systematically understated.

Failure 3: ratings depended on the model. Structured product AAA ratings were themselves outputs of Gaussian-copula-like models. When the model failed, the rating failed with it; AIG's super-senior CDS book required a \$180bn US rescue in 2008 (Hull, 2014).

The formula that killed Wall Street

David X. Li's 2000 JP Morgan paper *On Default Correlation: A Copula Function Approach* brought the Gaussian copula to credit. Within five years every rating agency, structured-product desk, and risk system used it or a variant. Felix Salmon's 2009 *Wired* piece "Recipe for Disaster: The Formula That Killed Wall Street" (Salmon, 2009) named it after the 2008 collapse of the structured-credit market. The paper is one of the most-cited and most-vilified derivative-pricing notes in history.

The Gaussian copula is wrong in crisis

The single-factor Gaussian copula prices tranches with a constant ρ from a static snapshot. In a crisis:

- asset correlations rise ($\rho \to 1$): tails are far heavier than the model assumes;
- the Gaussian has no tail dependence: extreme co-movements are near-zero-probability by construction;
- implied correlation across tranches becomes a surface, not a constant — the credit analogue of the vol smile.

Practitioners now use base-correlation models, t-copulas with tail dependence, and stressed add-ons. The Gaussian copula survives as a *quoting convention*, not a valuation truth.

33.6 Prosperity and desk connections

CVA: credit risk in every derivative

Every OTC derivative has *bilateral counterparty risk*: if your swap is \$5M in the money and the counterparty defaults, you become an unsecured creditor recovering at most \$5M $\times R$. In plain English, *CVA* is the market value of that counterparty-default risk — the amount of value you would lose today if the counterparty defaulted right now, weighted by how likely they are to default and how far in the money the trade will be when they do.

Definition 33.5 (Credit Valuation Adjustment)**.** *CVA* is the expected discounted loss on a derivative from counterparty default. For a single swap against counterparty C:

$$\mathrm{CVA} \;=\; (1 - R_C) \int_0^T D(t)\, \mathbb{E}_{\mathbb{Q}}[\mathrm{EE}(t)]\, h_C(t)\, e^{-\int_0^t h_C(u)\,du}\, dt,$$

with $\mathrm{EE}(t) = \max(V(t), 0)$ the expected positive exposure (loss only when in our favour) and h_C the counterparty hazard.

CVA is structurally a CDS on your own derivative exposure, using the same machinery (stripped hazards, recovery, $\int D\, d[1 - S]$ integrals) layered on Monte Carlo exposure paths. Mechanics are deferred to Chapter 36; the point is that every desk implicitly runs a CDS pricer, because every desk contributes to the firm's counterparty-risk book.

Intuition

Sell a \$10M IRS to a BB-rated hedge fund and you are long the swap and *short* CDS protection on the fund — a CDS on the expected positive exposure, not on fixed notional. CVA is the price of that implicit CDS, and a bank's CVA desk is the internal CDS-hedging desk the firm trades against.

The symmetric adjustment is *DVA* (debit value adjustment): the expected discounted gain to us from the possibility that *we* default on the counterparty, computed with the same integral but using our own hazard h_{self} and the expected *negative* exposure $\max(-V(t), 0)$. Bilateral CVA from our perspective is then CVA − DVA. DVA is controversial because it credits a bank's P&L when its own credit deteriorates — a "benefit from our own default risk" that is impossible to hedge without buying protection on yourself.

Mark-to-market of a seasoned CDS

A CDS is costless at inception but acquires MtM as spreads move. Enter a 5Y protection *buyer* at $s_0 = 180$ bp; three months later the residual 4.75Y reprices at $s_1 = 250$ bp. The MtM is (Hull, 2014)

$$V_{\text{MtM}} \;=\; (s_1 - s_0) \cdot A(T - 0.25) \cdot N,$$

where $A(T - 0.25)$ is the residual risky annuity of a 4.75-year CDS at today's hazard curve. Using the worked-example numbers and a residual annuity of roughly 4.00, the MtM on \$10M notional is

$$V_{\text{MtM}} \;\approx\; (0.025 - 0.018) \cdot 4.00 \cdot \$10\text{M} \;=\; \$280{,}000,$$

a profit for the buyer (insurance now worth more). If spreads tighten to 120 bp, MtM is −\$240,000.

Why annuity-weighted. $(s_1 - s_0)A$ follows from $\mathrm{PL}(s) = s \cdot A$: the buyer pays $s_0 AN$ per annum while the fair price is $s_1 AN$, so the buyer is AN better off per unit of widening. This is the long-bond-rally mechanism — A is the credit analogue of duration — and why CS01 $= A \cdot N \cdot 10^{-4}$.

Intuition

A CDS is an annuity-weighted short position on the spread. Every bp of widening is worth $A \cdot N \cdot 10^{-4}$ to the buyer — \$4,200/bp on a 5Y \$10M CDS with $A \approx 4.2$. A credit trader's P&L report is "$\sum_i \mathrm{CS01}_i \cdot \Delta s_i$" over 250 names.

Credit-curve stripping mirrors the zero-curve bootstrap

Prosperity has no defaulting instruments, but CDS stripping is structurally identical to the Treasury-strip bootstrap (Chapter 29): a sequence of 1-D root-finds across maturities converting observed prices into calibrated latent rates. Bootstrap a Prosperity coupon-bond curve and you can bootstrap a CDS curve; only the instrument differs.

The credit Strat's job description

The credit-derivatives Strat owns four systems, in order of centrality: (i) a nightly/intraday *PD stripping pipeline* turning broker CDS quotes into per-entity hazard curves; (ii) a *single-name CDS pricer* with upfront/coupon conversion and full Greeks (CS01, JTD); (iii) a *CDO/tranche pricer*, Monte Carlo under a Gaussian copula for sanity, base-correlation or t-copula overlay for production; (iv) a *CVA engine* (Chapter 36) pricing counterparty-default risk on every trade. Interviewers test (i)–(ii) deeply and (iii)–(iv) conceptually.

33.7 Key facts to memorise

- A **CDS** is default insurance: the buyer pays a running premium s on notional; the seller pays $(1-R)\times$ notional on a credit event.
- **PD and survival**: $S(T) = 1 - \mathrm{PD}(T)$. Risk-neutral PD (implied by CDS) exceeds physical PD (historical agency default frequency) by a credit risk premium.
- **Recovery**: $R = 40\%$ for senior unsecured is the market convention. $\mathrm{LGD} = 1 - R$.
- **CDS par spread**: $s^\star = (1-R)\int_0^T D(t)\,d[1-S(t)] \,/\, \sum_i \delta_i D(t_i)S(t_i)$ — expected discounted loss over risky annuity.
- **Quick approximation**: $s^\star \approx h(1-R)$. A 300 bp CDS with $R = 40\%$ implies a hazard rate of about 500 bp.
- **Hazard rate**: $h(t)\,dt = \mathbb{P}(\tau \in [t, t+dt] \mid \tau > t)$. Survival is $S(T) = \exp\!\left(-\int_0^T h(u)\,du\right)$.
- **Stripping** is the credit analogue of zero-curve bootstrapping (Chapter 29): piecewise-constant hazard rates solved maturity-by-maturity, one root-find per step.
- **CDX.NA.IG** (125 IG names, fixed coupon 100 bp), **CDX.NA.HY** (100 HY names, 500 bp), **iTraxx Main** (125 European IG) are the flagship indices.
- **Tranches** carve a pool into seniority layers: equity (0–3%), mezz, senior, super-senior (15%+). First losses hit the equity tranche.
- **Li's single-factor Gaussian copula**: $X_i = \sqrt{\rho}M + \sqrt{1-\rho}Z_i$, default times via $\tau_i = F_i^{-1}(\Phi(X_i))$. One ρ for the whole pool.
- **2008 failure**: regime-dependent ρ rises in crisis, so AAA super-senior tranches priced at low ρ were catastrophically mispriced when defaults clustered — "the formula that killed Wall Street" (Salmon, 2009).
- **Credit Greeks**: CS01 (1 bp credit sensitivity), JTD (jump-to-default), recovery delta.

Chapter 34

Commodities

Commodities are physical goods with storage costs, seasonality, and the tightest link to the real economy. A barrel of crude has to sit somewhere and somebody pays the rent, which warps every derivative price. This chapter covers futures-curve mechanics, the cost-of-carry formula extended with *storage cost* u and *convenience yield* y (the hidden dividend earned by whoever has physical inventory when a refinery runs dry), and Schwartz's one-factor mean-reverting model.

Prerequisites

From earlier chapters:

- Chapter 14: the cash-and-carry no-arbitrage argument for forward prices on storable assets, $F = Se^{rT}$ for a pure-investment asset.
- Chapter 15: call and put payoffs, risk-neutral expectation, the Black–Scholes pricing machinery.
- Chapter 29: the discount factor $D(t) = e^{-rt}$ under a flat zero curve, used to PV future cash flows.

If the cost-of-carry argument of Chapter 14 feels hazy, re-read that chapter first — this one builds directly on it.

34.1 Futures curves: contango and backwardation

A *futures curve* (or *forward curve*) is the function $T \mapsto F(T)$ mapping each maturity to today's price for delivery at T. For a stock it is an upward-sloping exponential dictated by the risk-free rate. For a commodity it has shape: it can slope up, slope down, bend, or wiggle seasonally, encoding everything the market believes about storage, supply, demand, and inventory.

Definition 34.1 (Contango)**.** A futures curve is in *contango* if longer-dated futures trade above shorter-dated futures:

$$T_2 > T_1 \implies F(T_2) > F(T_1).$$

The curve slopes upward.

Definition 34.2 (Backwardation)**.** A futures curve is in *backwardation* if longer-dated futures trade below shorter-dated futures:

$$T_2 > T_1 \implies F(T_2) < F(T_1).$$

The curve slopes downward.

Real curves can be contangoed at the front and backwardated at the back, or vice versa. Traders therefore speak of the *front- month spread* $F(T_2) - F(T_1)$ between the two nearest contracts as the local measure of curve shape: positive means local contango, negative means local backwardation.

A worked contango curve

Consider West Texas Intermediate (WTI) crude with spot $S = \$70$ per barrel and a net cost-of-carry $r + u - y$ varying mildly with maturity. A plausible contangoed curve:

Maturity T	Net carry $r + u - y$	Futures price $F(T)$
1 month	3.0%	\$70.18
3 months	3.5%	\$70.62
6 months	4.0%	\$71.41
12 months	5.0%	\$73.59

Each row is $F(T) = 70 \cdot e^{(r+u-y)T}$. The 12-month future trades \$3.59 above spot. This is the *normal* shape for storable commodities with ample inventories and no scarcity premium.

A worked backwardation curve

Now suppose a refinery outage spikes demand for *prompt* barrels. Spot rises but distant futures do not. Model this as large negative net carry (y dominates $r + u$):

Maturity T	Net carry $r + u - y$	Futures price $F(T)$
1 month	-8.0%	\$69.54
3 months	-6.0%	\$68.96
6 months	-4.0%	\$68.61
12 months	-2.0%	\$68.61

The curve slopes downward at the front: "hold a barrel today and it will be worth more than a promise of one in a month."

Contango vs backwardation

- **Contango**: $F(T)$ increases in T. Normal for storable commodities with loose inventories and no scarcity. *Long futures bleeds*: as time passes the contract rolls down the curve toward the lower spot price.

- **Backwardation**: $F(T)$ decreases in T. Typical for supply-constrained or expiring stockpile regimes. *Long futures earns a roll yield*: the contract rolls up the curve toward the higher spot price as it ages.

The sign of the curve slope is the single most-watched indicator in a commodity dealing room.

Intuition

Backwardated = market pays you to hold physical now: you earn an implicit dividend (the *convenience yield*) that shows up nowhere on P&L but reveals itself in the curve's shape. Contango = market pays you to defer; warehouses are full and holding costs storage and financing with no offsetting benefit.

Roll yield and commodity index returns

A commodity index fund (e.g. an ETF tracking the Bloomberg Commodity Index or S&P GSCI) does not own physical barrels. It holds front-month futures and on a scheduled monthly "roll window" sells the expiring contract and buys the next. The fund's return decomposes into three pieces:

$$r_{\text{index}} = \underbrace{r_{\text{spot}}}_{\text{spot return}} + \underbrace{r_{\text{roll}}}_{\text{roll yield}} + \underbrace{r_{\text{collateral}}}_{\text{T-bill rate}}.$$

Collateral yield comes from investing the notional cash in short T-bills. *Spot return* is the percentage change in the commodity spot (proxied by the nearest-futures price). *Roll yield* is the kicker:

$$r_{\text{roll}} \approx \frac{F(T_1) - F(T_2)}{F(T_2)},$$

negative in contango (sell low, buy high), positive in backwardation. Contango is a persistent drag on index returns: the crude-oil sub-index lost 5–10% per year to negative roll yield during the 2014–2016 contango.

Intuition

An index fund tracking a contangoed commodity pays the carry embedded in the term structure every month. This is the biggest reason naive "long commodity" strategies have underperformed over the last two decades: not because spot went nowhere, but because they paid carry continuously.

Contango $\neq$ expected price path

A common beginner error reads a contangoed curve as "the market expects prices to rise." It does not. $F(T)$ is a risk-adjusted delivery price set by carry economics, not a forecast of S_T. The expected spot $\mathbb{E}[S_T]$ can equal, exceed, or fall below $F(T)$ depending on the risk premium. Keynes's *normal backwardation* theory argues producers hedging forward supply push $F(T) < \mathbb{E}[S_T]$, making backwardation the structurally "normal" state. Do not confuse curve shape with directional view.

34.2 Cost-of-carry with storage and convenience yield

Chapter 14 derived $F = Se^{rT}$ for a pure- investment asset by cash-and-carry replication. For commodities this breaks in two ways. First, holding the physical barrel is not free: you pay *storage costs* (tank rental, insurance, boil- off for LNG). Second, holding the barrel confers a *benefit* — if your refinery runs dry you can feed it, whereas a futures contract cannot be cracked into gasoline. That benefit is the *convenience yield.*

Definition 34.3 (Storage cost rate)**.** The *storage cost rate* u is the continuously-compounded rate at which a holder of one unit of physical commodity pays to store it. A typical crude storage cost of \$0.50/barrel/month on a \$70 barrel corresponds to $u \approx 8.6\%$ per year; realised values are smaller because tanks are not full. The modelling idealisation is continuous payment proportional to spot value.

Definition 34.4 (Convenience yield)**.** The *convenience yield* y is the continuously-compounded implicit benefit rate of holding physical inventory versus a futures contract on it. It captures the optionality to meet a supply shock, avoid a stockout, or feed a downstream process, and is applied per-year to the spot value.

Deriving the cost-of-carry formula

Extend the Chapter 14 cash-and-carry argument. At time 0 choose between two strategies delivering one unit of commodity at T:

Strategy A: hold futures. Buy one long futures contract at price F. Invest Fe^{-rT} in the risk-free asset; at T the bond pays F, you pay F to take delivery. Net: one unit of commodity at T, cost today Fe^{-rT}.

Strategy B: cash-and-carry. Borrow S at rate r, buy spot, pay storage continuously at rate u applied to the value of the carried physical, and enjoy convenience yield y also applied to the value of the carried physical. At T:

- cost of carried physical (spot + storage - convenience) grows at rate $r + u - y$, so the equivalent time-T accumulated cost is $Se^{(r+u-y)T}$;
- the physical unit itself is delivered to you.

Net: one unit of commodity at T, accumulated cost $Se^{(r+u-y)T}$.

Both strategies replicate "one unit of commodity at T." No-arbitrage equates their costs: $F = Se^{(r+u-y)T}$.

Commodity forward pricing

For a storable commodity with continuously-compounded storage cost rate u and convenience yield y,

$$\boxed{F(T) = S\,e^{(r+u-y)\,T}.}$$

- $u > y$ (loose inventories, storage dominates): $F > Se^{rT}$, curve in contango above the pure financing curve.
- $y > u+r$ (tight inventories, convenience dominates): $F < S$, curve in backwardation below spot.
- Equities and currencies are special cases: set $u = 0$ and y equal to the dividend or foreign rate.

The sign of $r + u - y$ is the sign of the curve slope at T — exactly the continuously-compounded front-to-back roll rate.

Worked example: crude oil forward

Let spot crude $S = \$80$, risk-free rate $r = 5\%$, storage cost $u = 2\%$, convenience yield $y = 4\%$, horizon $T = 1$ year. Net carry:

$$r + u - y = 0.05 + 0.02 - 0.04 = 0.03.$$

Forward price:

$$F(1) = 80 \cdot e^{0.03\cdot 1} = 80 \cdot 1.03045 = \$82.44.$$

(Verified: $80e^{0.03} = 82.4364$.) Mild contango: the forward trades \$2.44 above spot as storage plus financing slightly exceeds convenience. If y rose to 8%, net carry flips to -1% and the one-year forward drops to $80e^{-0.01} = \$79.20$ (backwardation).

Where does convenience yield come from? The theory of storage

The convenience yield has economic content beyond a fudge factor. The Kaldor–Working *theory of storage* provides the clearest picture. A firm chooses each period whether to hold one extra barrel. The benefit is avoiding a stockout: if demand spikes or supply is disrupted, the firm feeds its refinery from inventory rather than scrambling in the spot market at punitive prices.

Formally, let $C(I)$ be the expected cost of a stockout per unit time when inventory level is I (decreasing and convex in I: more inventory, lower stockout cost). The marginal convenience benefit of the I-th barrel is $-C'(I)$, which is positive and decreasing in I. In competitive equilibrium, a firm holds inventory up to the point where this marginal benefit equals the marginal cost of storage:

$$\underbrace{-C'(I^*)}_{\text{marginal convenience}} \quad = \quad \underbrace{r+u}_{\text{cost of carry}} \quad - \quad \underbrace{\mathbb{E}[\Delta \ln S]}_{\text{expected spot drift}} \quad .$$

The left-hand side is the convenience yield y. Low inventories (near stockout) make $-C'(I)$ large, y is large, and the curve backwardates. Full tanks make $-C'(I) \approx 0$, so $y \approx 0$ and the curve defaults to contango driven by $r + u$.

This explains the empirical regularity: *curve shape is a near-monotonic function of the inventory-to-use ratio.* The "storage arbitrage line" plots y against days-of-supply as a smooth downward-sloping curve that shifts seasonally.

Intuition

Convenience yield is the *shadow price of an empty tank*: large when inventories are scarce (the next barrel has optionality to be sold into a shortage at a huge mark-up), near zero when tanks are full. A curve move from contango to backwardation is almost always a statement about *inventory tightening*, not about expected future price paths.

Inferring convenience yield from the curve

y is not quoted; the market quotes $\{F(T_i)\}$ and $\{r(T_i)\}$, and storage u is known from physical contracts. Inverting cost-of-carry,

$$y(T) = r(T) + u(T) - \frac{1}{T} \ln \frac{F(T)}{S}.$$

This *implied convenience yield* is treated as a market- observable state variable — the commodity analogue of extracting dividend yields from equity forward markets.

Calendar spreads and the curve as a tradeable instrument

The futures curve is itself a tradeable object. A *calendar spread* is a simultaneous long-one-maturity / short-another position:

$$\text{CL Dec–Jun spread} \;=\; F(T_{\text{Dec}}) - F(T_{\text{Jun}}).$$

A long spread profits if the curve steepens; short profits if it flattens. Spread volatilities are an order of magnitude below outright futures vols because both legs share the systematic level risk — what remains is *curve-shape* risk, driven by inventories and convenience yield.

Exchanges list calendar spreads as explicit products (e.g. ICE Brent "BZU" month-to-month spreads) with their own order books, because synthesising them from outright legs carries execution risk. Inter-commodity spreads — WTI vs Brent ("trans-Atlantic"), heating oil vs crude ("heat crack"), gasoline-crude ("crack 3:2:1") — dominate physical refinery hedging. The *crack spread* is the refiner's margin: the price of refined products (gasoline, heating oil) minus the crude input, e.g. the 3:2:1 crack $= \frac{1}{3}(2\,\mathrm{RBOB} + \mathrm{HO}) - \mathrm{WTI}$. The *spark spread* is the analogous power-generator margin: the price of electricity minus the fuel cost of generating it, typically Power $- H_R \cdot$ NatGas where H_R is the plant's heat rate (MMBtu per MWh).

Relation to convenience yield

The one-period log calendar spread equals the forward-forward net cost-of-carry:

$$\ln F(T_2) - \ln F(T_1) = (r + u - y)(T_2 - T_1) \quad \text{(flat-carry case)}.$$

Calendar-spread trades are *directional bets on changes in convenience yield y*. When inventory draws faster than expected, y rises at the front, $F(T_1)$ rises relative to $F(T_2)$, and the spread $F(T_2) - F(T_1)$ compresses.

Curve construction on the commodities desk

A Goldman commodities Strat maintains three curves per product: (i) the *futures curve* $F(T)$ bootstrapped from listed contracts (e.g. 36 monthly CL contracts on NYMEX); (ii) the *spot/prompt curve* for deliverable physical plus near-term basis; (iii) the *implied convenience yield curve* $y(T)$ backed out from the above and the Treasury zero curve. Equities need only spot and the dividend curve; here $y(T)$ is where all the commodity-specific alpha and risk live — it moves sharply with inventory data, OPEC announcements, and pipeline disruptions. Option Strats shock y by ± 100 bp and mark the P&L; it is a first-class risk factor alongside the futures price.

34.3 The Schwartz one-factor model

Cost-of-carry is a relation between spot and forward; it is silent on spot dynamics. For risk management, options pricing, and curve evolution we need a model of S_t. Geometric Brownian motion fits commodities poorly: it implies unbounded drift, whereas crude, copper, and natural gas all *mean-revert* to a long-run production-cost level. When WTI hit $-\$37$ in April 2020 or \$140 in 2008 it did not stay there; supply and demand pulled it back.

Schwartz (1997) proposed the canonical one-factor mean-reverting model: log-spot follows an Ornstein–Uhlenbeck process, so spot follows a geometric mean-reverting SDE.

Definition 34.5 (Schwartz one-factor model)**.** Let $X_t := \ln S_t$. Under the real-world measure,

$$dX_t = \kappa\,(\alpha - X_t)\,dt + \sigma\,dW_t, \tag{34.1}$$

where $\kappa > 0$ is the mean-reversion speed, α the long-run log-mean, and $\sigma > 0$ the volatility of log-spot. Equivalently, in the spot variable,

$$dS_t = \kappa\,(\mu - \ln S_t)\,S_t\,dt + \sigma\,S_t\,dW_t, \tag{34.2}$$

with $\mu = \alpha + \sigma^2/(2\kappa)$ after the Itô correction from log to level.

X_t is Gaussian Ornstein–Uhlenbeck: reverts to α at rate κ with steady-state variance $\sigma^2/(2\kappa)$ and shock half-life $\ln 2/\kappa$. Typical WTI calibration: $\kappa \approx 1$/yr (half-life $\approx$ 8 months), $\sigma \approx 30\%$. Electricity reverts in days, not months.

The futures price under Schwartz

Under the risk-neutral measure $\mathbb{Q}$, log-spot still follows an OU process but with risk-adjusted long-run mean α^*:

$$dX_t = \kappa\,(\alpha^* - X_t)\,dt + \sigma\,d\widetilde{W}_t.$$

The futures price is $F(t,T) = \mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t]$. X_T is Gaussian under $\mathbb{Q}$ with

$$\mathbb{E}^{\mathbb{Q}}[X_T \mid \mathcal{F}_t] = e^{-\kappa(T-t)}X_t + (1-e^{-\kappa(T-t)})\alpha^*,$$

$$\mathrm{Var}^{\mathbb{Q}}[X_T \mid \mathcal{F}_t] = \frac{\sigma^2}{2\kappa}(1-e^{-2\kappa(T-t)}),$$

and $S_T = e^{X_T}$ is lognormal, we use $\mathbb{E}[e^{X_T}] = \exp(\mathbb{E}[X_T] + \frac{1}{2}\,\mathrm{Var}[X_T])$ to obtain the Schwartz closed-form futures price.

Schwartz closed-form futures

Under the one-factor Schwartz model with risk-neutral long-run log-mean α^*, reversion speed κ, and volatility σ, the time-t futures price for delivery at T is

$$F(t,T) = \exp\Big[e^{-\kappa(T-t)}\ln S_t + (1-e^{-\kappa(T-t)})\,\alpha^* + \frac{\sigma^2}{4\kappa}(1-e^{-2\kappa(T-t)})\Big]. \tag{34.3}$$

As $T \to \infty$, $F(t,T) \to \exp(\alpha^* + \sigma^2/(4\kappa))$, a constant independent of current spot — the model's long-run equilibrium price.

Intuition

The long end of the futures curve is anchored at the equilibrium level α^*, independent of spot. Only the front moves with spot — the empirical stylised fact: when WTI spot spikes, one- month futures spike almost 1:1 while two-year futures barely move. A random-walk (Black–Scholes) spot model cannot reproduce this, which is why Schwartz is preferred for commodities.

Half-life, stationarity, and what the parameters mean

The log-spot's stationary distribution is $X_\infty \sim \mathcal{N}(\alpha, \sigma^2/(2\kappa))$. Two interpretable summary statistics follow.

Half-life of shocks. If $X_0 = \alpha + \delta$, then $\mathbb{E}[X_t - \alpha] = \delta e^{-\kappa t}$, half-life $\tau_{1/2} = \ln 2/\kappa$. For $\kappa = 1$/yr, $\tau_{1/2} \approx 8.3$ months; for electricity with $\kappa \sim 50$/yr, $\tau_{1/2} \approx 5$ days.

Steady-state std of log-spot. $\sqrt{\sigma^2/(2\kappa)}$ gives the one-sigma range around α. For $\sigma = 0.3$, $\kappa = 1$: ± 0.21 ($\pm 21\%$ around equilibrium). Large κ means strong reversion and a tight range.

Risk-neutral drift and the market price of risk

Switching $\mathbb{P} \to \mathbb{Q}$ adjusts the long-run mean by the market price of commodity risk λ^{MPR}:

$$\alpha^* = \alpha - \frac{\lambda^{\mathrm{MPR}}\sigma}{\kappa}.$$

Risk-averse investors demand a premium for long commodity exposure, so the $\mathbb{Q}$ reversion target is *lower* than the real-world target. This is the same Girsanov adjustment as in Black–Scholes, applied to the mean-reversion target. Calibration fits $(\kappa, \alpha^*, \sigma)$ by least-squares to the closed-form curve (34.3) against observed quotes.

Extensions: two-factor and three-factor models

The one-factor model has a critical flaw: α^* is constant, so the long end of the curve never moves — empirically it does, slowly. Gibson and Schwartz (1990) make the convenience yield y_t a second mean-reverting factor, letting the curve twist. Schwartz and Smith (2000) recast this as a short-term mean- reverting deviation plus a long-term equilibrium random walk ("S-S two-factor"), often easier to calibrate. Three-factor extensions add stochastic rates. For this book the one-factor model suffices. (Schwartz, 1997)

34.4 Commodity options on futures: Black's model

Listed commodity options are almost universally *options on futures*. A CL American call struck at \$80 delivers a CL futures position on exercise, not a barrel.

Black (1976) showed that options on futures can be priced with the Black–Scholes machinery by substituting F for S and treating the futures as if it paid dividend yield r. Intuition: a futures requires no up-front cash, so to make the BS PDE balance one endows F with dividend yield r.

Black's formula for European options on futures

For a European call and put on a futures with today's futures price F_0, strike K, maturity T, flat risk-free rate r, and futures volatility σ,

$$\begin{aligned} C &= e^{-rT}\big[F_0\Phi(d_1) - K\Phi(d_2)\big], \\ P &= e^{-rT}\big[K\Phi(-d_2) - F_0\Phi(-d_1)\big], \\ d_{1,2} &= \frac{\ln(F_0/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}. \end{aligned}$$

The volatility σ is the lognormal vol of the futures price itself, not of spot.

Worked example: 6-month WTI option

Let $F_0 = \$82$, $K = \$80$, $T = 0.5$ years, $\sigma = 30\%$, $r = 5\%$. Then

$$\begin{aligned} d_1 &= \frac{\ln(82/80) + \frac{1}{2}(0.30)^2(0.5)}{0.30\sqrt{0.5}} = \frac{0.02469 + 0.0225}{0.21213} = 0.2225, \\ d_2 &= d_1 - 0.30\sqrt{0.5} = 0.0103. \end{aligned}$$

Reading $\Phi(d_1) = 0.5880$, $\Phi(d_2) = 0.5041$,

$$\begin{aligned} C &= e^{-0.025}\big[82 \cdot 0.5880 - 80 \cdot 0.5041\big] \\ &= 0.9753 \cdot \big[48.219 - 40.325\big] = 0.9753 \cdot 7.894 = \$7.69. \end{aligned}$$

The put from put-call parity (which for options on futures reads $C - P = e^{-rT}(F_0 - K)$) gives $P = 7.69 - e^{-0.025}(82 - 80) = 7.69 - 1.95 = \$5.74$. (Verified numerically: $C = 7.6934$, $P = 5.7428$.)

Python reference implementation

```
import math

def _phi(x):
    # standard normal CDF via erf
    return 0.5 * (1.0 + math.erf(x / math.sqrt(2.0)))

def black76(F0, K, T, sigma, r, kind="call"):
    """
    Price a European option on a futures contract (Black 1976).
      F0    : today's futures price
      K     : strike
      T     : time to expiry (years)
      sigma : futures lognormal vol (per-year)
      r     : continuously-compounded risk-free rate
      kind  : "call" or "put"
    """
    d1 = (math.log(F0 / K) + 0.5 * sigma * sigma * T) / (sigma * math.sqrt(T))
    d2 = d1 - sigma * math.sqrt(T)
    disc = math.exp(-r * T)
    if kind == "call":
        return disc * (F0 * _phi(d1) - K * _phi(d2))
    else:
        return disc * (K * _phi(-d2) - F0 * _phi(-d1))

# Worked example: 6M WTI call, F=82, K=80, sigma=30%, r=5%
print(black76(82, 80, 0.5, 0.30, 0.05, "call"))  # 7.6934
print(black76(82, 80, 0.5, 0.30, 0.05, "put"))   # 5.7428
```

Listing 34.1: Black's 1976 formula for European options on futures.

Options on spot vs options on futures: Samuelson's effect

Not all commodity options are on futures. LME metals options reference the 3-month forward; OTC options on physical crude reference a cash spot index. For spot-referenced options the BS formula uses $r - y$ in place of the dividend yield — the Garman–Kohlhagen formula from Chapter 32 with y as the foreign rate.

For options on futures there is the *Samuelson effect*: futures vol rises as expiry approaches. Long-dated futures are anchored at the long-run mean; near-expiry futures converge to spot and inherit its vol. Under Schwartz,

$$\sigma_F(t,T) = \sigma\, e^{-\kappa(T-t)},$$

growing from $\sigma e^{-\kappa T}$ at option inception to σ at futures expiry. Feeding front-month vol into options on long-dated futures systematically overprices them.

Don't plug spot vol into Black's model

A frequent error calibrates σ from spot returns and feeds it into Black's formula on futures. These are different vols. Under Schwartz dynamics long-dated futures are *less* volatile than spot (mean-reversion anchors the long end to α^*). Always calibrate σ from the futures return series you are pricing against, or from implied vols of liquid options on that same contract.

34.5 Seasonality

Many commodities exhibit a periodic component in both their futures curve and implied vol, repeating yearly. This is a structural feature of physical supply/demand cycles, not a statistical artefact.

- **Natural gas**: Northern winter heating demand lifts Dec–Feb Henry Hub futures above summer months; January is the winter peak, August the trough. Storage injects all summer and withdraws all winter.
- **Agriculturals** (corn, soybeans, wheat): harvest brings a glut; old-crop (pre-harvest) trades at a premium to new-crop. The *crop year* creates a discontinuity at the harvest month.
- **Gasoline** (RBOB): summer driving demand and the costlier summer-grade blend peak in May–June.
- **Electricity**: diurnal and weekly seasonality on top of annual; peak vs off-peak blocks quoted separately.

To incorporate seasonality, write

$$\ln F(t,T) = \ln F_{\text{nonseasonal}}(t,T) + s(T),$$

where $s(T)$ is a deterministic periodic function (a sum of sinusoids fit to historical log-futures residuals after removing curve-level effects). Implied vol surfaces inherit the periodic component — winter gas options carry higher vol than summer.

A Fourier-style seasonal overlay

A simple parametric form uses annual and semi-annual sinusoids:

$$s(T) = a_1 \cos(2\pi T) + b_1 \sin(2\pi T) + a_2 \cos(4\pi T) + b_2 \sin(4\pi T),$$

with T in years. The four coefficients are fitted by OLS to log-futures residuals after stripping the Schwartz curve. For natural gas this captures most of the month-of-year effect; add a third harmonic if residuals show a third peak.

Seasonality as a model layer

Seasonality is treated as an *additive deterministic layer* on top of a non-seasonal stochastic model (Schwartz, Schwartz–Smith, etc.). It does not require new random factors; it shifts the curve's mean and the vol surface's level but preserves the stochastic dynamics. Calibration: fit the periodic function to residuals, then fit the stochastic model to the deseasonalised series.

Seasonality is not alpha

A rookie mistake: January gas trades above August, so buy August / sell January and declare victory. The spread is priced into the curve at all times; you are not earning a risk premium but paying inter-seasonal storage on the short leg. Seasonality is a deterministic feature the market has already priced. Tradeable edges exist only in *deviations* from the expected pattern (unusually warm winter, wet harvest).

34.6 Physical vs financial settlement

A futures contract settles one of two ways. *Physical settlement* delivers the underlying: the short delivers barrels, the long pays the final settlement price times contract size. *Financial (cash) settlement* pays P&L in cash against a published reference and terminates without physical transfer.

- **CL (WTI crude)**: physical, delivery at Cushing, OK. Long holders not closing out before expiry must take delivery of 1,000 barrels.
- **BZ (Brent crude)**: financial, against ICE Brent Index (Platts).
- **NG (Henry Hub gas)**: physical, at the Henry Hub pipeline junction, Erath, LA.
- **HG (copper)**: physical, at LME warehouses.
- **Agriculturals (ZC, ZS, ZW)**: physical, at approved silos.
- **Power/electricity**: financial; no economic way to deliver 10 MWh into the grid on a specific day.

The distinction matters *near expiry*. A fund long physical- settled oil must roll or arrange delivery (store, resell, or transport). The 2020 WTI event (20 April 2020, May contract at −$37.63) happened because long holders unable to take delivery paid shorts to take the contract away.

Intuition

Financial settlement is a bet on the index; physical is a promise to move real stuff. Physical settlement creates a hard deadline for speculators: be flat or hire a tanker. Fund-level commodity exposure is held through index replication or rolled front-month futures, with rolls executed mechanically 5–10 business days before First Notice Day to avoid physical assignment.

34.7 Case studies from history

Schwartz 1997: the canonical reference

Schwartz (1997) compared three stochastic models for commodity spot: a one-factor mean-reverting model ("Schwartz 1"), a two-factor with stochastic convenience yield, and a three-factor adding stochastic rates. The paper is why commodity options desks still structure pricing libraries around mean-reverting log-spot plus optional stochastic y_t. Schwartz later collaborated with Smith on the short-term/long-term reformulation that became the industry standard.

Metallgesellschaft 1993: stack-and-roll blowup

Metallgesellschaft AG sold long-dated fixed-price oil contracts to US customers and hedged by *stack-and-roll*: a large stack of near-month futures rolled forward each month. When WTI flipped from backwardation to steep contango in late 1993, the roll turned punitive (sell cheap expiring, buy expensive next month). Accounting forced MG to mark the futures leg while leaving the offsetting long-dated customer contracts unmarked; the board liquidated the hedge at the worst moment, crystallising ∼$1.3 billion of loss. Lesson: a theoretically correct hedge can destroy a firm when funding cannot survive the interim cash outflows.

Amaranth 2006: $6B on a gas-spread bet

Amaranth Advisors, via trader Brian Hunter, built an enormous long position in winter gas calendar spreads (long March, short April), betting winter scarcity would widen them. When spreads collapsed in September 2006 the fund lost ~$6.5 billion in a week and closed. Their position was $\sim 40\%$ of NYMEX March–April open interest — they were both the market and its victim. Lesson: seasonality trades are seductive but already priced, edges require genuine divergence forecasts, and size must stay a fraction of open interest.

34.8 Commodity playbooks

Schwartz-like dynamics in Prosperity

KELP (Prosperity 3) and its successors show the Schwartz one-factor signature: mean-reverting log-spot with a few-hundred-tick half- life and a well-defined long-run mean. The Frankfurt Hedgehogs' winning KELP approach estimated κ and α from a rolling sample and skewed quotes toward $\hat{\alpha}$: buy below, sell above. This is a textbook Schwartz-style trade: mean-revert log-spot, collect spread on noise-trader deviations, flatten inventory as price reverts. Recognise a mean-reverting log-spot and the Schwartz toolkit applies — no commodities needed.

Commodities Strats on the trading floor

A Goldman commodities Strat spends the day on four things. (i) *Curve construction*: bootstrapping daily futures curves for WTI, Brent, Henry Hub, corn, soy, and power hubs; QC against stale quotes, rolls, and holiday discontinuities. (ii) *Schwartz or Schwartz–Smith calibration*: refitting $(\kappa, \alpha^*, \sigma)$ each morning, with overrides on inventory data, OPEC, or weather. (iii) *Options-on-futures pricer*: a Black–76 library with American exercise, vol surface interpolation (seasonally adjusted), and Greeks including the commodity-specific $\partial V/\partial y$. (iv) *Seasonality* overlay, recalibrated quarterly. All four auto-run overnight; the trader inspects a curve-diff summary at the 7am run-through.

34.9 Key facts to memorise

- **Commodity cost-of-carry:** for a storable commodity, $F(T) = Se^{(r+u-y)T}$, where u is the storage cost rate and y is the convenience yield.
- **Contango** = upward-sloping curve ($F(T_2) > F(T_1)$ for $T_2 > T_1$); typical of loose inventories. **Backwardation** = downward-sloping; typical of scarcity.
- **Contango is not a price forecast.** $F(T)$ is a risk-adjusted forward price, not $\mathbb{E}[S_T]$. A contangoed market can still see spot fall.
- **Implied convenience yield** $y(T) = r(T) + u(T) - T^{-1}\ln(F(T)/S)$ backs out the per-barrel scarcity signal from observable quotes.
- **Schwartz one-factor:** log-spot is OU with reversion speed κ and long-run log-mean α. The long end of the futures curve is anchored at a constant equilibrium level $\exp(\alpha^* + \sigma^2/(4\kappa))$.

- **Schwartz futures price:** given in closed form by equation (34.3). Long-maturity futures do not move 1:1 with spot.
- **Black 1976:** price options on futures by the Black–Scholes formula with F in place of S and an extra discount factor e^{-rT} in front. Use futures vol, not spot vol.
- **Seasonality** is a deterministic periodic overlay on top of the stochastic model, both for the curve and for the vol surface. Tradeable edges are *deviations* from the pattern, not the pattern itself.
- **Physical vs financial settlement:** CL/NG/agris settle physically; Brent, power, and most vol products settle financially. Physical settlement matters near expiry — roll or be ready for delivery.
- **Metallgesellschaft, Amaranth:** a mathematically correct hedge can destroy a firm if the curve's shape moves against the funding path. Always stress-test the cash-outflow profile, not just the terminal P&L.

Chapter 35

Numerical Methods for Pricing

Most derivatives lack closed-form prices. Black–Scholes (Chapter 16) is the famous exception; American options, path-dependent exotics, multi-asset baskets, callable bonds, CDOs, and structured notes must be priced numerically. This chapter covers the three pillars a Strat uses daily — **Monte Carlo** (with variance reduction), **finite-difference PDE**, and **lattices (trees)** — plus the **Longstaff–Schwartz** algorithm, which converts the American-option optimal-stopping problem into a sequence of linear regressions. The aim is to let you *read* production code, understand its failure modes, and pick the right method for a given product; a decision table at the end summarises the choice.

Prerequisites

From earlier chapters:

- Chapter 10: stochastic differential equations (SDEs), Brownian motion, the Euler–Maruyama scheme.
- Chapter 16: the Black–Scholes PDE $\partial_t V + \frac{1}{2}\sigma^2 S^2 \partial_{SS} V + rS\partial_S V - rV = 0$, and the closed-form European call $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$.
- Chapter 17: Greeks as partial derivatives of the price; why accuracy in Greeks (not just price) matters.
- Chapter 29: discount factors $D(t) = e^{-rt}$ and the PV-of-future-cashflow mantra.

Linear algebra (matrix–vector products, tridiagonal solves) is assumed. Numerical PDE and lattice sections use only elementary Taylor expansions — if you have seen the heat equation and a forward difference, you have enough.

35.1 Monte Carlo pricing

Risk-neutral pricing (Chapter 16) gives the European-claim price

$$V_0 \;=\; e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[g(S_T)]\,, \tag{35.1}$$

under the risk-neutral measure $\mathbb{Q}$. Monte Carlo takes this at face value: simulate thousands of random price paths under $\mathbb{Q}$, evaluate the payoff at each terminal (or along each path), average, and discount. When the payoff depends on the whole path — Asian averages, barrier knock-outs, lookback maxima — no closed form exists, and this path-by-path averaging is often the only tool that works.

The algorithm

For a European payoff on a single asset under geometric Brownian motion (GBM),

$$S_T \;=\; S_0 \exp\Big[\Big(r - \tfrac{1}{2}\sigma^2\Big)\, T + \sigma\sqrt{T}\, Z\Big], \quad Z \sim \mathcal{N}(0,1), \tag{35.2}$$

the procedure is four lines:

1. Draw $Z_1, \ldots, Z_N \text{i.i.d.} \mathcal{N}(0,1)$.
2. Set $S_T^{(k)} = S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\, Z_k]$.
3. Compute payoff $g_k = g(S_T^{(k)})$.
4. Estimate $\hat{V}_0 = e^{-rT}\, \frac{1}{N}\sum_k g_k$.

For path-dependent payoffs (Asian, lookback, barrier), step 2 simulates the whole path by Euler–Maruyama on $0 = t_0 < t_1 < \cdots < t_M = T$:

$$S_{t_{n+1}} = S_{t_n} \exp\Big[(r - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z_{n,k}\Big]. \tag{35.3}$$

The exponential form is exact for GBM and removes time-step bias.

Convergence and standard error

The MC estimator $\hat{V}_0$ is unbiased, with standard error

$$\mathrm{SE}(\hat{V}_0) \;=\; e^{-rT}\, \frac{\sigma_g}{\sqrt{N}}, \qquad \sigma_g^2 := \mathrm{Var}[g(S_T)]\,. \tag{35.4}$$

Halving error quadruples path count — the defining number about MC:

Monte Carlo convergence rate

The root-mean-square error of a plain-vanilla Monte Carlo estimator is $O(1/\sqrt{N})$ regardless of dimension. This dimension-independence is what makes MC the only practical method for high-dimensional problems (basket options, CDOs, multi-factor rates), but $O(1/\sqrt{N})$ also means brute force alone is slow — hence variance reduction.

Pseudo-random versus quasi-random: a brief aside

The $O(1/\sqrt{N})$ rate assumes the Z_k are independent draws. *Quasi-Monte Carlo* (QMC) replaces pseudo-random uniforms with a low-discrepancy sequence (Sobol, Halton, Faure) that covers $[0,1]^d$ more uniformly than random noise. Its error is $O(\log^d(N)/N)$ — effectively $O(1/N)$ — and dominates MC for moderate dimensions ($d \lesssim 50$). Sobol with Brownian-bridge construction is the production choice for low-to-moderate dimensional European exotics. A naive sequential time-grid fill concentrates the first few coordinates on “cost-heavy” early steps and destroys the QMC benefit, so paths are always constructed by Brownian bridge or principal-components decomposition.

Worked example: European call by MC

Let $S_0 = 100$, $K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ year. Black–Scholes gives

$$C^{\mathrm{BS}} \;=\; 100\,\Phi(d_1) - 100\, e^{-0.05}\Phi(d_2) \;=\; 10.4506. \tag{35.5}$$

Drawing $N = 10^6$ standard normals (seed 42) yields

$$\hat{C}^{\mathrm{MC}} = 10.4342, \quad \mathrm{SE} = 0.0147, \quad |\hat{C} - C^{\mathrm{BS}}| = 0.0164. \tag{35.6}$$

The absolute error is within 1.1 SE of the true price, consistent with the estimator's Gaussian sampling distribution (Glasserman, 2004). Pushing the error below one cent needs $(0.0147/0.005)^2 \approx 9$ times more paths — where variance reduction earns its keep.

The $O(1/\sqrt{N})$ rate is visible in a convergence table for the same setup (seed 42):

N (paths)	$\hat{C}^{\mathrm{MC}}$	SE	$\lvert\hat{C} - C^{\mathrm{BS}}\rvert$
10^3	10.25	0.46	0.20
10^4	10.47	0.15	0.02
10^5	10.44	0.046	0.009
10^6	10.434	0.015	0.016

Each tenfold path increase shrinks SE by $\sqrt{10} \approx 3.16$ — one significant digit. Six-digit accuracy from 10^6 paths would need 10^{12} paths (weeks of compute); variance reduction replaces that brute force.

Intuition

Monte Carlo *is* the Law of Large Numbers applied to (35.1): any European payoff is priced by replacing the integral against an unknown density with an empirical average along a simulation. Variance reduction, Longstaff–Schwartz, and antithetics all make that empirical average a better estimator.

35.2 Variance reduction

Since MC error scales as $1/\sqrt{N}$, cutting *variance* by a factor k is equivalent to running k times more paths for free. Three techniques dominate production: antithetic variates, control variates, and importance sampling, each exploiting payoff structure brute force ignores.

Antithetic variates

If $Z \sim \mathcal{N}(0,1)$, so does $-Z$. Pairing each draw with its sign-flipped twin gives the estimator

$$\hat{V}^{\mathrm{AV}} = e^{-rT} \frac{1}{N} \sum_{k=1}^{N/2} \tfrac{1}{2}\big[g(S_T(Z_k)) + g(S_T(-Z_k))\big], \tag{35.7}$$

which uses N payoff evaluations organised into $N/2$ correlated pairs. Its variance is

$$\mathrm{Var}(\hat{V}^{\mathrm{AV}}) = \frac{1}{2N}\big[\mathrm{Var}(g(Z)) + \mathrm{Cov}(g(Z), g(-Z))\big]. \tag{35.8}$$

If g is monotone in Z (e.g. a call on GBM: higher $Z \to$ higher $S_T \to$ higher payoff), then $\mathrm{Cov}(g(Z), g(-Z)) < 0$ and variance drops. For antithetics to help, the payoff must be (anti-)symmetric enough in the driving shock.

Empirical check. For the European call above, plain MC with $N = 10^6$ gives $\mathrm{SE} = 0.0147$; the same paths as 5×10^5 antithetic pairs give $\mathrm{SE} = 0.0104$, a $\sim 2\times$ variance cut. For a linear payoff the reduction is infinite; the ITM cut-off drags the gain down.

Control variates

Want $\mu_Y = \mathbb{E}[Y]$ for a complicated payoff Y, and have a correlated X with known $\mu_X = \mathbb{E}[X]$ in closed form. The control-variate estimator is

$$\hat{\mu}_Y^{\text{CV}} \;=\; \bar{Y} - b\,(\bar{X} - \mu_X), \quad \text{with variance} \quad \text{Var}(\hat{\mu}_Y^{\text{CV}}) = \tfrac{1}{N}\big[\text{Var}(Y) - 2b\ \text{Cov}(X,Y) + b^2\,\text{Var}(X)\big]. \tag{35.9}$$

Minimising over b gives $b^\star = \text{Cov}(X,Y)/\,\text{Var}(X)$, at which

$$\text{Var}(\hat{\mu}_Y^{\text{CV}}) \;=\; \tfrac{1}{N}\,\text{Var}(Y)\,(1 - \rho_{XY}^2), \tag{35.10}$$

where ρ_{XY} is the correlation between X and Y. Variance reduction is $(1-\rho^2)^{-1}$; $\rho = 0.9$ already buys $\sim 5\times$.

Worked example: lookback call controlled by vanilla. A floating-strike lookback call on GBM pays $(\max_{0\le t\le T} S_t - K)^+$. With the same parameters, $N = 10^5$ paths, $M = 100$ Euler steps, use the terminal vanilla payoff $(S_T - K)^+$ as control X (expectation: BS price 10.4506). Empirically,

- plain MC: $\widehat{LB} = 17.867$, SE $= 0.0483$;
- with CV: $\widehat{LB} = 17.904$, SE $= 0.0209$;
- $\rho(Y,X) = 0.901$, $b^\star = 0.938$, variance ratio ≈ 5.3.

The control variate cut the path count $5\times$ for the same accuracy at negligible extra cost.

Variance reduction cheatsheet

Side-by-side comparison on the same ATM call ($S_0 = 100, K = 100, r = 5\%, \sigma = 20\%, T = 1$, $N = 10^5$):

Method	**SE**	**Comment**
Plain MC	0.046	Baseline, $N = 10^5$ paths.
Antithetic	0.033	$\approx 2\times$ variance cut; free.
Control variate (geom Asian)	0.002	$\approx 500\times$ variance cut for arithmetic-Asian; massive for products with tight-correlated closed-form cousins.
Importance sampling, $\mu = 0.6$	0.010	Useful only deep OTM; neutral here.
Quasi-MC (Sobol)	0.006	$\approx 60\times$ variance cut; error scales as $\sim 1/N$ rather than $1/\sqrt{N}$.

The standout is always a good control variate. Antithetic is a $2\times$ freebie, importance sampling shines only in the tail, and QMC is near-universal but needs careful path construction. Production libraries apply antithetics and Sobol by default and add a product- specific control variate when one exists.

Stratified sampling and Latin hypercube

Stratified sampling partitions the sample space into K equal- probability strata and draws a fixed number of samples from each. For GBM, split the Z axis into bins $\Phi^{-1}(k/K)$ and draw N/K samples per bin. The variance is

$$\text{Var}(\hat{V}^{\text{strat}}) \;=\; \frac{1}{N}\sum_{k=1}^{K} \pi_k\,\text{Var}(g \mid \text{stratum } k) \;\le\; \frac{1}{N}\,\text{Var}(g), \tag{35.11}$$

with equality only if g is constant within each stratum. For a near-ATM call, stratifying on the terminal shock gives 2–5× variance reduction. Latin hypercube extends this to multiple dimensions by stratifying each coordinate and reshuffling.

Stacking. Production pricers stack techniques: a typical autocallable pricer uses Sobol numbers + Brownian-bridge paths + antithetics + control variate on the digital component. The reductions compound multiplicatively and can exceed 100×, turning a three-hour MC into two minutes.

Importance sampling

For deep-OTM options, most paths contribute zero; the signal lives in the rare tail. Importance sampling shifts the sampling measure so tail paths become typical, then re-weights via the Radon–Nikodym derivative:

$$\mathbb{E}^{\mathbb{Q}}[g(S_T)] \;=\; \mathbb{E}^{\widetilde{\mathbb{Q}}}\left[g(S_T)\,\frac{d\mathbb{Q}}{d\widetilde{\mathbb{Q}}}\right], \tag{35.12}$$

the trick is to pick $\widetilde{\mathbb{Q}}$ so the integrand is as flat as possible. For GBM, drift the Brownian by μT: sample $Z \sim \mathcal{N}(\mu, 1)$ and multiply the payoff by $\exp(-\mu Z + \frac{1}{2}\mu^2)$. Choosing μ so the typical path lands near the strike can reduce variance by orders of magnitude for deep-OTM options. See (Glasserman, 2004), chapter 4, for the rate-function heuristic.

Three variance reduction techniques

1. **Antithetic variates.** Reuse Z_k and $-Z_k$. Free for any payoff with monotone sensitivity to the driving shock.

2. **Control variates.** Subtract a closed-form-priced correlated payoff. Variance reduction $= (1-\rho^2)^{-1}$; works whenever the product has a simpler cousin whose price is known.

3. **Importance sampling.** Sample from a tilted measure and re-weight. Essential for deep OTM, large-loss CVaR, rare- default simulation, and any payoff dominated by a thin tail.

Intuition

Variance reduction buys paths. A 4× cut = 4× fewer paths for the same accuracy, or the same paths for half the error. Production pricers stack antithetics, a control variate, and stratified sampling; the savings compound multiplicatively.

Antithetic variates can backfire

If the payoff is not monotone in the shock — a straddle, a butterfly, anything with a kink in both directions — antithetics can *increase* variance. Always measure the post-antithetic standard error empirically rather than assume the technique helps.

35.3 Longstaff–Schwartz for American options

American options (exercisable any time before expiry) are the default in equity derivatives. A single-stock American put is worth strictly more than its European counterpart because of early exercise. The pricing problem is *optimal stopping*: the optimal exercise time depends on the future distribution conditional on the current state — exactly what simulations do not give path-by-path.

Before 2001, practical American pricers were limited to lattices and low-dimensional PDEs. Longstaff and Schwartz (2001) changed that: *replace the unknown continuation value with its regression on the current state*, then compare the regression's prediction to the immediate-exercise payoff. The whole method is a sequence of OLS regressions.

The problem statement

Let $\{S_t\}_{t=0}^{T}$ be simulated under $\mathbb{Q}$ with exercise dates $t_0 < t_1 < \cdots < t_M$. The American option value is

$$V_0 \;=\; \sup_{\tau \in \mathcal{T}} \mathbb{E}^{\mathbb{Q}}\left[e^{-r\tau} g(S_\tau)\right], \tag{35.13}$$

where the supremum is over stopping times τ taking values in $\{t_0, \ldots, t_M\}$. By dynamic programming,

$$V(t_m, S) \;=\; \max\Big\{g(S),\; e^{-r\Delta t}\, \mathbb{E}^{\mathbb{Q}}\big[V(t_{m+1}, S_{t_{m+1}}) \mid S_{t_m} = S\big]\Big\}. \tag{35.14}$$

The conditional expectation inside is the *continuation value* $C(t_m, S)$. A lattice knows C exactly (it is a two-point average); a Monte-Carlo simulation does not.

The regression idea

Longstaff–Schwartz approximate $C(t_m, \cdot)$ on the in-the-money paths by a linear combination of basis functions $\{\psi_1, \ldots, \psi_L\}$:

$$C(t_m, S) \;\approx\; \sum_{\ell=1}^{L} \beta_\ell \psi_\ell(S). \tag{35.15}$$

Typical bases are monomials $\{1, S, S^2\}$ or Laguerre/Hermite polynomials. The coefficients β_ℓ come from regressing the realised discounted future cash flow against $\psi_\ell(S_{t_m})$.

Longstaff–Schwartz algorithm

Backward sweep for an American put with $M + 1$ exercise dates and N simulated paths:

1. At t_M: set the realised cash flow on each path to the terminal payoff $(K - S_{t_M})^+$.
2. For $m = M - 1, M - 2, \ldots, 1$:
   a) Identify in-the-money paths at t_m ($K > S_{t_m}^{(k)}$).
   b) On those paths, let $Y^{(k)}$ be the realised future cash flow discounted back to t_m.
   c) Regress Y on $\{\psi_\ell(S_{t_m})\}$ to get $\hat{C}(t_m, S_{t_m}^{(k)})$.
   d) Exercise on path k if $g(S_{t_m}^{(k)}) > \hat{C}(t_m, S_{t_m}^{(k)})$; otherwise carry the existing future cash flow. If exercising, zero out any later CF on that path.
3. Discount every path's cash flow back to t_0 and average.

The regression uses only in-the-money paths because out-of-the-money exercise is never optimal: the regression needs to learn the continuation value exactly where the exercise decision is non-trivial.

Worked example: the Longstaff–Schwartz 2001 paths

We reproduce the original paper's hand-worked example exactly. Take $N = 8$ paths, $M = 3$ annual exercise dates (so four time points $t = 0, 1, 2, 3$), strike $K = 1.10$, risk-free rate $r = 6\%$, discount factor per step $e^{-0.06} = 0.9418$, and basis $\{1, S, S^2\}$. The paths are

path	$t = 0$	$t = 1$	$t = 2$	$t = 3$
1	1.00	1.09	1.08	1.34
2	1.00	1.16	1.26	1.54
3	1.00	1.22	1.07	1.03
4	1.00	0.93	0.97	0.92
5	1.00	1.11	1.56	1.52
6	1.00	0.76	0.77	0.90
7	1.00	0.92	0.84	1.01
8	1.00	0.88	1.22	1.34

Step 1 (terminal payoff). $(K - S_3)^+ = (0,\ 0,\ 0.07,\ 0.18,\ 0,\ 0.20,\ 0.09,\ 0)$.

Step 2 ($t = 2$). In-the-money paths are $\{1, 3, 4, 6, 7\}$ with $S_2 = (1.08, 1.07, 0.97, 0.77, 0.84)$. Their discounted future cash flows are $Y = (0, 0.0659, 0.1695, 0.1884, 0.0848)$ (multiply the $t = 3$ payoff by $e^{-0.06}$). Regressing Y on $\{1, S, S^2\}$ gives coefficients $\hat{\beta} = (-1.070,\ 2.983,\ -1.814)$ and continuation estimates $\hat{C} = (0.037, 0.046, 0.118, 0.152, 0.156)$. Immediate exercise pays $K - S_2 = (0.02, 0.03, 0.13, 0.33, 0.26)$. Exercise now on paths $\{4, 6, 7\}$; hold on $\{1, 3\}$.

Step 3 ($t = 1$). ITM paths are $\{1, 4, 6, 7, 8\}$ with $S_1 = (1.09, 0.93, 0.76, 0.92, 0.88)$. Discounted future cash flows (taking into account the new exercise times from $t = 2$) are $Y = (0, 0.1224, 0.3108, 0.2449, 0)$. Regression gives $\hat{\beta} = (2.038,\ -3.335,\ 1.356)$ and $\hat{C} = (0.013, 0.109, 0.286, 0.117, 0.153)$. Immediate exercise pays $(0.01, 0.17, 0.34, 0.18, 0.22)$. Exercise on $\{4, 6, 7, 8\}$; hold on $\{1\}$.

Step 4. Discount each path's exercise cash flow back to $t = 0$ and average over all 8 paths:

$$\hat{V}_0 = \frac{1}{8}\left(e^{-0.18} \cdot 0.07 + e^{-0.06} \cdot 0.17 + e^{-0.06} \cdot 0.34 + e^{-0.06} \cdot 0.18 + e^{-0.06} \cdot 0.22\right) = 0.1144. \quad (35.16)$$

This is exactly the Longstaff–Schwartz paper's headline result.

Why in-the-money paths only?

At $t = 2$ in the worked example, OTM paths ($S > K$) contribute zero to the put payoff because $(K - S)^+ = 0$ rules out immediate exercise. Including them contaminates the fit: the regression wastes capacity over a region where the exercise decision is moot, hurting accuracy near the boundary where it matters. The original paper proves restriction to ITM paths is asymptotically consistent (Longstaff and Schwartz, 2001).

Dimensionality: where LS genuinely wins

For a single-stock American put, CRR trees with $N = 5000$ steps deliver four-decimal accuracy in ~ 100 ms — LS is slower. LS wins decisively once the state space has two or more dimensions:

- **Bermudan swaption** on a two-factor LMM model with 40 exercise dates and a dozen forward rates;
- **Callable range accrual** whose coupon depends on the whole path of a reference rate;
- **Multi-asset American basket option** on ten stocks with a 10×10 correlation matrix;

- **American convertible bond** driven by stock, credit, and rates jointly.

The state S_t becomes a vector $\boldsymbol{X}_t$ and the basis becomes a tensor product or low-order polynomial of its components; the rest of the algorithm is unchanged. A tree explodes as 2^d per step; LS scales linearly in paths. This is why LS displaced trees as the industry default for exotic Americans.

Code listing: Longstaff–Schwartz in ∼30 lines

```python
import numpy as np

def longstaff_schwartz_put(S0, K, r, sigma, T, M, N, seed=0):
    """Price an American put on GBM by Longstaff-Schwartz MC."""
    rng = np.random.default_rng(seed)
    dt = T / M
    df = np.exp(-r * dt)
    # Simulate GBM paths under Q (exact log-Euler).
    Z = rng.standard_normal((N, M))
    logS = np.log(S0) + np.cumsum(
        (r - 0.5*sigma**2)*dt + sigma*np.sqrt(dt)*Z, axis=1)
    S = np.hstack([np.full((N,1), S0), np.exp(logS)])  # shape (N, M+1)

    # Terminal payoff.
    CF = np.maximum(K - S[:, -1], 0.0)
    # Backward induction: for m = M-1 down to 1.
    for m in range(M-1, 0, -1):
        itm = S[:, m] < K
        if not itm.any():
            CF = CF * df
            continue
        X = S[itm, m]
        Y = CF[itm] * df   # realised discounted continuation
        A = np.column_stack([np.ones_like(X), X, X**2])
        beta, *_ = np.linalg.lstsq(A, Y, rcond=None)
        C_hat = A @ beta
        exer = K - X
        ex = exer > C_hat
        # Update cashflow for exercisers.
        CF_new = CF * df  # default: carry & discount
        idx = np.where(itm)[0][ex]
        CF_new[idx] = exer[ex]
        CF = CF_new
    return df * CF.mean()
```

Listing 35.1: Longstaff–Schwartz American put pricer.

Run on $S_0 = 36$, $K = 40$, $r = 6\%$, $\sigma = 20\%$, $T = 1$ year, $M = 50$ exercise dates, $N = 100\,000$ paths reproduces Longstaff and Schwartz's Table 1 price of 4.478 to three decimal places.

Intuition

LS converts the optimal-control problem (when to exercise?) into a sequence of regressions (what is the expected continuation value?). The regressed $\hat{C}(\cdot)$ is biased — only as good as the basis — but the resulting exercise policy is *low-biased*: a suboptimal rule gives a lower bound on the true American price, useful for conservative pricing. Upper bounds from duality methods exist too, but LS low bounds are the industry workhorse.

Choosing the basis, and why the low-biased property matters

The original paper uses weighted Laguerre polynomials $L_k(S)$ of order two or three on ITM paths. We used monomials above and still reproduced the price because eight paths cannot distinguish two bases of the same rank. For larger N, Laguerre and Hermite polynomials are better conditioned than raw monomials and should be preferred once S spans more than an order of magnitude. Use three to five basis functions per state-variable dimension (Glasserman, 2004); beyond that, flexibility overfits.

The low-biased property matters. LS's stopping rule is suboptimal (the regression approximates), so the realised value $\hat{V}_0^{\mathrm{LS}}$ along chosen stopping times is a lower bound on the true American price. Combined with a dual-martingale upper bound (Rogers 2002, Andersen–Broadie 2004) one gets a tight confidence interval. In production the LS price is the point estimate; the dual bound is spot-checked on regression-risk-sensitive products to ensure the interval stays tight.

title

A common bug: regressing over all paths at each step instead of only ITM ones. OTM paths add noise where the exercise decision is moot and degrade the fit near the exercise boundary where it matters. Always filter to ITM paths before regressing.

35.4 Finite-difference PDE methods

Finite-difference methods solve the Black–Scholes PDE from Chapter 16 *directly on a grid*: discretise both time and asset price, replace derivatives with differences, step backward from the terminal payoff. This is the natural tool when you already have the PDE but no closed-form solution. For low-dimensional products (one or two state variables) with a PDE characterisation — vanilla European and American options, barriers, single-curve rates products — finite-difference methods win. They are deterministic, converge at $O(\Delta t + \Delta S^2)$ or better, and deliver high-quality Greeks as grid by-products.

The Black–Scholes PDE on a grid

Recall from Chapter 16 that the value $V(t, S)$ of a European derivative on a non-dividend asset satisfies

$$\frac{\partial V}{\partial t} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV \;=\; 0, \qquad V(T, S) = g(S). \tag{35.17}$$

Discretise on $S_i = i\Delta S$, $i = 0, \ldots, I$, and $t_n = T - n\Delta t$, $n = 0, \ldots, N$, with $V_i^n \approx V(t_n, S_i)$. Centred differences give

$$\alpha_i = \tfrac{1}{2}\Delta t(\sigma^2 i^2 - ri), \quad \beta_i = -\Delta t(\sigma^2 i^2 + r), \quad \gamma_i = \tfrac{1}{2}\Delta t(\sigma^2 i^2 + ri), \tag{35.18}$$

and the PDE becomes (after backward-in-time substitution $\tau = T - t$)

$$V_i^{n+1} - V_i^n \;\approx\; \alpha_i V_{i-1}^{?} + \beta_i V_i^{?} + \gamma_i V_{i+1}^{?}, \tag{35.19}$$

where the choice of $V_i^{?}$ (old time n, new time $n+1$, or average) defines the scheme.

Explicit scheme

Choosing $V_i^{?} = V_i^n$ gives the *explicit* update

$$V_i^{n+1} \;=\; V_i^n + \alpha_i V_{i-1}^n + \beta_i V_i^n + \gamma_i V_{i+1}^n. \tag{35.20}$$

It is trivial to implement (no linear solves) but suffers a CFL-type stability restriction: one can show (Hull, 2014) that stability requires

$$\Delta t \lesssim \frac{(\Delta S)^2}{\sigma^2 S_{\max}^2}, \tag{35.21}$$

so halving the space step forces a *quartering* of the time step. Unusable for fine grids.

Concrete illustration. For $S_{\max} = 400$, $I = 1000$ (so $\Delta S = 0.4$), $\sigma = 0.2$, the bound is $\Delta t \lesssim 2.5 \times 10^{-5}$: pricing a $T = 1$ option needs over 40 000 time steps, and halving ΔS quarters Δt. Explicit schemes are competitive only on coarse grids and rarely appear in production, but they are pedagogically clean.

Implicit scheme

Choosing $V_i^? = V_i^{n+1}$ gives the *implicit* update

$$(1-\beta_i)V_i^{n+1} - \alpha_i V_{i-1}^{n+1} - \gamma_i V_{i+1}^{n+1} \;=\; V_i^n, \tag{35.22}$$

a tridiagonal system solved in $O(I)$ per step by Thomas's algorithm. Unconditionally stable but first-order in time.

Crank–Nicolson: the industry default

Crank and Nicolson (1947) averaged explicit and implicit:

$$(I - \tfrac{1}{2}L)V^{n+1} \;=\; (I + \tfrac{1}{2}L)V^n, \tag{35.23}$$

where L is the tridiagonal matrix with rows $(\alpha_i, \beta_i, \gamma_i)$. The resulting scheme is

- unconditionally stable for any $\Delta t, \Delta S$;
- second-order accurate in *both* Δt and ΔS (locally $O(\Delta t^2 + \Delta S^2)$);
- one tridiagonal solve per time step, i.e. as cheap as implicit;
- the canonical choice for single-factor European and American options in practice.

Crank–Nicolson summary

Crank–Nicolson on the Black–Scholes PDE is *unconditionally stable* and *second-order accurate in space and time*. It is the default scheme for any single-factor equity or rates derivative that has a PDE characterisation. For American options, supplement each time step with an elementwise $V_i^{n+1} \leftarrow \max(V_i^{n+1}, g(S_i))$ projection to enforce the exercise constraint.

Worked example: European call via Crank–Nicolson

Same call ($S_0 = 100$, $K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$), grid with $S_{\max} = 400$, $I = 1000$, $N = 100$. Terminal $V_i^0 = (S_i - K)^+$; boundary $V(0,t) = 0$ and $V(S_{\max}, t) \approx S_{\max} - K\,e^{-r\tau}$.

Running Crank–Nicolson and interpolating at $S_0 = 100$ gives

$$V_0^{\mathrm{CN}} = 10.4502, \qquad \text{vs. } C^{\mathrm{BS}} = 10.4506, \qquad |\mathrm{err}| = 3.8 \times 10^{-4}. \tag{35.24}$$

Four-decimal accuracy in a PC-second.

Greeks for free from the grid

A bonus of finite-difference: the full Greek surface falls out of the grid without bumping. Since the grid stores $V(t, S_i)$ at every node,

$$\Delta(t, S_i) \approx \frac{V(t, S_{i+1}) - V(t, S_{i-1})}{2\Delta S}, \tag{35.25}$$

$$\Gamma(t, S_i) \approx \frac{V(t, S_{i+1}) - 2V(t, S_i) + V(t, S_{i-1})}{(\Delta S)^2}, \tag{35.26}$$

$$\Theta(t, S_i) \approx \frac{V(t + \Delta t, S_i) - V(t, S_i)}{\Delta t}, \tag{35.27}$$

all from a single grid solve. MC needs one price per bumped parameter per Greek — a $2N_{\text{paths}}$ penalty per Greek, mitigated only by pathwise or likelihood-ratio estimators ((Glasserman, 2004), ch. 7).

American options on the FD grid

For an American option, add one line per step: after the linear solve, project onto the exercise payoff,

$$V_i^{n+1} \leftarrow \max(V_i^{n+1},\ g(S_i)). \tag{35.28}$$

This is the discrete analogue of the free-boundary condition $V(t, S) \geq g(S)$ and delivers American prices at the same $O(NI)$ cost. The projection can be formulated as a linear- complementarity problem (LCP) solved by projected SOR for higher boundary accuracy; simple projection is fine for single-stock puts.

Intuition

For a single-asset product with a PDE, finite difference annihilates MC: an $O(IN)$ grid gives four decimals where 10^6 MC paths give two. The dimension curse bites at 2–3 state variables; by 4–5 factors the tensor grid is dead and MC wins.

35.5 Trees: the CRR binomial model

Before FD libraries and Longstaff–Schwartz, American options were priced on *lattices*. The Cox et al. (1979) binomial tree (CRR) is the simplest, still taught everywhere and occasionally used in production for simple contracts (small American options, discretely-monitored single-asset barriers). The idea: approximate continuous-time lognormal GBM by a discrete up/down lattice, which makes early-exercise decisions trivial to check at each node (exactly what BS closed-form cannot do) and converges to BS as the step count $N \to \infty$.

Construction

Discretise $[0, T]$ into N steps of length $\Delta t = T/N$. At each step the underlying goes up by u or down by d:

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = 1/u = e^{-\sigma\sqrt{\Delta t}}, \qquad p = \frac{e^{r\Delta t} - d}{u - d}. \tag{35.29}$$

This matches the first two moments of $\log S$ under $\mathbb{Q}$ and converges to GBM as $N \to \infty$. The tree is *recombining* (up-down = down-up), giving $n + 1$ nodes at time n, total $O(N^2)$.

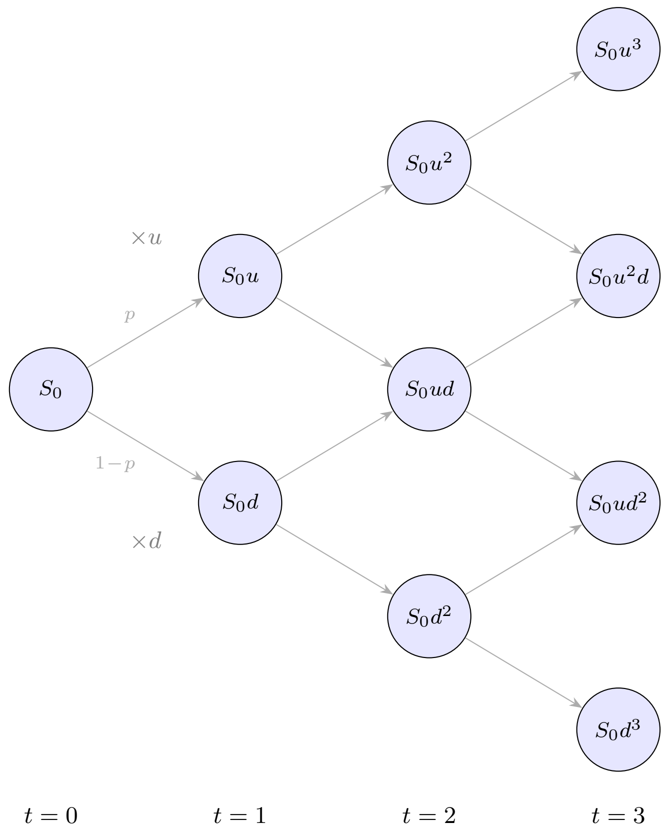


Figure 35.1: Three-period CRR binomial tree. At each step the price moves up by factor $u = e^{\sigma\sqrt{\Delta t}}$ or down by $d = 1/u$, with risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. The recombining property ($ud = du = S_0$) means the tree has $N+1$ terminal nodes after N steps, not 2^N. Option prices are obtained by *backward induction*: compute payoffs at $t = 3$, then take risk-neutral expectations one step at a time back to $t = 0$.

Backward induction

At expiry, the option value at each terminal node equals the payoff. Moving back one step, the value at an internal node (n, j) is

$$V_{n,j} \;=\; e^{-r\Delta t}\big[p\,V_{n+1,j} + (1-p)\,V_{n+1,j+1}\big]. \tag{35.30}$$

For an American option, take $V_{n,j} = \max(g(S_{n,j}), \text{the above})$. This is exactly (35.14) executed on a tree.

Convergence. For the same call parameters, CRR gives

N	CRR price	error vs BS
10	10.2534	−0.1972
50	10.4107	−0.0399
200	10.4406	−0.0100
1000	10.4486	−0.0020

Error scales as $1/N$, matching CRR's known $O(1/N)$ convergence (Hull, 2014). For 10^{-4} accuracy, a 1000×100 CN grid is about as fast as a CRR tree with $N = 5000$.

Intuition

CRR is pedagogically beautiful — it visualises risk-neutral valuation as a discrete conditional expectation — and cheap for simple single-factor Americans. For serious work, Crank–Nicolson (better accuracy, same cost) or Longstaff–Schwartz (path-dependence, multi-factor) supersedes it.

Greeks from a CRR tree

Like the FD grid, CRR delivers Greeks from stored node values:

$$\Delta_0 \;\approx\; \frac{V_{1,0} - V_{1,1}}{S_0(u-d)}, \qquad \Gamma_0 \;\approx\; \frac{(V_{2,0} - V_{2,1})/(S_0u^2 - S_0) - (V_{2,1} - V_{2,2})/(S_0 - S_0d^2)}{\frac{1}{2}(S_0u^2 - S_0d^2)}. \tag{35.31}$$

Θ needs two trees with different T or a pairwise time- differencing argument. Vega and rho require a full rebuild with bumped σ or r, so tree Greeks are cheap for spot-dependent quantities and expensive for others.

Tree variants worth knowing

Black–Derman–Toy (BDT) trees extend the binomial to short- rate models with time-varying vol, still used for simple callable-bond pricing. **Trinomial trees** add a middle branch, eliminating CRR's $O(\sqrt{\Delta t})$ odd/even oscillation near strike and accelerating convergence to $O(1/N^2)$. **Implied trees** (Rubinstein 1994, Derman–Kani 1994) calibrate transition probabilities from market option prices to construct a smile-consistent lattice; they were the local-vol workhorse before Dupire's PDE superseded them.

35.6 Greeks in Monte Carlo: pathwise and likelihood-ratio

Unlike FD grids, Monte Carlo does not give Greeks for free. The naive *bump and reprice* estimator

$$\hat{\Delta} \;=\; \frac{\hat{V}_0(S_0+\epsilon) - \hat{V}_0(S_0-\epsilon)}{2\epsilon} \tag{35.32}$$

has two problems: (i) fresh paths give independent noise with variance $\propto 2/\sqrt{N}$, amplified by $1/\epsilon$; (ii) bump discretisation error is $O(\epsilon^2)$. The fix — *common random numbers* across bumped prices — kills independent noise and cuts Greek variance by 10–100×.

Two more sophisticated estimators dominate production:

Pathwise (adjoint) Greeks. If the payoff g is almost-surely differentiable in the parameter, one can swap differentiation and expectation,

$$\frac{\partial}{\partial S_0}\,\mathbb{E}[g(S_T(S_0))] \;=\; \mathbb{E}\!\left[\frac{\partial g}{\partial S_T}\cdot\frac{\partial S_T}{\partial S_0}\right]. \tag{35.33}$$

For GBM, $\partial S_T/\partial S_0 = S_T/S_0$, and for a European call, $\partial g/\partial S_T = \mathbb{1}_{S_T>K}$, so the pathwise delta is

$$\hat{\Delta}^{\mathrm{pw}} \;=\; e^{-rT}\,\frac{1}{N}\sum_{k=1}^{N}\mathbb{1}_{S_T^{(k)}>K}\,\frac{S_T^{(k)}}{S_0}. \tag{35.34}$$

No bump, no bias. Pathwise fails when the payoff has a kink or jump (digitals, barrier knock-outs): differentiating the indicator gives a delta function an expectation cannot swallow.

Likelihood-ratio (LR) Greeks. LR differentiates the density instead of the payoff:

$$\frac{\partial}{\partial\theta}\,\mathbb{E}[g(S_T)] \;=\; \mathbb{E}\!\left[g(S_T)\,\frac{\partial\log p(S_T;\theta)}{\partial\theta}\right]. \tag{35.35}$$

The score $\partial_\theta \log p$ is smooth, so LR works even for discontinuous payoffs at the cost of higher variance for smooth ones. The rule: *pathwise for smooth, LR for discontinuous*; when both apply, pathwise usually has smaller variance. (Glasserman, 2004) chapter 7 is canonical.

Monte Carlo Greek estimators

Bump-and-reprice with common random numbers is the universal default. **Pathwise** is better when the payoff is differentiable. **Likelihood-ratio** is necessary when the payoff has jumps (digitals, barriers). A production Monte Carlo engine typically implements all three, dispatched by payoff type.

35.7 Method selection: the decision table

The flowchart below is what every derivatives-group Strat internalises by the end of their first year.

Situation	**Recommended method**
Closed-form formula exists (vanilla European on BS, forward, cash-and-carry bound, etc.).	Use the closed form. No numerical error, constant-time.
European payoff, 1–2 underliers, PDE known.	Crank–Nicolson finite difference. Second-order accurate, unconditionally stable, Greeks free from the grid.
European payoff, ≥ 3 underliers, or complicated path-dependence.	Monte Carlo with antithetic + control variate. Curse-of-dimension- free.
American or Bermudan exercise, 1–2 state variables.	Crank–Nicolson with per-step projection onto the exercise payoff, or CRR tree if you need to ship it this afternoon.
American or Bermudan exercise with path-dependence, or ≥ 3 factors.	Longstaff–Schwartz Monte Carlo. The only scalable option.
Deep out-of-the-money / rare event (credit default, CVaR tail, autocallable KO).	MC with importance sampling. Plain MC wastes 99% of paths.
Characteristic function known (Heston, VG, Kou, affine models).	Fourier inversion (Carr–Madan FFT). $O(N \log N)$ per strike slice, closed-form accuracy.

Numerical method decision tree

1. Closed form $\to$ use it.
2. Low-dim European $\to$ Crank–Nicolson.
3. High-dim European $\to$ MC with variance reduction.
4. Low-dim American $\to$ CN with projection, or CRR.
5. High-dim or path-dependent American $\to$ Longstaff–Schwartz MC.
6. Characteristic function known $\to$ Fourier.

When in doubt, implement two methods and cross-check: agreement to 10^{-4} is the standard test that neither pricer has a bug.

35.8 Back-references: where each method shows up

Earlier Goldman-oriented chapters use these methods without naming them:

- **Chapter 30 (rates).** Hull–White caps have a closed form under the HJM measure change; the Bermudan swaption alongside does not and uses Longstaff– Schwartz MC on the short-rate SDE. Single-factor Hull–White is also cheap on a 1D Crank–Nicolson PDE in r, which gives crisper short-rate Greeks.
- **Chapter 31 (equity).** Heston and rough-vol have known characteristic functions, so Carr–Madan FFT prices Europeans in milliseconds. Path-dependent exotics (autocallables, lookbacks, cliquets) fall back on MC with antithetics, strata, and a BS control variate.
- **Chapter 33 (credit).** Vanilla CDS prices are closed-form PVs of survival-weighted cashflows. Gaussian-copula CDO tranches on 120 names are simulated by MC with importance-sampling tilt on the common factor (Glasserman, 2004).

A derivatives book typically wraps all four pricers behind one `Pricer.price(product)` facade with a config flag. The engineering challenge is rarely the maths — it is wiring the right solver to the right product and making every solver return the same Greek.

Calibration: the outer loop

Pricers sit inside a calibration loop that tunes the model to liquid- vanilla market prices. For local-vol:

$$\min_{\sigma_{\text{loc}}(t,S)} \sum_i \left[C^{\text{model}}(K_i, T_i;\, \sigma_{\text{loc}}) - C^{\text{market}}(K_i, T_i)\right]^2, \tag{35.36}$$

with each C^{model} evaluation a PDE or MC run. A thousand quotes over a hundred LM iterations fires the inner pricer 10^5 times, so inner-pricer speed is the dominant library metric. When accuracy ties, the faster method wins — why Carr–Madan FFT (Heston) and Crank–Nicolson (local-vol) dominate their niches.

Numerical hazards in production

A partial list of bugs every derivatives library has hit once:

- **Off-by-one discount factor.** One extra step between exercise and today costs $\sim 0.02\%$ per basis point, surfacing as systematic PV bias under Chapter 36's P&L- explain scrutiny.
- **Antithetic on two-sided kink payoffs.** Variance silently increases, the price still converges — just slower. Regress empirical SE against $1/\sqrt{N}$ as a check.
- **LS regression on OTM paths.** Price biased low by 5–20 bp depending on moneyness, never flagged by unit tests.
- **Explicit FD blowing up on a vol bump.** User bumps vol from 20% to 40%; (35.21) is violated; prices explode. Fix: ban explicit schemes in production.
- **Basis conditioning.** Raw monomials $\{1, S, S^2, S^3\}$ give near-singular regression matrices when S has wide range; use orthogonal polynomials or rescale to $[-1, 1]$.

35.9 Benchmarks and cross-checks

Every derivatives library ships a benchmark suite that cross-checks each pricer against others at canonical parameter points:

1. **Vanilla European call, flat BS.** Every pricer matches (35.1) to 10^{-4} of notional, catching discount- factor and time-convention bugs.
2. **Put–call parity.** $C - P = S - K\,e^{-rT}$ holds path-by-path on shared paths.
3. **MC vs CN convergence curves.** Error vs $1/\sqrt{N}$ for MC, vs Δt for CN; slope deviations flag bugs.
4. **LS vs CRR for small-state American puts.** A tree with $N = 10^4$ gives $< 10^{-3}$ accuracy as the LS reference.
5. **Greek sanity.** Grid Δ, Γ, Θ agree with BS to $< 1\%$ and with MC CRN bumps within CI.

These run nightly in CI. On a Strats desk the rule is: *no pricer ships without a reference pricer that agrees to tolerance at three points.* Choosing the right reference is often harder than implementing the primary pricer.

Two-pricer cross-check

Never trust a single pricer in production. Every product family has at least two (closed form + MC, CN + tree, or MC + LS); convergence agreement at a benchmark point is the only reliable correctness test. Unit tests without cross-pricer agreement are theatre.

35.10 Summary

- The toolkit has four members: closed form, Monte Carlo, finite difference, trees. Pick whichever delivers required accuracy fastest given dimension and exercise style.
- **Monte Carlo**: $O(N^{-1/2})$, dimension-free, the only option for high-dim baskets or path-dependence.
- **Variance reduction** — antithetics, control variates, importance sampling — cuts MC variance by 5–100×.
- **Longstaff–Schwartz**: the canonical American pricer. Run paths forward; at each exercise date regress continuation on a polynomial basis. ∼30 lines price the universe.
- **Crank–Nicolson**: default 1D–2D PDE scheme. Unconditionally stable, second-order accurate, linear-time per step. A 100×1000 grid matches BS to four decimals in a second.
- **CRR trees**: $O(1/N)$, easy to reason about, legitimate for very small American jobs. Superseded by CN and LS otherwise.
- The flowchart in §35.7 is what every derivatives Strat carries in their head. When in doubt, cross- check two methods.

Prosperity example

The Prosperity backtester is itself a Monte Carlo simulator: each day replays historical order flow with pseudo-random counterparty arrivals, and your PnL is one draw from a stochastic distribution. Variance-reduction logic transfers: run many days (MC averaging), compute paired deltas when you tweak a parameter (antithetic: same shocks, new parameter), and use a simpler strategy as a control variate. The Frankfurt Hedgehogs credit "backtest on as many days as possible" as their single most important rule — variance reduction applied to algo trading research.

On a Goldman Sachs desk

On a Strats desk the numerical toolkit is productised as a library — call it `PricerLib` in SecDB dialect — that all desks consume. A structuring Strat rarely writes MC from scratch: the workflow is (i) check the product-to-pricer routing table, (ii) pick the default (CN for low-dim exotics, LS for Americans, Carr–Madan for Hestons), (iii) add a cross-check in the alternate pricer and diff-test, (iv) hand off to the risk book. The differentiator is knowing which solver is numerically reliable for a product family and owning the stability and convergence tests in CI.

Three papers that run derivatives desks

- Crank and Nicolson (1947): time-averaging scheme for the heat equation, Cambridge, decades before Black–Scholes. Almost every parabolic PDE in finance is the heat equation in disguise.
- Cox et al. (1979): binomial tree as a pedagogical device, which became the first tractable American pricer and the industry standard for 22 years.
- Longstaff and Schwartz (2001): replaced the tree with a regression on simulated paths, breaking the curse of dimension for Americans. The most-cited paper in computational finance — read the eight-path Section 1 example until you can reproduce it without looking.

Chapter 36

Risk in Production: Greeks, P&L Explain, and xVA

This chapter takes the single-option Greeks of Chapter 17 into production: *portfolio-scale* Greeks across thousands of positions, *P&L Explain* as the Strat's morning ritual, and the *xVA alphabet soup* (CVA, DVA, FVA, KVA) accounting for what textbook Black–Scholes ignores — counterparty default, own default, funding costs, regulatory capital. The new content is *scale* and *accountability*: a 10k-trade desk cannot reprice by hand; a Strat who cannot explain yesterday's P&L to four significant figures is unemployed by Friday; a pricing model valuing a derivative at $1,000,000 while ignoring $30k CVA, $10k FVA, and $8k KVA is wrong by 5% on a product whose bid–offer is 2%.

Prerequisites

From earlier chapters:

- Chapter 17: Black–Scholes Greeks for a single European option (delta, gamma, vega, theta, rho), and the first-order hedging identity $\Delta V \approx \Delta \cdot \Delta S$.
- Chapter 20: Sharpe ratio, drawdown, VaR, and the idea that risk is a *distribution* of P&L outcomes, not a point estimate.
- Chapter 29: zero curves and bucketed rate risk — the object we called a *key-rate duration* generalises to the KRD vector used here.
- Chapter 33: hazard rates $h(t)$, survival $S(t) = e^{-\int_0^t h(u)du}$, and the expected-loss formula $\mathrm{EL} = (1 - R)\,\mathrm{PD}$. CVA is the counterparty-exposure analogue.

36.1 Portfolio-scale Greeks

A real options book holds tens of thousands of positions across hundreds of underlyings, dozens of expiries, and a range of strikes. The risk system's job is to *aggregate* the Greeks into a handful of numbers a trader can act on.

Linear aggregation of Greeks

Greeks of a portfolio are additive in the *dollar* convention, not in the *per-share* convention. Let the book hold n_i contracts of option i on underlying $U(i)$, with per-contract delta δ_i, gamma γ_i, vega ν_i, and let m be the contract multiplier (100 shares for US equity options). The *dollar* *delta* of the book on underlying u is

$$\Delta_u^{\$} = \sum_{i:U(i)=u} n_i\, m\, \delta_i\, S_u,$$

i.e. dollars gained per 1% move in S_u. Dollar delta is additive across positions on the same underlying; aggregating across underlyings is meaningful only if they are assumed to move together (e.g. a beta-weighted aggregate).

Share-equivalent delta $\sum_i n_i m \delta_i$ gives shares to trade to hedge; *dollar delta* gives loss per % move. Desk convention picks one.

Portfolio risk representation

A book's risk is stored as a collection of *bucketed Greeks*, not a single number per Greek:

- **Delta** is bucketed by *underlying* (one number per stock / index / currency pair).
- **Gamma** is bucketed by underlying and often by *strike band* around spot.
- **Vega** is bucketed by underlying, *tenor* (maturity band: 1M, 3M, 6M, 1Y, 2Y, 5Y...) and often *strike band* (25-delta put, ATM, 25-delta call) — a full vega *surface*.
- **Rho** is bucketed by currency and by *tenor* on the rates curve (key-rate durations, KRDs).

Aggregation without bucketing discards hedging information. A "delta-neutral" book can be long \$50M AAPL and short \$50M MSFT — a pairs trade, not a hedged position.

Key-rate durations (KRDs)

For rate-sensitive books (swaps, bonds, swaptions), a single duration number is useless — you need to know *which part of the curve* you are exposed to. The answer is the *key-rate duration* vector from Chapter 29.

Let the zero curve be parameterised by pillar tenors $\{T_1, \ldots, T_K\} = \{1\mathrm{M}, 3\mathrm{M}, 6\mathrm{M}, 1\mathrm{Y}, 2\mathrm{Y}, 5\mathrm{Y}, 10\mathrm{Y}, 30\mathrm{Y}\}$. The key-rate duration at pillar k is

$$\mathrm{KRD}_k := -\frac{1}{V}\frac{\partial V}{\partial r_k},$$

with other pillars held fixed. Equivalently, bump r_k by 1 bp, reprice, divide by $V \cdot 10^{-4}$. The vector $(\mathrm{KRD}_1, \ldots, \mathrm{KRD}_K)$ is the book's exposure along the curve. A 5Y-receive, 10Y-pay swap has near-zero total duration but $\mathrm{KRD}_{5\mathrm{Y}} > 0$ and $\mathrm{KRD}_{10\mathrm{Y}} < 0$ — a pure curve-shape bet (steepener).

KRDs satisfy a sum rule: on a parallel shift, $\sum_k \mathrm{KRD}_k \cdot \Delta r_k = \mathrm{DV01}$ in appropriate units. They are additive across trades in the same currency.

Cross-gamma

On a book with multiple underlyings, a single gamma per underlying misses the *joint* convexity. For a portfolio $V(S_1, S_2, \ldots, S_n)$ the full second-order Taylor expansion is

$$\Delta V \approx \sum_i \Delta_i\, \Delta S_i + \tfrac{1}{2}\sum_i \sum_j \Gamma_{ij}\, \Delta S_i \Delta S_j,$$

where $\Gamma_{ij} = \partial^2 V/(\partial S_i \partial S_j)$. Diagonals Γ_{ii} are the usual single-underlying gammas; off-diagonals Γ_{ij}, $i \neq j$, are *cross-gammas*: the rate at which Δ_i changes as S_j moves.

Cross-gamma is nonzero when the portfolio contains instruments whose delta on one underlying depends on another's price. Examples:

- *Index option* decomposed into constituents: a 50-delta SPX call has delta ≈ 0.50 against SPX, but across the 500 constituents each delta depends on the others via the index weight — large cross-gammas.
- *Spread option* with payoff $\max(S_1 - S_2 - K, 0)$: $\partial^2 V/(\partial S_1 \partial S_2) < 0$ — rallying S_2 reduces delta to S_1.
- *Basket option* on $w_1 S_1 + w_2 S_2$: cross-gamma $\neq 0$ whenever $w_1 w_2 \neq 0$.

Single-name books can ignore cross-gamma; index market-making books cannot — ignoring it produces P&L Explain residuals of several percent.

Worked example: a 4-position mini-book

Four European options on four underlyings; $r = 4\%$, $m = 100$.

Underlying	Type	Qty	S	K	σ	Tenor
AAPL	Call	+10	150	150	28%	30d
MSFT	Put	+5	400	390	24%	60d
GOOG	Call	−8	140	145	30%	45d
SPX	Call	+20	5000	5000	15%	90d

Summing per-contract Greeks times quantity times multiplier:

Position	Share delta	Dollar delta ($)	Vega per 1 vol pt ($)
AAPL +10 C 150	+532	+79,847	+171
MSFT +5 P 390	−177	−70,661	+301
GOOG −8 C 145	−326	−45,654	−153
SPX +20 C 5000	+1,135	+5,673,609	+19,527
Total	+1,164	+5,637,141	(see below)

The share-delta total is misleading: 97% of the $1,164 delta is SPX at $5000 — AAPL/MSFT/GOOG are rounding error in dollar terms. Dollar delta makes this clear: almost $5.7M is SPX.

A single vega total is useless — 30-day AAPL vol and 90-day SPX vol respond to different moves. The desk representation is a *vega bucket table* (rows = underlying, columns = tenor):

Underlying	30d	45d	60d	90d
AAPL	+171.0	—	—	—
MSFT	—	—	+301.4	—
GOOG	—	−152.7	—	—
SPX	—	—	—	+19,527.0

Each cell is dollars per 1 vol point move on that $(\textit{underlying}, \textit{tenor})$ bucket; a real desk sees a heatmap across many more buckets. Trader's question: "if 30-day single-name vols rally 2 points while index vol stays flat, what do I make?" Answer: $2 \times (171 + 301.4 - 152.7) = +\$639$, dominated by MSFT. The SPX bucket is an order of magnitude larger than all single-name vega combined — the book's vol risk lives there.

Intuition

Aggregating Greeks without bucketing is like aggregating a multi-currency P&L without FX — the single number hides the risk. A "delta-neutral" book can be long one name and short another; a "vega-flat" book can be long front-month vol and short back-month — an enormous *term-structure* bet. The bucketed view is the hedgeable view.

36.2 P&L Explain: the Strat's daily proof-of-work

Every morning on every derivatives desk, a Strat takes yesterday's closing book plus yesterday's and today's market data, and *explains* the overnight change in mark-to-market (MTM) value using the book's Greeks. This report is *P&L Explain* (or *P&L Attribution*) — the Strat's daily proof-of-work. If it balances, pricing library and risk system agree, the trader starts the day, and Middle Office signs off on official P&L. If not, the Strat's job is to find out why, fast.

The Explain formula

The overnight MTM change is a second-order Taylor expansion in the market state variables:

P&L Explain formula

$$\Delta V \;\approx\; \underbrace{\Delta \cdot \Delta S}_{\text{delta P\&L}} \;+\; \underbrace{\tfrac{1}{2}\Gamma(\Delta S)^2}_{\text{gamma P\&L}} \;+\; \underbrace{\nu\,\Delta\sigma}_{\text{vega P\&L}} \;+\; \underbrace{\Theta\,\Delta t}_{\text{theta P\&L}} \;+\; \underbrace{\rho\,\Delta r}_{\text{rho P\&L}} \;+\; \underbrace{\varepsilon}_{\text{residual}}.$$

Healthy book: $|\varepsilon| < 1\%$ of $|\Delta V|$. A residual above a few percent flags a missing Greek, stale bump size, model misalignment, or a late-booked trade. Above 10% means *stop trading until fixed.*

Terms use *start-of-day* Greeks (not end-of-day) — the Explain is a predictor of what P&L *should have been* given the observed market moves.

For multi-underlying books, the delta, gamma, and cross-gamma terms expand:

$$\text{2nd-order spot P\&L} \;=\; \sum_i \Delta_i\,\Delta S_i \;+\; \tfrac{1}{2}\sum_{i,j}\Gamma_{ij}\,\Delta S_i \Delta S_j.$$

The vega term expands over the *vol surface*: $\sum_{u,k,j} \nu_{u,k,j}\,\Delta\sigma_{u,k,j}$ where (u, k, j) indexes (underlying, tenor, strike bucket). The rho term expands over KRD pillars: $\sum_k \rho_k\,\Delta r_k$.

Decomposition: the canonical five bucket Explain

The typical Explain groups the expansion into five buckets:

1. **Delta / Spot P&L.** First-order spot move. Dominates for directional positions.
2. **Gamma / Convexity P&L.** Second-order spot move. Positive for long-gamma books (market making), negative for short-gamma books (selling options).
3. **Vega / Vol P&L.** Change in implied vols. Bucketed over the vol surface.
4. **Theta / Time decay.** Passage of time. Usually negative for long-option books, positive for short-option.

5. **Rho / Rate P&L.** Change in zero rates. Usually small for options books; dominant for rates desks.

Residual = MTM change minus the sum of these five.

Worked example: ATM call, 1 day, S:100→101, σ:20%→21%

A single ATM call on a non-dividend stock, start of day:

$$S_0 = 100, \quad K = 100, \quad r = 2\%, \quad \sigma_0 = 20\%, \quad T_0 = 0.25 \text{ (3 months)}.$$

Black–Scholes price and Greeks (Chapter 17), to 6 d.p.:

$$C_0 = 4.232160, \quad \Delta = 0.539828, \quad \Gamma = 0.039695,$$

$$\nu = 19.847627 \text{ per 1.00 change in } \sigma, \quad \Theta = -8.934063 \text{ per year}, \quad \rho = 12.437656.$$

Equivalently, $\nu = 0.198476$ per 1 vol point, $\Theta = -0.024477$ per calendar day.

By close of day 1:

$$S_1 = 101, \quad \sigma_1 = 21\%, \quad r_1 = r = 2\%, \quad T_1 = T_0 - 1/365 = 0.247260.$$

Step 1 — reprice.

$$C_1 \;=\; 4.963545, \qquad \Delta V \;=\; C_1 - C_0 \;=\; 0.731386.$$

Actual MTM change: \$0.731386 per share, \$73.14 per contract.

Step 2 — contributions. Moves: $\Delta S = +1$, $\Delta\sigma = +0.01$, $\Delta t = +1/365$, $\Delta r = 0$. Plug in:

$$\begin{aligned}
\text{delta P\&L} &= \Delta \cdot \Delta S = 0.539828 \times 1 = +0.539828,\\
\text{gamma P\&L} &= \tfrac{1}{2}\Gamma(\Delta S)^2 = 0.5 \times 0.039695 \times 1 = +0.019848,\\
\text{vega P\&L} &= \nu \cdot \Delta\sigma = 19.847627 \times 0.01 = +0.198476,\\
\text{theta P\&L} &= \Theta \cdot \Delta t = -8.934063 \times (1/365) = -0.024477,\\
\text{rho P\&L} &= \rho \cdot \Delta r = 0.
\end{aligned}$$

Sum: $0.539828 + 0.019848 + 0.198476 - 0.024477 + 0 = +0.733675$.

Step 3 — residual.

$$\varepsilon \;=\; \Delta V - \text{sum} \;=\; 0.731386 - 0.733675 \;=\; -0.002289.$$

The residual is $-\$0.00229$ per share (-0.31% of ΔV) — well inside the 1% tolerance. Explain balances.

Step 4 — the residual. Sum of unmodelled third-order Taylor terms:

- *Speed* $\partial\Gamma/\partial S$: higher-order convexity.
- *Vanna* $\partial\Delta/\partial\sigma = \partial\nu/\partial S$: spot–vol cross-term. With both ΔS and $\Delta\sigma$ nonzero, this dominates.
- *Volga* $\partial^2 V/\partial\sigma^2$: second-order vol convexity.
- *Charm* $\partial\Delta/\partial t$: how delta ages overnight.

On a market-making book with large vanna and volga (deep OTM options), these terms contribute tens of basis points each and the 5-Greek Explain breaks down. Fix: add *second-order Greeks* (vanna, volga, charm, speed) — but that raises position cost from five numbers to fifteen. Trade-off: vanilla books stop at five; exotic books carry the full second-order expansion.

Summary table:

Contribution	P&L ($ per share)
Delta	+0.539828
Gamma	+0.019848
Vega	+0.198476
Theta	−0.024477
Rho	0.000000
Sum of Greeks	+0.733675
Actual ΔV	+0.731386
Residual	−0.002289
Residual / $\lvert\Delta V\rvert$	−0.31%

Intuition

Clean P&L Explain is the Strat's daily proof-of-work. Sum of Greeks matches MTM within 1%: risk system and pricing library are in sync, screen Greeks are trustworthy. Residual above 1%: a missing vol bucket, stale bump size, late-booked trade, or pricing change not propagated to risk. Healthy books have residuals near zero; unhealthy books have residuals drifting systematically — usually a missing Greek (vega on a new bucket, cross-gamma on a new basket). A persistent 2% residual eventually becomes a $50M mystery loss.

Reference implementation

A minimal single-option P&L Explain, matching the worked example above:

```python
from math import log, sqrt, exp, pi, erf

def ncdf(x): return 0.5*(1 + erf(x/sqrt(2)))
def npdf(x): return exp(-x*x/2) / sqrt(2*pi)

def bs_call_greeks(S, K, r, sig, T):
    d1 = (log(S/K) + (r + 0.5*sig*sig)*T) / (sig*sqrt(T))
    d2 = d1 - sig*sqrt(T)
    price = S*ncdf(d1) - K*exp(-r*T)*ncdf(d2)
    delta = ncdf(d1)
    gamma = npdf(d1) / (S*sig*sqrt(T))
    vega  = S*npdf(d1)*sqrt(T)              # per 1.00 in sigma
    theta = (-S*npdf(d1)*sig/(2*sqrt(T))     # per year
             - r*K*exp(-r*T)*ncdf(d2))
    rho   = K*T*exp(-r*T)*ncdf(d2)          # per 1.00 in r
    return price, delta, gamma, vega, theta, rho

def pnl_explain(S0, K, r, sig0, T0, dS, dsig, dr, dt_days):
    C0, d, g, v, th, rh = bs_call_greeks(S0, K, r, sig0, T0)
    C1, *_             = bs_call_greeks(S0+dS, K, r+dr, sig0+dsig,
                                        T0 - dt_days/365)
```

```
    dt = dt_days / 365
    contribs = {'delta': d*dS,    'gamma': 0.5*g*dS*dS,
                'vega':  v*dsig, 'theta': th*dt, 'rho': rh*dr}
    total    = sum(contribs.values())
    residual = (C1 - C0) - total
    return C1 - C0, contribs, residual
```

Listing 36.1: Single-option P&L Explain.

36.3 xVA: the alphabet soup of valuation adjustments

Black–Scholes assumes the counterparty always pays, the dealer always pays, the hedge is fundable at the risk-free rate, and no capital is tied up against the position. None of these is true in practice. The *valuation adjustments* ("xVA") are the price of reality:

Adjustment	What it prices	Sign
CVA	Counterparty default on positive exposures	− (reduces value)
DVA	Own default on negative exposures	+ (increases value)
FVA	Funding spread on uncollateralised positions	− (typically)
KVA	Regulatory capital cost over life of trade	−
MVA	Initial-margin funding (cleared trades)	−

The risk-adjusted price is

$$V_{\text{risky}} \;=\; V_{\text{BS}} \;-\; \text{CVA} \;+\; \text{DVA} \;-\; \text{FVA} \;-\; \text{KVA} \;-\; \text{MVA}.$$

On a long-dated swap with a weak counterparty, xVA can run to 5% of notional — larger than the bid–ask spread, and the difference between winning the trade and losing money.

CVA — credit valuation adjustment

CVA is the expected loss from counterparty default — the derivative analogue of the bond formula $\text{EL} = (1-R)\,\text{PD}$ from Chapter 33, with *exposure* replaced by the stochastic MTM and integrated over the trade's life.

CVA formula

Let $V(t)$ be the dealer's MTM at time t. On counterparty default at time $\tau \le T$ the dealer's exposure is $V(\tau)^+$ — you lose only if in the money when they default. Unilateral CVA:

$$\text{CVA} \;=\; (1-R)\int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}\big[V(t)^+ \,\big|\, \tau = t\big]\,dPD(t),$$

where R is recovery, $D(t)$ the risk-free discount factor, $\mathbb{E}[V(t)^+]$ the *expected positive exposure* (EPE) at time t, and $dPD(t) = -dS(t) = h(t)S(t)\,dt$ the marginal risk-neutral default probability (Chapter 33).

Discrete form on grid $0 = t_0 < \cdots < t_N = T$:

$$\text{CVA} \;\approx\; (1-R)\sum_{k=1}^{N} D(t_k)\cdot \text{EPE}(t_k)\cdot \big[S(t_{k-1}) - S(t_k)\big],$$

where the bracket is the marginal default probability in $(t_{k-1}, t_k]$ stripped from CDS quotes.

Worked example: CVA on a 5-year interest-rate swap. Dealer is fixed receiver on a 5-year at-par swap; counterparty's 5-year CDS trades at 150 bp ($R = 40\%$). Flat hazard $h = s/(1-R) = 0.025$ gives survival $S(t) = e^{-0.025t}$; risk-free rate $r = 3\%$. EPE on an at-par vanilla swap has the hump $\mathrm{EPE}(t) \approx N\sqrt{t(T-t)/T}$, peaking near $T/2$; take $N = 100$. Semi-annual steps:

t (yr)	EPE(t)	$D(t)$	ΔS_k	contribution
0.5	67.08	0.9851	0.01242	0.4925
1.0	89.44	0.9704	0.01227	0.6389
1.5	102.47	0.9560	0.01212	0.7121
2.0	109.54	0.9418	0.01196	0.7406
2.5	111.80	0.9277	0.01182	0.7354
3.0	109.54	0.9139	0.01167	0.7010
3.5	102.47	0.9003	0.01152	0.6379
4.0	89.44	0.8869	0.01138	0.5417
4.5	67.08	0.8737	0.01124	0.3953
5.0	0.00	0.8607	0.01110	0.0000
			Sum =	**5.60**

Each contribution is $(1-R)\cdot D(t_k)\cdot \mathrm{EPE}(t_k)\cdot \Delta S_k$. CVA is \$5.60 per \$100 of peak EPE, or 5.6% of peak exposure. On a \$1bn notional swap (with a calibrated EPE replacing the simplified hump), CVA runs into single-digit millions — a charge the salesperson must quote on top of the BS-implied swap rate.

DVA — debit valuation adjustment

DVA is the mirror of CVA: the benefit the *dealer* gets from their own default probability. If I owe you \$100M and default, I pay only $R \cdot 100\mathrm{M}$ — so ex ante my expected payout is less than \$100M, the liability is worth less than \$100M to me, and my balance sheet gains when the market perceives my credit deteriorating.

DVA uses the dealer's own survival curve and *negative* exposure:

$$\mathrm{DVA} \;=\; (1-R_{\mathrm{own}})\int_0^T D(t)\,\mathbb{E}^{\mathbb{Q}}\big[(-V(t))^+\big]\,dPD_{\mathrm{own}}(t).$$

DVA is accounting-real but not cash-real

DVA is the most controversial xVA. Under IFRS 13 and US GAAP (post-2007), banks must recognise DVA in fair-value accounting — so when a bank's own CDS widens in a crisis, it books a *gain* on its derivative liabilities. Hence the Q3 2011 headlines where Morgan Stanley, Goldman Sachs, and Citi reported multi-billion DVA gains as their own spreads blew out — gains they could never monetise unless they defaulted. Many banks now strip DVA from "core" earnings. Economic logic clean; political optics terrible.

FVA — funding valuation adjustment

When a dealer's uncollateralised derivative becomes an asset (positive MTM), the dealer must borrow cash to hedge at their unsecured rate, which exceeds the risk-free rate. The difference is the *funding spread*; FVA is its discounted integral over the trade's life.

Symmetrically, if the position becomes a liability, cash received can be invested at the unsecured rate — a funding benefit (FBA). FVA is often quoted net of FBA.

Approximate formula:

$$\text{FVA} \approx \int_0^T D(t) \cdot s_{\text{fund}}(t) \cdot \mathbb{E}^{\mathbb{Q}}[V(t)^+]\, dt,$$

where $s_{\text{fund}}(t)$ is the dealer's funding spread over OIS. Same EPE integral as CVA, different weight (funding spread vs. hazard × LGD).

FVA applies only to *uncollateralised* trades. Fully collateralised trades (CSA-perfect daily cash margining at OIS) have zero FVA because the collateral funds the hedge — which is why mandatory clearing post-Dodd–Frank (Chapter 38) dramatically reduced FVA on vanilla flow. (A *CSA*, Credit Support Annex, is the ISDA document specifying daily collateral exchange; *OIS*, the overnight-index-swap rate, is the near-risk-free rate paid on that posted cash.)

KVA — capital valuation adjustment

Under Basel III, banks hold regulatory capital against derivative exposures: CVA-risk capital (capital against CVA itself moving), default-risk capital (RWA × 8%), plus trading-book market-risk capital. Shareholders demand (say) 10% return on equity. Over a long-dated trade this return on capital is non-trivial and is passed to the client as KVA.

$$\text{KVA} \approx \int_0^T D(t) \cdot \text{ROE} \cdot \text{Cap}(t)\, dt,$$

where $\text{Cap}(t)$ is regulatory capital consumed at time t and ROE is target return on equity. On a 30-year uncollateralised swap with a low-rated counterparty, KVA can be the largest xVA.

MVA — margin valuation adjustment

MVA is the cost of funding *initial margin* posted to a CCP (central counterparty — the clearing house that interposes itself between trade counterparties) or bilaterally under uncleared margin rules. IM is typically funded at the dealer's unsecured rate but earns only OIS inside the CCP, producing a spread cost. Less commonly quoted than CVA/DVA/FVA, increasingly important post-UMR (Uncleared Margin Rules, phased 2016–2022, requiring two-way IM on bilateral derivatives).

xVA alphabet soup

- **CVA**: counterparty default. Negative to dealer; uses counterparty hazard, EPE.
- **DVA**: own default benefit. Reported in IFRS/GAAP, often stripped from adjusted earnings.
- **FVA**: funding cost on uncollateralised positions; zero for trades collateralised at OIS.
- **KVA**: regulatory capital × ROE over life; dominant on long-dated, low-rated, uncollateralised trades.
- **MVA**: initial-margin funding spread.

On a weak-credit 30y uncollateralised swap: CVA $\sim$ 100 bp, FVA $\sim$ 50 bp, KVA $\sim$ 200 bp. Total xVA can exceed the BS bid–ask by 5×.

xVA is not naively additive across the book

Novices compute per-trade CVA/FVA and sum. Wrong in two ways.
Netting. Under an ISDA Master with CSA, counterparty exposure is *netted* across all trades. A \$100M ITM swap and a \$90M OTM swap with the same counterparty net to \$10M, not \$100M — per-trade CVA over-estimates tenfold.
Collateral. Variation margin reduces CVA to the margin-period-of-risk tail (∼10 days). Fully collateralised in cash at OIS: CVA and FVA near zero; only MVA and KVA remain. xVA is computed at *netting-set* level; attributing to individual trades (for incremental pricing) is a separate calculation called *incremental xVA*.

36.4 Risk limits and VaR

Bucketed Greeks and xVA tell a desk where its risk lives; *limits* constrain how much is allowed. A typical desk operates under a hierarchy:

- **Greek caps.** Hard caps on dollar delta, share delta, and vega per underlying and vega bucket, e.g. "AAPL 30-day vega capped at \$50k per vol point."
- **Notional caps.** Gross and net notional by product and counterparty.
- **Scenario P&L caps.** Loss limit under a stress — e.g. "5% SPX drop with vols up 20% must not exceed \$10M."
- **VaR and Expected Shortfall.** Statistical tail measures of loss.

VaR and Expected Shortfall

Value-at-Risk (VaR) at confidence α and horizon h is the loss not exceeded with probability α over h:

$$\mathrm{VaR}_\alpha(h) \;=\; \inf\{\ell \geq 0 : \mathbb{P}(L_h \leq \ell) \geq \alpha\},$$

with L_h the loss. Typical: $\alpha = 99\%$, $h = 1$ day.

Expected Shortfall (ES, also CVaR) is expected loss given a tail event:

$$\mathrm{ES}_\alpha(h) \;=\; \mathbb{E}[L_h \mid L_h \geq \mathrm{VaR}_\alpha(h)].$$

ES is always at least as large as VaR at the same confidence and is a *coherent* risk measure (VaR is not — it can violate subadditivity, penalising diversification).

The Basel Fundamental Review of the Trading Book (FRTB) replaced VaR with 97.5% ES as the regulatory market-risk measure from 2023 (Chapter 38).

Why a single VaR number is not enough

VaR/ES is a *portfolio-level aggregate* — it collapses the risk distribution to one number. Too coarse for a trader running the book in real time: it does not say *which* Greek drives risk, or whether a breach is vega or delta. Desks run VaR *plus* a dashboard of bucketed Greeks. VaR answers the CRO's question ("how bad could tomorrow be?"); bucketed Greeks answer the trader's ("what do I hedge now?").

Intuition

Risk limits are the desk's exoskeleton. Greeks tell you *where* risk is; limits tell you *whether* you are allowed to have it. VaR/ES is the regulator's view, bucketed Greeks the trader's.

Both exist because neither alone suffices: ES without Greeks is un-hedgeable; Greeks without ES miss tail correlations. In production both are computed nightly and monitored intraday.

36.5 Key facts to memorise

- Portfolio Greeks are *bucketed* by underlying, tenor, strike band, and curve pillar — not aggregated into scalars.
- **Dollar delta** $= nm\delta S$ gives \$ per % move; **share delta** $= nm\delta$ gives shares-to-hedge.
- KRDs decompose rate risk by pillar; parallel shifts recover DV01 as $\sum_k \mathrm{KRD}_k \Delta r_k$.
- Cross-gamma $\Gamma_{ij} = \partial^2 V/(\partial S_i \partial S_j)$ is essential for multi-underlying books (index options, spreads, baskets).
- **Explain formula**: $\Delta V \approx \Delta \cdot \Delta S + \frac{1}{2}\Gamma(\Delta S)^2 + \nu\,\Delta\sigma + \Theta\,\Delta t + \rho\,\Delta r + \varepsilon$; healthy residual $|\varepsilon| < 1\%$ of $|\Delta V|$.
- Residual is third-order Taylor: speed, vanna, volga, charm. When vanna/volga matter, add second-order Greeks.
- **CVA** $= (1-R)\int_0^T D(t)\mathrm{EPE}(t)\,dPD(t)$, with hazard stripped from CDS (Chapter 33).
- **DVA** is the symmetric own-default adjustment — controversial because it books a gain when your own credit deteriorates.
- **FVA** is funding spread on uncollateralised exposures; zero for perfectly OIS-collateralised trades.
- **KVA** = ROE × regulatory capital over life; dominant on long-dated, low-rated, uncollateralised trades.
- xVA is computed at *netting-set* level; netting and collateral can reduce CVA by an order of magnitude.
- Risk-adjusted price: $V_{\text{risky}} = V_{\text{BS}} - \text{CVA} + \text{DVA} - \text{FVA} - \text{KVA} - \text{MVA}$.
- VaR is not coherent (violates subadditivity); ES is. FRTB uses 97.5% ES.
- A desk runs bucketed Greeks, stress scenarios, and VaR/ES in parallel — each answers a different question.

Prosperity example: per-round P&L Explain is trivial

Prosperity has no vol surface, rate curve, or counterparty, so the multi-Greek Explain collapses to one line: "end-of-round cash + end-of-round inventory at mid − start cash = round P&L." The discipline is still universal: after each round, split P&L into *executed-trade* (realised) and *mark-to-market inventory* (unrealised), attribute each line to a strategy (market making, stat arb, options scalping), and reconcile to the leaderboard. If attribution does not match the posted number, the local backtester is wrong — and an unexplained \$100/round gap becomes a \$2000-notional bug on submission day. The same instinct that chases a 50 bp Explain residual on a sell-side desk saves a Prosperity team from submitting a broken algo.

On a Goldman Sachs desk

On a Goldman Sachs equity-derivatives or rates-desk Strat seat, the 7am–9am workflow has been fixed for two decades. **07:00.** Overnight batch is done; pull previous-close book, previous-close data, today's open data. **07:15.** Run P&L Explain — the morning's most important job; report goes to Market Risk, Product Control, and trader by 08:30. **07:30.** If Explain balances ($< 1\%$ residual), move to xVA checks: any counterparty's CDS moved enough to shift CVA? Any new trade reprice the netting set? **08:00.** Investigate stale-trade or missing-Greek exceptions from the overnight batch. **08:30.** Send consolidated Explain and xVA report. **09:00.** Market open; pivot to trader support — pricing requests, scenario runs, hedge simulations. Same cadence on every derivatives desk in every major bank. An Explain that does not balance on Friday is why Strats work weekends.

Historical note: from accounting afterthought to trillion-dollar business

Pre-2008, CVA was a quarterly back-office adjustment with minimal trader involvement. Lehman's September 2008 default (and Bear Stearns' March rescue) made counterparty default a first-order risk overnight. Basel III (2010–2011) introduced *CVA-risk capital* — explicit capital against CVA itself moving — turning CVA into a first-class trading activity. Banks set up dedicated *CVA desks* that hedge counterparty credit by trading CDS, pricing CVA into every long-dated trade.

FVA was more contentious. Hull and White (2012) (Hull and White, 2012) argued FVA should not enter fair value because it is a financing cost, not a counterparty cost. The industry disagreed: JPMorgan took a $1.5bn FVA charge in 2014 and every major dealer followed. The accounting debate is technically open; market practice is settled.

Chapter 37

SecDB and Slang

SecDB is Goldman's risk/pricing database; Slang is the proprietary language that runs inside it. Together they price a trillion-dollar derivatives book overnight. Concretely: SecDB is a real-time, in-memory graph database holding the firm's entire position and risk state — every trade, curve, vol, Greek, and P&L — updated as bookings and market data arrive, so any desk can query live risk at any instant. A CS reader should not picture SQL: picture a reactive spreadsheet the size of a bank, where every cell recomputes automatically when any input changes. SecDB and Slang are internal to one firm — no vendor manual, no open-source repo. What we know comes from interviews with former Goldman engineers, Mike Dubno's InfoQ talk on the Slang VM, and trade-press coverage. Most claims here are **training-knowledge-based**, reflecting the public record as of May 2025; treat the chapter as an *informed sketch*, correct enough for an interview but expect details inside the firm to differ. SecDB is infrastructure, not a strategy: every Greek in Chapter 36, every P&L Explain, every trade ticket lives as a node in a graph database whose ancestor was written in the 1990s.

Prerequisites

From earlier chapters:

- Chapter 28: what a Strat does, the divisional structure of the Engineering division, and the observation that "Strats code in Slang against SecDB" is the single sentence most often repeated in Goldman job descriptions.
- Chapter 36: portfolio-scale Greeks, P&L Explain, and the idea that pricing a derivatives book is a *graph computation* — a curve changes, a million sensitivities must follow. SecDB is the object that makes this tractable.

This chapter needs no new mathematics. It is an engineering chapter about a database schema and a domain-specific language.

37.1 SecDB architecture: a graph that prices a bank

Consider Goldman's problem at 5pm New York on a Tuesday. The firm holds five million live derivative trades — swaps, swaptions, exotics, bonds, FX forwards, equity options — each with a mark-to-market value, Greeks, CVA, FVA. Those numbers depend on today's market data: zero curves in every currency, vol surfaces on every underlying, credit spreads for every counterparty. By the time Tokyo opens at 7pm New York, every trade must be repriced, its Greeks recomputed, and the risk distributed to every downstream system.

Repricing five million trades from scratch every night is hopeless: slow, and — more importantly — it gives the Strat no way to ask "*which* input moved this number?", the substance of P&L Explain (Chapter 36). SecDB's answer is to store the bank as a single *directed acyclic graph* (DAG) of dependencies: every number is a node, every pricing relationship an edge.

Nodes and edges

A SecDB *node* is a financial object with a value and a set of inputs. Typical nodes:

- An individual **trade** — a five-year USD swap, a six-month Brent Call. Its value depends on curves, vols, spot.
- A **curve** — the USD SOFR zero curve, bootstrapped from quoted swap rates. Its value depends on the market quotes that fed the bootstrap.
- A **quote** — the 2y swap rate, a leaf node fed by the market-data system.
- A **Greek** — the delta of a specific trade to a specific input bucket. Its value is itself a node, computed by bumping one input and re-evaluating the trade.
- A **portfolio aggregate** — the desk's total dollar delta to SOFR 2y, summed over every position whose pricer depends on that bucket.

An *edge* from A to B says "B depends on A". A trade's value node has edges from each curve and vol surface it consumes; a curve has edges from its bootstrapping quotes; a portfolio aggregate has edges from every trade it sums.

The graph is a DAG: cycles are refused at commit time, because they would produce undefined values.

SecDB graph architecture

- Every financial number in the bank — trade value, curve point, vol, Greek, aggregate — is a node.
- Every pricing dependency is a directed edge.
- The whole structure is a DAG; cycles are prohibited.
- When a leaf node (e.g. a market quote) changes, SecDB automatically re-evaluates every descendant node in topological order. No Strat writes "update everything that depends on SOFR 2y" — that is what the graph is *for*.
- The graph is persisted: a node is not just a value in memory, it is a row in the database with a history.

Propagation and sensitivities

The central operation is *graph propagation*: a leaf node changes; SecDB walks the DAG downstream, recomputing each affected node exactly once in topological order. Computation is lazy (recomputed only on query, unless marked eager) and cached.

Graph-native evaluation turns sensitivity calculation from a bespoke task into a one-liner. To compute $\partial V/\partial r$ for a trade whose value V depends on a rate input r:

1. Bump the leaf node $r \to r + \epsilon$.
2. Ask SecDB for V ("evaluate node V").
3. SecDB propagates $r + \epsilon$ downstream, recomputes every dependent node, and returns the new V.

4. Sensitivity $= (V_{\text{new}} - V_{\text{base}})/\epsilon$.

The Strat writes no plumbing: she does not enumerate which curves rebuild, which surfaces refit, which trades reprice. The graph does it. This is SecDB's most important property — *sensitivities are first-class*. A book-level dollar-delta-to-SOFR-2y has the same structure as a single-trade dollar delta; only the downstream nodes differ.

The naive approach — "bump one factor, reprice from scratch, repeat N times" — scales as $O(N \cdot M)$ where M is the work to price the portfolio. SecDB's cached graph reprices only the nodes actually affected, a 10–100x speed-up on a large book.

Intuition

A spreadsheet with formulas is a graph of cells; change an input cell, every formula downstream updates. SecDB is that spreadsheet scaled to fifty million rows, with the cells persisted to a database, the formulas written in a specialised language, and the recalculation engine optimised to remember what it has already computed. Everything else in this chapter is engineering consequences of that one idea.

Bi-temporal data

A plain graph database is insufficient: banks must answer two kinds of historical question, with different answers:

1. "What was the trade's value *as of* 31 December 2023?" — the question an auditor asks. This is **business date** (also called *valuation date* or *as-of date*).
2. "What was the value of that trade *as we believed it* on 2 January 2024 at 9am?" — the question a regulator asks after a fat-finger, or a Strat asks after a bug is fixed. This is **system date** (also called *knowledge date* or *transaction time*).

The two answers differ whenever a curve is re-bootstrapped, a trade amendment is booked with a past effective date, or a pricing-model bug is corrected. Business date is when the value *applies*; system date is when it was *entered*.

SecDB is *bi-temporal*: every node carries both timestamps, and every query specifies both. The default is "business = today, system = now", but an auditor can replay the graph as it looked on any historical system date, and a Strat debugging yesterday's P&L Explain can ask for the curves as they were fed into the batch at 6pm last night, not as they look now after an overnight correction.

Bi-temporal storage

Every node has two timestamps:

- **Business date**: the date the value refers to ("close-of-business 15 Mar 2024").
- **System date**: the date the value was recorded in SecDB ("15 Mar 2024 18:32:04").

A correction booked on 18 March, effective 15 March, creates a new row with business date 15 March, system date 18 March. The original row is not deleted; both are queryable. Bi-temporal storage is what makes SecDB *auditable*: every number ever reported to a regulator can be reconstructed exactly.

SecDB grows monotonically (nothing is overwritten), but the reproducibility is non-negotiable for a regulated bank.

A worked bi-temporal example

Consider a five-year USD interest-rate swap booked on 15 March. On the evening of the 15th, SOFR 2y fixes at 4.82% and the overnight batch marks the trade at \$123,456. The trade's value node is persisted with:

$$(\text{business date} = \text{15-Mar},\ \text{system date} = \text{15-Mar 18:32}), \qquad V = \$123{,}456.$$

On the morning of the 18th a correction comes in: the 15th's SOFR 2y fix was republished as 4.79% by the reference-rate provider. The batch reruns as of business date 15-Mar, pulling the corrected rate, and produces a new value \$124,010. SecDB writes:

$$(\text{business date} = \text{15-Mar},\ \text{system date} = \text{18-Mar 08:14}), \qquad V' = \$124{,}010.$$

The original row is *not* deleted. Three queries now behave as follows:

- "Value on 15-Mar as we knew it on the 15th": returns \$123,456. This is the number the trader saw in her Monday morning report.
- "Value on 15-Mar as we know it now": returns \$124,010. This is the corrected number, the one that appears in the audit trail.
- "P&L Explain for 18-Mar": reports a \$554 "market-data correction" line, explaining why the mark moved without any real market action.

If an auditor next year asks "what did Goldman believe this trade was worth on 16-Mar?", the query (business = 15-Mar, system = 16-Mar) returns \$123,456 — the value used in that day's regulatory capital report. Both answers are correct, both retrievable, neither overwrites the other. That is what bi-temporal storage *does*.

37.2 Slang as a language

Slang is the DSL that runs inside SecDB. It was designed in the 1990s by Mike Dubno and a small team around the question "what does the Strat need to write a pricer, and what can the VM do to make that writing productive?" The result is unlike any mainstream production language. The mental model: Slang is to SecDB what a cell formula is to a spreadsheet — the language in which pricing rules (option pricers, curve builds, Greek formulas) are expressed, executing in the same process as the risk data rather than in a separate client talking to a database.

Design choices

Dynamic typing, C-like syntax. Slang looks superficially like C (curly braces, semicolons, function declarations) but is dynamically typed: variables hold whatever they were last assigned; the VM checks types at use time. Exploratory pricing code writes quickly; refactoring a fifty-million-line codebase is harder because type errors appear at runtime.

No keywords — `if` and `while` are functions. Control-flow constructs are not built-in keywords but user-callable functions, and therefore *overridable*. A Strat wanting a specialised `if` that logs both branches while evaluating only the taken one writes `my_if(cond, t, f){...}`. Scheme does this; Python does not. The design reflects Goldman culture: give Strats the toolkit and trust them.

Spaced variable names. Slang allows spaces in identifiers: `average annual forward rate` rather than `averageAnnualForwardRate`. Code reads like financial quantities. The parser resolves ambiguity via context; the cost is that some mainstream constructs (juxtaposition for function application) are unavailable.

Functional style, limited side effects. Slang encourages pure functions. The graph engine caches results aggressively; a function with hidden state would return an incorrect cached output next time the same inputs appear. Pure functions compose cleanly with the cache. Side-effects exist (logging, persistence) but are boxed off.

Graph-aware VM. The Slang VM knows about SecDB: function-call results are keyed by the set of graph nodes the call read. Same call, unchanged nodes → cache hit. A changed node invalidates every cache entry whose key-set included it. This is how the bank reprices overnight: most results are already cached from last night; only the subset whose inputs moved recomputes.

Slang language features

- Dynamic typing, C-like syntax.
- Control-flow constructs (`if`, `while`, `for`) are functions, not keywords — and are overridable.
- Identifiers can contain spaces: `forward swap rate` is a legal variable name.
- Functional style — pure functions the VM can cache correctly.
- The Slang VM is graph-aware: cached results are keyed by the set of SecDB nodes read, and invalidated when any of those nodes changes.

A stylised example

The following is a stylised Slang-like pricer — pseudocode capturing the flavour, not real Slang syntax.

```
// Pure function: computes BS call given spot, strike,
// rate, vol, and maturity.  Graph-aware: the VM caches
// the result keyed by the five inputs.

black scholes call(spot, strike, rate, vol, maturity) {

    // d1 and d2 under the standard BS parameterisation.
    d1 = (log(spot / strike)
          + (rate + 0.5 * vol * vol) * maturity)
         / (vol * sqrt(maturity));
    d2 = d1 - vol * sqrt(maturity);

    // Normal CDF as an ordinary function call.
    return spot * norm cdf(d1)
         - strike * exp(-rate * maturity) * norm cdf(d2);
}

// Trade pricer reads SecDB nodes for each input and
// delegates to the BS pricer.  Because the nodes are
// inputs to a pure function, the VM knows which nodes
// this result depends on and will invalidate correctly.

```

```
price european call trade(trade) {
    spot     = market spot[trade.underlying];
    strike   = trade.strike;
    rate     = discount curve[trade.ccy].rate(trade.maturity);
    vol      = vol surface[trade.underlying]
                  .implied vol(trade.strike, trade.maturity);
    maturity = year fraction(today, trade.maturity);
    return black scholes call(spot, strike, rate,
                              vol, maturity);
}
```

Listing 37.1: Stylised Slang-like pricer for a European call (pseudocode)

A second snippet shows the purity *contract* for safe VM caching, contrasted with a subtly-wrong version:

```
// CORRECT: pure, graph-aware.  All reads go through
// declared SecDB inputs; no hidden state.

pv of swap(trade, valuation date, discount curve) {
    legs = generate cashflows(trade, valuation date);
    return sum over(legs,
                    leg -> discount curve.df(leg.pay date)
                         * leg.amount);
}

// WRONG: reads today from a global mutable clock.  The VM
// caches the result keyed only by (trade, valuation date,
// discount curve), but today is a hidden input.  Tomorrow
// the cache returns yesterday's answer.

pv of swap bad(trade, discount curve) {
    legs = generate cashflows(trade, global today());
    return sum over(legs,
                    leg -> discount curve.df(leg.pay date)
                         * leg.amount);
}
```

Listing 37.2: Pure vs impure — why the VM cache depends on purity

The fix: promote the hidden input to an explicit argument, and let the VM track it as a graph node. The Slang style guide reduces to "make your functions pure or you will poison the cache."

Three features of real Slang show up in the stylised form: spaced identifiers (`black scholes call`); function composition whose inputs are all SecDB nodes, so the VM tracks exactly which cache entries to invalidate; and no explicit types — all inferred from usage.

The VM's job: aggressive caching

Most of Slang's cleverness is in the VM, not the language surface. Without caching, overnight repricing would re-execute every function in every trade. With caching, the VM observes:

- `black scholes call(100, 100, 0.05, 0.2, 1.0)` has already been computed once today, returning 10.45. If queried again with identical inputs — return the cache.
- `price european call trade(trade_42)` previously depended on `market spot["AAPL"]`, `discount curve["USD"]`, and `vol surface["AAPL"]`. The SOFR 2y bucket moved; `discount curve["USD"]` invalidates; `price european call trade(trade_42)` invalidates; all dependents of that trade invalidate.

Memoisation at whole-language scale: every pure call is memoised, and invalidation is driven by graph edges, not annotations. The overnight batch recomputes only the numbers the market moved.

A day in the life of a graph change

A concrete scenario. At 5:02pm New York on Tuesday, the SOFR-curve build has just run against the 5pm swap-rate snap, and the 2y-bucket discount factor moved from 0.9100 to 0.9095. This single change triggers the following cascade, orchestrated by SecDB without Strat intervention:

1. The node `usd zero curve.discount(2y)` is marked dirty.
2. Every pricing function whose cached result read that node is invalidated. Call the invalidation set $\mathcal{I}_1$. Typically this includes every USD swap and swaption with a 2y-region sensitivity, every USD bond, every USD-funded exotic whose funding leg crosses the 2y bucket.
3. Every aggregate whose cached result read any node in $\mathcal{I}_1$ is marked dirty. The desk-level dollar-delta to SOFR 2y, the firm-level VaR, the USD-book CVA all enter a new invalidation set $\mathcal{I}_2$.
4. A batch scheduler walks the invalidation sets in topological order, dispatching recomputations across a compute cluster. Trades in $\mathcal{I}_1$ reprice; then aggregates in $\mathcal{I}_2$ are re-aggregated from the new trade values; then any reports, dashboards, or regulatory feeds that consume those aggregates rebuild.
5. By 5:07pm the downstream effects have propagated through perhaps a hundred thousand nodes; every number on every screen in the firm that depends on SOFR 2y now reflects the new curve. The Strat is not involved.

Nobody writes "if SOFR 2y moves, recompute the following ninety thousand trades." The graph *is* that specification; the propagation engine reads and acts.

37.3 Java and Python: the long migration

By the mid-2010s Slang's drawbacks were worsening with scale. Dynamic typing made refactoring painful across fifty million lines. The hiring pool was exactly one firm. New graduates came fluent in Python/Java, and the Slang transition cost months. Tooling lagged commercial ecosystems by a decade.

Goldman's response was gradual. Post-2015, *new* development moves to Java (performance-sensitive production) and Python (research, scripting), while the SecDB core — graph engine, persistence, the decades of production pricers — stays in Slang. The strategy is coexistence: Python and Java read/write SecDB nodes via bindings; heavy graph computation runs on the Slang VM.

Three practical shapes this takes:

- **Slang ↔ Python bindings.** Research Strats pull a curve from SecDB, manipulate it in NumPy/pandas, and write the result back. The Python code is not graph-aware — dependencies are not tracked for auto-invalidation — but for ad-hoc analysis that is acceptable.
- **Java services calling the graph.** Trading systems, risk dashboards, and regulatory reporting increasingly run as Java services that query SecDB. They are plumbing, not pricers. Java is chosen for ecosystem, performance, and ubiquity.

- **Slang stays where graph semantics matter.** A new exotic pricer, curve-build function, or xVA calculation — any code that must participate in overnight propagation with correct cache invalidation — stays in Slang. The VM's graph awareness is not cheaply replicated elsewhere.

Interop mechanics

Three interop patterns recur:

- **Pull a node into Python.** A binding reads a SecDB node as-of a (business, system) date pair and materialises it as a Python object — NumPy array for a curve, DataFrame for a blotter, scalar for a Greek. The value is a snapshot; if the underlying node moves, the Python copy goes stale silently. Fine for notebooks, catastrophic for production risk.
- **Write a node back from Python.** A derived curve or scenario written back enters the graph as a new node, versioned bi-temporally and attributed to the author. Writers face higher review standards than readers.
- **Call Slang from Java.** A Java service calls the Slang VM via RPC; the Slang side runs the pricer with full graph semantics including cache, and returns a value plus node-identity for later invalidation queries. Most Java services are thin clients, not pricing implementations.

Common mistake: assuming Python owns the state

A research Strat pulls a SOFR curve into Python, derives a stress scenario, runs backtests for a week, reports to the desk head. Between day 1 and day 7, SecDB corrected its 15-March SOFR 2y fix (Section 37.1). The Python copy still holds the *old* value — the pull was a snapshot — so the backtest silently refers to retracted market data.
The fix is discipline: tag every snapshot with its (business, system) pair, treat it as read-only, refresh explicitly when analysis crosses days. The graph engine tracks staleness for Slang; Python does not get that for free.

Industry reporting calls the migration "decades-long". A Strat joining Goldman in the mid-2020s meets an actively-maintained Slang codebase alongside growing Java and Python ecosystems, not a dying language. Slang, Python, and Java are all table stakes; the real skill is knowing *which* codebase to reach for.

37.4 Shipping code into SecDB

A Strat writing a new pricer does not "merge to master and deploy." SecDB is a shared production database holding a trillion dollars of risk; any graph change potentially affects every dependent trade.

Environments

Slang code moves through a sequence of environments:

1. **Local / sandbox.** The Strat develops against a SecDB snapshot; no production trades affected.
2. **Staging.** Promoted to a shared environment where QA, tests, and other desk Strats exercise the code against realistic data and compare outputs to expected values.

3. **Production.** After review and sign-off, the function joins the overnight batch; every dependent trade reprices on the new version.

Staging is where bi-temporal storage pays off: replay yesterday's batch with the new pricer substituted in, look at the P&L diff, and demonstrate the economic impact before touching production.

Versioning and auditability

Every Slang function change is versioned bi-temporally in SecDB itself — Slang code lives in the same graph it computes on. When a function updates:

- The previous version is preserved and tagged with its business-date range. Historical reproductions use the old version; new calculations use the new one.
- Reviewer identity, commit message, and diff are persisted; auditors walk back every pricer change in the bank's history.
- Fixing a bug does not erase its footprint: the old pricer's output remains retrievable as-of any past system date, and the correction shows up explicitly in P&L Explain.

Code and data are versioned the same way, so "show me what happened on this date" extends to "show me what *code* was running".

Testing and graph-replay

Testing a Slang pricer is unusual: the test harness is the same graph engine as production. Typical patterns:

- **Snapshot test.** Run the production and candidate pricers on a representative trade set against yesterday's business-date data; flag any price diff above threshold. Bi-temporal storage makes this reproducible — yesterday's data is frozen and queryable.
- **Bump-parity test.** Finite-difference Greeks should agree to leading order with any analytical Greeks the pricer exposes. Disagreement signals an analytical bug or a pricer discontinuity the bump stepped across.
- **Invalidation test.** The CI harness swaps in a mutable-global version of a function (the WRONG pattern above) and verifies the framework catches cache-pollution. This protects the purity contract the graph engine assumes.
- **Shadow run.** For major changes, run new and old pricers in parallel for days or weeks, reporting both into a P&L-diff feed. When cumulative drift is understood, promote and retire. Shadow runs are the industry's answer to "change an airplane's engine in flight".

Each pattern leans on graph-replay — rerun a calculation with business date fixed, system date stepped — which falls out of the same bi-temporal persistence that supports the audit.

Prosperity example

Prosperity's `traderData` JSON is a miniature SecDB. Each tick, the simulator hands your algorithm the previous `traderData` plus fresh market state and expects an updated string back. The string is persistent, versioned (each tick is a system-date snapshot), and readable after the round. It is the only state a Prosperity algorithm controls.
Three SecDB analogies:

- Treat `traderData` as a versioned state graph (rolling means, positions, signal caches, regime flags). Schema-version the JSON (`"schema": "v3"`) so you can evolve without breaking mid-round.

- Cache expensive computations (OU fits, vol estimates) keyed by inputs; update only on material moves. SecDB-style memoisation, a thousand times simpler.
- Debug post-hoc by replaying ticks with the stored `traderData` — the bi-temporal-replay analogue.

The scale is laughable (kilobytes vs. petabytes) but the *shape* — persistent, versioned, auditable state — is the same.

On a Goldman Sachs desk

Four talking points for an interview that turns to SecDB/Slang. The interviewer is typically an ex-Strat or a Java/Python-forward Engineer; either way, the bar is "understands SecDB is a graph and Slang exploits it", not "recites the language spec".

- *Memoisation and invalidation.* "The VM caches pure functions by the set of SecDB nodes they read. When a node changes bi-temporally — new system-date entry for the same business date — every cache entry whose read-set includes that node invalidates, and dependents recompute on next access. The correctness hinges on functions being pure; a function with hidden mutable state would return stale cached results."
- *Circular dependencies.* "The graph is a DAG; cycles are refused at commit time. A subtle cycle case: calibrating a vol surface to option prices that themselves depend on the surface. You break the cycle by making the calibration a batch step that runs against yesterday's surface and outputs today's, rather than a same-date recursion."
- *Bi-temporal debugging.* "To debug a P&L Explain discrepancy, replay the trade's valuation with business date fixed at yesterday and system date walked forward through the overnight corrections; the point at which the value jumps isolates the corrected input. This is the Slang debugger's most-used feature."
- *Why not rewrite in Python?* "Because the graph engine, the cache, the bi-temporal persistence, and the auditability are not a language feature — they are a fifty-million-line ecosystem, and the VM's integration with the database is load-bearing. Python and Java are layered on top; Slang is not going away."

37.5 What the competition looks like

Every major sell-side bank faces the same problem — price a huge derivatives book, propagate sensitivities, reproduce historical valuations — and the responses converge on the same architectural idea with different implementations. A Strat interviewing at Goldman benefits from knowing how the firm's stack compares.

Peer systems

JPMorgan's Athena is the most-cited peer. Built by a team including ex-Goldman engineers, Athena adopts SecDB-like ideas (graph dependencies, bi-temporal storage) but uses *Python* as its in-system language. The design bet: Python's ecosystem productivity beats a DSL's graph-awareness advantage, and framework code can recover most of the missing graph semantics.

Athena is a plausible counterexample to the claim that SecDB-style systems *require* a bespoke language.

Bank of America Merrill's Quartz and **Barclays's BARX / Jupiter** fall into the same family: Python-centric, graph-structured, bi-temporal. The common pattern across Athena/Quartz/Jupiter is that the language choice is the single biggest divergence from SecDB; the deeper architectural ideas are the same.

Deutsche Bank's dbAnalytics and Morgan Stanley's Merlin occupy a middle ground: partially proprietary, partially off-the-shelf, with various tiers of graph-awareness. The details matter less than the generalisation: every large sell-side derivatives desk runs on a system that you can describe, to first approximation, as "graph database plus pricer library plus bi-temporal persistence."

Why the industry converged here

Why graph-plus-bi-temporal rather than relational-plus-batch? Three forcing functions:

- **Sensitivity requests are ad-hoc.** "What is my P&L if EUR 10y spreads 1bp?" — a relational system needs a purpose-built job per question; a graph system bumps one node and queries another.
- **Regulators audit history.** Bi-temporal storage is a prerequisite for Basel and SEC reconstruction, not a nice-to-have. Firms that bolted audit onto single-timestamp databases regretted it.
- **Pricer reuse is the biggest development lever.** Pricing the same cashflow-discount logic in ten systems costs ten times and produces ten answers. A single graph-aware pricer store is a one-time investment that pays back every year in consistency.

Every sell-side derivatives business ends up at a SecDB-shaped solution, whether or not the firm calls it that.

Why SecDB is distinctive

Three things set SecDB/Slang apart. *Language*: no other major bank kept a proprietary DSL at Goldman's scale; productivity arguments pushed peers to Python or Java. *Coverage*: SecDB prices every derivative across every asset class, an unusual centralisation — some banks run separate systems per asset class. *Age*: the graph design was in place by the mid-1990s, before commercial vendors had anything comparable, and Goldman has run a proprietary graph database for three decades while peers moved to vendor stacks — a cultural choice as much as a technical one.

Intuition

A spreadsheet shows the core idea Goldman generalised to bank scale: cells depend on other cells via formulas, and updates propagate automatically. Everything SecDB adds — persistence, bi-temporality, graph-aware caching, purity-focused language — is engineering to make the spreadsheet idea survive five million trades and a regulator. The interview takeaway: Goldman invested early in a graph-database model the industry later converged on, and stayed with the language-level integration others backed away from.

SecDB, Slang, and the J. Aron heritage

SecDB predates Slang; its origins trace to J. Aron & Company, a commodity-trading firm Goldman acquired in 1981. J. Aron's traders needed a system to track commodity

positions, prices, and P&L across a heterogeneous book, and early SecDB grew out of that tool. The graph-of-financial-objects emphasis reflects trading-desk pragmatism: built by people who needed risk answers in seconds.
Slang was designed in the 1990s by Mike Dubno and a small team to replace an earlier SecDB-pricer language. Dubno's InfoQ talk remains the best public description of the VM. Three decades on, both remain in active production: Strats joining Engineering today commit code to systems whose DNA traces to a 1980s commodity-trading desk.

37.6 Key facts to memorise

Chapter summary

- **SecDB** is Goldman's proprietary risk/pricing database; **Slang** is the proprietary language that runs inside it. Together they price the firm's trillion-dollar derivatives book overnight.
- SecDB stores the bank as a DAG. Every financial number — trade value, curve, vol, Greek, aggregate — is a node; every pricing dependency is an edge.
- A leaf node changes; SecDB propagates downstream in topological order, recomputing exactly the affected descendants. This graph-native evaluation is why sensitivities are a one-liner and overnight batches are feasible.
- Storage is *bi-temporal*: every node has a business date (the date the value applies to) and a system date (the date the value was entered). Corrections are new rows, not overwrites. Bi-temporal storage is the basis of reproducibility and audit.
- Slang is dynamically typed, C-like in syntax, but has no keywords — `if` and `while` are functions and overridable. Identifiers can contain spaces. Functions are expected to be pure; the Slang VM caches their results keyed by the set of SecDB nodes read, and invalidates via the graph.
- Post-2015 Goldman direction: new development in Java (services) and Python (research), with Slang retained for core pricing logic. The migration is explicitly decades-long; SecDB itself stays Slang.
- Shipping Slang goes local → staging → production, with every change versioned bi-temporally in SecDB itself. Code and data live in the same graph; both are audit-replayable to any historical system date.
- Prosperity's `traderData` JSON is a miniature SecDB: persistent, versioned state-passing at a kilobyte-scale analogue of the bank's petabyte graph.
- Everything in this chapter is training-knowledge-based. Public documentation of SecDB and Slang is fragmentary; expect details inside the firm to differ.

Chapter 38

Regulatory Context

Every Strat's code runs inside a regulatory frame. A pricer that returns a fair value also returns numbers that feed capital ratios, stress-test submissions, transaction reports, and an auditor's workpaper. A Strat who does not know which regulation is generating the deadline on the calendar, or why a risk-engine input is flagged "non-modellable," spends the first year of the job nodding at acronyms in meetings.

This chapter is a vocabulary tour. The aim is to make four acronyms — Basel III, FRTB, CCAR, MiFID II — intelligible the first time a senior Strat uses them in a sentence, and to give a crisp account of the LIBOR-to-SOFR transition, the largest rates-market project of the 2020s, which still shapes every curve in fixed income (Chapter 29). We aim at literacy, not at computing regulatory capital.

Prerequisites

From earlier chapters:

- Chapter 20: VaR and Expected Shortfall. FRTB's move from VaR to ES presupposes the reader knows ES.
- Chapter 29: zero curves, the USD SOFR curve, and unsecured vs secured overnight rates. The LIBOR-to-SOFR transition sits on top of that foundation.
- Chapter 36: xVA and capital charges. KVA (capital valuation adjustment) is meaningful only once the reader knows what "capital" regulators mean.

This chapter introduces no new mathematics. It is a glossary: acronyms expanded, deadlines dated, operational meaning explained.

38.1 Basel III: capital adequacy

The **Basel Accords** are international banking regulations written by the Basel Committee on Banking Supervision at the Bank for International Settlements (BIS) in Basel, Switzerland. They are not themselves law: they are *standards* which member-jurisdiction regulators (US Fed, European Banking Authority, UK PRA, Swiss FINMA) transpose into national rulebooks. Banks comply with the local transposition.

Basel III is the third iteration, published in 2010 as the response to the 2007–2008 global financial crisis. Basel I (1988) introduced credit-risk capital ratios. Basel II (2004) added a market-risk framework and is widely regarded as having failed the crisis: capital buffers were too thin, too low-quality, too weakly enforced. Basel III raised the required capital, tightened what counts as capital, and added two new liquidity ratios.

The three pillars

Basel frameworks are organised around *three pillars*. The language is standard and appears in every Basel document.

1. **Pillar 1 — minimum capital requirements.** A numerical rulebook: compute risk-weighted assets (RWA) from credit, market, and operational exposures; hold capital of at least a specified fraction of RWA.
2. **Pillar 2 — supervisory review.** The local regulator examines the bank's internal capital assessment (the ICAAP in Europe) and may impose bank-specific add-ons over the Pillar 1 minimum.
3. **Pillar 3 — market discipline.** The bank publishes standardised disclosures (the "Pillar 3 report") so that counterparties, investors, and analysts can price the bank's risk. Transparency lets the market discipline banks that take on too much.

Basel III three pillars

- **Pillar 1**: quantitative minimum capital ratios. Credit-, market-, and operational-risk RWA summed, with capital as a percentage of the total.
- **Pillar 2**: the regulator's qualitative overlay. Internal capital assessment, stress testing, bank-specific add-ons.
- **Pillar 3**: public disclosure. Standardised tables on capital composition, RWA, and key risk metrics.

Every regulatory conversation about capital fits into one of these pillars; the pillar tells the reader whether a requirement is a hard ratio, a bilateral judgement, or a disclosure obligation.

Key ratios

The point of holding regulatory capital is loss absorption: when the bank's trades, loans, or operations lose money, equity and equity-like instruments take the hit before depositors or counterparties are touched. A higher capital ratio means more losses can be absorbed before the bank fails. Basel III tightened three families of ratio; the acronyms below appear on every bank's quarterly investor presentation.

- **Common Equity Tier 1 (CET1) ratio.** CET1 capital is the highest-quality regulatory capital: common shares, retained earnings, and little else. The CET1 ratio is CET1 divided by RWA. Basel III requires a minimum of 4.5% plus a capital conservation buffer of 2.5%, giving an effective floor near 7% before any additional buffers.
- **Tier 1 capital ratio.** Tier 1 capital is CET1 plus Additional Tier 1 (certain loss-absorbing hybrid instruments). The minimum Tier 1 ratio is 6% of RWA.
- **Total capital ratio.** Tier 1 plus Tier 2 (subordinated debt and similar), with a minimum of 8% of RWA. Tier 2 absorbs losses only in liquidation.
- **Leverage ratio.** Tier 1 capital divided by total exposure (*not* risk-weighted). The minimum is 3%. The leverage ratio exists because RWA-based ratios can be gamed by risk-weighting optimisation; a non-risk ratio is a backstop.

- **Liquidity Coverage Ratio (LCR).** High-quality liquid assets divided by net cash outflow over a 30-day stress scenario. Minimum 100%. LCR says: "survive a month of stress without selling illiquid assets."
- **Net Stable Funding Ratio (NSFR).** Available stable funding divided by required stable funding over a one-year horizon. Minimum 100%. NSFR says: "do not fund long-dated assets with overnight money."

The LCR and NSFR are the Basel III innovations most directly caused by 2008, when several firms had plenty of notional capital but ran out of cash in days. Capital and liquidity are different problems; the post-crisis framework treats them separately.

Intuition

Basel III's slogan: "more capital, better capital, more liquidity." More: higher minimum ratios. Better: a larger fraction must be common equity, not hybrid instruments that failed to absorb losses when needed. More liquidity: LCR and NSFR pin down cash survival at 30 days and funding stability at one year.

38.2 FRTB: the Fundamental Review of the Trading Book

FRTB stands for the *Fundamental Review of the Trading Book*. It is the BCBS market-risk capital standard that replaces the Basel II.5 framework (itself a 2009 patch on Basel II). FRTB was finalised in January 2019; the original 2022 go-live slipped with pandemic delays, and jurisdictional implementations are staggered 2024–2026.

Where Basel III is cross-cutting (credit, market, and operational risk), FRTB is narrowly about *market risk*: capital a bank holds against its trading-book positions moving adversely. Market-risk capital is the number that most directly bites a derivatives Strat, because it is computed from their pricers' sensitivities. Fundamentally FRTB did two things: it replaced Value-at-Risk with Expected Shortfall as the risk measure, and it drew a much harder line between the trading book (marked to market daily) and the banking book (held to maturity) to close long-standing arbitrage between the two.

Trading book versus banking book

Banks carve the balance sheet in two. The **banking book** holds assets the bank intends to hold to maturity — most obviously loans. The **trading book** holds positions to be traded — market-making inventory, hedges, proprietary positions. Banking-book assets attract credit-risk capital; trading-book positions attract market-risk capital. A long-standing problem was regulatory *arbitrage* between the two: a bank could park a position in whichever book gave a lower charge. FRTB imposes hard rules on book classification and restricts movement between them.

IMA and SA

FRTB offers two capital approaches per desk:

- **Internal Models Approach (IMA).** The bank uses its own risk model, subject to regulatory approval. Approval is granted *per desk*: each desk passes a battery of model-performance tests, and approval can be withdrawn if the desk fails backtesting.
- **Standardised Approach (SA).** A prescribed sensitivities-based formula: deltas, vegas, and curvatures per risk factor, with regulator-set weights and correlations. No model needed, but capital is generally higher than IMA on a well-run desk.

Every bank computes SA for every desk. IMA is a premium option: lower capital in exchange for the cost of building and defending internal models to the regulator.

From VaR to Expected Shortfall

The most visible change from Basel II.5 to FRTB is the *risk measure*. The old framework used **Value-at-Risk** at 99% over a 10-day horizon. FRTB replaces this with **Expected Shortfall** (Chapter 20) at 97.5%, over a liquidity-horizon-weighted 10-day basis.

The change is conceptual. VaR at 99% is a quantile: "the smallest loss I exceed with probability at most 1%," silent on tail severity. ES at 97.5% is the *expected* loss given that the loss exceeds the 97.5% quantile — a tail mean. For a Gaussian loss distribution the two numbers are similar; for fat tails — the case that matters — ES is substantially larger, and the capital charge scales with tail severity. FRTB picked 97.5% rather than 99% so that on Gaussian distributions the new and old numbers are comparable, while on fat tails the new number is strictly more conservative.

Liquidity horizons

A second FRTB innovation is the *liquidity horizon* overlay. Not all risk factors can be hedged in 10 days: a one-week FX position can, an illiquid exotic credit correlation trade cannot. FRTB assigns each risk factor a liquidity horizon from a menu (10, 20, 40, 60, 120 days) and scales the shock accordingly, so illiquid risks attract capital as if held unhedged for their full horizon.

Non-modellable risk factors

The most operationally significant piece of FRTB, and the one Strats hear about first, is the **non-modellable risk factor (NMRF)** regime. A risk factor counts as *modellable* under IMA only if it has real observable price history: broadly, 24 or more observable prices in the preceding year, with no gap longer than a month. Anything failing that test is *non-modellable*.

NMRFs do *not* enter the ES calculation. Instead they attract a separate punitive stressed-capital charge, computed per NMRF with no diversification benefit across NMRFs. An IMA desk running many NMRFs has its capital dominated by the NMRF charge and derives little benefit from the internal model. Keeping risk factors modellable becomes an ongoing infrastructure discipline: Strats negotiate with market-data teams over whether an extra vendor feed tips a factor into modellable status and whether the cost is worth the capital saving.

FRTB key changes

- **ES replaces VaR.** Expected Shortfall at 97.5% confidence, not VaR at 99%. Captures tail severity, not just a quantile.
- **Per-desk IMA approval.** Internal-model approval is granted and revoked desk by desk, not bank-wide. A failed backtest can push a desk onto the standardised approach overnight.
- **Liquidity horizons.** Shocks scale with how long it takes to hedge a given risk factor; illiquid factors see 40–120-day shocks rather than 10.
- **NMRF.** Risk factors with insufficient price observations attract punitive stressed-capital charges outside the internal model. Modellability becomes an infrastructure concern.
- **Trading-banking book boundary.** Hard rules on book classification and movement between books, to close the old arbitrage.

Intuition

FRTB makes illiquid and data-poor risks expensive. The consequence is a bias towards *liquid* risk factors: prefer the vanilla swap curve over a bespoke basis spread, listed-option vols over OTC exotic correlations, deep-market tenors over sparse ones. Either a desk concentrates risk in liquid buckets (lower NMRF burden, lower capital) or accepts a structurally higher capital footprint for the exotics franchise. The regime is a deliberate financial incentive to centralise risk in liquid instruments.

38.3 CCAR: the Fed stress test

CCAR is the *Comprehensive Capital Analysis and Review*, the Federal Reserve's annual stress test for large US bank holding companies (BHCs), introduced in 2011. Basel is international; CCAR is US-specific. Both ask whether a bank has enough capital to survive a bad scenario, but differ in mechanism.

CCAR applies to BHCs above a total-consolidated-assets threshold; since the 2018 EGRRCPA reforms this sits around $100 billion. The largest US banks and the US subsidiaries of large foreign banks are in scope; Goldman Sachs is a CCAR firm.

The scenario-based mechanism

Early each year the Fed publishes three scenarios: baseline, adverse, severely adverse. Each is a path for several dozen macro and market variables — GDP, unemployment, the Dow, the 10-year Treasury, VIX, house prices, credit spreads — over a nine-quarter horizon. The severely adverse scenario is deliberately deeper than the historical norm: a plausible worst-case recession.

Each in-scope bank is required to:

1. Project its balance sheet, revenue, losses, and capital ratios quarter by quarter under each Fed-mandated scenario.
2. Submit those projections along with a capital plan (dividends, buybacks, new issuance).
3. Show that, under the severely adverse scenario, the firm's post-stress capital ratios remain above the regulatory minima.

The Fed also runs its own **supervisory stress test** (DFAST — Dodd-Frank Act Stress Test) using Fed-built models. CCAR is the bank's exercise; DFAST is the Fed's. Together they form the annual stress-test cycle.

The Strat's role

For a markets-division Strat, the visible piece is the trading-book loss projection. The market-risk engine is re-run under each Fed scenario to produce a **stressed P&L**: what the trading book loses, path by path, quarter by quarter. This feeds the bank's projected losses and through them the projected capital ratios.

Contributing to CCAR is a recurring annual project on most trading-desk Strat teams: scenarios land, the team maps Fed macro variables onto pricing inputs, pricers re-run, results are aggregated, reconciled, and submitted. If the Fed disagrees, the bank either reconciles or accepts the Fed's DFAST numbers.

Failing CCAR is expensive. Before 2020 the Fed could issue qualitative objections blocking a bank's planned capital distributions for a year. Since 2020 the framework has evolved into the **Stress Capital Buffer**, in which CCAR stress losses feed directly into the required capital buffer, making the test an ongoing constraint rather than an annual pass/fail.

CCAR at a glance

- US-specific Federal Reserve stress test, annual, in scope for BHCs above roughly \$100 billion in total assets.
- Fed publishes baseline, adverse, and severely adverse macro scenarios. Banks project nine quarters of balance-sheet, P&L, and capital under each.
- Trading-book losses are generated by re-running the market-risk engine with desk-level pricer output; Strats own the inputs that feed the submission.
- Post-2020: stress losses feed the Stress Capital Buffer directly, making the test a continuous capital constraint rather than an annual pass/fail.

38.4 MiFID II: European market structure

MiFID II is the Markets in Financial Instruments Directive II, the EU's wholesale revision of securities-market regulation. It came into force on 3 January 2018, replacing the original MiFID (2007). Unlike Basel and FRTB (capital), MiFID II is about *market structure*: how venues operate, how trades execute, what must be reported.

Four pieces show up most often in a Strat's work.

- **Pre- and post-trade transparency.** Venues and systematic internalisers publish pre-trade quotes and post-trade execution details, with narrow instrument-specific deferral windows for large or illiquid trades. The goal: make European markets as transparent as US equity markets.
- **Best execution.** Firms must take *all sufficient steps* to obtain the best result for clients across price, costs, speed, likelihood of execution, and settlement. Annual execution-quality reports (RTS 27 and RTS 28) are required, subsequently scaled back by the "Quick Fix" amendments.
- **Research unbundling.** Historically sell-side research was paid implicitly out of trading commissions. MiFID II requires asset managers to pay explicitly, either from P&L or a Research Payment Account funded by disclosed client charges. The effect was a sharp contraction in sell-side research and a shift toward the largest names.
- **Transaction reporting.** Every in-scope transaction is reported T+1 with typically 65 fields per trade, including LEIs, buyer/seller designations, and decision-maker identifiers. The reporting systems are a significant engineering surface for any EU-facing sell-side firm.

MiFID II also introduced the **Systematic Internaliser (SI)** regime in modern form: a firm dealing on its own account, on an organised, frequent, and systematic basis, outside a regulated venue. SIs have quoting and transparency obligations; the SI designation is the main legal status under which large dealers internalise client flow in Europe.

MiFID II is directive-plus-regulation: the MiFIR regulation is directly applicable EU law, while the MiFID II directive is transposed by each member state. Post-Brexit the UK retains a largely parallel regime with its own tweaks.

MiFID II essentials

- EU regulation, 3 January 2018 go-live. Scope is market structure and conduct, not capital.

- Transparency, best execution, research unbundling, and T+1 transaction reporting are the four pillars most visible to a markets-division Strat.
- Introduces the Systematic Internaliser regime as the primary legal status for bilateral dealer flow.

38.5 LIBOR to SOFR

For decades the most important interest-rate benchmark in global finance was **LIBOR** — the London Interbank Offered Rate — a daily-published set of rates intended to represent the cost at which major banks could borrow unsecured from one another in London, across five currencies (USD, GBP, EUR, JPY, CHF) and seven tenors (overnight out to 12 months). At its peak LIBOR was embedded in an estimated $300 trillion of contracts: swaps, floating-rate notes, mortgages, student loans, corporate loans, structured products.

Why LIBOR was replaced

LIBOR was produced by a daily *submission* from a small panel of banks. Each submitted the rate it believed it could borrow at; the top and bottom quartiles were trimmed and the mean of the rest published as that day's LIBOR per currency and tenor. Submissions were not tied to observable transactions, because true unsecured interbank borrowing had become thin after 2008. LIBOR was largely judgement, and judgement is manipulable.

Between roughly 2005 and 2012 several panel banks manipulated submissions to profit from LIBOR-linked positions and to disguise their funding costs during the crisis. Investigations produced multi-billion-dollar fines and criminal convictions. The scandal convinced regulators that a judgement-based benchmark with this much economic weight was unsustainable.

LIBOR scandal 2012

Investigations in 2011–2013 revealed coordinated manipulation of LIBOR submissions at several panel banks. Emails between traders and submitters — "an extra basis point would be much appreciated" — became enforcement exhibits. Cross-jurisdiction fines reached roughly $9 billion; several individuals were convicted. In 2017 the UK FCA announced it would no longer compel LIBOR submissions after end-2021, triggering the global transition. Most tenors ceased publication at end-2021; the final USD settings ceased on 30 June 2023.

The replacement: risk-free rates

Each LIBOR-currency regulator designated a **risk-free rate (RFR)** as successor. All RFRs share two features: they are *overnight*, and based on *observable transactions* rather than submissions.

- **SOFR** (Secured Overnight Financing Rate) — USD. A transaction-based overnight rate backed by US Treasury repo volume. Published by the Federal Reserve Bank of New York.
- **SONIA** (Sterling Overnight Index Average) — GBP. A transaction-based overnight rate from unsecured sterling deposits. Published by the Bank of England.
- **ESTR** (Euro Short-Term Rate) — EUR. Published by the ECB, based on unsecured overnight borrowing by euro-area banks.

- **TONA** (Tokyo Overnight Average Rate) — JPY. Published by the Bank of Japan, unsecured overnight call market.
- **SARON** (Swiss Average Rate Overnight) — CHF, secured, published by SIX Swiss Exchange.

Secured (SOFR, SARON) versus *unsecured* (SONIA, ESTR, TONA) is a jurisdictional judgement. US regulators concluded that the Treasury repo market was the deepest USD overnight market; UK, EU, and Japanese regulators preferred unsecured overnight markets that better mirrored LIBOR. The consequence: USD SOFR contains no bank credit component, while SONIA and ESTR contain a small one. Basis markets exist between each RFR and the defunct LIBOR curves.

Backward-looking versus forward-looking

A crucial shift: RFRs are *backward-looking*; LIBOR was *forward-looking*.

LIBOR was a term rate: 3-month USD LIBOR fixed at the start of a 3-month interest period, so the borrower knew the interest amount on day one. The rate incorporated a 3-month forward view of unsecured bank funding.

RFRs are overnight. A 3-month rate is obtained by compounding overnight observations daily, so it is known only at the *end* of the period. This is operationally different: treasurers, lenders, and borrowers who planned around knowing their interest payment in advance had to adapt cashflow systems, invoicing, and debt covenants to a rate fixed in arrears. Term RFR products exist (Term SOFR, Term SONIA), computed from futures or OIS, but compounded-in-arrears dominates derivatives markets.

Fallback language

Trillions of notional in legacy contracts referenced LIBOR and matured past cessation. Bilateral rewrites were infeasible. The industry solution, driven by ISDA (International Swaps and Derivatives Association), was a **fallback protocol**: a standard contractual amendment which, once signed, rolls every in-scope LIBOR reference between two adhering parties to an RFR-based fallback at cessation.

The fallback rate has two components: a compounded-in-arrears RFR plus a **credit-adjustment spread**, set as the historical median spread between the RFR and the relevant LIBOR tenor over a five-year lookback window. The spread is a one-time constant, fixed at cessation announcement. Its purpose: ensure the fallback rate is economically equivalent to LIBOR at transition, not a step change in coupon.

Loan markets took a slightly different path with contract-specific "hardwired" or "amendment" fallback language, but the derivatives market was dominated by the ISDA protocol.

LIBOR to SOFR essentials

- LIBOR: submission-based, forward-looking, unsecured term rate. Ceased end-2021 (most tenors) and 30 June 2023 (remaining USD settings).
- Successor RFRs: SOFR (USD, secured), SONIA (GBP), ESTR (EUR), TONA (JPY), SARON (CHF). All are transaction-based, overnight, and backward-looking.
- ISDA fallback protocol: compounded RFR plus fixed credit-adjustment spread, applied automatically to adhering parties at cessation.
- Basis markets between RFRs and ex-LIBOR curves persist because USD SOFR is secured and the old USD LIBOR was unsecured; small but not zero basis everywhere.

See Chapter 29 for the curve-construction mechanics of SOFR discounting and Chapter 30 for SOFR OIS pricing.

Intuition

LIBOR's problem: it tried to measure something real (bank funding cost) using a method (quotes) that stopped being tethered to reality after 2008. RFRs fix this by measuring overnight markets that still have volume, at the cost of losing LIBOR's forward-looking term structure. Term RFRs and fallback spreads are the industry's compensation for that loss.

38.6 How regulation touches a Prosperity algorithm

Rules before code

Prosperity has no regulatory layer: no capital requirement, no stress test, no transaction report, no best-execution obligation. A competitor printing sixty thousand orders in a round is bounded only by the matching engine.

The Prosperity practice that *does* mirror the regulated world is "know the rules before you code." Each round publishes a position limit per product, enforced by the engine. Each round imposes a per-round order-count limit; exceed it and the algorithm fails to submit. Tariffs, import penalties, and conversion constraints (Chapter 25) are local analogues of regulatory constraints. A strategy optimal in backtest but violating a rule is worth zero; the first hour of each round is spent ensuring the algorithm lives inside the rulebook.

The transferable habit: before any simulation, write the hard constraints in the same file as the pricing logic. In Prosperity those are position and order limits. On a Goldman desk they are Pillar 1 capital, FRTB NMRF lists, CCAR stress sensitivities, MiFID best-execution evidence. Same activity, different rulebook.

38.7 How regulation touches a Goldman Strat

Regulation in a Strat's calendar

A Strat in Securities Engineering inherits regulatory calendars as a background fact of life.

FRTB IMA approval is a multi-year project. A desk seeking IMA status submits an application, runs its model in parallel with SA for at least a year to build a track record, defends backtesting to the regulator, then continues defending modellability and NMRF classifications indefinitely. Strats on IMA desks spend meaningful time on regulator-facing artefacts: NMRF evidence files, PLA (profit-and-loss attribution) reports, backtesting diagnostics. A desk failing backtesting three times in a 12-month window is de-approved and falls back to SA until it re-applies.

LIBOR to SOFR was a five-year coordinated push. From roughly 2018, every major bank's rates franchise ran an enterprise-wide LIBOR transition programme touching trading systems, pricing libraries, risk engines, client documentation, and legal. Discount curves switched from LIBOR to OIS (SOFR, SONIA, ESTR) discounting; every hard-coded LIBOR pricer was audited and rewritten; ISDA fallback adherence was coordinated firm-wide; non-ISDA client contracts were renegotiated bilaterally. The transition is not fully finished: legacy contracts with unusual fallback language remain, basis risk between term and compounded-in-arrears RFRs is live, and cross-currency basis swaps carry LIBOR-era residuals.

CCAR is an annual drumbeat. In Q1 the Fed scenarios land; February through April the firm's trading-book loss projections are produced, reconciled, and submitted; June the Fed publishes results. During submission season, market-risk Strats spend significant time mapping macro-scenario variables into desk-specific shocks, re-running pricers, and explaining reconciliation differences with DFAST.

MiFID II is an ongoing reporting tax. Every EU trade produces a T+1 regulatory report from post-trade infrastructure. Strats rarely own the reporting pipeline but feel it through required trade-metadata fields the pricing system must emit: LEIs, decision-maker codes, algo flags, best-execution evidence.

The regulatory apparatus is slow-moving and relentless. The junior Strat who internalises the acronyms in their first year spends the rest of the career not relearning them.

38.8 Key facts to memorise

Chapter summary

- **Basel III** (2010) is the international capital-adequacy framework. Three pillars: minimum capital, supervisory review, market discipline. Key ratios include CET1, Tier 1 $\geq$ 6%, total capital $\geq$ 8%, leverage $\geq$ 3%, and the liquidity ratios LCR and NSFR.

- **FRTB** (finalised 2019) is the Basel market-risk framework. Replaces VaR at 99% 10-day with **Expected Shortfall at** 97.5%, introduces per-desk IMA approval versus the standardised approach, adds liquidity horizons, and penalises non-modellable risk factors (NMRFs) punitively.

- **CCAR** is the Fed's annual stress test for US BHCs above roughly $100 billion. Macro scenarios drive nine-quarter projections; trading-book stressed P&L feeds the Stress Capital Buffer.

- **MiFID II** (EU, 2018) is about market structure: pre- and post-trade transparency, best execution, research unbundling, T+1 transaction reporting, and the Systematic Internaliser regime.

- **LIBOR ceased** at end-2021 for most tenors and 30 June 2023 for the final USD settings, driven by the 2012 manipulation scandal. Replaced by backward-looking transaction-based RFRs: SOFR, SONIA, ESTR, TONA, SARON.

- The **ISDA fallback protocol** rolled legacy LIBOR derivative contracts to compounded RFR plus a fixed credit-adjustment spread at cessation.

- RFRs are overnight and backward-looking; LIBOR was term and forward-looking. Interest is known only at period end under the new convention.

- The Strat's daily rhythm is shaped by regulation: FRTB modellability, CCAR submission season, MiFID reporting metadata, and the long tail of the LIBOR transition.

Chapter 39

Interview-Grade Problems

Quant interviews test probability, stochastic processes, linear algebra, dynamic programming, and the ability to talk through all four under pressure. This chapter walks **fifteen problems end-to-end** using a five-field template — problem statement, first attempt, what goes wrong, correct approach, interviewer variations — to expose the handful of patterns that recur across Goldman Sachs and Jane Street questions. The canonical industry reference is *A Practical Guide to Quantitative Finance Interviews* by Xinfeng Zhou, the "Green Book" (Zhou, 2020); we follow its conventions and cross-reference it where a problem appears nearly verbatim.

Prerequisites

The reader should be comfortable with:

- **Probability** (Chapter 1 Appendix A and Chapter 6): conditional expectation, Bayes' rule, independence, expectation and variance of Bernoulli, Binomial, Geometric, Poisson, Normal.
- **Stochastic processes** (Chapter 10): the Brownian motion SDE dW_t, the Ornstein–Uhlenbeck process $dX_t = -\theta X_t\,dt + \sigma\,dW_t$, and martingales.
- **Itô calculus and Black–Scholes** (Chapter 16): Itô's lemma on $f(W_t, t)$, the GBM SDE, and $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$.
- **Linear algebra**: eigenvalues/eigenvectors, the spectral theorem for symmetric matrices, SVD, and that PCA diagonalises a covariance matrix.
- **Algorithms**: memoised recursion, the 2D DP table, and $O(nm)$ analysis.

Every result is derived here or pointed to its source chapter.

Each problem uses the same five-field format; read them as interview transcripts.

39.1 Probability and combinatorics

Problem 1: the Monty Hall door switch

Problem statement. Three doors; behind one is a car, behind the other two are goats. You pick door 1. The host, who knows where the car is, opens door 3 to reveal a goat. He offers you the chance to switch to door 2. Should you switch? What is the probability of winning the car under each strategy?

First attempt. "Two doors remain, one has a car, so it is 1/2 either way." This is the answer almost every candidate gives first.

What goes wrong. Treating the host's action as if it carried no information. The door the host opens is not random: he must open a goat door and must not open yours. That conditioning changes the posterior.

Correct approach. Condition on the initial pick. With probability 1/3 you initially picked the car; conditional on that, the host opens either remaining door and switching loses. With probability 2/3 you initially picked a goat; conditional on that, the host is *forced* to open the other goat door, leaving the car as the door you could switch to. Hence

$$\mathbb{P}(\text{win} \mid \text{switch}) = \tfrac{2}{3}, \qquad \mathbb{P}(\text{win} \mid \text{stay}) = \tfrac{1}{3}.$$

A cleaner way: the initial pick is correct with probability 1/3, and the host's reveal is a deterministic function of your pick and the true location, so the posterior on "your pick is the car" stays at 1/3. The remaining 2/3 concentrates on the unopened door you did not pick.

Interviewer variations. (i) Host picks a door uniformly and happens to reveal a goat: by Bayes the posterior is 1/2 on each remaining door. (ii) n doors, host opens $n-2$ goat doors: switching wins with probability $(n-1)/n$. (iii) Pick *before* the host reveals anything: both strategies give 1/2.

Intuition

The host is your accomplice. Whenever you pick wrong (two thirds of the time), he is forced to open a goat door and thereby *tells you which remaining door has the car.* The paradox dissolves once you see his action is constrained, not random.

Problem 2: two children, at least one boy

Problem statement. A family has two children. You are told that at least one of them is a boy. What is the probability that both are boys? Separately: you are told that the *elder* child is a boy. What is the probability both are boys?

First attempt. "One child is a boy, so the other is independently a boy or girl, giving 1/2" for both.

What goes wrong. The two questions condition on different events: treating them as the same discards the difference between "at least one" and "the elder is."

Correct approach. The four equally likely birth orderings are $\{BB, BG, GB, GG\}$. Conditioning on "at least one boy" eliminates GG and leaves three outcomes $\{BB, BG, GB\}$, of which only BB is two boys; the answer is 1/3. Conditioning on "the elder is a boy" eliminates GB and GG and leaves $\{BB, BG\}$; the answer is 1/2. The difference is that the second piece of information distinguishes an ordered pair, while the first does not.

Interviewer variations. (i) "At least one boy born on Tuesday": posterior $13/27 \approx 0.481$ (the "Tuesday boy" paradox); extra distinguishing labels push toward 1/2. (ii) Three children, at least one boy: 4/7, by enumerating eight orderings minus GGG.

The "unspecified child" trap

"One" is ambiguous in English but precise in probability. "At least one is a boy" conditions on a set event; "a specific one (the elder) is a boy" conditions on a coordinate. On the ambiguous version, pin down *which* conditional is meant.

Problem 3: the 100 prisoners and the boxes

Problem statement. One hundred prisoners are numbered 1 to 100. In a room are 100 closed boxes; inside each is a slip bearing one of the numbers 1 to 100, with each number appearing exactly once in some uniformly random permutation. Each prisoner in turn enters the room, opens at most 50 boxes, and must find the slip bearing their own number. They may not communicate during the game and cannot leave evidence behind. All 100 must succeed for the group to win. Find a strategy that wins with probability at least 30%.

First attempt. "Each prisoner opens 50 random boxes: individual success $1/2$, group success 2^{-100}." Correct under random opening, which makes any 30% bound sound impossible.

What goes wrong. The naive analysis assumes independence. A coordinated strategy correlates the searches so almost all succeed or almost all fail.

Correct approach. Prisoner k opens box k; if the slip reads j, they open box j, then the box labelled by its slip, forming a cycle through the permutation. Prisoner k wins iff their cycle has length at most 50; *all* win iff the permutation has no cycle longer than 50.

For a uniform random permutation of n elements, the probability of a cycle of length $\ell > n/2$ is exactly $1/\ell$: such a long cycle is unique and placeable in $\binom{n}{\ell}(\ell-1)!(n-\ell)!$ ways out of $n!$. Hence

$$\mathbb{P}(\text{fail}) = \sum_{\ell=51}^{100} \frac{1}{\ell}, \qquad \mathbb{P}(\text{win}) = 1 - \sum_{\ell=51}^{100} \frac{1}{\ell} \approx 0.3118.$$

As $n \to \infty$ the failure probability approaches $\ln 2 \approx 0.693$, so the winning probability approaches $1 - \ln 2 \approx 0.307$.

Interviewer variations. (i) Open 60 boxes each: failure $\sum_{\ell=61}^{100} 1/\ell \approx 0.502$, win ≈ 0.498. (ii) Can you beat cycle-following? No: an information-theoretic argument. (iii) Verify $1/\ell$ by Monte Carlo.

Cycle lemma for random permutations

For a uniform random permutation of n symbols and any $\ell > n/2$, the probability the permutation contains a cycle of length exactly ℓ is $1/\ell$. Consequently the cycle-following strategy in the 100-prisoner problem wins with probability $1 - \sum_{\ell=51}^{100} 1/\ell \approx 0.3118$.

Problem 4: the secretary problem

Problem statement. n candidates are interviewed in uniformly random order. After each interview you must either hire the candidate on the spot or dismiss them forever. You can rank any two you have seen but not candidates you have not. What strategy maximises the probability of hiring the single best candidate?

First attempt. "Hire the first candidate better than everyone so far." Right in spirit, but leaves open when to *start* applying the rule.

What goes wrong. The rule triggers too early: the best-so-far among the first few is not a strong signal about the global best. A *warm-up* window is needed.

Correct approach. Fix threshold $r \in \{1, \ldots, n\}$. Interview the first $r-1$ (hire none), then hire the first candidate better than everyone so far. The success probability is

$$P(r) = \frac{r-1}{n} \sum_{i=r}^{n} \frac{1}{i-1}.$$

$(r-1)/n$ is the probability the best appears after position $r-1$; the sum accounts for the best-so-far at position $i-1$ lying in the warm-up. Write $P(r) \approx (r/n)\ln(n/r)$ for large n (harmonic sum as an integral); setting $dP/dr = 0$ gives $\ln(n/r) = 1$, i.e. $r = n/e$, and $P \to 1/e \approx 0.3679$.

For $n = 10$ the optimum is $r = 3$ (reject the first two, then hire the next new best), success ≈ 0.3987.

Interviewer variations. (i) Minimise *expected rank* instead: limiting value ≈ 3.87. (ii) "Postdoc" variant: hire the *second* best. (iii) Two hires — a staple follow-up.

The 1/e law

The secretary problem and its $1/e$ threshold date from the 1950s; Martin Gardner popularised it in his February 1960 *Scientific American* column as "the Googol game." It reappears in interviews because it teaches *optimal stopping under no prior*, which generalises to bidding, exercise, and liquidation.

Problem 5: the coupon collector

Problem statement. Cereal boxes contain coupons, each of which is independently one of n equally likely types. How many boxes do you expect to open before you have collected at least one of every type?

First attempt. "With n types I expect roughly n boxes."

What goes wrong. The first coupon is certain to be new, but the last is rare. As the collection nears completion the waiting time stretches out: the k-th new coupon, given $k-1$ already collected, is geometric with success probability $(n-k+1)/n$. Expected waiting time is $n/(n-k+1)$.

Correct approach. Summing over $k = 1, \ldots, n$,

$$\mathbb{E}[T_n] = \sum_{k=1}^{n} \frac{n}{n-k+1} = n \sum_{j=1}^{n} \frac{1}{j} = nH_n,$$

where H_n is the n-th harmonic number. For large n the expected time is $n \ln n + \gamma n + O(1)$ with $\gamma \approx 0.5772$ the Euler–Mascheroni constant. For a deck of $n = 52$ cards, $\mathbb{E}[T] \approx 235.98$.

The variance is $\mathrm{Var}(T_n) = n^2 \sum_{k=1}^{n} 1/k^2 - nH_n \sim \pi^2 n^2/6$, so the standard deviation scales linearly in n, of the same order as the mean.

Interviewer variations. (i) Unequal coupon probabilities: expected time grows; use inclusion–exclusion. (ii) Variance derivation: decompose T_n as a sum of independent geometrics and sum variances. (iii) Concentration: Chernoff-style bounds give $\mathbb{P}(T_n > n \ln n + cn) \sim e^{-c}$.

Intuition

The harmonic sum is where the time goes. The first half of types costs $\approx n \ln 2 \approx 0.69n$; the last few cost another $\approx 0.5n \ln n$. The tail dominates.

39.2 Stochastic processes

Problem 6: first hitting time of a Brownian motion

Problem statement. Let W_t be a standard Brownian motion starting at $W_0 = 0$. Let $T_a = \inf\{t \geq 0 : W_t = a\}$ for $a > 0$. What is $\mathbb{P}(T_a < \infty)$? What is $\mathbb{E}[T_a]$?

First attempt. "The Brownian motion is a martingale with mean zero; if I apply optional stopping at T_a I get $\mathbb{E}[W_{T_a}] = 0$, but $W_{T_a} = a$, contradiction. So T_a must be infinite with positive probability."

What goes wrong. The optional-stopping theorem has hypotheses. Applying it to an unbounded stopping time without checking uniform integrability is the classic mistake here.

Correct approach. *Step 1: hitting is certain.* By recurrence of one-dimensional Brownian motion, $\mathbb{P}(T_a < \infty) = 1$ for any finite a. Equivalently, W_t is recurrent in dimension 1 so every level is visited. *Step 2: mean is infinite.* Apply Itô's lemma to $e^{\lambda W_t - \lambda^2 t/2}$ to obtain a martingale, evaluate at T_a, and use optional stopping on the bounded stopping time $T_a \wedge N$. Taking $N \to \infty$ yields

$$\mathbb{E}[e^{-\lambda^2 T_a/2}] = e^{-\lambda a}, \qquad \lambda > 0,$$

the Laplace transform of the hitting-time density. Inverting, the density is

$$f_{T_a}(t) = \frac{a}{\sqrt{2\pi t^3}}\, e^{-a^2/(2t)}, \qquad t > 0,$$

a *Lévy distribution.* Its mean is $\int_0^\infty t f_{T_a}(t)\, dt = \infty$ because of the heavy $t^{-3/2}$ tail. So T_a is almost surely finite but has infinite expectation. *Step 3: scaling.* By the scaling $W_t \overset{d}{=} a^{-1} W_{a^2 t}$, one reads off $T_a \overset{d}{=} a^2 T_1$: first passage scales with the square of the level. Dimensional analysis alone gives this: $[a^2/t]$ has units of volatility squared.

Interviewer variations. (i) Hit either a or $-b$ first with $a, b > 0$: the probability of hitting a first is $b/(a+b)$ and the expected time is ab. (ii) Add a drift μ: the Laplace transform picks up a drift term and the hitting distribution becomes inverse Gaussian. (iii) Prove the reflection principle and use it to derive $\mathbb{P}(\max_{s \leq t} W_s \geq a) = 2\,\mathbb{P}(W_t \geq a)$.

Brownian hitting time

For $W_0 = 0$, $a > 0$:

$$\mathbb{P}(T_a < \infty) = 1, \quad \mathbb{E}[T_a] = \infty, \quad T_a \overset{d}{=} a^2 T_1,$$

with density $f_{T_a}(t) = \frac{a}{\sqrt{2\pi t^3}} e^{-a^2/(2t)}$. Finite in probability, infinite in mean.

Problem 7: half-life of an OU process

Problem statement. Let X_t satisfy the Ornstein–Uhlenbeck SDE $dX_t = -\theta(X_t - \mu)\, dt + \sigma\, dW_t$ with $\theta > 0$. Derive the *half-life* of mean reversion: the expected time for a displacement from μ to decay to half its initial value.

First attempt. "Half-lives are $1/\theta$."

What goes wrong. The time constant is $1/\theta$; the half-life differs by a factor of $\ln 2$. Confusing the two costs full credit.

Correct approach. Take expectations of the SDE; the noise term is zero-mean, so

$$\frac{d\,\mathbb{E}[X_t]}{dt} = -\theta\left(\mathbb{E}[X_t] - \mu\right),$$

with solution $\mathbb{E}[X_t] - \mu = (X_0 - \mu)e^{-\theta t}$. The half-life $t_{1/2}$ satisfies $e^{-\theta t_{1/2}} = 1/2$:

$$t_{1/2} = \frac{\ln 2}{\theta} \approx \frac{0.693}{\theta}.$$

For $\theta = 0.1$/day, $t_{1/2} \approx 6.93$ days.

Interviewer variations. (i) Stationary distribution: Gaussian, mean μ, variance $\sigma^2/(2\theta)$. (ii) Pair-trade holding horizon with residual half-life $t_{1/2}$: a few half-lives. (iii) Discrete time: $\rho = e^{-\theta \Delta t}$, so $t_{1/2} = -\ln 2 \cdot \Delta t / \ln \rho$.

Intuition

"Time constant" is when the deviation falls to $1/e \approx 0.368$; "half-life" is when it falls to 0.5. The two differ by $\ln 2 \approx 0.693$. Use half-lives with traders; use time constants in the calculus.

Problem 8: gambler's ruin and martingale optional stopping

Problem statement. Start with a chips at a fair game; each round win or lose one chip with probability 1/2. The game ends when the stack hits either 0 or N. What is the probability of exit at N? What is the expected number of rounds until exit?

First attempt. "Random walk, so maybe a/N?" The guess is correct; justifying it earns the credit.

What goes wrong. The "distance-weighted" argument leaves two holes: why is it exactly linear in a, and why may we use optional stopping?

Correct approach. *Probability of reaching N.* Let X_t denote the chip stack. X_t is a martingale because the game is fair, so by optional stopping (the stopping time τ is bounded in expectation and the martingale is bounded) $\mathbb{E}[X_\tau] = X_0 = a$. Writing $\mathbb{E}[X_\tau] = 0 \cdot \mathbb{P}(\text{ruin}) + N \cdot \mathbb{P}(\text{hit } N)$ gives

$$\mathbb{P}(\text{hit } N) = \frac{a}{N}.$$

Expected duration. Apply optional stopping to the second martingale $X_t^2 - t$. This is a martingale because $\mathbb{E}[X_{t+1}^2 - X_t^2 \mid X_t] = 1$ for a fair ± 1 random walk. So $\mathbb{E}[X_\tau^2] = \mathbb{E}[X_0^2] + \mathbb{E}[\tau]$.

$$\mathbb{E}[X_\tau^2] = 0 \cdot \mathbb{P}(\text{ruin}) + N^2 \cdot \frac{a}{N} = aN,$$

so $\mathbb{E}[\tau] = aN - a^2 = a(N - a)$. With $a = 10$, $N = 100$: $\mathbb{P}(\text{win}) = 10\%$, $\mathbb{E}[\tau] = 900$ rounds.

Unfair coin. If the win probability is $p \neq 1/2$ with $q = 1 - p$ and $r = q/p$, the martingale r^{X_t} gives $\mathbb{P}(\text{hit } N) = (1 - r^a)/(1 - r^N)$. Taking $r \to 1$ recovers a/N.

Interviewer variations. (i) Asymmetric barriers $[-b, a]$ in a Brownian setting. (ii) Why finite expected duration? Recurrent walk, absorbing barriers. (iii) Kelly connection: in a biased game, the gambler's-ruin formula justifies sizing below Kelly to control ruin probability (Chapter 20).

Problem 9: Itô's lemma on W_t^2 and e^{W_t}

Problem statement. Use Itô's lemma to derive the SDEs for W_t^2 and e^{W_t}, and identify an explicit martingale built from each.

First attempt. "By the chain rule, $d(W^2) = 2W\,dW$ and $d(e^W) = e^W\,dW$."

What goes wrong. That is the ordinary chain rule. Itô's lemma adds $\frac{1}{2}f''(W_t)\,dt$ from the quadratic variation of Brownian motion, since $(dW)^2 = dt$. Forgetting this is the definitional error of stochastic calculus.

Correct approach. Apply Itô's lemma $df(W_t, t) = f_t\,dt + f_W\,dW_t + \frac{1}{2}f_{WW}\,dt$.

Case 1: $f(W) = W^2$. Here $f_W = 2W$, $f_{WW} = 2$, so

$$d(W_t^2) = 2W_t\,dW_t + dt.$$

Integrating, $W_t^2 - t = 2\int_0^t W_s\,dW_s$, the stochastic-integral form of the Lévy martingale. Check: $\mathbb{E}[W_t^2 - t] = 0$, and increments $W_t^2 - W_s^2 - (t - s)$ have zero conditional mean given $\mathcal{F}_s$.

Case 2: $f(W) = e^W$. Here $f_W = f_{WW} = e^W$, so

$$d(e^{W_t}) = e^{W_t}\,dW_t + \tfrac{1}{2}e^{W_t}\,dt.$$

The drift is nonzero, so e^{W_t} is *not* a martingale. To kill the drift, multiply by $e^{-t/2}$: Itô's product rule gives

$$d(e^{W_t - t/2}) = e^{W_t - t/2}\,dW_t,$$

which is a pure stochastic integral and therefore a martingale. This is the *exponential martingale*, the building block of risk-neutral pricing.

Interviewer variations. (i) Apply Itô to W_t^3: $d(W^3) = 3W^2dW + 3W\,dt$, and $W_t^3 - 3tW_t$ is a martingale. (ii) For geometric Brownian motion $S_t = S_0e^{(\mu-\sigma^2/2)t+\sigma W_t}$, show $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$. (iii) Use the exponential martingale as the Radon–Nikodym density in Girsanov's theorem.

Itô lemma and two canonical martingales

$$W_t^2 - t \quad \text{and} \quad e^{W_t - t/2}$$

are martingales: the first from $d(W^2) = 2W\,dW + dt$, the second from the drift correction $\frac{1}{2}e^W\,dt$. These two drive every derivation in Chapter 16.

39.3 Linear algebra

Problem 10: the PCA question

Problem statement. You have n vectors of returns $x_1, \dots, x_n \in \mathbb{R}^d$, one per day, stacked into a matrix $X \in \mathbb{R}^{n\times d}$. Explain what principal component analysis does, derive the first principal component, and say what the eigenvalues mean.

First attempt. "PCA finds the top eigenvectors of the covariance matrix." True but not a derivation.

What goes wrong. Quoting the answer fails the test. The real question: how do you go from *maximise variance* to *top eigenvector*?

Correct approach. Assume data are centred: column means of X are zero. The sample covariance matrix is $\Sigma = \frac{1}{n-1}X^\top X$. We seek a unit vector $w \in \mathbb{R}^d$ that maximises the variance of the projection Xw:

$$\max_w \; w^\top \Sigma w \quad \text{subject to} \quad \|w\| = 1.$$

Lagrangian: $\mathcal{L} = w^\top \Sigma w - \lambda(w^\top w - 1)$. Setting $\nabla_w \mathcal{L} = 0$ gives $\Sigma w = \lambda w$, so w is an eigenvector of Σ and λ is the eigenvalue. Among all eigenvectors, the one maximising $w^\top \Sigma w = \lambda$ is the top one. Subsequent components are obtained by maximising variance subject to orthogonality to all prior components, which yields the next eigenvector in order.

The k-th eigenvalue is the variance along the k-th principal component; total variance is $\operatorname{tr}(\Sigma) = \sum_k \lambda_k$, so the variance explained by the first K components is $\sum_{k=1}^K \lambda_k / \sum_{k=1}^d \lambda_k$. Numerically, PCA is done via SVD $X = USV^\top$: principal directions are columns of V, squared singular values are scaled eigenvalues.

Interviewer variations. (i) Different feature scales? Standardise first, or large-unit features dominate. (ii) When does PCA fail? When variance is a poor proxy for signal — rare important risk factors may have low variance, noise may have high variance without predictive content. (iii) Factor models: top PCs of a rates curve are level, slope, curvature (Chapter 30).

Problem 11: eigenvalues of a rank-1 update

Problem statement. Let $u \in \mathbb{R}^n$ be a nonzero vector. What are the eigenvalues of the $n \times n$ matrix $I + uu^\top$?

First attempt. "Expand the characteristic polynomial." Feasible for small n but does not generalise.

What goes wrong. Determinant expansion is $O(n!)$ and hides the structure. The problem is trivial once you notice the 2D structure.

Correct approach. Split $\mathbb{R}^n$ into $\operatorname{span}(u)$ and its orthogonal complement $u^\perp$. For any $v \in u^\perp$, $(I + uu^\top)v = v$, so every vector in the $(n-1)$-dimensional space $u^\perp$ is an eigenvector with eigenvalue 1. For $v = u$, $(I + uu^\top)u = u + u(u^\top u) = (1 + \|u\|^2)u$, so u is an eigenvector with eigenvalue $1 + \|u\|^2$. The spectrum is therefore

$$\{\underbrace{1, 1, \ldots, 1}_{n-1}, 1 + \|u\|^2\}.$$

Check the trace: $\sum \lambda_i = (n-1) + (1 + \|u\|^2) = n + \|u\|^2 = \operatorname{tr}(I) + u^\top u = \operatorname{tr}(I + uu^\top)$. ✓

For $u = (1, 2, 3)^\top$, $\|u\|^2 = 14$, eigenvalues are $\{1, 1, 15\}$, sum 17 equals trace.

Interviewer variations. (i) "Eigenvalues of $A + uv^\top$ for general u, v (rank-1 update of general A)?" Use the matrix determinant lemma $\det(A + uv^\top) = \det(A)(1 + v^\top A^{-1}u)$. (ii) "Sherman–Morrison formula for $(A + uv^\top)^{-1}$?" $(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}$. (iii) "How does this relate to Kalman filters?" Kalman updates covariance by rank-1 additions; Sherman–Morrison is the $O(n^2)$ update path.

Intuition

$I + uu^\top$ stretches by $1 + \|u\|^2$ along u and leaves $u^\perp$ unchanged. See the geometry and the algebra writes itself — the template for every low-rank update problem.

39.4 Dynamic programming

Problem 12: longest common subsequence

Problem statement. Given two strings s of length m and t of length n, compute the length of their longest common subsequence (not substring). A subsequence preserves order but allows gaps.

First attempt. "Try every subset of s and test for occurrence in t." $O(2^m n)$; infeasible for $m = 50$.

What goes wrong. Exponential blow-up ignores the subproblem structure: the LCS of two prefixes depends only on LCSes of slightly shorter prefixes.

Correct approach. Let $L(i, j)$ denote the LCS length of $s[1..i]$ and $t[1..j]$. The recurrence is

$$L(i,j) = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0, \\ L(i-1, j-1) + 1 & \text{if } s_i = t_j, \\ \max(L(i-1,j), L(i,j-1)) & \text{otherwise.} \end{cases}$$

Filling the table bottom-up requires $O(nm)$ time and space. Reconstructing the actual subsequence requires a backtrace from (m, n) to $(0, 0)$.

For `ABCBDAB` and `BDCAB`, LCS length is 4 (e.g., `BCAB`). The Python below computes it in eleven lines.

```python
def lcs(s, t):
    m, n = len(s), len(t)
    dp = [[0] * (n + 1) for _ in range(m + 1)]
    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if s[i-1] == t[j-1]:
                dp[i][j] = dp[i-1][j-1] + 1
            else:
                dp[i][j] = max(dp[i-1][j], dp[i][j-1])
    return dp[m][n]

print(lcs("ABCBDAB", "BDCAB"))    # 4
```

Listing 39.1: Longest common subsequence by dynamic programming.

Interviewer variations. (i) Reduce space to $O(\min(m, n))$ by keeping two rows. (ii) Longest *common substring*: reset to zero on mismatch, answer is max cell, not $L(m, n)$. (iii) Edit distance (Levenshtein): 3 operations, similar $O(nm)$ DP. (iv) LCS on compressed inputs with $O(\log)$-alphabet is NP-hard.

Problem 13: 0/1 knapsack

Problem statement. Given items with weights w_i and values v_i for $i = 1, \dots, n$, and a knapsack of capacity W, choose a subset maximising total value subject to total weight $\le W$. Each item is either taken or left (no fractions).

First attempt. "Take items in decreasing value per unit weight." This is the *fractional* knapsack greedy; for 0/1 it is a heuristic, not optimal.

What goes wrong. Greedy by ratio may miss a combination that trades off smaller, worse-ratio items for a better fit. Counter-example: weights $[2, 3, 4, 5]$, values $[3, 4, 5, 6]$, capacity $W = 5$. Greedy by ratio picks item 1 (ratio 1.5) then item 3 (ratio 1.25) for weight 6, which exceeds capacity. The optimum picks items 1 and 2 (weight 5, value 7).

Correct approach. Let $K(i, w)$ denote the maximum value using a subset of the first i items with total weight $\le w$. Recurrence:

$$K(i, w) = \begin{cases} K(i-1, w) & \text{if } w_i > w, \\ \max\big(K(i-1, w),\ K(i-1, w - w_i) + v_i\big) & \text{otherwise.} \end{cases}$$

Base case $K(0, w) = 0$. Answer $K(n, W)$. Complexity $O(nW)$ time and space — *pseudopolynomial* because W is encoded in $\log W$ bits; the problem is NP-hard when W is allowed to grow exponentially in the input size.

```python
def knapsack(weights, values, W):
    n = len(weights)
    dp = [[0] * (W + 1) for _ in range(n + 1)]
    for i in range(1, n + 1):
        for w in range(W + 1):
            dp[i][w] = dp[i-1][w]
            if weights[i-1] <= w:
                take = dp[i-1][w - weights[i-1]] + values[i-1]
                dp[i][w] = max(dp[i][w], take)
    return dp[n][W]

print(knapsack([2,3,4,5], [3,4,5,6], 5))    # 7
```

Listing 39.2: 0/1 knapsack by dynamic programming.

Interviewer variations. (i) Unbounded knapsack: $K(i, w) = \max(K(i-1, w), K(i, w - w_i) + v_i)$. (ii) Fractional knapsack: greedy is optimal. (iii) Subset-sum: a special case. (iv) FPTAS via scaled values.

title

$O(nW)$ sounds polynomial but is exponential in the input bit-length since W is written in $\log W$ bits. For $n = 100$, $W = 10^{12}$, the table has 10^{14} cells. Under large W: branch-and-bound, FPTAS, or a problem-specific reformulation.

39.5 Options and market intuition

Problem 14: derive $\Delta = \Phi(d_1)$

Problem statement. Under Black–Scholes, the price of a European call is $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$ where $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)T]/(\sigma\sqrt{T})$. Derive the delta $\Delta = \partial C/\partial S$.

First attempt. "C is linear in S through the first term, so $\Delta = \Phi(d_1)$." Right answer, incomplete derivation: the Φ arguments also depend on S.

What goes wrong. Candidates miss that d_1, d_2 are functions of S. The result survives only via a non-trivial cancellation.

Correct approach. Both d_1 and d_2 depend on S through $\ln S$ with $\partial d_1/\partial S = \partial d_2/\partial S = 1/(S\sigma\sqrt{T})$. Differentiating,

$$\begin{aligned}\frac{\partial C}{\partial S} &= \Phi(d_1) + S\phi(d_1)\cdot\frac{1}{S\sigma\sqrt{T}} - Ke^{-rT}\phi(d_2)\cdot\frac{1}{S\sigma\sqrt{T}}\\ &= \Phi(d_1) + \frac{1}{\sigma\sqrt{T}}\left[\phi(d_1) - \frac{Ke^{-rT}}{S}\phi(d_2)\right].\end{aligned}$$

The square bracket vanishes by the identity $S\phi(d_1) = Ke^{-rT}\phi(d_2)$, which is a direct consequence of $d_1^2 - d_2^2 = 2\ln(S/K) + 2rT$ and the form of the standard normal density. (Plugging $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ into the ratio $\phi(d_1)/\phi(d_2) = e^{-(d_1^2-d_2^2)/2} = e^{-\ln(S/K)-rT} = (K/S)e^{-rT}$ gives the identity directly.) So $\Delta = \Phi(d_1)$.

Numerical check: $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$. $d_1 = (0.05 + 0.02)/0.2 = 0.35$, $\Phi(0.35) \approx 0.6368$. Call price ≈ 10.45.

Interviewer variations. (i) "Derive Gamma." Answer $\Gamma = \phi(d_1)/(S\sigma\sqrt{T})$. (ii) "Vega." Vega $= S\phi(d_1)\sqrt{T}$. (iii) "Put delta." $\Delta_P = \Phi(d_1) - 1$ via put–call parity. (iv) "Why is delta also the risk-neutral probability of expiring in the money?" It is not exactly — that is $\Phi(d_2)$. $\Phi(d_1)$ is the probability in a different measure (the stock numeraire). Confusing d_1 and d_2 here is a common pitfall.

title

$\Phi(d_2)$ is the risk-neutral probability the call finishes in the money. $\Phi(d_1) = \Delta$ is the same probability under the stock-as-numeraire measure. They differ by $\sigma\sqrt{T}$ in the argument. Interviewers use this to test numeraire change, not formula recall.

Problem 15: straddle P&L under an implied-realised mismatch

Problem statement. An ATM straddle (long one call + one put, both struck at the spot) on a $S = 100$ underlying has one year to expiry. You pay Black–Scholes prices using an implied volatility of $\sigma_{\text{imp}} = 20\%$. The realised volatility over the next year turns out to be $\sigma_{\text{real}} = 80\%$. Estimate the expected P&L.

First attempt. "Realised vol is four times implied, so payoff is four times cost." The ratio is roughly right but cost and payoff sit under different functions of σ.

What goes wrong. Under GBM, $\mathbb{E}|S_T - S_0| \approx S_0\sigma\sqrt{T}\sqrt{2/\pi}$ — linear in realised volatility — while the straddle price is $\approx S_0\sigma_{\text{imp}}\sqrt{T}\sqrt{2/\pi}$, also linear. The payoff scales as $\sigma_{\text{real}}/\sigma_{\text{imp}}$ but P&L is payoff minus cost.

Correct approach. Under GBM with zero rates and zero drift for simplicity, the expected straddle payoff is

$$\mathbb{E}[|S_T - K|] = S_0\sigma_{\text{real}}\sqrt{T/(2\pi)}\cdot 2 = S_0\sigma_{\text{real}}\sqrt{T}\sqrt{2/\pi}$$

(the factor 2 comes from either side of the strike). The ATM straddle price at implied vol σ_{imp} is approximately $2 \cdot S_0\sigma_{\text{imp}}\sqrt{T/(2\pi)} = S_0\sigma_{\text{imp}}\sqrt{T}\sqrt{2/\pi}$ (using the ATM approximation $C \approx 0.4S_0\sigma\sqrt{T}$). Expected P&L is therefore

$$\mathbb{E}[\text{P\&L}] \approx S_0\sqrt{T}\sqrt{2/\pi}(\sigma_{\text{real}} - \sigma_{\text{imp}}).$$

With $S_0 = 100$, $T = 1$, $\sigma_{\text{real}} = 0.8$, $\sigma_{\text{imp}} = 0.2$: $100 \cdot 0.7979 \cdot 0.6 \approx 47.87$. Expected payoff ≈ 63.8, cost ≈ 15.96, P&L ≈ 47.9 per straddle. A $4\times$ vol mismatch gives a roughly $4\times$ P&L because the cost is a small fraction of the realised payoff.

Interviewer variations. (i) "What if realised is only 30%?" P&L ≈ 7.98, still positive but modest. (ii) "What is the *variance* of P&L?" Large: the straddle payoff is heavy-tailed; in 2020 March VIX spikes the ratio exceeded 10x for weekly straddles. (iii) "Why would anyone sell a straddle?" The seller is short gamma and collects theta; profits when realised < implied. (iv) "Delta hedge the straddle — what is the P&L?" The classic "gamma P&L" identity: daily P&L $\approx \frac{1}{2}\Gamma S^2[(\Delta S/S)^2 - \sigma_{\text{imp}}^2\Delta t]$, summed over the life of the option.

Intuition

Option P&L is the integral of squared returns minus squared implied. Realised 80% vs implied 20% is a 16x difference in squared volatility — enormous for a long straddle. The intuition "vol is mispriced by 4x" is right; gamma P&L accrues at the difference of *squared* vols.

39.6 Meta-patterns across the fifteen problems

The problems span probability, stochastic processes, linear algebra, DP, and pricing but share four moves. Re-read this section after a first pass: the pattern names are the vocabulary you will use to think aloud.

Pattern 1: change of variables and dimensional analysis. Pass to log variables for GBM (Problem 9), to squared levels for variance martingales (Problem 8), to units of a^2 or $\sigma^2 T$ for hitting times (Problem 6). The right change of variables linearises a nonlinear problem.

Pattern 2: look for a martingale first. Gambler's ruin (Problem 8) and Brownian hitting times (Problem 6) reduce to optional stopping on X_t or $X_t^2 - t$; Itô (Problem 9) produces the two canonical martingales. Spot the symmetry, write the martingale, check integrability, apply optional stopping.

Pattern 3: bound before compute. Before computing $\mathbb{E}[T_a]$, ask finite or infinite. Before enumerating the LCS table, ask min and max possible answers. In the two-children problem, three outcomes with at least one boy give an instant upper bound of $1/3$. Bounding first catches algebra errors and focuses the derivation.

Pattern 4: recurrence to closed form. Every DP problem (LCS, knapsack) and random walk problem (coupon collector, secretary, ruin) has a recurrence. Seek closed form when possible (coupon $\to nH_n$; ruin $\to a/N$); accept the $O(nm)$ table when not (LCS, knapsack). Knowing which you are in — and asking — often decides the interview.

Four meta-patterns

Every Green-Book-style quant problem yields to one of:

1. **Change of variables** (log, squared, dimensional).
2. **Martingale first** (symmetry, optional stopping).
3. **Bound before compute** (finite vs infinite, min vs max).
4. **Recurrence to closed form** (or an $O(nm)$ table).

If none fires, the problem is a direct consequence of a definition not yet invoked; re-read the statement for the name of a distribution, martingale, or recurrence.

39.7 How to practise and what to expect

The routine used by the Frankfurt Hedgehogs (Frankfurt Hedgehogs, 2025b) and other top Prosperity finishers:

- Work through the entire Green Book (Zhou, 2020) in *writing*, not in your head. Budget about a month.
- For each problem write the five-field template: problem / first attempt / what goes wrong / correct / variations. Writing the five fields reveals whether you understood or merely copied.
- Tag each problem with its meta-pattern. The distribution exposes weak areas: 80% "change of variables" and no "martingale" tells you where to focus.
- Do mock interviews out loud. Whiteboard under time pressure is a different skill from written prep; a partner asking follow-ups exposes the gaps the real interview will.

Prosperity skills are interview skills

The manual rounds of IMC Prosperity (Chapter 26) are a live, adversarial quant interview: pattern classification of an unfamiliar product, mental arithmetic under uncertainty, order-of-magnitude estimation, Kelly sizing. Top-ten teams typically treat manual rounds as interview practice, with one teammate reading and another thinking aloud. A candidate who has done five Prosperity manual rounds handles a "how much would you pay for this coin-flip game" question with visible ease.

On a Goldman Sachs desk

The Goldman Strats interview pipeline, 2024–2025 cycle (Engineering, off-cycle):

- **CV screen**, then HireVue behavioural video.
- **First technical phone screen**: 45–60 min, one probability and one coding question. Typical: Monty Hall, two-children, derive BS delta; code LCS or knapsack.
- **Superday**: three or four 45-minute technical rounds with Strats; one is sometimes a 30-min desk shadow followed by a live pricing or hedging question. At least one round tests stochastic calculus at the level of Chapter 16; at least one tests mental arithmetic and market intuition (Problem 15 style).

- **Fit / behavioural**: final round with a senior Strat or MD.

The Green Book (Zhou, 2020) covers ∼70% of the probability/stochastic content. The rest splits between numerical problems (Chapter 35), linear algebra, and desk/product questions (Chapter 28). The Green Book alone is necessary but not sufficient.

The Green Book

Xinfeng Zhou's *A Practical Guide to Quantitative Finance Interviews* appeared in 2008; Zhou, then a hedge-fund quant, compiled questions he had asked, been asked, and collected from colleagues. The 2020 second edition added ML, more stochastic calculus, and sharper options coverage. It remains the de facto reference for junior quant prep. A 2026 trading-floor interviewer almost certainly worked through it, and a candidate can occasionally turn that to advantage: "this is close to Green Book Problem 5.12 but with drift added" signals preparation without being obnoxious.

Key facts to memorise

- **Consistency over tricks.** The five-field template applies to every quant interview question; use it in every written practice session.
- **Monty Hall $2/3$; two children $1/3$ (ambiguous) vs $1/2$ (ordered).** The difference is what you condition on, not the words.
- **100 prisoners win with probability $1 - \sum_{51}^{100} 1/\ell \approx 0.312$.** Cycle-following is optimal.
- **Secretary $1/e \approx 0.37$.** Reject the first n/e, accept the next best-so-far.
- **Coupon collector $nH_n \approx n \ln n$.** The tail dominates.
- **Brownian T_a is a.s. finite, infinite mean, $T_a \overset{d}{=} a^2 T_1$.** Dimensional analysis gives the scaling.
- **OU half-life $\ln 2/\theta$.** Time constant $1/\theta$; half-life shorter by $\ln 2$.
- **Gambler's ruin (fair, start a, barrier N): win prob a/N, duration $a(N-a)$.** Two martingales X_t and $X_t^2 - t$ give both in two lines.
- **Itô martingales $W_t^2 - t$ and $e^{W_t - t/2}$.** The engines behind Black–Scholes.
- **PCA = max variance = top eigenvector of Σ; rank-1 update $I + uu^\top$ has spectrum $\{1, \ldots, 1, 1 + \|u\|^2\}$.**
- **LCS and knapsack are $O(nm)$ DP.** Knapsack's $O(nW)$ is pseudopolynomial, not polynomial.
- **$\Delta = \Phi(d_1)$.** Cancellation via $S\phi(d_1) = Ke^{-rT}\phi(d_2)$. Not $\Phi(d_2)$, the risk-neutral exercise probability.
- **Straddle expected P&L**: $S_0\sqrt{T}\sqrt{2/\pi}(\sigma_{\text{real}} - \sigma_{\text{imp}})$. At 80% vs 20%, $S_0 = 100$, $T = 1$: P&L ≈ 48.
- **Meta-patterns.** Change of variables, martingale first, bound before compute, recurrence to closed form — one fires on almost every quant interview problem.

Part IX

Appendices

Appendix A

Math Prerequisites Refresher

This appendix is a refresher, not a textbook. If any of the concepts below are unfamiliar, consult a dedicated source; the "Further reading" section at the end points to the standard references. The goal here is to collect, in one place, the mathematical tools the body of the book assumes — so that when Chapter 6 writes down a conditional expectation, Chapter 10 invokes an Ornstein–Uhlenbeck SDE, or Chapter 16 appeals to Itô's lemma, the reader can flip here and recover the definitions in a few seconds.

Nothing in this appendix is proved. Everything is stated precisely enough to use. The rigour level matches that of a graduate-level cheat sheet: state, don't derive.

How to use this appendix

Read nothing here in sequence. Skip entire sections if you know the material. Treat the appendix as a lookup table: when the body cites "Itô's lemma" or "Feynman–Kac" or "the spectral theorem for symmetric matrices," come here, read the one paragraph that states the result, and go back. If after reading the relevant paragraph the result still does not click, the further-reading list at the end of this appendix is your next stop — not a re-read of this appendix.

A.1 Probability

A probability model is a triple $(\Omega, \mathcal{F}, \mathbb{P})$. The *sample space* Ω is the set of possible outcomes of a random experiment. The *σ-algebra* $\mathcal{F}$ is a collection of subsets of Ω closed under complement and countable union; its elements are *events*. The *probability measure* $\mathbb{P} : \mathcal{F} \to [0,1]$ assigns a number to each event with $\mathbb{P}(\Omega) = 1$ and $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$ for disjoint A_i.

A *random variable* is a measurable map $X : \Omega \to \mathbb{R}$. Its *distribution* is the pushforward measure $\mathbb{P} \circ X^{-1}$ on $\mathbb{R}$; in practice one specifies it by a *cumulative distribution function* (CDF) $F_X(x) = \mathbb{P}(X \le x)$ or, when X is absolutely continuous, by a *probability density function* (PDF) f_X satisfying $F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$.

Expectation, variance, moments

The *expectation* (or mean) of X is

$$\mathbb{E}[X] = \int_\Omega X\, d\mathbb{P} = \int_\mathbb{R} x\, f_X(x)\, dx \quad \text{(continuous case)}.$$

Expectation is linear: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$. The *variance* measures spread:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

and the *standard deviation* is $\sigma_X = \sqrt{\mathrm{Var}(X)}$. Covariance and correlation of two variables are

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)], \qquad \mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y} \in [-1,1].$$

The *k-th moment* is $\mathbb{E}[X^k]$; the *k-th central moment* is $\mathbb{E}[(X-\mathbb{E}X)^k]$. Skewness and kurtosis are the standardised third and fourth central moments:

$$\mathrm{Skew}(X) = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}, \qquad \mathrm{Kurt}(X) = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4}.$$

A Normal distribution has skewness 0 and kurtosis 3; *excess kurtosis* is $\mathrm{Kurt}(X) - 3$. Financial return distributions (equities, crypto, most derivatives-linked series) have excess kurtosis that is strongly positive — "fat tails." A naive risk calculation that uses only mean and variance understates the probability of large moves; see Chapter 20.

The *moment generating function* (MGF) of X is $M_X(t) = \mathbb{E}[e^{tX}]$ when finite in a neighbourhood of 0. MGFs, when they exist, uniquely determine the distribution and linearise sums of independent variables: $M_{X+Y}(t) = M_X(t)M_Y(t)$ for independent X, Y. The *characteristic function* $\varphi_X(t) = \mathbb{E}[e^{itX}]$ always exists and plays the same role — it underpins the Fourier-transform option-pricing methods used in Chapter 18.

Linearity and variance identities

- $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$ always.
- $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$.
- $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.
- If X, Y are independent, $\mathrm{Cov}(X,Y) = 0$ and hence $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.
- Zero covariance does *not* imply independence; it does for jointly Gaussian variables.

Conditional probability and Bayes' rule

For events $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)}.$$

Events A, B are *independent* if $\mathbb{P}(A\cap B) = \mathbb{P}(A)\mathbb{P}(B)$, equivalently $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Bayes' rule inverts the conditioning:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)},$$

where $\{A_i\}$ is a partition of Ω. This is the engine behind the Glosten–Milgrom market-maker update in Chapter 6 and any informed-trader posterior.

Example. A coin is biased: $\mathbb{P}(\text{heads}) = 0.7$. We flip it three times; what is the probability that the first flip was heads given that exactly two of the three flips came up heads? Let $A = \{\text{first is heads}\}$ and $B = \{\text{exactly two heads}\}$. Then $\mathbb{P}(B) = \binom{3}{2}(0.7)^2(0.3) = 0.441$, while $\mathbb{P}(A\cap B) = \mathbb{P}(\text{first H, one of next two H}) = 0.7\cdot 2\cdot 0.7\cdot 0.3 = 0.294$. So $\mathbb{P}(A \mid B) = 0.294/0.441 \approx 0.667$. Knowing that two of three are heads has raised the posterior on the first flip from 0.7 to 0.667 — wait, it *lowered* it, because observing "only two heads" is mild evidence against the first being heads. Compute before intuiting.

Conditional expectation of X given a σ-algebra $\mathcal{G} \subseteq \mathcal{F}$ is the $\mathcal{G}$-measurable random variable $\mathbb{E}[X \mid \mathcal{G}]$ satisfying $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbf{1}_G] = \mathbb{E}[X\mathbf{1}_G]$ for all $G \in \mathcal{G}$. Two facts used repeatedly in the body:

- *Tower property*: $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$, and more generally $\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]$ when $\mathcal{G} \subseteq \mathcal{H}$.
- *Taking out what is known*: if Y is $\mathcal{G}$-measurable, $\mathbb{E}[YX \mid \mathcal{G}] = Y\,\mathbb{E}[X \mid \mathcal{G}]$.

A *martingale* (with respect to a filtration $(\mathcal{F}_t)$) is a process (M_t) with (i) M_t is $\mathcal{F}_t$-measurable, (ii) $\mathbb{E}[|M_t|] < \infty$, and (iii) $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for all $s \leq t$. A *submartingale* replaces (iii) with $\geq$; a *supermartingale* with $\leq$. Martingales are the natural model for "no drift conditional on information," hence their role in the fundamental theorem of asset pricing (Chapter 16).

Two martingale properties we rely on repeatedly:

- *Doob's optional stopping.* Under integrability conditions (for instance, if the stopping time τ is bounded), $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$. A strategy with bounded duration against a martingale price has expected payoff zero.
- *Brownian motion and driftless Itô integrals are martingales.* If X_t is adapted and square-integrable, then $\int_0^t X_s\,dW_s$ is a martingale. This is why the risk-neutral expectation $\mathbb{E}_{\mathbb{Q}}[\cdot]$ of a discounted payoff is itself the no-arbitrage price.

Inequalities

A handful of inequalities are used enough to deserve naming:

- *Markov:* for $X \geq 0$ and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$.
- *Chebyshev:* $\mathbb{P}(|X - \mu| \geq k\sigma) \leq 1/k^2$. (Markov applied to $(X-\mu)^2$.)
- *Cauchy–Schwarz:* $\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$. Equality iff $X = cY$ almost surely. This is what makes $|\operatorname{Corr}(X, Y)| \leq 1$.
- *Jensen:* for a convex function ϕ and integrable X, $\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$; for concave ϕ the inequality flips. Corollary: $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ (positive variance), and $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$ (geometric mean $\leq$ arithmetic mean, the mathematical origin of the volatility drag $-\frac{1}{2}\sigma^2$ in log-returns and a running theme in Chapter 16 and Chapter 20).

Intuition

Jensen is a statement about curvature. A convex-up function averages up over any spread of its argument, so taking the expectation first loses some upside relative to averaging the function values. For the concave logarithm, this is why log-returns have a drag: you cannot replace "expected return" with "expected log return" without losing $\frac{1}{2}\sigma^2$, even in the absence of any trading cost.

Common distributions

The distributions below are used throughout the book. The notation $X \sim D$ means "X has distribution D."

Distribution cheat sheet

- **Bernoulli**(p): $\mathbb{P}(X = 1) = p, \mathbb{P}(X = 0) = 1 - p$. Mean p, variance $p(1-p)$.
- **Binomial**(n, p): sum of n i.i.d. Bernoulli(p). Mean np, variance $np(1-p)$.
- **Poisson**(λ): $\mathbb{P}(X = k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$. Mean and variance both λ. Limit of Binomial$(n, \lambda/n)$ as $n \to \infty$; models independent arrivals in continuous time (order-book events, trade arrivals).

- **Exponential**(λ): density $\lambda e^{-\lambda x}$ for $x \geq 0$. Mean $1/\lambda$, variance $1/\lambda^2$. The inter-arrival time of a Poisson process. *Memoryless*: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$.
- **Normal** $\mathcal{N}(\mu, \sigma^2)$: density $\frac{1}{\sqrt{2\pi}\sigma} \exp\big(-(x-\mu)^2/(2\sigma^2)\big)$. Mean μ, variance σ^2. Sums of independent Normals are Normal. Standard normal CDF denoted Φ, PDF ϕ.
- **Log-Normal**: if $\log X \sim \mathcal{N}(\mu, \sigma^2)$ then $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$ and $\mathrm{Var}(X) = (e^{\sigma^2} - 1)\, e^{2\mu+\sigma^2}$. The Black–Scholes terminal stock price is log-normal (Chapter 16).

Modes of convergence

Four modes of convergence for a sequence of random variables X_n to a limit X matter in what follows. From strongest to weakest:

- *Almost sure:* $X_n \xrightarrow{\text{a.s.}} X$ if $\mathbb{P}(\{\omega : X_n(\omega) \to X(\omega)\}) = 1$. "All sample paths eventually settle."
- *In probability:* $X_n \xrightarrow{p} X$ if $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$. Weaker than almost-sure convergence (which implies it).
- *In L^p:* $X_n \xrightarrow{L^p} X$ if $\mathbb{E}[|X_n - X|^p] \to 0$. For $p = 2$ this is "mean-square convergence"; implies convergence in probability.
- *In distribution:* $X_n \xrightarrow{d} X$ if $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X. The weakest of the four.

The LLN below comes in two flavours: the *weak law* (convergence in probability) and the *strong law* (almost-sure convergence). The CLT is a statement about convergence in distribution.

The multivariate normal

A random vector $\boldsymbol{X} \in \mathbb{R}^n$ is *multivariate normal* with mean $\boldsymbol{\mu} \in \mathbb{R}^n$ and covariance $\Sigma \in \mathbb{R}^{n\times n}$ (PSD, often PD) if its density is

$$f(\boldsymbol{x}) = \frac{1}{(2\pi)^{n/2}\, \det(\Sigma)^{1/2}} \exp\Big(-\tfrac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\boldsymbol{x}-\boldsymbol{\mu})\Big).$$

We write $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$. Key properties:

- *Affine transforms stay normal:* if $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ and A is $m \times n$, $\boldsymbol{b} \in \mathbb{R}^m$, then $A\boldsymbol{X} + \boldsymbol{b} \sim \mathcal{N}(A\boldsymbol{\mu} + \boldsymbol{b}, A\Sigma A^T)$.
- *Marginals are normal*, with the appropriate sub-vector of $\boldsymbol{\mu}$ and sub-matrix of Σ.
- *Conditional distributions are normal.* Partitioning $\boldsymbol{X} = (\boldsymbol{X}_1, \boldsymbol{X}_2)^T$ with corresponding block structure $(\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$ and $\Sigma = \left(\begin{smallmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{smallmatrix}\right)$,

$$\boldsymbol{X}_1 \mid \boldsymbol{X}_2 = \boldsymbol{x}_2 \;\sim\; \mathcal{N}\Big(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\boldsymbol{x}_2 - \boldsymbol{\mu}_2),\; \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Big).$$

  This formula is the entirety of the Glosten–Milgrom conditional-update in Chapter 6, Kalman filtering for latent-state estimation, and factor-model residual decomposition.
- *Sampling.* To sample $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{0}, \Sigma)$, Cholesky-factor $\Sigma = LL^T$ and set $\boldsymbol{X} = L\boldsymbol{Z}$ for $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0}, I)$. The standard recipe for simulating correlated risk factors.

Limit theorems

Two statements dominate classical probability, and both underpin backtesting and Monte Carlo in this book.

Law of Large Numbers. Let $X_1, X_2, \ldots$ be i.i.d. with finite mean $\mu = \mathbb{E}[X_1]$. Then the sample average converges to the mean:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\text{a.s.}} \mu \quad \text{as } n \to \infty.$$

Central Limit Theorem. With the same assumptions and additionally $\mathrm{Var}(X_1) = \sigma^2 < \infty$,

$$\sqrt{n}\,(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2) \quad \text{as } n \to \infty.$$

In words: sample averages are approximately Normal, with standard error $\sigma/\sqrt{n}$.

Rate of convergence in CLT

The CLT gives the asymptotic *scaling* — $1/\sqrt{n}$ — but not a distribution at finite n. The Berry–Esseen theorem sharpens this: the Kolmogorov distance between the CDF of $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ and the standard normal is bounded by $C\mathbb{E}[|X_1 - \mu|^3]/(\sigma^3\sqrt{n})$, with C a universal constant (≤ 0.4748 in the i.i.d. case). For backtest Sharpe-ratio inference with daily returns and $n \gtrsim 500$ observations, the Gaussian approximation is usually adequate; for intraday strategies with hundreds of thousands of trades it is almost exact.

Finite-variance assumption matters

The CLT requires $\sigma^2 < \infty$. Financial returns often exhibit heavy tails (kurtosis $\gg 3$); in the extreme (Pareto tail index $\alpha < 2$) the variance is infinite and the CLT fails — sample averages converge to a non-Gaussian stable law, or do not converge at all. Backtests that assume Gaussian standard errors on Sharpe ratios (Chapter 20) silently mis-size their confidence intervals when returns are that heavy-tailed.

A.2 Linear algebra

Vectors $\boldsymbol{x} \in \mathbb{R}^n$ are columns; matrices $A \in \mathbb{R}^{m\times n}$ act on them by left multiplication. The *inner product* is $\langle \boldsymbol{x}, \boldsymbol{y}\rangle = \boldsymbol{x}^T\boldsymbol{y} = \sum_i x_i y_i$ and the Euclidean norm is $\|\boldsymbol{x}\| = \sqrt{\langle \boldsymbol{x}, \boldsymbol{x}\rangle}$.

Matrix arithmetic

Matrix multiplication is associative but not commutative: in general $AB \neq BA$. The *transpose* A^T flips rows and columns; $(AB)^T = B^T A^T$ and $(AB)^{-1} = B^{-1}A^{-1}$ when invertible. The *trace* $\mathrm{tr}(A) = \sum_i A_{ii}$ is linear and cyclic: $\mathrm{tr}(AB) = \mathrm{tr}(BA)$, $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$.

Three useful factorisations beyond eigen/SVD:

- **LU**: $PA = LU$ where P is a permutation, L lower triangular (unit diagonal), U upper triangular. Standard general-purpose linear-system solver. $\mathcal{O}(n^3)$ to compute, $\mathcal{O}(n^2)$ to back-substitute.
- **Cholesky**: $A = LL^T$ when A is symmetric PD. About twice as fast as LU and numerically stable. The default for covariance matrices and normal equations.
- **QR**: $A = QR$ where Q has orthonormal columns and R is upper triangular. The preferred way to solve least-squares — avoids forming the ill-conditioned A^TA.

Rank, inverse, determinant

The *rank* of A is the dimension of its column space (equivalently its row space). $A \in \mathbb{R}^{n\times n}$ is *invertible* iff $\operatorname{rank}(A) = n$, iff $\det(A) \neq 0$, iff $A\boldsymbol{x} = \mathbf{0}$ has only the trivial solution.

The *determinant* satisfies $\det(AB) = \det(A)\det(B)$ and $\det(A^{-1}) = 1/\det(A)$. Geometrically, $|\det(A)|$ is the factor by which A scales n-dimensional volume.

The *inverse*, when it exists, satisfies $AA^{-1} = A^{-1}A = I$. For 2×2 matrices,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Avoid inverting explicitly in code — solve the linear system $A\boldsymbol{x} = \boldsymbol{b}$ directly (LU, QR, or tridiagonal; see Chapter 35).

Eigenvalues and eigenvectors

A scalar $\lambda \in \mathbb{C}$ and non-zero $\boldsymbol{v} \in \mathbb{C}^n$ are an *eigenvalue–eigenvector* pair of $A \in \mathbb{R}^{n\times n}$ if $A\boldsymbol{v} = \lambda\boldsymbol{v}$. Equivalently, λ is a root of the *characteristic polynomial* $\det(A - \lambda I) = 0$. An $n \times n$ matrix has n eigenvalues counted with multiplicity.

Trace and determinant relations: $\operatorname{tr}(A) = \sum_i \lambda_i$ and $\det(A) = \prod_i \lambda_i$.

Spectral theorem for symmetric matrices

If $A \in \mathbb{R}^{n\times n}$ is symmetric ($A = A^T$), then:

1. all eigenvalues λ_i are real;
2. eigenvectors corresponding to distinct eigenvalues are orthogonal, and one can choose an orthonormal basis of eigenvectors;
3. consequently $A = Q\Lambda Q^T$ where Q is orthogonal ($Q^TQ = I$) and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

This is the *eigendecomposition*. It is the mathematical skeleton behind PCA, portfolio variance minimisation, and the asymptotic distribution of OU processes in Chapter 10.

Diagonalisation. If A has n linearly independent eigenvectors (always the case for symmetric A, sometimes for non-symmetric), then $A = PDP^{-1}$ with D diagonal. Powers are trivial: $A^k = PD^kP^{-1}$. Matrix exponentials are $e^A = Pe^DP^{-1}$, and e^D is just the diagonal of scalar exponentials. This is what makes the solution of a linear homogeneous ODE $\dot{\boldsymbol{x}} = A\boldsymbol{x}$ trivial in the diagonalisable case: $\boldsymbol{x}(t) = e^{At}\boldsymbol{x}(0) = Pe^{Dt}P^{-1}\boldsymbol{x}(0)$.

Worked example. Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The characteristic polynomial is $\det(A - \lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3$, with roots $\lambda_1 = 3$ and $\lambda_2 = 1$. Eigenvectors: for $\lambda = 3$, $(A - 3I)\boldsymbol{v} = 0$ gives $\boldsymbol{v} \propto (1,1)^T$; for $\lambda = 1$, $\boldsymbol{v} \propto (1,-1)^T$. After normalisation the orthogonal matrix of eigenvectors is $Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $A = Q\operatorname{diag}(3,1)Q^T$. Sanity check: $\operatorname{tr}(A) = 4 = 3 + 1$ and $\det(A) = 3 = 3 \cdot 1$.

A symmetric matrix A is *positive semi-definite* (PSD) if $\boldsymbol{x}^TA\boldsymbol{x} \geq 0$ for all $\boldsymbol{x}$, equivalently all eigenvalues ≥ 0. It is *positive definite* (PD) if the inequality is strict for $\boldsymbol{x} \neq \mathbf{0}$. Covariance matrices are always PSD. A PD matrix has a *Cholesky factorisation* $A = LL^T$ with L lower triangular — the default way to sample from a multivariate Normal $\mathcal{N}(\mathbf{0}, A)$ in code.

Singular value decomposition

For any $A \in \mathbb{R}^{m\times n}$ (not necessarily square), the *singular value decomposition* (SVD) is

$$A = U\Sigma V^T$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal, and $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal with non-negative entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ (the *singular values*), where $r = \text{rank}(A)$. The columns of U and V are the *left* and *right singular vectors.*

The SVD generalises eigendecomposition to rectangular matrices and to non-symmetric square matrices. Numerically it is the most stable way to diagnose near-singularity (look at the smallest singular value relative to the largest; the ratio σ_1/σ_r is the *condition number*). A condition number above, say, 10^8 in double precision is a strong signal that small perturbations of the inputs produce large perturbations of the outputs — a common failure mode when regressing almost-collinear features (two highly-correlated price series, the classic pairs-trading hazard of Chapter 11).

The SVD also gives the Moore–Penrose *pseudoinverse* $A^+ = V\Sigma^+U^T$, where Σ^+ is formed by inverting the non-zero singular values and transposing. The least-squares solution of $A\boldsymbol{x} = \boldsymbol{b}$ (the projection of $\boldsymbol{b}$ onto the column space of A, minimising $\|A\boldsymbol{x} - \boldsymbol{b}\|$) is $\boldsymbol{x}^* = A^+\boldsymbol{b}$. When A has full column rank, $A^+ = (A^TA)^{-1}A^T$ — the formula behind every ordinary-least-squares regression, including the hedge-ratio estimation in Chapter 11.

PCA as eigendecomposition of the covariance

Given data matrix $X \in \mathbb{R}^{n\times p}$ (rows are observations, columns are features) with column means subtracted, the sample covariance is $\hat{\Sigma} = \frac{1}{n-1}X^TX \in \mathbb{R}^{p\times p}$. $\hat{\Sigma}$ is symmetric PSD, so by the spectral theorem $\hat{\Sigma} = Q\Lambda Q^T$.

Principal component analysis (PCA) is: order the eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$; the k-th *principal component* is the direction $\boldsymbol{q}_k$ (the k-th column of Q); the variance explained by the k-th component is λ_k and the fraction of total variance is $\lambda_k/\sum_j \lambda_j$. Keeping the top $k < p$ components gives the best rank-k approximation to X in Frobenius norm (Eckart–Young).

Equivalently, the PCA directions are the right singular vectors of the (centred) data matrix X, and the singular values satisfy $\sigma_i^2 = (n-1)\lambda_i$.

Intuition

PCA rotates the feature axes to line up with the directions of greatest variance. In trading: run PCA on a panel of stock returns; the first component is typically “the market,” the second is “long one sector, short another,” and so on. The eigenvectors give the portfolio weights, the eigenvalues give the variance of each resulting portfolio. This is the foundation of statistical-arbitrage factor models in Chapter 11.

Scale matters for PCA

PCA on raw data is dominated by whichever feature has the largest variance, regardless of signal. Standardise (subtract mean, divide by standard deviation) each column before running PCA unless the features are already on a comparable scale. For returns panels this is usually automatic; for mixed-unit features (a price and a volume, say) it is mandatory.

A.3 Stochastic calculus sketch

This section is a one-page tour of the machinery used in Chapter 6, Chapter 10, Chapter 16, Chapter 19, and Chapter 35. For actual study, use Øksendal (2003) or Shreve (2004).

Brownian motion

A *standard Brownian motion* (or *Wiener process*) W_t is a continuous-time stochastic process with:

1. $W_0 = 0$;
2. *continuous paths* $t \mapsto W_t$ (almost surely);
3. *independent increments*: for $s < t$, $W_t - W_s$ is independent of $(W_u : u \le s)$;
4. *Gaussian increments*: $W_t - W_s \sim \mathcal{N}(0, t-s)$.

Brownian motion is the unique (up to scaling) process satisfying these properties. Its paths are almost surely nowhere differentiable, but they have a very precise second-order behaviour, quantified by quadratic variation.

Quadratic variation and the rule $(dW)^2 = dt$

The *quadratic variation* of W on $[0, t]$ is

$$[W]_t \;=\; \lim_{\|\pi\|\to 0} \sum_k (W_{t_{k+1}} - W_{t_k})^2 \;=\; t$$

almost surely, where the limit is over partitions $\pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ with mesh $\|\pi\| \to 0$. This is the content of the heuristic

$$(dW_t)^2 = dt, \qquad dW_t\, dt = 0, \qquad (dt)^2 = 0,$$

which is the arithmetic rule that makes all of stochastic calculus "work." Squared Brownian increments do not vanish in the limit the way $(dt)^2$ does — they accumulate deterministically to dt.

Itô's lemma

For a smooth function $f(W_t, t)$, the ordinary chain rule would give $df = \partial_t f\, dt + \partial_W f\, dW_t$. The correction is the $(dW)^2 = dt$ term of quadratic variation:

Itô's lemma (scalar form)

For $f \in C^{1,2}(\mathbb{R} \times [0, T])$ and W_t a standard Brownian motion,

$$df(W_t, t) \;=\; \partial_t f(W_t, t)\, dt \;+\; \partial_W f(W_t, t)\, dW_t \;+\; \tfrac{1}{2}\, \partial_{WW} f(W_t, t)\, dt.$$

More generally, if X_t satisfies $dX_t = \mu_t\, dt + \sigma_t\, dW_t$, then for $f(X_t, t)$,

$$df = \Big(\partial_t f + \mu_t\, \partial_x f + \tfrac{1}{2}\sigma_t^2\, \partial_{xx} f\Big)\, dt \;+\; \sigma_t\, \partial_x f\, dW_t.$$

The extra $\frac{1}{2}\partial_{xx} f$ term is the *Itô correction*: it comes from the fact that W has non-trivial second-order variation. For deterministic x, it vanishes ($\sigma_t \equiv 0$) and one recovers ordinary calculus.

Stochastic differential equations

A *stochastic differential equation* (SDE) is an equation of the form

$$dX_t = \mu(X_t, t)\, dt + \sigma(X_t, t)\, dW_t, \qquad X_0 = x_0,$$

interpreted as the integral equation

$$X_t = x_0 + \int_0^t \mu(X_s, s)\, ds + \int_0^t \sigma(X_s, s)\, dW_s,$$

where the second integral is an *Itô stochastic integral*. Under mild Lipschitz and linear-growth conditions on μ and σ, the SDE admits a unique strong solution.

An SDE is a probabilistic generalisation of an ODE in which the velocity field acquires a random component. If $\sigma \equiv 0$, the SDE reduces to the ODE $\dot{X}_t = \mu(X_t, t)$. The new ingredient is that the Brownian differential dW_t is $\mathcal{O}(\sqrt{dt})$, not $\mathcal{O}(dt)$ — which is why the second-order term in Itô's lemma is not negligible.

Multi-dimensional Itô. For $X_t \in \mathbb{R}^n$ with $dX_t = \boldsymbol{\mu}(X_t, t)\, dt + \Sigma(X_t, t)\, dW_t$ (here $W_t \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{n\times d}$), and $f(\boldsymbol{x}, t) \in \mathbb{R}$,

$$df = \Big(\partial_t f + \boldsymbol{\mu}^T \nabla f + \tfrac{1}{2}\operatorname{tr}(\Sigma\Sigma^T \nabla^2 f)\Big) dt + (\nabla f)^T \Sigma\, dW_t.$$

The generator becomes $Lf = \boldsymbol{\mu}^T \nabla f + \frac{1}{2}\operatorname{tr}(\Sigma\Sigma^T\nabla^2 f)$, a linear second-order elliptic operator; this is what appears in multi-asset Black–Scholes and in basket/spread options pricing (Chapter 12).

Three SDEs recur throughout the body:

- *Arithmetic Brownian motion* $dX_t = \mu\, dt + \sigma\, dW_t$. Solution $X_t = x_0 + \mu t + \sigma W_t$. Normal at every horizon. Used as a toy model and for latent fair values in Kyle-style microstructure (Chapter 6).
- *Geometric Brownian motion* $dS_t = \mu S_t\, dt + \sigma S_t\, dW_t$. By Itô applied to $f = \log S_t$, the solution is $S_t = S_0 \exp\big((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\big)$, which is log-normal. Black–Scholes assumes GBM (Chapter 16).
- *Ornstein–Uhlenbeck process* $dX_t = \theta(\mu - X_t)\, dt + \sigma\, dW_t$. Mean-reverts to μ with speed θ; stationary distribution $\mathcal{N}(\mu, \sigma^2/(2\theta))$; half-life $\log 2/\theta$. The workhorse model of Chapter 10 and Chapter 11.

Solving the OU SDE explicitly. Apply Itô to $f(X_t, t) = X_t e^{\theta t}$. The partial derivatives are $\partial_t f = \theta X_t e^{\theta t}$ and $\partial_x f = e^{\theta t}$, with $\partial_{xx} f = 0$. Hence

$$df = \theta X_t e^{\theta t} dt + e^{\theta t} dX_t = \theta X_t e^{\theta t} dt + e^{\theta t}\big[\theta(\mu - X_t)dt + \sigma dW_t\big] = \theta\mu e^{\theta t}\, dt + \sigma e^{\theta t}\, dW_t.$$

Integrating from 0 to t,

$$X_t e^{\theta t} - X_0 = \mu(e^{\theta t} - 1) + \sigma \int_0^t e^{\theta s}\, dW_s,$$

so

$$X_t = X_0 e^{-\theta t} + \mu(1 - e^{-\theta t}) + \sigma \int_0^t e^{-\theta(t-s)} dW_s.$$

The last term is a Gaussian with mean zero and variance $\sigma^2 \int_0^t e^{-2\theta(t-s)} ds = \sigma^2(1 - e^{-2\theta t})/(2\theta)$, which tends to $\sigma^2/(2\theta)$ as $t \to \infty$ — recovering the stationary distribution. This is the exact solution used to simulate OU processes without discretisation error in Chapter 10.

Feynman–Kac formula

The Feynman–Kac formula is the bridge between PDEs and expectations under a diffusion. It is what lets Black–Scholes be stated either as a PDE (Chapter 16, equation for option price) or as a discounted expectation under the risk-neutral measure — both views are the same equation read in two directions.

Feynman–Kac (statement only)

Let X_t satisfy $dX_t = \mu(X_t, t)\, dt + \sigma(X_t, t)\, dW_t$, let L be the infinitesimal generator

$$L = \mu(x,t)\, \partial_x + \tfrac{1}{2}\sigma(x,t)^2\, \partial_{xx},$$

and let $r(x,t), g(x)$ be (bounded, sufficiently smooth) functions. If $u(x,t)$ solves the backward PDE

$$\partial_t u(x,t) + Lu(x,t) - r(x,t)\, u(x,t) = 0, \qquad u(x,T) = g(x),$$

then, subject to standard integrability conditions,

$$u(x,t) \;=\; \mathbb{E}\Big[\exp\Big(-\int_t^T r(X_s, s)\, ds\Big)\, g(X_T) \;\Big|\; X_t = x\Big].$$

The takeaway: *backward PDEs with terminal conditions are expectations of terminal payoffs under the diffusion, optionally discounted.* The Black–Scholes PDE with payoff $g(x) = (x - K)^+$ and discount r is exactly such an expectation under the risk-neutral dynamics — giving the closed-form call price of Chapter 16.

Girsanov and change of measure

Pricing under the risk-neutral measure $\mathbb{Q}$ (Chapter 16) requires a tool to move between the physical measure $\mathbb{P}$ (under which the stock drifts at some real-world rate μ) and $\mathbb{Q}$ (under which it drifts at the risk-free rate r).

Girsanov's theorem, in its most-used form, says: let W_t be a $\mathbb{P}$-Brownian motion and θ_t a suitably integrable adapted process. Define

$$Z_t = \exp\Big(-\int_0^t \theta_s\, dW_s - \tfrac{1}{2}\int_0^t \theta_s^2\, ds\Big),$$

and a new measure $\mathbb{Q}$ by $d\mathbb{Q}/d\mathbb{P} = Z_T$. Then $\widetilde{W}_t = W_t + \int_0^t \theta_s\, ds$ is a $\mathbb{Q}$-Brownian motion. In other words: changing the drift of a diffusion by θ_t is equivalent to reweighting the probability measure by Z_t.

The practical content: under the risk-neutral measure $\mathbb{Q}$, all discounted tradeable asset prices are martingales. The *fundamental theorem of asset pricing* states that absence of arbitrage is equivalent to the existence of such a measure; completeness of the market is equivalent to its uniqueness. Everything that follows in Black–Scholes, Heston, local vol, and the LIBOR Market Model is Girsanov applied cleverly.

$\mathbb{P}$ versus $\mathbb{Q}$

Parameters calibrated from historical data (realised vol, realised drift) live under $\mathbb{P}$; parameters backed out from option prices (implied vol, implied forward) live under $\mathbb{Q}$. You cannot mix them in a single formula without explicitly writing down the Radon–Nikodym derivative. A common error in junior-strat code is to price an option with a risk-neutral BS formula while plugging in the historical drift of the underlying — the wrong measure, and the result is neither a fair value nor a forecast.

A.4 Stochastic control sketch

Stochastic control asks: given a controlled diffusion and a cost or reward over $[0, T]$, what control α_t (chosen dynamically, using only information available at time t) maximises expected reward? This is the framework of Chapter 9 (Avellaneda–Stoikov market-making) and Chapter 19 (optimal execution).

Setup and the value function

Given a controlled SDE

$$dX_t = \mu(X_t, \alpha_t)\, dt + \sigma(X_t, \alpha_t)\, dW_t,$$

a running reward $r(x, \alpha)$, and a terminal reward $g(x)$, define the *value function*

$$V(x,t) \;=\; \sup_{\alpha} \,\mathbb{E}\left[\int_t^T r(X_s, \alpha_s)\, ds + g(X_T) \,\Big|\, X_t = x\right],$$

where the supremum is over *admissible* (progressively measurable, integrable) controls. $V(x,t)$ is "the best expected future reward achievable starting from state x at time t and playing optimally from there."

The guiding principle is *dynamic programming*: an optimal trajectory is optimal over any sub-interval. Formally (Bellman principle), for any stopping time $\tau \in [t, T]$,

$$V(x,t) = \sup_{\alpha} \mathbb{E}\left[\int_t^{\tau} r\, ds + V(X_\tau, \tau) \,\Big|\, X_t = x\right].$$

Hamilton–Jacobi–Bellman equation

Passing Bellman's principle to the infinitesimal limit $\tau \downarrow t$ and applying Itô gives the *Hamilton–Jacobi–Bellman* (HJB) partial differential equation:

HJB equation (statement only)

Under standard regularity assumptions, the value function $V(x,t)$ satisfies

$$\partial_t V(x,t) + \sup_{\alpha} \Big\{ L^{\alpha} V(x,t) + r(x,\alpha) \Big\} = 0, \qquad V(x,T) = g(x),$$

where L^{α} is the generator of the controlled diffusion under control α:

$$L^{\alpha} f = \mu(x,\alpha)\, \partial_x f + \tfrac{1}{2}\sigma(x,\alpha)^2\, \partial_{xx} f.$$

The optimal feedback control at (x,t) is any $\alpha^*(x,t)$ achieving the supremum inside the braces.

HJB is the stochastic-control analogue of Bellman's equation in discrete-time dynamic programming. Solving it — either in closed form (Avellaneda–Stoikov, Almgren–Chriss, Merton's portfolio problem), by reduction to a linear PDE via a clever transformation, or numerically on a grid — is how we pin down optimal quoting (Chapter 9) and optimal execution (Chapter 19).

A first-pass reader should leave this section with three facts: (i) the value function captures best-attainable future reward; (ii) HJB is a backward PDE with a sup over the control inside; (iii) the optimal control is whichever argument attains that sup. The body of the book applies HJB to two cases where it can be solved in closed form and one where it is solved numerically on a finite grid.

A tiny worked HJB: Merton's portfolio problem

To make the machinery concrete, consider Merton's canonical problem: invest wealth X_t in a risky asset (GBM with drift μ, vol σ) and a risk-free asset (rate r), with control α_t = fraction of wealth in the risky asset. Consume nothing; maximise expected utility of terminal wealth $U(X_T) = X_T^{1-\gamma}/(1-\gamma)$ (CRRA with coefficient $\gamma > 0$, $\gamma \neq 1$).

The wealth SDE is $dX_t = X_t[(r + \alpha_t(\mu - r))\,dt + \alpha_t\sigma\,dW_t]$. The HJB equation is

$$\partial_t V + \sup_{\alpha} \Big\{ x[r + \alpha(\mu - r)]\,\partial_x V + \tfrac{1}{2}\alpha^2\sigma^2 x^2 \partial_{xx} V \Big\} = 0.$$

Optimising over α (take $\partial/\partial\alpha$):

$$\alpha^*(x,t) = -\frac{(\mu - r)\,\partial_x V}{\sigma^2 x\,\partial_{xx} V}.$$

For CRRA utility the value function separates as $V(x,t) = f(t)\,x^{1-\gamma}/(1-\gamma)$, plugging in gives α^* constant:

$$\alpha^* = \frac{\mu - r}{\gamma\,\sigma^2}.$$

This is the *Merton ratio*, identical to the Kelly fraction of Chapter 20 up to the risk-aversion scaling γ. The log-utility limit $\gamma \to 1$ recovers full Kelly. The whole apparatus of Avellaneda–Stoikov market-making (Chapter 9) is an HJB in a richer state space (inventory plus cash), solved by the same ansatz-and-reduce technique.

A.5 Numerical tools

A handful of low-level numerical primitives appear throughout the derivatives-pricing and calibration parts of the book. Each is sketched here; full treatments with stability analysis and Python references are in Chapter 35.

Bisection finds a root of a continuous function on $[a, b]$ given $f(a)f(b) < 0$. Repeatedly halve the interval, keeping whichever half still brackets a sign change. Linear convergence; one guaranteed bit of accuracy per iteration; robust. Used for implied-volatility inversion in Chapter 18 when a root bracket is easy.

Newton–Raphson improves by using the derivative: $x_{n+1} = x_n - f(x_n)/f'(x_n)$. Quadratic convergence near a simple root. Fast but fragile: diverges for poor initial guesses or near inflection points. The standard implied-vol solver uses Newton with a safety bisection fallback (Chapter 35).

Worked Newton. Solve $f(x) = x^2 - 2 = 0$ (so the true root is $\sqrt{2} \approx 1.41421356$). With $f'(x) = 2x$ the iteration is $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$. Starting from $x_0 = 1.5$:

$$x_1 = 1.41666\ldots,\ x_2 = 1.41421568\ldots,\ x_3 = 1.41421356\ldots$$

Three iterations, full double-precision accuracy. Error roughly squares each step — quadratic convergence in action. An implied-vol Newton solve on a well-conditioned option starts from a Brenner–Subrahmanyam initial guess and typically converges in 3–5 iterations.

Secant method, a derivative-free cousin of Newton, estimates $f'(x_n)$ from $(f(x_n) - f(x_{n-1}))/(x_n - x_{n-1})$. Superlinear convergence (golden-ratio rate ≈ 1.618), cheaper per iteration when f' is expensive. The default choice in SciPy's `brentq` / `brenth` solvers combines it with bisection for robustness.

Tridiagonal solve (*Thomas algorithm*) solves a system $A\boldsymbol{x} = \boldsymbol{d}$ where A is tridiagonal in $\mathcal{O}(n)$ time, not $\mathcal{O}(n^3)$. This is what makes implicit finite-difference schemes for parabolic PDEs (Crank–Nicolson for Black–Scholes) viable at realistic grid sizes.

Monte Carlo basics: estimate $\mathbb{E}[f(X)]$ by drawing n i.i.d. samples $X^{(i)}$ and computing $\hat{\mu}_n = \frac{1}{n}\sum_i f(X^{(i)})$. By the LLN this converges; by the CLT the standard error is $\hat{\sigma}/\sqrt{n}$, giving the infamous $\mathcal{O}(1/\sqrt{n})$ rate — to halve the confidence interval requires quadrupling the samples. Variance-reduction techniques (antithetics, control variates, importance sampling, stratification) are the only practical way to cut compute; see Chapter 35.

Finite differences approximate derivatives of a smooth function by

$$f'(x) \approx \frac{f(x+h)-f(x-h)}{2h} \quad (\text{central}, \mathcal{O}(h^2)), \qquad f''(x) \approx \frac{f(x+h)-2f(x)+f(x-h)}{h^2}.$$

Greeks (Chapter 17) are computed this way when a closed form is unavailable or cumbersome. Choice of bump size h trades off truncation error (grows with h) against cancellation error (grows as h shrinks, as subtraction of nearly-equal numbers loses significant digits); a double-precision sweet spot is typically $h \sim 10^{-4}$ for first derivatives and 10^{-2} for seconds.

Optimisation. Unconstrained minimisation of a differentiable $f : \mathbb{R}^n \to \mathbb{R}$ reduces to solving $\nabla f = 0$. Standard methods: gradient descent (simple, slow); Newton's method (quadratic convergence, but requires Hessian); quasi-Newton (BFGS, L-BFGS: Hessian approximated from gradient history, the workhorse of `scipy.optimize.minimize`). For convex problems every local minimum is global; most calibration problems in quant finance (implied-vol fitting, SABR parameter fitting) are non-convex and benefit from good initial guesses.

Do not invert matrices

"Compute $A^{-1}\boldsymbol{b}$" in code should almost never be written `np.linalg.inv(A) @ b`. Use `np.linalg.solve(A, b)` (LU with partial pivoting), or `scipy.linalg.cho_solve` for PSD A, or Thomas for tridiagonal A. Explicit inverses are both slower and numerically worse.

A.6 Further reading

The following are the standard references. Anything in this appendix that was unfamiliar is best re-learned from one of them, not from the internet.

- **Probability.** Grimmett and Stirzaker (2001), *Probability and Random Processes* (Oxford). The standard undergraduate–graduate reference; clean proofs, wide coverage, good exercises. Chapters 1–7 cover everything in Section A.1 and more.
- **Linear algebra.** Strang (2016), *Introduction to Linear Algebra*. Concept-first, with the computational story never far from the theory. The classic complement is Trefethen and Bau (1997), *Numerical Linear Algebra*, for SVD, conditioning, and stable algorithms — what to use in production.
- **Stochastic calculus.** Øksendal (2003), *Stochastic Differential Equations*. The most readable graduate introduction; Itô, SDEs, Feynman–Kac, and an extremely clean optional-stopping chapter. Shreve (2004) *Stochastic Calculus for Finance II* is the finance-specialised alternative and is the text that sits on most derivatives strats' desks.
- **Stochastic control.** Pham (2009), *Continuous-time Stochastic Control and Optimisation with Financial Applications*. HJB, viscosity solutions, Merton-type problems. Cartea et al. (2015) applies the machinery directly to optimal market making and execution and is the single best companion to Chapter 9 and Chapter 19.
- **Numerical methods for finance.** Glasserman (2004), *Monte Carlo Methods in Financial Engineering* — the definitive reference for variance reduction and quasi-Monte Carlo in derivatives pricing. For PDE methods, Duffy (2006) is a standard desk reference.

Minimal prerequisite stack

If starting from scratch and pressed for time, a viable one-semester self-study path is: Strang (linalg, 8 weeks) $\to$ Grimmett–Stirzaker chapters 1–5 (probability, 4 weeks) $\to$ Øksendal chapters 1–5 (Brownian motion, Itô, SDE, Feynman–Kac, 4 weeks). Everything else in this appendix is either consolidation or an application of those three books.

Appendix B

Notation and Glossary

This appendix is a reference, not a chapter to read. It has two halves. The first is the *master notation table*: every symbol used in the book, what it means, and the chapter where it first appears. The second is the *trading glossary*: roughly a hundred and seventy terms in alphabetical order, each with a one or two sentence definition and a pointer back to the chapter that develops it properly. Entries here are deliberately terse. Where they feel too brief, they are doing their job: sending you back to the chapter where the concept is built up properly.

B.1 Master notation table

All symbols below are declared once in `latex/conventions.tex` and no chapter is permitted to redefine them. A handful of additional symbols are introduced locally (for example, the Ornstein–Uhlenbeck half-life τ in Chapter 10, or the hazard rate $h(t)$ in Chapter 33). Those are collected here alongside the global symbols so that this table serves as a single reference.

Probability, measure theory, and statistics

Symbol	Meaning	First use
$\mathbb{E}[\cdot]$, $\mathbb{E}^{\mathbb{P}}[\cdot]$	Expectation (physical measure by default)	Chapter 4
$\mathbb{E}^{\mathbb{Q}}[\cdot]$	Expectation under the risk-neutral measure $\mathbb{Q}$	Chapter 16
$\mathbb{P}(\cdot)$	Probability of an event under the physical measure	Chapter 4
$\mathbb{Q}$	Risk-neutral (equivalent martingale) measure	Chapter 16
$\mathrm{Var}(\cdot)$	Variance of a random variable	Chapter 4
$\mathrm{Cov}(\cdot,\cdot)$	Covariance of two random variables	Chapter 4
$\mathrm{Corr}(\cdot,\cdot)$	Pearson correlation coefficient	Chapter 11
$\mathbb{1}\{A\}$	Indicator of event A: 1 if A holds, 0 otherwise	Chapter 6
$\mathbb{R}$	The real numbers	Chapter 1
$\mathbb{R}_+$	Non-negative reals $[0,\infty)$	Chapter 16
$\mathbb{Z}$	The integers	Chapter 1
$\mathbb{N}$	The natural numbers $\{1,2,\dots\}$	Chapter 1
i.i.d.	Independent and identically distributed	Chapter 4
$\mathcal{N}(\mu,\sigma^2)$	Normal distribution, mean μ, variance σ^2	Chapter 4
$\Phi(\cdot)$	Standard normal CDF	Chapter 16

Symbol	Meaning	First use
$\phi(\cdot)$	Standard normal PDF	Chapter 16
$\Omega, \mathcal{F}, \mathbb{P}$	Probability space; sample space, σ-algebra, measure	Chapter 16
$\mathcal{F}_t$	Filtration: information available up to time t	Chapter 16
W_t, B_t	Standard Brownian motion / Wiener process	Chapter 16
dW_t	Brownian increment (used inside Itô SDEs)	Chapter 16
$\sim$	"Is distributed as"	Chapter 4
$\xrightarrow{p}$, $\xrightarrow{d}$, $\xrightarrow{a.s.}$	Convergence in probability, distribution, almost sure	Chapter 22
$\hat{\theta}$	Estimator of parameter θ	Chapter 11
$\bar{x}$, s^2	Sample mean, sample variance	Chapter 11
ρ	Autocorrelation / correlation coefficient (context-dependent)	Chapter 10
χ^2_k, t_k, F_{k_1,k_2}	Chi-squared, Student-t, and F distributions	Chapter 22
$\mathrm{KL}(p\|q)$	Kullback–Leibler divergence	Chapter 22

Microstructure, order books, and execution

Symbol	Meaning	First use
p^{bid}, p^{bid}_t	Best bid price (highest buy quote)	Chapter 2
p^{ask}, p^{ask}_t	Best ask price (lowest sell quote)	Chapter 2
p^{mid}	Mid price, $(p^{\text{bid}} + p^{\text{ask}})/2$	Chapter 2
p^{micro}	Microprice, size-weighted fair value	Chapter 4
p^{wall}	Wall mid (Prosperity-specific fair value)	Chapter 4
s	Bid–ask spread, $p^{\text{ask}} - p^{\text{bid}}$ (ticks or currency)	Chapter 2
Q^b_k, Q^a_k	Quantity at the k-th bid / ask level	Chapter 2
L^b, L^a	Number of price levels on bid / ask side	Chapter 2
τ_{queue}	Time to front of queue at a given price	Chapter 8
Δp	Tick size (minimum price increment)	Chapter 2
Δt	Sample period between observations	Chapter 2
r_t	Log return, $\log(p_t/p_{t-1})$	Chapter 5
VWAP	Volume-weighted average price	Chapter 19
TWAP	Time-weighted average price	Chapter 19
IS	Implementation shortfall	Chapter 19
η, ϵ	Almgren–Chriss temporary / permanent impact params	Chapter 19
λ_{AC}	Almgren–Chriss risk-aversion parameter	Chapter 19
OFI_t	Order-flow imbalance at time t	Chapter 8
VPIN	Volume-synchronized PIN toxicity metric	Chapter 8
PIN	Probability of informed trading	Chapter 6

Market making (Avellaneda–Stoikov and extensions)

Symbol	Meaning	First use
q, q_t	Inventory (signed position held by the maker)	Chapter 9
$r(s, q, t)$	Reservation / indifference price	Chapter 9
δ^b, δ^a	Bid / ask quote distance from reference	Chapter 9
γ	Risk-aversion parameter (A–S γ)	Chapter 9
κ	Order-arrival intensity (A–S κ)	Chapter 9
A	Arrival-rate scale parameter (A–S)	Chapter 9
σ_q	Inventory variance	Chapter 9
$Q_{\max}$	Hard position limit	Chapter 9

Kyle / Glosten–Milgrom / adverse-selection models

Symbol	Meaning	First use
λ	Kyle's lambda: linear price-impact coefficient	Chapter 6
u	Noise-trader order flow	Chapter 6
x	Informed (insider) order flow	Chapter 6
v	Fundamental value known to insider	Chapter 6
Σ_0	Prior variance of fundamental value	Chapter 6
σ_u^2	Variance of noise-trader flow	Chapter 6
α, μ	PIN model: informed probability, arrival rate	Chapter 6
$\varepsilon_b, \varepsilon_s$	PIN model: uninformed buy / sell arrival rates	Chapter 6

Options, Black–Scholes, and the vol surface

Symbol	Meaning	First use
S, S_t	Spot price of the underlying	Chapter 15
K	Strike price of an option	Chapter 15
r	Risk-free rate (continuously compounded)	Chapter 15
σ	Volatility (standard deviation of log returns)	Chapter 10
σ_{imp}	Implied volatility	Chapter 16
σ_{rv}	Realised volatility	Chapter 18
T, $T - t$	Maturity; time to expiry	Chapter 15
$C(S, t)$	European call option price	Chapter 15
$P(S, t)$	European put option price	Chapter 15
F, F_t	Forward price of the underlying	Chapter 15
d_1, d_2	Black–Scholes auxiliary quantities	Chapter 16

Symbol	Meaning	First use
$m = \log(K/F)$	Log-moneyness	Chapter 18
q_{div}	Continuous dividend yield	Chapter 16
β_{SABR}, α_{SABR}, ν_{SABR}	SABR parameters	Chapter 18
ρ_{SABR}	SABR spot/vol correlation	Chapter 18
$w(k,T)$	Total implied variance, $\sigma^2_{\text{imp}}T$	Chapter 18

Greeks

Symbol	Meaning	First use
Δ	Delta: $\partial V/\partial S$	Chapter 17
Γ	Gamma: $\partial^2 V/\partial S^2$	Chapter 17
ν, $\mathcal{V}$	Vega: $\partial V/\partial \sigma$	Chapter 17
Θ	Theta: $\partial V/\partial t$ (time decay)	Chapter 17
ρ_{opt}	Rho: $\partial V/\partial r$	Chapter 17
vanna	$\partial^2 V/\partial S\,\partial \sigma$	Chapter 17
volga	$\partial^2 V/\partial \sigma^2$	Chapter 17
charm	$\partial^2 V/\partial S\,\partial t$	Chapter 17
speed	$\partial^3 V/\partial S^3$	Chapter 17
colour	$\partial^3 V/\partial S^2 \partial t$	Chapter 17
DDE	Dual delta: $\partial V/\partial K$	Chapter 17

Mean reversion, stat arb, and time series

Symbol	Meaning	First use
θ_{OU}	OU mean-reversion speed	Chapter 10
μ_{OU}	OU long-run mean	Chapter 10
σ_{OU}	OU diffusion coefficient	Chapter 10
$\tau_{1/2}$	Half-life of mean reversion, $\log 2/\theta$	Chapter 10
z_t	Standardised spread (z-score)	Chapter 11
β_{coint}	Cointegrating coefficient	Chapter 11
ADF	Augmented Dickey–Fuller test statistic	Chapter 11
H	Hurst exponent	Chapter 11
MA(q), AR(p), ARMA(p,q)	Time-series model orders	Chapter 22

Risk, performance, and capital allocation

Symbol	Meaning	First use
SR	Sharpe ratio, $(\mu - r_f)/\sigma$	Chapter 20
SortR	Sortino ratio (downside-deviation denominator)	Chapter 20
Calmar	Calmar ratio, return over max drawdown	Chapter 20
IR	Information ratio, active return over tracking error	Chapter 20
MDD	Maximum drawdown	Chapter 20
VaR_α	Value-at-risk at confidence α	Chapter 36
ES_α, CVaR_α	Expected shortfall / conditional VaR	Chapter 36
f^*	Optimal Kelly fraction	Chapter 20
L	Leverage	Chapter 20

Fixed income and rates

Symbol	Meaning	First use
$z(T)$	Zero / spot rate for maturity T	Chapter 29
$D(T)$	Discount factor to maturity T	Chapter 29
y	Yield to maturity	Chapter 29
$f(T_1, T_2)$	Forward rate from T_1 to T_2	Chapter 29
$f(0, T)$	Instantaneous forward rate	Chapter 29
MacD	Macaulay duration	Chapter 29
ModD	Modified duration	Chapter 29
Conv	Convexity	Chapter 29
DV01, BPV	Dollar value of a basis point	Chapter 29
s^*	Par swap rate	Chapter 30
$A(0; T_1, T_n)$	Annuity / PV01 factor over swap tenor	Chapter 30
$\kappa_{\mathrm{HW}}, \theta_{\mathrm{HW}}, \sigma_{\mathrm{HW}}$	Hull–White mean reversion, level, vol	Chapter 30
α, β (SABR-rates)	Short-rate SABR skew parameters	Chapter 30
$\phi(t)$	HJM drift adjustment	Chapter 30

Credit

Symbol	Meaning	First use
PD	Probability of default (over horizon)	Chapter 33
LGD	Loss given default, $1 - R$	Chapter 33
R	Recovery rate	Chapter 33
$S(t)$	Survival probability to time t	Chapter 33
$h(t), \lambda_c(t)$	Hazard rate (credit intensity)	Chapter 33
s_{CDS}	CDS par spread (running coupon)	Chapter 33

Symbol	Meaning	First use
CVA, DVA, FVA, KVA	Valuation adjustments (credit, debit, funding, capital)	Chapter 36
$\mathrm{EE}(t)$, $\mathrm{PFE}_\alpha(t)$	Expected / potential future exposure	Chapter 36

FX, commodities, and equities

Symbol	Meaning	First use
X_t	FX spot rate (domestic per foreign)	Chapter 32
r_d, r_f	Domestic / foreign risk-free rates	Chapter 32
RR, BF	Risk reversal, butterfly (FX vol quotes)	Chapter 32
y_{conv}	Convenience yield	Chapter 34
$\kappa_S, \mu_S, \sigma_S$	Schwartz one-factor model parameters	Chapter 34
basis	Futures minus spot (or near minus far)	Chapter 34
β_{CAPM}	Equity beta to benchmark	Chapter 31
q_{div}	Continuous dividend yield (equity)	Chapter 31

Miscellaneous notation and conventions

Symbol	Meaning	First use
$:=$	LHS defined as RHS ($:=$)	Chapter 1
$=:$	RHS defined as LHS ($=:$)	Chapter 1
$\mathcal{O}(\cdot)$	Big-O asymptotic notation	Chapter 1
$\lfloor x \rfloor$, $\lceil x \rceil$	Floor, ceiling	Chapter 26
x^+, x^-	Positive and negative parts, $\max(x, 0)$, $\max(-x, 0)$	Chapter 15
$\lVert x \rVert_p$	ℓ_p norm	Chapter 22
$\langle \cdot, \cdot \rangle$	Inner product	Chapter 22

B.2 Trading glossary

Terms are arranged strictly alphabetically. Cross-references to other glossary entries are in *italics*. Where a term has multiple meanings, the trading meaning is given first. Seeded from `prosperity-wiki/trading-glossary.md` and extended from all body chapters.

Adverse selection Losses a liquidity provider incurs because counterparties with superior information trade against them. The core cost that widens bid–ask spreads in the Glosten–Milgrom model. See Chapters 6 and 9.

Almgren–Chriss Optimal execution framework balancing expected trading cost against variance of cost, yielding a characteristic exponentially-decaying trajectory. See Chapter 19.

Amendment Modification of a resting order's price or size; many venues treat an amendment to a more aggressive price as a new order for queue-priority purposes. See Chapter 2.

Ask (offer) Price at which a seller is willing to sell. The *best ask* is the lowest such price currently resting on the book. See Chapter 2.

At-the-money (ATM) An option whose strike equals (or is closest to) the current spot or forward price. See Chapter 15.

Augmented Dickey–Fuller (ADF) test Hypothesis test for a unit root in a time series. The null hypothesis is that the series has a unit root (is non-stationary). Rejecting the null at the chosen p-value is necessary (but not sufficient) for a mean-reverting or cointegrated spread; the Engle–Granger test applies ADF to regression residuals to test for cointegration. See Chapters 10 and 11.

Authorised participant (AP) Large financial institution (typically a bank or broker-dealer) with the contractual right to create and redeem ETF shares directly with the fund in exchange for the underlying basket of securities. AP arbitrage between the ETF market price and its net asset value keeps the two tethered. See Chapter 12.

Autocallable Structured equity product that redeems early if the underlying is above a barrier on observation dates. See Chapter 31.

Avellaneda–Stoikov Stochastic-control market-making framework that derives the optimal bid and ask quotes for a dealer managing inventory risk. The key outputs are the *reservation price* $r = s - q\gamma\sigma^2(T-t)$ (mid adjusted for inventory) and the symmetric quote distance $\delta^* \approx \gamma\sigma^2(T-t) + \frac{2}{\gamma}\ln(1+\gamma/\kappa)$. See Chapter 9.

Back-month A futures contract with a more distant expiry than the *front month*. See Chapter 34.

Backtest Simulation of a strategy on historical data. Must avoid look-ahead bias and survivorship bias. See Chapter 21.

Backwardation Futures curve state where near contracts trade above far contracts. Typically signals scarcity or high convenience yield. See Chapter 34.

Basel III Post-2008 banking capital and liquidity framework (CET1, leverage ratio, LCR, NSFR). See Chapter 38.

Basis Difference between two related prices — spot vs futures, cash bond vs CDS, or near vs far contracts. See Chapters 14 and 34.

Basis point (bp) One one-hundredth of a percent. $100\,\text{bp} = 1\%$. See Chapter 29.

Bid Price at which a buyer is willing to buy; the *best bid* is the highest resting buy quote. See Chapter 2.

Binomial tree Discrete-time lattice model for option pricing in which the underlying moves up by factor $u = e^{\sigma\sqrt{\Delta t}}$ or down by $d = 1/u$ each step, with risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. The recombining property ($ud = du$) gives $N+1$ terminal nodes after N steps. Option prices follow by backward induction. The Cox–Ross–Rubinstein (CRR) choice of u recovers Black–Scholes as $\Delta t \to 0$. See Chapter 35.

Black–Scholes equation PDE $\partial_t V + \frac{1}{2}\sigma^2 S^2 \partial_{SS} V + rS\partial_S V - rV = 0$ for European options under log-normal spot dynamics. See Chapter 16.

Book (i) The full set of resting orders for an instrument; (ii) the portfolio a trading desk risk-manages ("running the book"). See Chapters 2 and 28.

Brownian motion Continuous-time stochastic process with independent Gaussian increments; the driver of most continuous-time pricing models. See Chapter 16.

Butterfly (i) Option spread: long two wings, short twice the body — bets on low realised movement around the body strike; (ii) FX vol quote convention, measuring smile curvature. See Chapters 15 and 32.

Calendar spread Long one maturity, short another on the same underlying and strike. Used to trade term structure of vol or of the forward curve. See Chapters 14 and 15.

Calmar ratio Annualised return divided by maximum drawdown. See Chapter 20.

Call option Contract giving the holder the right (not obligation) to buy the underlying at strike K by expiry T. Payoff $(S_T - K)^+$. See Chapter 15.

Cap / floor An interest-rate *cap* is a portfolio of call options (caplets) on a floating rate: each caplet pays $\max(\ell_t - K, 0) \cdot$ notional for period t. A *floor* is the equivalent with puts. Caps are used to hedge floating-rate borrowing; floors to hedge floating-rate lending. Priced via Black's model on the forward rate. See Chapter 30.

Carry Return from simply holding a position (funding rate, dividend, convenience yield). "Positive carry" strategies earn yield while waiting. See Chapters 29 and 34.

Cash-and-carry arbitrage Long spot, short futures (or vice versa) to lock in the cost-of-carry relationship. See Chapter 14.

CCAR Comprehensive Capital Analysis and Review — the Federal Reserve's annual stress test of large US banks. See Chapter 38.

CCP (central counterparty) Clearing house that interposes itself between buyer and seller, becoming the legal counterparty to both sides of a trade. Eliminates bilateral counterparty risk via multilateral netting and daily variation margin. Examples: LCH, CME Clearing, Eurex Clearing. See Chapters 3 and 36.

CDO (collateralised debt obligation) Structured security backed by a pool of debt instruments (loans, corporate bonds, or CDS contracts). Cash flows are split into tranches with different seniority: equity tranche absorbs first losses and earns highest spread; senior tranche absorbs last losses and earns lowest spread. Default correlation is the key risk driver. See Chapter 33.

CDS (credit default swap) Bilateral contract where the buyer pays a running premium and receives par-minus-recovery on a reference credit event. See Chapter 33.

CDX, iTraxx Standardised credit-default-swap indices for North America (CDX) and Europe / Asia (iTraxx). See Chapter 33.

Charm Second-order Greek $\partial^2 V/\partial S \partial t$; rate at which delta decays. Matters for overnight hedge resets. See Chapter 17.

Cointegration Property where two or more non-stationary series have a stationary linear combination. Foundation of pairs trading. See Chapter 11.

Contango Futures curve state where far contracts trade above near contracts — the "normal" shape for storable assets with positive cost of carry. See Chapter 34.

Convenience yield Implicit benefit of holding physical commodity rather than a futures contract (optionality of consumption, supply chain). See Chapter 34.

Convexity (i) Second derivative of bond price wrt yield; (ii) any second-order effect that makes a P&L curve smile up or down. See Chapters 17 and 29.

Counterparty The other side of a trade. On a central exchange the CCP interposes itself; OTC leaves bilateral counterparty risk. See Chapter 3.

Covered interest parity (CIP) No-arbitrage condition linking the spot FX rate S, forward rate F, and interest rates in two currencies: $F = S \cdot e^{(r_d - r_f)T}$. Violations (CIP deviations) indicate funding stress, regulatory constraints, or segmented capital markets. Forms the basis of FX forward pricing. See Chapters 14 and 32.

Crack spread The price difference between crude oil and its refined products (gasoline, heating oil, diesel). Represents the refiner's gross margin. A positive crack spread makes refining profitable. Traded via futures spreads on crude and products; also called the *3-2-1 crack* (3 barrels crude → 2 gasoline + 1 heating oil). See Chapter 34.

CVA (credit valuation adjustment) Market-value adjustment to a derivative to reflect the counterparty's default risk. See Chapter 36.

Dark pool Non-displayed venue where orders are not visible pre-trade; used to execute large orders with reduced market impact. See Chapter 3.

Delta Sensitivity of option value to underlying price, $\partial V / \partial S$. Approximate hedge ratio. See Chapter 17.

Delta-hedging Trading the underlying to keep portfolio delta at target (usually zero); exposes the book to gamma and vega. See Chapter 17.

Depth Total resting quantity on one side of the book, or at a particular price level. See Chapter 2.

Desk A trading team sharing a book and mandate (e.g., "cash equities desk", "rates vol desk"). See Chapter 28.

Drawdown Peak-to-trough decline in equity curve. See Chapter 20.

DV01 (BPV) Dollar value of a one-basis-point parallel shift in yields. See Chapter 29.

DVA (debit valuation adjustment) The mirror image of CVA: reflects the *own-credit* effect in OTC derivative valuation. If you default, your liability disappears, so a higher own-credit spread *increases* derivative value to you. DVA is controversial because it creates a perverse incentive and is difficult to hedge. See Chapters 33 and 36.

Engle–Granger Two-step cointegration test: estimate the cointegrating regression, then ADF-test its residuals. See Chapter 11.

Error correction model (ECM) Time-series representation of a cointegrated pair. Each variable's change is driven partly by the previous period's deviation from the long-run equilibrium (the *error-correction term*). In pairs trading, the ECM formalises how the spread is pulled back to its mean: a large positive deviation implies a negative drift in the next period. See Chapter 11.

ETF Exchange-traded fund — a basket instrument creating/redeeming in kind with authorised participants, enabling basket-vs-constituent arbitrage. See Chapter 12.

Exchange Central venue matching buy and sell orders in a regulated, transparent way. See Chapter 2.

Expected shortfall (ES, CVaR) Average loss conditional on being beyond VaR. Coherent risk measure. See Chapter 36.

Fair value Best estimate of the true price of an instrument, usually blending mid, microprice, and model inputs. See Chapter 4.

Feynman–Kac Theorem representing solutions of certain parabolic PDEs as expectations of stochastic-process functionals; underpins risk-neutral pricing. See Chapter 16.

Fill The execution of all or part of an order. See Chapter 2.

FOK (fill-or-kill) Order that must execute in full immediately or be cancelled entirely. See Chapter 2.

Forward OTC bilateral contract to transact at a fixed price on a future date. See Chapter 15.

Forward rate Implied rate of borrowing/lending between two future dates, derived from the zero curve. See Chapter 29.

FRTB Fundamental Review of the Trading Book; post-crisis market-risk capital regime replacing VaR with expected shortfall. See Chapter 38.

Funding leg In a CDS or structured trade, the leg that pays or receives the funding cost. See Chapters 33 and 36.

Futures Exchange-traded, margined, standardised forward. Daily mark-to-market is the key difference from a forward. See Chapter 34.

FVA (funding valuation adjustment) Adjustment for the funding cost of uncollateralised derivative exposures. See Chapter 36.

Gamma Second derivative of option value wrt underlying, $\partial^2 V/\partial S^2$. Long gamma profits from realised movement. See Chapter 17.

Gamma scalping Delta-hedging a long-gamma position to monetise realised vs implied volatility differential. See Chapter 17.

Glosten–Milgrom Microstructure model where the spread arises purely from adverse selection on informed traders. See Chapter 6.

Greek Partial derivative of derivative value wrt a model input (delta, gamma, vega, theta, rho, and higher-order). See Chapter 17.

Half-life Time for an OU process (or AR(1)) to revert half-way to its mean, $\log(2)/\theta$. See Chapter 10.

Hazard rate Instantaneous conditional default intensity, $h(t) = -S'(t)/S(t)$. See Chapter 33.

Hedge Offsetting position taken to neutralise a specific risk exposure. See Chapter 17.

HJB (Hamilton–Jacobi–Bellman) Continuous-time dynamic-programming PDE for optimal control; the tool that produces Avellaneda–Stoikov and Almgren–Chriss closed forms. See Chapters 9 and 19.

Iceberg order Resting order displaying only a small tip of the true quantity. See Chapter 2.

Implementation shortfall (IS) Difference between execution price and decision-time price, including opportunity cost. See Chapter 19.

Implied volatility (IV) Volatility input that, put into Black–Scholes, matches the market price. The market's implicit forecast plus a risk premium. See Chapters 16 and 18.

Informed trader Market participant with a private signal about the fundamental value of an asset. In the Kyle and Glosten–Milgrom models, informed order flow is the source of adverse selection and drives bid–ask spread widening. Contrast with *noise trader*, who trades for liquidity or portfolio reasons and provides cover for the informed. See Chapter 6.

In-the-money (ITM) Option with positive intrinsic value. See Chapter 15.

Inventory Signed position carried by a market maker; drives reservation-price skewing. See Chapter 9.

IOC (immediate-or-cancel) Order that executes whatever can fill immediately, the remainder cancelled. See Chapter 2.

Information ratio (IR) Active return over tracking error of a portfolio vs its benchmark. See Chapter 20.

Intrinsic value The immediate exercise payoff of an option: $(S-K)^+$ for calls, $(K-S)^+$ for puts. See Chapter 15.

Itô's lemma Chain rule for stochastic calculus, adding a second-order $\frac{1}{2}\sigma^2$ term. The workhorse for deriving SDE-based PDE prices. See Chapter 16.

Johansen test Multivariate cointegration test producing rank and cointegrating vectors. See Chapter 11.

KELP Prosperity 4 product; semi-mean-reverting with moderate volatility — classic market-making candidate. See Chapter 25.

Kelly criterion Optimal log-utility bet-sizing rule $f^* = \mu/\sigma^2$ for normal returns. Maximises long-run growth. See Chapter 20.

KVA (capital valuation adjustment) Adjustment for the lifetime capital cost of a derivative. See Chapter 36.

Kyle's lambda Linear price-impact coefficient in the Kyle (1985) model. See Chapter 6.

Latency End-to-end delay from market event to trading response. Measured in microseconds for competitive venues. See Chapter 3.

Level (book level) Aggregated quantity at a single price in the order book. See Chapter 2.

LGD (loss given default) Fraction of exposure lost on default, $1-R$. See Chapter 33.

LIBOR Legacy interbank offered rate, now replaced by risk-free rates (SOFR, SONIA, ÉSTR). See Chapters 30 and 38.

Limit order Resting order to buy/sell at a specified price or better. See Chapter 2.

Liquidity Ability to transact in size without moving the price much. See Chapters 2 and 9.

Long A position that profits when the price rises. See Chapter 3.

Look-ahead bias Backtest flaw in which future information leaks into past decisions. See Chapter 21.

MAGNIFICENT_MACARONS Prosperity 4 product with imported variant, sunlight / sugar inputs and a transport cost — a locational-arbitrage product. See Chapter 25.

Maker / taker Exchange fee structure in which limit-order providers (*makers*) receive a rebate and market-order takers pay a fee. Inverted at some venues (*inverted markets*). Maker/taker economics drive order routing decisions: brokers often route to venues that pay the highest maker rebate, creating potential conflicts of interest. See Chapters 2 and 9.

Mark-to-market (MTM) Valuation of a position at current market prices rather than historical cost. Daily MTM drives realised P&L, margin calls, and risk limits. Futures are marked-to-market daily with cash variation margin exchanged. See Chapters 3 and 36.

Market impact Adverse price movement caused by one's own trading. Temporary impact recovers; permanent impact does not. See Chapter 7.

Market maker Participant continuously quoting two-sided prices and earning the spread. See Chapter 9.

Market order Immediate execution order at the best available prices, walking the book if necessary. See Chapter 2.

Matching engine Exchange software that pairs compatible buy and sell orders according to the venue's priority rules. See Chapter 2.

Mean reversion Tendency of a time series to pull back toward a long-run mean. See Chapter 10.

MiFID II European regulation on market transparency, best execution, and pre- and post-trade reporting. See Chapter 38.

Microprice Size-weighted fair value, $\frac{Q^a p^{\text{bid}}+Q^b p^{\text{ask}}}{Q^a+Q^b}$. Tilts mid toward the thinner side of the book. See Chapter 4.

Momentum Tendency of recent winners to continue to outperform over some horizon. See Chapter 13.

Moneyness How far an option is from the money. Usually expressed as K/S, $\log(K/F)$, or in delta. See Chapters 15 and 18.

Monte Carlo simulation Numerical method that estimates option prices, risk measures, or path-dependent payoffs by simulating many random paths of the underlying process and averaging the discounted payoffs. Standard errors shrink as $1/\sqrt{N_{\text{paths}}}$. Exact for path-dependent products (barriers, Asian options) that resist closed-form solution. See Chapter 35.

NAV (net asset value) The fair value of an ETF's underlying basket per share, computed from current constituent prices. The authorised participant arbitrage mechanism keeps the ETF market price close to NAV: if the ETF trades at a premium, the AP creates new shares to sell; if at a discount, it redeems shares. See Chapter 12.

No-arbitrage Assumption that no portfolio exists today that costs nothing and delivers a non-negative payoff tomorrow with strictly positive probability. The no-arbitrage condition is the foundation of derivative pricing: any instrument whose cash flows can be replicated by a traded portfolio must sell at the replication cost. Violations are called arbitrage opportunities. See Chapters 4, 14 and 15.

Noise trader Market participant who trades for liquidity or non-informational reasons (hedging, index rebalancing, portfolio adjustment). In Kyle and Glosten–Milgrom models, noise-trader flow provides cover for informed traders and is the source of market-making revenue. Contrast with *informed trader*. See Chapter 6.

Notional Face amount of a contract; for derivatives, the reference amount not exchanged at inception. See Chapter 29.

OIS (overnight index swap) Interest-rate swap in which one leg pays a fixed rate and the other leg pays the compounded daily overnight rate (SOFR for USD, SONIA for GBP, ÉSTR for EUR). OIS discounting became the standard risk-free discount curve after the 2008 crisis revealed LIBOR's credit risk. CSA-collateralised derivatives are discounted at the OIS rate. See Chapters 30 and 36.

Olivia Prosperity 4 bot trader whose trading-ID pattern gave away informed flow in past editions. A "trader-ID" exploit example. See Chapters 25 and 27.

Order book (LOB) Collection of all resting buy and sell limit orders for an instrument. See Chapter 2.

Order-flow imbalance (OFI) Signed net change in resting quantity at the top of the book; a short-horizon predictor. See Chapter 8.

Ornstein–Uhlenbeck (OU) Continuous-time mean-reverting SDE: $dX_t = \theta(\mu - X_t)dt + \sigma dW_t$. See Chapter 10.

OTC (over-the-counter) Bilateral trade negotiated directly between two parties outside a central exchange. OTC derivatives (swaps, forwards, swaptions) are customisable to any size, maturity, or structure but carry bilateral counterparty credit risk and reduced pre-trade transparency. The CSA and ISDA Master Agreement govern the legal relationship. See Chapter 3.

Out-of-the-money (OTM) Option with zero intrinsic value. See Chapter 15.

Out-of-sample Data not used to fit the model; performance on it is the only honest measure. See Chapter 21.

PICNIC_BASKET Prosperity 4 product constructed as a weighted bundle of underlying items — the canonical ETF-arbitrage example. See Chapters 12 and 25.

PD (probability of default) Probability that a credit entity defaults over a horizon. See Chapter 33.

Price-time priority Matching rule: better price first; among equal prices, older order first. See Chapter 2.

Put option Right to sell the underlying at strike by expiry. Payoff $(K - S_T)^+$. See Chapter 15.

Put–call parity Identity $C - P = S - Ke^{-rT}$ (no dividends). See Chapter 15.

POV (percent-of-volume) Execution algo that trades a target fraction of realised market volume. See Chapter 19.

Prosperity3bt Community backtester for the Prosperity platform. See Chapter 24.

QR (quantitative researcher) Buy-side / prop research role focused on developing signals and models. See Chapter 28.

Quote A displayed resting price with its size; a market maker's current bid or ask. See Chapter 2.

RAINFOREST_RESIN Prosperity 4 product with a very stable fair value around 10 000; textbook market-making target. See Chapter 25.

Realised volatility Sample standard deviation of log returns over a window. See Chapter 18.

Recovery rate Fraction of notional recovered on default. See Chapter 33.

Reservation price Market maker's risk-adjusted mid, tilted by inventory. See Chapter 9.

RFR (risk-free rate) Overnight benchmark replacing LIBOR: SOFR (USD), SONIA (GBP), €STR (EUR). See Chapters 30 and 38.

Rho Option's sensitivity to the risk-free rate, $\partial V / \partial r$. See Chapter 17.

Risk limit Hard bound on a risk metric (gross, net, VaR, greeks) enforced by risk management. See Chapter 36.

Roll Act of closing a near-expiry position and opening the equivalent in a later expiry. See Chapter 34.

Roll's spread estimator Estimator of the effective spread from return autocovariance: $s = 2\sqrt{-\mathrm{Cov}(r_t, r_{t-1})}$. See Chapter 5.

SABR Stochastic-$\alpha\beta\rho$ volatility model; industry standard for interest-rate smile fitting. See Chapters 18 and 30.

Schwartz model One-factor mean-reverting log-price model for commodities. See Chapter 34.

SecDB Goldman Sachs' proprietary securities database; the canonical object store for trades, prices, risks. See Chapter 37.

Settlement Exchange-of-cash-and-asset step that completes a trade. See Chapter 3.

Sharpe ratio Risk-adjusted return, $(\mu - r_f)/\sigma$, annualised. See Chapter 20.

Short Position profiting from price declines; in equities, often requires borrowing the asset. See Chapter 3.

Skew Asymmetry of the implied-vol smile. Typically OTM puts trade above OTM calls in equities. See Chapter 18.

Slang Goldman Sachs' internal domain-specific language used on top of SecDB. See Chapter 37.

Slippage Execution price minus expected price; combines impact, timing, and discretisation losses. See Chapter 19.

SOFR Secured Overnight Financing Rate; the USD risk-free rate replacing LIBOR. See Chapters 30 and 38.

Sortino ratio Sharpe variant using only downside deviation in the denominator. See Chapter 20.

Spread Ask minus bid (book spread), or the quoted difference for a basket / relative-value trade. See Chapters 2 and 5.

Squid Ink Prosperity 3 product whose highly-mean-reverting signature made OU/z-score strategies dominant. See Chapter 27.

Stat arb Statistical arbitrage — systematic relative-value trading on cross-sectional mean-reversion signals. See Chapter 11.

Stop order Order activated when the market trades through a trigger price; converts to a market or limit order. See Chapter 2.

Straddle Long call plus long put at the same strike; payoff $|S_T - K|$. Buys volatility. See Chapter 15.

Strangle Long OTM call plus long OTM put; cheaper than a straddle, needs bigger move. See Chapter 15.

Strat Goldman Sachs role combining quant modelling, software engineering, and trader-adjacent work. See Chapters 28 and 37.

Strike Price at which an option can be exercised. See Chapter 15.

Survival probability Probability a credit has not defaulted by t: $S(t) = \exp(-\int_0^t h(u)\,du)$. See Chapter 33.

Swap OTC contract to exchange cash flows. Most common: fixed vs floating interest rate (IRS). See Chapter 30.

Swaption Option to enter a fixed/floating interest-rate swap at a specified future date. A *payer swaption* is the right to pay fixed (a call on rates); a *receiver swaption* is the right to receive fixed. The vol surface for swaptions is indexed by option expiry and underlying swap tenor. Priced via Black's or Bachelier's model. See Chapter 30.

Theta Option's time-decay Greek, $\partial V/\partial t$. See Chapter 17.

Tick size Minimum price increment on a venue. Constrains spreads and quoting. See Chapter 2.

Toxic flow Counterparty flow that systematically predicts short-term price moves against you; adverse selection made concrete. See Chapter 8.

Tracking error Standard deviation of portfolio return minus benchmark return. See Chapter 20.

Trend-following Strategy family buying winners and selling losers at medium horizons. See Chapter 13.

Triangular arbitrage FX arbitrage across three currency pairs when implied cross-rates misalign. See Chapter 14.

TWAP Time-weighted average price; also an execution algo slicing uniformly in time. See Chapter 19.

Uncrossed book Book state where best bid $<$ best ask (no immediate matching possible). See Chapter 2.

Value-at-Risk (VaR) Loss threshold exceeded with probability $1 - \alpha$ over a horizon. See Chapter 36.

Vanna Cross-Greek $\partial^2 V / \partial S \partial \sigma$. See Chapter 17.

Variance swap OTC contract paying the difference between realised variance $\hat{\sigma}^2_{\mathrm{rv}}$ (measured daily over the swap's life) and a fixed strike K_{var}, scaled by notional: P&L $= N_{\mathrm{var}} \cdot (\hat{\sigma}^2_{\mathrm{rv}} - K_{\mathrm{var}})$. Provides pure exposure to realised volatility, independent of delta. Replication by a log-contract links fair variance to the implied vol surface. See Chapter 31.

Vega Option's sensitivity to implied volatility, $\partial V / \partial \sigma$. See Chapter 17.

VOLCANIC_ROCK_VOUCHERS Prosperity 4 option-like product tied to an underlying rock price; canonical volatility / Greeks case study. See Chapter 25.

Volatility surface Implied-vol as a function of strike and maturity. Shape encodes skew, term structure, and higher-order moments. See Chapter 18.

Volga Second vega: $\partial^2 V / \partial \sigma^2$. See Chapter 17.

VPIN Volume-synchronized probability of informed trading; a flow-toxicity metric. See Chapter 8.

VWAP Volume-weighted average price; both a benchmark and a common execution algo. See Chapter 19.

Walk-forward Out-of-sample evaluation in which the model is re-estimated on an expanding or rolling window. See Chapter 21.

Wall mid Prosperity-specific fair-value construct using the deepest price levels as more robust reference points. See Chapter 4.

Yield Discount rate implied by a bond's price. See Chapter 29.

Yield curve Plot of zero (or par) rates against maturity. Key shapes: normal, inverted, flat, humped. See Chapter 29.

Z-score Standardised deviation $z = (x - \mu)/\sigma$. The workhorse signal for pairs/mean-reversion strategies. See Chapter 11.

Zero rate Continuously-compounded yield on a zero-coupon bond to maturity T. See Chapter 29.

B.3 Further reading by topic

The bibliography in Chapter C collects every source cited in the book with full publication details. This short appendix of pointers is intended as a fast-access map from a topic to the one or two sources that, in the author's opinion, repay the most effort.

Microstructure, orders and spreads (Harris, 2003) for the breadth and institutional detail; (Bouchaud et al., 2018) for the statistical-physics angle; (Hasbrouck, 2007) for empirical technique.

Market making (Cartea et al., 2015) for the stochastic-control treatment from first principles; (Avellaneda and Stoikov, 2008) for the original paper.

Stat arb and pairs (Chan, 2013) and (Chan, 2022) for worked implementations; (Avellaneda and Lee, 2010) for a clean exposition of the mean-reverting residual idea.

Options and Greeks (Hull, 2014) as the canonical textbook; (Natenberg, 2015) for practitioner intuition about the surface and trading greeks; (Sinclair, 2013) for vol arbitrage in practice.

Execution (Almgren and Chriss, 2000) for the foundational framework; (Cartea et al., 2015) for modern extensions.

Machine learning and validation (López de Prado, 2018) for feature design and cross-validation traps; (Jansen, 2020) for a broader applied treatment.

Risk and capital (Thorp, 2006) on Kelly sizing; (Glasserman, 2004) on Monte Carlo for risk; (Derman, 2004) for the culture.

Rates and credit (Hull, 2014) again, complemented by (Black, 1976), (Hull and White, 1990), and (Brace et al., 1997) for primary references.

Commodities (Schwartz, 1997), (Gibson and Schwartz, 1990), and (Schwartz and Smith, 2000) for the mean-reverting log-price framework.

Interview prep (Zhou, 2020) for classic market-making and probability puzzles.

End of Chapter B. Return to the chapter where you encountered the symbol or term, or proceed to Chapter C for full source details.

Appendix C

Bibliography and Reading Order

This appendix reorganises the bibliography as three suggested reading orders, depending on your goal. If you have read cover to cover, you have used the default; the orderings below are for readers who want to focus. It also serves as an annotated bibliography: every source cited anywhere in the book is collected here, grouped by topic, with a one-sentence note on why you would read it and which chapters draw on it. The final two sections list the primary sources that are free online and the topics this book deliberately does not cover.

How to use this appendix

There are three kinds of reader this book is written for: someone preparing for the IMC Prosperity competition, someone starting an internship or graduate role on a bank Strats desk (Goldman Sachs or otherwise), and a completionist who wants the full picture. The *Three reading orders* below (Section C.1) give the first two a fast route through the material that matters for their goal. The *Annotated bibliography* (Section C.2) is the reference side: fifteen topic clusters, each listing the specific books, papers, and blog posts that feed the relevant chapters, with a short note on why the source is worth reading. The *Free online sources* (Section C.3) section lists the material you can legally read tonight without paying for anything. Finally, *What's not in this book* (Section C.4) is an honest list of topics this book avoids, with pointers for where to go next.

C.1 Three reading orders

Reading 635 pages cover to cover is a real commitment. If you picked this book up with a specific goal in mind, one of the three tracks below will get you there faster. Each track is a sequence of chapters with the pages that would otherwise be in the way skipped. The tracks overlap substantially in the foundations; they diverge in what they emphasise after Chapter 15.

Track 1 — Prosperity competition

The Prosperity track is about producing a working algorithm in five rounds. You need enough microstructure to understand why the simulator behaves the way it does, the small handful of strategies that keep winning writeups (market making, pairs/basket arbitrage, options scalping, voucher stat arb), and a playbook for diagnosing which strategy to apply to each product. You do *not* need rates, credit, commodities, SecDB, FVA, or regulatory capital.

1. **Orientation** (Ch 0–3). How to read the book, what a market is, price and value, the mechanics of an order book and matching engine. Skipping this is the single most common reason Prosperity writeups are full of confused terminology.

2. **Microstructure minimum** (Ch 4–7). Bid–ask spreads, the Roll/Kyle/Glosten–Milgrom models, inventory dynamics, and market impact at an intuitive level. You do not need the full Hasbrouck machinery for Prosperity — a few hundred ticks is not enough data for it anyway — but the vocabulary matters.

3. **Core strategies** (Ch 8–13). Market making and Avellaneda–Stoikov, mean reversion and Ornstein–Uhlenbeck, pairs trading and cointegration, basket/ETF stat arb, momentum, arbitrage zoo. These six chapters are the *meat* of what wins Prosperity rounds.

4. **Options minimum** (Ch 14–17). Just enough to trade the volcanic rock vouchers: option payoffs, Black–Scholes, the Greeks, and delta hedging. You can skip the full vol surface chapter on a first pass and come back to it for Round 5.

5. **Which-strategy-when** (Ch 22). The pattern-recognition chapter is the keystone for Prosperity: given a new product on day one of a round, how do you decide what to do with it?

6. **Prosperity playbook** (Ch 23–26). Round-by-round analysis, the Frankfurt Hedgehogs' 2nd-place writeup as a case study, trader-ID exploitation, and manual trading tricks. This is where all the theory cashes out into leaderboard points.

Roughly 250 pages. Prerequisite maths: comfortable with probability at the level of expected value, variance, and the normal distribution; first-year calculus; and enough linear algebra to invert a 2×2 matrix. No stochastic calculus required.

Track 2 — Goldman Sachs Strats

The GS track is about walking onto a trading floor and not being useless in week one. That means knowing the business (what equities, rates, credit, commodities, FX desks actually trade), the quant machinery (Black–Scholes rigorously, vol surface, Greeks, numerical methods), execution and risk in production (limits, VaR, liquidity risk), and the institutional vocabulary (SecDB, Slang, FVA, Volcker, MiFID). You *do* want the interview and culture material; you do not need the full Prosperity playbook.

1. **Foundations** (Ch 1–3). Markets, price formation, order books. Same as Track 1.

2. **Microstructure** (Ch 5–7). The Kyle, Glosten–Milgrom and market-impact chapters. On a rates or credit desk you will not be quoting in a limit order book, but the information-asymmetry framework is part of how every quant thinks about markets.

3. **Options and volatility** (Ch 14–17). Black–Scholes derived from first principles, the Greeks, delta and gamma hedging, and the full volatility surface (skew, term structure, sticky-delta vs sticky-strike). This is the *minimum* a bank Strat must know.

4. **Execution, risk, and methods** (Ch 18–22). Execution algorithms (VWAP, Almgren–Chriss), slippage, risk management (VaR, stress, Kelly), backtesting hygiene, ML for trading, and the pattern-recognition chapter.

5. **Asset-class depth** (Ch 27–33). Rates and fixed income, credit derivatives, FX, commodities, structured products, and a broad survey of how desks are organised.

6. **GS-specific** (Ch 34–38). Numerical methods in the Strats toolkit, production risk systems, SecDB and Slang, regulatory capital and Volcker, and the interview-problem chapter.

Roughly 400 pages. Prerequisite maths: the whole of Chapter A, including stochastic calculus to the level of Itô's lemma and SDEs. If your probability is rusty, spend a weekend with Grimmett and Stirzaker before Chapter 14.

Track 3 — Full cover-to-cover

The default. Read in order. 635 pages plus appendices. The only reason to pick a track over this one is time. Everything in Tracks 1 and 2 is in here; the extra chapters you pick up on the full path are the ones you will be glad of the *second* time you need a reference.

Three reading tracks at a glance

Track	Chapters	Pages	Maths level
Prosperity	0–17 (core), 22, 23–26	∼250	Prob + calc
GS Strats	1–3, 5–7, 14–22, 27–38	∼400	Prob + stoch calc
Full cover-to-cover	0–38 + App A–C	∼635	All of App A

Prob + calc means you are comfortable with expectation, variance, normal distributions, and single-variable calculus. *Prob + stoch calc* adds Itô's lemma, stochastic differential equations, and risk-neutral pricing at the level of Shreve's book. *All of App A* means all of the above plus the PDE and numerical linear algebra material.

C.2 Annotated bibliography by topic

This section reorganises the references by topic rather than alphabetically. Every source given a `\cite` macro in `conventions.tex` is covered, and every bibliographic entry in `references.bib` is cited at least once in one of the topic clusters below. Each entry lists the source, a one-sentence note on why you would read it, and pointers to the chapters that cite it.

Market microstructure

Bouchaud et al. (2018) — *Trades, Quotes and Prices.* The modern book on limit order book dynamics, written by the Capital Fund Management team; empirically grounded, with chapters on square-root market impact, order flow autocorrelation, and Hawkes processes. Cited in Chs 4–7, 12, 18.

Harris (2003) — *Trading and Exchanges.* The single best "how markets actually work" textbook, aimed at practitioners; no equations, exhaustive on institutional detail (makers, takers, fees, tick sizes, exchange rulebooks). Cited in Chs 1–4, 5, 7.

Hasbrouck (2007) — *Empirical Market Microstructure.* The econometrician's companion to Harris; teaches you how to estimate spread decompositions, VAR models of trade and quote dynamics, and Hasbrouck information shares. Cited in Chs 4, 5, 7.

Foucault et al. (2024) — *Market Liquidity.* Foucault, Pagano and Röell, 2nd edition. Theoretical treatment of liquidity and market design, covering fragmentation, tick-size reforms, and dark pools. Cited in Chs 5, 7, 18.

Roll (1984) — Roll's implicit spread estimator. A beautifully simple paper: estimate the effective bid–ask spread from the serial covariance of returns. Cited in Chapter 5 (Ch 4).

Kyle (1985) — Kyle's continuous auction model. The foundational paper on strategic informed trading and price impact; where "Kyle's lambda" comes from. Cited in Chs 5, 18.

Glosten and Milgrom (1985) — Glosten and Milgrom. The information-asymmetry dealer model: why spreads exist and how market makers protect themselves against adverse selection. Cited in Chapter 6 (Ch 5).

Hawkes (1971) — Hawkes processes. Self-exciting point processes; the modern tool for modelling order-arrival clustering. Cited in Chs 7, 18.

Pelgrims (2020) — Pelgrims, *Matching Engines.* A clear, engineering-focused blog post on how a price-time priority matching engine is actually implemented. Cited in Chapter 2 (Ch 3).

Market making

Avellaneda and Stoikov (2008) — Avellaneda and Stoikov. The paper behind the market-making quoting formula that every modern HFT shop's textbook starts from; inventory-aware bid/ask skew under exponential arrival intensity. Cited in Chs 8, 22.

Ho and Stoll (1981) — Ho and Stoll. The original inventory-based dealer pricing model; A–S is the continuous-time refinement of this. Cited in Chapter 9.

Hummingbot Foundation (2021) — Hummingbot's A–S guide. A practitioner's walk-through of Avellaneda–Stoikov with the parameter intuition laid out in plain English; the best first read before tackling the original paper. Cited in Chs 8, 23.

Mean reversion and pairs trading

Chan (2013) — Chan, *Algorithmic Trading.* A compact, practitioner-focused book with working Python for mean-reversion, momentum, and stat-arb strategies; the single best bridge between academic theory and a deployable algo. Cited in Chs 9–11, 19–22.

Chan (2022) — Chan, *Quantitative Trading.* The earlier, more business-facing Chan book; more on infrastructure, less on strategy details. Cited in Chs 11, 19.

Cartea et al. (2015) — Cartea, Jaimungal and Penalva. The rigorous textbook for execution and market making under stochastic control; includes the full Avellaneda–Stoikov derivation and extensions. Cited in Chs 8, 9, 18.

Basket and statistical arbitrage

Avellaneda and Lee (2010) — Avellaneda and Lee. The canonical PCA-residual stat-arb paper; how to isolate idiosyncratic mean-reverting signals from a large cross-section. Cited in Chapter 12 (Ch 11).

Frankfurt Hedgehogs (2025b) — Frankfurt Hedgehogs, Prosperity 3. The 2nd-place 2025 Prosperity writeup; the clearest public description of how to win the basket and voucher arbitrage rounds. Cited in Chs 11, 23–26.

Frankfurt Hedgehogs (2025a) — Frankfurt Hedgehogs polished algorithm. The companion code repository to the writeup above. Cited in Chs 23–26.

Momentum and trend following

Moskowitz et al. (2012) — Moskowitz, Ooi and Pedersen. The "time-series momentum" paper — the empirical backbone of modern CTA strategies. Cited in Chapter 13 (Ch 12).

Jegadeesh and Titman (1993) — Jegadeesh and Titman. The classic cross-sectional momentum paper; together with MOP it forms the evidence base for systematic trend strategies. Cited in Ch 12.

Daniel and Moskowitz (2016) — Daniel and Moskowitz, *Momentum Crashes*. The complement to MOP: documents when momentum fails catastrophically and why. Cited in Chs 12, 19.

Arbitrage

Damodaran (2012) — Damodaran's options arbitrage notes. A free NYU Stern note that collects put–call parity, conversion/reversal, and box spreads in one accessible document. Cited in Chs 13, 15.

Stoll (1969) — Stoll's put–call parity paper. The original derivation of put–call parity. Cited in Chapter 14 (Ch 13).

Merton (1973) — Merton's rational option pricing. Extends Black–Scholes to a proper no-arbitrage framework; contains the American-option arbitrage bounds. Cited in Chs 13, 14.

Options and Black–Scholes

Hull (2014) — Hull, *Options, Futures, and Other Derivatives*. The industry-standard derivatives textbook; encyclopaedic, readable, and the reference for anything you need to revise before a GS interview. Cited throughout Chs 14–17, 27–32.

Natenberg (2015) — Natenberg, *Option Volatility and Pricing*. The options-trader's bible; everything about how vol actually trades (skew, term structure, events), with almost no equations. Cited in Chs 14–17.

MIT OpenCourseWare (2013) — MIT 18.S096, lectures 15–21. Free MIT OCW lectures that rigorously derive Itô calculus and Black–Scholes from first principles; the best free option if Shreve feels too heavy. Cited in Chs 14, 16, 34.

Black (1976) — Black's 1976 commodity-options paper. Introduces the Black-76 variant used for futures options and interest-rate derivatives. Cited in Chs 28, 30.

Cox et al. (1979) — CRR binomial trees. The discrete-time binomial pricing model; the pedagogical counterpart to Black–Scholes. Cited in Chs 14, 34.

Volatility surface, gamma, and vol trading

Bennett (2014) — Bennett, *Trading Volatility*. A 250-page free PDF from a sell-side vol desk; correlation, term structure, skew, dispersion, variance swaps. Absurdly practical. Cited in Chs 16, 17.

Sinclair (2013) — Sinclair, *Volatility Trading*. The retail-trader's companion to Bennett; smaller, more opinionated, excellent on dispersion and vol arbitrage. Cited in Ch 17.

Dupire (1994) — Dupire's local vol. Derives the unique local volatility surface consistent with observed option prices. Cited in Ch 17.

Derman and Kani (1994) — Derman and Kani. The parallel development of local vol; the "sticky-strike vs sticky-delta" intuition originates here. Cited in Ch 17.

Heston (1993) — **Heston stochastic volatility.** The closed-form stoch-vol model; the baseline for every more complex vol model used on a desk. Cited in Chs 17, 34.

Demeterfi et al. (1999) — **Demeterfi et al., GS QS research note on variance swaps.** The paper that made variance swaps standard; replicate variance with a weighted strip of options. Cited in Chs 17, 34.

Execution

Almgren and Chriss (2000) — **Almgren and Chriss.** The foundational paper on optimal execution under a linear impact model; where VWAP-style schedules come from. Cited in Chs 18, 22.

Almgren et al. (2005) — **Almgren, Thum, Hauptmann, Li.** Empirical calibration of the square-root market-impact law on Citigroup equity data; the paper you cite when a PM asks "how big can I trade?". Cited in Ch 18.

Risk management and Kelly

Thorp (2006) — **Thorp's Kelly survey.** Ed Thorp's own summary of Kelly sizing across blackjack, sports betting, and the stock market; the clearest introduction to fractional-Kelly practice. Cited in Chs 19, 22.

Kelly (1956) — **Kelly's original paper.** The 1956 Bell Labs paper that started it all; short, derives $f^* = p - q/b$ from information-theoretic first principles. Cited in Ch 19.

Lowenstein (2000) — **Lowenstein, *When Genius Failed*.** The LTCM narrative history; essential reading for why leverage and correlation can kill even Nobel laureates. Cited in Chs 19, 35.

Backtesting and machine learning

López de Prado (2018) — **López de Prado, *Advances in Financial Machine Learning*.** The rigorous book on why naive ML for finance fails (leakage, overfitting, non-IID data) and what to do about it (triple-barrier labels, purged CV, fractional differentiation). Cited in Chs 20, 21.

Bailey and Lopez de Prado (2014) — **Bailey and López de Prado, deflated Sharpe.** Shows how Sharpe ratios lie when you select from many trials; essential reading before publishing any backtest. Cited in Chs 20, 21.

Jansen (2020) — **Jansen, *Machine Learning for Algorithmic Trading*.** The 900-page how-to; more code, less rigour than López de Prado; great for feature engineering recipes. Cited in Ch 21.

Chan (2020) — **Chan's conditional parameter optimisation blog.** A short note on adapting strategy parameters to market regimes. Cited in Chs 21, 22.

QuantStart (2016) — **QuantStart HMM regime detection.** A hands-on blog post on fitting Hidden Markov Models to equity returns. Cited in Ch 22.

Rabiner (1989) — **Rabiner's HMM tutorial.** The standard reference for HMMs; forward–backward, Viterbi, Baum–Welch. Cited in Ch 22.

Prosperity writeups and tooling

Frankfurt Hedgehogs (2025b) — **Frankfurt Hedgehogs.** 2nd-place 2025 writeup (see Section C.2); central case study throughout Chs 23–26.

IMC Trading (2026) — **IMC Prosperity 4 wiki.** The primary source of truth for the current competition's rules, mechanics, and round schedule. Cited in Chs 23–26.

Merle (2025a) — **Merle's Prosperity 3 backtester.** An open-source local backtester for Prosperity algorithms; indispensable for iterating between rounds. Cited in Chs 23, 24.

Merle (2025b) — **Merle's Prosperity 3 visualiser.** The companion visualiser; lets you debug why your algorithm got filled where it did. Cited in Ch 23.

Nash (1950) — **Nash bargaining.** The game theory paper behind the Prosperity "Shells from Sea Turtles" manual-trading round. Cited in Ch 26.

Thaler (1988) — **Thaler on the winner's curse.** Explains why sealed-bid auctions systematically overpay; relevant to Prosperity's manual rounds. Cited in Ch 26.

Rates and fixed income

Hull and White (1990) — **Hull and White, 1990.** The short-rate model that is still the front-office workhorse for Bermudan swaption pricing. Cited in Chs 28, 29.

Hagan et al. (2002) — **Hagan, Kumar, Lesniewski, Woodward, *Managing Smile Risk.*** The foundational SABR paper; every rates vol trader knows the asymptotic expansion by heart. Cited in Chs 17, 29.

Brace et al. (1997) — **Brace, Gątarek and Musiela.** The LIBOR Market Model (BGM/LMM) paper; how to price exotics consistent with the full swaption surface. Cited in Ch 29.

Hull and White (2012) — **Hull and White, FVA debate.** The original "funding is not a valuation adjustment" argument; you must know both sides for any Strats interview. Cited in Chs 33, 37.

Credit

Li (2000) — **Li's Gaussian copula.** The paper that industrialised CDO pricing and, two decades later, was blamed for the financial crisis. Cited in Chapter 33 (Ch 29/30).

Salmon (2009) — **Salmon, Wired article.** The "Formula That Killed Wall Street" — an accessible and deliberately provocative retelling of the Li copula story. Cited in Ch 29/30.

FX and commodities

Garman and Kohlhagen (1983) — **Garman and Kohlhagen.** The FX version of Black–Scholes (two rates instead of one). Cited in Ch 30.

Clark (2011) — **Clark, *FX Option Pricing.*** The practitioner's FX vol reference; covers premium-adjusted deltas, smile conventions, and barrier pricing. Cited in Ch 30.

Castagna (2007) — **Castagna, *FX Options and Smile Risk.*** The Vanna–Volga construction for FX smile; the market standard for interpolation. Cited in Ch 30.

Borio et al. (2016) — **BIS, Covered Interest Parity Lost.** Documents the post-2008 breakdown of CIP; essential modern FX reading. Cited in Chs 13, 30.

Schwartz (1997) — Schwartz 1997. The three-factor commodity pricing paper; where mean reversion in convenience yields was established. Cited in Ch 31.

Gibson and Schwartz (1990) — Gibson and Schwartz. The earlier two-factor oil model; foundation of Ch 31's commodity curve material.

Schwartz and Smith (2000) — Schwartz and Smith. The short-term/long-term decomposition used on commodity desks today. Cited in Ch 31.

Numerical methods

Glasserman (2004) — Glasserman, *Monte Carlo Methods in Financial Engineering.* The standard MC reference; variance reduction, LSM for Americans, sensitivities. Cited in Chs 34, 17.

Longstaff and Schwartz (2001) — Longstaff and Schwartz. The LSM regression method for pricing American options by simulation; the standard approach on every rates and equities exotics desk. Cited in Ch 34.

Crank and Nicolson (1947) — Crank and Nicolson. The 1947 paper that gave the finite-difference scheme used for pricing any PDE-based option model. Cited in Chs 34, App A.

Cox et al. (1979) — CRR 1979. See options section; also a cornerstone numerical method.

Interviews

Zhou (2020) — Zhou, the "Green Book". Hundreds of interview problems with full solutions; the book every quant candidate has actually read, and the source of Chapter 39's problem selection. Cited in Ch 38.

GS culture, institutional context, and careers

Derman (2004) — Derman, *My Life as a Quant.* Emanuel Derman's memoir of physics-to-quant; the flavour of a 1990s trading floor, and where the BDT and local-vol models came from. Cited in Chs 17, 27, 36.

Mathematical background

Grimmett and Stirzaker (2001) — Grimmett and Stirzaker. The standard UK undergraduate probability book; measure-theoretic from Ch 5 onwards, with excellent exercises. Cited in Chapter A (App A).

Strang (2016) — Strang, linear algebra. The best introductory linear-algebra text for intuition; pairs with MIT OCW 18.06 lectures. Cited in App A.

Trefethen and Bau (1997) — Trefethen and Bau. The gold-standard book for numerical linear algebra; QR, SVD, and conditioning done properly. Cited in App A, Ch 34.

Øksendal (2003) — Øksendal. The standard first book on SDEs and Itô calculus; readable, complete proofs, exercises. Cited in App A, Chs 14, 16, 34.

Shreve (2004) — Shreve Vol. II. The de facto Masters-in-Financial-Mathematics textbook; rigorous continuous-time mathematical finance. Cited in App A, Chs 14–17, 34.

Pham (2009) — Pham. Stochastic control and HJB equations applied to finance; the heavy machinery behind Avellaneda–Stoikov and Almgren–Chriss. Cited in Chs 8, 18, 34.

Duffy (2006) — Duffy. The practical guide to finite-difference methods for derivatives pricing; Crank–Nicolson, operator splitting, American options. Cited in Ch 34.

C.3 Primary sources that are free online

A non-trivial fraction of the best quant-trading material is free. This section collects the items from the bibliography above that can be read legally without paying for a textbook or a journal subscription. For a self-funded reader this is the shortest path to covering most of the book's territory.

MIT OpenCourseWare — 18.S096 *Topics in Mathematics with Applications in Finance.* Free video lectures and PDF notes from 2013, covering measure theory, Itô calculus, Black–Scholes, stochastic volatility, and portfolio theory. Lectures 15–21 map almost one-to-one onto Chapter 16–Chapter 18 of this book. Available at `https://ocw.mit.edu/courses/18-s096-topics-in-mathematics-with-applications-in-finance-fall-2013/`.

Bennett, *Trading Volatility.* A 250-page sell-side research document distributed free as a PDF at `https://www.trading-volatility.com/`. The single best free reference for equity vol trading. A long-standing mirror also exists via Google Scholar searches for the title.

Rosenthal, UIC lecture slides on market microstructure. Hosted on the University of Illinois Chicago course pages for the Financial Engineering master's programme; each lecture is a self-contained PDF on one microstructure topic (matching engines, spread estimation, market impact).

Damodaran's option arbitrage and investment fables notes. NYU Stern hosts Aswath Damodaran's entire library of teaching materials at `https://pages.stern.nyu.edu/~adamodar/`. The options arbitrage note (Damodaran, 2012) is a particularly clean, free summary of put–call parity and related bounds.

QuantStart, Hummingbot, Letian Wang, and Reasonable Deviations blogs. Practitioner-level tutorials with working code. QuantStart covers backtesting and HMM regime detection (QuantStart, 2016); Hummingbot publishes the best free walk-through of Avellaneda–Stoikov (Hummingbot Foundation, 2021); Letian Wang's blog has a long series on pairs trading and cointegration; Reasonable Deviations summarises López de Prado's *AFML* chapter by chapter. The URLs are in the `.bib` file.

arXiv and SSRN. Most of the cited journal papers (Avellaneda and Stoikov, 2008, Almgren and Chriss, 2000, Avellaneda and Lee, 2010, Bailey and Lopez de Prado, 2014, Moskowitz et al., 2012) are on arXiv or SSRN as author preprints. Search for the paper title; the author's institutional page is usually the first hit and legally hosts a free copy.

Pelgrims, *Matching Engines.* A clear blog-post walk-through of a price-time-priority matching engine's internals (Pelgrims, 2020) at `https://jellepelgrims.com/posts/matching_engines`; the best free companion to Chapter 2.

Prosperity 4 wiki. The official competition wiki (IMC Trading, 2026) is free and is the canonical source for rules, mechanics, and the tutorial-round simulator. The Merle backtester and visualiser (Merle, 2025a; Merle, 2025b) are open-source on GitHub.

MIT OpenCourseWare — 18.06 Linear Algebra (Strang). Free companion lectures to Strang (2016); the best way to rebuild linear-algebra fluency from scratch before Chapters 11, 17, and 34.

C.4 What's not in this book

No single book can cover every part of quantitative finance. This one deliberately stops where depth would cost the reader more than it gave back, or where the material is so specific to a single firm or a single piece of hardware that a textbook would be the wrong vehicle for it. Here are the conscious omissions, with a pointer to where you would go next if you need them.

Microsecond-latency execution engineering. This book covers execution cost models (Almgren and Chriss, 2000, Almgren et al., 2005) and order-book dynamics, but it does not teach you how to build an exchange gateway that quotes at 500 ns. That territory is FPGA design, kernel bypass (Solarflare OpenOnload, DPDK), and colocation-rack physics. Start with Donald MacKenzie's sociology-of-HFT papers, the Optiver/Jane Street engineering blogs, and the talks on the CppCon YouTube channel under "low latency". Nothing on the GS Off-Cycle placement requires this.

Regulatory compliance at operational depth. Chapter 37 covers the capital-regulation frameworks (Basel III, Volcker, Dodd–Frank, MiFID II) at the level a Strat needs to understand a trading restriction. It does not teach you how to *be* the compliance officer who implements them. For that, the Financial Conduct Authority's FCA Handbook and the Office of the Comptroller of the Currency's guidance are the primary sources. Read ICAAP and ILAAP documentation from any large UK bank's pillar 3 disclosures for practical flavour.

Case-by-case firm history beyond LTCM, MG, and Amaranth. The blow-ups covered in Chapter 19/35 are LTCM (Lowenstein, 2000), Metallgesellschaft, and Amaranth, because they illustrate different failure modes (leverage/correlation, hedging-rollover, concentration). Knight Capital 2012, Archegos 2021, and the 2020 WTI-goes-negative episode are all worth reading up on. Gregory Zuckerman's *The Man Who Solved the Market* (on Renaissance) and Patterson's *The Quants* fill in the firm-by-firm picture.

Crypto-specific microstructure. Centralised-exchange crypto (Binance, Coinbase, FTX-before-2022) behaves *almost* like traditional equity markets, with tick sizes and limit order books. Decentralised-exchange (AMM) microstructure — Uniswap v2/v3, constant-product curves, impermanent loss, MEV, sandwich attacks — is genuinely different and is not covered here. The Paradigm Research and a16z crypto research blogs are the right next step.

Specific firm stacks beyond SecDB/Slang. Chapter 36 deliberately drills into SecDB and Slang because they are Goldman-specific and directly relevant to the Off-Cycle role. The equivalent stacks at JP Morgan (Athena/Python), Morgan Stanley (MSS-inhouse), Bank of America (Quartz/Python), Deutsche Bank (gRisk/Java), and Citadel (C++/Python) get at most a sentence. If you move firm, your first two weeks will replace this chapter with an internal wiki.

Tail-risk and catastrophe modelling at insurance depth. VaR and expected shortfall (Chapter 19/35) are as far as this book goes. Full tail-risk modelling — extreme value theory, peaks-over-threshold, copulas for dependence in the tail — is covered in McNeil, Frey and Embrechts, *Quantitative Risk Management*, which is the standard next step.

Behavioural finance beyond a single Thaler reference. The winner's curse (Thaler, 1988) gets a citation; prospect theory, herding, disposition effect, and the full Kahneman–Tversky programme do not. Kahneman's *Thinking, Fast and Slow* is the accessible starting point.

Don't treat this book as comprehensive

This book is a *starting point* for two specific goals: the IMC Prosperity competition and a Goldman Sachs Off-Cycle placement. It is not an encyclopaedia of quantitative finance and it does not try to be. Two failure modes to avoid:

1. *Assuming that what's in here is all there is.* If you reach a production desk and find yourself saying "the textbook didn't mention FVA/XVA/cross-currency basis/repo squeezes", that is the textbook working as intended. The next step is the pillar 3 disclosures, Hull's Chapter 33 on XVAs, and your desk's internal documentation — none of which would fit in a single book.

2. *Assuming that a cited result is still current.* Market microstructure papers from the 1980s (Kyle, 1985, Glosten and Milgrom, 1985) remain conceptually correct but describe a world of floor-traded specialists; their empirical magnitudes (spread sizes, impact coefficients) are out of date by orders of magnitude. Always cross-check with Bouchaud et al. (2018) or the latest BIS Quarterly Review for modern numbers.

When in doubt, follow the citation trail. Every topic in this book has at least one primary source listed in the bibliography above; the primary source will always be deeper, and usually more current, than the summary offered here. The point of this book is to make that primary source readable — not to replace it.

That is the end of the book. If you have reached this page having worked through one of the three reading orders above, you are now at the level where the bibliography, not the chapters, is the next step. Good luck in Prosperity, and good luck on the desk.

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